TrioDocs

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https://triodocs.org

What is Trio?

What is Trio?

Trio is an open-source automated insulin delivery system (OS-AID) for iOS based on the OpenAPS algorithm with adaptations for Trio.

This system considers your user-entered settings, carbohydrates, and historical insulin use to automate insulin delivery to reduce the time spent managing your diabetes.

Before starting with Trio, you should evaluate all your closed-loop options, including:

- Commercial systems
- Other open-source applications
  - Loop
  - AndroidAPS

🚧 Documentation Under Construction 🚧

This documentation is being released while still under construction. Please keep in mind the following information:

- Any page with a ✏️ indicates the information has not yet been updated. The information on those pages should not be referenced at this time.
- We welcome assistance with proofreading, editing, and updating the information in this documentation. If you would like to contribute to the documentation, please submit issues or PRs here
- New educational videos are coming soon.

To use Trio, you are required to build the application from the source code. This does not require substantial technical know-how but is a time-consuming process. You may need several sessions to complete your first build.

Upon installation, you will follow an onboarding wizard in the app to configure your settings optimally. We strongly recommend using the New User Setup Guide as you walk through onboarding so you are able to easily reference any additional information you may need.

If you find yourself stuck, the community is here to help on Discord or Facebook.

Contribution

Trio is built by a volunteer community. If you are interested in helping as a programmer, you can contribute to the Trio or Trio-Oref code base. Trio is written in Swift, and the Trio-Oref/OpenAPS algorithm is currently written in JS and is in the process of being converted into Swift.

You can also provide support in online support groups by sharing your own success and troubleshooting common errors.

Usage

Welcome to the Usage homepage! Here you will find a variety of topics to learn about that will help you in using Trio to manage your diabetes.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

User Interface

A detailed look at Trio's user interface and features

Core Concepts

Information on Basal Rates, ISF, and CR

Features

Information on Overrides, Temp Targets, Statistics, Bolus Calculator, and Using Dynamic ISF

User Interface

Trio User Interface

Light Mode

Dark Mode

Main Screen

The main screen of Trio can be divided into four sections:

Use the links below to learn more about each section:
Current Status | Graph & History | Adjustment Status | Action Buttons

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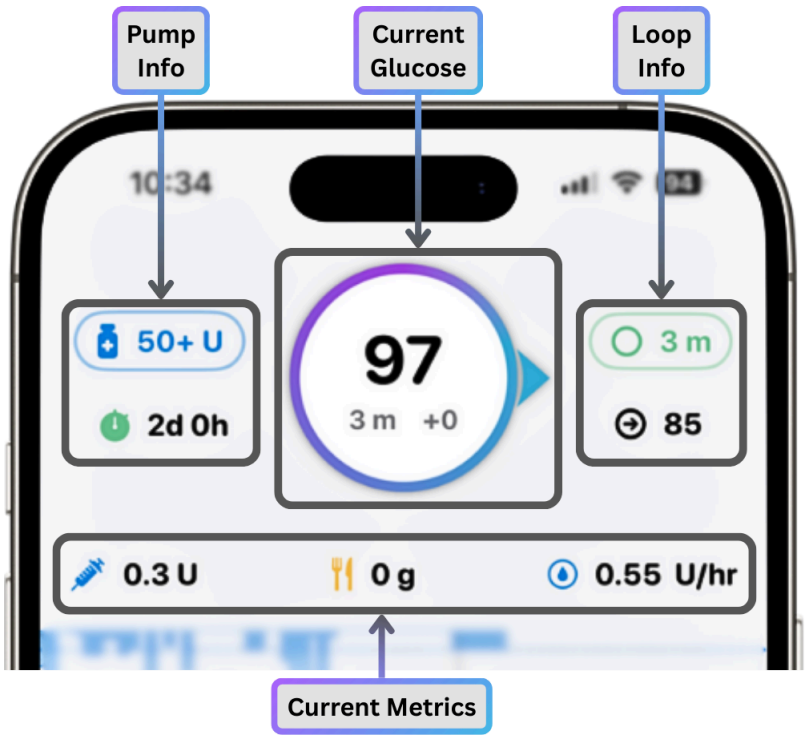
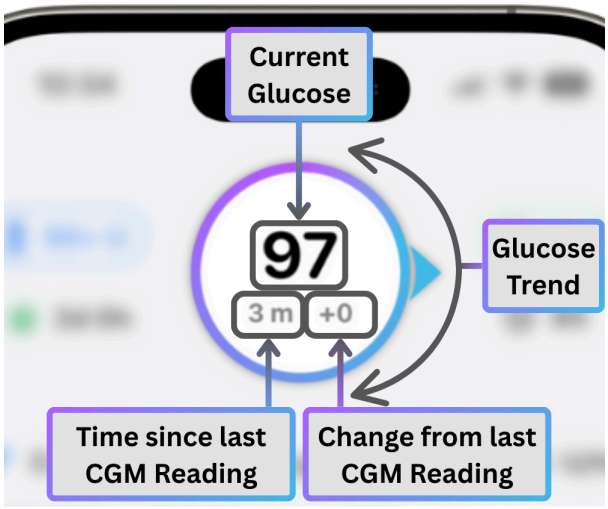
Current Status

The Current Status section functions as an easy reference point for a variety of important data points.
Use the tabs below to learn more about each section:



Current Glucose

Here you will find a centered bobble containing your current glucose reading, the time since the last glucose reading, the change in glucose from the previous reading, and an arrow that indicates the trending direction of your glucose.



Pro Tip

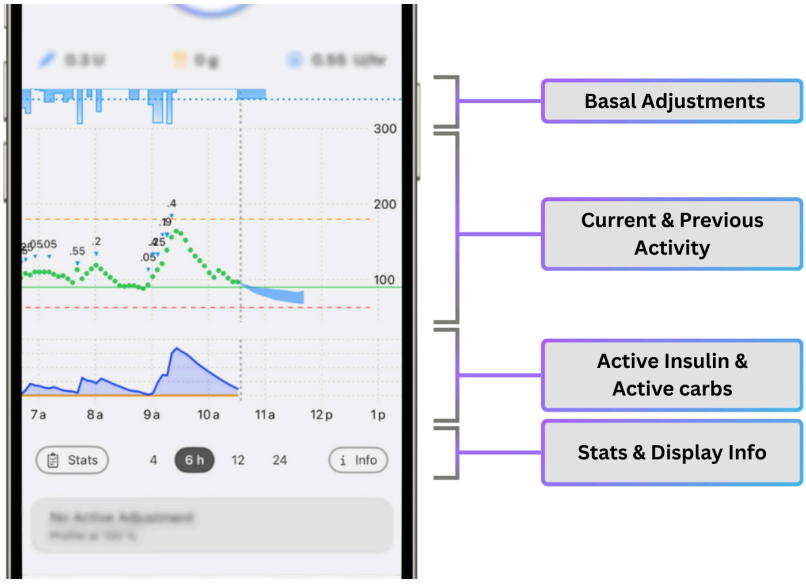
If you long press the bobble, it will snooze CGM alerts

Pump Info

Omnipod Users

Graph and History

The middle of the screen graphs your blood sugar data and entered carbs overlaid with Trio's insulin delivery and glucose forecasting.
Use the tabs below the image to learn more about each section.



Basal Adjustments

Here you will find the adjustments made to your basal delivery. See below for more details:

Therapy Basal Reference Line: The rate you entered in settings. (In Trio 0.2.x, this was called "Profile Basal Rate")

No Basal Adjustment Basal is being delivered at your current therapy basal rate.

Basal Increase Basal increased above your current therapy basal rate.

Basal Decrease Basal decreased below your current therapy basal rate.

Current & Previous Activity

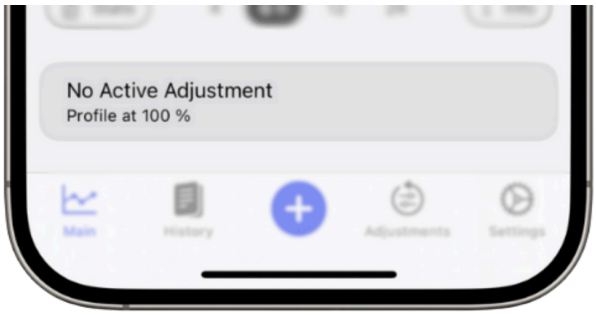
This section shows a combined view of your previous, current, and forecasted activity.

Cone of Uncertainty

Adjustment Status

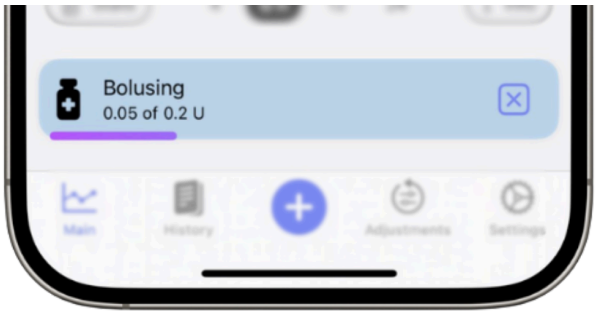
Under the graph you'll find the current status of any adjustment currently active. Adjustments include boluses in progress, temp targets, and overrides.
Use the tabs below to learn more:

No Active Adjustment



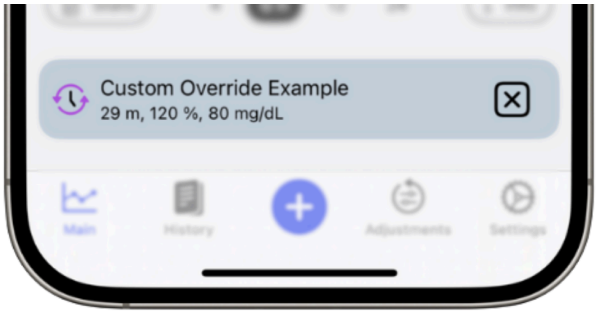
- This is the default view when there is no bolus-in-progress, active override, or active temp target

Bolus in Progress



- Shows the status of current bolus
- Press the [X] to cancel the current bolus

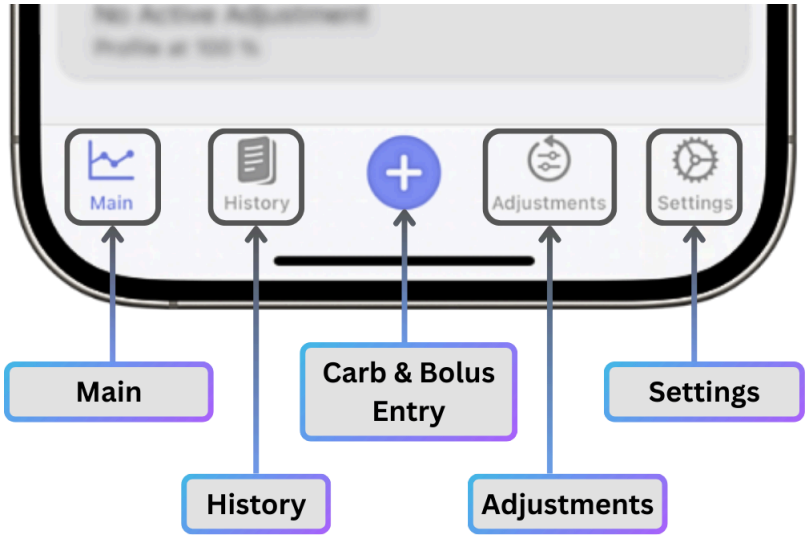
Active Override



- Shows details of the current override running
- Press the [X] to cancel the override

Action Buttons

At the bottom of the screen, you have five action buttons. Use the tabs below to learn more about each section:



Main

This button takes you back to the home screen no matter where you are in the app.

History

This button brings up the History screen. Here you will find previous treatments, meal entries, glucose readings & manual glucose entries, and previous use of adjustments. Here you can add a finger stick glucose entry by tapping the "+" at the top right.

More information about each tab:

Treatments

- Scroll to see previously enacted Temp Basals, Boluses, and SMBs
- Swipe left to delete inaccurate bolus and SMB entries. You cannot delete Temp Basal entries

Meals

- Scroll to see previously entered carb, fat, and protein entries
- Swipe left to edit or delete inaccurate entries
- Tap "Show Future" or "Hide Future" to expose and remove future carb entries (This does not delete them, only allows you to see or not see them)

Glucose

- Scroll to see previous glucose readings

Core Concepts

Welcome to the Core Concepts homepage! Here you will find a variety of topics to learn about that will help you in using Trio to manage your diabetes.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Basal Rates

How Trio uses your *basal rates* to manage your continuous delivery throughout the day.

Testing Basal Rates

How to test your basal rates

Insulin Sensitivities

How *Insulin Sensitivities* adjust your insulin based on your glucose levels.

Testing ISF

How to test your Insulin Sensitivity Factor (ISF)

Carb Ratios

How *carb ratios* determine the insulin you need for the carbs you eat.

Testing CR

How to test your Carb Ratio (CR)

Basal Rates

Highlights

- Basal rates define your background insulin delivery throughout the day
- Different basal rates can be set for different times of day (up to 48 entries with 30-minute intervals)
- Trio uses your basal profile as a baseline but adjusts it every loop cycle based on your current glucose
- Accurate basal rates are crucial for correct insulin on board (IOB) calculations

What Are Basal Rates?

Basal rates are the foundational insulin delivery settings in Trio. They represent the amount of rapid-acting insulin delivered continuously throughout the day to manage your glucose levels when you're not eating. Think of basal insulin as the background insulin your body needs to maintain steady glucose levels during fasting periods.

Basal rates are measured in **units per hour (U/hr)**. For example, a basal rate of 0.8 U/hr means you receive 0.8 units of insulin spread out over the course of that hour.

Why Basal Rates Matter

Basal is Key

Basal rates are the foundation of the Trio algorithm. Before testing any other core settings, ensure your basal rates are set appropriately.

Your basal rates serve several critical functions in Trio:

1. **Baseline for Algorithm Adjustments**: Trio uses your scheduled basal profile as a starting point, then adjusts it every loop cycle based on your current glucose, trend, and other factors. Your basal rate should cover what your body needs during times of fasting and low activity.
2. **Insulin on Board (IOB) Calculations**: Your scheduled basal rate is treated as the "zero point" for IOB. When Trio delivers a temporary basal rate higher than your scheduled rate, your IOB increases. When it delivers a lower temporary basal rate, your IOB decreases (and can even become negative).
3. **Foundation for Autosens and Dynamic ISF**: Autosens and Dynamic ISF modify your basal rates (along with ISF and targets) based on your sensitivity ratio.
4. **Total Daily Dose (TDD) Calculations**: Your basal insulin contributes to your TDD, which Trio uses for Dynamic ISF and other algorithm features.
5. **Basal Rates as a Safety Net**: If your pump has recieved no communication from Trio because your phone is out of range or otherwise unavailable, it will revert back to your programmed basal rates after the last temp basal expires. Your profile basal rates are stored on your pump. This ensures insulin delivery is present even when your phone is out of contact.

How Trio Uses Basal Rates

> **Basals are the Foundation**
>
> Trio uses your basal profile as the baseline of your insulin needs. When SMBs are enabled, you may see a wide variation in the temp basal rates set. *This is normal.* This is how Trio operates. It is including your baseline basal rates in the full insulin required calculation and giving that basal through SMBs and/or temp basal rates.

Every 5 minutes, Trio's algorithm:

1. Reads your current scheduled basal rate
2. Applies Autosens adjustments
3. Calculates the optimal temporary basal rate based on:
    - Current glucose level
    - Glucose trend
    - Insulin on board (IOB)
    - Carbs on board (COB)
    - Target glucose
    - SMB Settings
4. Sets a temporary basal rate (within your safety limits) and/or delivers an SMB

This means your actual insulin delivery will vary throughout the day, even if you're fasting, as Trio responds to your glucose patterns in real time.

Time-Based Basal Schedules

Because insulin needs vary throughout the day due to circadian rhythms, hormone fluctuations, and activity patterns, Trio supports **time-based basal schedules**:

- You can define different basal rates for different times of day
- Time slots are in 30-minute increments
- Maximum of 48 entries (covering 24 hours)
- The first entry must start at 00:00 (midnight)
- Each rate applies from its start time until the next scheduled rate

Example Basal Schedule:

Time	Basal Rate (U/hr)	Why It Might Vary
00:00	0.6	Lower overnight needs

03:00	0.9	Dawn phenomenon (early morning glucose rise)
09:00	0.7	Daytime baseline
17:00	0.8	Evening increase

Basal Rate Ranges

The allowed basal rate range depends on your insulin pump:

Pump Type	Minimum (U/hr)	Maximum (U/hr)	Increment (U/hr)
Omnipod 5 & DASH	0.0	30.0	0.05
Omnipod Eros	0.05	30.0	0.05
Dana	0.0	3.0	0.05
Medtrum Nano	0.0	25.0	0.05
MiniMed x15 & x22	0.0	35.0	0.05 (0-10) 0.1 (10-35)
MiniMed x23 & x54	0.0	35.0	0.025 (0-1) 0.05 (1-10) 0.1 (10-35)

Max Basal Safety Limit

In addition to your pump's maximum, Trio has a **Max Basal** safety setting (default: 2.0 U/hr) that prevents the algorithm from setting temporary basal rates above this value. Make sure your Max Basal setting is appropriate for your needs.

Testing Your Basal Rates

Accurate basal rates are essential for Trio to function optimally. Here are two methods to verify and adjust your basal rates:

1. Fasting Tests

The traditional method for testing basal rates:

1. Fast for 4-6 hours (no food at least 2 hours before starting)

2. Avoid exercise during the test
3. Monitor your glucose:
    - **Steady glucose**: Basal rate is accurate
    - **Rising glucose**: Basal rate is too low (increase it)
    - **Falling glucose**: Basal rate is too high (decrease it)

Test different times of day separately, as your basal needs vary throughout the day.

2. Monitor Your IOB Patterns

Review your insulin on board (IOB) throughout the day when you're not eating and have no residual COB:

- **Consistently negative IOB** at certain times → Basal rate may be too high for those hours
- **Consistently positive IOB** at certain times → Basal rate may be too low for those hours
- **IOB hovering around zero** → Basal rate is appropriate

> **Important**
>
> This general rule does not apply when food is consumed. Basal testing and IOB evaluation can only be completed during times of fasting.

Starting Point for New Users

If you're coming from pump therapy, you can start with your pump's basal settings, but be prepared to adjust them:

- Trio may require different basal rates than traditional pump therapy
- Monitor your glucose patterns during fasting periods
- Make gradual adjustments (typically 10-20% changes)
- Wait several days between adjustments to see the full effect

If you don't have existing basal rates, work with your healthcare provider to establish safe starting values based on your total daily insulin requirements.

Relationship to Other Settings

Your basal rates interact with other Trio settings:

- **Autosens**: Multiplies your basal rates by the sensitivity ratio (default range: 70-120%)
- **Dynamic Settings**: When "Adjust Basal" is enabled, Dynamic ISF adjustments also affect basal needs
- **SMB (Super Micro Bolus)**: SMB calculations reference your current basal rate for SMB limits (e.g., "50% of current basal rate")
- **Total Daily Dose**: Basal insulin is included in your TDD calculation along with bolus insulin

Summary

Basal rates are your baseline insulin delivery schedule that Trio uses as a foundation for its automated insulin delivery. While Trio adjusts these rates dynamically every loop cycle, having accurate scheduled basal rates ensures:

- Correct IOB calculations
- Effective Autosens adjustments
- Appropriate algorithm behavior
- Safe insulin delivery within your physiological needs
- Accurate safety net is in place when Trio cannot communicate with pump

Take the time to test and refine your basal rates. They're one of the most important settings for successful automated insulin delivery.

Insulin Sensitivity Factor (ISF)

Highlights

- ISF determines how much one unit of insulin lowers your glucose
- It's the most common setting that needs adjustment for new Trio users
- A lower ISF number means you're more resistant to insulin (need more insulin)
- A higher ISF number means you're more sensitive to insulin (need less insulin)
- ISF can vary throughout the day and is set as a time-based schedule

What Is ISF?

Insulin Sensitivity Factor (ISF), also called Insulin Correction Factor (ICF), is the amount your blood glucose drops when you receive one unit of insulin. It's expressed as:

- **mg/dL per unit (mg/dL/U)**
- **mmol/L per unit (mmol/L/U)**

For example, an ISF of 50 mg/dL/U means that 1 unit of insulin will lower your glucose by 50 mg/dL (2.8 mmol/L).

Example Calculation

Bill has an ISF of 50 mg/dL/U. His current glucose is 250 mg/dL and his target is 100 mg/dL. How many units of insulin does Bill need to reach his target?

Here is the formula:

$$\frac{Current\ Glucose - Target\ Glucose}{ISF}$$

Calculate Bill's Insulin Dose:

$$\frac{250 - 100}{50} = \frac{150}{50} = 3\ units$$

Answer

Bill needs 3 units of insulin to reach his target glucose of 100 mg/dL.

Why ISF Matters

ISF is one of the most critical settings in Trio because it determines:

1. **SMB and Temp Basal Delivery**: How much insulin is needed to bring high glucose back to target or keep glucose at target
2. **Algorithm Calculations**: Trio uses ISF every loop cycle to calculate optimal insulin delivery
3. **Glucose Predictions**: ISF affects how Trio predicts future glucose levels

> ⚠ **Common Issue for New Users**
>
> Incorrect ISF is the #1 cause of problems when starting Trio. If you experience:
>
> - Frequent lows after corrections
> - Rollercoaster glucose patterns
> - Too much or too little insulin from SMB/UAM
>
> Your ISF likely needs adjustment.

How Trio Uses ISF

Like basal rates, Trio does **not** use your ISF verbatim. Instead, your scheduled ISF is modified dynamically based on:

1. Autosens Adjustments

When Autosens is enabled (which it is by default), Trio analyzes 8-24 hours of glucose data to calculate a Sensitivity Ratio. This ratio adjusts your ISF:

- **Sensitivity Ratio > 1.0**: You're more insulin-resistant → ISF is decreased (more insulin)
- **Sensitivity Ratio < 1.0**: You're more insulin-sensitive → ISF is increased (less insulin)
- **Limits**: Default range is 70-120% (controlled by Autosens Min/Max settings)

Example:

Bill has an ISF of 50 mg/dL/U. Autosens detects resistance and has set his sensitivity ratio to 1.15 (115%). What will Bill's new ISF be?

Here is the formula:

$$\frac{Current\ ISF}{Autosens\ Ratio}$$

Calculate Bill's new ISF:

$$\frac{50}{1.15} = 43.5 mg/dL/U$$

Answer

Bill's ISF is adjusted from 50 mg/dL/U to 43.5 mg/dL/U. Bill requires 15% more insulin.

2. Dynamic ISF (Optional)

Dynamic ISF is a more aggressive way to adjust your ISF based on current glucose and/or historical insulin dosing:

- **Requires**: 7 days of TDD data with 85% coverage (minimum 1,715 data points)
- **Types**: Logarithmic (default) or Sigmoid
- **Output**: Dynamic Ratio that replaces the Autosens Ratio each loop cycle
- **Temporary**: The adjustment changes every loop cycle; it's not a permanent ISF change

Info

There are 2 algorithms available for Dynamic ISF. Logarithmic Dynamic ISF adjusts your ISF based on current glucose levels and Total Daily Dose (TDD). Sigmoid Dynamic ISF adjusts your profile ISF based on how far from your target glucose you currently are.

Key Parameters:

- **Adjustment Factor**:
  - **Sigmoid**: Adjustment Factor controls how aggressively ISF changes. Learn more about how changes to Sigmoid Adjustment Factor influences ISF adjustments here. The default for Sigmoid Adjustment Factor is 50%.
  - **Logarithmic**: Controls what ISF is used at which glucose. It should be manually set to approximately $0.02 \times TDD$. Learn more about how changing Adjustment Factor influences Logarithmic dynamic ISF here.

- **Autosens Maximum**: Controls the upper limit of the autosens ratio (default: 120%)
- **Autosens Minimum**: Controls the lower limit of the autosens ratio (default: 70%)
- **Weighted Average of TDD**: Blends recent (24h) and historical (10-day) insulin use (default: 35%)

Dynamic ISF is designed to make your ISF more aggressive (lower) when glucose is high and less aggressive (higher) when glucose is near target.

Time-Based ISF Schedules

Your insulin sensitivity varies throughout the day due to hormones, activity, and circadian rhythms. Trio supports time-based ISF schedules:

- Time slots in 30-minute increments
- Maximum of 48 entries (covering 24 hours)
- First entry must start at 00:00 (midnight)
- Each ISF applies from its start time until the next entry

Example ISF Schedule:

Time	ISF (mg/dL/U)	Why It Might Vary
00:00	60	Overnight baseline
03:00	45	Dawn phenomenon (more resistant)
09:00	55	Morning sensitivity increase
17:00	50	Evening decrease

ISF Value Ranges

- **Minimum**: 9 mg/dL/U (0.5 mmol/L/U)
- **Maximum**: 540 mg/dL/U (30 mmol/L/U)
- **Default**: 200 mg/dL/U (11.1 mmol/L/U)
- **Step**: 1 mg/dL/U (0.1 mmol/L/U)

> **Units**
>
> Trio stores ISF internally in mg/dL but displays it based on your glucose unit preference. Conversions are automatic.

Testing and Adjusting Your ISF

Starting Point

If you're coming from pump therapy, you can transfer your pump's ISF as a starting point. However, be prepared to adjust it based on Trio's behavior.

Testing Method

To test if your ISF is accurate:

1. **Choose a testing time**: When you're "stuck" above target (around 150 mg/dL or 8.3 mmol/L) or by bringing yourself to a higher glucose with glucose tabs
2. **Prepare Trio for testing**: Set Max IOB to 0 (*Optional*: Set Autosens Max and Min to 100% to avoid Autosens adjustments)
3. **Give a correction bolus** based on your current ISF
    - To determine your testing dose, follow the formula $\frac{Current\ Glucose - Target\ Glucose}{ISF}$
4. **Wait 4 hours** without eating or exercising
5. **Evaluate the result**:
    - **Still above target after 4 hours**: ISF is too high (too conservative) → DECREASE the ISF value
    - **Below target after 4 hours**: ISF is too low (too aggressive) → INCREASE the ISF value
    - **At target after 4 hours**: ISF is accurate
6. **Return Max IOB to your previous setting** after you've completed the test

> **Understanding ISF Adjustments**
>
> - **Lower ISF number** = More insulin per unit of glucose correction = More aggressive
> - **Higher ISF number** = Less insulin per unit of glucose correction = Less conservative
>
> This can be counterintuitive at first!

Relationship to Other Settings

ISF interacts with many Trio settings:

Autosens Integration

- **Autosens Min/Max**: Limits how much Autosens can adjust your ISF (default: 70-120%)
- **Sensitivity Ratio**: Applied to ISF and basal rates simultaneously

Dynamic ISF Integration

- **Replaces Autosens**: When Dynamic ISF is enabled, it calculates a Dynamic Ratio instead of using the Autosens Ratio
- **Bounded by Autosens Limits**: Dynamic Ratio is still constrained by Autosens Min/Max settings
- **Affects Basal** (optional): When "Adjust Basal" is enabled, Dynamic Ratio also modifies the baseline basal rate
- **Learn how your Profile ISF is used in Dynamic ISF by reviewing the page on Using Dynamic ISF

Temp Target Behavior (when enabled)

- **High Temp Targets** (>100 mg/dL): Increases effective ISF (less aggressive insulin delivery)
- **Low Temp Targets** (<100 mg/dL): Decreases effective ISF (more aggressive insulin delivery)
- **Half Basal Exercise Target**: Default 160 mg/dL; affects ISF scaling during exercise targets
- Temp Target Behavior settings will replace the Autosens or Dynamic Ratio calculations when enabled.

Sensitivity Target Behavior

- Adjusts target glucose based on the Sensitivity Ratio calculated by the algorithm
- `Sensitivity Raises Target` Increases your target glucose when Sensitivity Ratio is calculated less than 100%.
- `Resistance Lowers Target` decreases your target glucose when Sensitivity Ratio is calculated greater than 100%.

Bolus Calculator

ISF is used in the bolus calculator formula:

$$Total\ Insulin = \frac{Current\ BG - Target\ BG}{ISF} + \frac{Carbs}{CR} + Trend\ Correction - IOB$$

Carbs on Board (COB) Calculations

ISF is used to determine how quickly carbs are absorbed in your body and how fast Trio reduces the COB amount.

- ISF is used to determine your Carb Sensitivity Factor (CSF)
  - **Formula**: $CSF = \frac{Carb\ Ratio}{ISF}$
  - CSF shows how much 1g of carbohydrate will raise your glucose
- Carb Sensitivity Factor is used to determine how many carbs have been absorbed during that loop cycle
  - **Formula**: $Increase\ in\ glucose \times CSF$
  - Your COB will be reduced by this amount*

*If no change is indicated in your glucose readings, your `Min 5m Carb Impact` setting will be used for COB determination

Common ISF Mistakes

1. ISF Too Low (Too Aggressive)

Symptoms:

- Frequent hypoglycemia after corrections
- Rollercoaster patterns (high → over-correction → low → high)
- Excessive SMB delivery resulting in lows

Solution:

- **Test your ISF**
- **Autosens or Sigmoid Dynamic ISF**: Increase your profile ISF value (make it less aggressive)
- **Logarithmic Dynamic ISF**: Re-evaluate your Adjustment Factor and other Dynamic settings using the desmos graphs

2. ISF Too High (Too Conservative)

Symptoms:

- Persistent high glucose despite corrections
- Slow return to target
- Insufficient SMB delivery

Solution:

- **Test your ISF**
- **Autosens or Sigmoid Dynamic ISF**: Decrease your profile ISF value (make it more aggressive)
- **Logarithmic Dynamic ISF**: Re-evaluate your Adjustment Factor and other Dynamic Settings using the desmos graphs

3. Ignoring Time-of-Day Variations

Symptoms: - Good control at some times of day, poor at others - Dawn phenomenon not adequately addressed

Solution: Set different ISF values for different times of day based on need

Best Practices

1. **Start conservatively**: Begin with a higher (less aggressive) ISF and adjust downward if needed

2. **Make small changes**: Adjust by 5-10% at a time
3. **Wait between changes**: Give each adjustment 2-3 days to see the full effect
4. **Test systematically**: Test one time period at a time
5. **Consider patterns**: Look for consistent trends over multiple days, not single events
6. **Monitor Autosens Ratio**: Watching how your Autosens Ratio changes or doesn't change can indicate an issue with your Dynamic ISF settings
7. **Work with your healthcare provider**: Especially when making significant changes

Summary

ISF is a foundational setting that determines how much insulin is needed to correct high glucose. Getting your ISF right is crucial for Trio to function effectively. While Trio adjusts your ISF dynamically through Autosens or Dynamic ISF, having accurate baseline ISF values ensures:

- Safe and effective corrections
- Appropriate SMB delivery
- Smooth glucose control without rollercoasters
- Effective algorithm predictions and decisions

Take the time to test and refine your ISF carefully. It's one of the most important settings for successful automated insulin delivery.

Carbohydrate Ratios (CR)

Highlights

- Carb ratio (CR) tells you how many grams of carbs one unit of insulin covers
- Unlike basal rates and ISF, CR is used more directly by Trio's algorithm
- A lower CR number means you need more insulin per gram of carbs (more insulin-resistant)
- A higher CR number means you need less insulin per gram of carbs (more insulin-sensitive)
- CR can vary throughout the day and is set as a time-based schedule

What Is Carb Ratio?

Carbohydrate Ratio (CR), also written as Insulin-to-Carb Ratio (I:C or ICR), is the amount of carbohydrates (in grams) that one unit of rapid-acting insulin can "cover" or neutralize. It's expressed as:

grams per unit (g/U)

For example, a CR of 10 g/U means that 1 unit of insulin will cover 10 grams of carbohydrates.

Example Calculation

Bill has a CR of 10 g/U. If Bill eats a meal containing 50 grams of carbs, how much insulin does Bill need?

Here is the formula:

$$\frac{grams\ of\ carb\ eaten}{Carb\ Ratio}$$

Calculate Bill's Insulin Dose:

$$\frac{50}{10} = 5\ units$$

Answer

Bill needs 5 units of insulin for his 50g meal.

Why Carb Ratio Matters

CR is essential for Trio because it determines:

1. **Carbs on Board (COB) Corrections**: How much insulin is needed for unabsorbed carbs still affecting your glucose
2. **Bolus Calculator Recommendations**: The carb coverage component of total insulin needed
3. **Automatic Bolusing**: When SMB (Super Micro Bolus) is enabled, CR affects insulin delivery for meals
4. **Carbs on Board (COB) Calculations**: Combined with your current ISF, your CR is used to calculate how many carbs have been absorbed and no longer need to be accounted for in corrections

> **Direct Usage**
>
> Unlike basal rates and ISF, which are modified by Autosens or Dynamic ISF, **carb ratio is used directly** by Trio's algorithm.

How Trio Uses Carb Ratio

In the Bolus Calculator

Trio's bolus calculator uses your CR as part of the total insulin dose calculation:

$$Total\ Insulin = \frac{Current\ BG - Target}{ISF} + \frac{Carbs}{CR} + Trend - IOB$$

The carb coverage portion specifically uses:

$$Carb\ Insulin = \frac{Carbs\ Entered + Current\ COB}{Carb\ Ratio}$$

Where:

- **Carbs Entered**: New carbs you're adding in the treatment screen
- **Current COB**: Carbs on board from previous meals still being absorbed
- **Max COB Limit**: Default 120g; prevents excessive insulin when carbs are not yet digested.

Time-Based CR Schedules

Your carbohydrate sensitivity can vary throughout the day due to hormones, activity levels, and circadian rhythms. Trio supports time-based CR schedules:

- Time slots in 30-minute increments
- Maximum of 48 entries (covering 24 hours)

- First entry must start at 00:00 (midnight)
- Each CR applies from its start time until the next entry

Example CR Schedule:

Time	CR (g/U)	Why It Might Vary
00:00	12	Overnight - more sensitive
06:00	8	Morning - dawn phenomenon, more resistant
12:00	10	Midday baseline
18:00	9	Evening - slightly more resistant

CR Value Ranges

- **Minimum**: 1.0 g/U
- **Maximum**: 50.0 g/U
- **Default**: 30.0 g/U
- **Step**: 0.1 g/U

The available values range from 1.0 to 50.0 in 0.1 increments, giving you fine-grained control over your carb ratios.

Testing and Adjusting Your CR

Starting Point

If you're coming from pump therapy, you can transfer your pump's CR as a starting point. However, verify its accuracy through testing, especially if you experience:

- High glucose peaks after meals
- Lows 3-4 hours after meals
- Sharp drops after carbs when SMB is enabled (may also indicate ISF issues)

Testing Method: Meal Experiment

The standard way to test your CR is a controlled meal experiment:

1. **Choose testing conditions**:
    - Start with glucose in range (near target)

- Zero or minimal IOB
- Zero or minimal COB from previous meals
- No recent exercise
- Use a meal with known, accurately counted carbs

2. **Perform the meal bolus**:
   - Calculate insulin: $\frac{Carbs}{Current\ CR}$
   - Deliver the full bolus manually
3. **Monitor for 3-4 hours**:
   - Avoid additional food or corrections during this time
   - Track your glucose pattern
4. **Evaluate the result**:
   - **Return to starting glucose**: CR is accurate
   - **End up high**: CR is too high (too conservative) → DECREASE the CR value
   - **End up low**: CR is too low (too aggressive) → INCREASE the CR value

> **Understanding CR Adjustments**
>
> - **Lower CR number** = More insulin per gram of carbs = More aggressive
> - **Higher CR number** = Less insulin per gram of carbs = Less conservative
>
> Example: Changing from CR 10 to CR 8 means you'll get MORE insulin for the same carbs.

Common CR Mistakes

1. CR Too Low (Too Aggressive)

Symptoms: - Post-meal glucose drops too quickly - Hypoglycemia 2-3 hours after meals - Need to eat snacks to prevent lows after normal-sized meals

Solution: Increase your CR value (make it less aggressive)

2. CR Too High (Too Conservative)

Symptoms: - Persistent high glucose after meals - Large glucose spikes that take hours to come down - Need frequent corrections after eating

Solution: Decrease your CR value (make it more aggressive)

3. Confusing CR with ISF Issues

Important: High post-meal glucose can be caused by incorrect CR, incorrect ISF, or a combination of both:

- **CR problem**: Affects the initial meal bolus and SMBs when COB > 0
- **ISF problem**: Affects corrections and SMB once COB = 0

If you're unsure which setting needs adjustment:

- Test CR with controlled meal experiments (as described above)
- Test ISF with correction boluses when fasting (no COB)

4. Not Accounting for Meal Composition

Your actual carb coverage needs can vary based on:

- **Fat content**: High-fat meals may need reduced upfront insulin (Reduced Bolus Option)
- **Protein content**: Large amounts of protein may require additional coverage
- **Glycemic index**: Fast-absorbing carbs vs. slow-absorbing carbs
  - Fast-absorbing: Utilize the Super Bolus Option
  - Slow-Absorbing: Utilize the Reduced Bolus Option

These factors affect meal timing and absorption, not necessarily your CR itself.

Relationship to Other Settings

COB (Carbs on Board)

- CR is used to calculate how much insulin is needed for unabsorbed carbs
  - **Formula**: $\frac{(Current\ COB + New\ Carbs)}{Carb\ Ratio}$
  - Max COB setting (default: 120g) prevents excessive insulin from carb stacking
- CR is used to determine your Carb Sensitivity Factor (CSF)
  - **Formula**: $CSF = \frac{Carb\ Ratio}{ISF}$
  - CSF shows how much 1g of carbohydrate will raise your glucose
- Carb Sensitivity Factor, derived from CR, is used to determine how many carbs have been absorbed during that loop cycle
  - **Formula**: $Increase\ in\ glucose \times CSF$
  - Your COB will be reduced by this amount*

*If no change is indicated in your glucose readings, your `Min 5m Carb Impact` setting will be used for COB determination

---

Best Practices

1. **Start conservatively**: Begin with a higher (less aggressive) CR and adjust downward if needed
2. **Make small changes**: Adjust by 1-2 g/U at a time
3. **Test one meal at a time**: Don't change multiple CR entries simultaneously
4. **Wait between changes**: Give each adjustment several days and multiple meals to evaluate
5. **Track Carbs during testing**: Track meals, carb counts, boluses, and results
6. **Account for accuracy**: Carb counting accuracy matters when testing CR

CR vs. ISF: Key Differences

Understanding the difference between these two settings helps you adjust the right one:

Aspect	Carb Ratio (CR)	Insulin Sensitivity Factor (ISF)
What it measures	Carbs covered per unit	Glucose drop per unit
Used for	Meal coverage	Corrections
Units	grams/unit	mg/dL/unit or mmol/L/unit
When to test	During controlled meals	During fasting/corrections
Symptoms if too aggressive	Post-meal lows	Correction lows, rollercoasters
Symptoms if too conservative	Post-meal highs	Persistent highs despite corrections

---

Summary

Carbohydrate ratio determines how much insulin you need to cover the carbs you eat. Getting your CR right is essential for:

- Accurate meal bolus calculations
- Appropriate COB corrections
- Accurate COB determinations
- Effective SMB delivery for meals
- Overall glucose control after eating

Work with Your Healthcare Provider

Always consult with your diabetes healthcare team when making significant changes to your CR or when you're unsure about adjustments. They can provide personalized guidance based on your complete medical history.

Features

Welcome to the Features homepage! Here you will find a variety of topics to learn about that will help you in using Trio to manage your diabetes.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Temporary Targets

How and when to use Temporary Targets

Overrides

How and when to use Overrides

Statistics

A detailed look at Trio's Statistics interface

Bolus Calculator

Information on what's possible with Trio's Bolus Calculator

Using Dynamic ISF

A series of tips, tricks, and best practices for using Dynamic ISF in Trio

Overrides

What is an Override?

An Override is a set of parameters you can place on Trio temporarily. With an Override, you can alter a variety of variables to increase or decrease insulin delivery:

- Basal Rate Adjustment
- Inversely change ISF, CR, or both
- Override Target
- Disable SMBs
- Override Max SMB Minutes
- Set a time frame for the override or enable indefinitely until you cancel

> **Good To Know**
>
> Overrides adjust your core profile settings. They do not influence the Oref algorithm adjustments to the sensitivity ratio used.

There are multiple cases when an Override may be useful. Below you will find a few examples. This list is not all-inclusive and you may find additional uses for Overrides as you use Trio.

In addition to Overrides, Trio also has a more automated, less customizable feature called Temporary Targets.

How Do I Set an Override?

Overrides can be found under the Adjustments tab at the bottom of the Trio app screen.

You can also find your saved Overrides listed here.

Overrides can be set remotely using Loop Follow or automated using Shortcuts.

Examples of Use

Temporary Targets

What is a Temporary Target?

Temporary Targets, also called Temp Targets, are glucose targets set by the user for a short period of time. Temp targets can be used to increase or decrease insulin delivery.

There are multiple cases when a temp target may be useful. Below you will find a few examples. This list is not all-inclusive and you may find additional uses for temp targets as you use Trio.

In addition to temp targets, Trio also has a more customizable feature called Overrides.

How Do I Set a Temporary Target?

Temp Targets can be found under the Adjustments tab at the bottom of the Trio app screen.

You can also find your saved Temp Targets listed here.

Temp targets can be set remotely using Loop Follow or automated using Shortcuts.

Examples of Use

Cardio Exercise

Tips

- Set a high temp target 1-2 hours before your exercise
- Enable exercise-related settings to reduce insulin delivered while exercising
- Pack a snack when you plan to exercise

Cardiovascular exercise can have immediate and extended effects on glycemic control. Generally, individuals will find their blood sugar drops during and continue to drop for 3-4 hours after. They may also experience increased insulin sensitivity for 24 hours after this type of exercise.

In preparation for exercise, you can use the following temp target features to reduce the occurence of lows:

- Set a temp target >100 mg/dL (>5.5 mmol/L) approximately 1-2 hours before you start exercising.
- *Enable* High Temp Target Raises Sensitivity.
- If you want SMBs automatically disabled while exercising with a temp target above 100 mg/dL (>5.5 mmol/L), ensure Allow SMB With High Temptarget is *OFF*.
- If you want SMBs to continue even when you set a temp target above 100 mg/dL (>5/5 mmol/L), ensure Allow SMB With High Temptarget is *ON*.

Having access to fast-acting carbs during this type of exercise is essential. Pack a snack in case you need a source of simple carbohydrates during your routine. Some find they need a slow acting carb before they exercise and others find having carbs on board leads to insulin on board, which leads to lows. Your diabetes may vary, so follow what you need.

Pre-Meal or Eating Soon

Tips

- Set a low temp target 1-2 hours before meals to help with spikes
- Enable "Low Temptarget Lowers Sensitivity" to allow for greater insulin delivery

Some users don't want to pre-bolus, but still would like a small boost of insulin before their meal. You can achieve this by setting a lower temp target than your usual glucose target approximately 1-2 hours before you plan to eat. How much you lower the target glucose will correlate to how much additional insulin will be calculated.

In preparation for meals, you can use the following temp target features to boost insulin delivery and reduce post-meal spikes:

- For a small boost in insulin delivery, set a temp target approximately 20-40 mg/dL (approx 1-2 mmol/L) below your target, 1-2 hours prior to meals.
- Enable Low Temptarget Lowers Sensitivity to give yourself more insulin for the upcoming meal and to bring your blood sugar down faster to the pre-meal range. Note: If you use this option, your temp target must be below 100 mg/dL (5.5 mmol/L) to trigger this feature.

Recovery From a Low Blood Sugar

Tips

- Set a high temp target for 30-60 minutes following a low
- Disable "Allow SMBs with High Temp Target"
- Save this as a pre-set Temp Target so you can enable it easily and have the option to enable it remotely

When you are recovering from a low blood sugar, you especially do not want Trio to over-correct for the rapid rise in glucose readings. To prevent this, users often set a temp target above 100 mg/dL (>5.5 mmol/L) for a duration of 30-60 minutes following a low blood sugar. This will reduce the correction given for the rapid rise in glucose if you do the following:

- Set a temp target >100 mg/dL (>5.5 mmol/L) that is also higher than your usual target glucose for between 30-60 minutes
- Turn *OFF* Allow SMB With High Temptarget
- Turn *ON* High Temp Target Raises Sensitivity

Statistics

Trio's Statistics feature provides comprehensive data analysis and visualization of your diabetes management metrics. View detailed glucose patterns, insulin usage, loop performance, and meal data across multiple time periods.

Accessing Statistics

To view statistics:

a. Open Trio

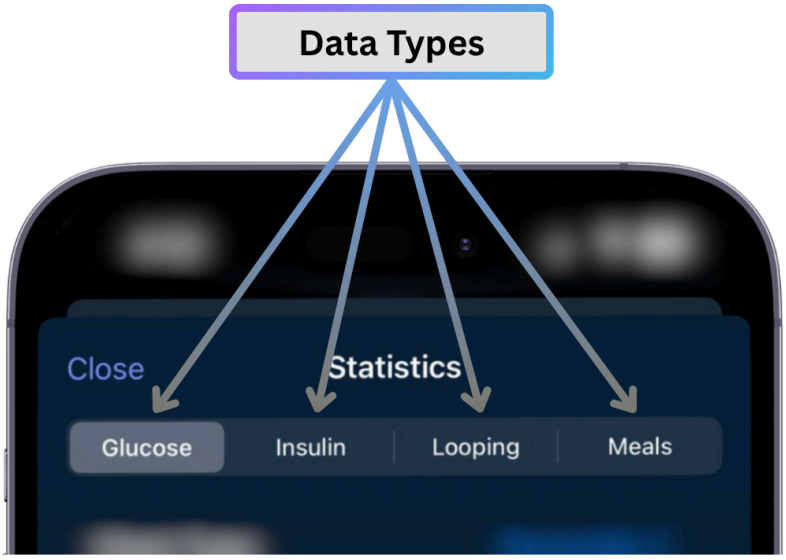
b. Navigate to the Statistics screen (icon in the navigation menu)

c. Select a category tab (Glucose, Insulin, Looping, or Meals)

d. Choose your preferred chart type and time period

Data Types

The Statistics screen is organized into four main categories, each accessible via tabs at the top:



- **Glucose** - CGM & blood sugar metrics, time-in-range, and glucose distribution
- **Insulin** - Total daily dose and bolus distribution
- **Looping** - Loop cycle performance and reliability metrics
- **Meals** - Macronutrient tracking and meal analysis

Each category offers multiple chart types and time period options to help you understand patterns and make informed adjustments to your therapy settings.

Glucose Statistics

Available Metrics

The Glucose section displays seven key metrics:

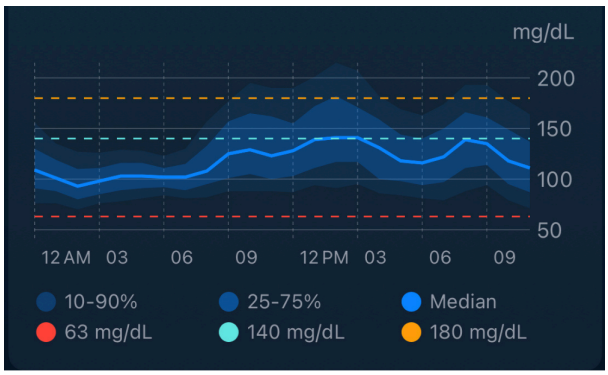
Metric	Description	Formula
eA1c	Estimated A1c from CGM data	mg/dL: $\frac{Avg\ Glucose+46.7}{28.7}$
GMI	Glucose Management Indicator	$3.31 + (0.02392 \times Avg\ Glucose)$
Average	Mean of all glucose readings	$Sum/Count$
Median	50th percentile glucose value	Middle value when sorted
SD	Standard Deviation (glucose variability)	$\sqrt{(\frac{\Sigma(x-mean)^2}{(n-1)})}$
CV	Coefficient of Variation	$\frac{SD}{Mean} \times 100\%$
Days	Number of days with glucose data	Count of unique dates

Display Units

- **eA1c**: Available as % (NGSP) or mmol/mol (IFCC)
- **GMI**: Available as % or mmol/mol
- **Glucose values**: Displayed in mg/dL or mmol/L based on your settings

Chart Types

1. Percentile by Time (AGP - Ambulatory Glucose Profile)



The default view shows hourly glucose percentiles across 24 hours:

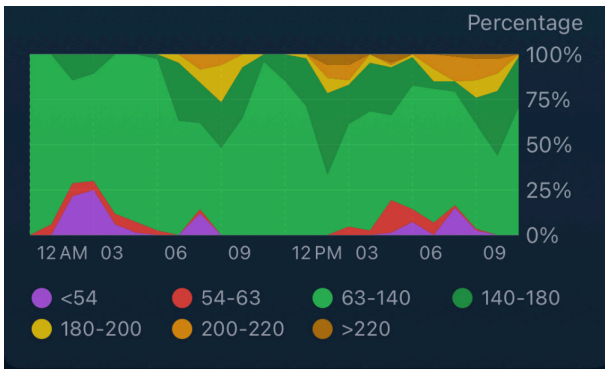
- **Dark band**: 25th-75th percentile (middle 50% of values)
- **Light band**: 10th-90th percentile (middle 80% of values)
- **Center line**: Median (50th percentile)
- **Reference lines**: Your configured low, target, and high thresholds

This chart helps identify:

- Time-of-day patterns (dawn phenomenon, overnight lows, etc.)
- Glucose variability at specific times
- How tightly controlled your glucose is

2. Distribution by Time

Shows the percentage of time spent in each glucose range for every hour of the day:



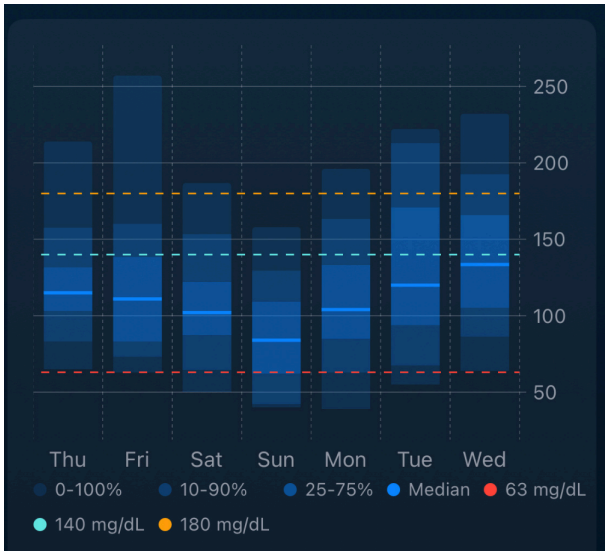
Ranges displayed:

- Purple: < 54 mg/dL (< 3 mmol/L)
- Red: 54-70 mg/dL (3-3.9 mmol/L)
- Bright Green: 70-140 mg/dL (3.9-7.8 mmol/L)
- Dark Green: 140-180 mg/dL (7.8-10 mmol/L)
- Yellow: 180-200 mg/dL (10-11.1 mmol/L)
- Orange: 200-220 mg/dL (11.1-12.2 mmol/L)
- Dark Orange: > 220 mg/dL (> 12.2 mmol/L)

Use this to identify times when you're most likely to be out of range.

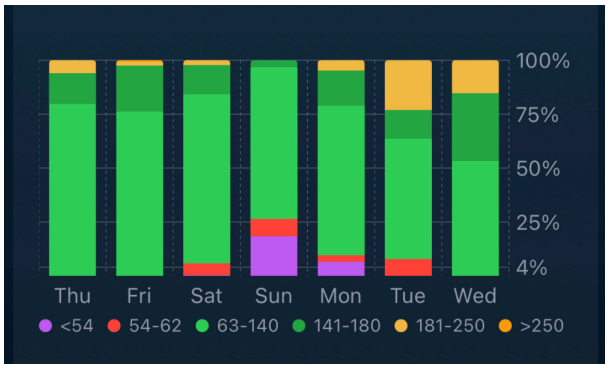
3. Percentile by Day

Box plot showing daily glucose distribution:



- Each day shows a box-and-whisker plot
- Box represents 25th-75th percentile
- Whiskers extend to 10th and 90th percentiles
- Center line is the median
- Compare day-to-day variability

4. Distribution by Day



Stacked bar chart showing the percentage of time in each range per day:

- See which days had optimal control
- Identify patterns across multiple days
- Useful for reviewing weekly trends

Distribution by Day uses the same color chart as the Distribution by Time chart.

TITR vs. TING

Trio supports two tighter TIR calculation methods:

Time in Tight Range (TITR)

- **In Range**: 70-140 mg/dL (3.9-7.8 mmol/L)
- **Low**: 54-70 mg/dL (3-3.9 mmol/L)

- **Very Low**: < 54 mg/dL (< 3 mmol/L)

Time in Normoglycemia (TING)

- **In Range**: 63-140 mg/dL (3.5-7.8 mmol/L)
- **Low**: 54-63 mg/dL (3-3.5 mmol/L)
- **Very Low**: < 54 mg/dL (< 3 mmol/L)

You can select your preferred method in the Statistics settings.

> **TIR Goals**
>
> The American Diabetes Association recommends:
>
> - **> 70%** time in range (70-180 mg/dL or 3.9-10 mmol/L)
> - **< 25%** time above range (>180 mg/dL or >10 mmol/L)
> - **< 4%** time below 70 mg/dL or 3.9 mmol/L
> - **< 1%** time below 54 mg/dL or 3 mmol/L

Time Periods

Select from five time period options:

- **Today**: From midnight to current time
- **Day (D)**: Last 24 hours
- **Week (W)**: Last 7 days
- **Month (M)**: Last 30 days
- **3 Months (3M)**: Last 90 days

Insulin Statistics

Chart Types

Total Daily Dose (TDD)

View your insulin usage broken down by type:

Components:

- **Manual Bolus**: Insulin you delivered manually for meals/corrections
- **SMB**: Super Micro Boluses delivered automatically by Trio
- **External**: Insulin injections logged but not delivered by pump
- **Temp Basal**: Insulin from temporary basal rates (above scheduled)

- **Scheduled Basal**: Insulin from your scheduled basal profile

The TDD chart displays:

- Bar graph of total daily insulin
- Statistics panel showing totals and averages for the visible range
- Scrollable timeline to review historical data

Bolus Distribution

Detailed breakdown of bolus insulin types:

- Manual bolus totals
- SMB totals
- External insulin totals
- Average bolus amounts per period

Use this to understand:

- How much insulin comes from automatic vs. manual delivery
- SMB effectiveness and contribution to total insulin
- Bolus patterns over time

Time Periods

Choose from four time periods:

- **Day (D)**: Last 24 hours (shows hourly breakdown)
- **Week (W)**: Last 7 days
- **Month (M)**: Last 30 days
- **3 Months (3M)**: Last 90 days

Looping Statistics

Loop Performance Metrics

Track how reliably Trio is functioning:

Metric	Description
Loop Count	Total number of loop cycles executed
Successful	Number of successful completions

Failed	Number of failed cycles
Success %	Percentage of successful loops
Median Interval	Typical time between loops (minutes)
Median Duration	Typical loop execution time (seconds)
Glucose Count	Number of CGM readings received
Days	Days tracked

Chart Types

Loop Performance Chart

Bar chart showing loop execution over time:

- **Green bars**: Successful loops
- **Red bars**: Failed loops
- Hover/tap for detailed counts per time period
- Scrollable timeline

Use this to identify:

- Times when loops frequently fail
- CGM connection issues
- Overall system reliability

Time Periods

Same as glucose statistics:

- Today, Day (D), Week (W), Month (M), 3 Months (3M)

Good Loop Performance

You should typically see:

- Loop interval: 5-6 minutes (Trio loops every 5 minutes normally)
- Success rate: > 95%
- Failures usually indicate CGM or pump connectivity issues

Meal Statistics

Macronutrient Tracking

View your nutritional intake over time:

Macros tracked:

- **Carbohydrates**: Total grams per period
- **Fat**: Total grams per period
- **Protein**: Total grams per period

The meal chart displays:

- Bar graph showing daily or hourly totals
- Breakdown by macronutrient type
- Scrollable timeline
- Date range label

FPU (Fat Protein Units) Support

If FPU conversion is enabled in settings, Trio tracks fat and protein to calculate their insulin impact for low-carb meals.

Time Periods

Same as insulin statistics:

- Day (D), Week (W), Month (M), 3 Months (3M)

> **Note**
>
> Meal statistics only include carbs, fat, and protein that you've logged in Trio. Accurate logging is essential for useful meal statistics.

Customization Options

Time-in-Range Type:

- Select TITR (Time in Tight Range) or TING (Time in Normoglycemia)
- Changes the range definitions for TIR calculations

Found under User Interface in Features settings

eA1c/GMI Display:

- Percentage (NGSP format)
- mmol/mol (IFCC format)

Found under User Interface in Features settings

Glucose Units:

- mg/dL
- mmol/L

Found under Units and Limits in Therapy Settings

Target Ranges:

- High Limit: Default 180 mg/dL (customizable)
- Low Limit: Default 70 mg/dL (customizable)

These limits affect the color coding and range calculations in all glucose charts.

Found under User Interface, Low and High Thresholds, in Features Settings

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Understanding Your Statistics

Glucose Variability

Standard Deviation (SD):

- **Low SD** (< 30 mg/dL): Stable, consistent glucose
- **Moderate SD** (30-50 mg/dL): Some variability
- **High SD** (> 50 mg/dL): Significant swings

Coefficient of Variation (CV):

- **Target**: < 36%
- **CV < 36%** indicates well-controlled glucose variability
- **CV > 36%** suggests unpredictable glucose patterns

Improving CV and SD

High variability often indicates:

- Inaccurate carb counting
- Incorrect ISF or carb ratios
- Inconsistent meal timing
- Exercise without appropriate adjustments

GMI vs. eA1c

Both provide estimates of your average glucose management:

- **eA1c**: Calculated from average glucose using the ADAG formula
- **GMI**: Similar calculation, recommended by international consensus

These are **estimates** based on CGM data, not laboratory A1c measurements. They correlate well but may differ from lab results.

TDD Trends

Monitoring your Total Daily Dose helps you understand:

- **Increasing TDD**: May indicate insulin resistance, weight gain, or illness
- **Decreasing TDD**: May indicate increased sensitivity, weight loss, or more activity
- **Stable TDD**: Consistent insulin needs

Significant TDD changes should prompt review of your basal rates, ISF, carb ratios, and Dynamic ISF Settings.

Loop Reliability

High loop success rates (> 95%) indicate:

- Good CGM connectivity
- Reliable pump communication
- Stable Trio operation

Frequent loop failures warrant investigation:

- Check CGM sensor placement and transmitter battery
- Verify pump is in range and functioning
- Review phone battery and background app settings

Data Availability

Statistics are calculated from data stored in Trio's database:

- **Glucose**: Requires CGM data
- **Insulin**: Requires pump event history
- **Looping**: Requires loop execution records
- **Meals**: Requires logged carb entries

If you've recently started using Trio or reset your data, some statistics may not be available until sufficient data accumulates.

Minimum data for meaningful statistics:

- **1 day**: Today and Day views
- **7 days**: Week view
- **30 days**: Month view
- **90 days**: 3 Month view

Performance Notes

Statistics calculations can be intensive for large datasets. Trio uses several optimizations:

- **Caching**: Frequently accessed calculations are cached
- **Background processing**: Heavy calculations run in background threads
- **Pagination**: Charts load data in scrollable segments
- **Concurrent calculations**: Multiple statistics computed in parallel

You may notice a brief delay when first opening Statistics or switching time periods on older devices. This is normal and ensures the UI remains responsive.

Troubleshooting

"No data available" Message

If you see this message:

1. **Check data source**: Ensure you have CGM data (for glucose), logged carbs (for meals), etc.
2. **Verify time period**: Try a shorter time period if you recently started using Trio
3. **Restart Trio**: Sometimes refreshing the app resolves data loading issues

Statistics Don't Match Other Apps

Small differences are normal due to:

- Different calculation methods (mean vs. median)
- Different time period boundaries
- Different range definitions
- Data filtering (Trio may exclude some invalid readings)

Large discrepancies should be investigated—verify your data is syncing correctly.

Slow Performance

If statistics load slowly:

- **Reduce time period**: Use Day or Week instead of 3 Months
- **Clear app cache**: Settings > Advanced (if available)
- **Update Trio**: Ensure you're running the latest version
- **Device storage**: Free up space on your iPhone

Summary

Trio's Statistics feature provides powerful insights into your diabetes management:

- **Glucose Statistics**: Understand your time-in-range, variability, and patterns
- **Insulin Statistics**: Monitor your TDD and bolus distribution
- **Loop Statistics**: Track system reliability and performance
- **Meal Statistics**: Review your macronutrient intake

Use these statistics to:

- Identify patterns and trends
- Make informed adjustments to therapy settings
- Share data with your healthcare team
- Track progress toward your diabetes management goals

Regularly reviewing your statistics helps ensure Trio is working optimally and your settings remain accurate as your insulin needs change.

Bolus Calculator

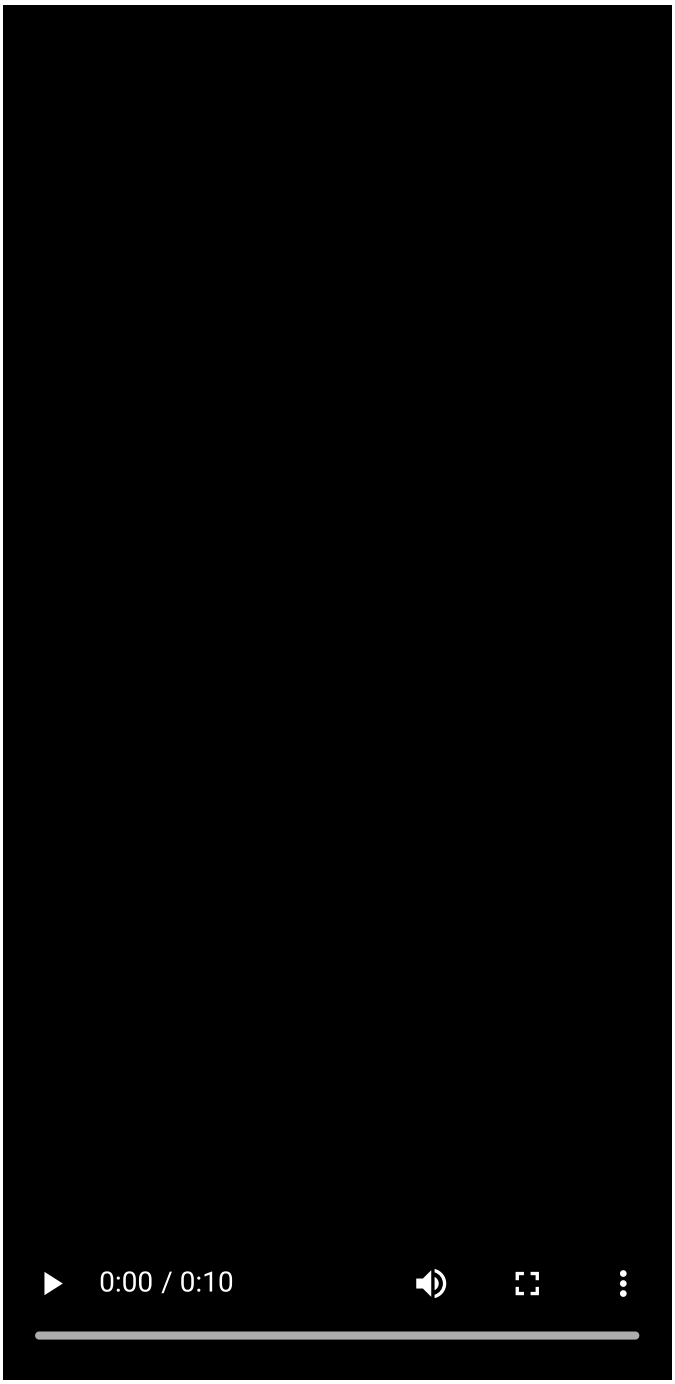
Bolus Calculator Interface

You can access the bolus calculator by tapping the + icon

There are 4 main sections of the Bolus Calculator Interface

Dynamic Glucose Forecast

As you enter your carbs and insulin, this dynamic graph at the top of the bolus calculator will adjust the forecast lines or cone to account for your future data.
Watch the video below to see this in action.



Meal Entry

In the Meal Entry section, you will enter your carbs. If you have FPU's enabled, you can also enter your fat and protein.

Below this, there is a place to enter the time for future or past carbs.

You can also add a note for this meal.

Dosing Calculation

In the Dosing Calculation section, it starts with the Reduced Bolus and Super Bolus options at the top, if you have those enabled.

Below that is the suggested bolus amount that Trio has calculated. Tap the "i" icon for more details and read the section below for more details on the information shared when you tap on this icon.

If you check the "External Insulin" box, the bolus will be added to your IOB, but it will not be delivered by Trio.

Enact Button

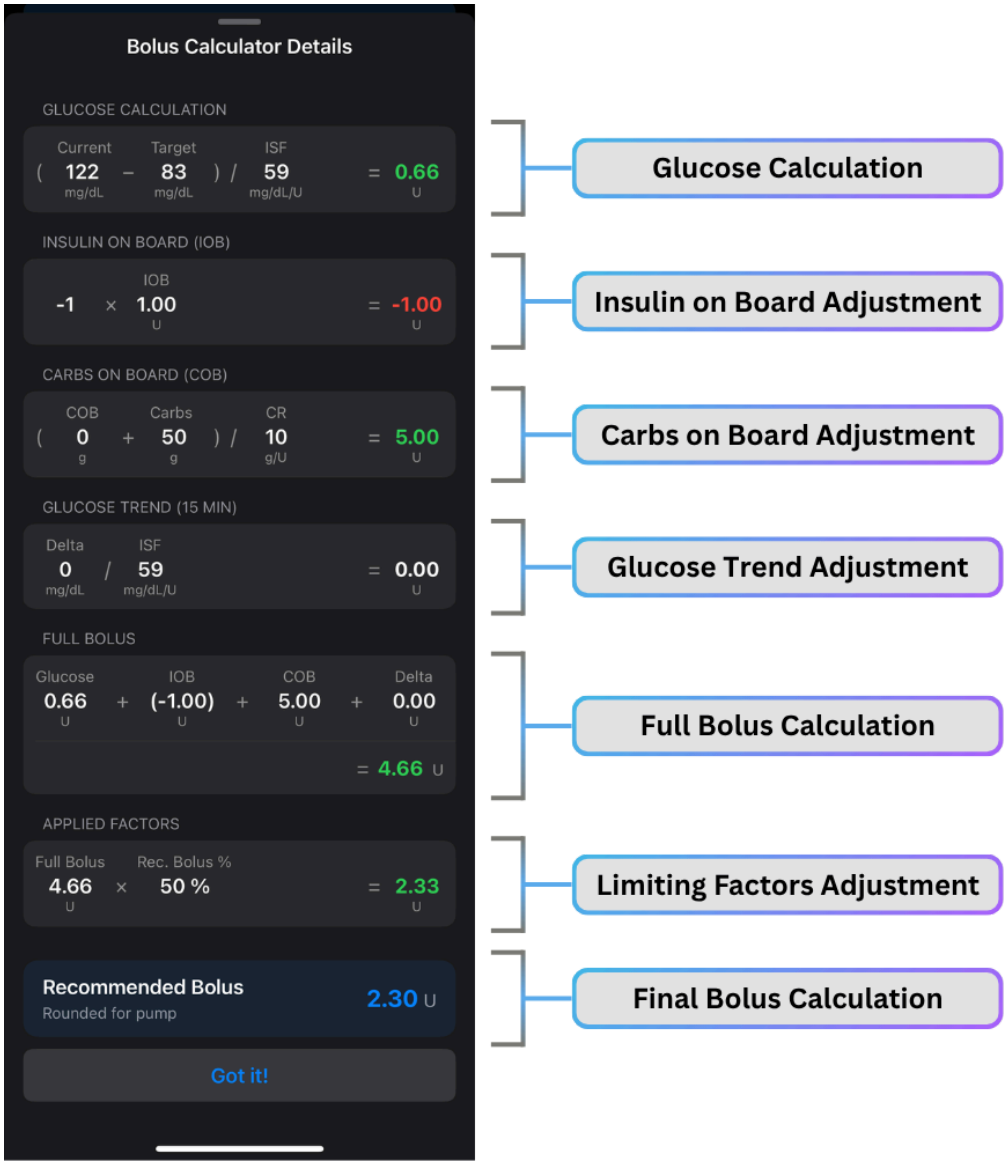
This button will show one of the following messages:

- **Continue Without Treatment**: This means no COB or IOB will be added
- **Log Carbs**: Carbs will be logged, but no insulin will be delivered
- **Log FPU**: FPUs will be logged, but no insulin will be delivered
- **Log Meal**: Both carbs and FPUs will be logged, but no insulin will be delivered
- **Log Meal/FPU/Carbs and Enact Bolus**: Carbs and/or FPUs will be logged and insulin will be delivered
- **Log External Insulin**: Insulin will be added to IOB, but it will **not** be delivered

How Does the Bolus Calculator Determine Dosage?

There are multiple calculations that are used to determine the bolus recommendation in Trio.

A positive calculation will be shown in green and a negative calculation will be shown in red.



Glucose Calculation

The bolus calculator in Trio starts with your basic dosage needed to address your current glucose reading. If your glucose is below your target, it will be a negative number. If your glucose is above your target, it will be a positive number.

Let's walk through Bill's current bolus calculation, starting with the first step, **Glucose Calculation**.

Glucose Above Target: Bill's current glucose reading is 122 mg/dL. His target glucose is 83 mg/dL. Trio is currently using a calculated ISF of 59 mg/dL/U. How much insulin does Bill need to reach his target glucose?

Here is the formula you will need:

$$\frac{(Current\ Glucose - Target\ Glucose)}{ISF}$$

Calculate Bill's Glucose Calculation:

$$\frac{(122 - 83)}{59} =$$

$$\frac{39}{59} =$$

$$0.66\ units$$

Answer

Bill needs **0.66 units** to reach his glucose target without accounting for any other factors.

Glucose Below Target: Bill's current glucose reading is 70 mg/dL. His target glucose is 83 mg/dL. Trio is currently using a calculated ISF of 90 mg/dL/U. How much insulin does Bill need to reach his target glucose?

Here is the formula you will need:

$$\frac{(Current\ Glucose - Target\ Glucose)}{ISF}$$

Calculate Bill's Glucose Calculation:

$$\frac{(70 - 83)}{90} =$$

$$\frac{-13}{90} =$$

$$-0.14\ units$$

Answer

Because Bill was below target, the bolus calculator estimates he has the equivalent of 0.14 units already present based on his current ISF. Therefore, Trio will **subtract 0.14 units** from his insulin needed in the bolus calculator.

Trio takes many additional factors into account. Next, it will adjust for the amount of insulin already in his system.

Insulin on Board Adjustment

The next step in the bolus calculation is the adjustment for current insulin on board (IOB). To prevent the calculator from giving you insulin already given, it subtracts your current IOB.

This will also replace any missing insulin that may have occurred after basal reduction or suspension. If you are trending low before a meal, this will add any negative IOB to return you to baseline basal in preparation for addressing your insulin needed for the incoming carbs.

Let's look at how Trio addresses both positive and negative IOB:

Postive IOB: Bill currently has 1.0 units of insulin on board (IOB) from previous SMBs, manual boluses, and/or increased temp basal rates. How will the bolus calculator adjust for this IOB?

Here is the formula you will need:

$$-(IOB)$$

Calculate Bill's Insulin on Board Adjustment:

$$-(1.0) =$$

$$-1.0\ units$$

Answer

The bolus calculator will **subtract 1.0 units** from the bolus recommendation.

Negative IOB: Bill currently has -1.0 units of insulin on board (IOB) from previous reduced or suspended basal rates. How will the bolus calculator adjust for this IOB?

Here is the formula you will need:

$$-(IOB)$$

Calculate Bill's Insulin on Board Adjustment:

$$-(-1.0) =$$

$$+1.0\ units$$

Answer

The bolus calculator will **add 1.0 units** from the bolus recommendation.

There are still more factors for Trio to take into account, so we aren't done yet! Next, let's see how your current COB and the new carbs entered will influence the bolus recommendation.

Carbs on Board Adjustment

The previous step took into account the insulin you already have in your system. This next step will add the insulin needed for both the carbs already on board and the new carbs you just entered, which serves to negate any IOB that was due to previous meals (COB).

Let's look at how Trio addresses times when there are carbs on board and times when there are no carbs on board:

Without COB: Bill currently has no carbs on board (COB), but he is entering 50g into the bolus calculator. His current carb ratio (CR) is 10 g/U. How much insulin does he need to counter both the existing and new carbs in his system?

Here is the formula you will need:

$$\frac{(COB + Carbs\ Entered)}{CR}$$

Calculate Bill's Glucose Calculation:

$$\frac{(0 + 50)}{10} =$$

$$\frac{50}{10} =$$

$$5.00\ units$$

Answer

Based on his current core settings, Bill needs **5.00 units** to counter the carbs he's entered.

With COB: Bill currently has 20g carbs on board (COB), but he is entering 50g into the bolus calculator. His current carb ratio (CR) is 10 g/U. How much insulin does he need to counter both the existing and new carbs in his system?

Here is the formula you will need:

$$\frac{(COB + Carbs\ Entered)}{CR}$$

Calculate Bill's Glucose Calculation:

$$\frac{(20 + 50)}{10} =$$

$$\frac{70}{10} =$$

$$7.00\ units$$

Answer

Based on his current core settings, Bill needs **7.00 units** to counter the carbs he's entered and his current COB.

We aren't done yet! Aren't you glad Trio is doing all of this for you and you only need to follow along to understand the reasoning behind it? Next, it will look at how our glucose is trending to ensure it's not over or under treating you.

Glucose Trend Adjustment

Your glucose trend (Delta) takes into account how much your glucose readings have changed over the last 15 minutes. If you are experiencing a steep rise or steep fall, this helps Trio counter that activity and prevent an over- or under-dose.

Let's look at how Trio accounts for an increasing trend and a decreasing trend in glucose:

Increasing Trend: Bill's last 3 glucose readings were steadily increasing for a combined change of +29 mg/dL. His current ISF is 59 mg/dL/U How will Trio treat this trend in glucose?

Here is the formula you will need:

$$\frac{Delta}{ISF}$$

Calculate Bill's Glucose Trend Calculation:

$$\frac{29}{59} =$$

$$0.49\ units$$

Answer

Based on his current ISF, Bill needs **0.49 units** to counter the current trend in his glucose readings.

Decreasing Trend: Bill's last 3 glucose readings were steadily decreasing for a combined change of -15 mg/dL. His current ISF is 100 mg/dL/U How will Trio treat this trend in glucose?

Here is the formula you will need:

$$\frac{Delta}{ISF}$$

Calculate Bill's Glucose Trend Calculation:

$$\frac{-15}{100} =$$

$$-0.15\ units$$

Answer

Based on his current ISF, Trio will **subtract 0.15 units** to counter the current trend in his glucose readings.

Next, Trio combines all of the dosage components that it has calculated into one full bolus recommendation.

Full Bolus Calculation

Trio starts your bolus recommendation at the full bolus amount, then adjusts it in the next step. Before it can adjust, it needs to know what the full bolus needed is. To do this, it takes all the previous calculations and combines them.

Let's look at the combination used in the image to see what the full bolus recommendation is:

In the image above, you can find the steps of each calculation. Let's combine them to see what Bill's full bolus calculation will be.

Here is the formula you will need:

$$Glucose\ Calc + IOB\ Calc + COB\ Calc + Delta\ Calc$$

Calculate Bill's Full Bolus Calculation:

$$0.66 + (-1.00) + 5.00 + 0.00 =$$

$$0.66 - 1.00 + 5.00 + 0.00 =$$

$$4.66\ units$$

Answer

Based on all the important factors that make up his bolus calculation, Trio determines Bill needs 4.66 units for this bolus.

Now that Trio has the full bolus determined, Trio reduces this based on the settings you've put in place.

Limiting Factors Adjustment

Trio will reduce your bolus amount by the Recommended Bolus Percentage you've set.

Let's look at how Trio will adjust Bill's bolus based on the default setting of 50%:

Bill has his Recommended Bolus Percentage set to the default of 50%. In the previous step, we calculated he needed 4.66 units. How will Trio adjust his bolus recommendation based on this setting?

Here is the formula you will need:

$$Full\ Bolus \times Recommended\ Bolus\ Percentage$$

Calculate Bill's Limiting Factors Adjustment:

$$4.66 \times 50\% =$$

$$2.33\ units$$

Answer

Trio determines Bill needs 2.33 units for this bolus.

But, wait! Trio can't administer 2.33 units because he's using an Omnipod pump. Trio addresses this in the final step.

Final Bolus Calculation

Now that Trio has the final recommended bolus determined, it must ensure it provides the pump with a bolus command that it can execute. Meaning, a bolus amount that the pump can give.

This is pretty simple. Trio will round up or down the bolus calculation. If the calculated amount is greater than 0.06, it will round up. If it is less than 0.05, it will round down. If it is exactly 0.00 or 0.05, it will not change the amount.

Trio has determined Bill needs 2.33 units. Bill is using an Omnipod pump, so boluses can only be delivered in increments of 0.05 units. How will Trio adjust Bill's final bolus recommendation?

Answer

Because Bill's recommended bolus ends in a number less than 0.05, it will be **rounded down to 2.30 units**.

Congratulations! You've completed the full bolus calculation. We realize this is an extensive and lengthy process, so we encourage you to ensure your core settings are tested and accurate for the Trio algorithm and your algorithm settings are set appropriately so that you do not need to constantly second guess these determinations.

Tip

- While it is **always** wise to use your best judgement rather than trusting the bolus recommendation blindly, accurate settings will prevent the need for you to manually intervene and override the bolus recommendation.
- If your glucose updates in the middle of your meal entry, Trio will update your calculation in real time.

Using Dynamic ISF

Important

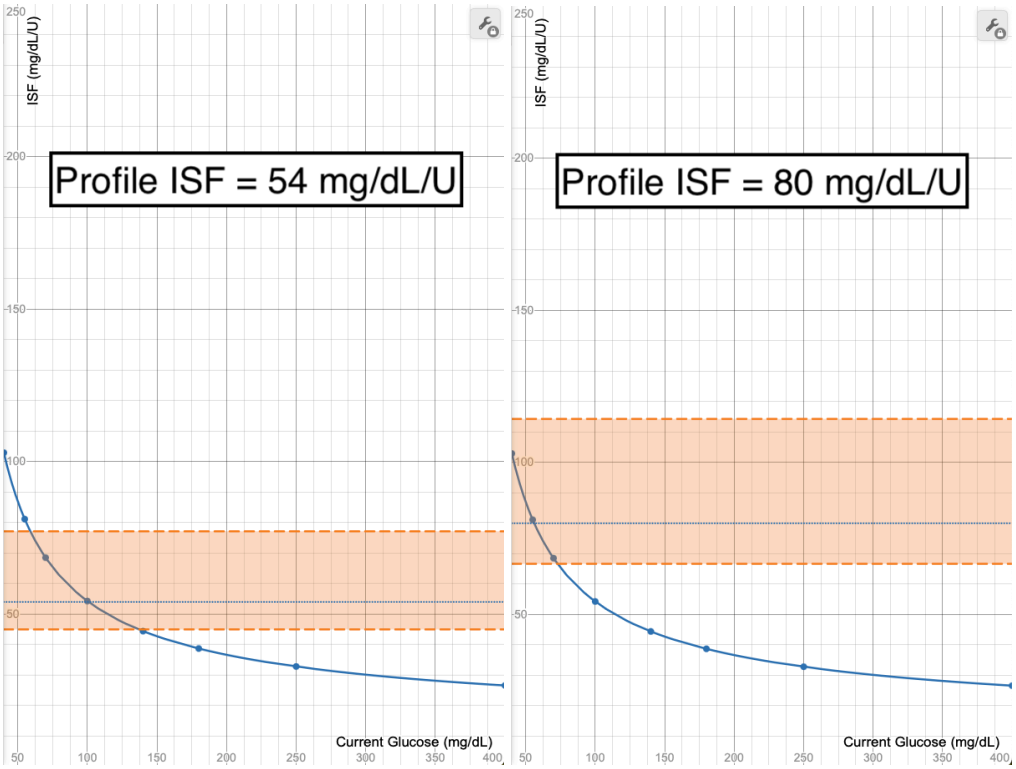
- The images used are hypothetical graphs. Where the lines and values actually fall for you rely heavily on *your* personal settings. The focus of these illustrations is to show what shifts rather than what values will be used for your specific settings.
- The blue line in these images shows what ISF is used (Y-axis) when your glucose is at a certain value (X-Axis). Each point on the graph (X,Y) means at a glucose of "X", an ISF of "Y" is used.
- The orange shaded area is what portions of that blue line Trio is allowed to use. The lower orange line is the ISF determined by your Autosens Maximum and the upper orange line is the ISF determined by your Autosens Minimum. Trio cannot use any values outside of the orange shaded area.

Logarithmic Dynamic ISF

If you've spent any time in the Desmos Graphs, you may have noticed certain adjustments do not have the impact you'd expect. Below are a few that you should know about:

Profile ISF

- Your profile ISF is not directly used to determine your ISF when Logarithmic Dynamic ISF is enabled. It influences the limits of what ISF is allowed.
- Adjusting your profile ISF when using Logarithmic Dynamic ISF will not have a direct impact on your sensitivity calculations. It will only have an impact on the Maximum and Minimum values allowed.



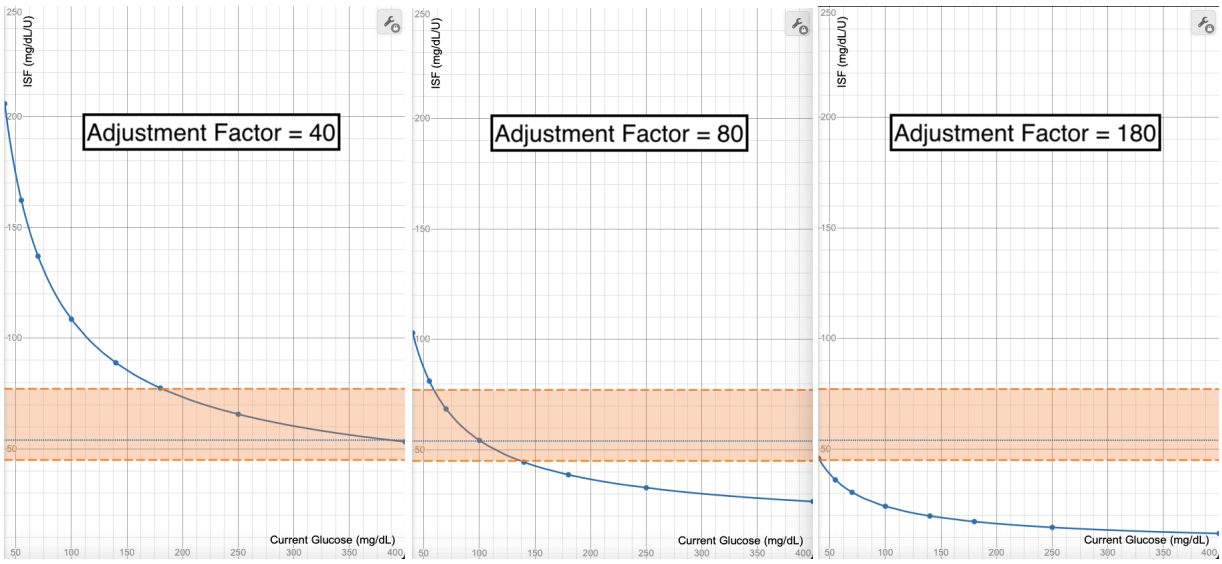
Profile ISF Tip

- This is why the Desmos graphs are invaluable whenever adjustments are needed with Logarithmic Dynamic ISF.

Adjustment Factor

- Changing your Adjustment Factor will have a huge impact on what sensitivity ratio and thus what ISF is used.
- It is strongly advised to enter any planned changes to Adjustment Factor into the Desmos Graph before changing it in your settings.

Notice how the graph shifts towards (0,0) to increase the rate of change as Adjustment Factor increases. While this does increase how fast ISF adjusts, if Autosens Maximum and Autosens Minimum aren't also adjusted, it can have the unintended consequence of always using a much too high or much too low ISF.



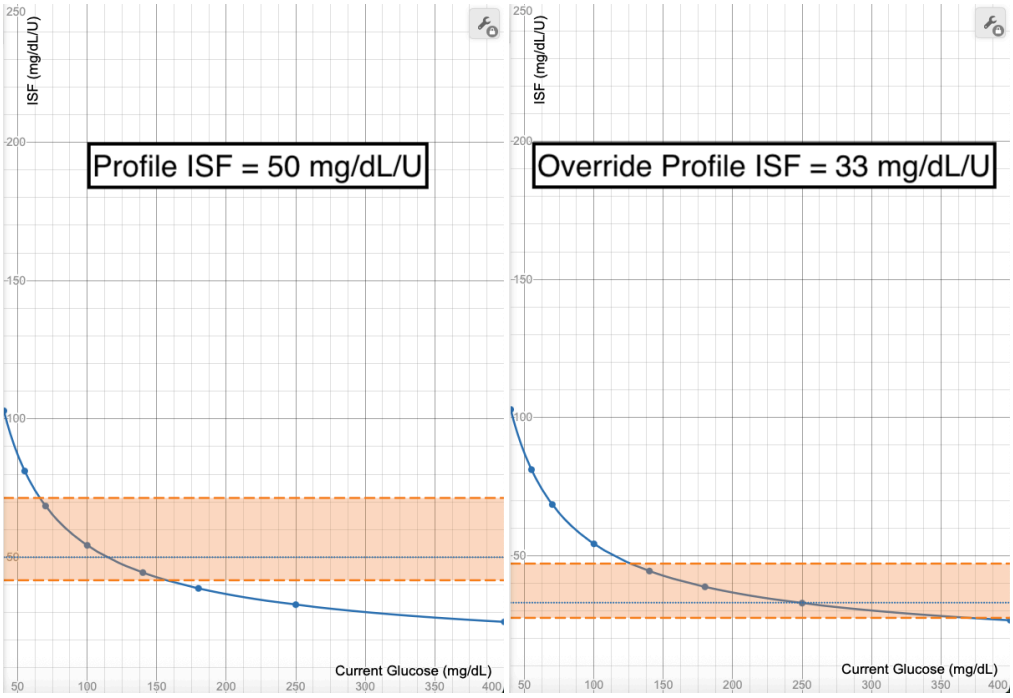
Using Overrides With Logarithmic Dynamic ISF

Overrides change your profile settings before they are sent to the Oref algorithm for adaptation. As noted above, because Logarithmic Dynamic ISF does not utilize your profile ISF for the sensitivity calculation, adjusting your profile ISF using an override will not have a significant impact on your dosage calculations. For this reason, it is advised to use Temp Targets instead of or in conjunction with Overrides when using Logarithmic Dynamic ISF.

What effect will using a >100% Override have on Logarithmic Dynamic ISF?

If you start with a Profile ISF of 50 mg/dL/U and set an override of 150%, this will change your Profile ISF to 33 mg/dL/U.

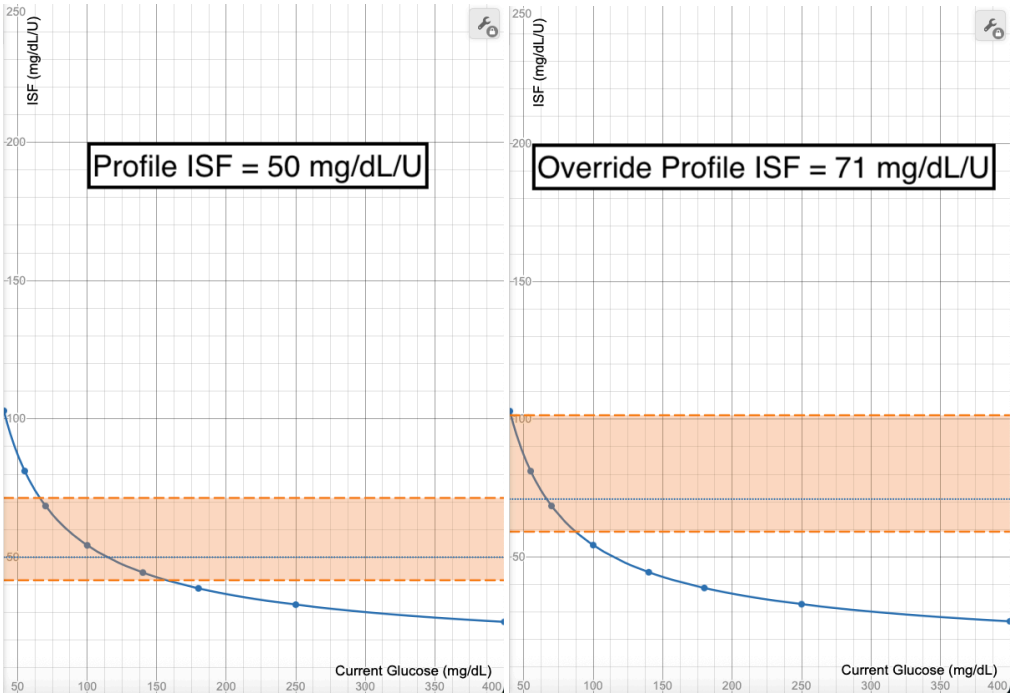
Notice how the calculated ISF line does not change with this adjustment to the Profile ISF, only the limits of what section of the curve are allowed. It shifts the limits to only allow the lower ISF portions of the graph, meaning the ISF allowable range will be 27-47 mg/dL/U and the adjustments stop at 47 mg/dL/U if glucose is reading below than 126 mg/dL.



What effect will using a <100% Override have on Logarithmic Dynamic ISF?

If you start with a Profile ISF of 50 mg/dL/U and set an override of 70%, this will change your Profile ISF to 71 mg/dL/U.

Notice how the calculated ISF line does not change with this adjustment to the Profile ISF, only the limits of what section of the curve are allowed. It shifts the limits to only allow the higher ISF portions of the graph, meaning the ISF allowable range will be 59-100 mg/dL/U and the adjustments are capped if glucose is reading higher than 87 mg/dL.



Using Temp Targets With Logarithmic Dynamic ISF

When you utilize a Temp Target AND have Target Behavior settings enabled, this will disable Logarithmic Dynamic ISF and utilize a new formula for your sensitivity ratio.

Temp Targets Tips

- Worth noting that your Autosens Maximum and Autosens Minimum are still respected when this is utilized.
- If setting a Temp Target adjusts your Sensitivity Ratio to 60%, but your Autosens Minimum is 70%, the Autosens Minimum will be used.

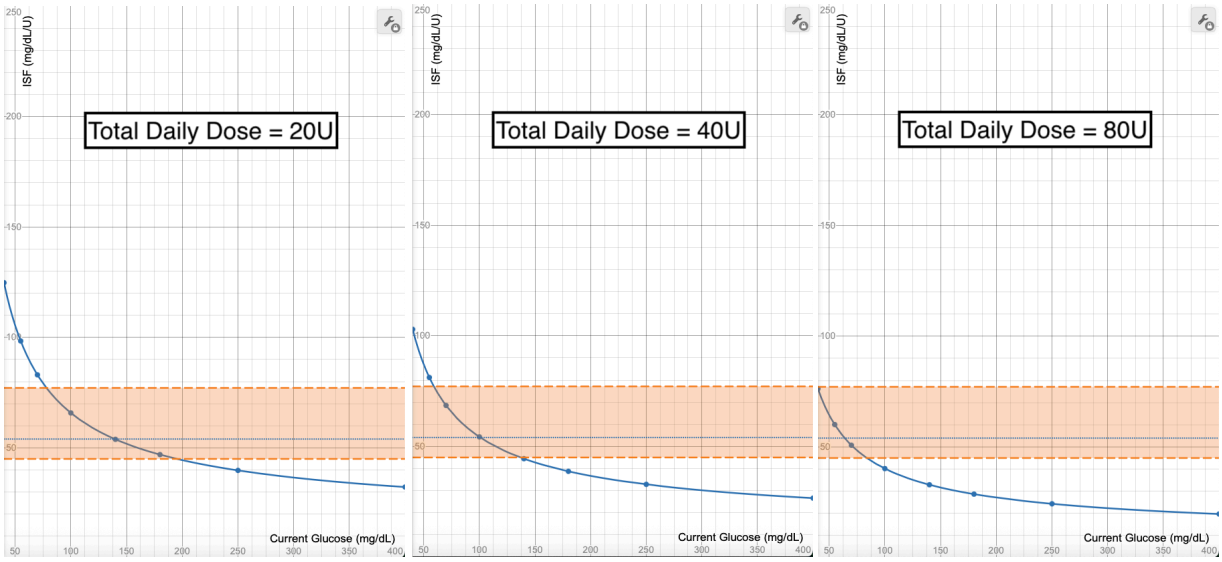
Using Both Overrides and Temp Targets with Logarithmic Dynamic ISF

Yes, you can enable both an Override and a Temp Target at the same time. When you do, Trio will first adjust your profile settings based on the Override set, then those profile settings will be sent to the Oref algorithm for the Sensitivity Ratio and insulin required calculations. The Target Behavior settings will influence the Sensitivity Ratio used as explained above. Your other override-adjusted profile settings will then be utilized to determine the next insulin dosing decision.

Influence of Total Daily Dose on Logarithmic Dynamic ISF

It's not unheard of for there to be occasions when your daily insulin use can change drastically. It can increase significantly if you need a dose of steroids or are coming down with an illness. It can decrease significantly if you start a new exercise regimine. It's important to know how those changes will influence Trio's calculation in Logarithmic Dynamic ISF.

Using a Total Daily Dose (TDD) of 40 units as a baseline, you can see how this changes the ISF calculations when TDD is halved (20 units) or doubled (80 units).



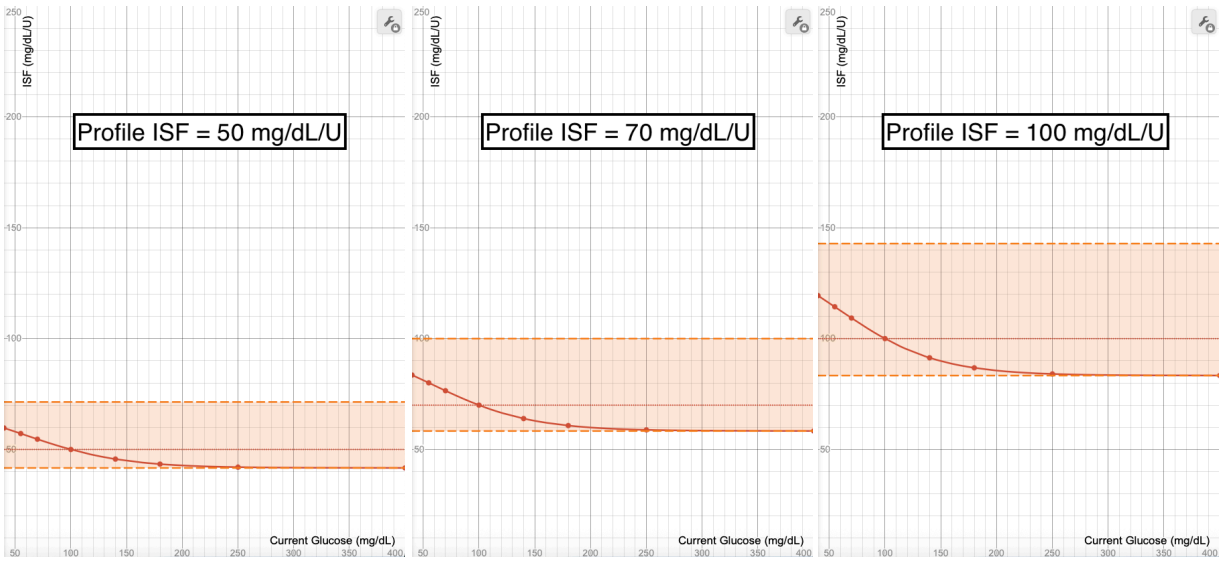
TDD & Logarithmic Tip

- You may want to temporarily adjust your Autosens Minimum or Maximum if you notice a significant change in your TDD.
- Before making changes, remember to check them in the Desmos Graph to ensure your new insulin needs are being met.

Sigmoid Dynamic ISF

Profile ISF

When you make changes to your Profile ISF, either as a profile setting change or through an Override, it will shift your Sigmoid Dynamic ISF curve to ensure your Profile ISF is always used when you are at your target glucose.



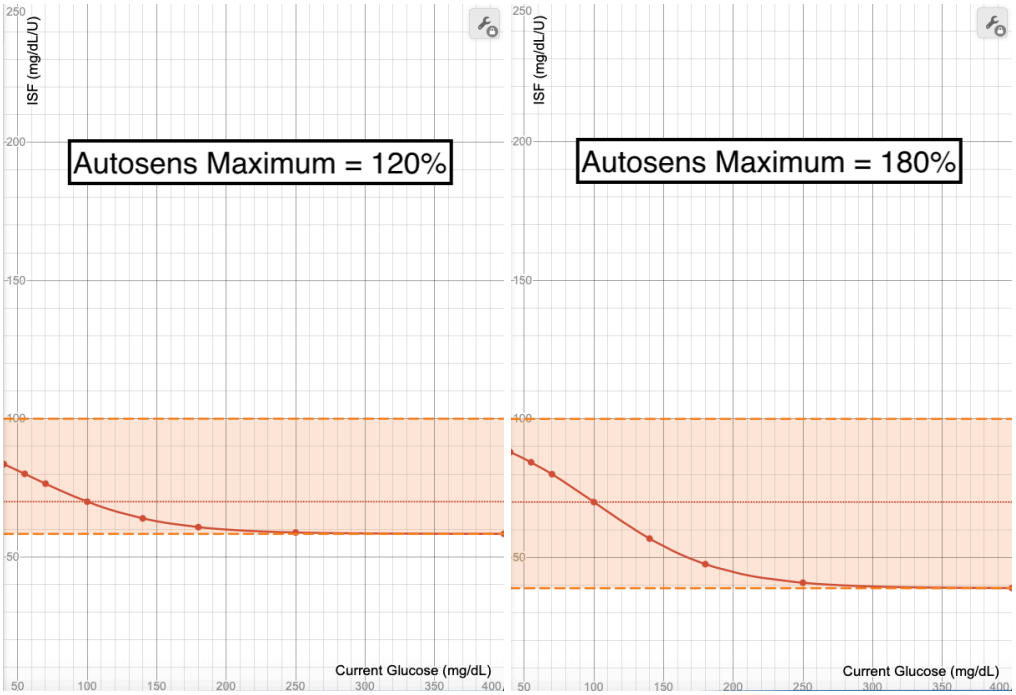
Though it can also shift how steep the curve is, and thus how quickly and to what degree adjustments change, changing your ISF should be done with caution and not as a means to change the curve. Rather, adjusting your Adjustment Factor is the preferred way to adjust how quickly values adjust.

Profile ISF Tips

- If you have tested your ISF at a glucose of 150 mg/dL (8.3 mmol/L) or greater, you may find your Profile ISF is too strong when you enable Sigmoid especially if your target glucose is 100 mg/dL (5.5 mmol/L) or lower.
- To resolve this, you can either increase your target glucose to 150 mg/dL (8.3 mmol/L) or increase your Profile ISF to account for the difference.

Autosens Maximum

Increasing your Autosens Maximum has the expected effect of raising the percentage in which ISF can be adjusted when using Sigmoid Dynamic ISF. What may not be as expected is the steepness of the curve also changes. If you increase your Autosens Maximum with Sigmoid, you'll see the maximum sensitivity ratio increase and also the speed and amount of ISF changes getting greater.



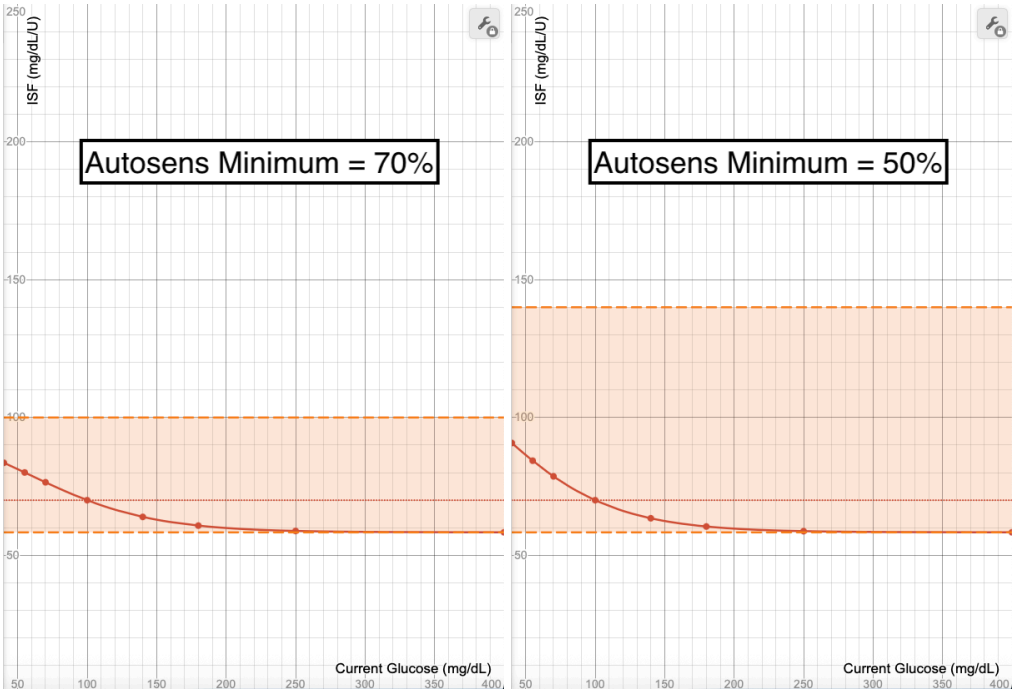
Autosens Maximum Tip

- If you want to avoid the steepness of the curve increasing while increasing your Autosens Max, reduce your Adjustment Factor to counter this effect.

Autosens Minimum

Decreasing your Autosens Minimum has the expected effect of lowering the percentage in which ISF can be adjusted when using Sigmoid Dynamic ISF. The same way that Autosens Maximum will change the steepness of the curve, adjusting Autosens Minimum will have the same effect.

Notice how Sigmoid is still locked at your Profile ISF being used when glucose is at target? And notice how there's a significant amount of empty space at the top of the orange section in the 50% graph? This means that lowering your Autosens Minimum may not actually lower the adjustments allowed to the extent that you hope, but the only noticeable result may be the curve becoming steeper.



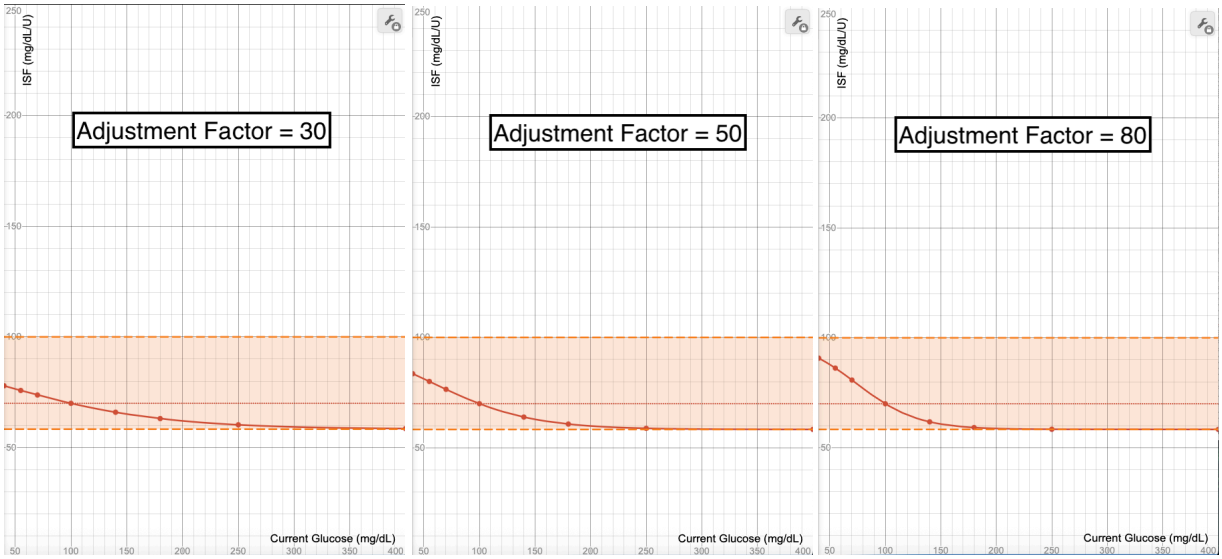
Autosens Minimum Tips

- Check your planned adjustment in the Sigmoid Desmos Graph to ensure it's having the intended result.
- If you are looking to decrease Autosens Min because of lows, the issue may be with your Profile ISF setting.

Adjustment Factor

When using Sigmoid, increasing your Adjustment Factor will cause your sensitivity ratio to increase and decrease at a faster pace as your glucose rises and falls.

You can see in the graphs below how the curve gets steeper as the Adjustment Factor increases.



Sigmoid Adjustment Factor Tips

- If you need Sigmoid to respond faster, increase the Adjustment Factor.
- If you need Sigmoid to respond slower, decrease the Adjustment Factor.
- This is a fine tuning tool, and should be adjusted only after your core settings have been established as accurate.

Target Glucose

Changing your Target Glucose either with an Override, Temp Target, or profile Target change, shifts the Sigmoid graph so that your Profile ISF is used at the new Glucose Target set.

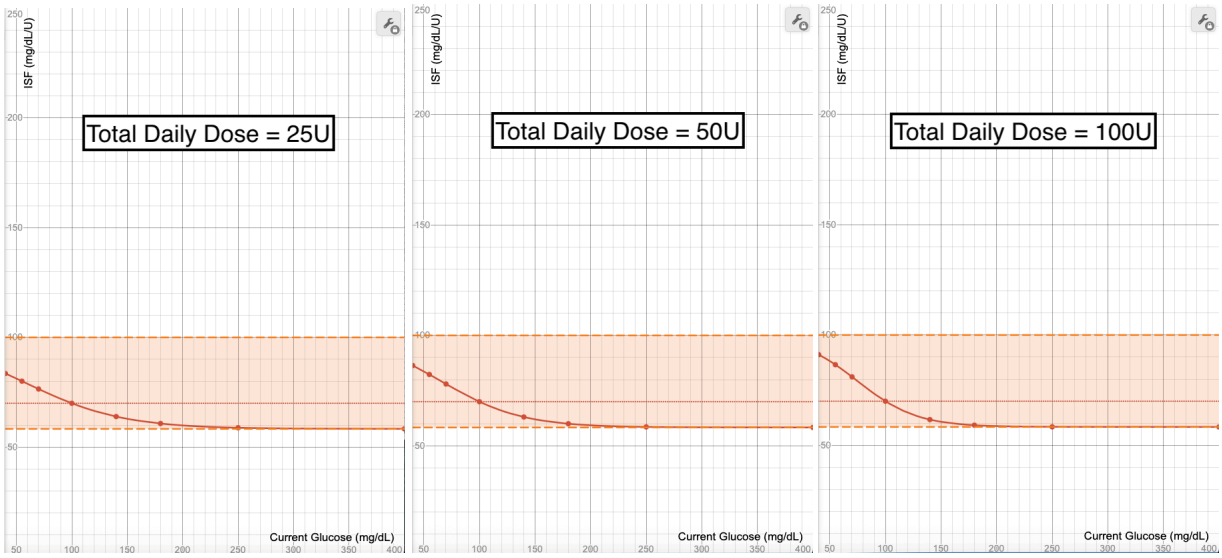
Target Glucose Tip

- Be aware that if you make a drastic change your Profile Target Glucose while using Sigmoid, you may find that your ISF suddenly seems too weak or too strong. Watch for a few days to ensure you don't need to increase or decrease your ISF to match the new Glucose Target.

Influence of Total Daily Dose on Sigmoid Dynamic ISF

It's not unheard of for there to be occasions when your daily insulin use can change drastically. It can increase significantly if you need a dose of steroids or are coming down with an illness. It can decrease significantly if you start a new exercise regimine. While this has a big impact on Logarithmic Dynamic ISF, it does not have as much of an impact on Sigmoid Dynamic ISF.

In the graphs below, you can see how the graph changes, but not by much as the Total Daily Dose (TDD) of 50 units is halved to 25 or doubled to 100 units. It does cause the steepness of the curve, thus the response time to adjust accordingly. It slows with a lower TDD and speeds up with a higher TDD.



Sigmoid & TDD Tip

- Most likely you won't need to make any adjustments to Sigmoid if you have a sudden increase or decrease in TDD.

Installation and Update

Welcome to the installation and update homepage! Here you will find information about building, installing, and updating your Trio app.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Attention Trio Users

The `main` branch is the preferred build for everyone except expert builders.

If you are coming from an older `main` version 0.2.x, please read the Migration Guide before updating.

App Requirements

- Compatible Devices
- Apple Developer

Build with Mac

Overview on building Trio using a Mac computer and Xcode

Build with Browser

Overview on building Trio using a web browser and TestFlight

Upgrade

How to upgrade Trio to the latest version

Companion Apps

Apps that work alongside Trio

Compatible Devices

Welcome to the Compatible Devices homepage! Here you will find information about the devices that will work with your Trio app.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Insulin Pumps

Information about insulin pumps compatible with Trio

CGM

Information about continuous glucose monitors (CGMs) that are compatible with Trio

iPhone

Answers the question: What phones will work with Trio?

Mac

What are the Mac requirements?

Compatibility Matrix

How do versions for iOS, macOS, and Xcode affect my ability to build with a Mac?

Insulin Pump

Compatible Pumps

The following pumps are compatible with Trio:

- Omnipod 5 (Open Beta))
- Omnipod DASH
- Omnipod Eros
- Dana-i
- DanaRS-v3
- Medtrum Nano
- Older Medtronic pumps

Developers are currently working on adding the following pumps to Trio:

- Tandem Mobi

No other pumps work with Trio at this time, but other open-source closed-loop options, such as AAPS: Android Artificial Pancreas System and OpenAPS, support other pumps.

Omnipod Pumps

Reminder and Disclaimer

The use of Omnipod pumps with Trio is not supported by Insulet, although they are aware it is happening. Do not call Insulet asking for help with your Trio build, setup, or operation. You are fully responsible for your use of Trio and do so at your own risk. Please read these documents and familiarize yourself with Trio before using the app.

Omnipod 5

***Open Beta**

- We encourage anyone interested in participating in the Omnipod 5 Open Beta to read these FAQs
- Omnipod 5 is currently available in the `dev` branch. See here for help on building dev with Browser Build.
- As this is an open beta, we encourage most people to wait while experienced users test this implementation

Internet Required for First Use

Internet access is required for online authentication the very first time you pair an Omnipod 5 pod with Trio. Once authenticated, all future Omnipod 5 pods will pair offline. If you switch to a new iPhone, however, Internet access will once again be required to authenticate the first Omnipod 5 pod.

Must have a CGM with a heartbeat

- Omnipod 5 code is experimental and does not provide a heartbeat at this time
- This means you rely on your CGM to wake up the app when it is in the background or the phone is locked
- If your CGM does not supply a heartbeat, the app with stop automatically running when it is not open

The Omnipod 5 pods were launched in 2022 and use Bluetooth, so they connect directly to the phone and don't require a RileyLink or similar device. The Omnipod 5 pods are designed to be used with SmartAdjust, an automated looping algorithm that runs on the pod and connects directly to a CGM. This algorithm is only available when used with the official Omnipod 5 app or Controller. When paired with Trio, the Omnipod 5 pod will not connect to a CGM and only receive commands (automated and manual) from the Trio app running on an iPhone. If you want to use the SmartAdjust algorithm on the pod, you have to pair a brand new pod with the Omnipod 5 app or Controller. To reiterate: your Omnipod 5 pod and CGM must both be in-range of and connected to your iPhone running Trio in order for automated looping to occur.

Omnipod DASH

The DASH Pods were launched in 2019 and use Bluetooth, so they don't require a RileyLink or similar device as they connect directly to the phone. DASH Pods are easy to identify with their blue tab instead of the clear tab used on Eros and Omnipod 5.

Omnipod Eros

Eros pods were launched in 2013 and continue to be sold by Insulet. As of December 2023, they are no longer available in the US but are obtainable in other countries for now. Eros pods are also referred to as "Omnipod System", "Omnipod Classic", or "Gen3". The Eros uses radio waves for communication between the Pod and the PDM. The iPhone does not support these radio waves, so in order to use Eros with Trio, you need to get a separate device that can translate radio waves to Bluetooth. These devices are called RileyLink, EmaLink, etc.

Sooil Dana pumps

Every Dana pump has built-in BLE communications. Therefore, no RileyLink-compatible device is needed to use Dana-i / DanaRS-v3 with the *Trio* app.

Dana-i

The Dana-i is the latest and greatest from the Korean pump manufacturer *Sooil*, released in 2020.

DanaRS

The DanaRS was first released in 2002, with firmware version v1, which is not supported at this time. Only firmware version v3 and onwards are supported with the *Trio* app. Check here to see how to check your firmware version. If you do have v1, though, please reach out to the Trio development team if you're interested in helping v1 get supported by running some tests.

Medtrum Nano

> **Medtrum Now Supported**
>
> Both the 200U (MD0201 & MD8201) and 300U (MD8301) version are supported in Trio 0.7 and later

Every Medtrum patch pump has built-in BLE communications. Therefore, no RileyLink-compatible device is needed to use Medtrum patch pump with the *Trio* app.

The Medtrum patch pump is the second tubeless pump option available to the *Trio* app. The patch pump is 13.8gram, 40.5mm X 31.5mm x 11.5mm, making it the smallest and most discreet patch pump on the market as of now. It uses a 90-degree steel cannula instead of the Teflon cannula of the Omnipods, with a virtually painless insertion.

Medtronic Pumps

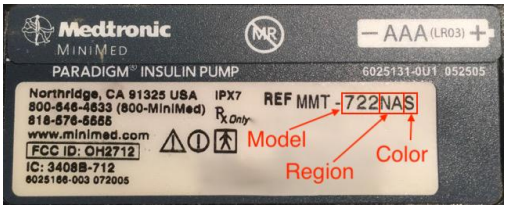
There are a number of Medtronic insulin pumps manufactured between 2006 – 2012 which are Trio compatible. Compatibility has two requirements: (1) pump model and (2) firmware.

- Medtronic 515 or 715 (any firmware)
- Medtronic 522 or 722 (any firmware)
- Medtronic 523 or 723 (firmware 2.4 or lower)
- Medtronic Worldwide Veo 554 or 754 (firmware 2.6A or lower)
- Medtronic Canadian/Australian Veo 554 or 754 (firmware 2.7A or lower)

If you have one of the pumps listed above, you are good to go on Trio! Congrats!

Medtronic Pump Model

To determine your pump model, look at the backside of your pump. There should be a sticker on the underside of the pump. On the right-hand side of the sticker, it says REF MMT-XXXXXX




```
* MMT ---> Pump Manufacturer Model (MiniMed Medtronic)
* 722 ---> Pump Model Number
* NA ---> Pump Region (NA=North America, CA=Canada/Australia, WW=Worldwide)
* S ---> Pump Color (S=Smoke, L=Clear/Lucite, B=Blue, P=Pink/Purple)
```

Some pumps may have an “L” or “S” or "R" before the pump region, e.g., a model number like MMT-722LNAS. This does not affect Trio compatibility.

Medtronic Pump Firmware

A pump’s firmware is the internal software that runs your pump. Older Medtronic firmware allows Trio to act as a “remote control” to set temp basals and report back pump data. Newer firmware has disabled that “remote control” access and, therefore, cannot be used with these DIY closed-loop systems. There is currently no ability to downgrade a pump’s firmware or replace it with older firmware. Before you buy a used pump, make sure you are getting one with compatible firmware. **You cannot change the firmware on a Medtronic pump.**

The firmware on all 515/715 and 522/722 model Medtronic pumps is compatible with Trio. You will only need to check the firmware version for 523/723 and 554/754 model Medtronic pumps.

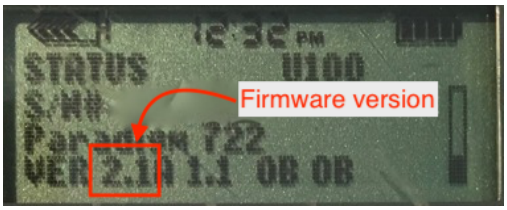
- Medtronic 515 or 715 (any firmware)
- Medtronic 522 or 722 (any firmware)
- Medtronic 523 or 723 (firmware 2.4 or lower)
- Medtronic Worldwide Veo 554 or 754 (firmware 2.6A or lower)
- Medtronic Canadian/Australian Veo 554 or 754 (firmware 2.7A or lower)

To find your pump's firmware, you will need to power it on. If the pump has not been powered on for some time (i.e., it has been in storage without a battery for a while), it will run through a start-up count, and the firmware version will appear on the bottom right of the pump's screen. Don't turn away, as the version number will only be displayed for a little while before the screen moves on to other information displays.

If the pump has been active recently or has a reservoir installed, follow these steps:

1. Press the

button on your pump.
2. Scroll down to the bottom of the status display by clicking the button.
3. Read the bottom line of the display.



Medtronic Pump Differences

If you are in the position of being able to shop around for different pump models, there are some slight differences between the Trio-compatible Medtronic pumps.

500 vs 700: The difference between the Medtronic 500 series and the 700 series pumps is the size of the insulin reservoirs. The 500 series pumps use a 180-unit reservoir, and the 700 series pumps use a 300-unit reservoir (or a smaller 180-unit reservoir, if you want).

x15/x22 vs x23/x54: The noteworthy differences between the x15 and x22 pumps versus the x23 and x54 series pumps are:

- The x23/x54 pumps will allow for smaller insulin deliveries in certain situations if the smaller scroll rate is selected in the Bolus>Setup>Scroll Rate menu. **Trio will have the insulin delivery automatically rounded by the pump to the units available in the pump model, and any smaller adjustments (to make up for the rounding) will be made through Trio's use of temp basals. If you want the smaller increments of basal rates, you can still enter those values in the Trio app's settings and Trio will use those values for the purposes of insulin delivery calculations.**

Pump Model	Basal increments	Bolus increments	Range
515/715 and 522/722	0.05 0.1	0.1 0.1	deliveries of less than 10 units greater than 10 units
523/723 and 554/754	0.025 0.05 0.1	0.025 0.05 0.1	between 0.025 to 0.975 units between 1 to 9.95 units greater than 10 units

- Additionally, because of the way Trio fetches information from the pump, the x23/x54 series of pumps are slightly better at conserving battery life through the use of the MySentry packets to collect information from the pump. x22 pumps do not use MySentry.
- The x23/x54 series pumps are also faster at delivering boluses greater than 10 units. On an x23 pump, a 13-unit bolus takes 5:00 minutes to complete. On an x22 pump, a 13-unit bolus takes 8:40 minutes to complete.

Finding a Medtronic Pump

Finding a compatible Medtronic pump is probably the most difficult part for most new Trio users. Our suggestions:

- Talk to friends in the diabetic community.
- Ask your endocrinologist.
- Ask at a local JDRF chapter meeting if someone has an old backup pump they'd be willing to donate to you.
- Join diabetic supply groups on Facebook, both for-trade and for-sale groups. Looping in a time of COVID is a group focused on DIY looping devices and supplies. Be sure to read their rules before joining.
- Check Craigslist often and be willing to expand your search area to include larger cities.
- Check out the **HelpAround, NextDoor, OfferUp, and LetGo** apps for pumps.

The most successful results appear to come from either one-on-one discussions with fellow diabetics/doctors or the use of apps (Craigslist, NextDoor, LetGo, HelpAround). If you are using Craigslist, you may use an app on your iPhone to make the searching easier. There are apps to search multiple cities for pre-set keywords and set up alerts.

Safe Purchasing

If you choose to purchase from a remote or unknown seller, here are some tips for safe purchasing:

- Use PayPal and purchase using the "Goods and Services" payment option. This costs nothing for the buyer, but the seller will lose 2.95% of the sale to PayPal fees. PayPal offers some protection for both buyer and seller in

the event of fraud.

- Ask for photos of the pump. Check the serial number on the backside of the pump and verify it matches the pump's serial number shown in the display menu. Ask for a short video of the pump or at least a photo of the pump turned on so that you can see the pump's firmware and model number. Cracks and some wear on these pumps are expected. These pumps are not usually free of marks. Many people are successfully looping on pumps with cracks and rub marks, but you may want to ask if you are concerned about any you see in photos.
- Beware if the bottom of the reservoir/motor sleeve has the drive support cap pushed out, as shown here. Those pumps will generally not work (or only work intermittently); however, some people have successfully repaired those pumps, as shown in that link. Just to make sure you know, it should be checked in advance.
- Repairs to cracks or missing bits of plastic on battery cap area and reservoir caps are possible and not very difficult in most situations. You can read more about how to repair those here.
- Ask for shipping that includes a tracking number. USPS Priority Mail's smallest box is a great option. It's only $10.40 domestically in the US and provides tracking. Ask the seller to add some packing protection around the pump, such as bubble wrap, to keep it safe during shipping. Ensure you get a tracking number within a reasonable period after you have paid.

Red flags that may indicate a scam:

- Asking for payment through "friends and family" on PayPal, especially if you don't know the person or have any solid references for them. Paying in that way offers you no buyer protection. It's just like giving the seller cash, so you had better trust the seller.
- Offering an "almost new" pump is a big red flag. These pumps were manufactured between 2006-2012. It is highly unlikely that a pump of that age is unused and sitting in shrink wrap. There are some out there, but they are scarce.
- The seller is not able to provide new pictures of the pump when requested. Sure, they posted some photos with the ad, but what if they downloaded them from other people's ads? The seller should be able to furnish a couple of "new" photos at your request. A good one to ask for is the battery and reservoir tops so you can see their condition.

Pump Supplies

Medtronic will not typically sell pump supplies directly to customers who have yet to purchase a registered Medtronic pump. Ask your insurance about purchasing pump supplies through a durable medical equipment (DME) provider. Typically, the DME provider will coordinate with your insurance and doctor's office to get the necessary insurance approval and prescriptions for the supplies. If you are brand new to Medtronic infusion sites, ask for help from friends to try a variety of infusion sets before purchasing an entire 90-day supply of any type in particular.

Tandem Pumps

Tandem Mobi

> **Tandem Mobi in Development!**
>
> Work is ongoing to bring more pumps to the OS-AID ecosystem. While nothing is certain, progress is being made on Tandem Mobi.

0,5
1,0
1,5
2,0

Compatible CGM

✅ The following CGMs can be used with Trio:

- Dexcom G5
- Dexcom G6 / ONE
- Dexcom G7 / ONE+
- FreeStyle Libre 1 / 2 / 2 Plus (limited models)
  - US Libre 1 10-day sensors (via transmitters)
  - US Libre 1 14-day sensors (via transmitters)
  - International Libre 1 sensors (via transmitters)
  - European Libre 2 / 2 Plus sensors (via transmitters)
  - European Libre 2 / 2 Plus sensors (direct via Bluetooth)
- FreeStyle Libre 3 / 3 Plus (Open-beta)
  - Libre 3 and Libre 3 Plus, all models (direct via Bluetooth)
- Eversense E3 / 365 (Open-beta)
- Accu-Chek SmartGuide (TESTING)
- xDrip4iOS
- (Nightscout)
- (Glucose Simulator)

Nightscout

While using Nightscout as a CGM is an option, it should be avoided if possible because it will not keep Trio running in the background like other CGM options. You will have to open Trio manually to make it run loop cycles.

xDrip4iOS

To use xDrip4iOS as a CGM source, you must build it yourself with the same Apple Developer account you used to build your Trio app. You cannot use Shuggah or a version distributed by someone else's TestFlight. Please see the following for instructions on how to build xDrip4iOS yourself: link

However, if you are using Dexcom G6 or ONE with xDrip4iOS, you can choose the Dexcom G6 option in Trio rather than xDrip4iOS, and Trio will intercept the glucose readings even if you're using Shuggah or someone else's TestFlight of xDrip4iOS.

Dexcom G5 / G6 / ONE

Trio can intercept glucose readings between the transmitter and the Dexcom app.
If you are using a Dexcom G5, G6, or ONE sensor, tap `Settings` > `Devices` > `Continuous Glucose Monitor`, add your CGM to enter your transmitter's 6-digit ID.
Dexcom Share Credentials are not necessary.

When switching transmitters, you must delete your current transmitter from Trio by tapping `Settings` > `Devices` > `Continuous Glucose Monitor` > `Dexcom G6 / ONE`, scrolling down, and tapping `Delete CGM`.
Once you do this, you can add the new transmitter with its Transmitter ID.
Remember to enable `Upload Readings` to have *Trio* send glucose to *Nightscout*.

Dexcom G7 / ONE+

Trio can intercept its glucose readings as long as the Dexcom G7 or ONE app is installed on the same phone. When a new G7 sensor is paired to the Dexcom G7 app, or a new ONE+ sensor is paired to the Dexcom ONE+ app, Trio will automatically start reading it.

Glucose Simulator

The Glucose Simulator adds artificial CGM readings to the screen so you can see how your readings might look in the app. When using this CGM option, you cannot manually influence the readings shown to reflect a desired glucose response. Actions taken by the Trio algorithm also do not affect the CGM readings in the Glucose Simulator. They are for visual purposes only. For this reason, using the Glucose Simulator will not help you understand how the algorithm influences blood sugars.

> **Warning**
>
> ***The Glucose Simulator should never be used in conjunction with a live pump connected to a person (or animal).***

FreeStyle Libre 1 / 2 / 2 Plus

This option pairs a compatible Libre 1, Libre 2, or Libre 2 Plus sensor directly with Trio without using a separate app like xDrip4iOS. For Libre 3 and Libre 3 Plus sensors, see FreeStyle Libre 3 / 3 Plus.

> **Libre 2 Direct not supported with a FREE Apple Developer Account**
>
> Pairing a Libre 2 / 2 Plus sensor directly (Libre 2 Direct) requires *Near Field Communication (NFC)* tag reading.
> ❌ This capability is not available when building Trio with a **free** Apple Developer Account.
> ✅ Libre 1 and Libre 2 / 2 Plus sensors read through a third-party transmitter do not use your phone's NFC and can be used with a free account.

Supported Sensors

- FreeStyle Libre 1 / 2 / 2 Plus (limited models)

- US Libre 1 10-day sensors (via transmitters)
- US Libre 1 14-day sensors (via transmitters)
- International Libre 1 sensors (via transmitters)
- European Libre 2 / 2 Plus sensors (via transmitters)
- European Libre 2 / 2 Plus sensors (direct via Bluetooth)

Unsupported Sensors

❌ Unsupported FreeStyle Libre sensors

- Non-European FreeStyle Libre 2 (US models not supported)
- FreeStyle Libre Pro
- FreeStyle Libre H

FreeStyle Libre 3 / 3 Plus

***Open beta**

FreeStyle Libre 3 / 3 Plus is currently only available in the dev branch. See here for help on building the `dev` branch with Browser Build.

Only One App Can Read the Sensor

A sensor started in Trio can no longer be read by the FreeStyle Libre 3 app, and a sensor started in the FreeStyle Libre 3 app cannot be paired to Trio. If you want to switch to a different Trio or Loop app, though, you can enter the Receiver ID in the new app from the Libre 3 driver in whichever app started the sensor (Trio or Loop). This also means the FreeStyle Libre 3 app's low and high glucose alarms are not active while the sensor is paired to Trio, so set up Trio's own Glucose Alarm Limits. LibreLinkUp and LibreView are also not accessible, so look to use Nightscout, Tidepool, and Loop Follow instead.

Trio supports both the FreeStyle Libre 3 (14 days) and the FreeStyle Libre 3 Plus (15 days) sensors, in all regions. The sensor is paired to Trio via NFC and then uses a direct Bluetooth connection to receive readings every minute, though it uses only one reading every five minutes. The one-minute readings are still viewable in the Libre 3 driver, though.

FreeStyle Libre 3 / 3 Plus not supported with a FREE Apple Developer Account

Pairing requires *Near Field Communication (NFC)* tag reading.
❌ This capability is not available when building Trio with a **free** Apple Developer Account.
You need a **paid** Apple Developer Account for that. See Apple Developer Account.

Eversense E3 / 365

> ⚠ ***Open beta**
>
> **Eversense is currently only available in the dev branch.**

> ⚠ **Internet Required for First Use**
>
> **Internet access is required only for the initial Eversense login when pairing the transmitter to Trio; afterward, an internet connection is no longer required for readings to populate.**

Trio supports both the Eversense E3 (90 days & 180 days) and the Eversense 365 (full year) transmitters.

Accu-Chek SmartGuide

> ⚠ ***Feature Branch**
>
> **Accu-Chek SmartGuide is currently only available in the `dev` branch. See here for help on building the `dev` branch with Browser Build.**

Trio suports the Accu-Chek SmartGuide CGM. This is a 14-day CGM from Roche. It requires 2 calibrations: one after 12h and one 0.5-2.5h after the first calibrations. These calibrations are required, but can be done within Trio.

You can use this CGM with one Trio (including calibrating the CGM) or use it along side the official SmartGuide app and SmartGuide Predict app.

Compatible Phones

Trio requires a relatively recent iPhone. The faster the phone, the better for the algorithm and battery life. You need a minimum version of the mobile operating software, called the phone's "iOS", to be installed on your iPhone. Trio is compatible with iPhones running on iOS 17 or newer.

iPhone

Trio is compatible with phones running on **iOS 17 or newer**.

These iPhones are compatible with iOS version 17 and newer:

- iPhone 17*, all variants
- iPhone 16*, all variants
- iPhone 15, all variants
- iPhone 14, all variants
- iPhone 13, all variants
- iPhone 12, all variants
- iPhone 11, all variants
- iPhone XS Max
- iPhone XS
- iPhone XR
- iPhone SE (3rd generation; 2022 first release)
- iPhone SE (2nd generation; 2020 first release)

iPhone 16 (all) and iPhone 17e with Atlas/InPlay DASH pods

Using an iPhone 16 (the entire line) or an iPhone 17e with Atlas/InPlay DASH pods can cause communication issues, but SAW/TWI DASH pods, Omnipod 5 pods, and all other pumps work fine. More info is available at LoopDocs.

iOS Version

If your iOS version is not listed, e.g., 26.4.1, choose the first row that is lower than your iOS version.

iOS Version	Minimum Xcode Version	Minimum macOS Version
26.4	16.4	15.3
18.1	16.1	14.5

18.0	15.4	14.5
17.5	15.4	14.0
17.4	15.3	14.0
17.0	15.0	13.5

iOS Version and Developer Mode

Find Your iOS Version

Your phone's iOS version can be found under the phone Settings -> General -> About display. The iOS number is found on the `Software Version` line. When you start looping, you should delay iOS updates until others have confirmed the software functions as normal on the newest update.

> **Warning**
>
> Using iOS Beta versions on the iPhone used for insulin delivery is **never** recommended.

Developer Mode - Mac Build

With iOS 16 and watchOS 9, Apple added a feature called "Developer Mode." This feature must be enabled on your iPhone and Apple Watch to allow the installation of Trio. To learn more, click on this link: Apple Link about Developer Mode.

When you build Trio on your phone from Xcode directly, you must enable Developer Mode. This is also required in order to use the Trio app on a watch paired to your phone.

> **Developer Mode with iOS 16, watchOS 9**
>
> If you already have Trio, built with Xcode on a Mac, on your phone/watch, when you update to iOS 16/watchOS 9, you will be told that Trio requires Developer Mode to run.
>
> You cannot run Trio on your phone (or watch) until you have enabled Developer Mode on the device(s).

Automatic Updates

Apple provides frequent updates to the iOS. These updates often include critical security patches and improved new features.

Updates to iOS *could* make Trio stop working.
- **Minor** updates (like 16.5 -> 16.5.1) are usually not an issue.
- **Major** updates (like 17.x -> 18.x) often contain changes that Trio developers need to include in the app.

Turn Off Automatic Update Installs!

- Once you accept an iOS update, you cannot go backward
- Some iOS updates require updates to Xcode and macOS before people can build Trio on that device again
- It is rare, but iOS updates have caused Trio to stop working until other updates were made and Trio was rebuilt on that phone
- Turn off automatic updates so you can choose when to update your phone and avoid being caught without a working Trio app
- Google the instructions for your device:
  1) Configure your phone to *download* the updates automatically
  2) Choose to perform the *installation* of the updates manually

Battery Health

Make sure your phone's battery is functioning optimally. Your phone will become a critical health device, and you want it to keep working.

- Make sure a charger and cord are in your diabetes supplies
- Consider buying a battery pack, keep it charged, and add it to your travel bag

Low Power Mode

Some people have reported that Trio continues working in the background (phone locked) even in Low Power Mode. Others have reported that they still get red loops. You can experiment to determine if your phone/iOS/app is able to maintain green loops in low-power mode. Otherwise, the best practice is to avoid Low-Power Mode.

Apple Watch

Trio has a companion app for the Apple Watch. The companion app shows the loop status and current insulin and carbs onboard. It also lets you add carbs, do a manual bolus, and set a temporary target. However, the companion app only serves as a remote control for the iPhone app; connection to the CGM and pump is still done through your iPhone, and you will have to bring it with you.

Apple Watch Series 3 and newer are supported. Newer versions of the Apple Watch have better battery life and support always-on display.

Mac

The Mac version requirements to support building with a Mac are found in the Compatibility Matrix. However - those requirements can change rapidly.

- If you plan to Build with a Mac, you should assume you need the most recent macOS and Xcode
- If you do not normally keep your Mac up to date, it is recommended you use Build with Browser

Version Compatibility Matrix

This page provides a compatibility matrix listing the supported versions of iOS, Xcode, and macOS for building Trio. It helps ensure you are using the correct combination of tools and operating systems when creating your own version of Trio.

Compatibility Matrix

iOS Version

If your iOS version is not listed, e.g., 26.4.1, choose the first row that is lower than your iOS version.

iOS Version	Minimum Xcode Version	Minimum macOS Version
26.4	16.4	15.3
18.1	16.1	14.5
18.0	15.4	14.5
17.5	15.4	14.0
17.4	15.3	14.0
17.0	15.0	13.5

Check your Software Versions

Click the links below to check your version of:

- iOS
- Xcode
- macOS

Apple Developer Account

Overview

Time Estimate

- 15-20 minutes to complete the enrollment forms.
- Up to 2 days to wait for a confirmation email that enrollment has been activated.

Summary

- There are two options: Paid ($99/year) or Free (rebuild weekly, Xcode only).
- Paid Developer Account: Go to the Apple Developer website to enroll in an individual account.
- Free: No action is required at this time. Continue to Prepare your Computer.

FAQs

- **How long does activating my Apple Development account after enrolling take?** After you enroll, make sure you look for a confirmation email. Apple says confirming and setting up a new Apple developer account can take 24-48 hours. However, some people have had the process take minutes. One sure way to make it take longer is to use a different credit card to pay for the Apple Developer account enrollment than is already associated with that Apple ID. When you do that, finishing the enrollment process may be a hassle.
- **Do I use my Apple ID or my child's Apple ID to enroll in the Apple Developer program?** The Apple ID you use to enroll in the developer program must be associated with an adult. It does not need to be the same Apple ID the PWD uses on their iPhone. For example, a parent installing the *Trio* app on their child's iPhone should configure a separate Apple ID for their child but use their own Apple ID to build the Trio app for that child.

Apple Developer Program

To build the Trio app on a phone, you must use an Apple developer account associated with an adult (minimum age of 18). The Apple Developer account is tied to the email address associated with your Apple ID. You can build apps on phones for everyone in your family with a single Apple Developer Account tied to the Apple ID of an adult.

You have two options for an individual account: free or paid.

Free Developer Account

If you decide to use a **FREE** developer account, here's what you need to know:

- You can only use the Build Trio with Mac method to build Trio.
- Apps signed with a free developer account will expire after 7 days. On the 7th day, your Trio app will turn white when you open it and immediately close. To rebuild Trio, you must find a computer and rebuild the app onto your

iPhone again. You cannot rebuild the app on day 5, hoping to reset the 7-day clock. The app will still expire on the 7th day from when it was first signed and created.

- If you switch to a paid account after trying out the free account, you must rebuild your Trio app to sign it with the new paid account. Furthermore, switching from a free account requires entering all the settings again and starting a fresh pod.
- You will have to do an extra step during the build process to remove Siri and Apple Push capabilities to build with free accounts. Because free accounts do not have access to Apple Push notifications, you will also not be able to use Remote Commands through Nightscout.

Most Libre Sensors Require a Paid Account

Pairing a sensor directly to *Trio* — Libre 2 / 2 Plus using Libre 2 Direct, or Libre 3 / 3 Plus — requires the Near Field Communication (NFC) capability, because your phone must scan the sensor itself to start it. That capability is not available with a free account. Each new sensor needs its own scan, not just the first one, so this is not a step you can work around.

Libre 1 and Libre 2 / 2 Plus sensors read through a third-party transmitter (Miao Miao, Bubble, etc. — now uncommon) do not use your phone's NFC and can be used with a free account. Some Libre models cannot be used with *Trio* at all — see Compatible CGM.

Live Activity Requires Paid Account

The Live Activity feature for *Trio* is not available with a free account.

Paid Developer Account

If you decide to use a **PAID** developer account, here's what you need to know:

- The paid developer account is $99 per year. The default setting is to auto-renew annually. You can change that selection in your developer account settings at any time.
- It can take up to two days for your Apple Developer account to become active.
- If your household has multiple Trio users, only one developer account is needed. This account can be used to build Trio on multiple phones; however, each phone must have its own Apple ID.
- If you use the Build Trio with Browser method you must update the build once every 90 days.
- If you use the Build Trio with Mac method, you must build it at least once a year.

Enrolling in the Apple Developer Program

To enroll, go to the Apple Developer Program website Apple Developer website.

Tip

Make sure you use the credit card already associated with the Apple ID. Switching credit cards can cause delays.

If you choose to use the free account, you don't have to do anything on that website.

Browser Build Overview

Build Trio with Browser - Overview

Congratulations on choosing to build Trio using your web browser, aka "Browser Build." This documentation provides all the necessary instructions to complete your first browser build and to update future builds.

Requirements:

- Compatible Phone ↗
- Compatible CGM ↗
- Compatible Insulin Pump ↗
- Apple Developer Account ↗
- GitHub Account ↗
- TestFlight ↗

Getting Ready to Build

The first step in getting ready to build Trio using Browser Build is to review the above requirements. These are considered the minimum requirements for Building Trio using this method.

The second step is to read over and become familiar with the build instructions in their entirety. This includes familiarizing yourself with common Build Errors. Most mistakes you can make while building Trio have already been made, and our dedicated community has documented necessary fixes.

When you are ready to proceed, take your time, read carefully, and complete the tasks on each page. Read the top three boxes on each page; icons for those boxes are displayed below for reference.

Summary of Steps

Time Estimate (click to open/close)

- If you have never built the *Trio* app (or any other OS-AID app) allow up to one week elapsed time
  - Request and get an *Apple Developer* Account: 1-2 days
  - Create and configure your *GitHub* account and `repositories`: 1-2 hours
  - Add `Secrets` (requires *Apple Developer* Account): 1 hour
  - Perform the *GitHub* `Action` steps: 30 minutes to 2 hours
- If you have previously built the *Trio* app with *Xcode* you have fewer steps and are probably familiar with some of the concepts
  - Expect 1 to 4 hours

Summary for Browser Build with Links (click to open/close)

You may be overwhelmed at first with all the new terms and the many steps. Take a deep breath and work through the steps one at a time.

- Use the template to Save Your Information ↗

If you prefer watching your instructions, a narrated video is available:

- How to Build the *Loop* App With a Web Browser
  - The steps for building *Loop* and *Trio* are similar, so this video may be helpful

Once you have your *GitHub* and *Apple Developer* accounts, the steps below are a high-level summary with links to the detailed `TrioDocs` section.

You can think of the first part as a scavenger hunt where you find or generate and save six `Secrets`. All six secrets should be saved in your `Secrets` reference file. You will need to be able to use them again.

- *GitHub*:
  - Create a GitHub account ↗ if you don't have one
  - Create and Save the `GH_PAT` Secret ↗
  - (Optional) Create a Free *GitHub* Organization ↗ (strongly recommended)
- Make up a Password ↗
- *Apple*:
  - Collect the four *Apple* `Secrets` ↗

Now it's time to use those `Secrets` to build the *Trio* app

- *GitHub*:
  - `Fork` the repository (make your copy of `Trio`) ↗
    - Using an organization? Add `Secrets` and `Variables` to your Organization and all your apps can use them
    - Not using an organization? You have to add `Secrets` and `Variables` to each repository individually
  - Add `Secrets` to your `organization` or a single `fork` ↗
  - `Action: 1. Validate Secrets` ↗
  - `Action: 2. Add Identifiers` ↗
- *Apple*:
  - Configure Identifiers for *Trio* ↗
  - Create your version of *Trio* in `App Store` (personal use only, not for distribution) ↗
  - Set up `Internal TestFlight Group for the App` ↗
- *GitHub*:
  - `Action: 4. Build Trio` ↗
    - Yes we can count, but Action 3 is included in Action 4
- Phone: Install the *Trio* app using the *TestFlight* app ↗

FAQs (click to open/close)

- **Do I need a Mac computer?** No. This can be done on any browser, although it will be easier using a computer or tablet than just using a phone.
- **Isn't it hard to build every 90 days?** The initial setup and installation take a lot of your focused time. But once you build, future updates and builds are really fast. You just need to keep your Apple Developer account up to date. Review any email that says a GitHub action failed - read the error and take action. For more information, see the next bullet.
- **Can you explain automatic building?** There's a whole page in LoopDocs on Automatic Update & Build with most of that information valid for Trio users
  - Successful Weekly Action
  - Successful Monthly Action
  - Automatic Certificate Renewal - annual event
  - Open-Source App Schedule for rebuilding
- **Can I use this for my child?** You, as the adult, can install using *TestFlight* on your child's phone. The explicit steps are provided at Install on Phone: *TestFlight* for a Child ↗ .
- **Can I still use my customizations?** Yes. Customize using Browser ↗
- **Is there a build video?** Yes. How to Build the *Loop* App With a Web Browser
  - You need to adjust when watching that because you are building *Trio* instead of *Loop* and the video is somewhat out of date

Tips and Tricks

The steps to configure for building with a browser require a lot of focused attention. Take it in steps and take breaks as needed. Some people prefer a video. Some people prefer more detailed steps on a webpage and others only high-level steps.

How to Ask for Help

First time setup should take several hours, but if you are having trouble don't get frustrated:

- Click here to find help ↗ .

Who should use a free GitHub Organization option (click to close/open)

There are several places where you see a recommendation to use a free GitHub Organization instead of building directly from your personal free GitHub account.

When you use an organization, you don't have to add 6 Secrets to each repository for every app you build. It can save a lot of time in the long run.

- Even if you just build Trio and nothing else, using an organization is pretty easy and your build works just the same as using only a personal account - but there is no particular advantage (until you decide maybe you do want to build another app)
- Who should definitely build with an organization:
  - Anyone who is a caregiver will want LoopFollow
  - Anyone who wants to use LoopFollow for the amazing alarm capabilties
  - Anyone who might want to try a different app such as xDrip4iOS

The Short Version

Complete instructions are found at this link for those comfortable with using *GitHub* and navigating the *Apple Developer* and *Apple App Connect* pages.

- Using GitHub Actions + FastLane to deploy to TestFlight

The Video Version

If you prefer watching your instructions, there is a narrated video of each step needed to build using a browser. This video was created for Loop version 3.2.x. So translate in your head - Trio instead of Loop and some steps are out-of-date.

- How to Build the *Loop* App With a Web Browser

> Do not decide to uncheck the box for Near Field Communications in the Identifer for `Trio` - that is required to build.

Click in the comments for a full index of topics. If you have issues with a particular section, use the index to advance to the relevant part of the video. Subtitles are in English. You can choose a different language but the automatic translation feature may provide translations that are not completely accurate.

The Detailed Version

If you prefer seeing step-by-step instructions, those are provided too.

Some sections have a Section Summary for those who just want high-level steps without a lot of graphics. It also provides a high-level view of what you are going to do before you do it.

- To view the summary, click on the summary header
- If the summary is all you need, use the `skip forward` symbol (⏭) to skip to the next instruction

- Or follow the detailed instructions below the summary

What if I get stuck?

- First, take some time to read over the instructions to ensure you have understood all the steps.
- Compare your screen's display with the images in the documents. Is there something different? Does yours have an error message? If so, does it guide you to the problem and solution?
- If you are still stuck, you can seek assistance via Facebook or Discord.

Introduction to Secrets

In order to build using the Browser Build method, you will need to collect and save six secrets from your free GitHub account and your paid Apple Developer account.

Save Your Information

Everyone needs to read this section!

You need to keep a digital copy of your 6 `Secrets`.

- You need to copy and paste those `Secrets` to build the app with a browser
- Make sure your editor does not change any characters in your `Secrets`; use a text-only editor like NotePad (PC) or TextEdit (Mac)
- Many people add other information to the `Secrets` file for easy reference

> **A Note about Capitalization and Spaces**
>
> In places, you use a name like "`FastLane API Key`" or "`FastLane Access Token`". Please copy from the docs to use those exact names.
>
> The `Secrets` that you add use names that are capitalized and use underscore `_` instead of spaces or hyphens. Be precise and careful.

> **Use a Text-Only Editor**
>
> **Be sure to use a Text-Only editor like NotePad (PC) or TextEdit (Mac) to archive your information.**
>
> If you use a "smart" editor, it may change small letters to capital letters at the beginning of a line when you paste items into your archive file.
>
> If even one character is capitalized when it should not be, you will get errors.

Make a `Secrets` Reference File

Please save yourself a lot of agony. Make a `Secrets` Reference folder and then create a `Secrets` Reference file in it.

You don't need to know anything about the `Secrets` to do this. Come up with your own names, but make sure you know where you put this file.

1. Decide where you want your `Secrets` saved and create a special folder
2. Open a text-only editor (do not use a Smart Editor, those might change capitalization or formatting and ruin your `Secrets`)
    - Create a blank `Secrets` Reference file using the text-only editor
    - Tap on the copy icon below to copy the template and paste it into your blank file

- Save the file

3. As you move through the steps of collecting the six secrets, fill in the file
    - The `##` characters are used to indicate sections or title (text only file)
    - There is a reference pattern for each SECRET: replace that with your SECRET or you can add your secret below the pattern to give yourself confidence you got the correct item
4. In addition to the `Secrets`, this template has places for other pieces of information you may find handy to save in the same file
5. At the very beginning is a place to indicate the last day you updated the file

Template to use for Secrets

```
MySecretsReferenceFile.txt

## This file updated on: DD MMM YYYY

Information needed to use Browser Build.
The same secrets are used for every app you build.

First some handy username and password information:

## Apple ID username for your Developer account is the associated email:
<your information here>
suggest you use your desired password tool to save the password for this account

## GitHub personal account username
<your information here>
suggest you use your desired password tool to save the password for this account

### GitHub organization name
<your GitHub organization name>
Using an organization is optional.
But it you do use one, add the name here for reference.
You always log in using your GitHub personal account to access your organization.

There are Six Secrets and One Variable:
 - the list below shows the expected format of each.
You can replace each prototype with your secret as you gather them,
 - or put your secret below a prototype.
The letter A indicates an alphanumeric character;
 - some may be small letter, some capital letters and some numerals.
The FASTLANE_KEY may have additional characters included.
  Be sure to copy the entire key including
  -----BEGIN PRIVATE KEY-----
  through
  -----END PRIVATE KEY-----

## SECRETS BELOW:

TEAMID
<looks like this>
AAAAAAAAAA

FASTLANE_ISSUER_ID
<looks like this>
AAAAAAAA-AAAA-AAAA-AAAA-AAAAAAAAAAAA

FASTLANE_KEY_ID
<looks like this>
AAAAAAAAAA

FASTLANE_KEY
<looks like this>
-----BEGIN PRIVATE KEY-----
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAA
-----END PRIVATE KEY-----

GH_PAT
<looks like this>
ghp_AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

MATCH_PASSWORD
<looks like this>
```

```
AnyThingYouWant-sameForEveryRepository

This Variable (not a secret) is required for the annual certificate renewal:

ENABLE_NUKE_CERTS = true

## Repository Names for my Fork:

If you use an organization (recommended)
https://github.com/my-name-org/Trio

or (if you don't use an organization):
https://github.com/my-name/Trio


Add additional apps here if you decide to build any
```

New Terms with `GitHub Browser Build`

The `GitHub Browser Build` may use new and unfamiliar **terms**.

These instructions can be used without knowing what any of these terms mean. But if you are curious:

Some of these **terms** have ToolTips, so hover your mouse over those - or review them in the Glossary ↗ .

- Some terms in the Glossary are not in alphabetical order to improve logical grouping. All the individual `Secrets` are listed after `Secrets` in the Glossary.
- `Actions`: available in your *GitHub* repositories to build your app
  - You must be logged in to *GitHub* to run the actions manually
  - The build action is automatically run for you once your repository is configured
- `Fork`: you make a copy of the code needed to build an app and store it in your *GitHub* account (using a free *GitHub* organization is recommended for storing your `forks`)
- `Secrets`: are required to enable *GitHub* to build an app from a repository using *GitHub* Actions
  - Six `Secrets` must be entered to enable your app (or apps) to build
  - These `Secrets` work for any branch in your fork (`main` or `dev`, for example) and if you use the recommended free *GitHub* organization to build, they work for any app you fork to your organization
  - If you build more than one app, they use the same `Secrets` and the use of a *GitHub* organization is even more strongly recommended
- Certificates: These are required to enable *GitHub* to build an app from a repository using *GitHub* Actions
  - They are manually created with the Create Certificates action
  - They expire at the end of one year
  - Certificates are automatically renewed as long as you add the required `Variable` ↗
- API `Key`: `Application Programming Interface Key`
  - You create an API key from the *Apple Developer* website
  - Doing this provides 3 of your `Secrets`
  - These secrets enable your *GitHub* account to interface with *Apple* to create your app

- `Identifiers`: are required to build the *Trio* app with `GitHub Browser Build` (these are automatically generated for you) when you run `Add Identifiers`
- `App Store Connect`: a website available for *Apple Developer*s to review apps build with your *Apple Developer* account
    - Once you purchase an *Apple Developer* annual account, you are an *Apple Developer* and have access to this site
    - Most users will not have an App until using the `GitHub Browser Build`
    - The instructions walk you through creating and naming your app: Create Trio App in App Store Connect ↗

Next Step

Ready to begin? Proceed to:

→ **Create GitHub Account**

Navigation: ← Back to Overview | Next: Create GitHub Account →

Create a GitHub Account

⏭ If you already have a *GitHub* Account, skip ahead to Create *GitHub* `Personal Access Token`.

New *GitHub* Account

If you do not already have a *GitHub* account, you need to create one. Be sure to record the email, password, and username for your *GitHub* account.

Decide on a couple of usernames that you will be happy with - this will get embedded into your *GitHub* URL. Your first choice might not be available, so be prepared with several candidates. Your personal URL will be: `https://github.com/username`.

- Click on this link to sign up for a free account: *GitHub* account signup
  - You will need to enter the **email** you want associated with your *GitHub* account
  - You will be asked to enter a **password**
  - You will be asked to enter a **username**
  - You will be asked if you want to receive email, ok to say `N` for no - you still get important account information with that email
  - Solve the puzzle to prove you're a person
  - Check the associated **email** to get the code and enter the code into github.com to confirm your account
- You should get the Welcome to *GitHub* screen
  - Indicate it is "Just me" on your team and Continue
  - Don't check anything on the next screen, just tap `Continue`
  - Select the `Free` option by selecting `Continue for Free`

The free level comes with plenty of storage and compute time to build the *Trio* app.

Create `GitHub Personal Access Token`

Section Summary (click to open/close)

Log into your *GitHub* account to create a personal access token, which you will save as `GH_PAT`.

Click to create a new `personal access token`:

- Enter a name for your token, use "`FastLane Access Token`"
- Change the Expiration selection to `No expiration`
- Select the `workflow` `permission scope` (`repo` will be automatically selected)
  - This step enables automatic building
- Click "Generate token"
- Copy the token and record it. It will be used below as `GH_PAT`

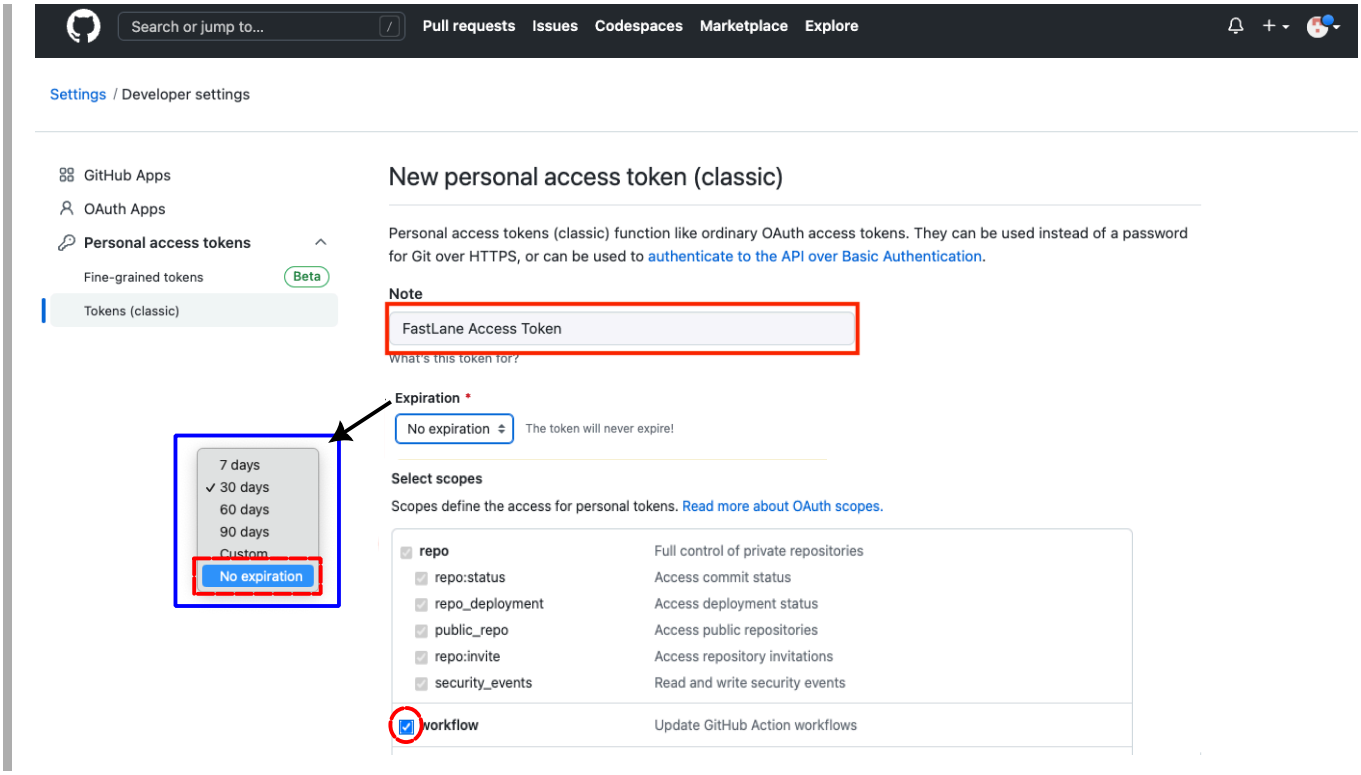
⏭ To skip the detailed instructions, click on Make up a Password.

Default Settings

These instructions are a prerequisite for automatic update and automatic build.

You must be logged into your *GitHub* account before starting this step. If you are continuing, you are already logged in.

1. You will be creating a new *GitHub* `Personal Access Token` and giving it the name "`FastLane Access Token`"
2. Open this link: https://github.com/settings/tokens/new
   - Referring to the graphic
     - Note that `Tokens (classic)` is highlighted
     - Most people use the `classic Token`
       - If you are a developer who needs to use fine-grained `tokens`, that is fine
     - Edit the note box to be `FastLane Access Token`
   - The default Expiration time is 30 days - but you should select `No expiration` (use the dropdown menu to select)
     - *GitHub* will show a yellow warning when you do this
     - It is ok to ignore the warning
   - Add a check beside the `workflow` permission scope
   - A check will automatically appear in the `repo` scope as well - this is normal
   - Scroll all the way to the bottom and click `Generate token` (it's a long way, ignore all other settings, do not check anything else)



What does `repo` and `workflow` do?

- `repo`: enables manual build actions
- `workflow`: enables `repo` and automatic build actions

3. A new screen appears showing your access token
    - Copy the `token` and record it - once you leave this screen you can't see it again
    - You will use this for `GH_PAT` when you set up your Secrets
    - You can regenerate a Personal Access Token for `GH_PAT` if you lose it, but then you have to update that in the `Secrets` for your organization or in all repositories using *GitHub* Build if you don't use an organization

Create a Free *GitHub* Organization

This step is optional but it can save you a lot of time and is strongly recommended.

Prerequisite: You need a personal *GitHub* account. If you do not already have a personal *GitHub* account, click on New *GitHub* Account.

In the instructions below, use your *GitHub* username instead of `my-name`.

1. Follow the directions below to create a new *GitHub* organization account with a username of `my-name-org` (of course naming is up to you)
    - *GitHub* provides documentation at this link, New *GitHub* Organization ↗, or you can follow the bullets below
    - Log into `my-name` and click on your icon (at upper right) and choose Settings
    - On the left side-bar, click on Organizations

- In the new view, click on New Organization and choose Free for the plan by clicking on `Create a free organization`.
- In the `Set up your organization` screen:
  - Enter `my-name-org` into the `Organization name` box
  - Enter the same email you use for `my-name` account
  - Select this organization belongs to `My personal account`
  - Check the box to accept the terms of service
  - Tap on the next button

2. You now see a Welcome screen
   - Unless you plan to collaborate with others, just tap `Complete setup`
   - You can always add others at a later time
3. Confirm access by entering the same password as you use for `my-name`

> **GH_PAT comes from personal `my-name` account**
>
> The *GitHub* personal access token used as one of the 6 `Secrets` is associated with your personal account (`my-name`); so if you already have one, you just keep using it.

Your personal *GitHub* account is still needed. The free organization points to your personal account as a member. Please do not delete your personal *GitHub* account after creating an organization. If you do that - you will no longer have access to your organization.

Make up a Password

If you have not already made up a password, do it now and record it as `MATCH_PASSWORD`. Note that if you build more than one app, you use the same `MATCH_PASSWORD` for all the apps. Every app needs to have the same 6 Secrets.

> **What is this Password used for?**
>
> The `MATCH_PASSWORD` is a special password called a *passphrase* that is used to encrypt and decrypt your certificates from *Apple* so they can be used when building with *GitHub*; it helps keeps your certificates secret and safe.
>
> A *passphrase* is a little different from a regular password. A password is what you usually use to log in to a website or game. A *passphrase* is a secret code that helps protect important files by adding an extra layer of security when files are encrypted.

Passphrase Suggestions

To make a *passphrase* that is hard to guess and keeps your certificates safe:

- Use at least 15 characters (20 is even better) and you can include spaces
- Mix in big letters, small letters, numbers, and symbols like $ or #

- Don’t use words or numbers that people might guess, like your name or birthday

If you already created a MATCH_PASSWORD that does not meet the criteria listed above, no worries, you are protecting an *Apple* developer certificate - not the family fortune. That information is provided for new builders who want guidance.

Navigation: ← Back to Introduction to Secrets | Next: Collect Apple Secrets →

Apple Secrets

You already generated two of the six required secrets when you created your GitHub account.

Now it's time to collect the other four. Remember to save the secrets in your `Secrets Reference` File ↗

You need to save your information digitally, so you can copy and paste. The information is created in one place and used in another. Refer to Configure `Secrets` ↗ for how the `Secrets` are used. In addition to the 6 `Secrets`, other important information to keep handy (like usernames and passwords) is listed below. Be sure to keep this file secure.

Created / used at developer.apple.com

- Email address (this is your username)
- password
- Four items used as `Secrets`
  - `TEAMID`
  - `FASTLANE_ISSUER_ID`
  - `FASTLANE_KEY_ID`
  - `FASTLANE_KEY`

Created / used at github.com

- Email address
- password
- username
- If your username is `my-name` then:
  - Your *GitHub* address (URL) will be: `https://github.com/my-name`
  - Your (optional but recommended) *GitHub* organization address will be : `https://github.com/my-name-org`
  - Your Trio repository address will be: `https://github.com/my-name-org/Trio`
- One item used as a Secret
  - *GitHub* Personal Access Token (`GH_PAT`)

Create yourself

- a password - make one up and save it (`MATCH_PASSWORD`)

Collect the Four `Apple Secrets`

Section Summary (click to open/close)

You will be saving 4 `Secrets` from your *Apple* Account in this step.

1. Sign in to the *Apple Developer* portal page.
2. If you need to accept a new agreement (happens at least twice a year), be sure to do so now
   - Scroll down to the Agreements section of your developer portal
3. The first *Apple* `Secret` is your Team ID.
   - Click on Membership Details on your developer portal
   - Copy the Team ID from this section. Record this as your `TEAMID`.
4. The final 3 *Apple* `Secrets` come from the creation of the "`FastLane API Key`".
   - Go to the App Store Connect interface, click the "Integrations" tab, and create a new key with "Admin" access. Give it the name: "`FastLane API Key`".
5. Record three more secrets
   - Record the issuer id; this will be used for `FASTLANE_ISSUER_ID`.
   - Record the key id; this will be used for `FASTLANE_KEY_ID`.
   - Download the `API Key` itself, and open it in a text editor. The contents of this file will be used for `FASTLANE_KEY`. Copy the full text, including the "-----BEGIN PRIVATE KEY-----" and "-----END PRIVATE KEY-----" lines.

This section provides detailed instructions for the four `Secrets` associated with your *Apple Developer* ID.

Name	Description
`TEAMID`	This 10-character identifier is associated with your *Apple Developer* ID and never changes
`FASTLANE_ISSUER_ID`	The issuer ID is associated with your *Apple Developer* ID and never changes
`FASTLANE_KEY_ID`	Key ID provided when you create an `API Key` in App Store Connect; it is associated with the `FASTLANE_KEY`

`FASTLANE_KEY`	Copy the full key from the text file you downloaded when generating the `API Key` - Filename has `FASTLANE_KEY_ID` value embedded in it.
	Include everything in the file from
	`-----BEGIN PRIVATE KEY-----`
	and ending in
	`-----END PRIVATE KEY-----`

New *Apple Developer* Account

⏭ If you have an *Apple Developer* Account, skip ahead to Find `TEAMID`.

If not, you need to purchase one ($99 annual fee). It may take a few days for the account to be enabled.

- TrioDocs has an *Apple Developer* Program ↗ page that explains in detail how to sign up for an account
- This link takes you straight to *Apple Developer* account to sign up

Find `TEAMID`

Sign in to your *Apple Developer* account at this link: *Apple Developer* portal page.

> If you need graphics for this part - use LoopDocs for reference. The secrets are all the same for all OS-AID apps.

1. Click `Account` in the top menu bar
2. If you need to accept a new agreement (at least twice a year), be sure to do so now
3. Click the `Membership Details` icon
4. Next to the `Team ID` field, is a 10-character ID number.

Record this for use as `TEAMID` in your `Secrets` file. You will also need it when you Create App Group ↗.

- Stop a moment and double-check
- If you get this wrong, you will have errors at the very end, which require you to delete some items and repeat some steps on this page

> **Do not "type" what you think you see**
>
> **Copy and paste** the `Team ID` from the webpage.
>
> - `TEAMID` must be 10 characters
> - Avoid typing an `8` when it should be a `B`

Generate `API Key`

> If you need graphics for this part - use LoopDocs for reference. The secrets are all the same for all OS-AID apps.

This step is used to create and save the final 3 `Secrets` you need from your *Apple Developer* account.

> **Paid *Apple Developer* Account is Required**
>
> To generate the `API Key`, you must have a paid *Apple Developer* account.

Click this link to open in a new tab: `App Store Connect/Access/Integrations/API`

- Click the `Integrations` tab

If this is your first time here, you will see:

- "Permission is required to access the App Store Connect API. You can request access on behalf of your organization."
- Click on `Request Access` and follow directions until access is granted

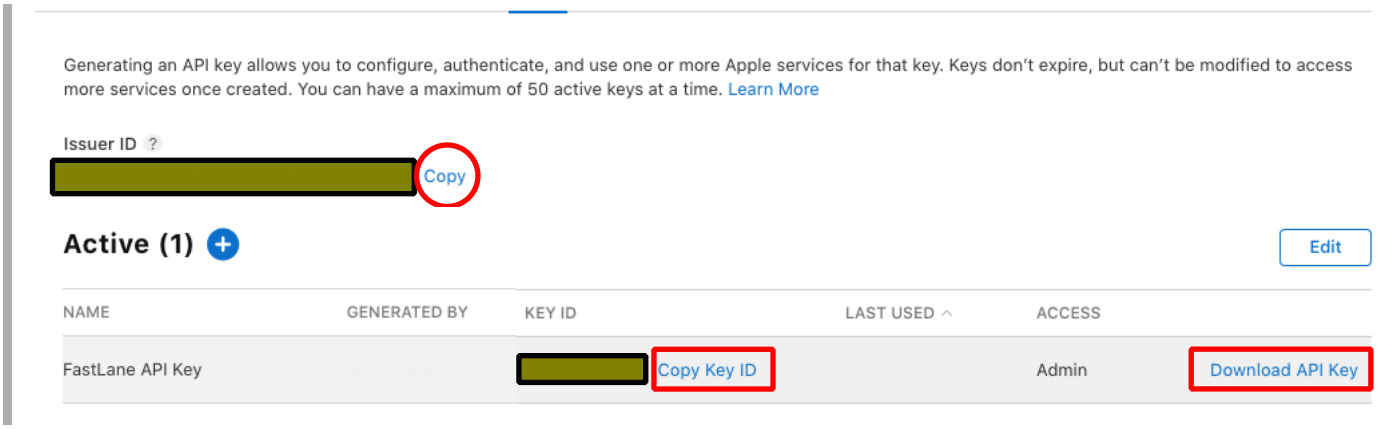
The numbered steps below correspond to the actions you take in the subsequent windows:

1. Click on the `Generate API Key` button or the blue + sign to open the `Generate API Key` dialog box
2. Enter the name of the key as "`FastLane API Key`"
3. Choose `Admin` in the access dropdown menu
4. Confirm the name and that "`Admin`" is selected and then click on the "`Generate`" button

Copy `API Key Secrets`

The `Integrations` screen appears again with content similar to the graphic below; the key information is blanked out for security.

Review the graphic and then follow the directions below to save more parameters you will need to configure `Secrets`.

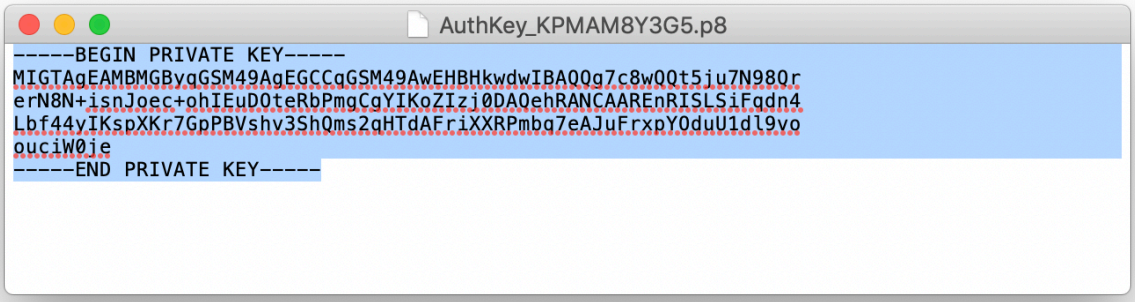


1. A button labeled Copy is always adjacent to the `Issuer ID` above the word Active (this is the same for all keys that you generate with this *Apple Developer* ID)
   - Tap on the `Copy` button - this copies the `Issuer ID` into your paste buffer
   - In the file where you are saving information, paste this with the indication that it is for `FASTLANE_ISSUER_ID`
2. Hover to the right of the `Key ID` and the `Copy Key ID` button shows up
   - Tap on the `Copy Key ID` button - this copies the `Key ID` into your paste buffer
   - In the file where you are saving information, paste this with the indication that it is for `FASTLANE_KEY_ID`
3. Click on the `Download API Key` button - you will be warned you can only download this once.
4. Find your `AuthKey` download in your downloads folder. The name of the file will be "`AuthKey_KeyID.p8`" where `KeyID` matches your `FASTLANE_KEY_ID`

   The next task is to rename the file so you can open it. Highlight the filename and choose rename, then add ".txt" after ".p8". In other words, modify `AuthKey_AAAAAAAAAA.p8` to `AuthKey_AAAAAAAAAA.p8.txt` and click on `Use`

`.txt` when questioned.

5. Double-click to open the `AuthKey_AAAAAAAAAA.p8.txt` file. It will look similar to the screenshot below. You need to highlight **ALL OF THE CONTENTS** of that file and copy it and then paste it into your Secrets Reference file as the `FASTLANE_KEY`.
   - **Click inside that file**
   - Highlight **all** the text, including the "`-----BEGIN PRIVATE KEY-----`" and "`-----END PRIVATE KEY-----`" lines and then
   - Copy **all** of the text to the clipboard (Cf. screenshot below).
     - On a *Mac*, use `⌘ Cmd` + `A`, then `⌘ Cmd` + `C` to copy all the contents
     - On a **PC**, use `^ Ctrl` + `A`, then `^ Ctrl` + `C` to copy all the contents
   - In the file where you are saving information, paste this with the indication that it is for `FASTLANE_KEY`



Organize your Key File

> **Pro Tip: Use the same folder as your Secrets Reference File**
>
> It's a good idea to keep all your important files in one place. You probably set up a folder for your Secrets Reference File ↗. Use the same folder for your API key - be sure to label the file so you know what the key is. For example, change the name from KPMAM8y3G5.p8 to API_KEY_KPMAM8y3G5.p8.

Do Not Confuse Your Keys

> **`API Key` vs `APN Key`**
>
> If you use Remote Commands with `Nightscout` or `Trio`, you may notice the Application Programming Interface (API) key has the same type of format as the *Apple* Push Notification (APN) key. The keys for both of these purposes are p8 keys, but they should not be confused with each other.
>
> The `Secrets` for building with *GitHub* use the `API Key`.
>
> The config vars for `Nightscout`, `Trio` and `Loop` use the `APN Key`.

Done with *Apple* Secrets

In summary, from this section, you have found or generated the following and saved copies for later use

- `TEAMID`
- `FASTLANE_ISSUER_ID`
- `FASTLANE_KEY_ID`
- `FASTLANE_KEY`

> **Time for a Break?**
>
> This is a good place to pause if you need to. Just note where you are on the page so you can return later.

Navigation: ← Back to Create a GitHub Account | Next: Fork and Prepare →

Fork and Prepare Trio Repository

Fork `Trio`

Important Decision: Personal Account or Organization?

Before you fork, decide whether to use a GitHub organization. This affects where you store your secrets.

Use a GitHub Organization if you:

- ✓ Plan to build multiple apps (Trio + LoopFollow)
- ✓ Want to enter secrets once for all apps
- ✓ Might try customized versions or development builds

Use Personal Account if you:

- Only building Trio (no other apps planned)
- Prefer simpler setup (fewer concepts to learn)

📖 **Haven't decided?** Review Create a Free GitHub Organization ↗ first, then come back here.

Changing Your Mind Later

You can switch to an organization later if needed, but it's easier to decide now.

Section Summary (click to open/close)

Fork https://github.com/nightscout/Trio into your account.

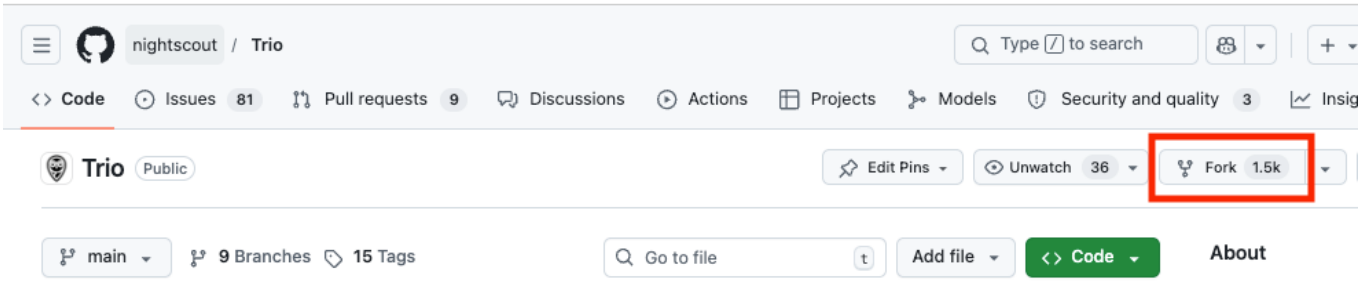
⏭ To skip the detailed instructions, click on Validate `Secrets`

Existing Fork

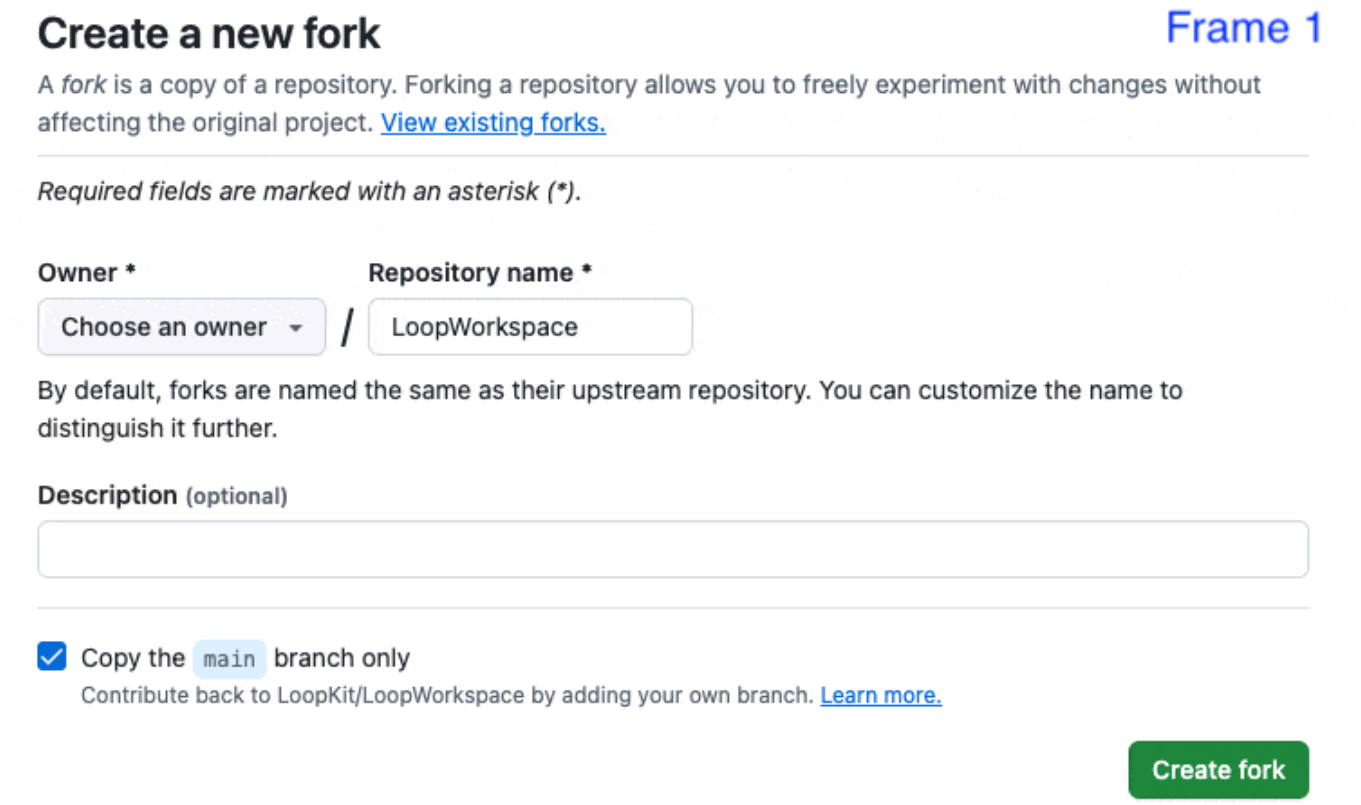
If you already have a fork of `Trio`, click on Already Have Trio to decide what to do. That section provides links to return you to these instructions.

Create the `Fork`

1. Click this link https://github.com/nightscout/Trio to open the `Trio` repository owned by `nightscout`
2. At the upper right side of the screen, click on the word `Fork`

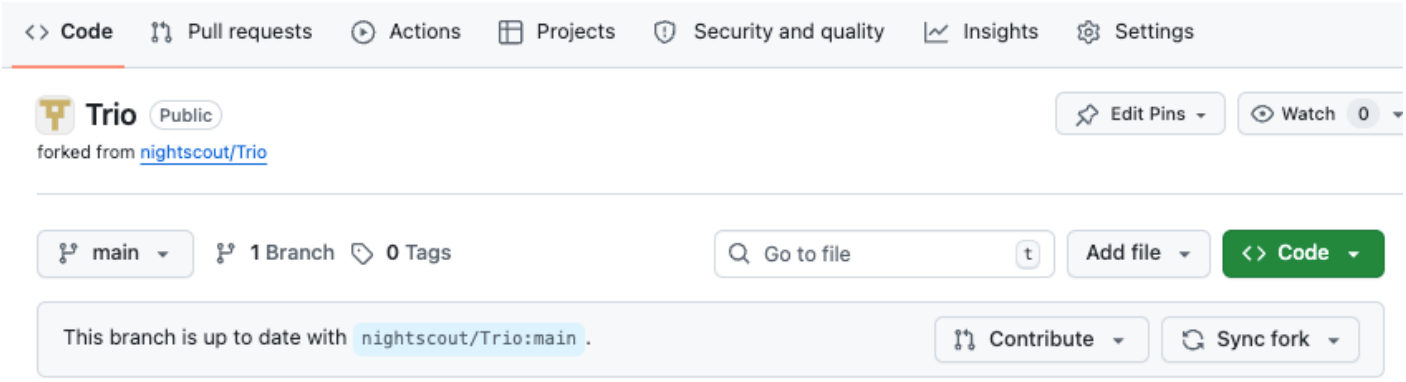


3. Refer to the GIF below - note this GIF is for LoopWorkspace - but the actions you take are the same for Trio:
    - If you created a free organization (strongly recommended), you will see the display shown in the GIF below where you will choose your organization as the owner
        - If you did not set up a free organization, `my-name` will be automatically filled in as the owner (`Owner`)
    - If you already have a fork, you should not proceed, see Already Have Trio
    - The repository name is already filled in
        - **Do not rename the repository to something else**
        - It needs to **match the original repository name** or **automatic building will not work**
    - Leave the selection that says "`Copy the main branch only`" checked
    - Click on the green `Create fork` button



Successful Fork

After creating the fork, your screen should be similar to the next graphic.



Carefully examine your screen paying attention to the bullets below.

- Note that your internet address (URL) is `https://github.com/my-name-org/Trio` or `https://github.com/my-name/Trio` where `my-name` is the name you chose
- The comment on the second row indicates where the fork came from (that is a clickable link)
- The branch that is selected is `main`
- The message says "`This branch is up to date with nightscout/Trio:main`"

Configure `Secrets`

If you set up a GitHub organization (strongly recommended), follow this set of instructions.

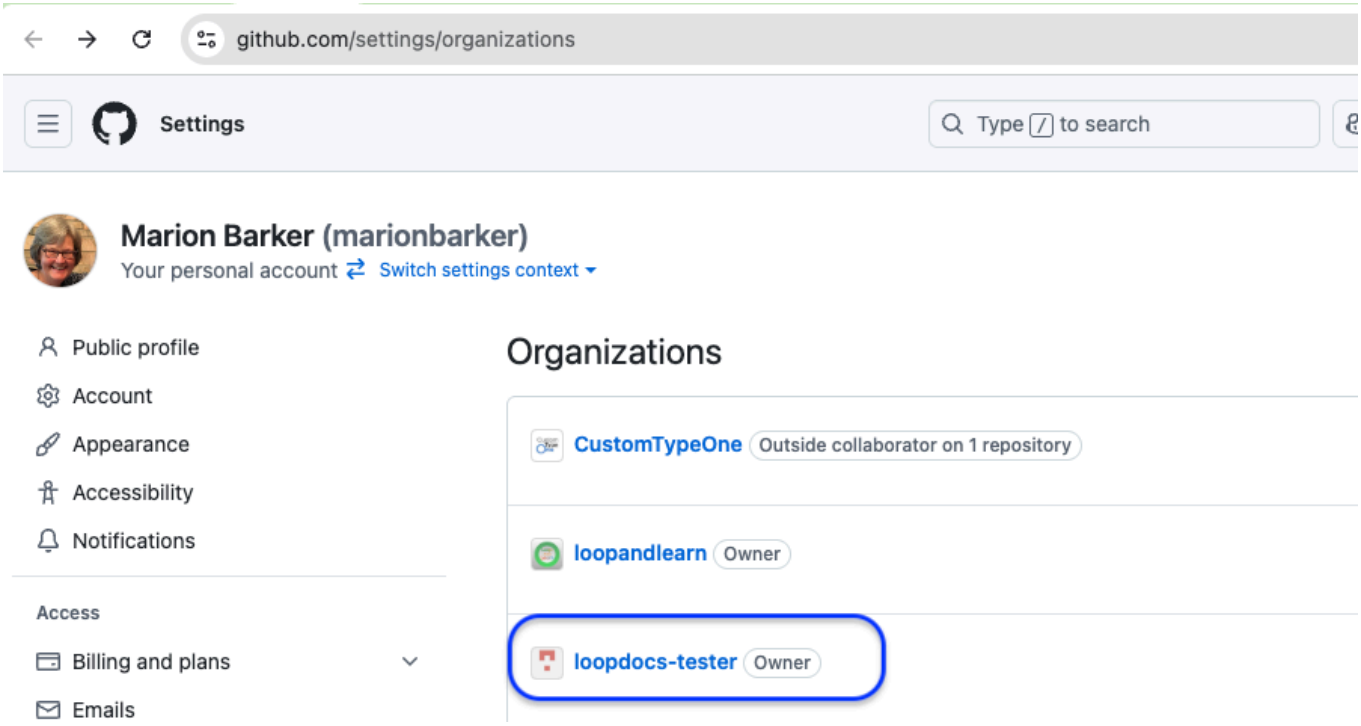
If you decided to not to use a GitHub organization, skip ahead to Personal Account: Prepare to Enter `Secrets`.

> **What if I already have a bunch of forks in my personal account?**
>
> You do not have to switch to an organization. But if you want to make the move, follow instructions here Switch to a GitHub Organization

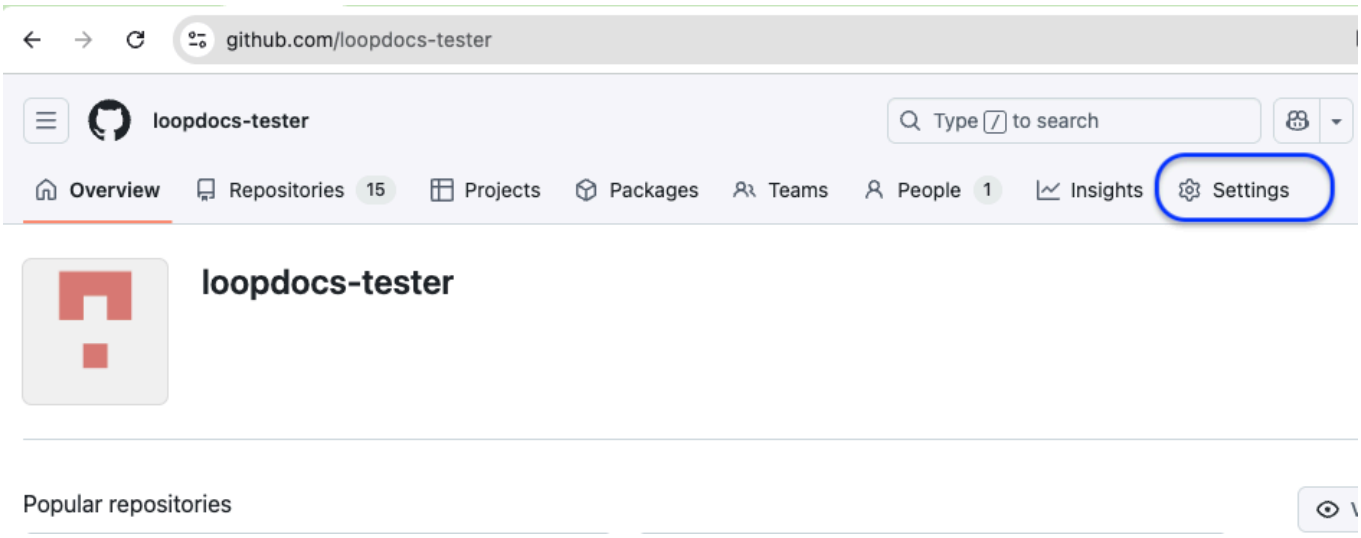
Prepare to Enter `Secrets`

You will be adding `Secrets` and `Variables` to your organization. This makes them available to any app you decide to build as long as you set up your free *GitHub* organization as the owner of the fork.

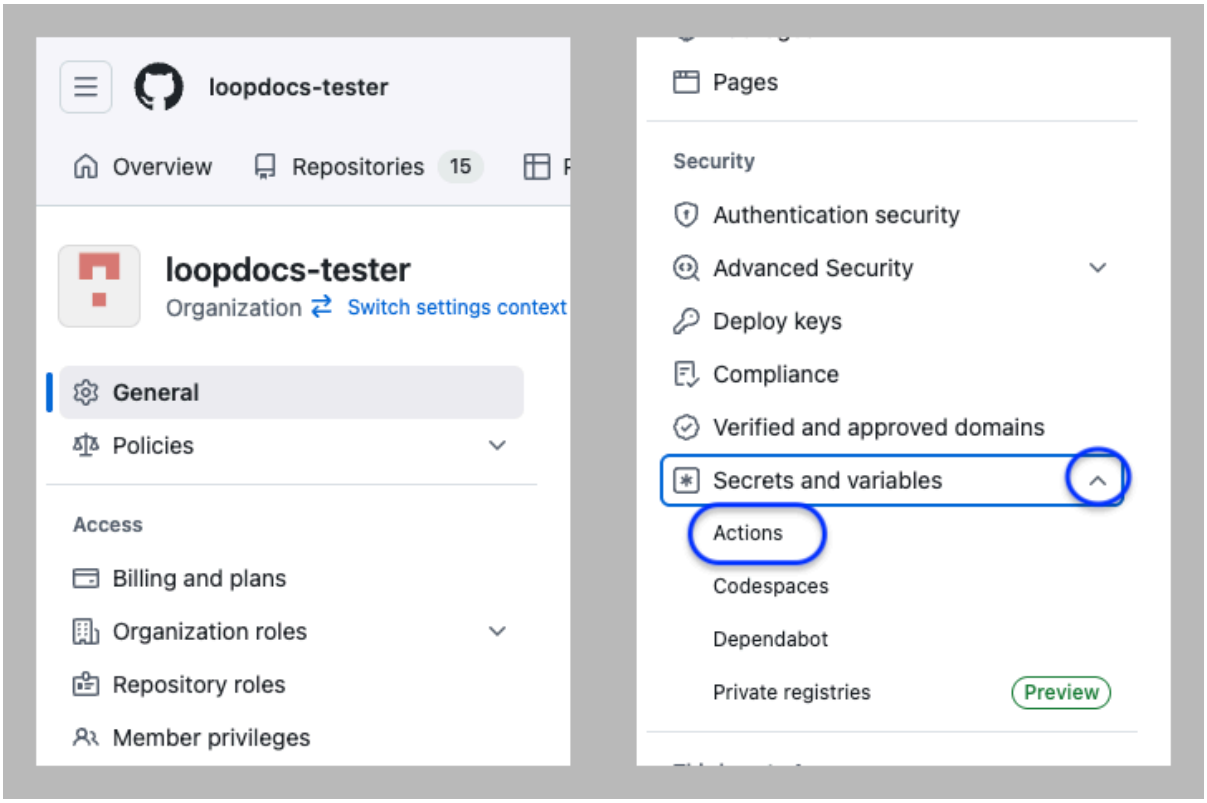
1. Tap on this *GitHub* link to see your organizations. (If you don't see a screen similar to the graphic below - you are not logged in to *GitHub*).



2. Choose your organization name from the list (most people will only see one organization)
    - I chose `loopdocs-tester` as my organization, so you will see that name in the URL for graphics in this section.



3. Click on the Settings Icon near the top right
    - If you don't see ⚙️ `Settings`, make your browser wider or scroll to the right
    - After you click on ⚙️ `Settings`, your screen will show a lot of menu items on the left side of the screen.
    - Scroll down until you can see the `Security` section with `Secrets and variables` drop down.
    - Click on the dropdown icon and then select `Actions`



The next steps are identical whether you are configuring your organizations `Secrets` and `Variables` or doing this for every repository in a personal account.

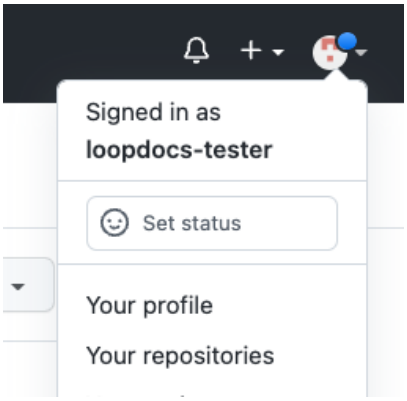
- Skip ahead to Enter the Secrets

Personal Account: Prepare to Enter `Secrets`

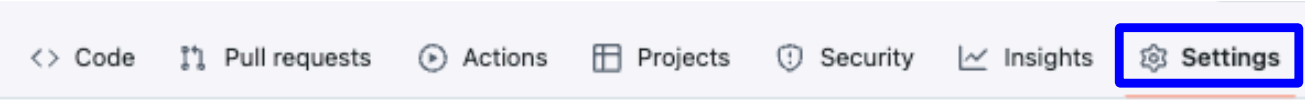
> This section is only if you are using a personal *GitHub* account to build. Using an organization is recommended. If you are using an organization, skip ahead to Enter the Secrets.

Log into *GitHub*.

1. Return to your forked copy of `Trio`
   - Click on your personal icon at the upper right to see the dropdown menu and select "`Your repositories`"

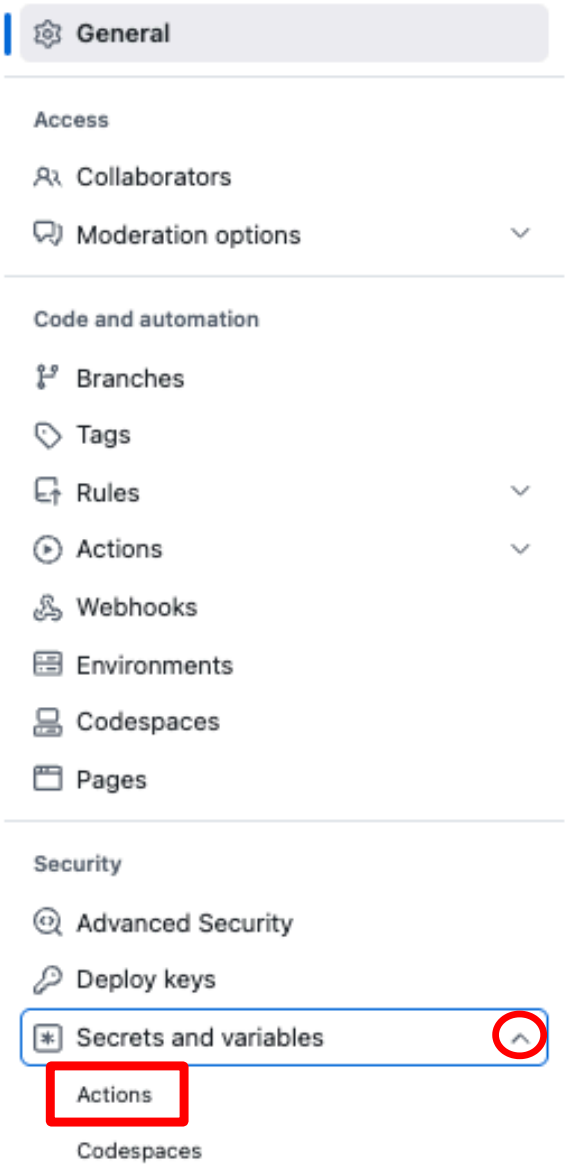


2. Click on `Trio` to open that repository
   - You should see a bar of icons across the top with Settings on the far right



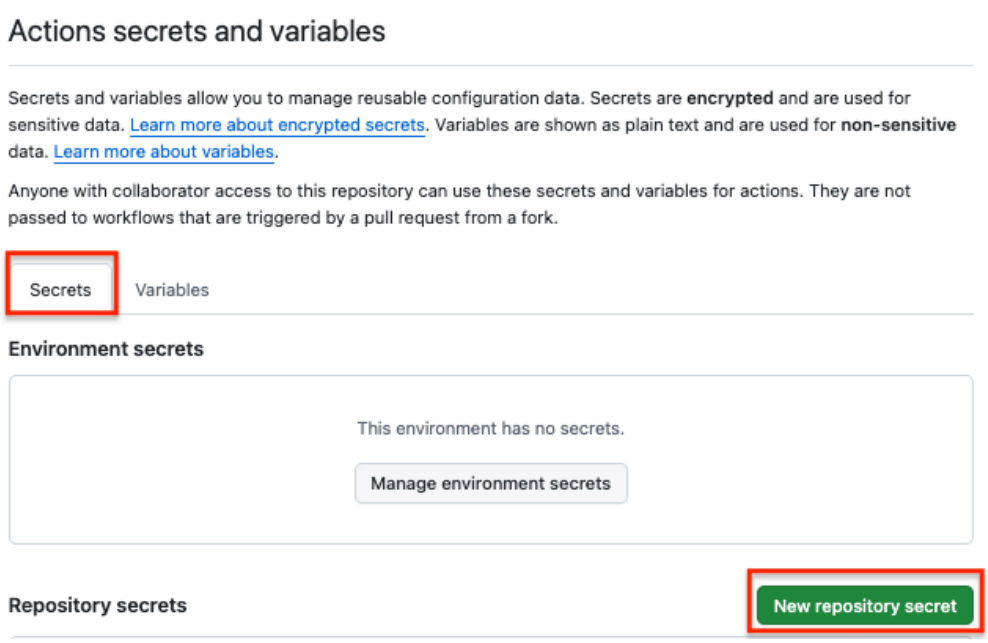
3. Click on the Settings Icon near the top right of your Trio

- If you don't see ⚙️ `Settings`, make your browser wider or scroll to the right
- If you still don't see ⚙️ `Settings`, then you are **not** on your fork or you need to sign in to your *GitHub* account
- After you click on ⚙️ `Settings`, the left side of your screen should resemble the graphic below



4. Refer to the graphic above:

- On the left side, find the `Secrets and variables` dropdown (red circle)
- Tap on `Actions` (red rectangle)

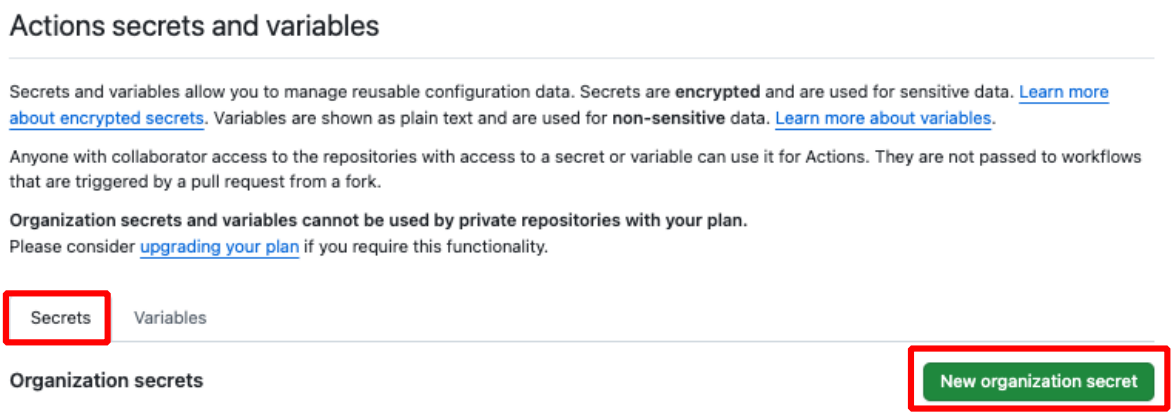


At this point the instructions are the same whether you are using an organization or a personal account.

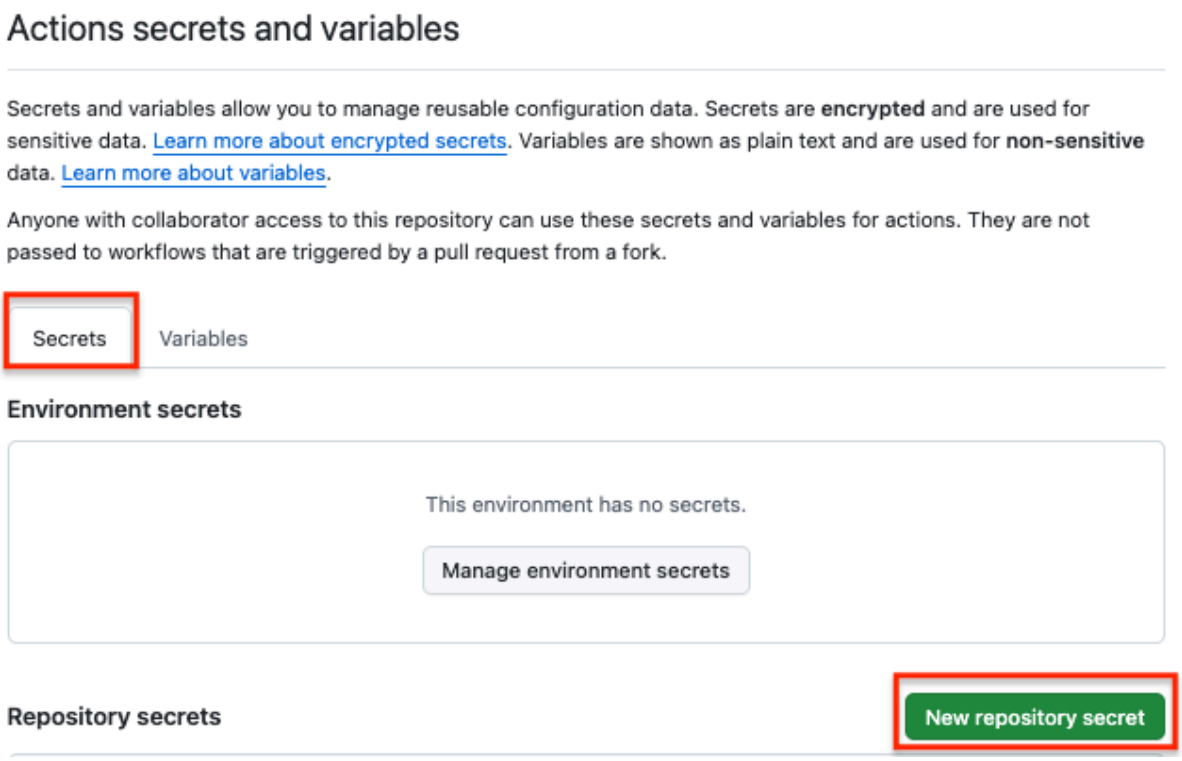
Enter the `Secrets`

The steps to enter the `Secrets` and `Variables` are identical whether you are configuring these in your organization account or repeating this for every repository in a personal account. Your screen should look like one of the graphics below. If not head back to Configure Secrets.

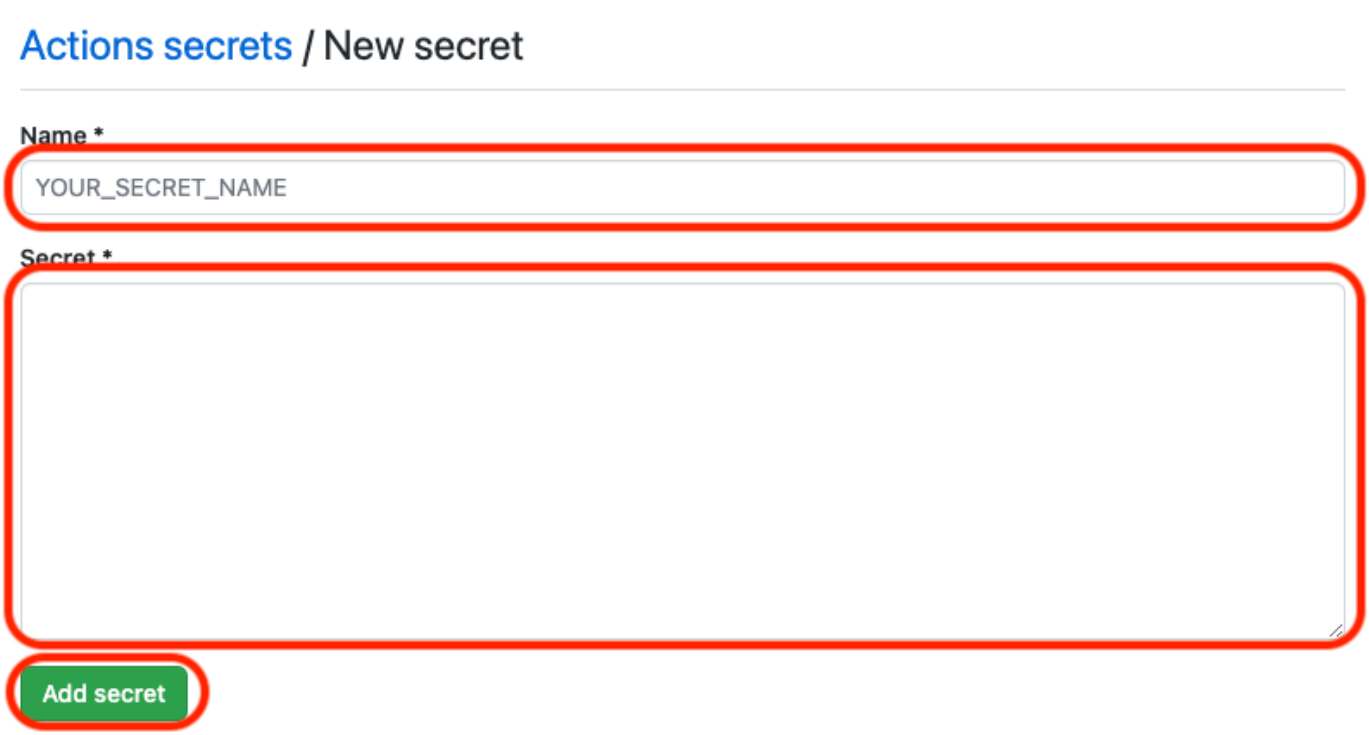
If you are using a *GitHub* organization, tap on the green button for `New organization secret`:

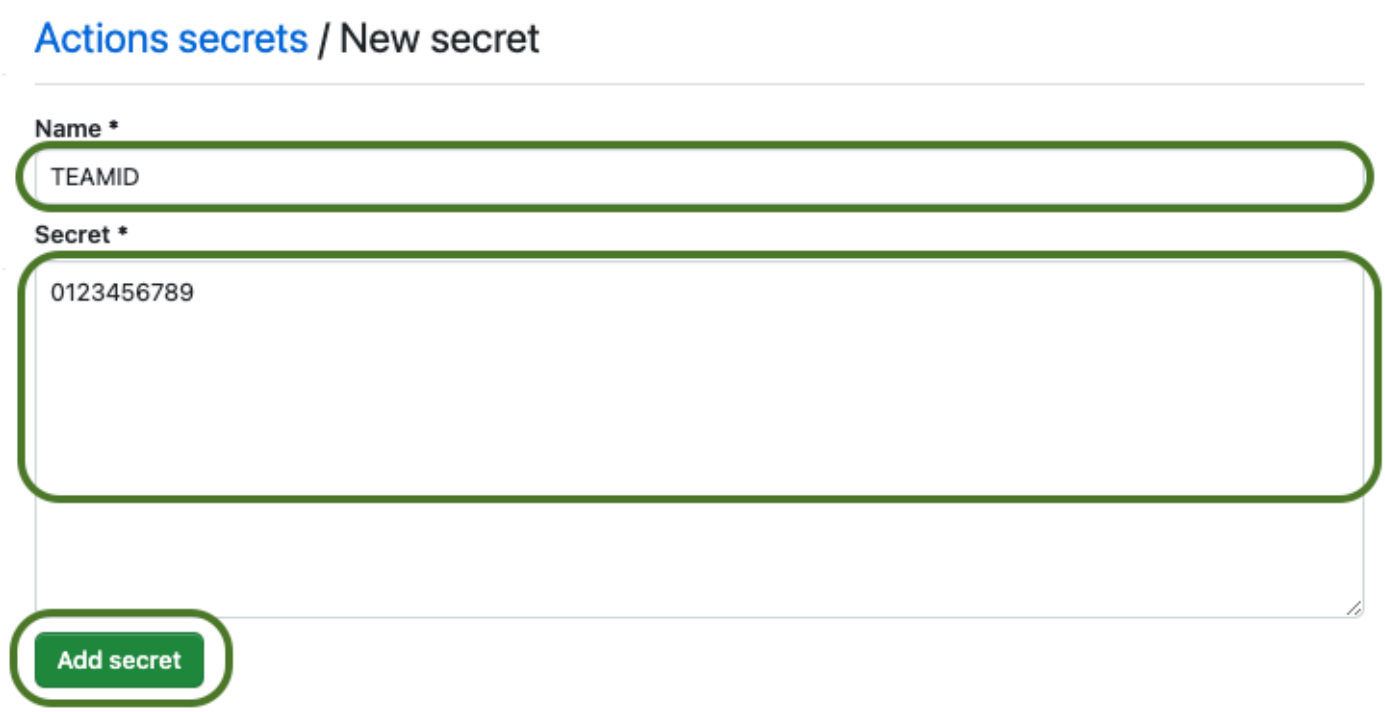


If you are using a personal account, tap on the green button for `New repository secret`:



1. After you tap on the `New secret` button
    - A new screen appears as shown in the first graphic below
    - Do not do anything until reading the sub-bullets, examining the graphics, and proceeding to the next section where each `Secret` name is provided for you to copy and paste
        - Under `Name *`, click on `YOUR_SECRET_NAME` and paste one of the 6 secret names, as directed in Enter Each Secret
        - Click inside the `Secret *` box and paste the value for that secret
        - Once you click on `Add Secret`, the secret will be added
        - The second graphic below shows `TEAMID` added and ready for save





Enter Each Secret

Enter the name of each `Secret` found in Save Your Information and your value for that `Secret`.

- Once you save a secret value, you will not be able to view what you entered, so check carefully before you hit `Add Secret`
  - You can replace the value for any secret later - but you can't view the saved value
- Be especially careful with your `TEAMID`

    - If `TEAMID` is incorrect, the initial `Actions` will succeed but `Build Trio` will fail and you will have some clean-up to do
- You can copy the names of the `Secrets` by hovering to the right of each word below until you see the copy button (⧉). Click on the button to copy the `Secret` name and paste it into *GitHub* where you see `YOUR_SECRET_NAME`. This avoids spelling errors.

    ```
    TEAMID
    ```

    ```
    FASTLANE_ISSUER_ID
    ```

    ```
    FASTLANE_KEY_ID
    ```

    ```
    FASTLANE_KEY
    ```

    ```
    GH_PAT
    ```

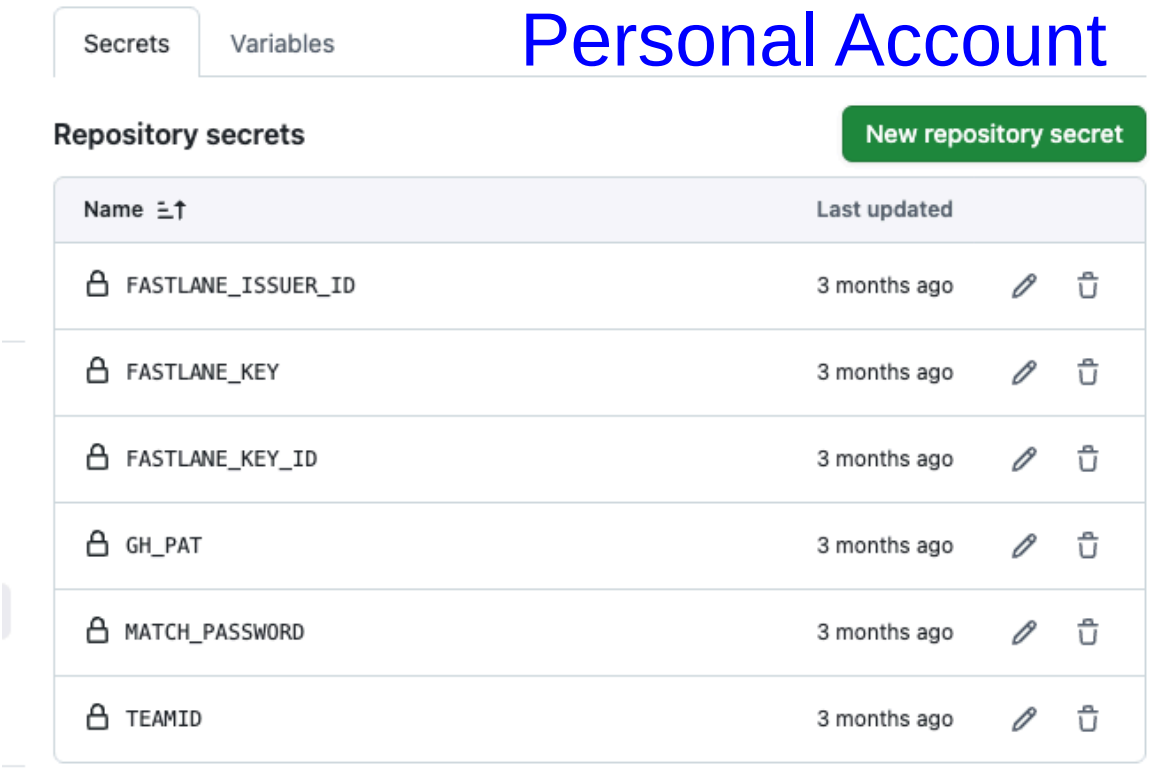
    ```
    MATCH_PASSWORD
    ```

- For the `FASTLANE_KEY` value, copy the entire contents from
  `-----BEGIN PRIVATE KEY-----`
  through
  `-----END PRIVATE KEY-----`
- For `MATCH_PASSWORD` value - if you did not already make up a password and save it with your other `Secrets`, do it now
    - The `MATCH_PASSWORD` must be the same for any repository using this method

Once you add all six `Secrets`, your screen should look similar to the one of the two examples in the graphic below.

> An organization account (top half of graphic) has a column for Visibility which is not seen in a personal account (bottom half of the graphic). The default setting for visibility is `Public repositories`. If yours says anything else, you should update the visibility by tapping on the pencil icon.

- Check that all of your `Secrets` are spelled correctly
- If one is misspelled, delete it and add a `New secret` with the correct name



Add Variable

This variable (**not** a secret) provides automatic renewal of certificates, which expire once per year.

1. While in the same screen where you enter the `Secrets`, click on the `Variables` tab to the right of the `Secrets` tab

- If you aren't at the screen:
  - ***GitHub*** **organization account**: go to your organization page and select Settings; scroll down, select `Secret and Variable` and then select `Actions`
  - ***GitHub*** **personal account**: go to your repository for the app you are building, select Settings; scroll down, select `Secret and Variable` and then select `Actions`

2. Select new variable and give it the name the `ENABLE_NUKE_CERTS` and enter `true` as the value

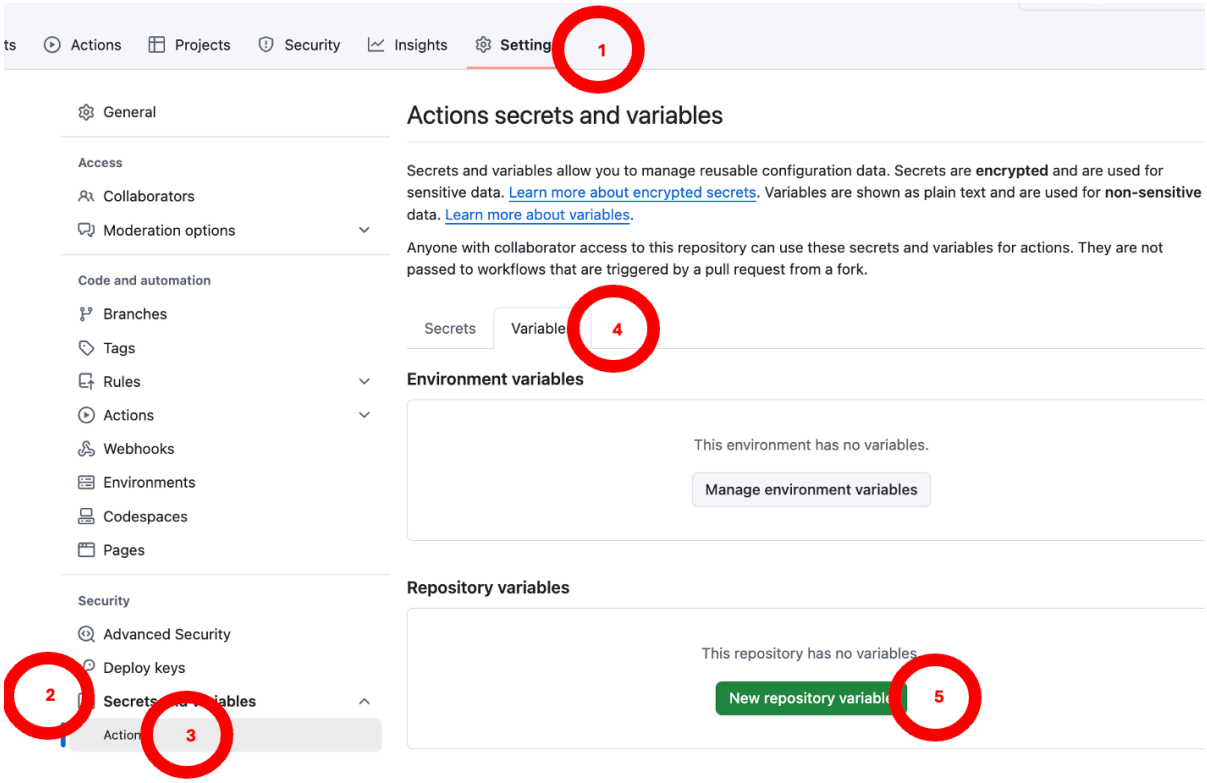
```
ENABLE_NUKE_CERTS
```

What will this accomplish?

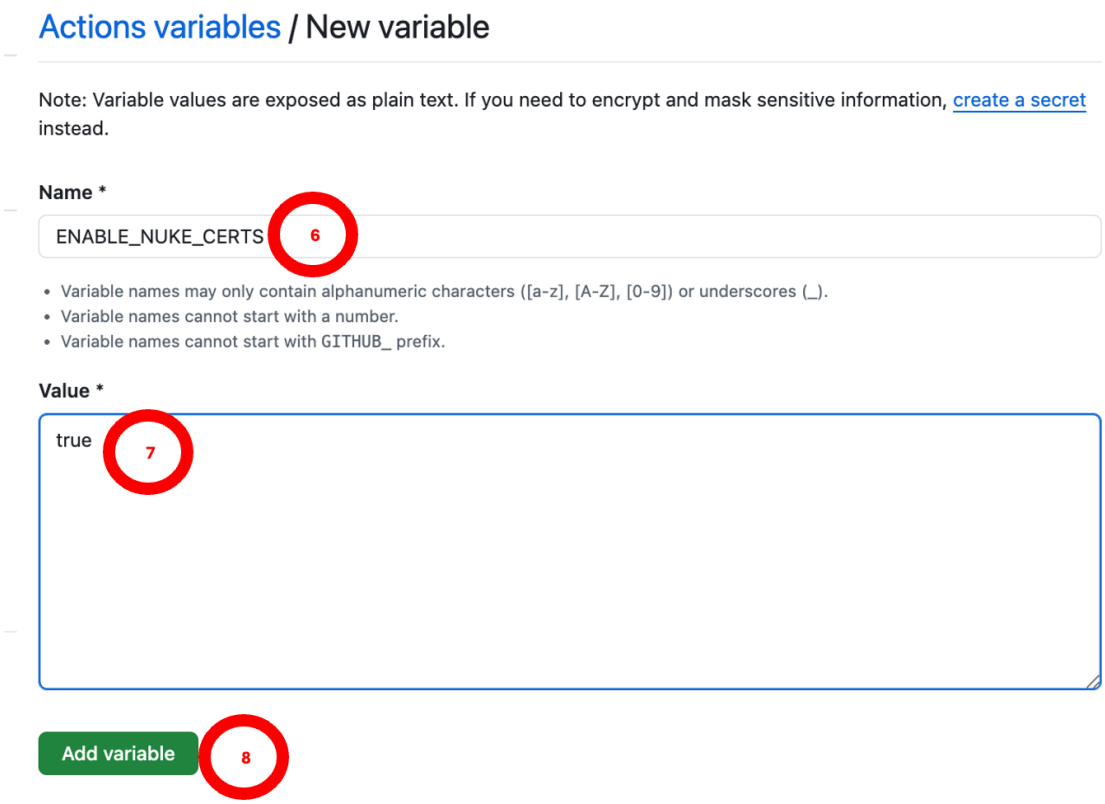
- Certificates will be automatically updated if you have configured this `Variable`
- As long as your *Apple* developers license is valid and all agreements are signed: **you can skip the steps to create or renew your certificates**!

Not sure how to get to the Variables screen? Review the two graphics.

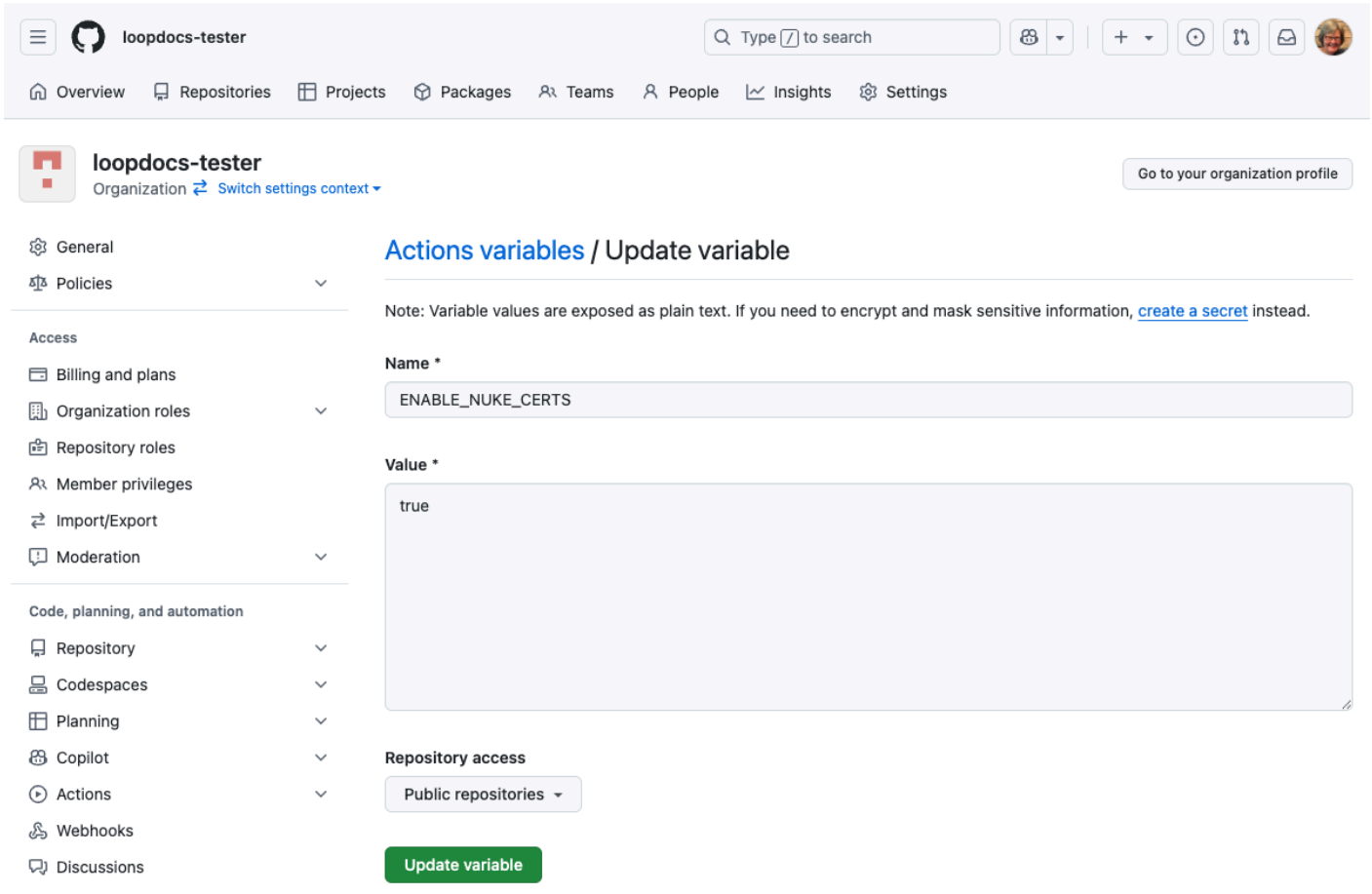
This graphic shows how to access Variables for a repository (similar steps required for an organization).



This next graphic shows filling out the variable for a repository that is not part of an organization.



This final graphic shows how to add the `ENABLE_NUKE_CERTS` to an organization. Be sure that Repository access is set to `Public repositories`. When adding this to a repository, that option is not shown.



Next Step

Secrets are configured! Now let's validate them:

→ **Validate Secrets**

Navigation: ← Back: Collect Apple Secrets | Next: Validate Secrets →

Switch to a *GitHub* Organization

If you are someone who already has a lot of forks in your personal account and want to switch to using a *GitHub* organization. Here's how:

1. Follow the steps to create your organization
    - Create a Free *GitHub* Organization
2. Add the `Secrets` and the `Variable` to your *GitHub* organization as explained in Prepare to Enter `Secrets`
3. Fork all the repos you normally use, but this time, set your organization as the owner
4. For each repository in your organization:
    - Tap on the Actions tab
    - Enable Actions
    - Run the Create Certificates Action and wait for success
    - Run the Build Action
        - WHAT??
        - That's right - all the setup is done on the *Apple* side and you already did that
        - If you successfully built with your private *GitHub* account, everything is already configured
        - The one thing you might want to do is copy customizations from your personal account fork to the organization fork
5. Return to your private *GitHub* account
    - Your choice: either delete the forks in your personal account or at least disable the building from your personal account

Important

> Your personal *GitHub* account is still needed. The free organization points to your personal account as a member. If you delete your personal *GitHub* account, you lose access to your organization account too.

Already Have Trio?

Some people may already have a copy (`fork`) of `Trio` .

If your copy (`fork`) is **not** from `nightscout` , follow the Delete and Start Fresh directions.

If your copy (`fork`) is from `nightscout` :

- Open your `Trio` repository (`https://github.com/my-name-org/Trio`) where you use your version of `my-name` in the URL
- Review the graphic in the Configure: Successful Fork section
    - Make sure all the items highlighted by red rectangles are correct with the possible exception of your fork being up to date
- If you see a message that your *fork* is not up to date - tap on the `Sync fork` button and follow the instructions
- Continue with Validate Secrets

Delete and Start Fresh

If your fork is not from `nightscout` :

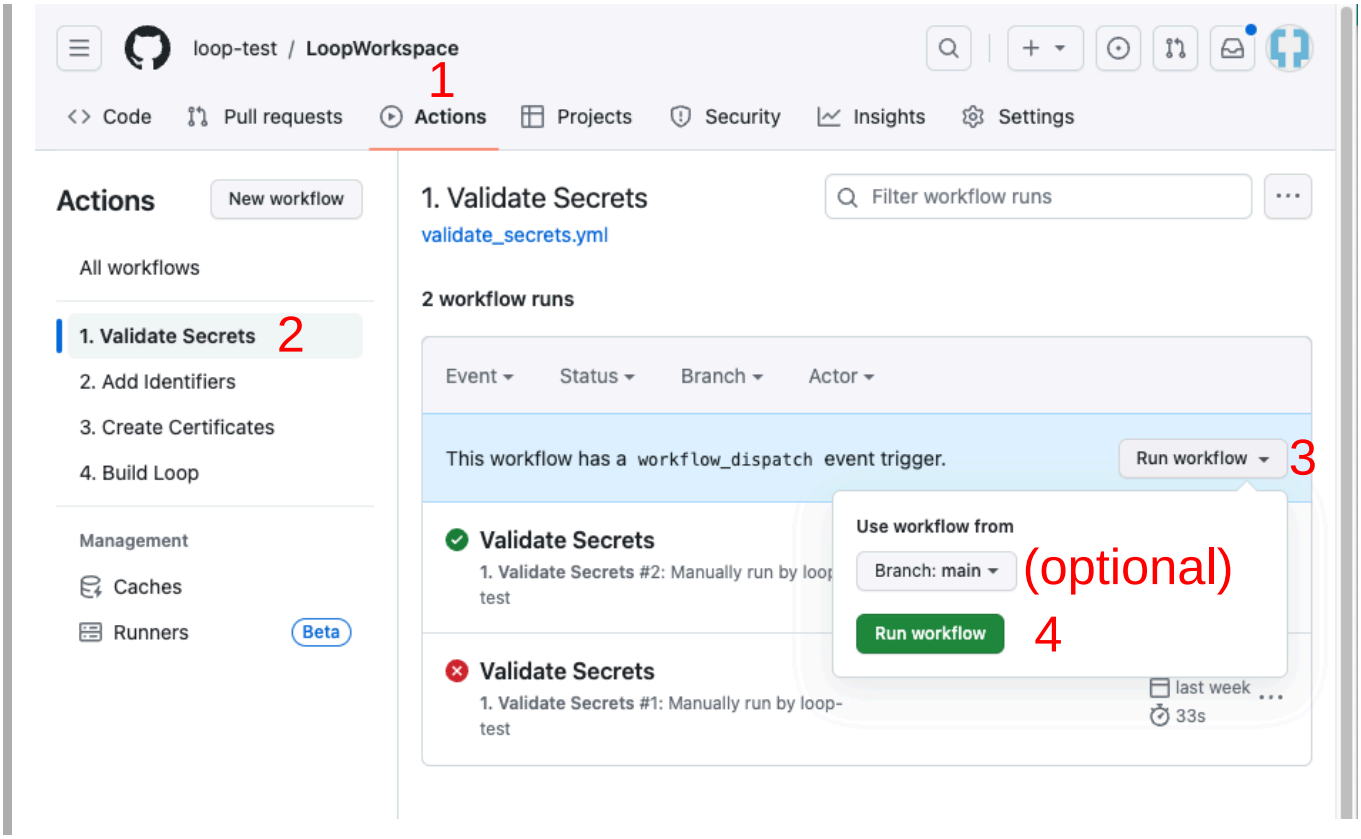
- Delete your Trio repository
    - Instructions to delete a repository are found at *GitHub* Docs
- Return to Fork Trio and follow all the instructions

Validate Secrets

Validate repository secrets

This step validates most of your six Secrets and provides error messages if it detects an issue with one or more. In addition, if you do not have a private Match-Secrets repository it creates one for you.

1. Click on the "Actions" tab of your Trio repository and enable workflows if needed
2. On the left side, select "1. Validate Secrets".
3. On the right side, click "Run Workflow", and tap the green `Run workflow` button.
4. Wait, and within a minute or two you should see a green checkmark indicating the workflow succeeded.
5. The workflow will check if the required secrets are added and that they are correctly formatted. If errors are detected, please check the run log for details.



Be Patient

- Refresh the browser if you are unsure if the action started
- Do not start a new action until the first one completes
- If it seems to take too long to finish - refresh your browser to see if it is done

---

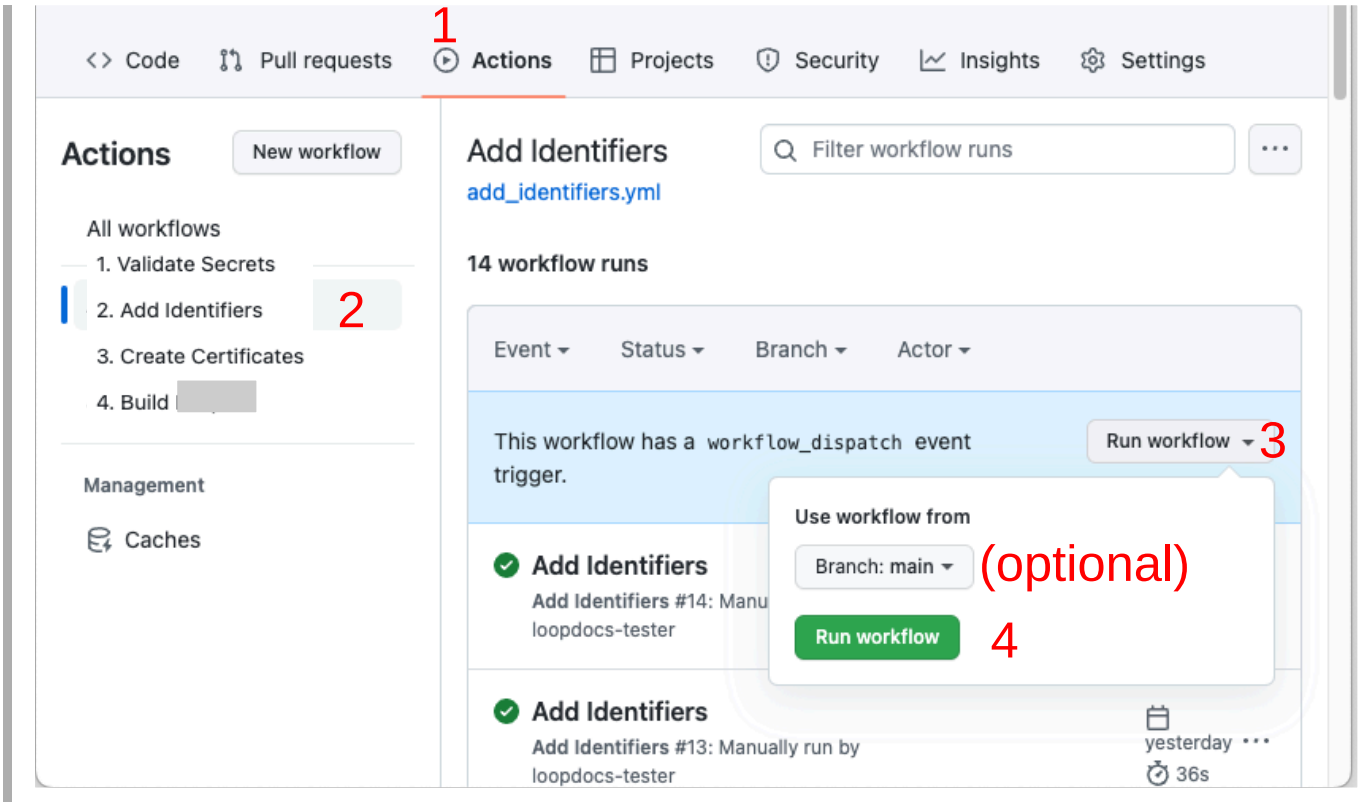
Navigation: ← Back: Fork and Prepare → | Next: Add Identifiers →

Add Identifiers

The `Add Identifiers` action creates the Identifiers needed to create the Trio app.

Refer to the graphic below for the numbered steps:

1. Click on the `Actions` tab of your Trio repository
2. On the left side, click on `2. Add Identifiers`
3. On the right side, click `Run Workflow` to show a dropdown menu
    - You will see your default branch (typically this is `main`)
4. Tap the green button that says `Run workflow`.



Be Patient

- Refresh the browser if you are unsure if the action started
- Do not start a new action until the first one completes
- If it seems to take too long to finish - refresh your browser to see if it is done

The `Add Identifiers` Action should succeed or fail in a few minutes. Do not continue to the next step until this one succeeds.

Navigation: ← Back: Validate Secrets | Next: Prepare App →

Prepare App

New Trio Builders

Create the `Trio App Group`

If you already have a `Trio App Group`

- You can skip this step - your existing App Groups are found at this link: App Group List
- If your `Trio App Group` was created from a Mac with Xcode, you may choose to edit the Description to make the **NAME** match

If you do not have a `Trio App Group`:

- Go to Register an App Group on the Apple developer site and use the table below to help you create one.
- Replace `TEAMID` with your *Apple* Developer ID.

NAME	Xcode version (NAME)	IDENTIFIER
Trio App Group	group org nightscout TEAMID trio trio-app-group	`group.org.nightscout.TEAMID.trio.trio-app-group`

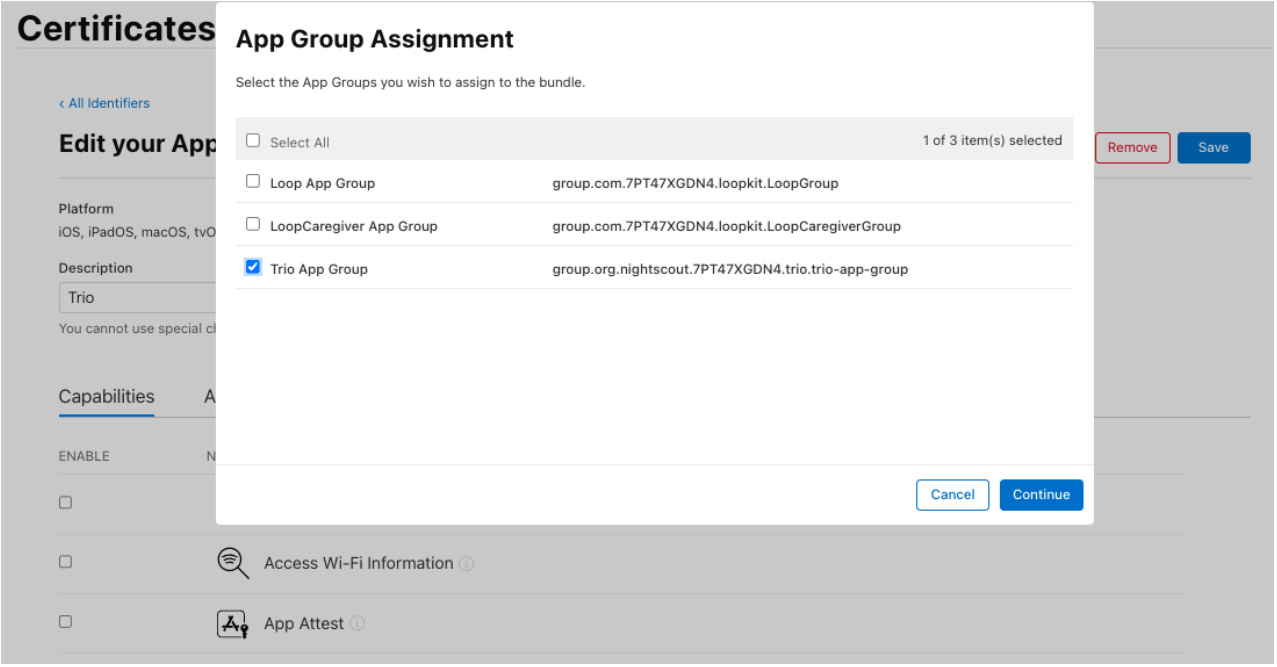
Table of `Identifiers`

These are the `Identifiers` created by running the *GitHub* action "`Add Identifiers`". Only 3 of them need to have the Trio App Group added. The `Trio` Identifier also needs the Time Sensitive Notifications capability enabled manually.

NAME	Xcode version (NAME)	IDENTIFIER
Trio	XC org nightscout TEAMID trio	`org.nightscout.TEAMID.trio`
Trio LiveActivity	-	`org.nightscout.TEAMID.trio.LiveActivity`
Trio Watch App	XC IDENTIFIER	`org.nightscout.TEAMID.trio.watchkitapp`
Trio Watch Complication	XC IDENTIFIER	`org.nightscout.TEAMID.trio.watchkitapp.TrioWatchComplication`

Add Trio App Group to `Identifiers`

- Open the App IDs Identifier page for your Apple Developer Account.
  - Click on the `Trio` Identifier and assign the `Trio App Group` to the Identifier - see graphic below.

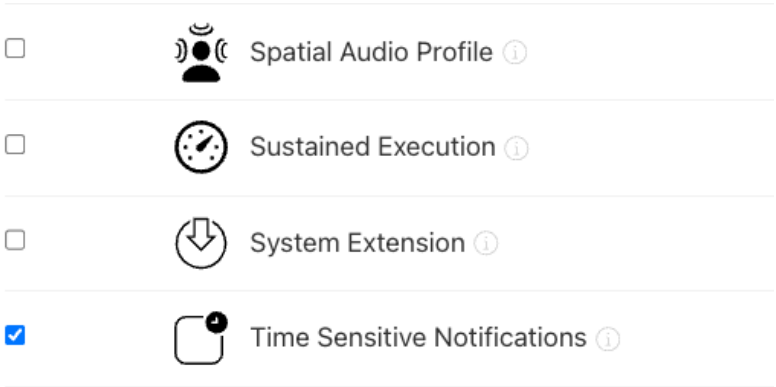


- Before saving the `Trio` Identifier, Enable Time Sensitive Notifications as described in the next section
- Repeat the App Group assignment for the other 2 `Identifiers` that need to have an App Group assigned: `Trio Watch App` and `Trio Watch Complication`

Enable Time Sensitive Notifications

There is one difference for the `Trio` Identifier compared to the other `Identifiers`. You must manually enable the Time Sensitive Notifications capability.

While editing the `Trio` Identifier to add the `Trio App Group`, keep scrolling down until you see Time Sensitive Notifications. Make sure that check box is enabled, as shown in the graphic below. Then tap `Save`.



This capability is only required for the `Trio` Identifier - you do not need to enable it for the other `Identifiers`.

Configure Trio App

Follow the directions in LoopDocs, but use the Trio `Bundle ID`

- In `App Store Connect`, the `Bundle ID` for Trio will be: `org.nightscout.TEAMID.trio`

If you want to take it a step at a time, continue to Create `Certificates`.

If you are feeling confident, you can go straight to Build App which includes the create certificates step.

Navigation: ← Back: Add Identifiers | Next: Prepare TestFlight Users →

Prepare TestFlight

TestFlight Overview

There are two parts to this. Step 1 is only done one time. Step 2 is done for each new app.

1. A user must be registered under your Your App Store Connect User List before they can be added to a *TestFlight* internal test group for any app
    - As a developer, you are already included as a user with the Role of Account Holder, Admin
    - You can skip this step for yourself and only need it to add another adult to your test group
2. For each new app, you must also create a *TestFlight* Internal Testing Group for that app
    - You must add yourself to the *TestFlight* Internal Testing Group for each app

Most builders can skip ahead to creating the *TestFlight* Internal Testing Group.

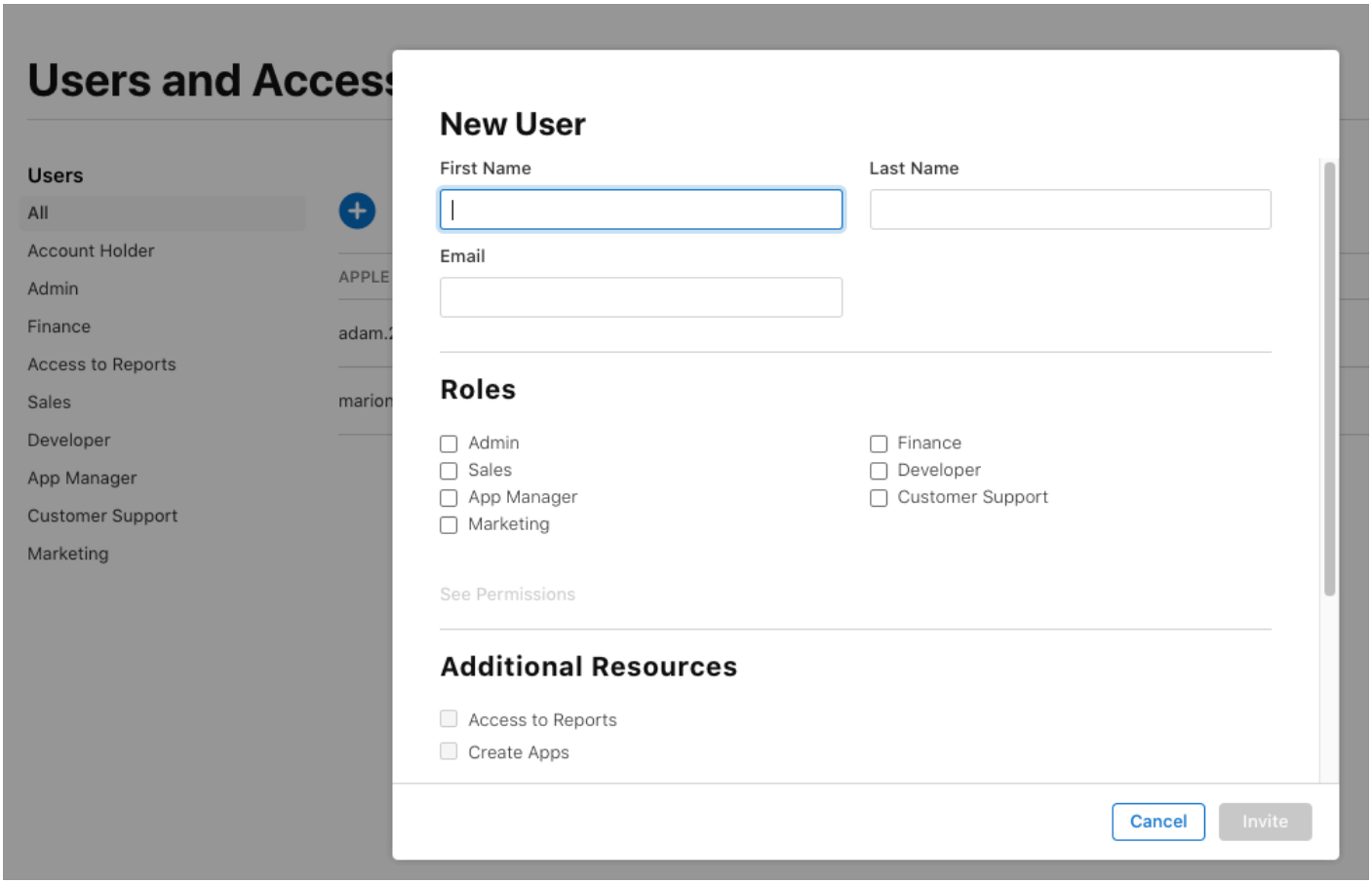
Your App Store Connect User List

> **Step 1: add user for access for any of your apps**
>
> Before you can select someone for the *TestFlight* Internal Testing group for any app, you must first add them to your App Store Connect User list.
>
> As a developer, you are already included as a user with the Role of Account Holder, Admin.

You are configuring a private capability for your family using an Internal Testing group. You need the *Apple ID* email address for each adult installing from your build. When building for a child, you will use an adult's *Apple ID*, not theirs. See *TestFlight* for a Child ↗ .

First you need to add the email address(es) to your *App Store Connect* Access Users list:

- Open this link: Users and Access
    - You must provide a role for each person - `Customer Support` is a good choice
    - Once you have added them here, you'll be able to select them in the `TestFlight` internal test group for each of your apps

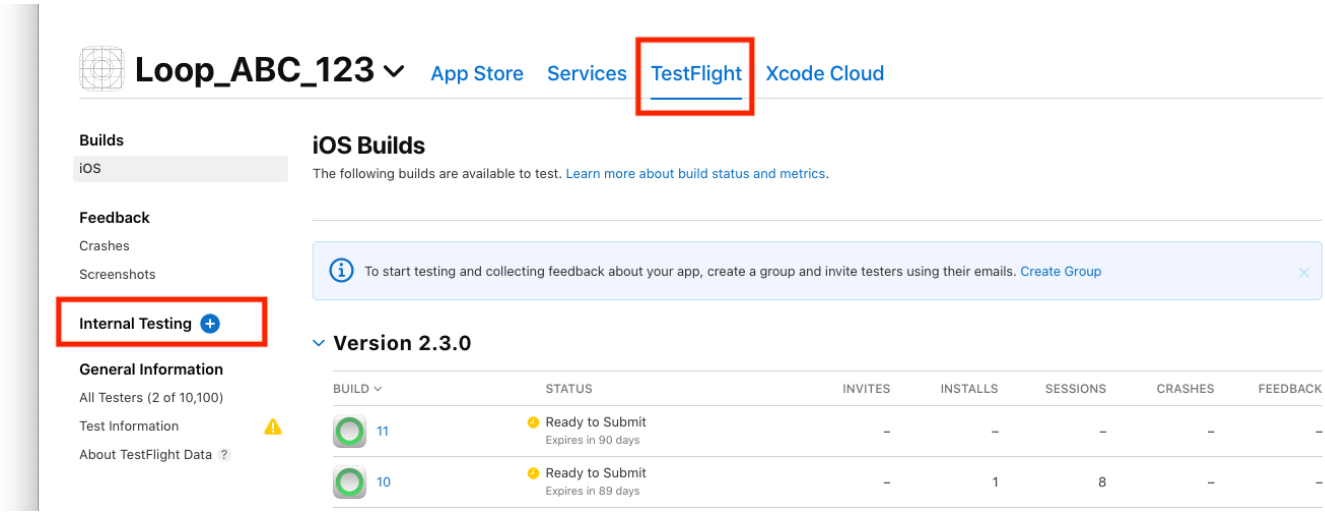


Configure *TestFlight* Group for the App

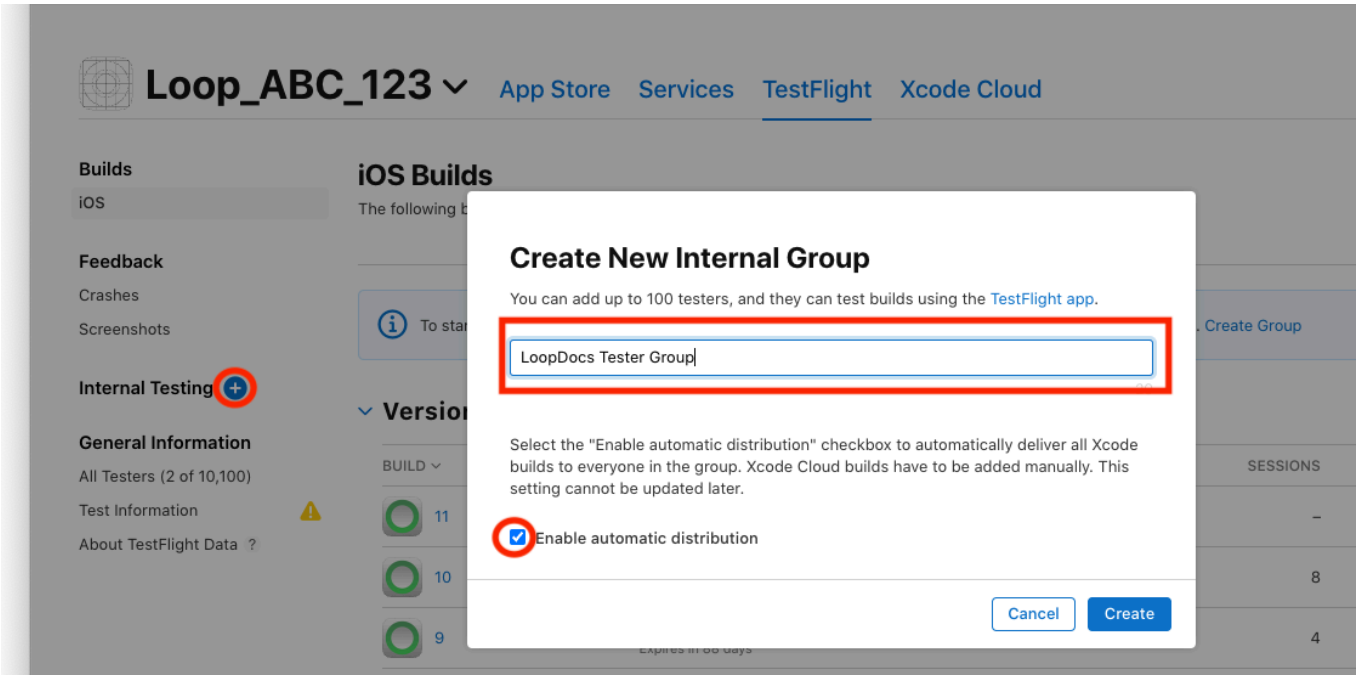
Add user for each app

Once you add a user to have access to your TestFlight internal test group for an app, you don't need to do it again - it remains available to them across rebuilds and different versions for that app.

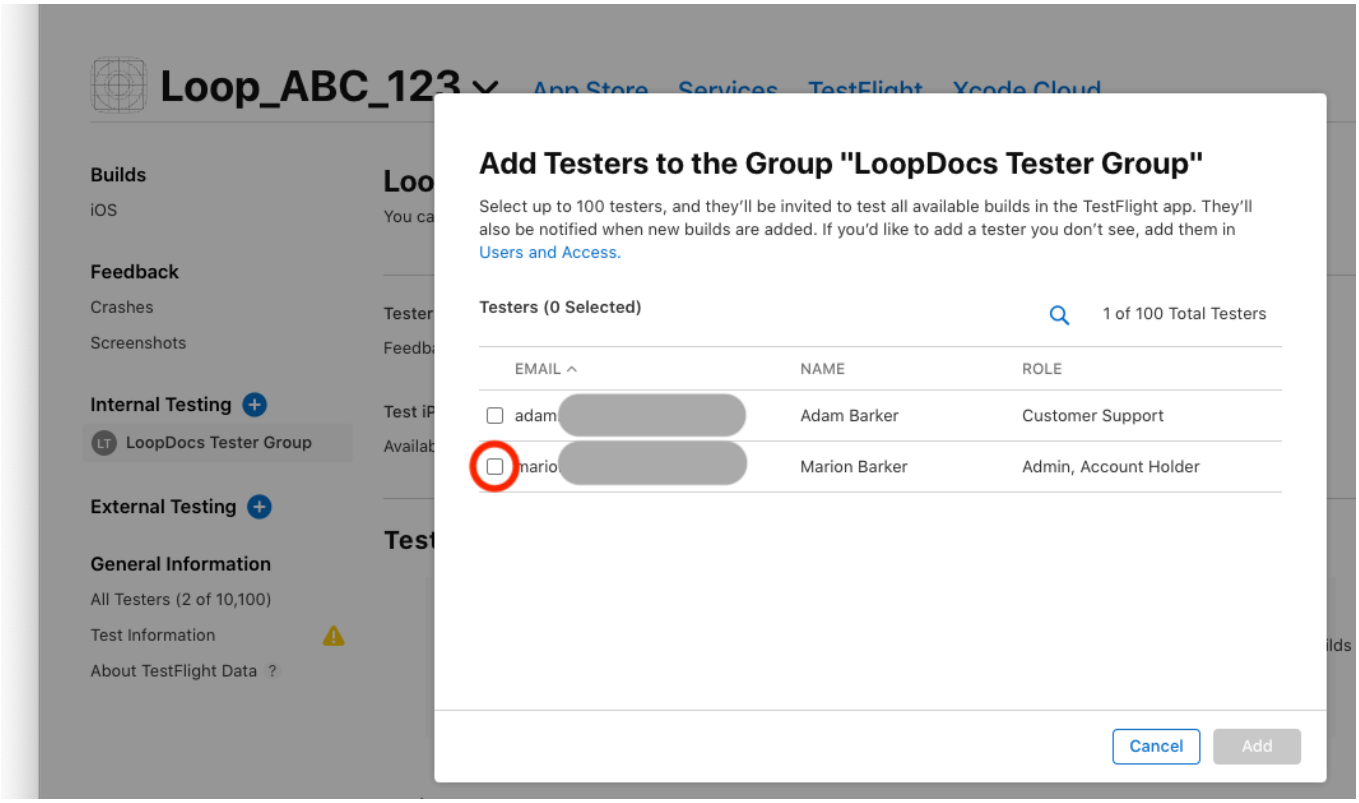
1. Open this link: App Store Connect / Apps to view your apps; log in if needed.
    - Select your *Trio* app
    - Click on the `TestFlight` tab
    - **If you already have an Internal Testing Group for this app, skip to Step 4**
2. Click the blue plus button (⊕) next to `Internal Testing` to add a group.



3. Fill out the name you want for the `Internal Testing` group
   - Be sure to check the box `Enable automatic distribution`
   - Click `Create` when done



4. You can add or remove emails to the Internal Test Group at any time
   - Click in the box beside each person you want to include
   - Each person in this group will get an email each time you update (build again) using the `GitHub Browser Build` method
   - Click `Add` when you are done
   - If building for a minor child, you will send the invitation to yourself or another person because a minor child is not authorized to use *TestFlight*: See TestFlight for a Child



Next Step

TestFlight group configured! Ready to build:

Navigation: ← Back: Prepare App | Next: Build Trio →

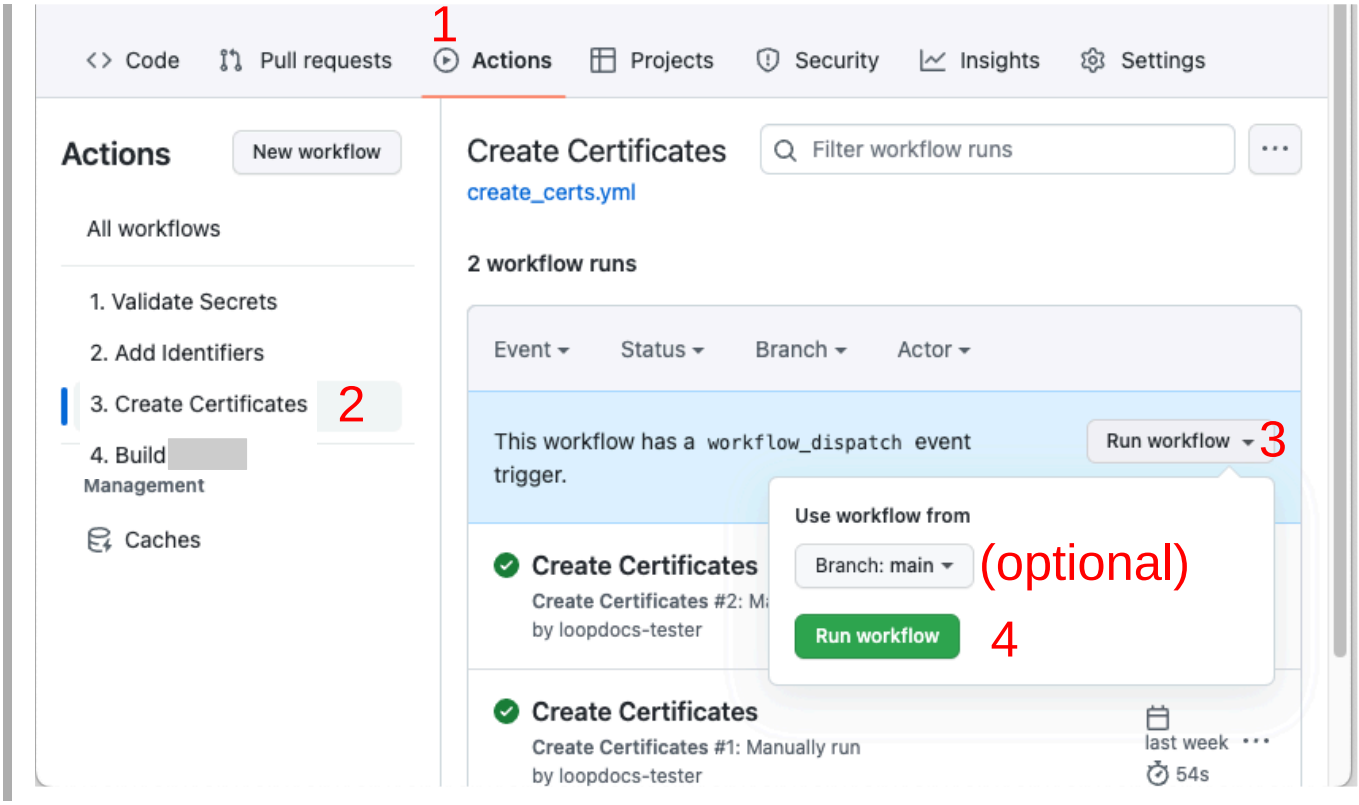
Create Certificates

Create `Certificates`

It is optional to run `Create Certificates` separately, but for a new builder, it helps to run it to make sure everything is working before you try the build. (The build action runs create certificates as the first step.)

Refer to the graphic below for the numbered steps:

1. Click on the `Actions` tab of your Trio repository
2. On the left side, click on `3. Create Certificates`
3. On the right side, click `Run workflow` to show a dropdown menu
    - You will see your default branch (typically `main`)
4. Tap the green button that says `Run workflow`



5. Wait a minute or two for the action to finish

> **Be Patient**
>
> - Refresh the browser if you are unsure if the action started
> - Do not start a new action until the first one completes
> - If it seems to take too long to finish - refresh your browser to see if it is done

Once you see the green check mark by `Create Certificates`, the next step is to Build the app.

This is an optional step.

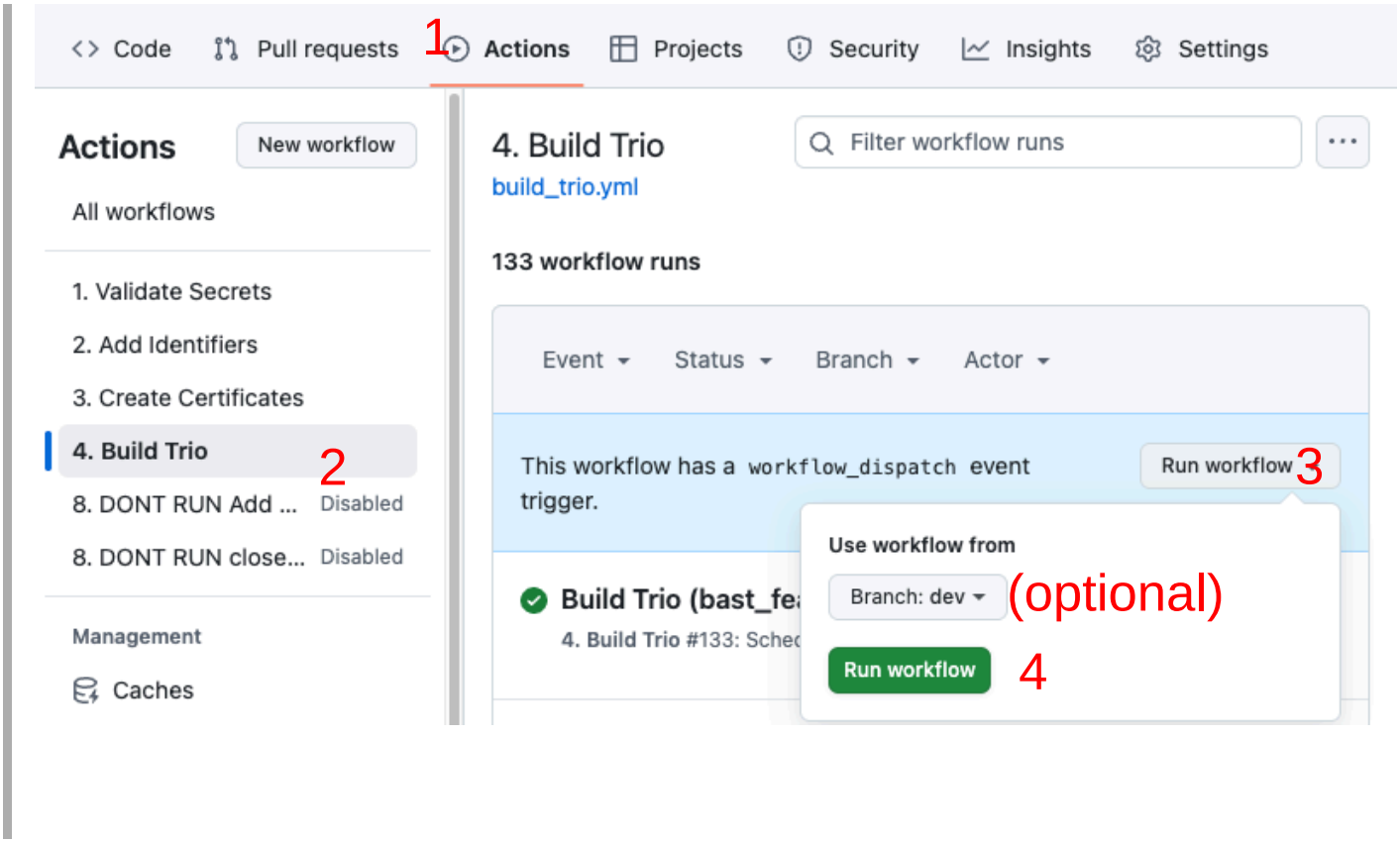
Navigation: ← Back: Prepare App | Next: Build App →

Build the App

Build the App

After you run Identifiers and Prepare the App, you are ready to Build.

1. Click on the `Actions` tab of your Trio repository
2. On the left side, click on `4. Build Trio`
3. On the right side, click `Run Workflow` to show a dropdown menu
    - You will see your default branch (typically this is `main`)
4. Tap the green button that says `Run workflow`.



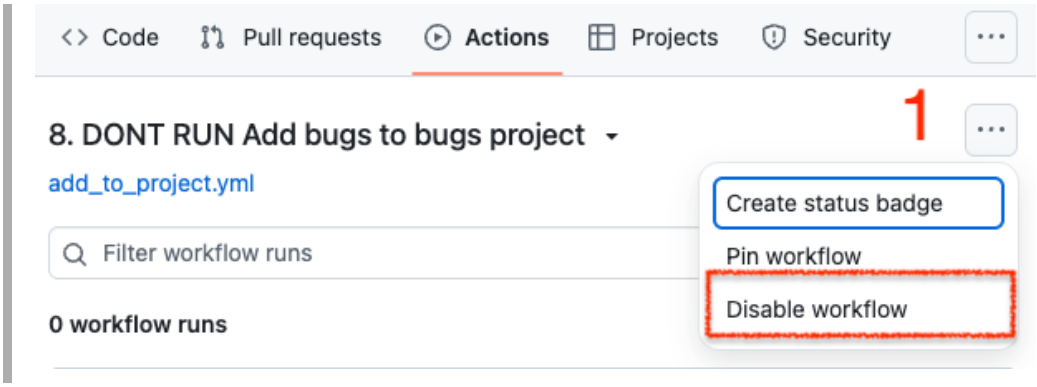
Refresh, Do Not Repeat

> **Hit Refresh**
>
> After you tap the green Run workflow button, *GitHub* can be slow to update.
>
> - Refresh the browser if you are unsure if the action started
> - Do not start a new action until the first one completes

Disable the DO NOT RUN Actions

You may notice some *GitHub* actions that have `DONT RUN` or `DO NOT RUN` in their names. Those are special actions that are used by the developers. They are skipped in your `Fork`, but you may see logs for them that indicate they failed.

If they already have the notation `disabled` beside them, you won't see them run - and fail. Otherwise, click on each one, click on the three dots at the upper right, and then select `Disable workflow`. Then you will avoid the annoyance of seeing logs that say the action was automatically run and then skipped.



Navigation: ← Back: Prepare App | Next: Install from TestFlight →

Install Trio from TestFlight

General Installation Information

The *Trio* app must be built at least every 90 days when using a browser to build.

> The automatic build actions should occur every Sunday.

It is recommended you manually install the new build using *TestFlight* at your convenience.

After you Build the *Trio* App ↗ with a browser and you get the email that it is available in *TestFlight*, you are ready to install on as many phones as you and your family members need.

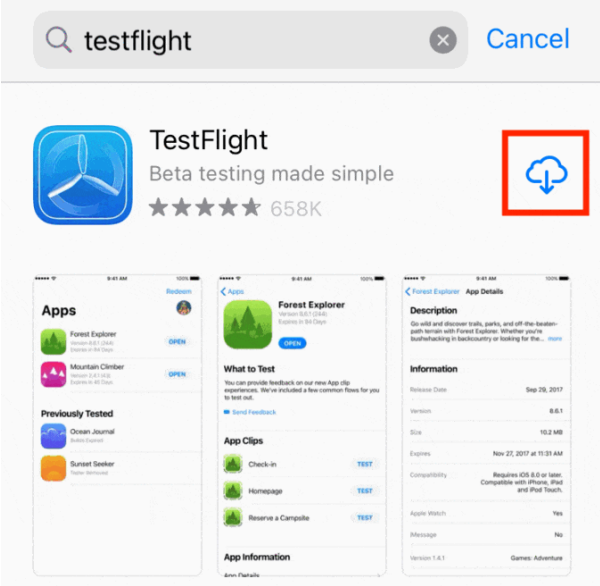
- If you later need to add an adult family member to your list, refer to *TestFlight* Overview ↗ .
- Children (under 13 in US, varies by country) cannot use *TestFlight* with their ID. When you use *TestFlight* for a Child, you will need to use your ID on their phone (not the whole phone - just the Media & Purchase portion), so send the *TestFlight* invitation to the email associated with your ID.

Install *TestFlight*

If you already have the *TestFlight* app installed on your phone, skip ahead to Install App with *TestFlight*.

To install *TestFlight*, refer to the GIF below:

- On the phone, open the App Store and Search for *TestFlight*
- Install or Download to that phone *TestFlight*
  - Hint: On child's phone, do this while logged in as yourself for Media & Purchase
  - Logging in as an adult is explained in *TestFlight* for a Child



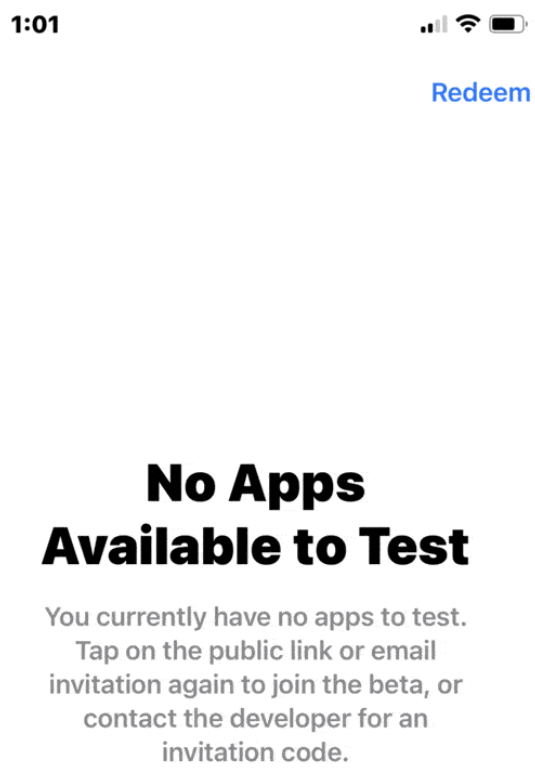
Install App with *TestFlight*

Once you get an email that your app is available to test on iOS and watchOS, you can install the app on your phone from *TestFlight*.

- Note this can be half-hour to an hour after the build displays the green check mark on your browser
- Once the app is on your phone, you can choose to install the watch version using your phone *Watch* app
- If you did not already add your email to your *TestFlight* group for your app, go and do it now using these instructions ↗

The first time you use *TestFlight* on any phone associated with a given email, you must `Redeem` the code sent to that email inviting you to test the app. The GIF below is for someone who has never used *TestFlight*; the example is for *Loop* but *Trio* acts the same.

- Initial screen indicates there are no Apps available to test, tap on Redeem
- Enter your code and tap redeem to enter it
- Click on OK to acknowledge
- Click on Install



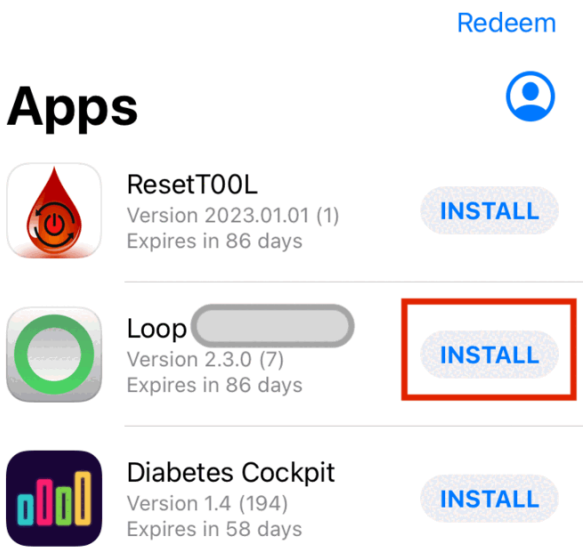
If you already have the *Trio* app on the phone, you'll see the warning about possible loss of data. Don't worry, all your settings remain. Go ahead with the installation.

Subsequent Times on Phone

- Open the *TestFlight* app and find the name you used for your *Trio* app in the Configure *Trio* App step
- If you have previously used *TestFlight* on this phone and don't see the latest build, sometimes quitting and restarting *TestFlight* will bring that new build into the list of available apps
- Most people will just tap install to get the most recent build
    - If you have more than one version number available in *TestFlight* you can choose which version to install by tapping on the `Previous Builds` row and then selecting the desired version
- Tap on Install
    - If you already have the *Trio* app installed on this phone, you will be warned that the app already exists on your phone and that you might lose data

    - Click Install again (your pump connection and all your data will be fine)
- Choose Open
- Make sure the *Trio* app is operating as expected

This GIF shows an example of installing *Loop* from *Testflight*.



Automatic Update, Build, Install

If you followed all the instructions for setting up Build with Browser, your GitHub account will automatically take the following actions:

- Update the version of your fork within a week of a new release
    - Automatically create a new build and upload it to *TestFlight*
    - This is only for the `default` branch, typically `main`
- Build the app at least once a month and upload it to *TestFlight*

You need to take this manual action for your *Trio* app in *TestFlight*, at every 90 days, but better to do it monthly.

- Install each new build from *TestFlight* on the phone as soon as it is detected
- That's fine for some apps, like *LoopFollow*, but you do not want an app that controls your insulin delivery to install when you are not paying attention

Recommendation

Recommended settings:

- Allow automatic update of your fork
- Allow automatic build and upload to *TestFlight*
- Disable automatic installation on phone from *TestFlight*

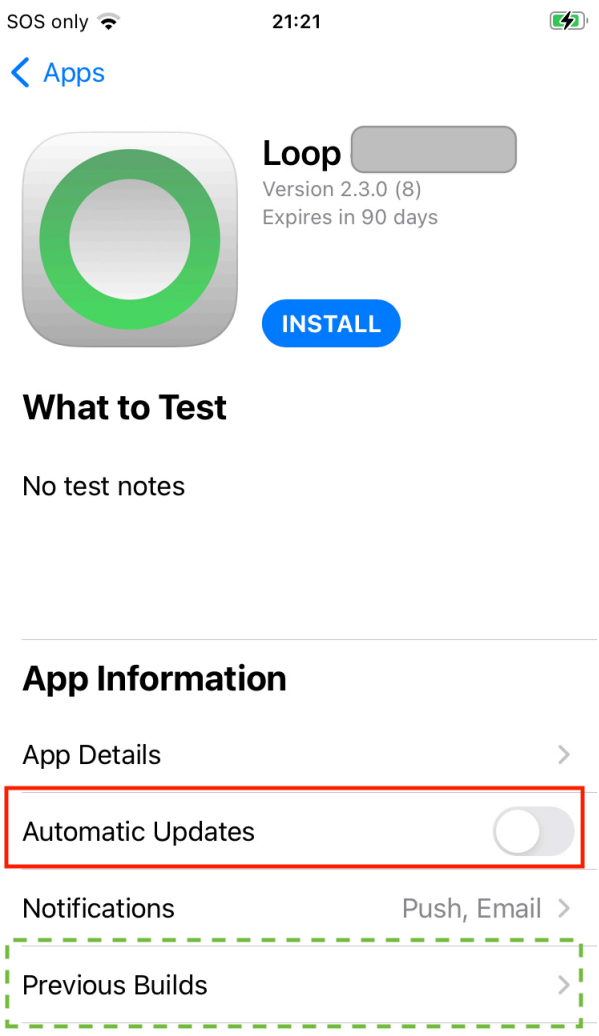
If you are running the development code, you may prefer to turn off the automatic update, but keep the automatic build. To read more about modifying automatic update and build options, please read LoopDocs: Automatic Update & Build.

Disable Automatic Install from *TestFlight*

Once the app is available in *TestFlight*, you can adjust whether it is automatically installed when updated versions are available. We recommend you disable automatic installation so you can choose when to switch to a newer build, which in some cases, may be a newer version.

Go back to the *TestFlight* app on your phone and tap on your app name in the list to see an expanded screen similar to the graphic below. The row to enable or disable automatic updates is highlighted in the graphic, which shows the feature disabled. This is recommended for all users.

- If you leave automatic update enabled (default), then whenever a new build is created and uploaded to *TestFlight* , it will be installed immediately (see Unexpected *TestFlight* Beta Expiration)
- WARNING: If you switch between Building with Browser and *Mac*, you must disable automatic update or Xcode will not be able to install to your phone



When you are ready to install, just open the *TestFlight* app and click Install to get the most recent build and then click Open when it completes the installation. All your settings and connections to CGM and Pump are maintained.

Previous Builds

If you tap on the row that says `Previous Builds`, highlighted by the dashed-green rectangle in the graphic above, you can view and choose an older (or lower version number) build (as long as it has not expired).

- In some cases, you need to do this to see the newest build if the most recent build has a lower version number than an older build

Unexpected *TestFlight* Beta Expiration

> **Apple *TestFlight* Bug**
>
> It doesn't happen to everyone but it has happened a few times over the years and only for people who have Automatic Update enabled for their *TestFlight* app.
>
> Symptom:
>
> - App has plenty of time before expiration
> - A new build becomes available, for example from the automatic monthly rebuild
> - User is told *Trio* Beta is not available
>
> Solution:
>
> - Open *TestFlight* and turn off automatic installation for your app
> - Install the previous version of the app on the phone manually (with monthly build, it should still be ok)
> - Manually start a new build (if there were updates you want to get)
> - Manually install the new update
>
> More information:
>
> - Link 1: Apple forum
> - Link 2: Stackoverflow thread

TestFlight for a Child

Minor children are not allowed to install or use the *TestFlight* app.

An adult, who is a member of the Internal *TestFlight* Group ↗ can log into Media & Purchase (see steps below) without affecting the *Apple* ID associated with a phone (and thus the health records used by the *Trio* app for the minor child). After the adult installs or updates the app using *TestFlight*, they probably should reverse those steps to remove their credentials from Media & Purchase.

Media & Purchase affects access to the App Store, Books, Music and Podcasts.

On the Child phone:

- Tap on Settings

- At the very top of Settings, tap on the Name of the phone, for example, `my kids phone`
- *Apple* ID Screen appears
    - Tap on Media & Purchases
    - Tap on Sign Out, and confirm
    - Sometimes the phone requires a reboot before you can sign in with a different ID
- Sign in with the adult *Apple* ID email and password
- Install or Update the app from *TestFlight* on child phone
- Repeat the process to sign out the adult and (if needed) sign back in the child

Change the App Store Connect Name

Suppose you really don't like the name you picked initially for the *Trio* app that shows up in the *TestFlight* app.

You can change it.

Open this link: App Store Connect Apps and log in as needed.

- Click on your app name.
- Click on App Information on the left side (make browser wider if you don't see this).
- Modify the Name under Localizable Information and click on the Save button (upper left)
- If you chose a name that is in use, you'll see the warning screen - try again

Browser Build Errors

Help with Errors

> **Get Help, Not Frustrated**
>
> For first-time builders, there are a lot of one-time steps that all have to be correct and many screens look similar.
>
> If you are updating your build, the process can be confusing and some error messages can be misleading.
>
> You are not alone - please reach out.

If you are having trouble:

Please DO:

- Use the TrioDocs search feature
- Use the Trio Discord search feature
  - Not a member? Here's the invite link: Invitation to Trio Discord
- Ask a mentor for help

Please do NOT:

- Search in a general internet browser to find your error
- Ask ChatGPT (or other AI) how to fix your error
- Spend hours in frustration
- Rename or delete a repository
- Delete your GitHub account - you then lose access to everything you've done
- Remove your App from App Store Connect

Where to Get Help with Browser Build

These are several places to ask for help, but please choose only one. You may get a faster response from Facebook, but there are mentors on all these sites:

- LoopandLearn Facebook Group
- Trio Facebook Group
- Trio Discord: Support: Browser Build

What Information is Needed for Help

1. Your GitHub name used for building (required):
   - Best to provide a link to your repository, like the example, but substitute your information

- If you are building an app other than Trio, replace Trio with the app repository name

If you use an organization

```
https://github.com/my-name-org/Trio
```

If you do not use an organization

```
https://github.com/my-name/Trio
```

2. A brief summary of the step you are on and what isn't working.

That's all we need to get started.

Mentors can go to your public *GitHub repository*, make sure your code is up to date, examine your customizations (if any), then click on the Action tab and view your log files directly.

- Do not copy from the log file and post the words
- Do not take a screenshot of what you think is an error

GitHub Address, Not Pictures

If you've been around the DIY community for a while, you know the mantra about screenshots. Well, when using a browser to build, **screenshots are close to useless**.

All that is needed to assist is your link to the *GitHub* repository where you are building.

Quick Reference for Browser Build Errors

Please make this your first stop if you get any kind of an error with Browser Build.

Link Styles

If you notice an arrow pointing up and to the right beside the link:

- This means a new tab or window (depending on your browser configuration) is opened when you click on the link
- Your original place in the docs is in the original tab, so you can return to your same position in the docs

If there is no upright arrow by a link, then the location is on the same page:

- Once you have read that section, hit the back arrow in your browser to return to your original location

Look at the Annotation

If you get an error in an Action, examine the display and scroll down to view the annotation. Some errors have a clear annotation telling you what to do. If the annotation is not clear to you, please ask a mentor for help.

Specific Trio Help

Time Sensitive Notifications Error

Trio version 0.8.4.3 added a requirement for Time Sensitive Notifications capability to be enabled for your `Trio` Identifier. If it is not enabled, the `Build Trio` action fails with an error similar to this:

```
error: Provisioning profile "match AppStore org.nightscout.***.trio" doesn't include the
com.apple.developer.usernotifications.time-sensitive entitlement.
```

Solution: Enable Time Sensitive Notifications for your `Trio` Identifier, then run the `Build Trio` action again.

Other Errors

```
🚧 Documentation Under Construction 🚧
```

Refer to LoopDocs Browser Build Errors to see if you find anything useful or please ask a mentor for help.

Customizations

🚧Documentation Under Construction🚧

Customize with Browser Build

Change the Branch for a Submodule

If you ever want to use a different branch or commit for a submodule than the one provided by your current Trio workspace, this is how to do it.

1. Go to the repository of interest - it does not need to be in the same repository that Trio uses
    - If you provide the full SHA, GitHub can find that specific code version while running the build_trio.yml action
2. Identify the exact commit you want to use and record the full SHA
3. Go into the `.github/workflows/build_trio.yml` file and edit it at or around line 217 (in the Customize Trio section)
4. Add 3 lines to bring in the desired commit
    - First line is a comment explaining what you are doing
    - Second line defines the submodule commit you want to use
    - Third line updates that submodule as you just specified

An example is shown below. While this particular example is for an older submodule which is no longer part of Trio, you can still use it as a prototype for changing the SHA for any submodule as desired.

Copy and Paste to Customize - Edit SHA and Submodule Name for your situation

```
# Point to the most recent SHA for OmniBLE branch pod-keep-alive
git update-index --cacheinfo 160000 dade6ed309eb72232a187d88179a367e34f800d9 "OmniBLE"
git submodule update
```

Update with Browser

🚧Documentation Under Construction🚧

With the automatic updates, this mostly requires you keep your Apple Developer account valid.

- Sign your developer agreements promptly
- Make sure you configured your Browser Build for automatic updates

Build Dev with Browser

Summary of Tasks to Build Trio `dev`

> **Use for `dev` or feature branch**
>
> This page is useful for building the `dev` branch or any other feature branch found in Trio. Where you see `dev` just replace with `branch-name` where `branch-name` is the branch you want.

These steps assume:

- You previously built Trio using *GitHub* Actions (Browser Build)
- You confirmed your *Apple* Developer license agreements are up to date

These are the new steps for you to follow:

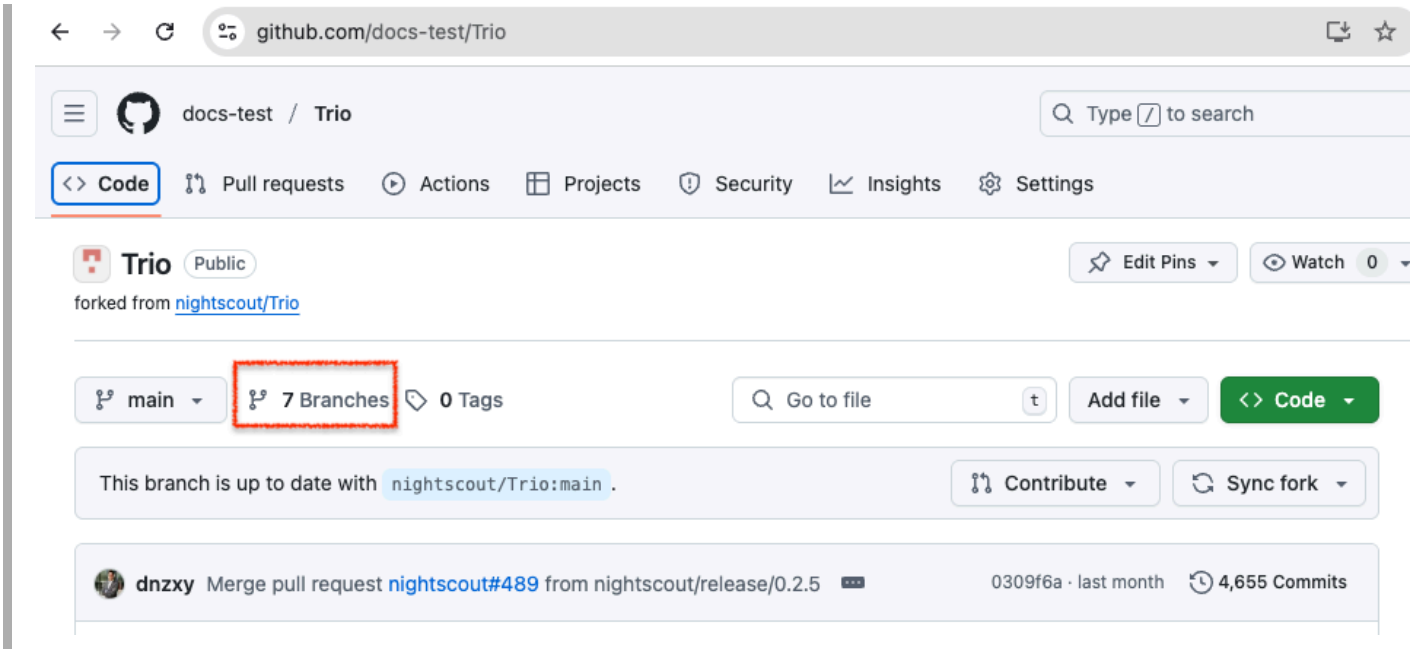
1. Configure `Fork` with `dev` branch
    - Optional: Configure the `dev` branch as default
2. If new Identifiers are required: Update `Identifiers`
3. If Time Sensitive Notifications hasn't been enabled yet: Enable Time Sensitive Notifications
4. Optional: Update `Certificates`; this is included in Build Trio
5. Build Trio `dev`

Configure `Fork` with `dev` branch

- If you do not have a `dev` branch you must first configure one following these directions.
- If you already have a `dev` branch, skip ahead to Update Branch and be sure to select the `dev` branch when you update your `fork`.

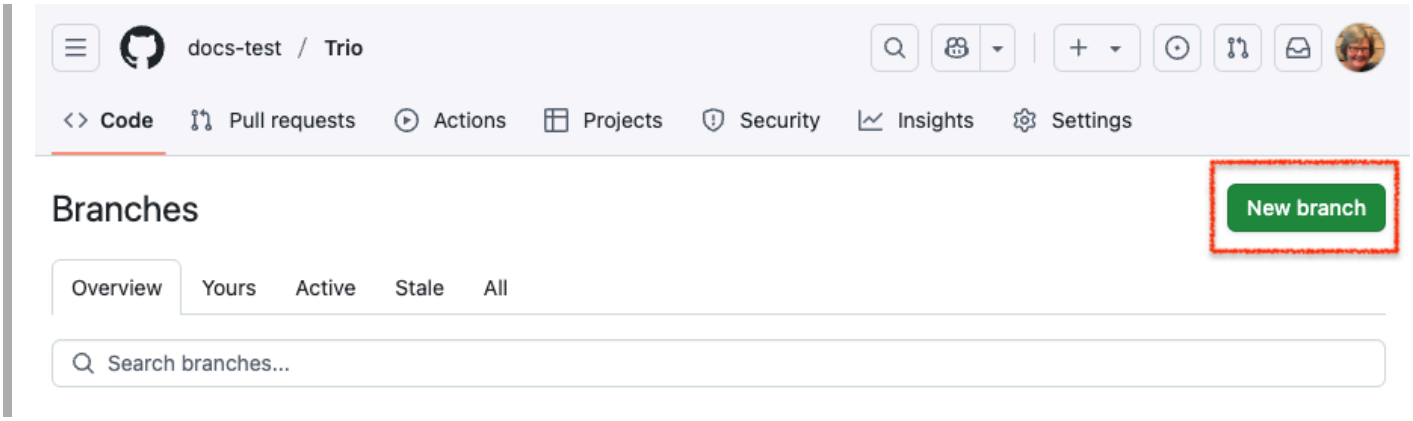
Add `Branch`

First, you need to get to your fork of the Trio repository and tap on the branch icon. (Do not worry about how many branches you have.):



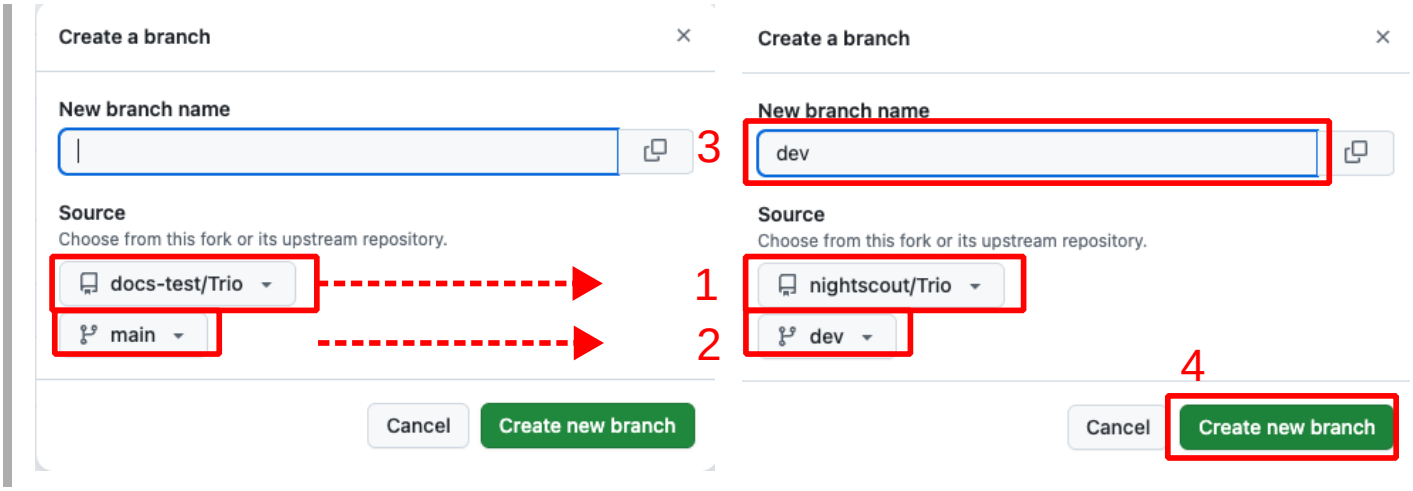
The next screen displays the branches you currently have.

- If the branch you want is already there - click on it so you can make sure it is in sync with Trio
- If the branch you want is not there, tap on the `New branch` button on upper right
  - If there is not a `New branch` button in the upper right, you are not logged in



Each step in the list below matches with the number in the graphic. On the left half of the graphic is the default selections for your `fork` when `main` is the default branch. The right side shows the display after making the indicated selections:

1. Click on the drop-down menu labeled 1 in the graphic and choose nightscout/Trio
2. Click on the drop-down menu labeled 2 in the graphic and choose `dev`
3. Click on the `Branch` name box labeled 3 in the graphic and type `dev`
   - The branch name in your `fork` should always match the branch name you are adding; check that you type it correctly
4. Review the dialog items to make sure everything is correct, and then tap on Create branch



Configure the `dev` branch as default

> **This recommendation is only for expert testers who are using `dev` or a feature branch for their personal use. Typically only developers configure the `dev` branch as their default branch.**

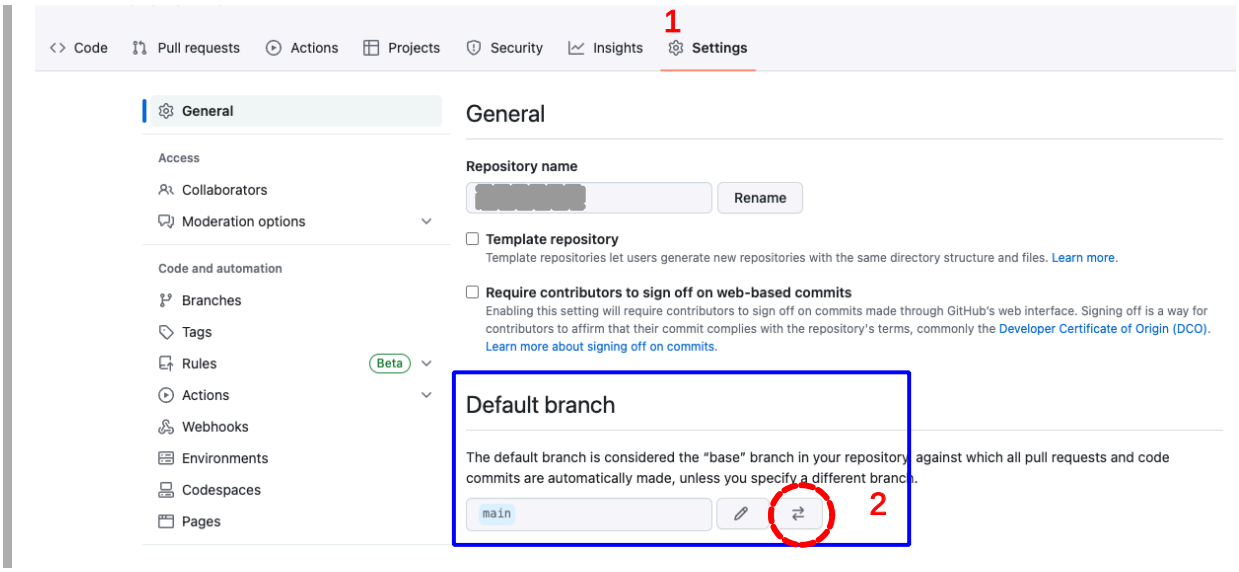
- By making the `dev` branch your default, you will get automatic build of any `dev` updates weekly
- You will no longer get automated builds of the `main` branch, but can run those manually

These are the steps to modify the default branch.

For this example, we show how to change from a default branch of `main` to a default branch of `dev`. Note — only the owner of the repository can take this action and they must be logged in. Otherwise the Settings tab does not appear.

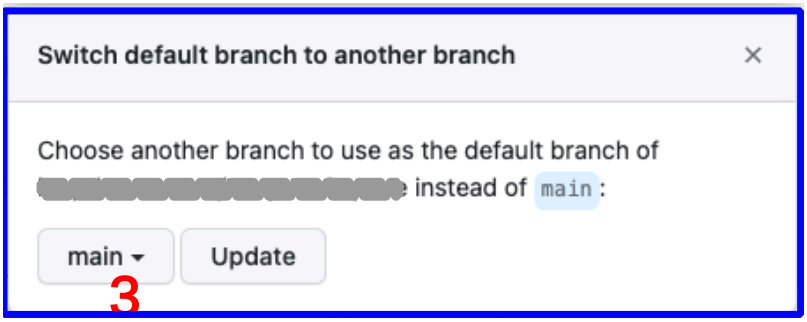
For the numbered steps below, refer to the graphic found under each group of steps.

1. Click on the Settings Icon near the top right of your Trio repository
    - You may need to scroll down to see the `Default Branch` as shown in the graphic
    - Do not tap on the Branches tab to the left under Code and Automation, that is not the correct menu



2. To the right of the default branch name there is a pencil and a left-right arrow icon (⇄)
    - Tap on the left-right arrow icon (⇄) to bring up the `Switch default branch to another branch` dialog

3. Click on the dropdown next to the current default branch, in this example, `main`



1. Select the desired default branch, in this example, `dev` was selected
2. Click on the `Update` button
3. You will be presented with an are-you-sure question.
    - Click on the red `I understand, update the default branch.` button

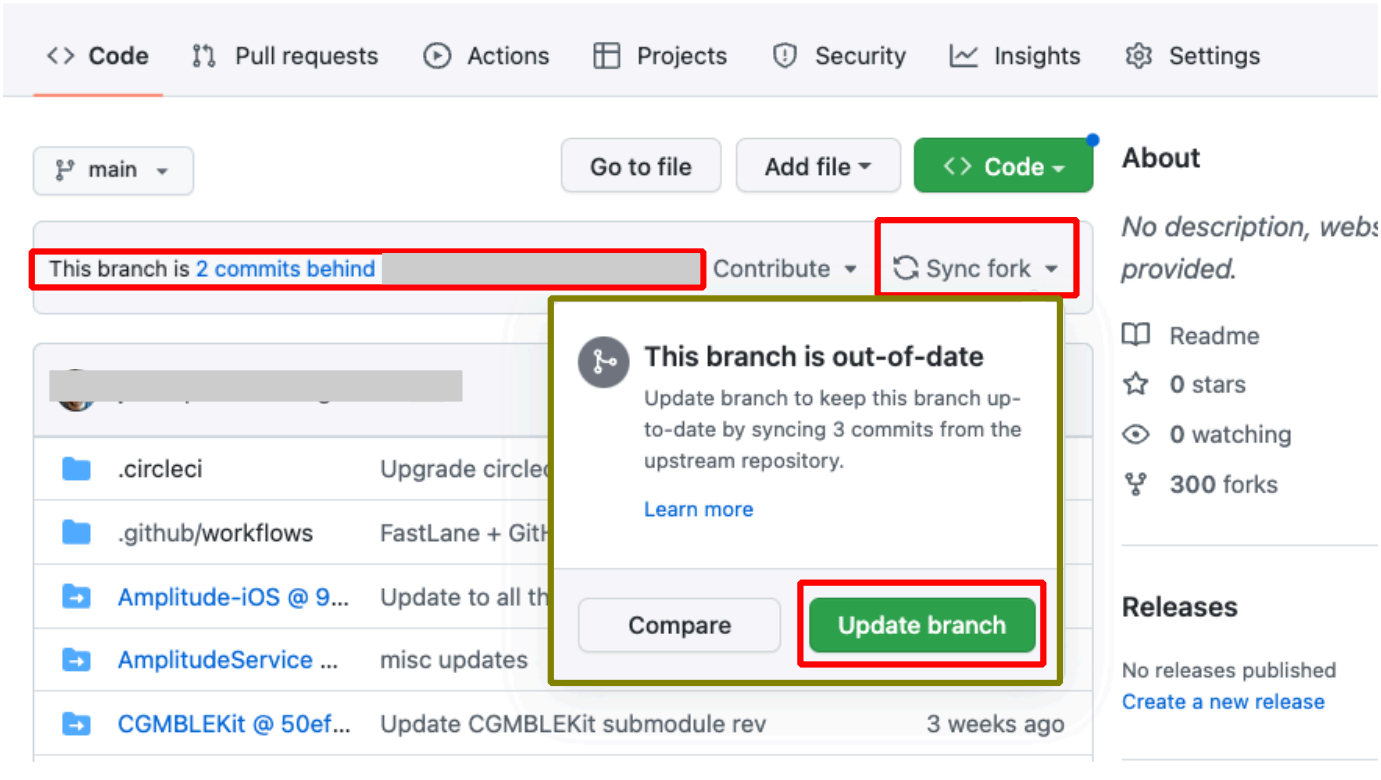
Your default branch has been changed.

Update `Branch`

The `dev` branch of the `nightscout/Trio` repository will be updated frequently. This is how you know if your `fork` needs to be updated as well. As long as the automatic runs are happening weekly for your repository — you will not need these manual instructions, unless you want to manually build an update sooner.

Tap the `Code` button (upper left) and ensure this branch in your `fork` is up to date.

- Select the desired branch in the dropdown menu (this graphic shows `main` branch, to get 0.5 (or newer), you must choose `dev` branch)
- If the message indicates this branch is "behind", tap on the sync `fork` button and then the Update branch button

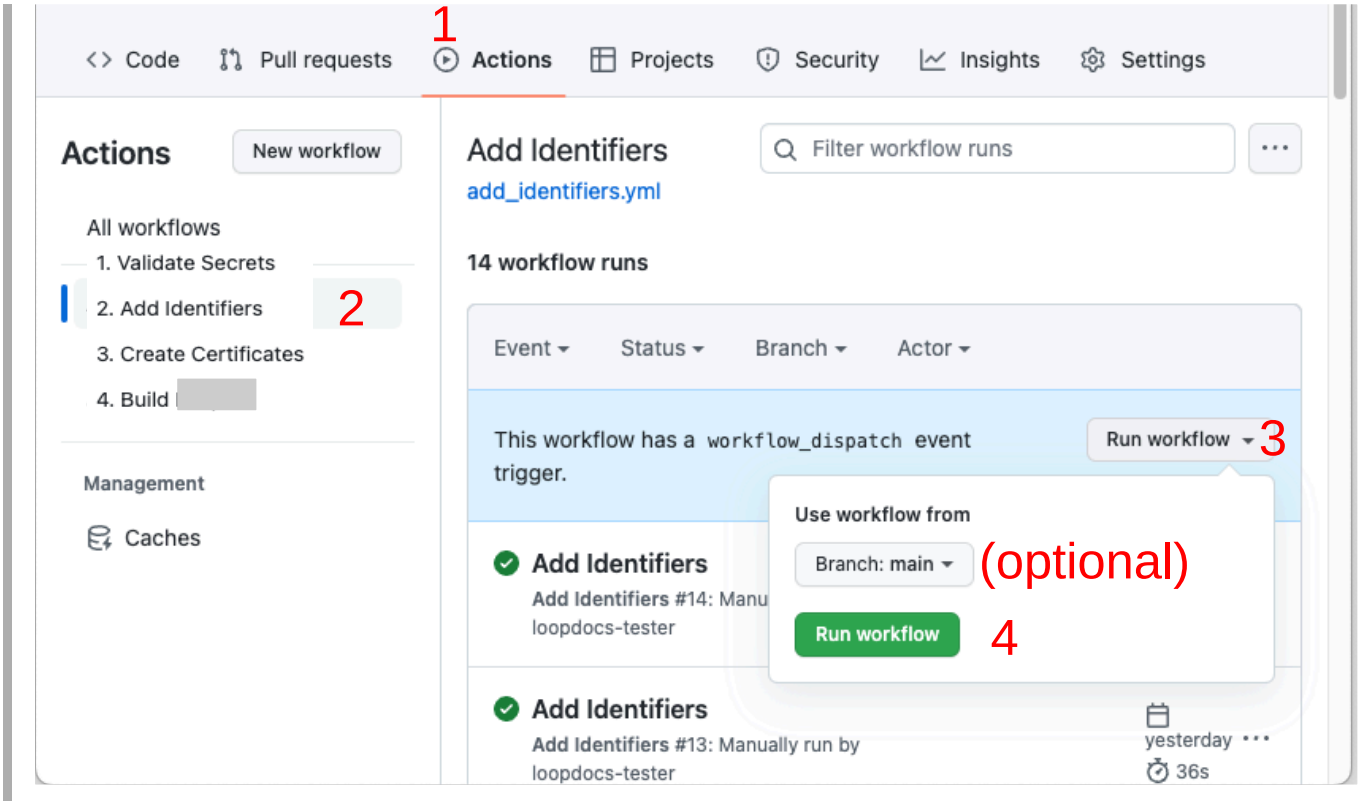


Update `Identifiers`

If identifiers are added to the `dev` or feature branch you are using, this will be required, otherwise you can skip this.

Refer to the graphic below for the numbered steps:

1. Click on the `Actions` tab of your Trio repository
2. On the left side, click on `2. Add Identifiers`
3. On the right side, click `Run Workflow` to show a dropdown menu
    - You will see your default branch (typically this is `main`)
    - **To update the `Identifiers` found in dev but not main, you must select `dev`**
4. Tap the green button that says `Run workflow`.



Be Patient

- Refresh the browser if you are unsure if the action started
- Do not start a new action until the first one completes

The `Add Identifiers` Action should succeed or fail in a few minutes. Do not continue to the next step until this one succeeds.

Enable Time Sensitive Notifications

Trio `dev` version 0.8.4.3 added the requirement for the Time Sensitive Notifications capability to be enabled for your `Trio` Identifier. If you already enabled it, you can skip this step.

- This is a one-time manual step in your *Apple Developer* account - the `Add Identifiers` action does not enable this capability for you
- If the capability is not enabled, the `Build Trio` action fails

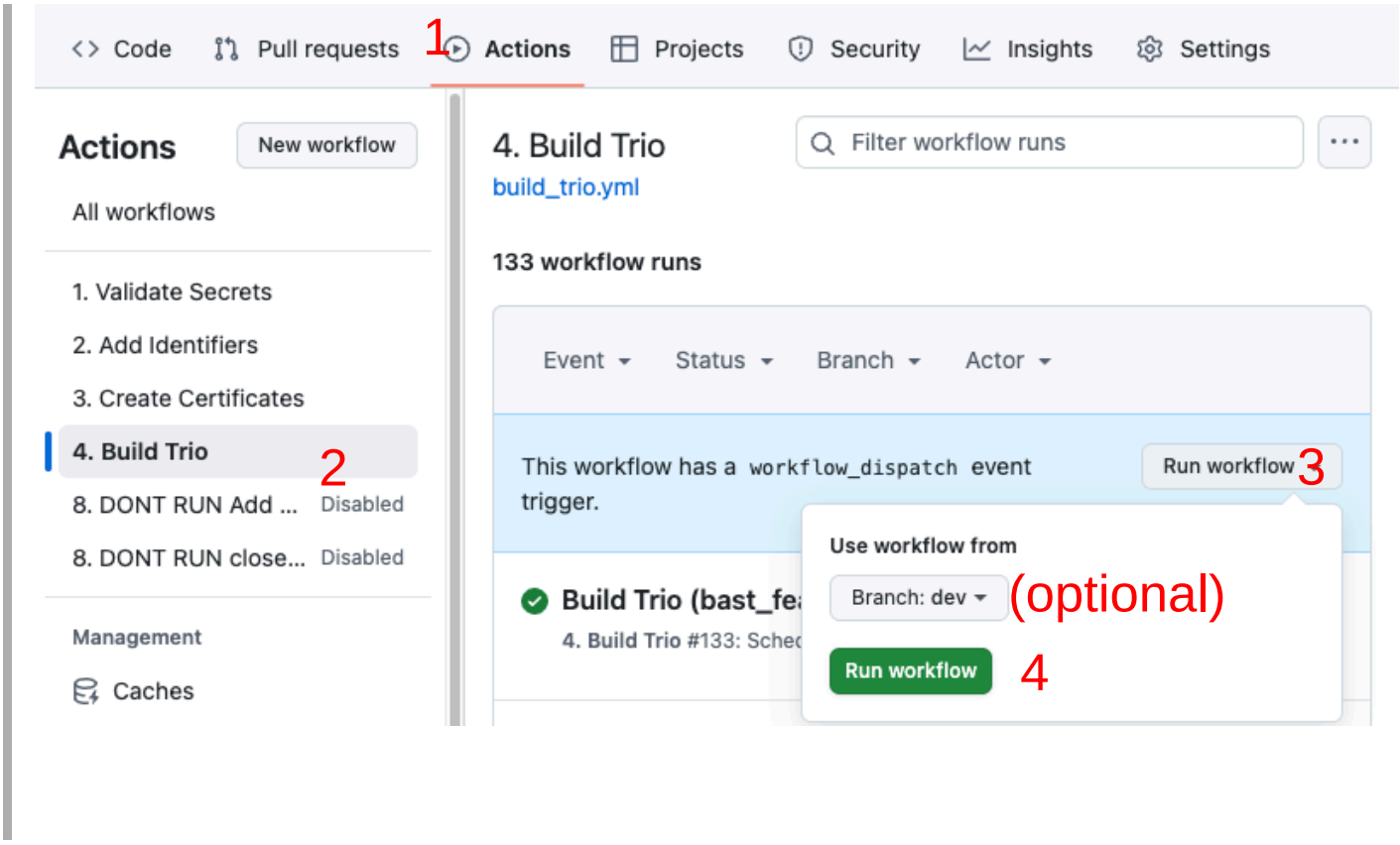
Follow the directions at Enable Time Sensitive Notifications, then return to this page and continue with Build the App.

- You do not need to run any extra actions after checking the box - `Build Trio` includes the Create Certificates step, which updates your provisioning profile to include the new capability

Build the App

If you completed all the steps on this page successfully (got a green checkmark ✅), you are ready to run Action: Build Trio.

If you choose to build a different branch than your default branch, there is an extra step when you `Build Trio`. In addition to the normal steps 1, 2, and 3 in the graphic below, you must also do the (optional) step. Select the `dev` branch in the branch dropdown menu before continuing to step 4 and tapping on the green Run workflow button:



Refresh, Do Not Repeat

> **Hit Refresh**
>
> After you tap the green Run workflow button, *GitHub* can be slow to update.
>
> - Refresh the browser if you are unsure if the action started
> - Do not start a new action until the first one completes

Update Build Errors

Read the **build action annotations** if you see an error.

Time Sensitive Notifications Error

If you did not Enable Time Sensitive Notifications for your `Trio` Identifier, the `Build Trio` action fails with an error similar to this:

```
error: Provisioning profile "match AppStore org.nightscout.***.trio" doesn't include the
com.apple.developer.usernotifications.time-sensitive entitlement.
```

Solution: Enable Time Sensitive Notifications and then run `Build Trio` again - make sure you select the `dev` branch.

Identifiers Error

This example is from the past when Push Notifications were added to the `dev` branch.

Example Error

> **Annotations**
> 2 errors and 1 warning
>
> **Build**
> `Process completed with exit code 1.`
>
> **Build:** Trio.xcodeproj#L0
> `Provisioning profile "match AppStore org.nightscout.***.trio" doesn't support the Push Notifications capability.  (in target 'Trio' from project 'Trio')`
> Show less

If you decide to look at the action **log**, instead of reading the annotations, you may see an **error** similar to the two graphics below.

```
518 [23:27:00]: ▸ ::error file=/Users/runner/work/Trio-dev/Trio-
    dev/Trio.xcodeproj::Provisioning profile "match AppStore org.nightscout.***.trio" doesn't
    support the Push Notifications capability.  (in target 'Trio' from project 'Trio')
519 [23:27:00]: ▸ /Users/runner/work/Trio-dev/Trio-dev/Trio.xcodeproj: error: Provisioning
    profile "match AppStore org.nightscout.***.trio" doesn't include the aps-environment
    entitlement. Profile qualification is using entitlement definitions that may be out of
    date. Connect to network to update. (in target 'Trio' from project 'Trio')
520 [23:27:00]: ▸ note: Disabling previews because SWIFT_VERSION is set and
    SWIFT_OPTIMIZATION_LEVEL=-O, expected -Onone (in target 'Trio' from project 'Trio')
521 [23:27:00]: ▸ note: Run script build phase 'Swiftformat' will be run during every build
    because the option to run the script phase "Based on dependency analysis" is unchecked.
```

Solution

1. Update `Identifiers` - make sure you select `dev` branch
2. Build the App - make sure you select `dev` branch

Mac Build Overview

Build Trio with Mac - Overview

Congratulations on choosing to build Trio using your Mac computer. This documentation provides all the necessary instructions to download a local copy of the Trio repository using the loopandlearn `TrioBuildSelectScript`. It will then guide you on using a compatible computer with the Xcode application to build and deploy the Trio application to a compatible device.

Requirements:

- Compatible Phone ↗
- Compatible CGM ↗
- Compatible Insulin Pump ↗
- Apple Developer Account ↗
- Compatible Computer ↗
- Xcode ↗

Getting Ready to Build

The first step in getting ready to build Trio using your Mac computer is to review the above requirements. These are considered the minimum requirements for Building Trio using your Mac. Don't stress if you are unable to meet the above requirements. Other options are available for building Trio, and I recommend you start with Build Trio with Browser ↗ .

The second step is to read over and become familiar with the build instructions in their entirety. This includes familiarizing yourself with common Build Errors ↗ . Most mistakes you can make while building Trio have already been made, and our dedicated community has documented necessary fixes.

When you are ready to proceed, take your time, read carefully, and complete the tasks on each page. Read the top three boxes on each page; icons for those boxes are displayed below for reference.

Overview

Time Estimate

- 5 minutes, if your computer meets minimum requirements.
- 30-60 minutes, if you need to install macOS updates.

Summary

To build the Trio app on a Mac, your computer, iPhone, and Xcode must have compatible versions. If you are buying a Mac to use the build with the Mac method, choose one that can be updated to the Tahoe (macOS 26) operating system and has at least 256 GB (512 GB is better). The Build with Browser method works on any computer or tablet.

FAQs

- **Do I need a Mac?** No, you can Build Trio with Browser on any computer.
- **How often do I need to use the computer to build with the Mac method?** Computer access is required when:
  - Initially installing the Trio app.
  - Trio will expire annually for a paid account or weekly for a free account.
  - Updating to a newer Trio release.
  - You do not need access to an Apple computer to update your phone's iOS, troubleshoot, or change Trio settings
- **What if my computer doesn't meet the minimum requirements** If your 'macOS' does not meet the minimum standards as your iOS version dictates, you will need to complete a software update on your computer to continue. Suppose you cannot update your computer to the required 'macOS' due to hardware limitations. In that case, your options are as follows:
  - Build Trio with GitHub using the Browser Build method.
  - Borrow an up-to-date Mac computer.
  - Purchase an up-to-date Mac computer.

What if I get stuck?

- First, take some time to read over the instructions to ensure you have understood all the steps.
- Compare your screen's display with the images in the documents. Is there something different? Does yours have an error message? If so, does it guide you to the problem and solution?
- If you are still stuck, you can seek assistance via Facebook or Discord.

Prepare your Computer

Prepare your Computer

Computer Compatibility

Let's confirm that your computer is compatible with and meets the requirements for building Trio.

The iOS of the phone on which you wish to build Trio dictates the required macOS of your computer. At a minimum, the current release of Trio requires iOS 16.3 or higher, Xcode version 15 or higher, and macOS 13.5 or higher.

The table below lists the **minimum** requirements to build the current release of Trio. If your macOS or Xcode version is higher, you can build on a Mac.

Find your iPhone iOS version in the table below.

iOS Version

If your iOS version is not listed, e.g., 26.4.1, choose the first row that is lower than your iOS version.

iOS Version	Minimum Xcode Version	Minimum macOS Version
26.4	16.4	15.3
18.1	16.1	14.5
18.0	15.4	14.5
17.5	15.4	14.0
17.4	15.3	14.0
17.0	15.0	13.5

Check your iOS version

1. On your iPhone, click on the 'Settings' application.
2. Next, select the 'General' option.
3. Then, select the 'About' option.
4. Review and take note of your 'iOS Version' at the top of the page.
5. Compare your 'iOS Version' with the above table to confirm what "macOS" your computer requires.

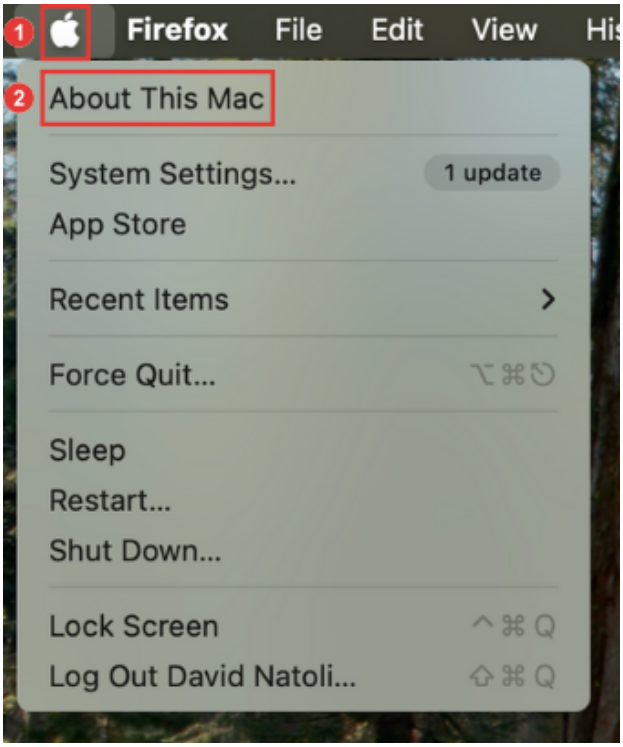
Check your macOS version

1. Select the 'Apple' icon at the top left corner of your computer screen.
2. Next, select the 'About This Mac' option.
3. This will open an application with information specific to your Mac computer. Take note of your 'macOS', and in combination with your 'iOS Version,' review the above table to confirm compatibility.

Check the Space Available on your Hard Drive

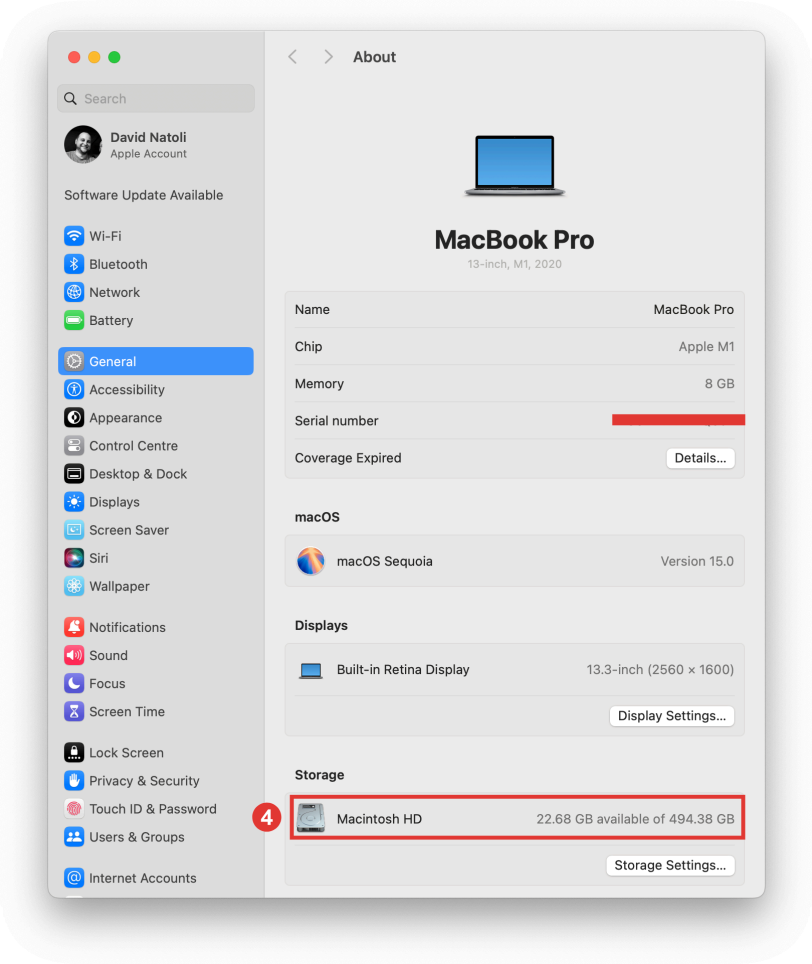
In order to install Xcode on your Mac computer, your hard drive must have at least 50GB of free space. You can check how much space you have by following the instructions below.

1. Select the 'Apple' icon at the top left corner of your computer screen.
2. Next, select the 'About This Mac' option.



3. This will open an application with information specific to your Mac computer. Select the 'More Information' button to open a new window with additional information about your computer.

4. Ensure the 'General' tab is open. Scroll down to the 'Storage' section. Your hard drive will be listed with the volume available out of the total capacity.
5. As noted above, a minimum of 50GB capacity is required to install Xcode and its components.



Prepare your iPhone

Overview

Time Estimate

- 5 minutes, to check your device and iOS.
- 20 minutes to update your compatible device to a new iOS.
- 0 minutes, if you own an Android and will not use Apple products; check out AndroidAPS Documentation

Summary

- Check your phone against the compatible device list.
- Make sure the phone has good battery life.
- Activate `Developer Mode`.
- Turn off automatic updates.

FAQs

- **Can I use an Android?** No. Check out AndroidAPS Documentation.
- **Does my phone need a cell plan?** No. Trio works using your phone's Bluetooth communications with your CGM and your insulin pump; no internet connection is required. However, an active internet connection is required if access to Dexcom Follow or Nightscout monitoring is a priority.
- **What watches work with Trio?** Trio has a companion application for the Apple Watch. The Trio watch app can add carbs and boluses, set temporary targets, and visualize your glucose, insulin, and carbs onboard. This application works only with the Apple Watch. Some Garmin watches are also compatible but only display data.

Compatible iPhones

Trio requires a relatively recent phone. The faster the phone, the better for the algorithm and battery life. You need a minimum version of the mobile operating software, called the phone's iOS, to be installed on your phone.

Trio is compatible with phones running on **iOS 17 or newer**.

These iPhones are compatible with iOS version 17 and newer:

- iPhone 17*, all variants
- iPhone 16*, all variants
- iPhone 15, all variants
- iPhone 14, all variants
- iPhone 13, all variants

- iPhone 12, all variants
- iPhone 11, all variants
- iPhone XS Max
- iPhone XS
- iPhone XR
- iPhone SE (3rd generation; 2022 first release)
- iPhone SE (2nd generation; 2020 first release)

iPhone 16 (all) and iPhone 17e with Atlas/InPlay DASH pods

Using an iPhone 16 (the entire line) or an iPhone 17e with Atlas/InPlay DASH pods can cause communication issues, but SAW/TWI DASH pods, Omnipod 5 pods, and all other pumps work fine. More info is available at LoopDocs.

iOS Version

If your iOS version is not listed, e.g., 26.4.1, choose the first row that is lower than your iOS version.

iOS Version	Minimum Xcode Version	Minimum macOS Version
26.4	16.4	15.3
18.1	16.1	14.5
18.0	15.4	14.5
17.5	15.4	14.0
17.4	15.3	14.0
17.0	15.0	13.5

Find your iOS Version

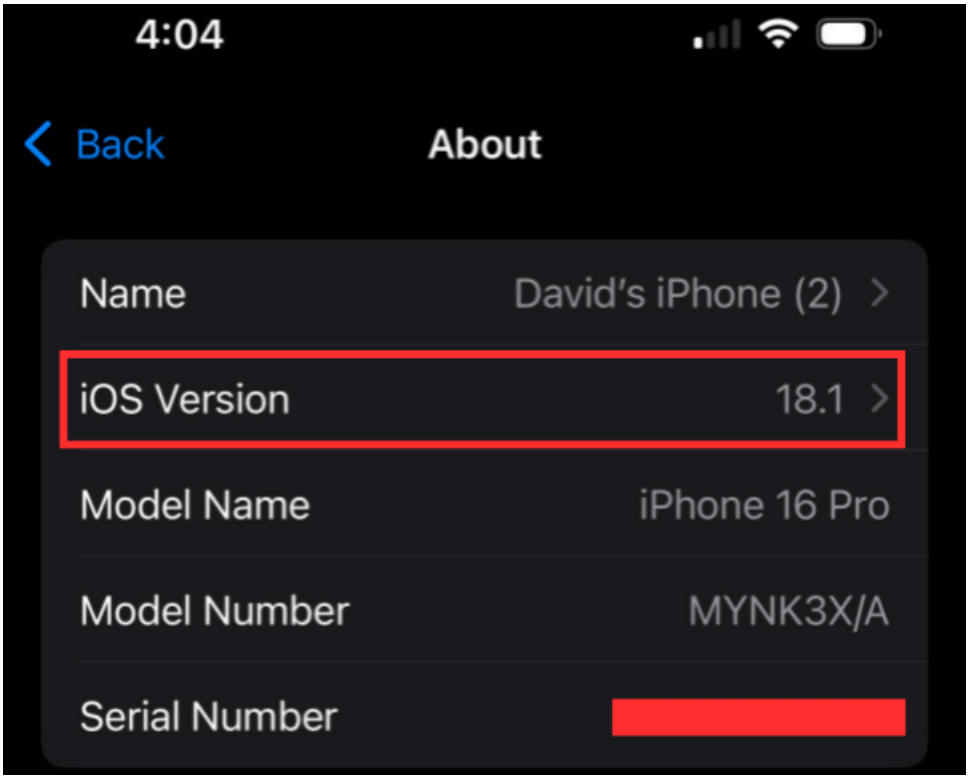
Warning

Your phone iOS dictates your MacOS and Xcode requirements - The more up-to-date you keep your phone iOS, the more up-to-date your computer and MacOS must be to build Trio with your Mac.

You are encouraged to update your iPhone as soon as you determine you can do so safely:

- Check to see if the mentors give the iOS version a thumbs up: Version Update
- Make sure your Mac and Xcode versions will enable you to build if you need to once you update your iOS
- If you cannot update your Mac, check out the Build with Browser ↗

1. On your phone, click on the 'Settings' application.
2. Next, select the 'General' option.
3. Then, select the 'About' option.
4. Review and take note of your 'iOS Version' at the top of the page.
5. Compare your 'iOS Version' with the above table to confirm what "macOS" your computer requires.



Warning

iOS Beta versions are untested with Trio. Therefore, they should never be on your device. Limited support will be available for beta version users.

Automatic Updates

Apple frequently updates iOS, often including critical security patches and improved new features. In some instances, iOS updates can make Trio stop working. **Minor** updates (like 16.5 -> 16.5.1) are usually not an issue. Whereas, **Major** updates (like 17.x -> 18.x) often contain changes that Trio developers need to include in the app.

Turn Off Automatic Updates!

- Once you accept an iOS update, you cannot go backward.
- Some iOS updates require updates to Xcode and macOS before people can build Trio on that device again.
- It is rare, but iOS updates have caused Trio to stop working until other updates were made and Trio was rebuilt on that phone.
- Turn off automatic updates so you can choose when to update your phone and avoid being caught without a working Trio app.
- Google the instructions for your device:
  1) Configure your phone to download the updates automatically.
  2) Choose to install the updates manually.

Developer Mode

All devices using Trio must have 'Developer Mode' activated. This change came with iOS 16 and watchOS 9. This feature must be activated to allow Trio to install on the device. To learn more, click on this link: Apple Developer Mode.

Activate Developer Mode - iPhone

1. To activate 'Developer Mode', your phone must be turned on and plugged into the computer with the Xcode application open.
2. Your phone will then prompt you to trust the computer. Select 'Trust'.
3. Next, open your phone's 'Settings' application.
4. In the 'Settings' application, scroll and choose 'Privacy & Security'.
5. Scroll to the bottom of the page, and under the 'Security' banner, you will see an option for 'Developer Mode';
   - If it says On - no further action is required.
   - If it says Off, tap on the row and slide the slider to the green (enabled) position.
6. Your phone will then prompt you to restart the device.
7. After the restart, you will be prompted to turn on 'Developer Mode'.
8. You are now ready to begin building from Xcode onto this phone.

Activate Developer Mode - Watch

> **Build, Enable, Build**
>
> Users' reports indicate that when building a new Apple Watch, you must first build the app with Xcode before the developer mode becomes available. So, plan to build with the watch paired, and then enable `Developer Mode` and build again.

In addition to your phone, if you plan on using an Apple Watch, then you must activate 'Developer Mode' on the watch itself. To determine if 'Developer Mode' is activated, follow the following instructions:

1. Open the 'Settings' application on the watch.
2. In the 'Settings' application, scroll and choose 'Privacy & Security'.
3. Scroll to the bottom of the page, and under the 'Security' banner, you will see an option for 'Developer Mode';
    - If it says On - no further action is required.
    - If it says Off, tap on the row and slide the slider to the green (enabled) position.
4. Your watch will then prompt you to restart the device.
5. After the restart, you will be prompted to turn on 'Developer Mode'.
6. You are now ready to begin building from Xcode onto this watch.

Enable Watch WidgetKit

There are now options that appear after you enable 'Developer Mode'. Go on and configure those now. 1. Open the 'Settings' application on your watch.

1. Next, scroll to the bottom of the page and tap on the 'Developer' option.
2. Scroll until you see WidgetKit Developer Mode and enable the option. This will allow faster updates of watch complications.

Set Up Xcode

Overview

Time Estimate

It will take 45 minutes to 2 hours, depending on your internet connection, but you don't need to babysit the download.

Summary

- Check the phone's iOS version
  - The iOS version determines the minimum Xcode version.
  - The minimum Xcode version determines the minimum macOS version.
  - If necessary, update your MacOS first and then return to this step.
- Download (or update) Xcode from your computer's 'App Store' application.

FAQs

- **Why isn't my Xcode installing?**
  The two most common reasons are:
  a. Lack of internet connection, or;
  b. Not enough space on your hard drive. Xcode is a large download that requires at least 50GB of space to unpack and install properly.
    - You will have installation failures if you do not have enough space on your hard drive. Although the download takes a long time, the good news is that you can walk away once it starts. If your laptop goes to sleep when you close the lid or when the screen saver starts, disable the screen saver and leave the computer open.
    - After Xcode has finished downloading (it looks like the progress bar is almost completed), it takes a long time to unpack and install; be patient.
- **Can I install Xcode on an external drive?** Unfortunately, no. Xcode needs to be on the Mac hard drive.

Download and Configure Xcode

What is Xcode?

Xcode is a free application for Apple computers. You will use Xcode to turn the "raw" Trio source code into an iOS application and install it onto your iPhone. Which version of Xcode you install on your computer depends on the iOS version you have on the iPhone you will be installing Trio on and the macOS version you have on your computer.

What version of Xcode will I need?

First, choose an appropriate Xcode version for your iOS device. Then, determine the minimum macOS version required for that Xcode version. Update to at least that minimum macOS version.

How do all the minimum versions relate to each other?

The current release of Trio requires iOS 16.3 or higher, Xcode version 15 or higher, and macOS 13.5 or higher.

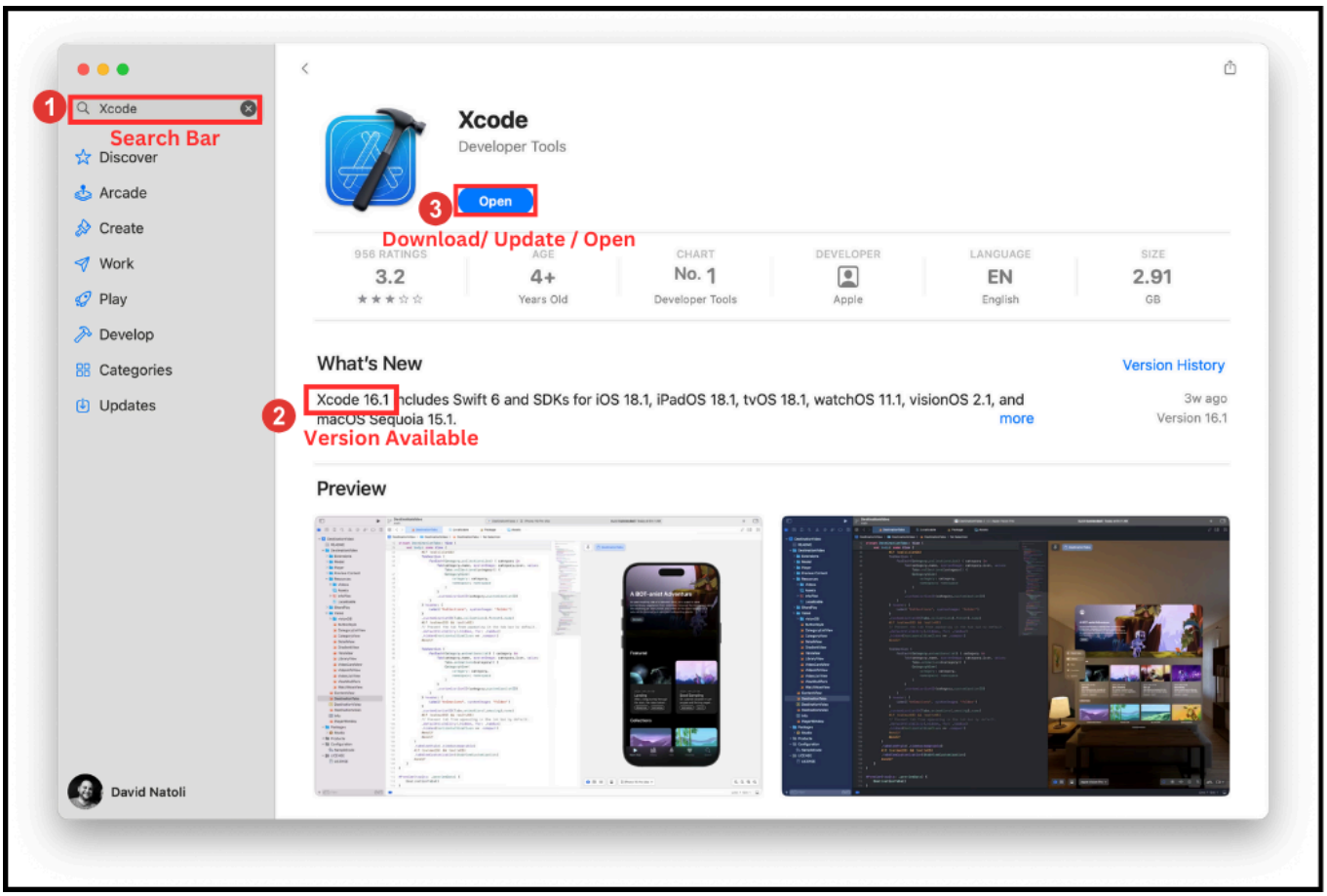
The table below lists the **minimum** requirements to build the current release of Trio. If your macOS or Xcode version is higher, you can build with a Mac.

> **iOS Version**
>
> If your iOS version is not listed, e.g., 26.4.1, choose the first row that is lower than your iOS version.

iOS Version	Minimum Xcode Version	Minimum macOS Version
26.4	16.4	15.3
18.1	16.1	14.5
18.0	15.4	14.5
17.5	15.4	14.0
17.4	15.3	14.0
17.0	15.0	13.5

Download Xcode

1. On your Mac computer, open the app store.
2. Next, in the top left corner of the App Store, there is a search engine. Search for Xcode.
3. Select the Xcode application from the search results.
4. Review the available Xcode version.
5. If the Xcode version you require is higher than the Xcode version available in the App Store, you need to update your computer.
6. Select download. Be patient; depending on your download speeds, this could take a couple of hours.



7. The Xcode application will be available from your Launchpad or Applications file once the download is complete.

Configure Xcode

Time Estimate

- Installing the Command Line Tools takes about 10-15 minutes.
- 5 minutes to add your Apple ID, assuming you remember your password.

Summary

- Verify that Command Line Tools are installed.
- Add your Apple Developer account to Xcode.

FAQs

- **I still only see an account with `(personal team)` beside it even though I enrolled in the paid Developer Account program...what should I do?** You should check your spam email box in case Apple sent you an email there. Ensure you've waited the 48 hours Apple says it may take to get your account approved. If it's been 48 hours and you still don't see anything in your email, contact Apple support and ask them about your enrollment status. It may be held up by something on their end.

Now that you have downloaded Xcode, you must configure some settings to streamline Trio's build process. Let's get started.

Check Xcode Version

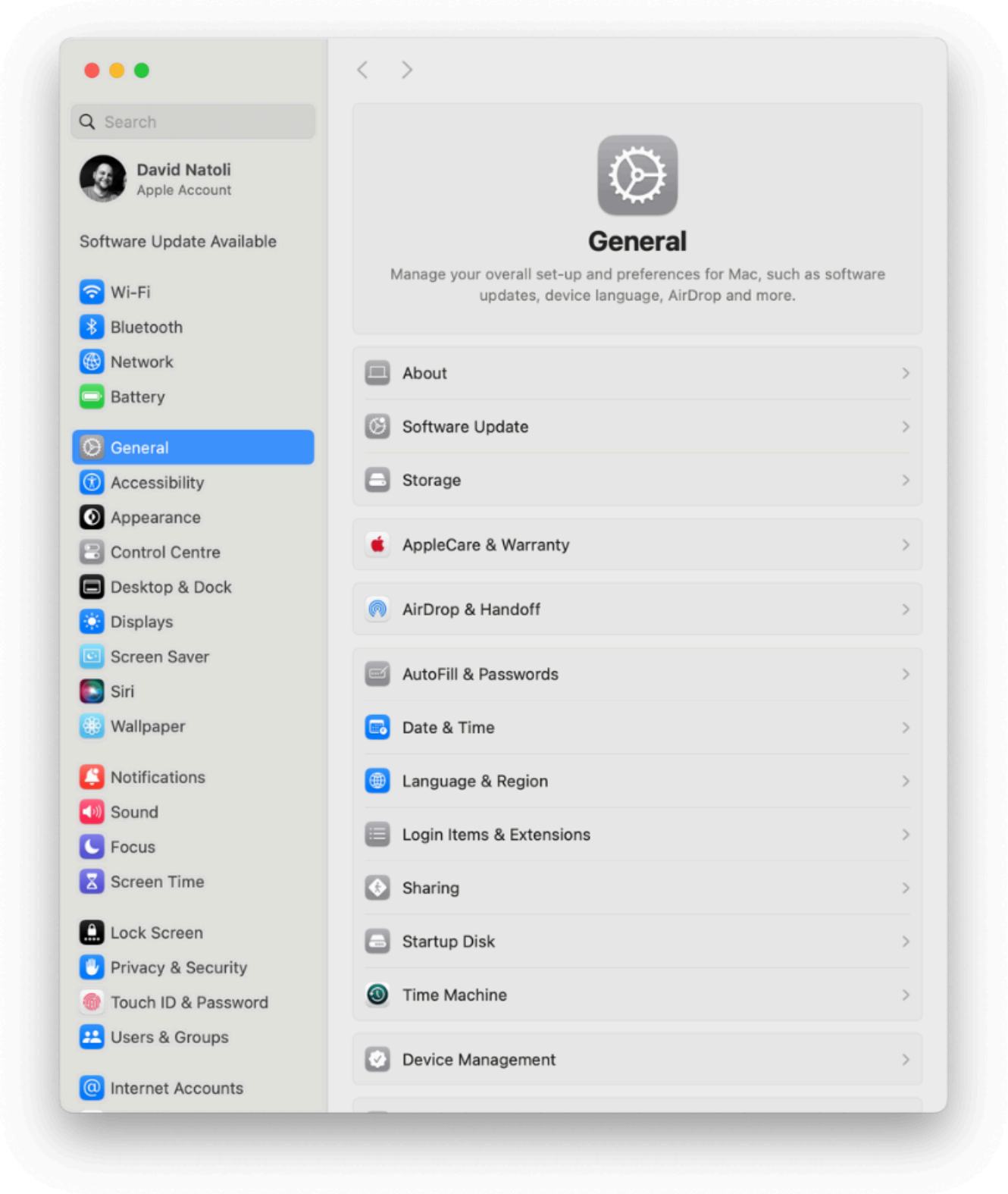
Let's check if the correct version of Xcode was installed.

1. Open Xcode from either Launchpad or Applications folder.
2. In the top left-hand corner of the window, select 'Xcode'
3. Next, select 'About Xcode' from the menu.
4. A window will open, and you can review the Xcode version installed on the computer.

Privacy Settings

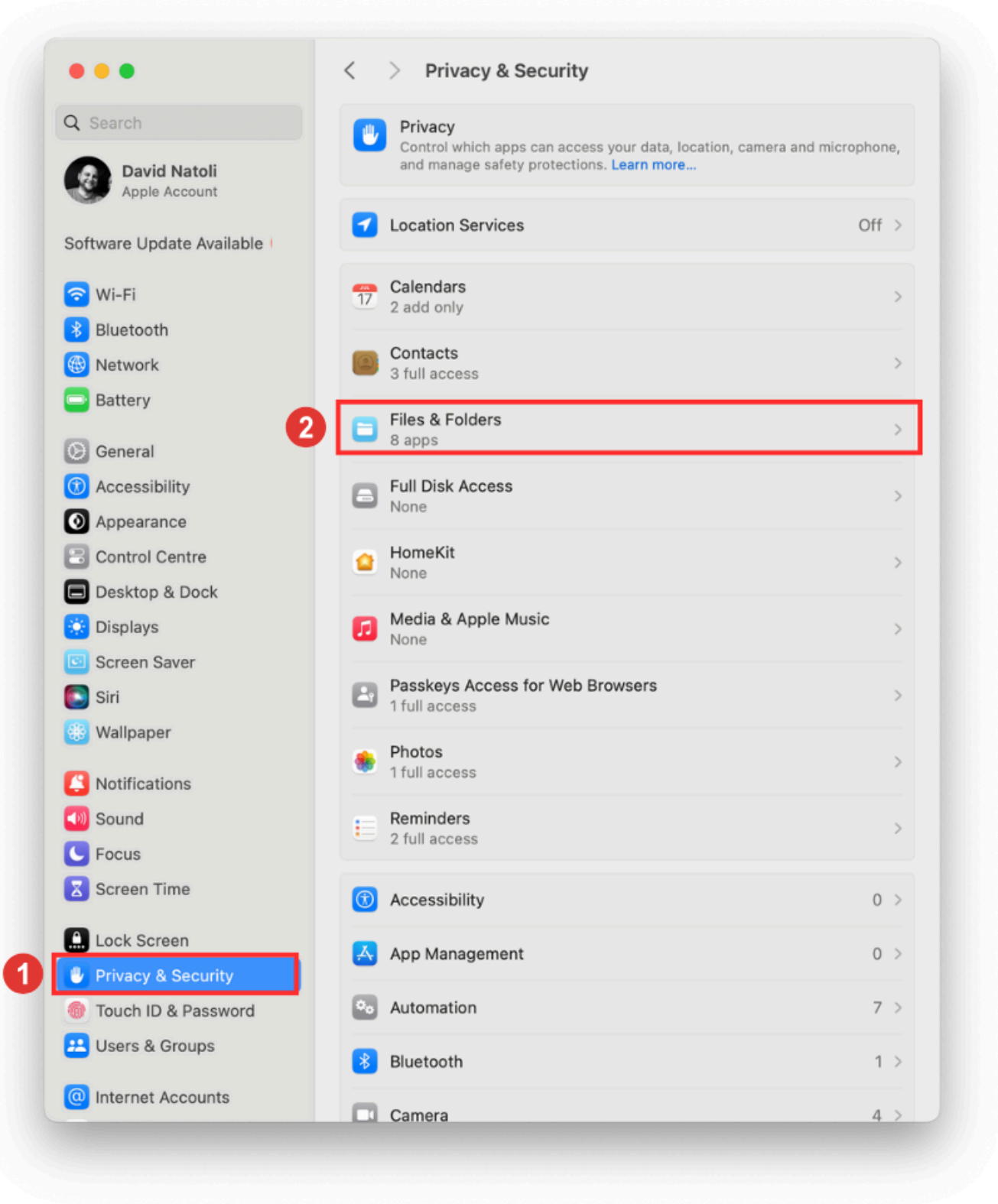
Sometimes, people have their macOS privacy settings configured so that Xcode does not have permission to access their computer's 'Downloads' folder. This can become problematic when using the build script to build an app with Xcode. To ensure access is allowed, follow these instructions:

1. Select the 'Apple symbol' in the top left-hand corner of the screen.
2. Next, select 'System Settings'.



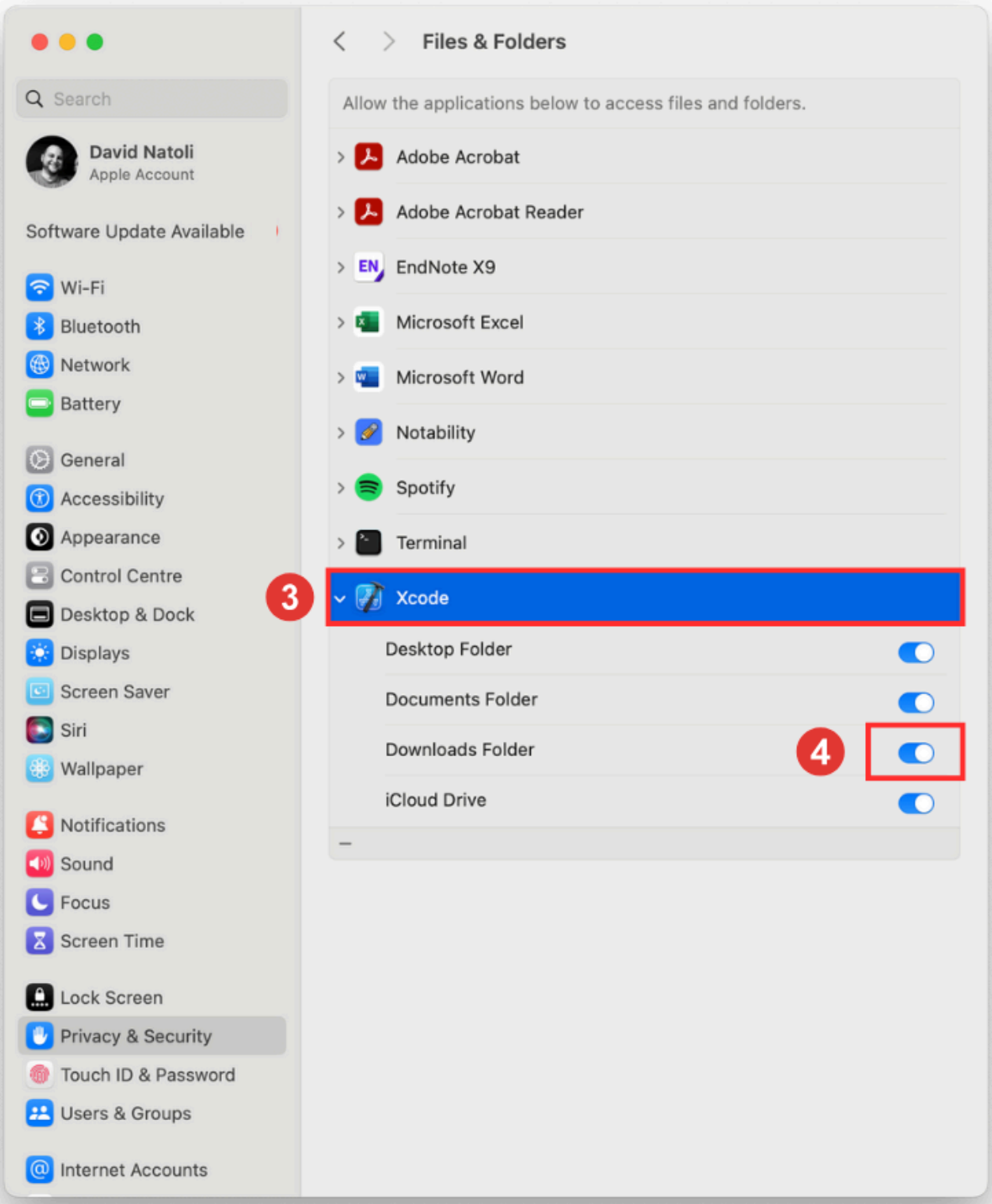
3. Next, we are going to select 'Privacy & Security".

4. Then, we are going to select the 'Files & Folders' option.



5. We can then select the Xcode application.

6. Once you select the Xcode application, ensure access to the downloads folder is toggled on.

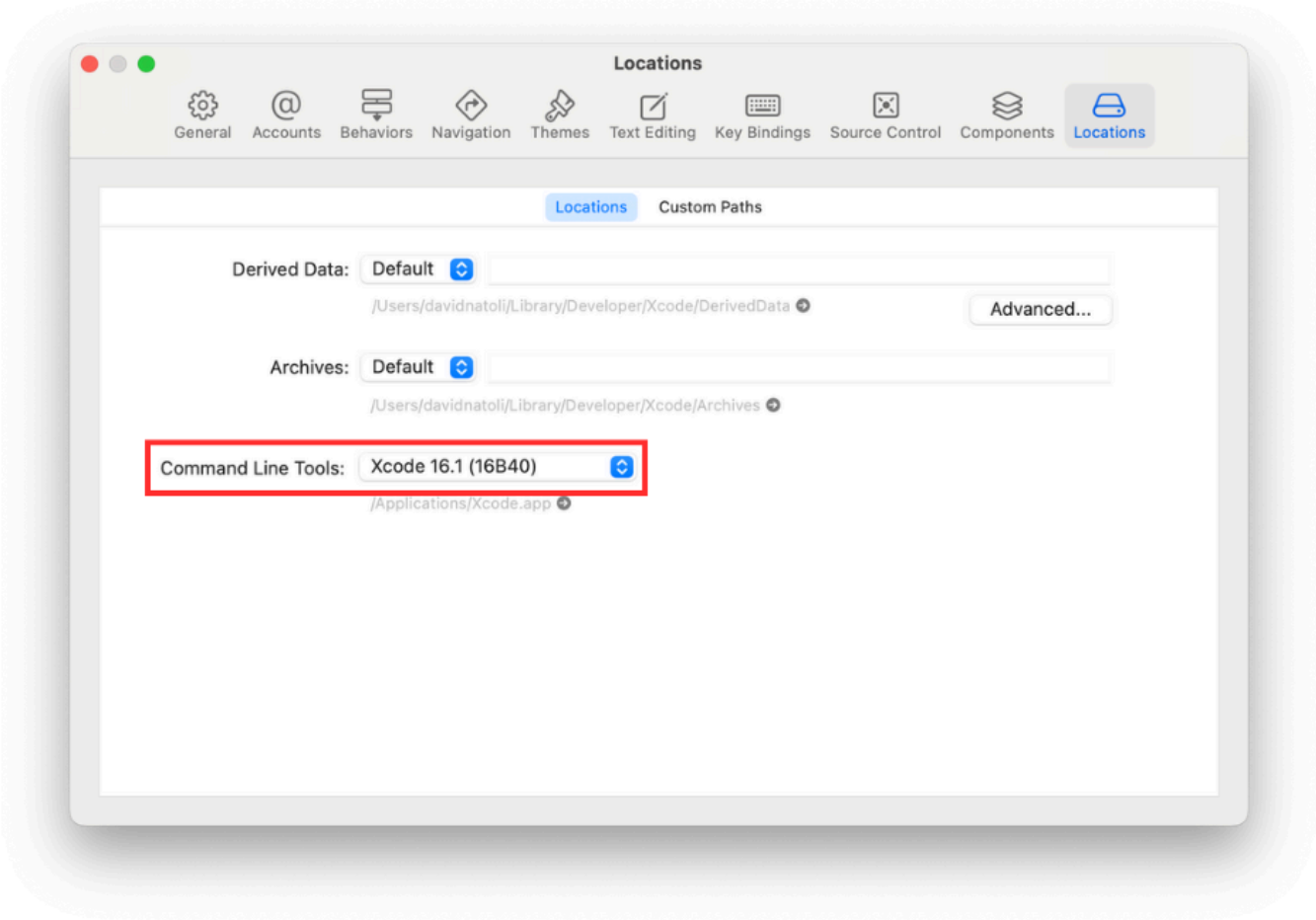


Command Line Tools

The first time you open Xcode, it will want to install a package of command-line tools. Please wait patiently for this to be completed. In some instances, the command-line tools may have been installed without asking. Before proceeding, please confirm that the tools have been installed correctly.

1. Open Xcode.
2. In the top left-hand corner of the screen, select 'Xcode'.

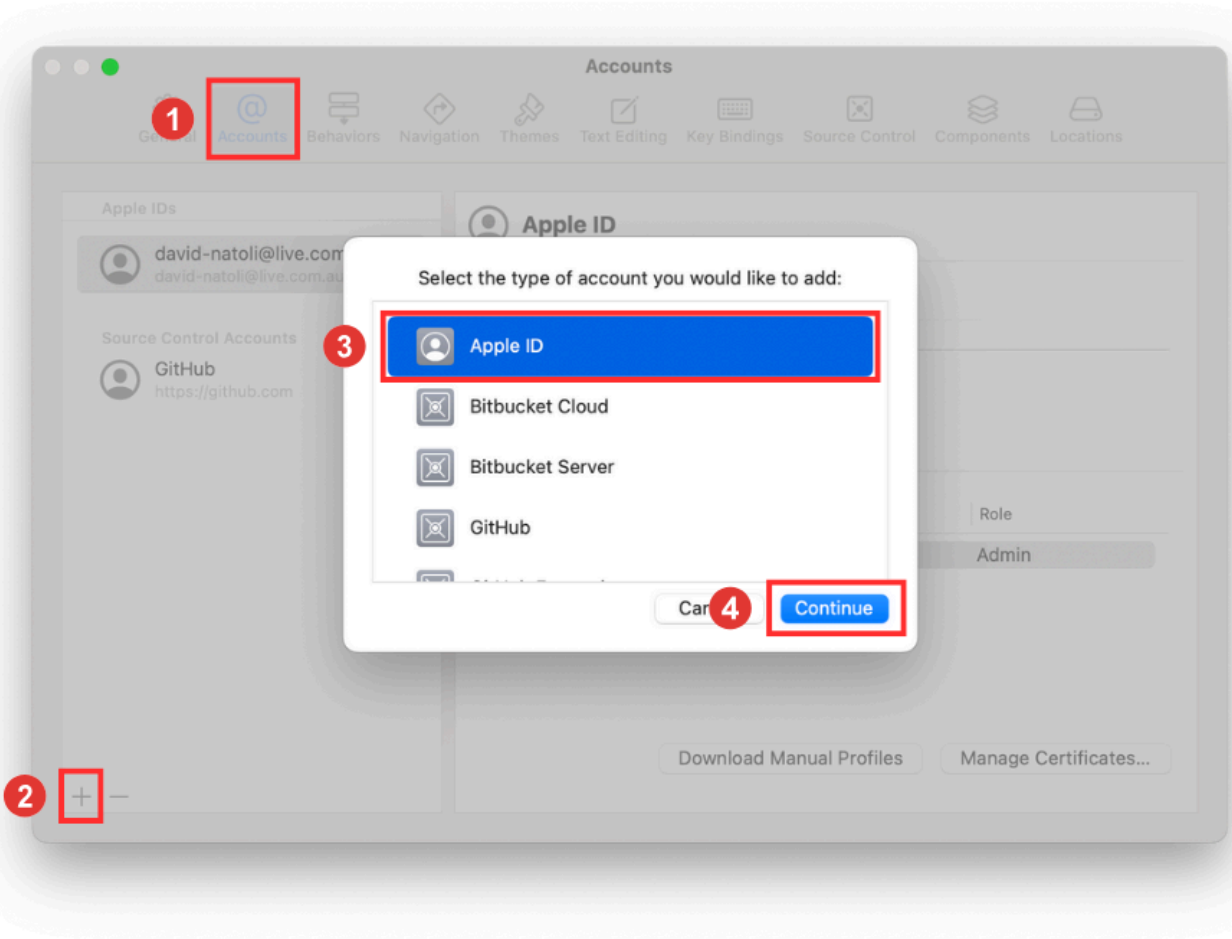
3. Next, select 'Settings'. This will open the Xcode 'Settings' window.
4. In the Xcode 'Settings' window, select 'Locations'
5. In the 'Locations' tab, you will observe a drop-down menu for 'Command Line Tools'. Ensure the Xcode version listed matches the version you just downloaded and installed.



Add Apple ID to Xcode

When you first use Xcode, you will need to add your Apple ID to the accounts within Xcode.

1. Open Xcode.
2. In the top left-hand corner of the screen, select 'Xcode'.
3. Next, select 'Settings'. This will open the Xcode 'Settings' window.
4. Next, ensure you are in the 'Accounts' tab.
5. Press the plus symbol in the bottom left-hand corner of the screen. Follow the prompts to add your Apple ID.


Accounts
1
Accounts
Behaviors
Navigation
Themes
Text Editing
Key Bindings
Source Control
Components
Locations
Apple IDs
david-natoli@live.com
Source Control Accounts
GitHub
https://github.com
Apple ID
Role
Admin
Select the type of account you would like to add:
3
Apple ID
Bitbucket Cloud
Bitbucket Server
GitHub
Can
4
Continue
Download Manual Profiles
Manage Certificates...
2

Build Trio with Xcode

Overview

Time Estimate

- 60-80 minutes for first-time builders.
- 10-15 minutes for repeat builders.

Summary

You will:

- Run the 'Build Select Script' to download Trio code.
- Follow the step-by-step instructions in the script.
- Press the Xcode 'Run' button to build Trio.
- Watch in awe as you build your very own app.

FAQs

- **Why does Xcode show a colorful spinning icon and not respond to me?** Unfortunately, sometimes Xcode gets confused, and you must force-quit the application.
- Many more FAQs for building Trio align with the steps that trigger the questions.

Build Trio with Xcode

Release 0.7.0 and newer

The main branch is the preferred build for everyone except expert builders.

If you are coming from an older main version 0.2.x, please read the Migration Guide before updating.

All Trio users are encouraged to follow along in Trio Discord.

Download Trio with 'Trio Build Select Script'

Trio uses the Trio Build Select Script to download Trio's source code, prepare your computer, and build Trio. Every attempt was made to put messages directly in the script for each step. The next few sections of this page walk you through what you will see when you run the script.

Open Terminal

Shortcut: Tap on the CMD and spacebar to open Spotlight search and type `Ter` and then return to open the Terminal

1. On your Mac computer, go to the 'Finder' app.
2. Next, select 'Applications' in the navigation pane on the left of the screen.
3. In the 'Applications' folder, scroll to locate and open the 'Utilities' folder.
4. Scroll to and open the 'Terminal' application.

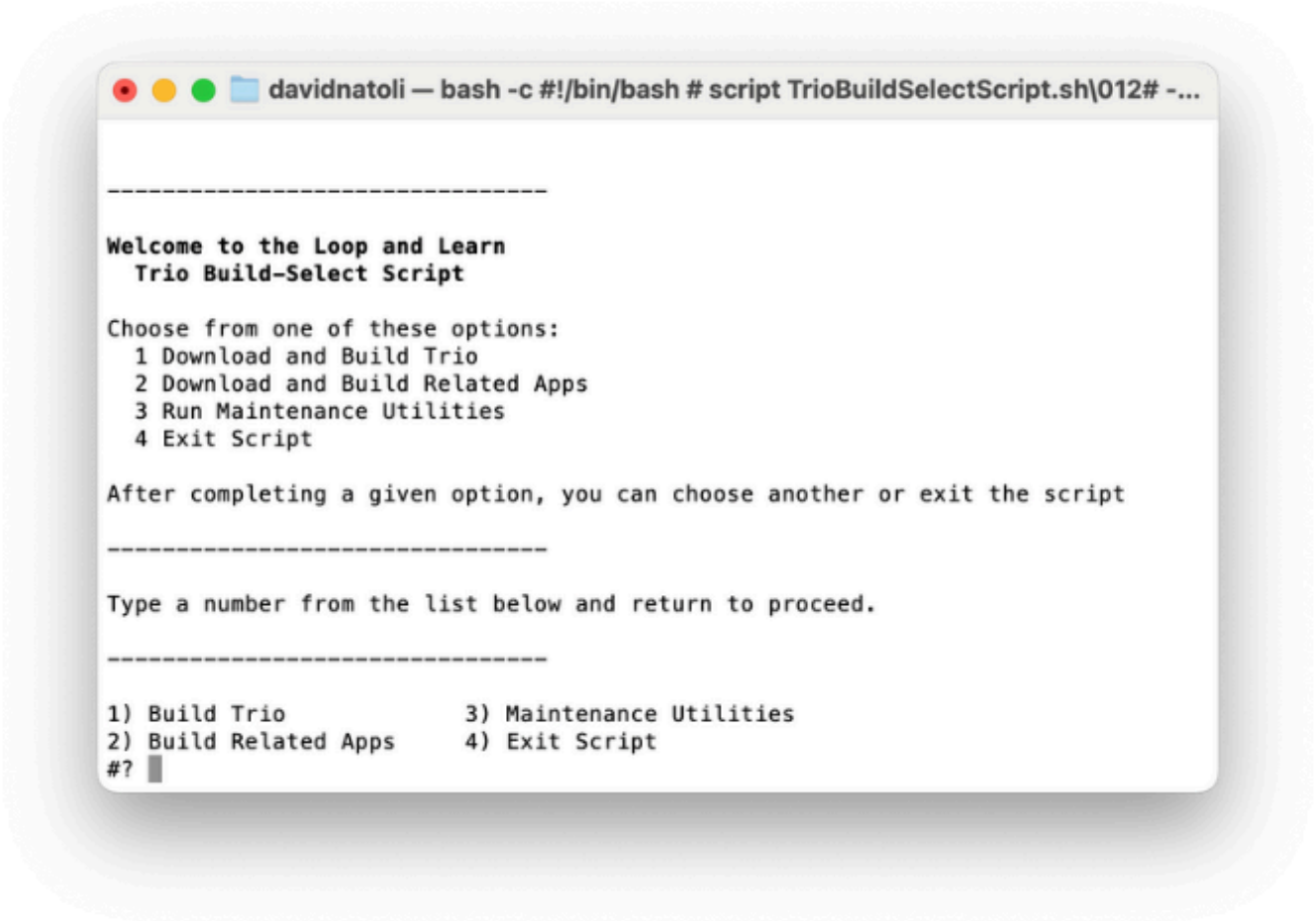
Run the 'Trio Build Select Script' in 'Terminal'

The Trio Build Select Script is designed to walk you through downloading Trio. Please take some time to read each step carefully.

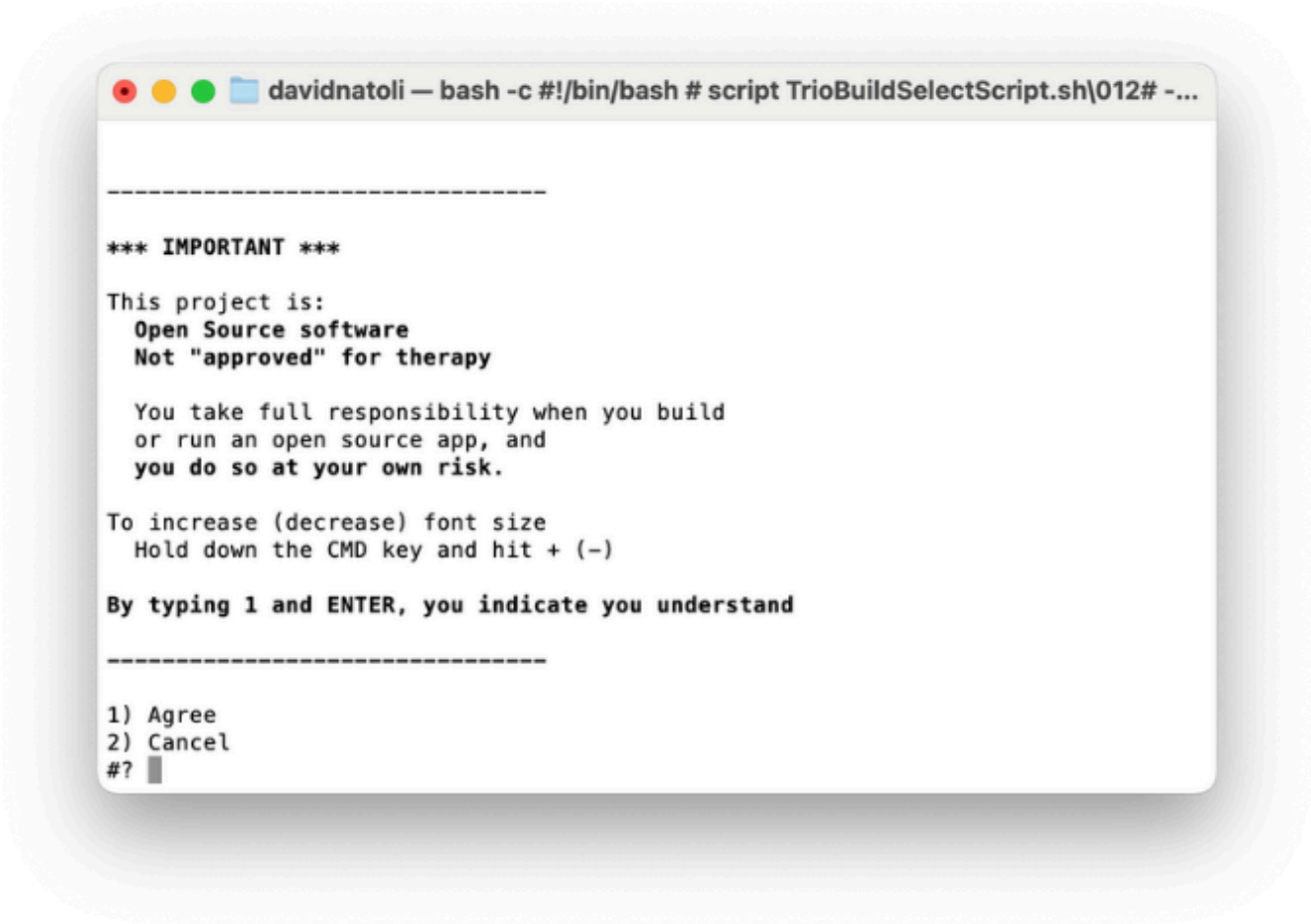
1. Copy the below script by hovering the mouse near the bottom right side of the text and clicking the copy icon.

```
  /bin/bash -c "$(curl -fsSL \
    https://raw.githubusercontent.com/loopandlearn/lnl-
scripts/main/TrioBuildSelectScript.sh)"
```

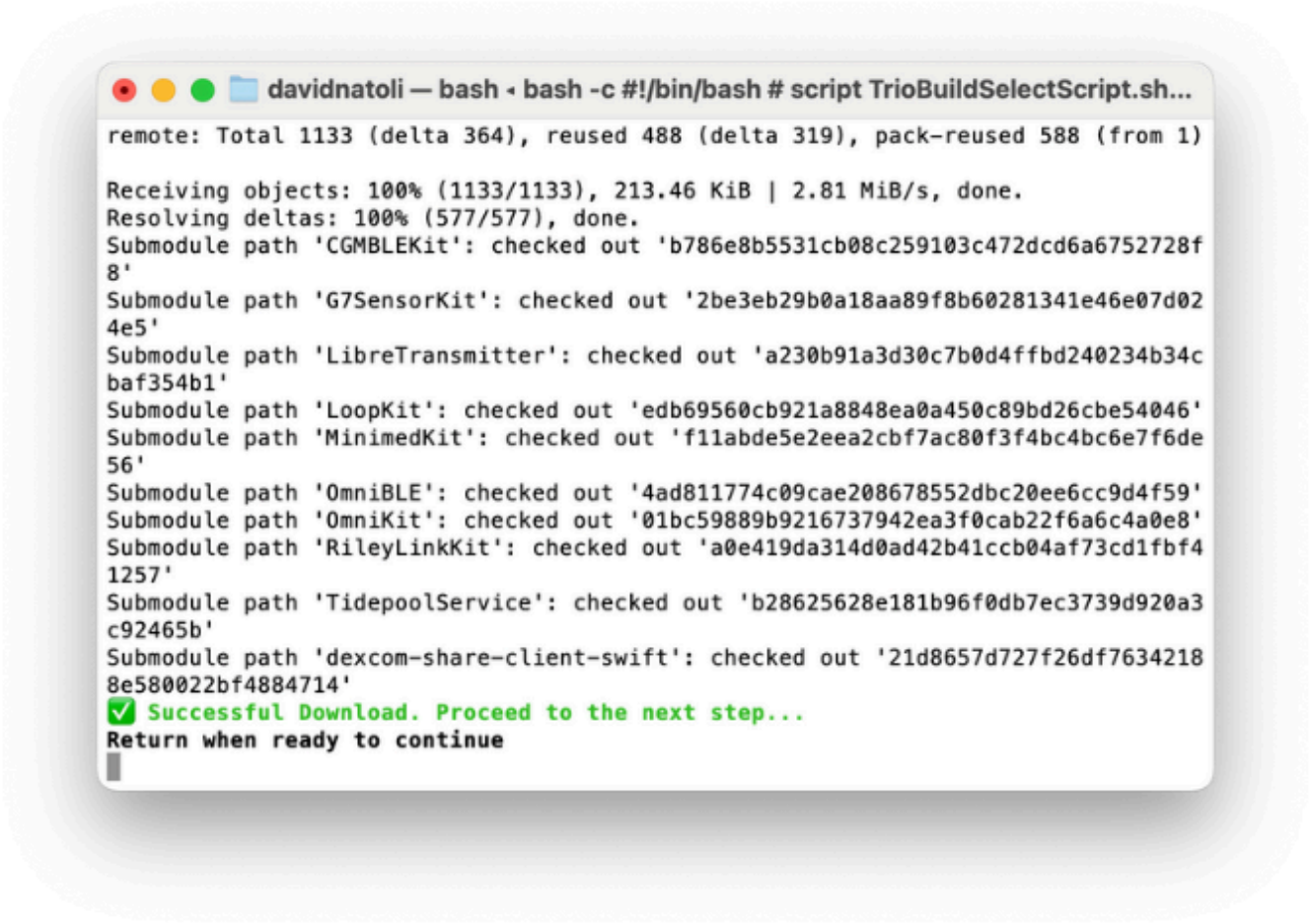
2. Paste the 'Build Select Script' into Terminal. Press return to run the script.
3. Running the script will display a series of menu options in the 'Terminal' window, as shown below. To build Trio, you will type 1, and then press return.



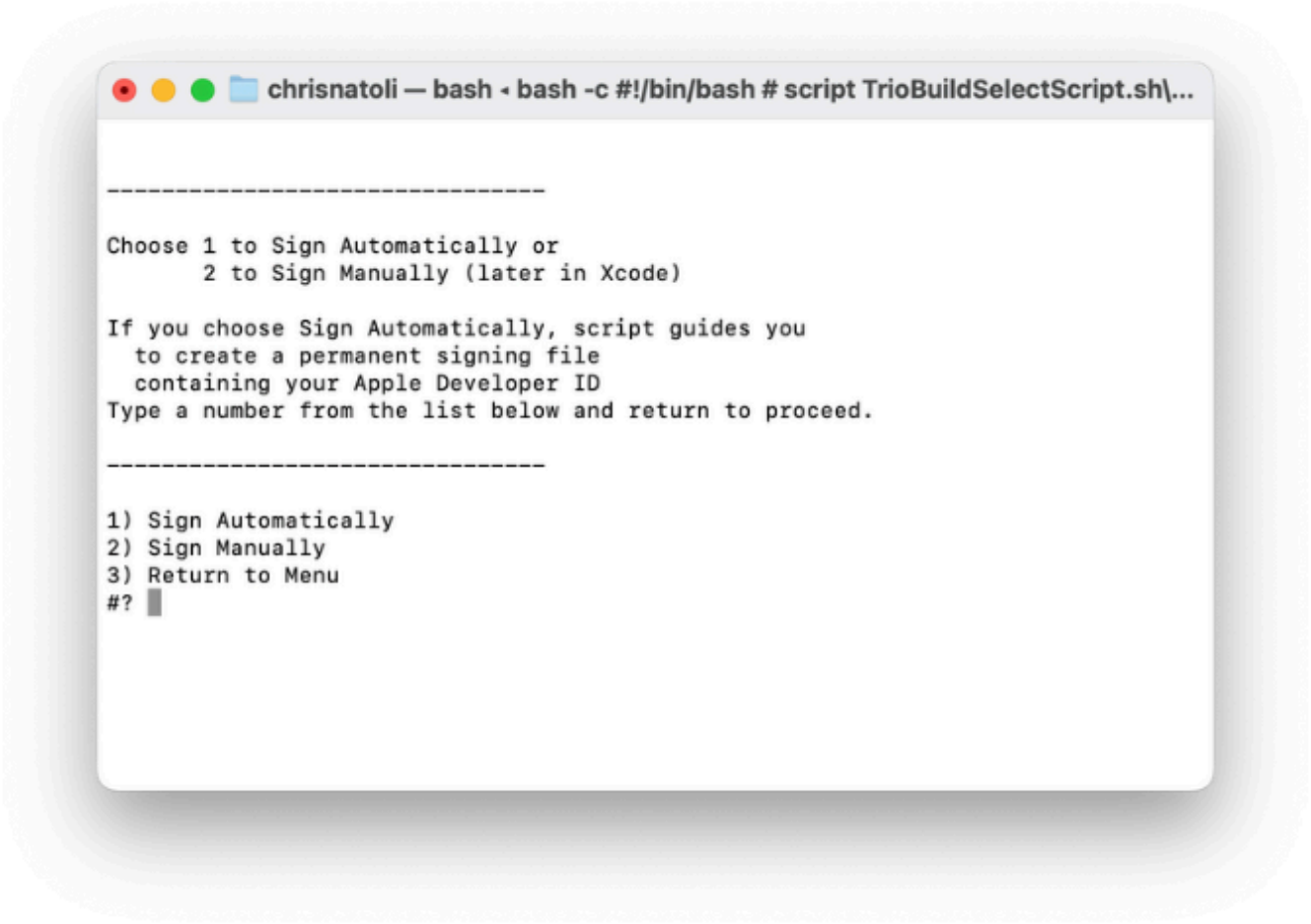
4. Next, the script will inform you that you are downloading open-source software. If you understand the warning, you will Type 1 and press return.



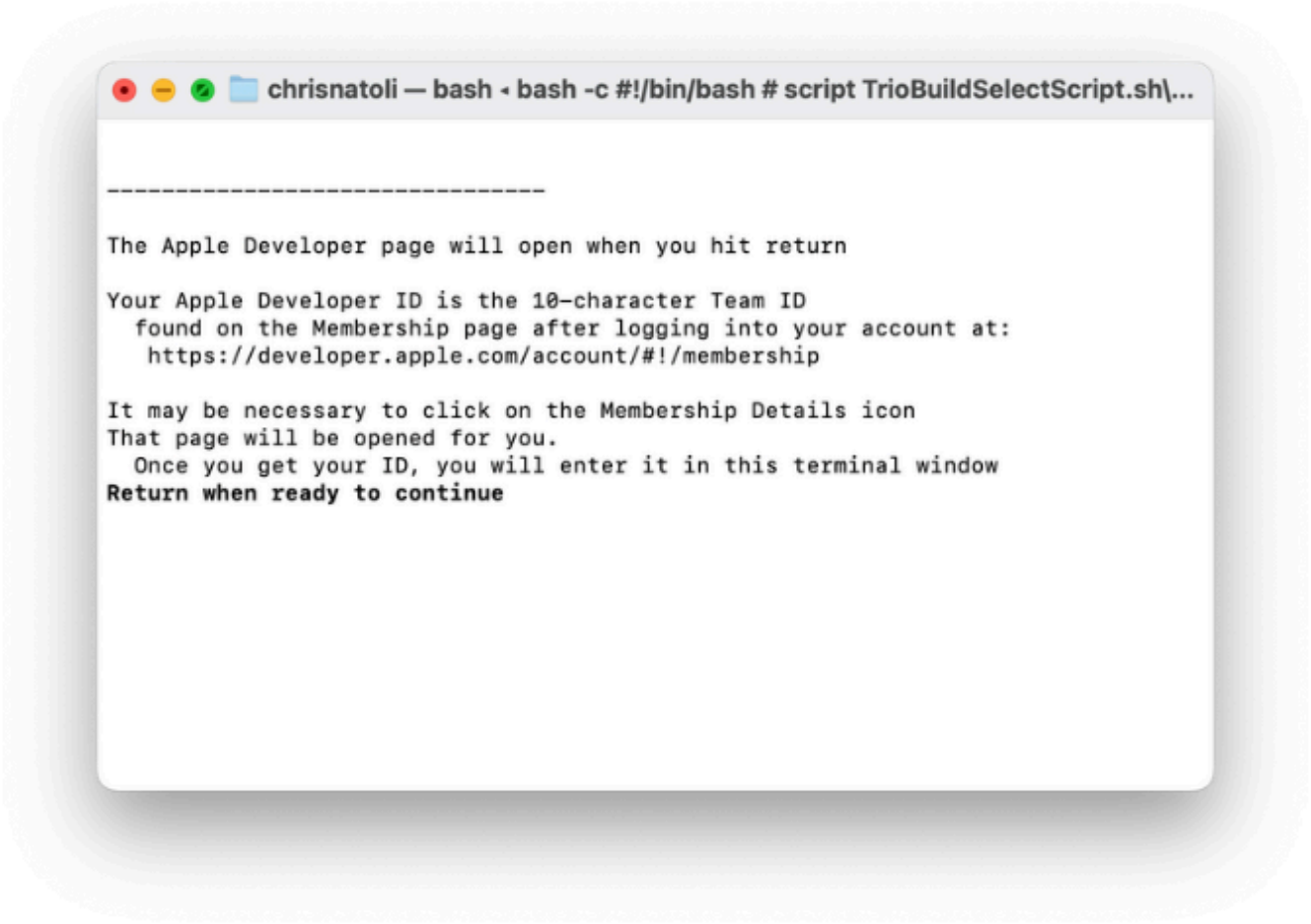
5. Next, the script will prompt you to select which Trio branch you want to download. It is recommended you Type 1 and press return to select the `main` branch.
6. Next, the script will begin downloading the Trio source code. Depending on your download speed, this can take 3 minutes to 30 minutes. While this happens, you may read words in the Terminal window that you do not understand. That is normal. If the download takes a while, you can leave the room and return later to check progress.
7. Once successfully downloaded, your terminal window will appear as below.



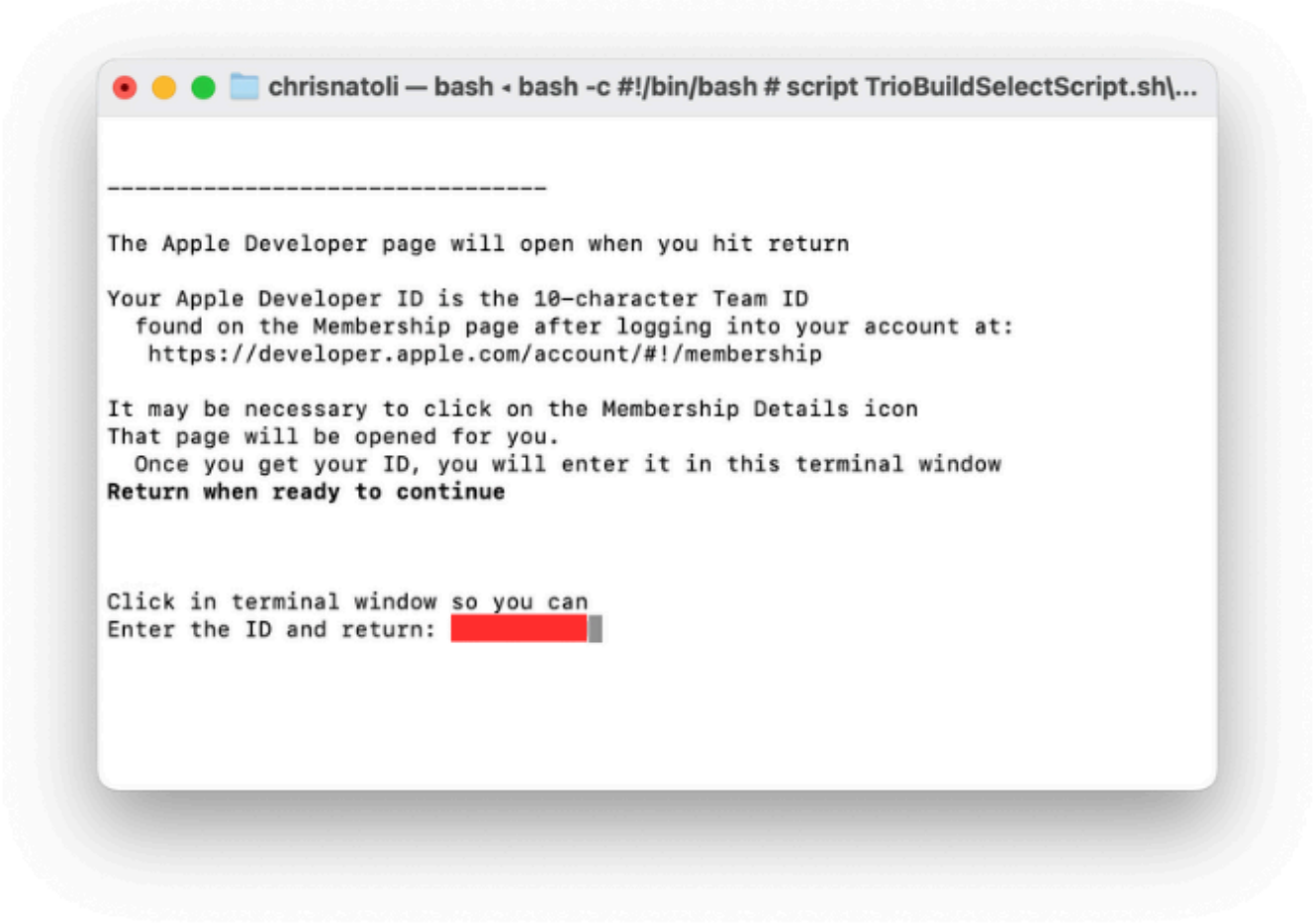
8. If you receive a failure message, scroll up through the window to find the error message(s). For assistance, visit Xcode Errors with Build Select.
9. You can hit return to continue if you did not receive errors.
10. Next, we will configure how we sign targets. If you have never built an Xcode app using your Apple Developer ID on this computer, you will be asked how you want to sign the targets the first time you use the script. If you have previously built with the build script, you can skip forward to step 14.
11. If you are building with a paid Apple Developer account, type 1 and press return. If you are building with a free Apple Developer Account, type 2 and press return. Free account users go to Free Apple Developer Account Build now.



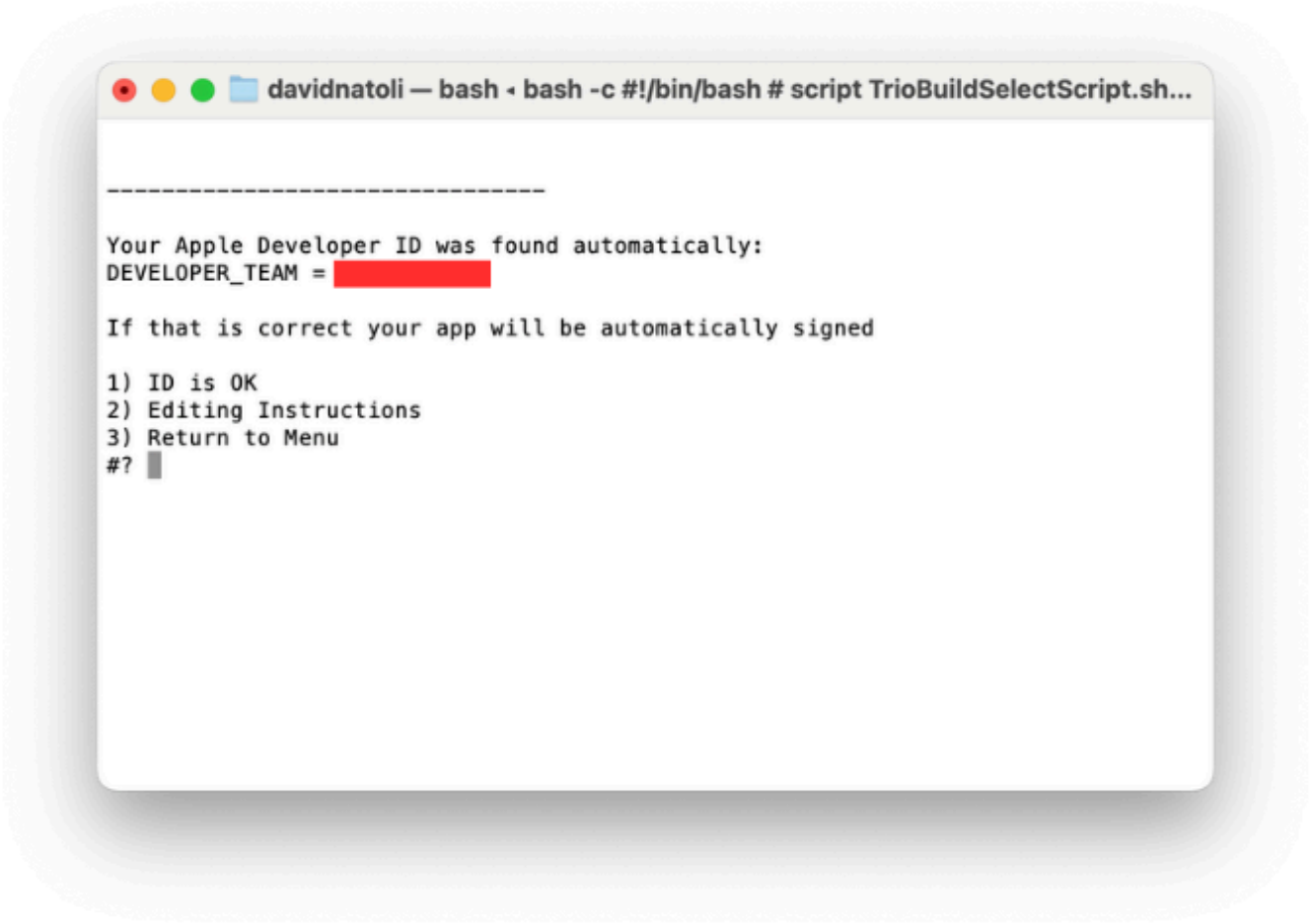
12. Once users select option 1, they will be prompted to gather their Apple Developer Team ID. To do so, follow the link provided in the Terminal window.



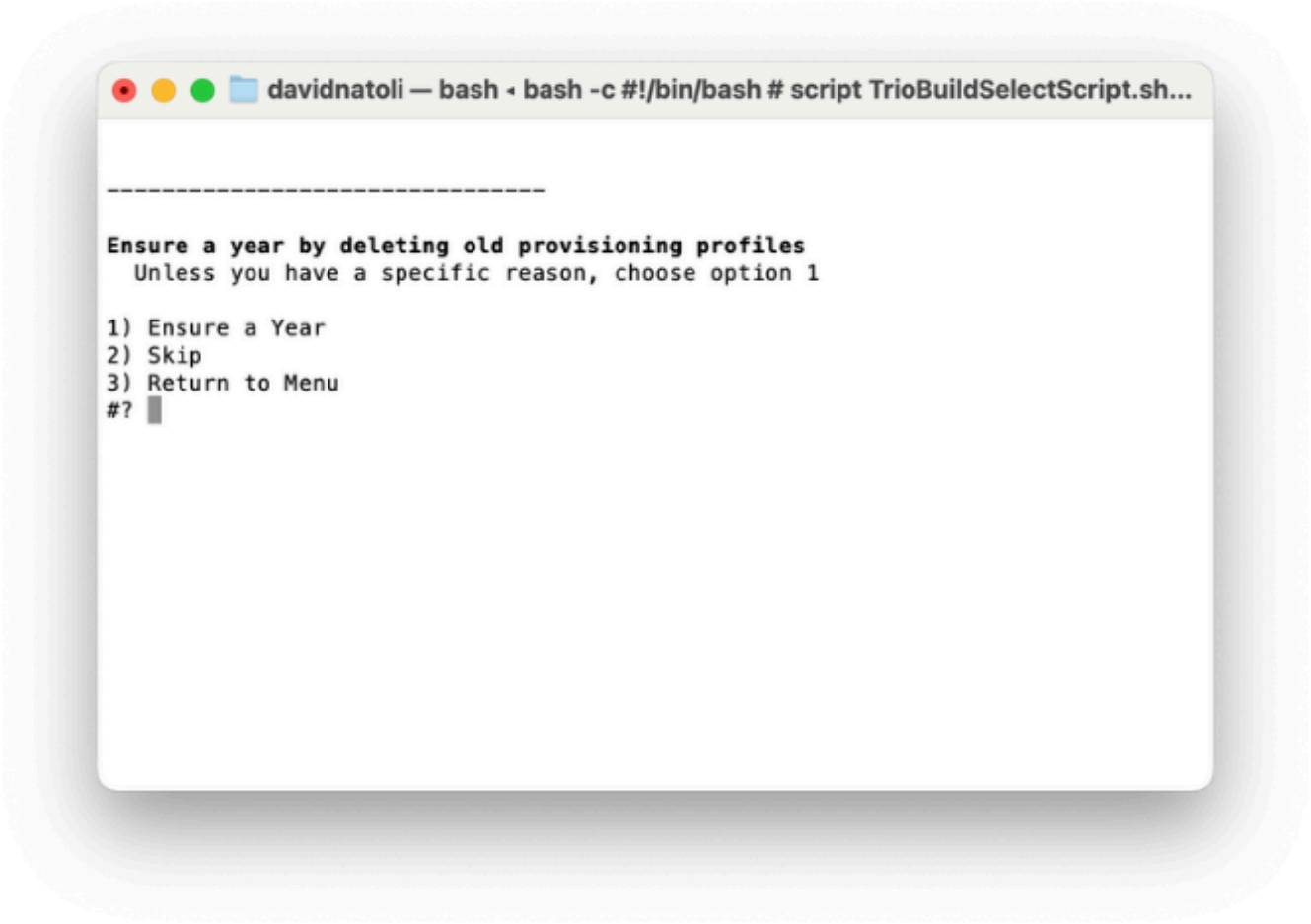
13. Once you have gathered your Team ID, you can return to Terminal and press return as prompted. The build script will prompt you to enter your Team ID into the Terminal window. Once you have entered the Team ID, you can press return.



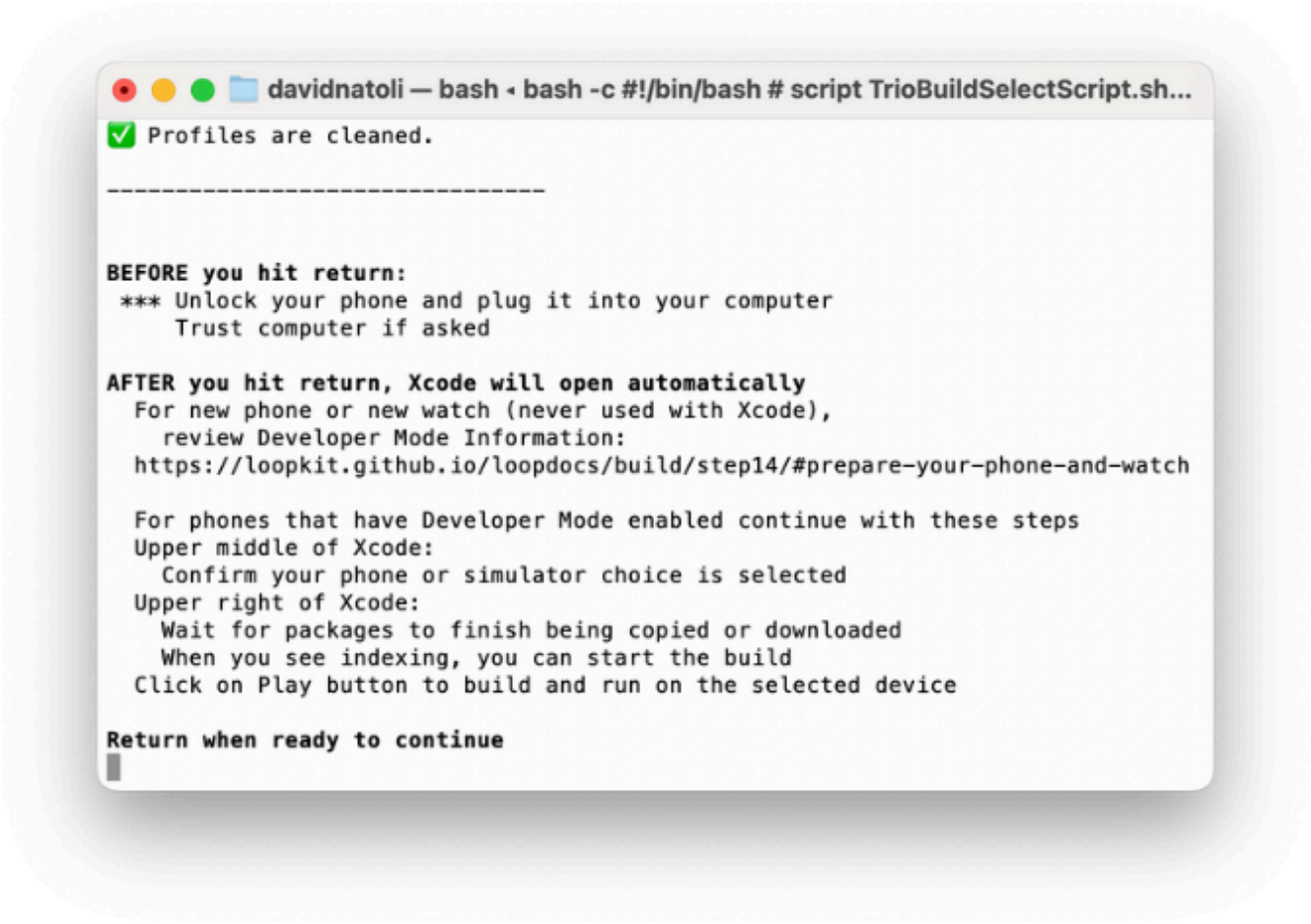
14. If you have previously built with the build script, you will be asked to confirm that your Team ID is correct. If it is, you can type 1 and press return.



15. The next question asks if you want to ensure a year with your new build. Unless you have a specific reason, type 1 and press return.

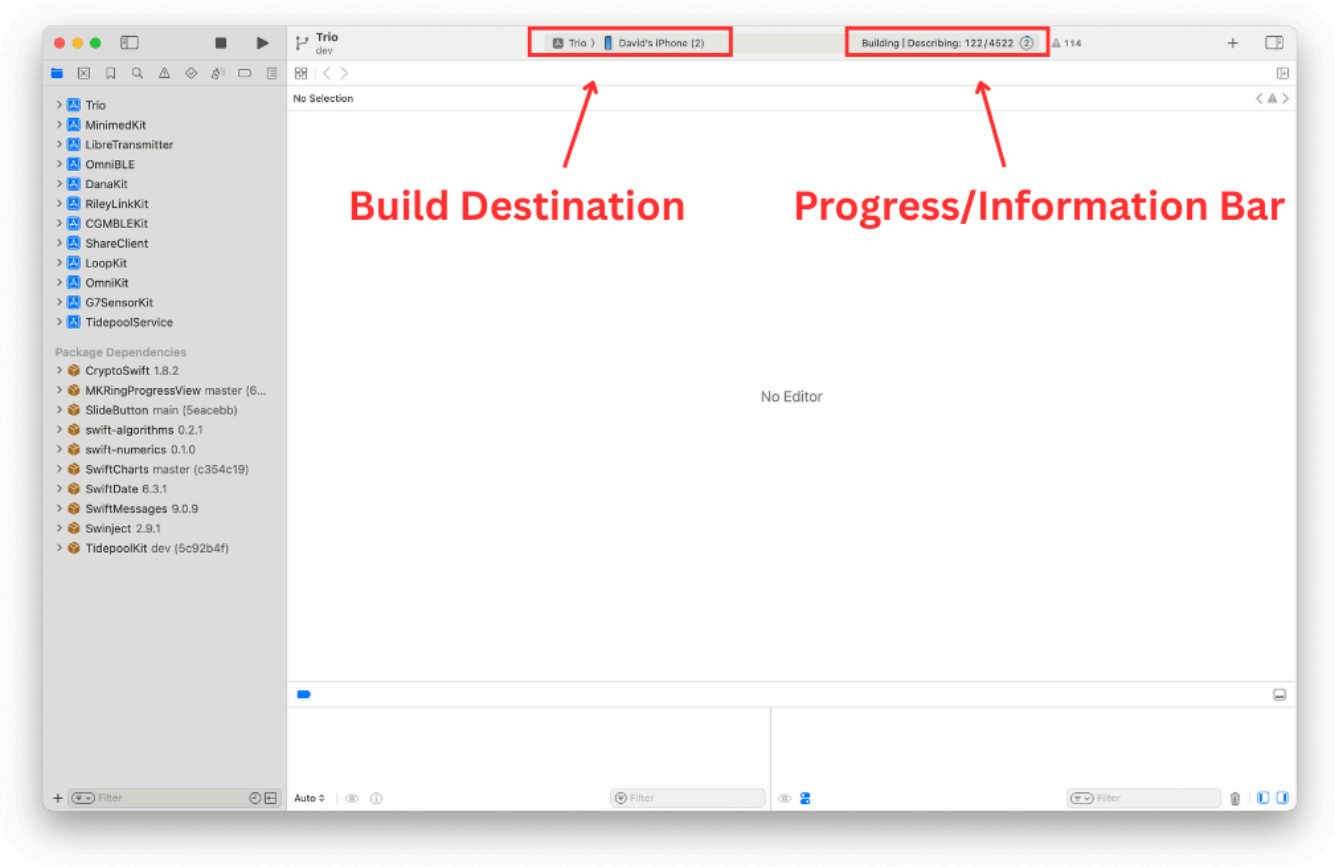


16. In the next Terminal window, you will be prompted to do the following:
    - Unlock your phone.
    - If you have an Apple Watch, ensure it is unlocked and on your wrist.
    - Plug your phone into the computer.
    - Press return to continue.

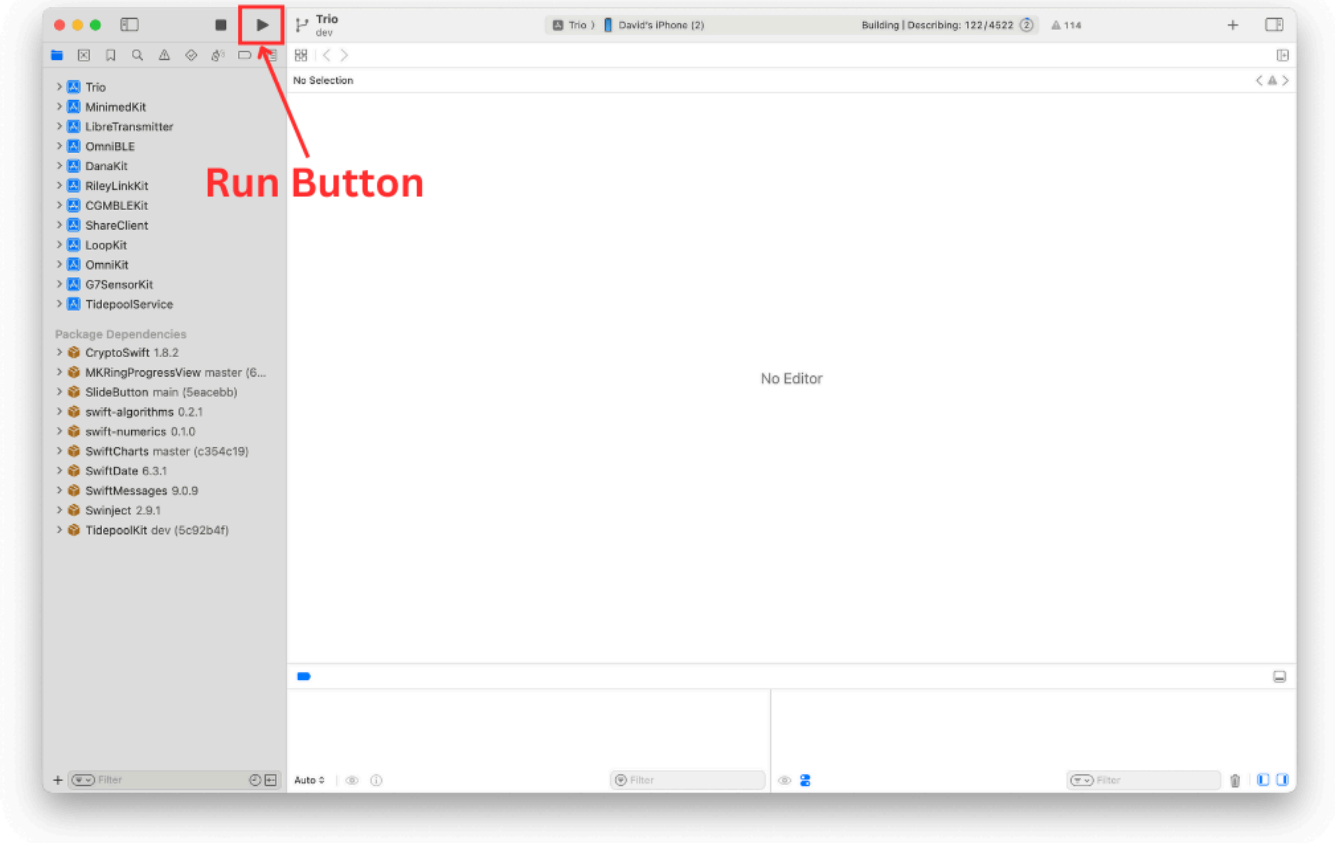


Build Trio

1. Next, the script will open the Xcode application. When Xcode first opens, it takes some time for the project and its associated packages to load. You might see a progress wheel with the words 'downloading', 'copying', or 'indexing'.
2. Next, we will set our build destination (a.k.a. your device). At the top of the screen, there is a progress/information bar. To the left of the bar, you will see 'Trio' and the logo. Next to this is your build location. Click this and select the device you would like to build to.



3. We will now build to the location we selected. You can now select the run button.



4. Once you press the run button, you can monitor progress in the information bar.
5. Once the build is complete, a transient message will pop up on the window saying "Build Succeeded." Trio will launch on your phone or build device. You can unplug the device from the computer. When you unplug the

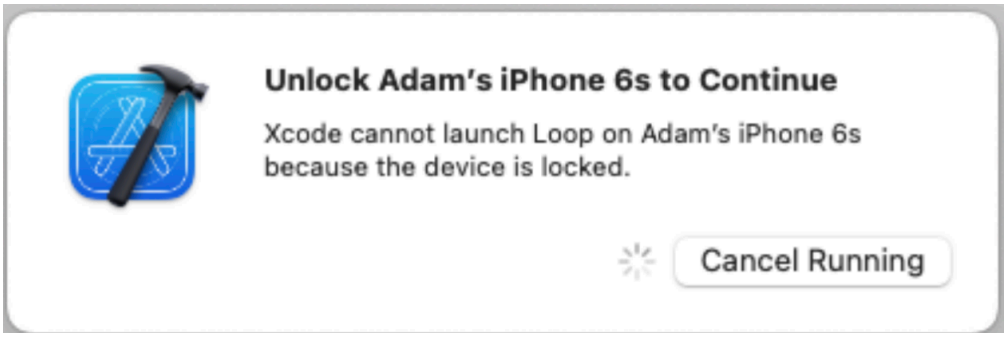
device, a message will pop up in Xcode noting the lost connection. Acknowledge this message. Once unplugged, the Trio app will close on the phone, and users must re-open the application.

First Build Messages

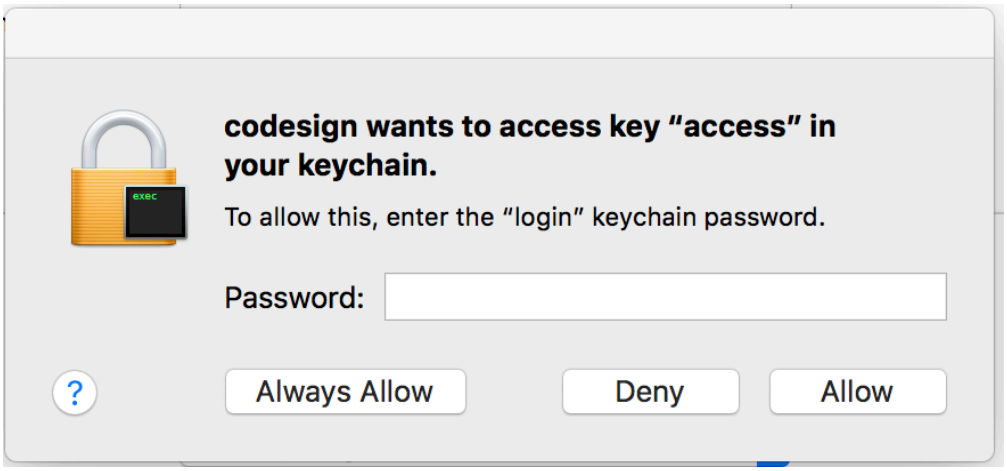
If this is your first time building an app with this device, you may get some of the following messages:

- "Device "YOUR DEVICE NAME" isn't registered in your developer account."
  - If this message occurs, select 'Register Device', and Trio will continue to build.

- "Unlock "YOUR DEVICE NAME" to Continue"
  - Unlock your device.



- "code sign wants to access key "access" in your keychain".
  - Enter your computer password and press 'Always Allow'.
  - Please note: This message may appear numerous times. **Do not get frustrated and press 'Deny'**. Just keep entering your computer password and press 'Always Allow'.



Build Errors

Yellow Errors

If you receive any yellow errors or warnings, don't freak out. This is normal, and Trio will function as normal. Please do not try to fix these errors, as you will likely do more harm than good.

Red Errors

If you receive any Red Errors, your application will fail to build. Please visit the Build Errors page to troubleshoot the errors. Once these errors have been addressed, you can return to the build documents and continue.

Clean Build Folder

Once you've resolved a build error and started the build process again, Xcode will continue to show a red indicator on the top line from the previous failure. If you don't like seeing that, clean the build folder to clear the error. Otherwise, Xcode will still work on the build if the steps show across the top line. When the build succeeds, the red circle will disappear.

Clean Build Folder

- In the Xcode menu, select Product, then Clean Build Folder
- Wait for cleaning to complete: you'll see a "Clean Finished" message

Successful Build

Congratulations on building Trio, and welcome to the community.
Please see the New User Guide to get started with Trio.

Build a Feature Branch

The 'Trio Build Select Script' only offers to build the `main` and `dev` branches, but you can use the 'Build Trio Script' to build a feature branch that was advertised in Discord but has not yet been merged into `dev`.

In the command (shown below), there is a space after the final quote, followed by a hyphen, another space and then the branch name. This example would build `other-branch`, if such a branch existed.

> This is just an example. You need to substitute the branch you desire for `other-branch`. The example below uses a continuation character to put the extra characters on a new line to make this easier to read.

Replace `other-branch` with the desired branch

```
/bin/bash -c "$(curl -fsSL \
  https://raw.githubusercontent.com/loopandlearn/lnl-scripts/main/BuildTrio.sh)" \
   - other-branch
```

Free Apple Developer Account Build

Prepare to Sign

You typically get here after choosing to Sign Manually after a successful download using the Build Select script. Normally this option is chosen by people building the app with a Free Apple Developer Account option or if you want to build to a simulator on your computer.

The instructions found on this page are for the first build. With the Free version, you need to build every week. You cannot build again on day five and hope for another seven days. Unfortunately, it doesn't work that way.

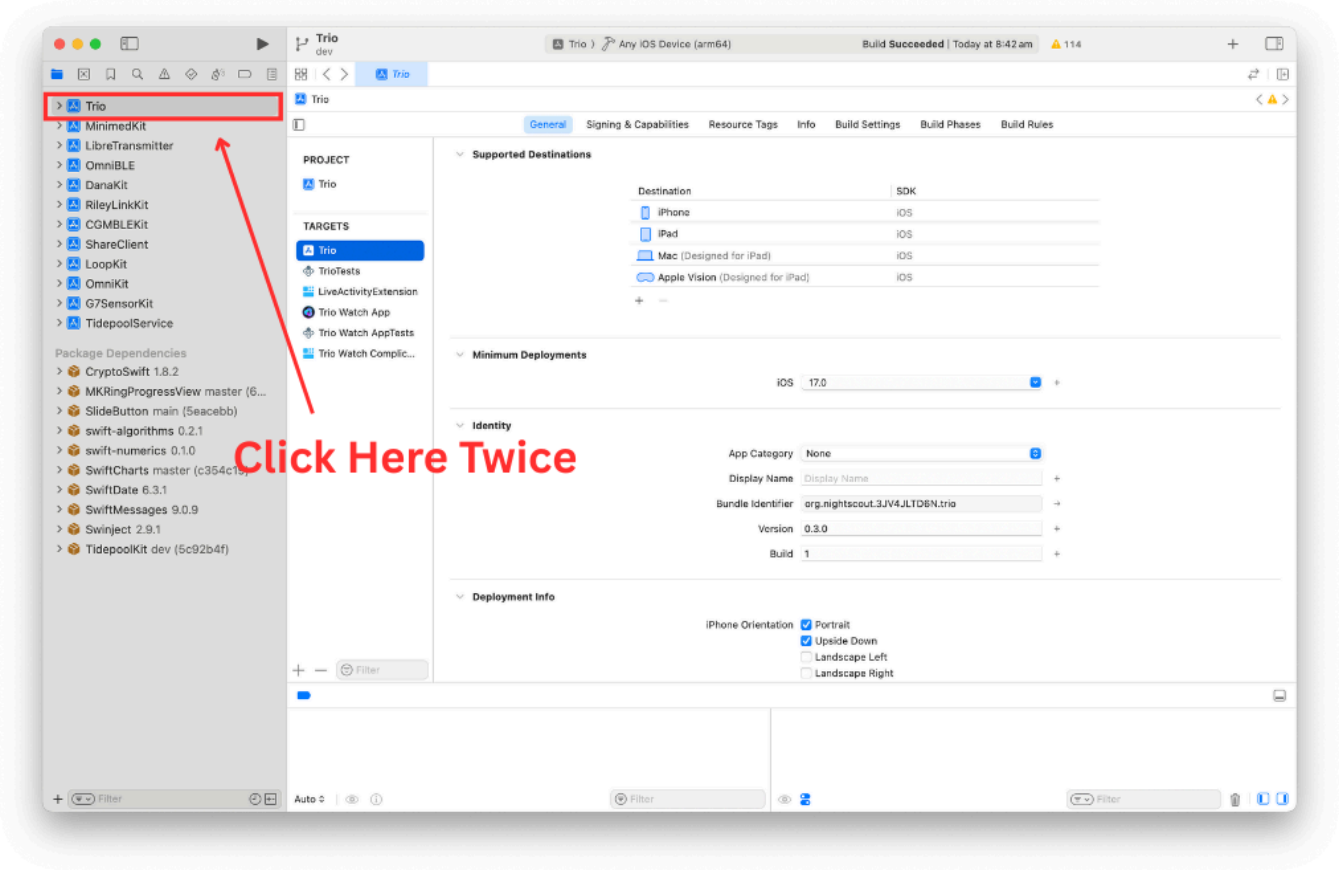
Select the Trio Folder

Don't touch that button!

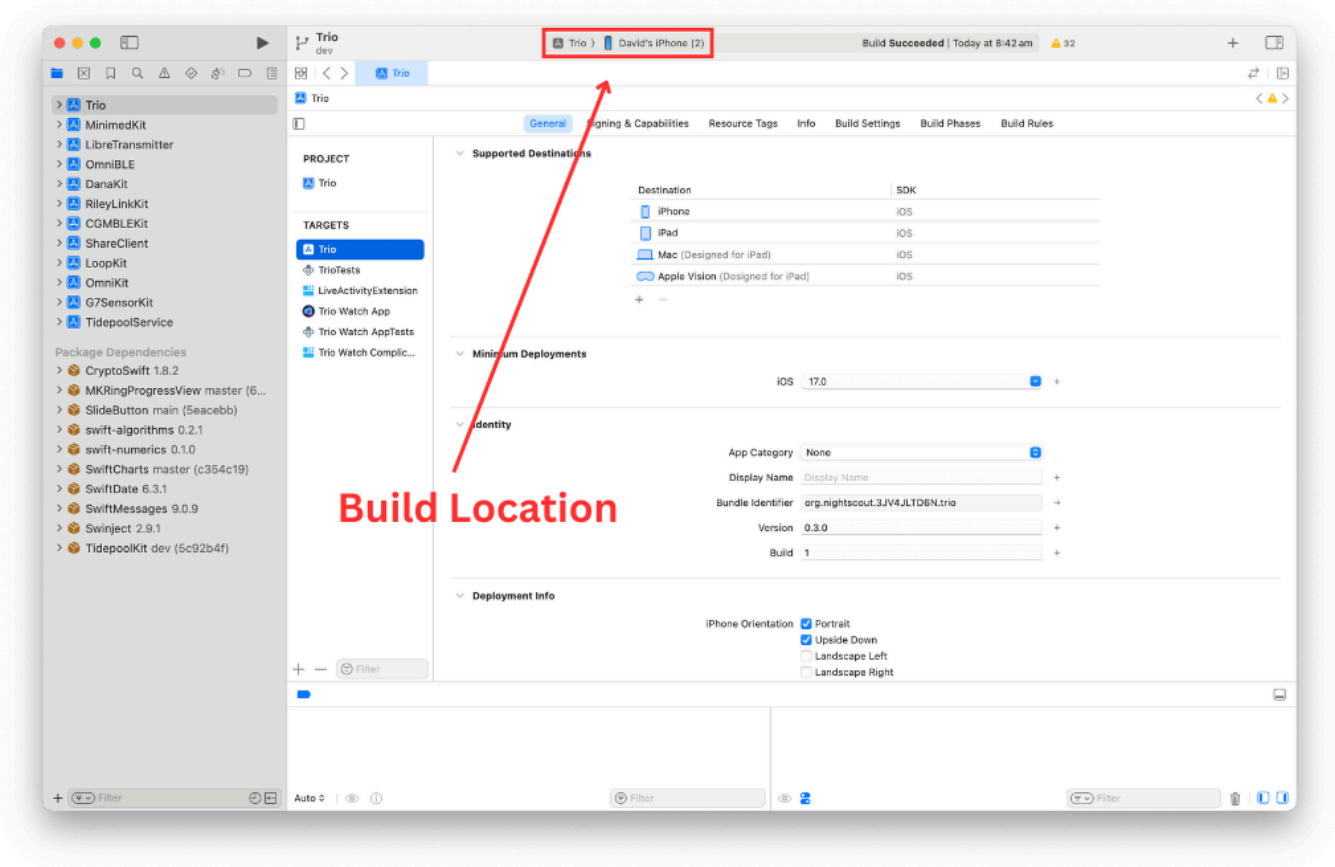
You will be told exactly where on each screen you should click. Please only click in the designated places.

Follow the directions and compare your Xcode screen to the graphics as you walk through the steps.

1. Follow the 'Build Select Script' prompts to open Xcode. Ensure you follow the instructions on each screen.
2. Once Xcode is open, in the navigation pane on the left side of the screen, double click on 'Trio'



3. Next, ensure your phone is set as the build location. If not, click on the build location and select your device from the drop down list.
    - If this is the first time your phone or watch has been connected to Xcode, you will need to tell the phone and watch to "Trust this Computer".



I don't see my phone

- If you don't see your phone in the Devices section, unplug and plug in again.
- Still don't see your phone - reboot the phone - and if that doesn't work - reboot the computer.
- Still don't see your phone - try a different cable or USB slot.

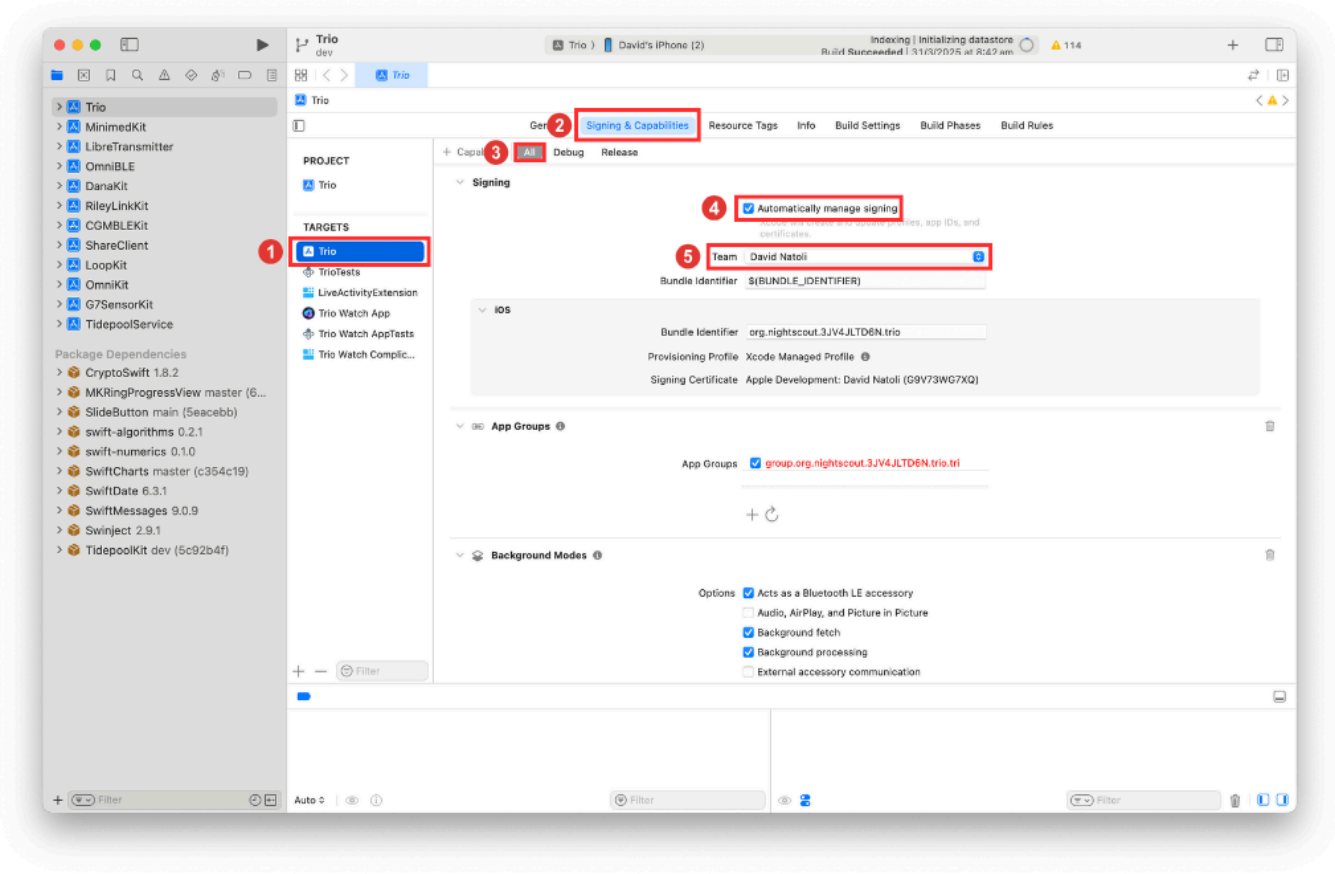
Sign Targets

What does Signing Targets Mean?

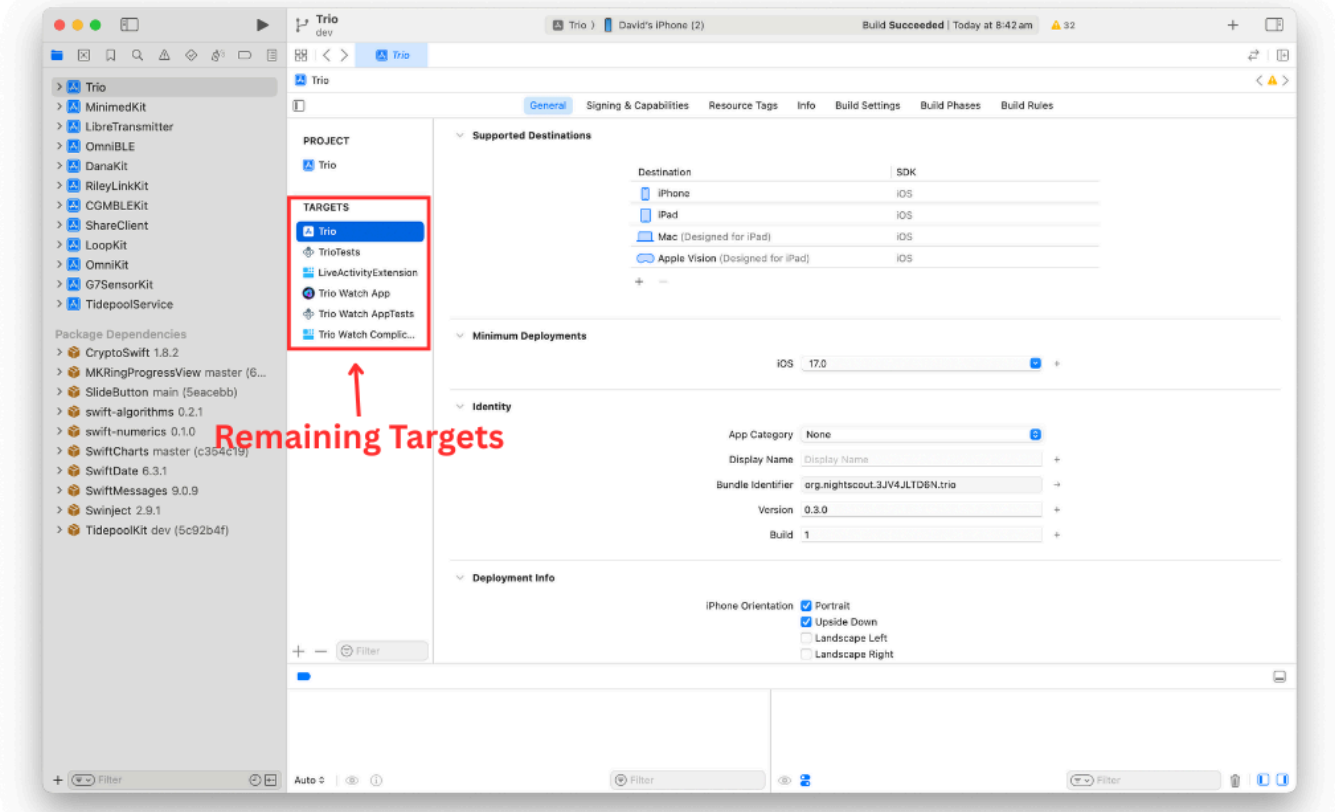
"Signing Targets" in Xcode identifies who built the app. You cannot deploy an app to a phone without signing each target associated with that app.

The graphic below indicates in red the places you need to click in order to begin signing targets.

1. First, ensure the target 'Trio' is selected.
2. Next, we are going to select the 'Signing & Capabilities' tab.
3. We will then filter the capabilities to 'All'
4. Next, please check 'Automatically manage signing' option.
5. You can now add your 'Team'. Click the drop down box and ensure you select your free team.



6. Repeat the same steps for the remaining targets.



Disable Features

This section is required if you are using the free account. Some features of Trio are not available with a free Apple Developer Account. You will need to remove features that are not supported.

1. You must remove unsupported capabilities from the **Trio Target**. This is best done when you sign the target:
    - Delete **Push Notifications**
    - Delete **NFC Tag Reading** (NFC Scan can stay)
    - For **Healthkit**: Unselect Healthkit background delivery and then reselect it.
        - The free build doesn't allow HealthKit Access (Verifiable Health Records) capabilities, which is unchecked by default, so shouldn't be an issue, but the only way that I've found to clear the Healthkit Access error is to unselect/reselect background delivery.

Details about removing unsupported capabilities:

- You must disable Push Notification, Near Field Communication Tag Reading, and unselect/reselect HealthKit
    - You will see an error message when you select (personal team) for the **Trio Target**
- The Xcode error message starts with "Cannot create . . ." with a list of all the attributes not supported.
    - Scroll down and click on the little trash can icon next to each unsupported attribute
- Scroll up and both the "Cannot create . . ." and "No profiles for . . ." error messages are gone
- You are done with this target
- Proceed to the next target

> **Direct Libre pairing not supported with a FREE Apple Developer Account**
>
> Pairing a sensor directly to Trio — Libre 2 / 2 Plus using Libre 2 Direct, or Libre 3 / 3 Plus — requires *Near Field Communication (NFC)*, because your phone must scan the sensor itself to start it. Each new sensor needs its own scan.
> ❌ This capability is not available when building Trio with a **free** Apple Developer Account.
> ✅ Libre 1 and Libre 2 / 2 Plus sensors read through a third-party transmitter (Miao Miao, Bubble, etc.) do not use your phone's NFC, so those work with a free account.
> You need a **paid** Apple Developer Account for direct pairing.

> **Live Activity / Dynamic Island not supported with a FREE Apple Developer Account**

End of Free Account Steps

You have now completed the additional steps required for building with a free Apple developer account. You can return to Build Trio and complete the remaining steps.

Build Errors

Build Errors

> **Important**
>
> - **This page is only relevant to building with Mac and Xcode.**

There are two types of build indications that may be seen: they are warnings (yellow or purple icons) and red errors. You'll see the warnings and errors in the left-hand column of the Xcode window.

⚠️ **Yellow and** ⚠️ **Purple warnings** do not cause the build to fail, those are just warnings. You will often see yellow and purple icons. **Ignore those**. Do not try to do anything to fix those.

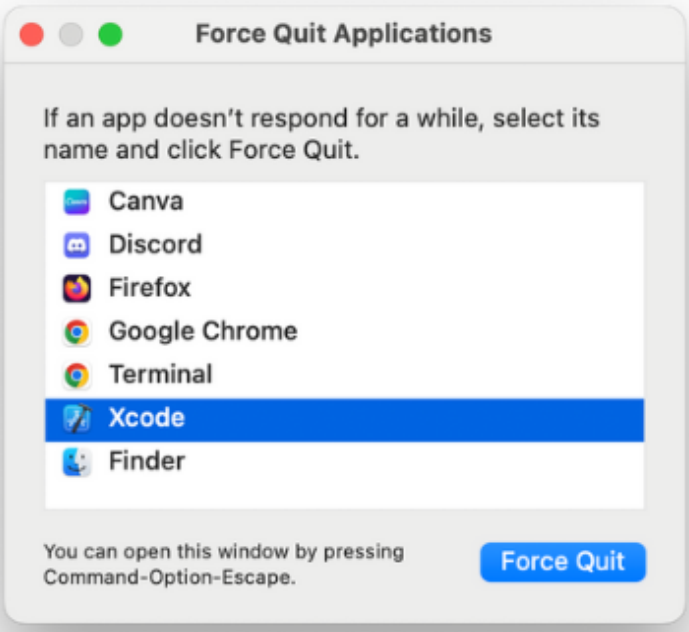
❌ **Red errors** will **have to be resolved** before you can successfully build the app. The steps below explain how to resolve them based on the messages you are seeing.

Xcode Not Responding

Sometimes, Xcode stops responding. You have to fix this before any of the other steps on this page will help. The signature is that Xcode shows a colorful spinning icon and does not respond to anything you do.

This can happen sometimes. You just need to force quit Xcode. Sometimes rebooting the Mac may be required, but start with force quit. Then just open up Xcode again and keep going.

1. Hold down these 3 keys `⌥ Option` + `⌘ Cmd` + `⎋ Esc` (or `⎇ Alt` + `⌘ Cmd` + `⎋ Esc`), until the `Force Quit` menu appears (should be fast)
2. Select `Xcode` and tap on the `Force Quit` button



Start with The Obvious Error Causes

Before you start trying to resolve your red errors, start with the most obvious things that can cause a red error message:

1. **Did you check that you have the minimum Xcode version for your iOS?** This is critical. Please review the iOS-driven requirements for the minimum version of macOS and Xcode. Please see Preparing your Computer.
2. **Did you check your Apple developer account for a new license agreement?** Periodically, Apple will release a new developer license agreement that you need to sign before you can build new apps. You will get a build failure if there is a pending license agreement to sign. Login to your Apple developer account to check if there is a new license agreement.
3. **Do you have a new computer, never used to build Trio?** Did you add your Apple ID to Xcode?
4. **Did you reboot, i.e., restart, your computer after updating Xcode?** You should reboot following Xcode installation or update and you must make sure your command line tools match the version of Xcode you just installed. Xcode Command Line Tools
5. **Did you get a fresh download of Trio code?** If you tried to build with an old download that you used a long time ago, that old version may not be compatible with the new iOS and Xcode versions. Check also, that you are actually using the new download in Xcode. When you use the Build Select Script, it automatically opens Xcode using the new download.
6. **Are you using a free developer account?** Make sure you finished the removal of Siri and Push Notification capabilities described in Free Developer Account Build.
7. **DO NOT USE BETA VERSIONS** If you are using an iOS beta version or an Xcode beta version, Trio might not build. Deleting iOS beta from a phone is a pain...so don't install it unless you know what you are doing.

Fix 95% of errors

If you have checked all those steps above and think you have a true build error, here's a tip that resolves 95% of all build errors when updating Trio code.

1. Open your project in Xcode as normal. Then go to the menu bar at the top of the screen and find the `Product` menu item. Use the drop down selection for `Clean Build Folder` or press `⇧ Shift` + `⌘ Cmd` + `K`. Either will work the same. Wait for the clean to finish before trying to build again.
2. On the far right, next to the name Full Path is the folder name that Xcode will be using to build. Make sure it is the new code you just downloaded and not an older folder.
3. If you are updating Trio and did not Delete Old Provisioning Profiles, do it now
4. Return to Xcode and try building your app again.
5. Still failing for phone or watch or both? Try the Unpair and Reboot procedure.

Unpair and Reboot

This is reported to fix a variety of watch building errors and `cannot prepare phone for development` errors:

1. Open Xcode.
2. Plug phone into computer and make sure it is unlocked.
3. Using the Xcode menu, select.
    - Windows.
    - Devices and Simulators.
    - On left side, Right-Click (or Control-Click) on your phone.
    - Choose Unpair Device.

It may not be necessary, but the suggestion is to reboot phone, watch and Mac - in other words, you can try to build without rebooting, but if that fails, repeat the steps and reboot before trying again.

The next time you plug this phone into your computer, you will be asked to trust the computer on the phone (and watch). Note, this is unpairing the device from Xcode and your computer, not the same as, and much faster than, unpairing your watch from your phone.

If the build fails again, look through the list below and see if you can match your error message with one of the error messages listed later in this page. If you really can't find your solution, then post for help. But help us help you.

- Ignore yellow and purple warning messages - those are not errors - do **not** try to fix them.
- Confirm it really is an error not already on this page; read this page carefully, including all the circled bits in the images in the Specific Error Messages section.
- Follow the steps in the Posting for Help section.
- WE CANNOT HELP without version numbers and screenshots.
- Do not take pictures of your computer screen with your phone, use screenshots.

Posting for help

STOP!! Read this section! Important!

The build errors listed below (and the checks listed above) will fix most of the problems you may encounter.

PLEASE READ THIS PAGE. The volunteers answering questions online would love to spend more time helping people use Trio and less time answering questions that can be addressed by using this page.

Therefore, try to resolve your build error yourself.
Then, if you need to post for help, please include enough information with the post so the volunteers know where you are in your troubleshooting attempts.

Your Post Must Include:

- The version of Xcode you are using.
- The version of Trio you are using.
- The version of iOS on your Trio device.
- Free or paid account, and if free, confirmation that you deleted Siri and Push Notification capabilities.
- Confirmation that you are not using an Xcode beta or iOS beta version (so we don't have to ask, please actually type "I am not using beta versions").
- Screenshots of your WHOLE Xcode window and/or Terminal window showing your error and any messages you've seen while working through the build errors/solutions. Do NOT use phone photographs of your computer screen. See below for instructions on how to capture a screenshot.
- State which fixes from the list below you have already tried AND post the screenshots of the results of those fix attempts.

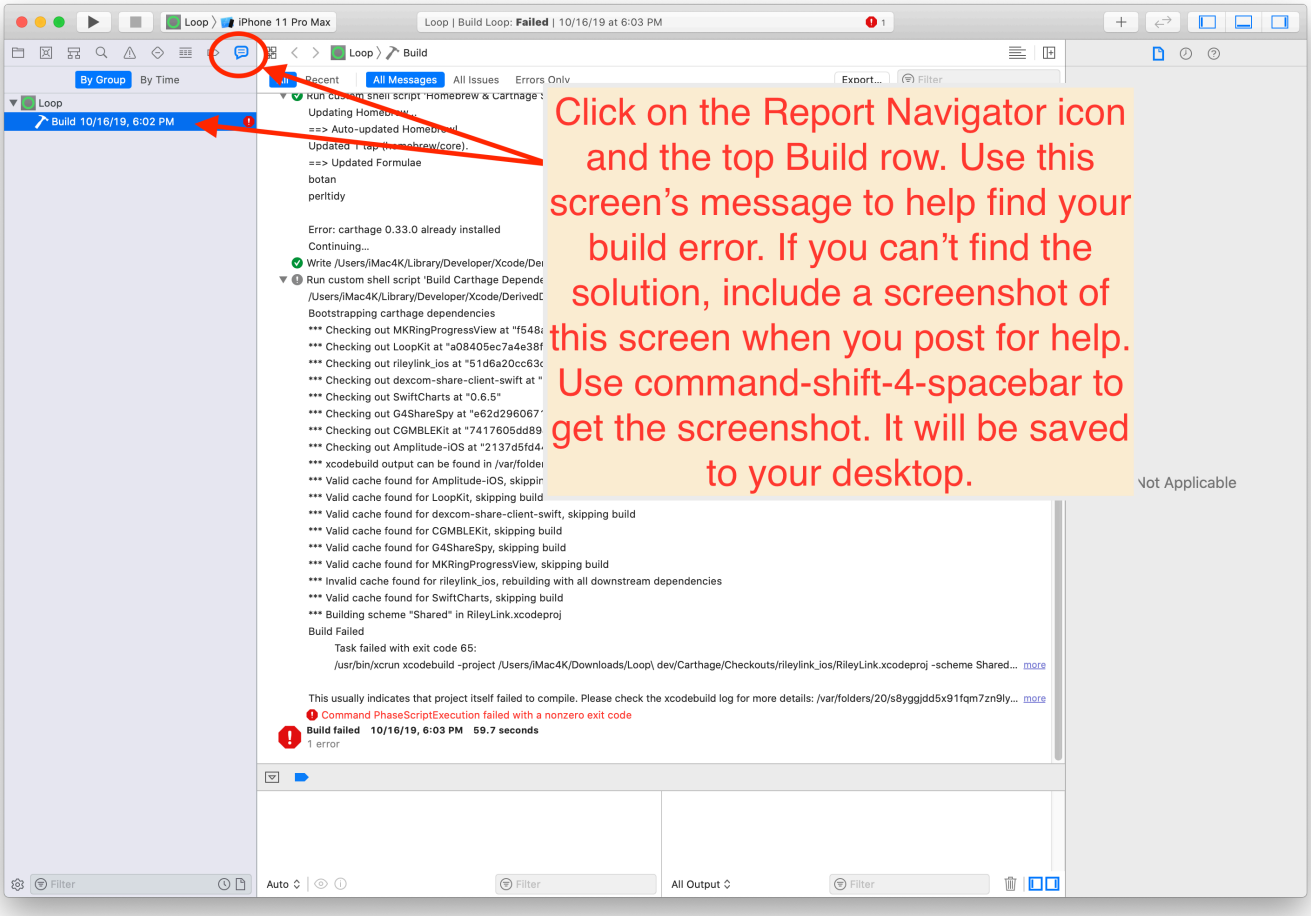
Screenshots

Please take screenshots of your issue and use them in your posts. On an Apple computer, press `⇧ Shift` + `⌘ Cmd` + `4` keys at the same time followed by pressing the space bar `Space` and then click on the window of interest. The screenshot will be saved to your desktop with a file name starting with the name "Screen Shot". Use screenshots instead of cell phone images or words whenever possible. Screenshots are higher resolution and easier to read.

Use the whole Xcode window screenshot when posting for build help.

Find Your Error Message(s)

To begin fixing the error, use the Report Navigator view to find your error message.



The key is to (1) **READ THE ERROR MESSAGE** and then (2) **FIND YOUR MESSAGE IN ONE OF THE TOPICS BELOW**.

Here's a super tip: Merely seeing the "exit code" in Xcode is not enough information to discern what error is causing your build to fail; some exit codes have multiple causes. Look at the detailed message to guide your search for the matching solution.

Notice the screenshots below have red circles highlighting certain error messages. Read your error message(s) from your screen, being guided by the red circles in the screenshots. Once you find your error message (hint: not "exit code"), you can either:

- Take the error message from your Xcode screen and use Trio's documentation search function to enter in some of that phrase to bring up the appropriate solution topic, or

- Take the error message from your Xcode screen and read through EACH OF THE TOPICS BELOW. Check each of the red circles to see if you have a match. Kind of like a matching puzzle.

For example, if you see "Invalid active developer path (/Library/Developer/CommandLineTools)" in your error message, use the search tool in Trio's documentation with "invalid active". You will get a couple of links and one is the Command Line Tools fix for that error message. Click on the link and you'll find the solution.

Specific Error Messages

Xcode Errors with Build-Select

The errors shown below should be prevented - the script will attempt to correct them automatically - follow the directions in the script.

If this is not successful, the script told you the download failed and exited. Scroll up in the terminal to find the error message(s):

- Read the error message
- Try to figure out the problem
- If you need help, reach out via Facebook or Discord

WARNINGS

If you see errors like these . . .

- `xcrun: error: invalid active developer path (/Library/Developer/CommandLineTools), missing xcrun at: /Library/Developer/CommandLineTools/usr/bin/xcrun`
- `xcode-select: Failed to locate 'git', requesting installation of command line developer tools`
- `xcode-select: error: tool 'xed' requires Xcode`

You missed one of these steps:

- Install Xcode
- Xcode command line tools

Update Trio with Xcode

Overview

Time Estimate

- 25 minutes if Xcode and macOS are already updated to support the current or desired iOS.
- Up to 2 days, if needed, to install macOS and Xcode update(s).

Summary

Summary of tasks to prepare for and update your app:

- Determine the required macOS and Xcode versions based on your phone's iOS.
  - If necessary, update first macOS and then Xcode.
- Check your Developer Account.
- Download the updated Trio code and build Trio.

In each of the sections below, follow links to sections of other build pages then hit the back button on your browser to return to this page.

FAQs

- **When do I update?** Anytime you want to change versions or if the app is about to expire.
- **Do I delete my old Trio app first?** No! If you keep your Trio app on your phone, your settings (and existing pod) will continue to work the same after the update.
- **Do I need to start a new pod when I update?** No. Your pod session will continue seamlessly if you use the same Apple Developer Account to sign the Trio app targets as you did the last time you built.
- **What if I'm using a new/different developer account?** If you aren't building with the same developer account used when your existing app was built (this includes going from free to paid), you will install a brand new (second) Trio app on your phone. Your existing pod won't work with the new app, so you should time this transition when you are due to change pods. Delete the old app once you get the new one all setup.
- **What if it is a new computer but the same developer account?** No big deal...use the Updating Steps to check that your new computer has the required compatible versions and then build your app. This will include installing Xcode, configuring Xcode Settings, and adding your Developer ID to Xcode.

Prepare to Update

Under ordinary circumstances, you only have to update Trio once it expires (1 year for a paid account). However, we encourage regular updates when new versions are released because they often contain bug fixes or improvements, which may increase operational stability.

iOS Updates

Before updating Trio, it is important to check for any iOS updates. It's good practice to keep your other devices updated wherever possible. If updates are available, check their compatibility with Trio before updating.

macOS and Xcode Version

The table below lists the minimum requirements to build the current release of Trio.
First, you need to know your iOS version, macOS version, and Xcode version.
Next, find your iOS version in the table below, and review the correlating minimum macOS and Xcode versions.
If required, update your macOS and Xcode.

> **iOS Version**
>
> If your iOS version is not listed, e.g., 26.4.1, choose the first row that is lower than your iOS version.

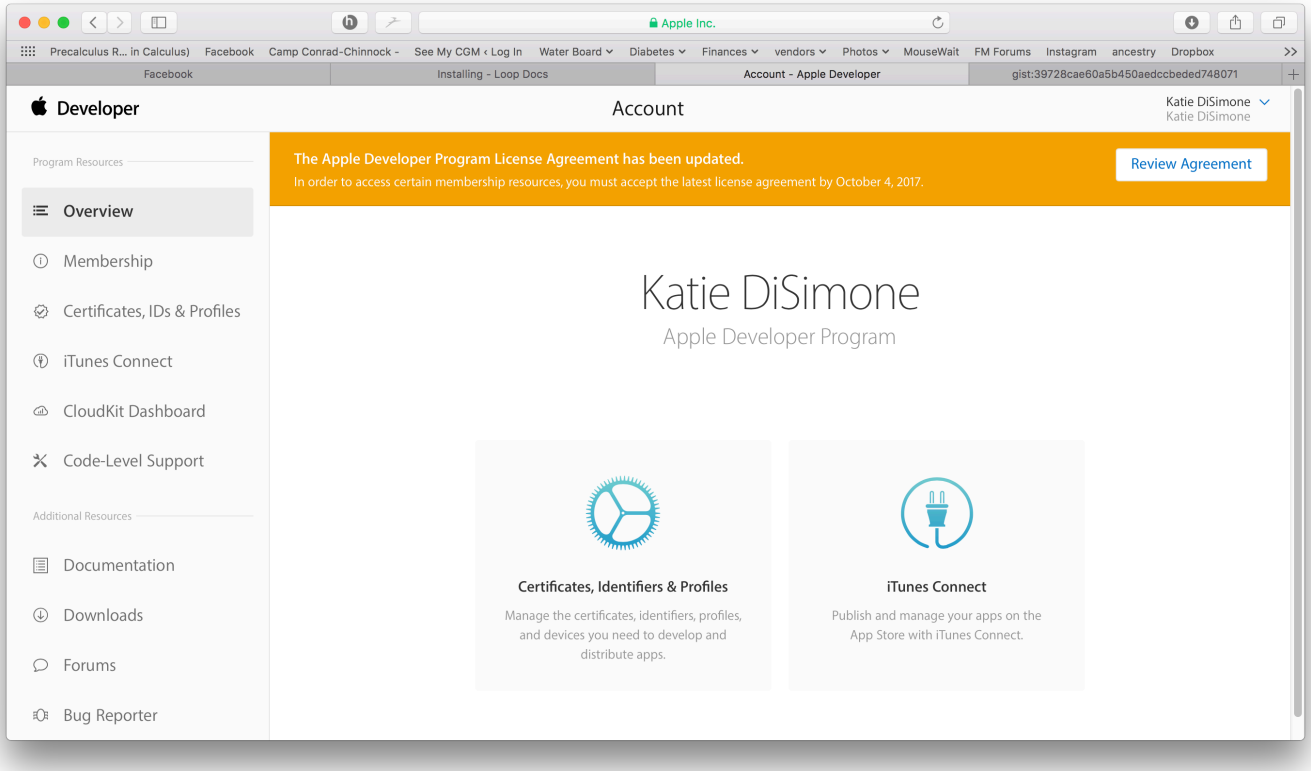
iOS Version	Minimum Xcode Version	Minimum macOS Version
26.4	16.4	15.3
18.1	16.1	14.5
18.0	15.4	14.5
17.5	15.4	14.0
17.4	15.3	14.0
17.0	15.0	13.5

> **Update MacOS first, Then Xcode**
>
> Your macOS must meet the minimum requirement for the Xcode version needed to support your current iOS, as detailed in the link above.
>
> - If the macOS is too old, the Xcode version will not appear in the App Store
> - You might think you don't need to update Xcode (but you do)
> - Your build will fail, and mentors might need to help you

Apple Developer - Check for updated agreements

Apple frequently updates its License Agreement for the Developer Program. You need to log in to your developer account to manually check if there is a new agreement to accept. If you see a big red or orange banner atop your

Developer Account announcing a new license agreement, like shown below, please read and accept it before building Trio.



Update Trio with Xcode

The easiest way to update *Trio* is to download a fresh copy.

Once you completed the Prepare to Update steps listed above, head over to Build with Mac and follow the directions.

Additionals

New Trio users do not need to read the rest of this page.

Frequent Builders

If you build frequently, you do not have to delete the profiles every time. When the build script asks if you want to "Ensure a Year?", you can skip that step.

On the other hand, you may need to delete the provisioning profiles or saved Xcode information about a version of currently on your computer. The maintenance utilities found in the BuildSelectScript can be run to delete your provisioning profiles or clear derived data. Or you can use the individual commands in the next sections to do the same thing.

Delete Provisioning Profiles

You can delete your provisioning profiles by copying this command and pasting it into any terminal. This does not affect any build you currently have on your phone - this just forces your current computer to generate a new one next time you build with *Xcode*.

- For those using *Xcode 16* or newer:

Copy and Paste to manually remove Xcode 16 Provisioning Profiles on your computer

```
rm ~/Library/Developer/Xcode/UserData/Provisioning\ Profiles/*.mobileprovision
```

- For those using *Xcode 15* or older:

Copy and Paste to manually remove Xcode 15 Provisioning Profiles on your computer

```
rm ~/Library/MobileDevice/Provisioning\ Profiles/*.mobileprovision
```

Delete Derived Data

If you build using the same clone on your computer and then update that clone, sometimes you want to remove derived information that Xcode remembers and force it to start fresh.

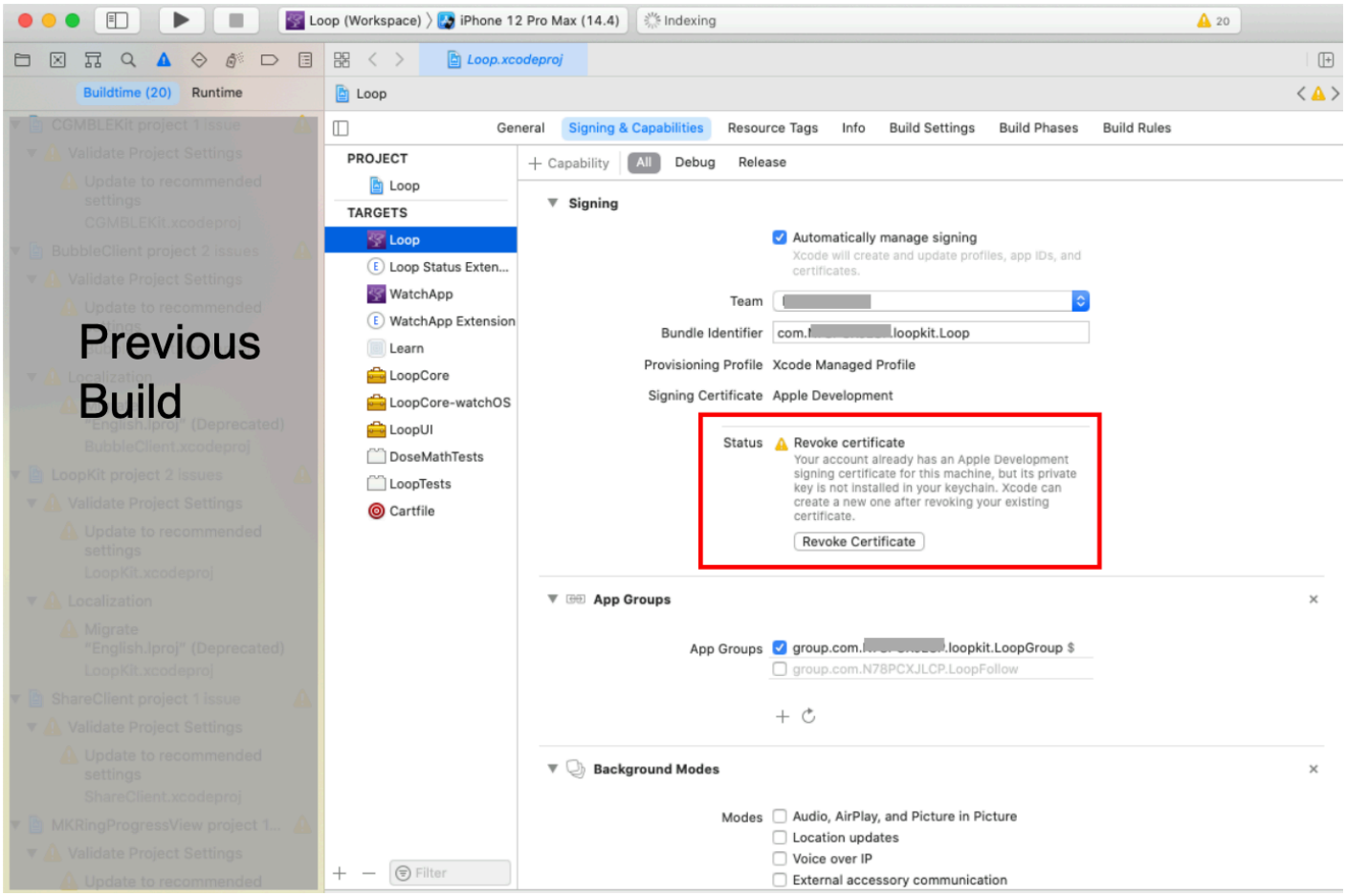
First quit out of *Xcode*. The following command will delete all derived information for all your clones, so next time you build any app from an existing clone on your computer, the build will take longer. All dependencies will download again. So wait until you see the "indexing" indication on *Xcode* before trying to build.

Copy and Paste to manually force Xcode on your computer to start fresh

```
rm -rf ~/Library/Developer/Xcode/DerivedData
```

Revoke Certificate Issue

What does it look like if you run into the Revoke Certificate message? When you prepare to Sign the Targets with Xcode, you'll see the message highlighted in the figure below.



More information is shown in the orange box below.

Revoke certificate

The important part of this message is:

- ". . . signing certificate . . . private key is not installed in your keychain . . ."

WAIT - You might not need to revoke your certificate

1. You might get this if you logged in as a different user, have a new computer or if your computer had to undergo a factory reset.
    - You can transfer your keychain to your new computer (or just revoke and keep going).
    - To transfer your keychain, check this Apple Documentation Link.
2. Your version of Xcode is way out-of-date
    - Mentors have seen this with people trying to build with Xcode 11.4 or earlier
    - Update Xcode to the most recent version

If you revoke and keep going:

- If you do hit Revoke Certificate, you'll be given a new one.
- Even with the new certificate, your Apple Developer ID is not affected.
- You can re-build on the existing Trio app on your device(s) and maintain all your settings with the new certificate.

Be aware that you will have to rebuild to every device that used the certificate you just revoked and if you have other apps built with this certificate, they will stop working too.

Direct Download of Xcode

Many people find updating Xcode from the App Store to be incredibly slow - especially when a new version has just been released. This method still takes time and enough space on your disk but is faster than going through the App Store. Depending on your internet speed, this download can be done in about an hour. Then once it is downloaded, expect another fifteen minutes to several hours (depending on the speed of your computer) for the "xip" file to "expand".

The instructions do not hold your hand.

- Your macOS must be at the minimum version (or newer) to support the version of Xcode you're about to download
- You need to know how to log into your Apple Developer account and navigate those menus
- You need to know how to use Finder to navigate to Downloads
- You need to know how to drag the Xcode icon into your Applications folder (after download and expand completes)
- After you have done a direct download, the App Store will not show you updates
    - Either repeat the Direct Download or
    - Delete Xcode from Applications folder

- Open the App Store and search for Xcode
- Install fresh
- After you use the App Store for a download, then Updates will show in the future

Here are the different steps you need to follow when doing the Direct Download instead of the App Store method:

1. Open the Apple Developer Download page
    - You may need to login
    - Examine the menus (on my computer there are buttons across the top)
    - Click on Applications
    - Look at the available applications, which should include one or more Xcode version
    - Scroll down until you find the item you want (for example, Xcode 15.4 or Xcode 16)
    - Click on View Details and click on the Download button for the "xip" file
2. Wait for Download to complete
3. Expand the file by clicking on it in Finder
4. Move the Xcode icon to Applications after the expansion completes
5. Check the Command Line Tools setting under Xcode->Settings
    - The selection cannot be blank or Build-Script will fail to open Xcode automatically
    - It should be the same version as your Xcode
6. Reboot the computer

Upgrade

Regardless of Build Method, you can only upgrade (build a newer version or different feature branch) or update (build the same version) if you have a valid Apple Developer account.

- Sign your developer agreements promptly

Other tasks required to upgrade to the latest version, or to rebuild to keep *Trio* working on your phone, depend on your build method.

- Browser Build Update ↗
- Mac Build Update ↗

Return to the Released Version

Trio keeps its released version on the `main` branch. Test versions live on the `dev` branch and on feature branches.

If *Trio* shows you a warning at startup saying it was built from `dev` or from another branch, this page tells you how to get back to the released version.

Why am I seeing a warning?

Trio checks which branch it was built from each time it starts, and warns you if that branch is not `main`. The warning appears once per start, so it comes back if you quit *Trio* and open it again.

- The `main` branch is the released version and is what almost everyone should run
- The `dev` branch is for expert testers — it changes often and contains features that have not been fully tested
- A feature branch contains work on a single change and is usually meant for the person developing it

Seeing this warning does not mean something is broken. It means you are running a test version.

Before You Switch

Check the Version Numbers

Moving from `dev` back to `main` is usually a **downgrade** — the version on `dev` is normally newer than the version on `main`.

An older Trio may not be able to open your existing data

Trio stores your history (glucose, pump events, carbs, and algorithm data) in a database on your phone. When a newer version of *Trio* changes the layout of that database, it upgrades your existing database to match. That upgrade only runs in one direction.

Going back to an older version can therefore fail to open the database you already have. If it does, *Trio* stops on the loading screen at startup and does not continue.

Before you switch:

- Find the version on `main` — see the Trio releases page ↗
- Find the version you are running now — open *Trio* and go to Settings, where the version is shown at the bottom
- If the version on `main` is **older** than the version you are running, ask on Facebook ↗ or Discord ↗ whether that particular downgrade is safe before you build

If your build does end up stuck on the loading screen, you have not lost your settings — reinstalling the version you were on before will let *Trio* open normally again.

Plan the Timing

- Your pod or pump session continues normally as long as you build with the **same** *Apple* Developer Account you used last time
- If you are using a different Developer Account, you will get a second, separate *Trio* app on your phone, and your existing pod will not work with it — plan that change for when you are due for a pod change anyway
- Do not delete your existing *Trio* app first; building over it keeps your settings

> **This is not medical advice**
>
> Changing app versions changes the software that automates your insulin delivery. If you are unsure whether or when to switch, discuss it with your diabetes care team.

Browser Build

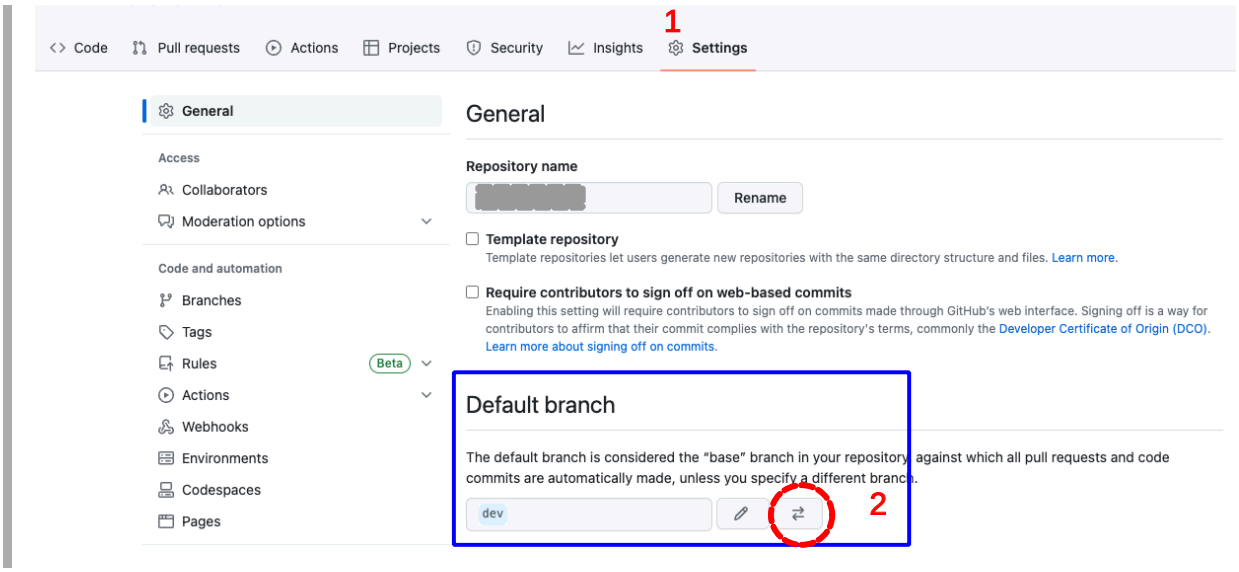
These steps are for people who build *Trio* using *GitHub* Actions in a browser.

Set `main` as Your Default Branch

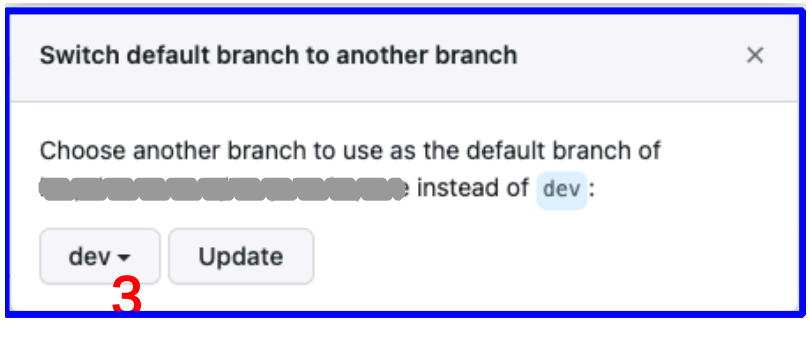
If you previously followed Configure the `dev` branch as default, you need to reverse it. If you never changed your default branch, skip to Update Your Fork.

Only the owner of the repository can do this, and you must be logged in — otherwise the Settings tab does not appear.

1. Click on the Settings icon near the top right of your *Trio* repository
   - You may need to scroll down to see the `Default Branch`
   - Do not tap on the Branches tab on the left under Code and Automation, that is not the correct menu



2. To the right of the default branch name there is a pencil and a left-right arrow icon (⇄)
   - Tap on the left-right arrow icon (⇄) to bring up the `Switch default branch to another branch` dialog
3. Click on the dropdown next to the current default branch, in this case `dev`



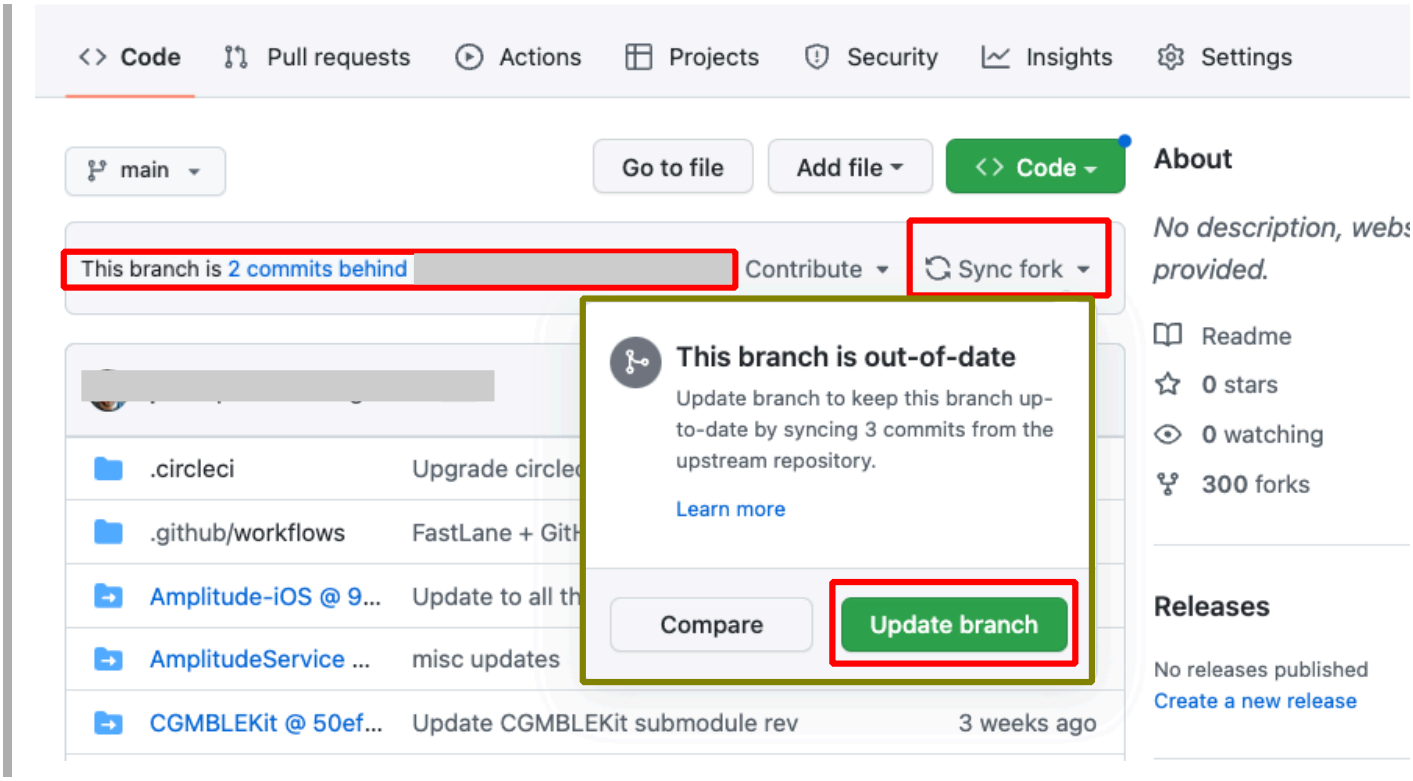
4. Select `main`
5. Click on the `Update` button
6. You will be presented with an are-you-sure question
    - Click on the red `I understand, update the default branch.` button

Your default branch is now `main`, and your weekly automatic builds will follow `main` from here on.

Update Your Fork

Tap the `Code` button (upper left) and make sure the `main` branch in your `fork` is up to date.

- Select `main` in the branch dropdown menu
- If the message says this branch is "behind", tap the sync `fork` button and then the `Update branch` button



Build Trio from `main`

Run the `Build Trio` action, making sure `main` is selected in the branch dropdown menu.

Follow the normal steps on Build the App, and install the resulting build from *TestFlight* as usual.

Hit Refresh

After you tap the green Run workflow button, *GitHub* can be slow to update.

- Refresh the browser if you are unsure if the action started
- Do not start a new action until the first one completes

Mac Build

These steps are for people who build *Trio* with *Xcode* on a *Mac*.

The easiest way is to download a fresh copy of the released code rather than changing the copy you already have.

1. Follow Build Trio with Xcode from the beginning
2. When the Build Select Script asks you which branch to build, **type 1 and press return to select `main`**
3. Continue with the normal build steps

Your settings and pod session are preserved as long as you sign with the same *Apple* Developer Account as before.

After You Switch

The startup warning stops appearing once you are running a build made from `main`.

If you still see it:

- Check that the build you installed is actually the new one — open Settings in *Trio* and look at the version and branch shown at the top
- For a browser build, check that you selected `main` in the `Build Trio` action, not just as your default branch
- Remember that *TestFlight* may still be offering you the older build — make sure you installed the newest one

Going Back to `dev` Later

If you want to test again in the future, follow Build Dev with Browser or select the `dev` branch in the Build Select Script when building with *Xcode*.

Companion Apps

Welcome to the Companion Applications homepage! These apps all work within the Trio ecosystem. They are separate apps that are interoperable with Trio and many other DIY AID systems.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Nightscout

LoopFollow

xDrip4iOS

Nightscout

Nightscout is an excellent tool for remotely viewing *Trio* actions. *Nightscout* can act as a stand-alone tool or be integrated with *Trio*. (*Nightscout* integrates with many open-source hybrid closed loop systems such as *Loop*, OpenAPS, AndroidAPS and *Trio*.)

When integrated with *Trio*, *Nightscout* provides monitoring of *Trio* activities such as viewing history of glucose, carbs, boluses, temp basals and overrides; troubleshooting *Trio* errors; and provides extensive reports for analyzing data trends and patterns. These reports assist when *Trio* Therapy Settings need to be adjusted.

For caregivers, *Nightscout* enables remote monitoring and even the ability to issue remote commands through *Nightscout* when both *Trio* and *Nightscout* are properly configured. If you add *LoopFollow*, those same remote commands can be issued from the caregiver's *LoopFollow* phone with better connectivity and feedback than using *Nightscout* alone.

For more information about *Nightscout*, please refer to Nightscout Documentation.

For more information about using remote control with a combination of *Nightscout* and *Trio*, please refer to the LoopFollowDocs.

LoopFollow

LoopFollow is an open-source DIY (do-it-yourself) app that displays information for a person with diabetes in an easy-to-use interface. It combines information from multiple other apps and provides custom alerts. It runs on iPhone, iPad or Mac devices. Some remote control features are available for the *Loop* and *Trio* apps to assist caregivers.

LoopFollow was created because using different apps as a parent or caregiver of a person with diabetes can be cumbersome. Add in the extra details from looping with an Open Source Automated Insulin Delivery (OS-AID) app and flipping back and forth between various sources of information can be a challenge. Many loopers use LoopFollow for themselves because of the flexible (and loud) alarm options.

LoopFollow works best when paired with *Nightscout*.

For more information, please refer to the LoopFollowDocs.

xDrip4iOS

xDrip4iOS (also known as "xDrip for iOS" or "xdripswift") is an open-source application to display real time blood glucose data.

If you prefer to use xDrip4iOS as a CGM source with *Trio* you must do several things:

- Build xDrip4iOS with the same Apple Developer ID as you use to build Trio
- In xDrip4iOS select use as a heartbeat for Trio (otherwise, your app will not loop properly).

For more information, please refer to the xDrip4iOS Documentation.

Configuration

Welcome to the Configuration homepage! Here you will find all the information needed to setup your app after you've built it. Configuration involves adjusting your settings. It is important that you do this in an educated way, so this section is here to guide you!

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

New User Setup Guide

Step by step guide on starting Trio

Trio Settings

A compilation of all settings found in Trio with explanations and definitions

Migration Guides

Guidance on locating and identifying the necessary settings for onboarding

New User Setup Guide

Welcome to the New User Setup Guide and congratulations on a successful Trio build! This guide walks you through how to set up your Trio app once you have installed it on your phone.

If you still need to install the app, head to the Build Instructions for Mac or Browser and come back here when you are ready to start the setup process!

Onboarding Wizard

Trio has an Onboarding Wizard that walks you through these steps when you first start the app. Each step contains a tab for **During Onboarding** and **After Onboarding**, explained below.

During Onboarding

Information under the **During Onboarding** tab provides additional information for that step of the Onboarding Wizard.

Use this documentation as your home base to refer back to as needed.

After Onboarding

Information under the **After Onboarding** tab shows you where to edit the relevant settings after onboarding is complete as well as the additional information on those settings.

Use this documentation as your home base to refer back to as needed.

> **Tip**
>
> If you've completed the Onboarding Wizard and need guidance on what to do next, jump to Step 7: Connect Your Devices

---

Jump to:

Step 1: Prepare Trio

Step 2: Therapy Settings

Step 3: Delivery Limits

Step 4: Algorithm Settings

Step 5: Notifications

Step 6: Bluetooth

Step 7: Connect Your Devices

Step 8: Enable Closed Loop

Step 9: Change App Icon (Optional)

Need help finding these settings in your current DIY app?

If you are coming from another Open Source Automated Insulin Dosing (OS-AID) app, these guides may help you with finding these settings in your previous app and inputing them into Trio.

- Trio 0.2.x
- iAPS
- Loop
- AndroidAPS

Step 1: Prepare Trio

In this step, you'll configure diagnostics sharing, optionally sync with Nightscout, and enter other essential setup information.

During Onboarding

Diagnostics

By default, Trio collects crash reports and other anonymized data related to errors, exceptions, and overall app performance. Sharing this data helps developers maintain and improve the app. This is completely anonymous and optional.

Tip

If you select `Enable Sharing`, you must also read and accept the Privacy Policy in the Onboarding Wizard before you are able to proceed.

Nightscout

Nightscout is a cloud-based platform that allows you to store your diabetes data.
Nightscout is not required and can be added at a later time under Services in the Settings menu of the app.

Units & Pump

In the Onboarding Wizard, you will select your Glucose Units and Pump Model, but you will not pair your pump yet. You will pair your pump and CGM after completing the Onboarding process (See Step 7).

After Onboarding

Step 2: Therapy Settings

The next step is to enter your Therapy Settings. These include:

- **Glucose Targets**: Trio's dosing will aim for this glucose when calculating insulin dosage
- **Basal Rates**: Used as a baseline for increasing or decreasing insulin needs
- **Carb Ratios**: How many grams of carbohydrate are countered by 1 unit of insulin
- **Insulin Sensitivities**: How much 1 unit of insulin will lower your blood glucose

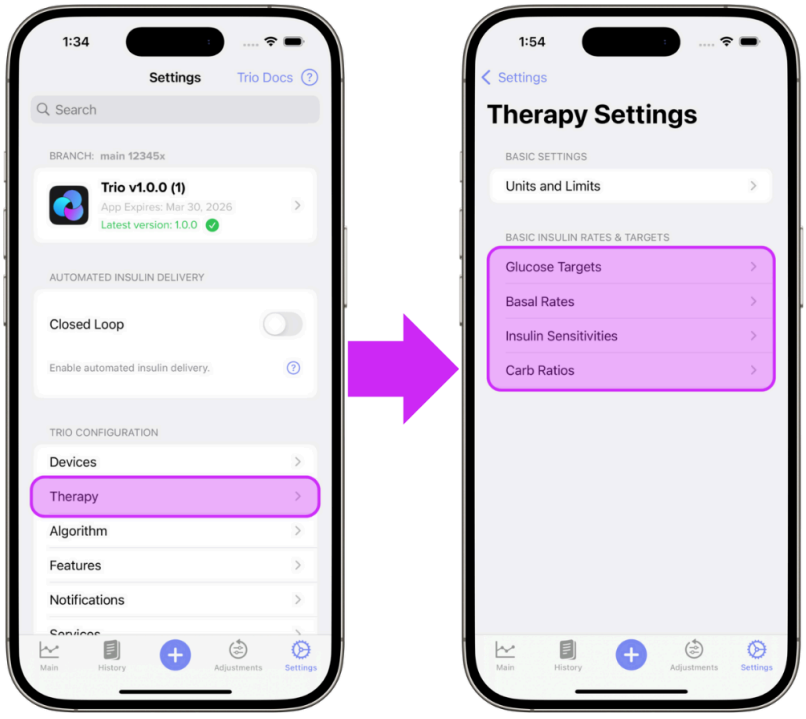
During Onboarding

For more information about each therapy setting in this step, use the links below:

- Glucose Targets
- Basal Rates
- Carb Ratios (CR)
- Insulin Sensitivities (ISF)

After Onboarding

Here is how to locate these settings after onboarding:



Below you'll find a step by step guide to edit each of these settings:

- Glucose Targets
- Basal Rates
- Carb Ratios (CR)

Step 3: Delivery Limits

In this step you will set the boundaries for insulin delivery and carb entries to help Trio keep your insulin dosing safe, yet effective.

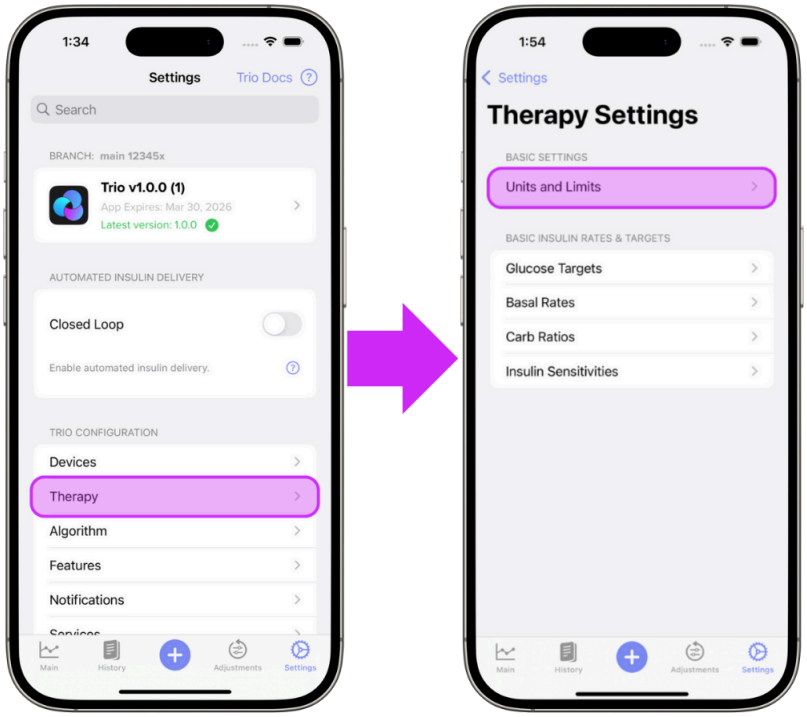
During Onboarding

Below you'll find more information on each of the settings in order of appearance in the Onboarding Wizard:

- Max IOB
- Max Bolus
- Max Basal Rate
- Max COB
- Minimum Safety Threshold

After Onboarding

Here is how to locate these settings after onboarding:



Below you'll find more information on each of these settings:

- Max IOB
- Max Bolus
- Max Basal Rate
- Max COB
- Minimum Safety Threshold

Step 4: Algorithm Settings

Trio includes several algorithm settings that allow you to customize the Oref algorithm behavior to suit your specific needs.

To configure the algorithm, you'll define the settings for Autosens, Super Micro Bolus (SMB), and Target Behavior.

Go Slow

- This step may feel overwhelming, so take it slow and use the links shared below to help guide your choices.
- Our strong recommendation is to leave everything on the default settings as a new user.

Important information for this step:

- DynamicISF requires at least 7 days of data and is not yet configurable
- Even if you're an updating user, you'll be guided through this step-by-step. It is important to read each step as some things may have changed
- **All additional "advanced settings" have been reset**
- The duration of insulin action (DIA) is now locked to Trio's new default of *10 hours*.
  - We strongly recommend ***not*** changing DIA as it is essential for an accurate IOB calculation and necessary for safe and stable operation.

During Onboarding

Autosens

Autosensitivity, or Autosens, adjusts insulin delivery based on observed sensitivity or resistance.

Step 1: Set Autosens Min
This is the lower limit of the Autosens Ratio.

Step 2: Set Autosens Max
This is the upper limit of the Autosens Ratio.

Super Micro Bolus (SMB)

SMB (Super Micro Bolus) is an oref algorithm feature that delivers small, frequent boluses instead of temporary basal adjustments, creating a more responsive system. The onboarding options are different depending on what you choose for Step 3.

Step 3: Enable/Disable SMB Always

Step 5: Notifications

In this step you will allow Trio to send you notifications. These include alerts, sounds, and icon badges of your choosing. Notifications give you important Trio information without requiring you to open the app. It is essential that these are allowed in your iPhone system settings. Once you complete onboarding, you can customize your notifications.

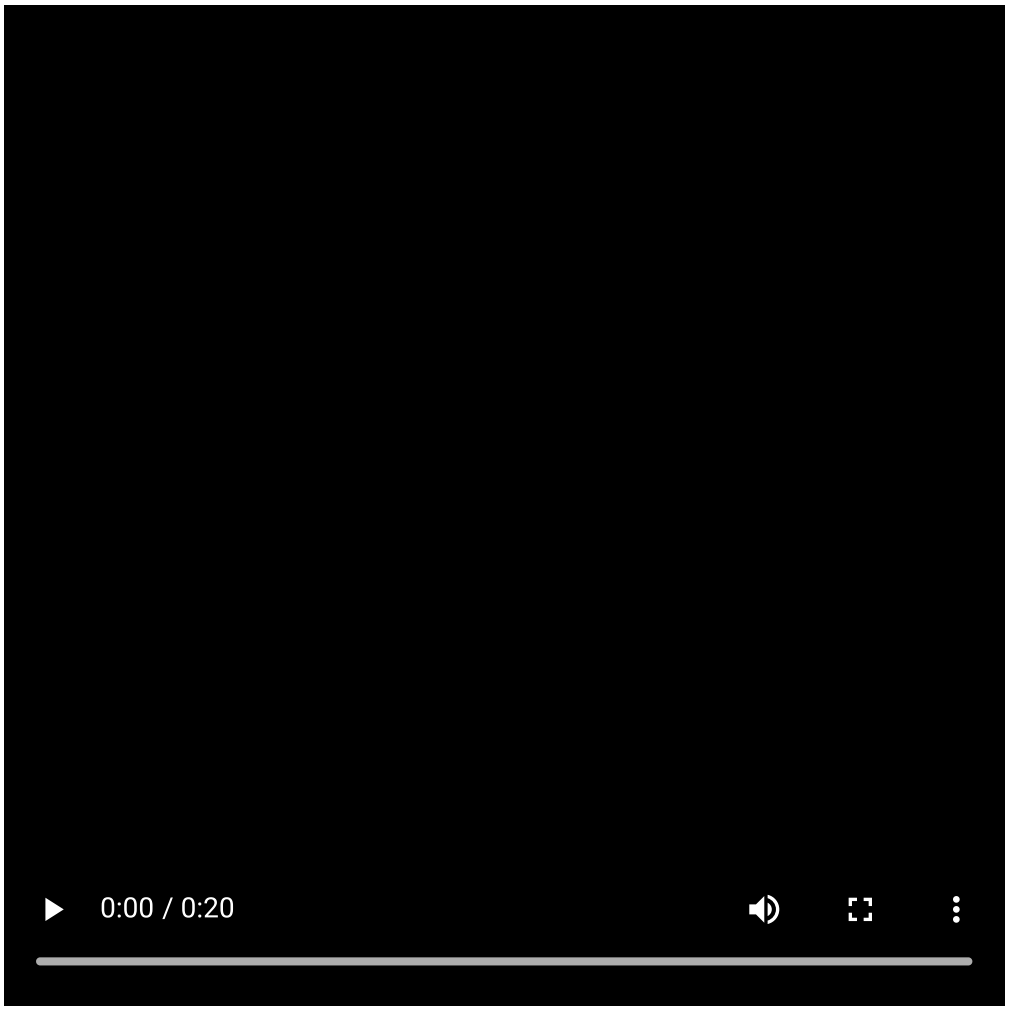
To edit Notifications after you've completed onboarding, head over to the Notifications section for more information on each type of notification and how to edit them.

Step 6: Bluetooth

Trio requires Bluetooth to function as a (hybrid) closed-loop system. If you do not have Bluetooth enabled in your iOS settings, you will not be able to connect your phone to your insulin pump or CGM. A pop up will appear that makes this easy to enable if it hasn't been already.

Congratulations!

But, you're not done yet!



Steps 7 & 8 are not in the Onboarding Wizard
These cannot be completed until you've finished onboarding, but *must* be completed before you can start using Trio.

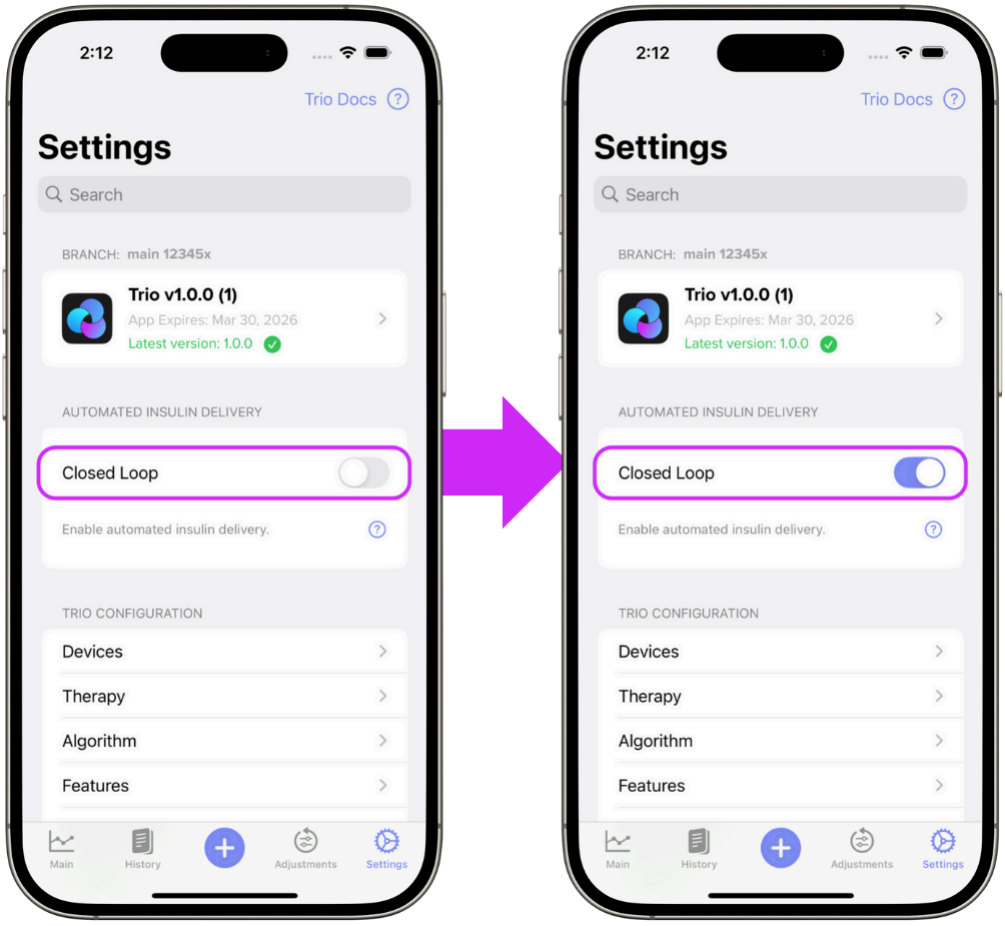
Step 7: Connect Your Devices

Connect your Insulin Pump
Connect your CGM
Connect your Watch

Step 8: Enable Closed Loop

> **Warning**
>
> Trio works best as a closed loop system. If you need to test your settings or are concerned about trying a new algorithm, it's best to close the loop and follow the configuration instructions here.

Closed loop functionality is turned **OFF** by default. This means Trio cannot make any adjustments automatically. The system relies solely on you to make any adjustments while Closed Loop is **OFF**. You can control your pump and manually bolus with the Trio app, but nothing can be done without your approval. This is often referred to as running in open loop. You will ***not receive any Trio-initiated protection from lows or highs while in open loop***.



More on closing the loop

Dynamic Settings

Trio will not allow you to enable Dynamic Settings until it has accumulated 7 days of data. This is essential so Trio can make sound dosing recommendations. Once it has enough data and you can enable Dynamic Settings, it is still recommended that you have already...

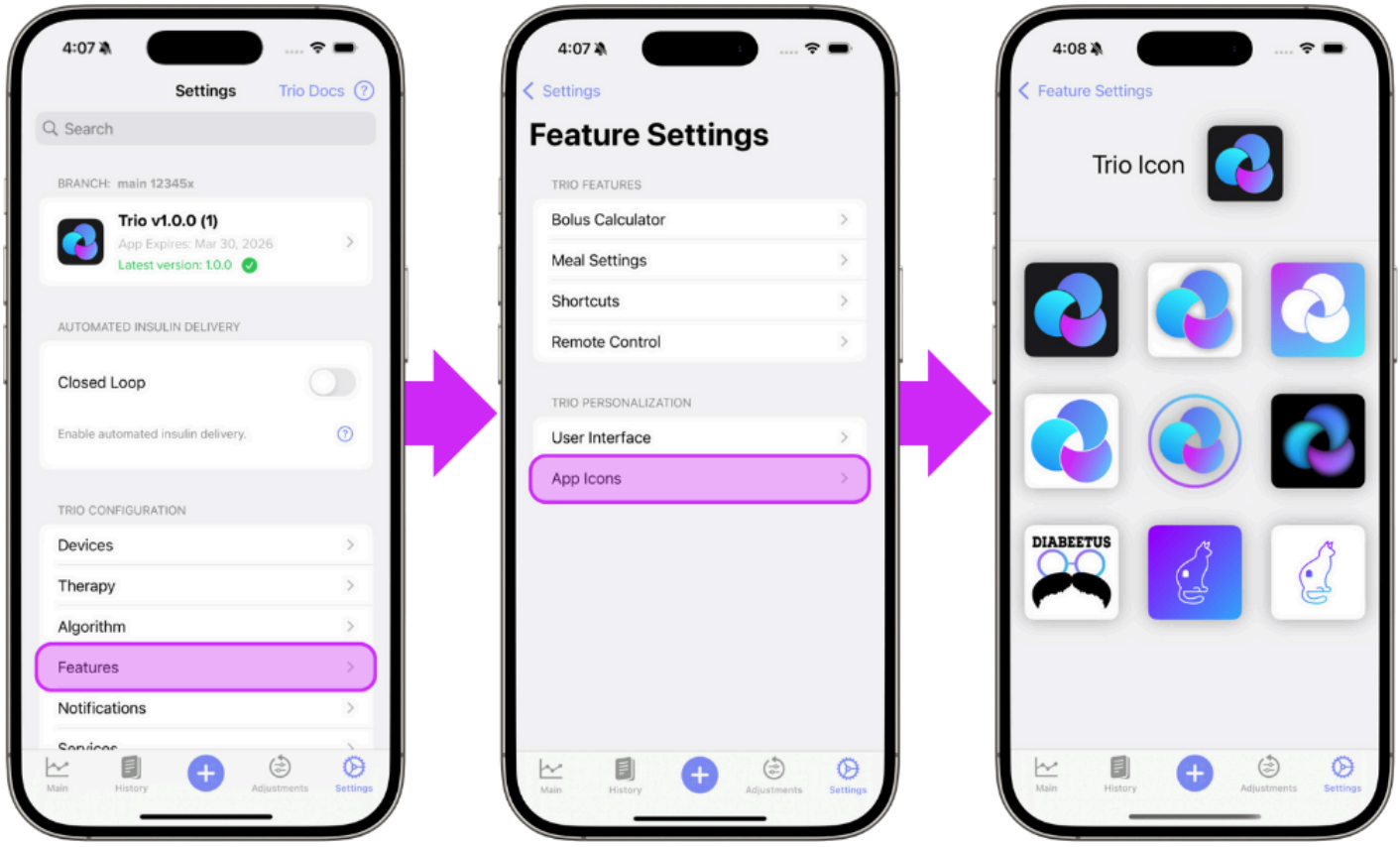
- ...tuned your core Trio settings (ISF, CR, and Basal Rates) for use in the Oref algorithm.
- ...used Trio with a *real* CGM and *real* pump (not simulators) for the recommended minimum of **7 consecutive days**.
- ...read and understand the Dynamic Settings and how they interact with each other.

DO NOT ALTER ANYTHING ELSE UNTIL YOU'VE TESTED AND VERIFIED YOUR SETTINGS

- If you are coming from another Oref-based system, like Trio 0.2, iAPS, or AAPS, you will want to watch your current settings to ensure they are performing as they did previously. You may need to make some adjustments, but you also may find your settings work as well as they did in the previous system.
- If you are coming from Loop, a commercial system, or multiple daily injections (MDI), you will want to first test your settings to ensure they do not require any adjustments. You've been using a completely different algorithm to manage your insulin dosing and that means you've tailored your settings to be optimized using that method of dosing. Because Trio uses a different algorithm, your settings will almost assuredly need to be adjusted to work optimally within the Oref algorithm.

Step 9: Change App Icon (Optional)

Under "App Icons" in the Settings Menu, you can find various icons for your Trio app.



Congratulations! You've completed the New User Setup for Trio!

Migration Guides

Welcome to the Migration homepage! Here you will find information on what to expect when starting on Trio. Here you will also find guidance on locating and identifying settings that can be moved to your new Trio app. Select the app you are moving from in the list below to get started.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Trio 0.2.x

What to expect when upgrading from Trio 0.2.x to Trio 0.5 (or higher)

iAPS

What to expect when switching from iAPS to Trio 0.5 (or higher)

Loop

What to expect when switching from Loop to Trio 0.5 (or higher)

AndroidAPS

What to expect when switching from AndroidAPS to Trio 0.5 (or higher)

Coming from Trio 0.2.x

Please Read This Page BEFORE Updating

Do not update without reading this documentation or you risk finding many unexpected changes that can easily be resolved with prior planning.

What to Expect After Install

- **Brand New Interface**
  A completely redesigned UI across the app, including the Watch app — clean, modern, and easier to use. Learn more in the User Interface Walk-Through.
- **All-New Onboarding**
  Step-by-step setup for new and returning users with smart defaults and safety checks. ***All users* will need to complete the new Onboarding Wizard after upgrading**. Learn more in the New User Setup Guide.
- **Rewritten Backend & Data Storage**
  Entire app logic and data storage rewritten for speed, stability, and future flexibility.
- **Remote Commands & iOS Shortcuts**
  Trigger actions like carb entry, bolus, overrides or temporary targets remotely or through shortcut automation.
- **Smarter Settings**
  Helpful hints, guidance sheets, and a built-in search function — just like iOS Settings. Information on the updated Trio Settings can be found here.
- **New Bolus Calculator**
  More accurate and simple dosing with clearer breakdowns and improved safety logic. Learn more about the Bolus Calculator.
- **Reworked Statistics**
  All in-app statistics and graphs have been overhauled and extended. Learn more about the New Statistics Screen.
- **Improved Live Activity Widget**
  Real-time loop data with optional chart — configurable and tailored to your preferences — visible on lockscreen, Watch, and CarPlay. Learn more about the Live Activity Widget.
- **Live Data on Watch**
  Highly customizable “contact widgets” for glucose, IOB, COB, and more, always visible. Learn more about the Contacts Configuration on Apple Watch.
- **All-New Integrations**
  Nightscout, Tidepool, and Apple Health connections have been rebuilt from scratch — faster and more stable. More on integrations here.
- **Safety Improvements**
  Dynamic ISF guarding (7-day required data), safer edge-case behavior, and improved support for high-insulin-resistance setups.

- **Updated Language & Translations**
  Clear, friendly, and easy-to-understand. Fully localized with ongoing help from our Crowdin contributors (translators still welcome).
- **Now Requires Loop Follow v4.0 or Later**
  More info at LoopFollowDocs
- **Autotune Removed**
  Autotune has not performed well with the addition of Dynamic ISF and many users using multiple ISF and CR in Therapy settings. For this reason, we have removed it until an improved Autotune-like feature is built.

What *Will* Transfer from Trio 0.2 to Trio 0.5 (or higher)?

- Current pump and CGM sessions (but always a good idea to have a backup pump ready whenever updating, just in case)
- 24 hr treatment history
- Core therapy settings (glucose targets, basal rates, ISF, CR)

What *Will Not* Transfer from Trio 0.2 to Trio 0.5 (or higher)?

- Algorithm settings, Dynamic ISF, and some integrations (like Nightscout) will reset.
- Treatment history older then 24 hours will not transfer
- Historical data used for Dynamic ISF and Statistics
- Override, Temp Target, and Meal Presets

> **Upgrading is a one-way street**
>
> Once you upgrade to v0.5.0 (or higher), going back to v0.2.x is not supported.

What to Expect in Your First 3 Hours

Here are some things to keep in mind after upgrading from Trio 0.2 to Trio 0.5 (or higher) in the first 3 hours:

- **Dynamic ISF & Autosens**
  Dynamic ISF is disabled for the first **7 days**. *You may need to enter carbs and bolus during this time if you aren't already doing so.* Autosens will be used instead (but there's a long standing issue with autosens where it will likely be stuck at 1 when there are carb entries or SMBs.)
- **Onboarding Wizard**
  Take this time to go through the Onboarding Wizard. Reference the New User Setup Guide or ask questions on Facebook or Discord if you have any questions.
- **Test Settings**
  This is a great time to test your settings so you can start fresh on solid footing.
- **New User Interface**
  Use this time to learn the New User Interface.

Do not use a pump or cgm simulator

- Doing so will result in false data being stored and you will have to delete the app with all the data and reinstall before you can use it on a live pump.
- This will also reset your 7-day waiting period for Dynamic ISF.
- More information on simulator use

What to Expect After 24 Hours

A full day on Trio 0.5 (or higher)! Congratulations! Here's what to look for:

- **Autosens**
  Autosens now has enough data to make adjustments
- **Review & Test Settings**
  This is a good time to review your last 24 hours of results to see if your core settings performed as expected. If you had unexpected challenges with your settings, look at how to test them over the next 7 days
  - Basal Testing
  - ISF Testing
  - Carb Ratio Testing
- **Review Dynamic ISF Documentation**
  Review the documentation on Using Dynamic ISF in preparation for Day 7.

What to Expect After 7 Days

You've completed a week on Trio 0.5 (or higher)! Here's what you should see:

- **Dynamic ISF**
  Dynamic ISF now has enough data. You now have the option to enable it if you would like to.

Can't Enable Dynamic ISF?

If Dynamic ISF does not give you the option to enable, you may have experienced one or more of the following:

- You were not in closed loop for the full 7 days.
  - *Trio needs 7 days of closed loop to safely enable Dynamic ISF*
- You had significant loss of connection with your CGM and/or pump.
  - *Check your last week of Looping Statistics in the Stats tab in the Trio app. You need both **7 days** and an **85% success rate** to enable Dynamic ISF.*
- You enabled a significant number of manual Temp Basals with long run times.
  - *When a manual Temp Basal is set, Trio is unable to complete a loop cycle for the duration of the temp basal. This will cause a reduction in your success rate. If you do not have **85% success rate** for **7 days**, you cannot enable Dynamic ISF.*

- **Dynamic CR** Dynamic CR has been removed due to no known scientific validity to CR changing with an increase or decrease in glucose.

Changes from Trio 0.2 Settings to Trio 0.5 (or higher) Settings

When you update from Trio 0.2, some of your settings might look different even if they have the same or similar names. That's because we changed some of the numbers from decimals to percentages to make them easier to understand.

Example

In Trio 0.2 you might set "Autosens Maximum" to 1.2. In Trio 0.5 (or higher), we show "Autosens Max" as 120%, which means the same thing, but is easier to understand.

Some setting names have also changed. We did this to make things less confusing. To help you out, we added definitions right in the app. Just tap the question mark icon next to any setting to see what it means. Healthcare Professionals and users can also find those settings and explanations in this section of the docs.

The charts below will help you see which setting names or formats have changed. The settings with changes are in bold.

Tip

- To convert a value from a decimal to a percentage, multiply it by 100.
- To convert from a percentage to a decimal, divide by 100.

Therapy Settings

Trio 0.5+ Name	*Entry Type*	*Format (example)*	Trio 0.2 Name	*Entry Type*	*Format (example)*	Change
Glucose Targets	*schedule*	*decimal (100 mg/dL / 5.5 mmol/L)*	**Target Glucose**	*schedule*	*decimal (100 / 5.5)*	**Name**
Basal Rates	*schedule*	*decimal (1.0 U/hr)*	Basal Profile	*schedule*	*decimal (1.0 U/hr)*	--
Carb Ratios	*schedule*	*decimal (10 g/U)*	Carb Ratios	*schedule*	*decimal (10 g/U)*	--
Insulin Sensitivities	*schedule*	*decimal (54 mg/dL/U / 3.0 mmol/L/U)*	Insulin Sensitivities	*schedule*	*decimal (54 / 3.0)*	--
Maximum Insulin on Board (IOB)	***picker***	*decimal (2 U)*	**Max IOB**	***typed value***	*decimal (2)*	**Name, *Entry Type***
Maximum Bolus	***picker***	*decimal (10 U)*	**Max Bolus**	***typed value***	*decimal (10)*	**Name, *Entry Type***
Maximum Basal Rate	***picker***	*decimal (2 U/hr)*	**Max Basal**	***typed value***	*decimal (2)*	**Name, *Entry Type***
Maximum Carbs on Board (COB)	***picker***	*decimal (120 g)*	**Max COB**	***typed value***	*decimal (120)*	**Name, *Entry Type***
Minimum Safety Threshold	***picker***	*decimal (60 mg/dL / 3.3 mmol/L)*	**Threshold Settings**	***typed value***	*decimal (60)*	**Name, *Entry Type, mmol/L entry***

Algorithm Settings

Autosens

Trio 0.5+ Name	*Entry Type*	*Format (example)*	Trio 0.2 Name	*Entry Type*	*Format (example)*	Change
Sensitivity Ratio	*N/A*	***percentage (105%)***	Autosens Ratio	*N/A*	***decimal (1.05)***	***Format***
Calculated Sensitivity	*N/A*	*decimal (45 mg/dL/U / 2.5 mmol/L/U)*	Calculated Sensitivity	*N/A*	*decimal (45 mg/dL/U / 2.5 mmol/L/U)*	--
Autosens Max	***picker***	***percentage (120%)***	**Autosens Maximum**	***typed value***	***decimal (1.2)***	**Name,** ***Entry Type, Format***
Autosens Min	***picker***	***percentage (70%)***	**Autosens Minimum**	***typed value***	***decimal (0.7)***	**Name,** ***Entry Type, Format***
Rewind Resets Autosens	*toggle*	*(On/Off)*	Rewind Resets Autosens	*toggle*	*(On/Off)*	--

SMB (Super Micro Bolus)

Trio 0.5+ Name	*Entry Type*	*Format (example)*	Trio 0.2 Name	*Entry Type*	*Format (example)*	Change
Enable SMB Always	*toggle*	*(On/Off)*	Enable SMB Always	*toggle*	*(On/Off)*	***Now disables unused SMB settings***
Enable SMB With COB	*toggle*	*(On/Off)*	Enable SMB With COB	*toggle*	*(On/Off)*	***Now only appears if Enable SMB Always is OFF***
Enable SMB with Temptarget	*toggle*	*(On/Off)*	Enable SMB with Temptarget	*toggle*	*(On/Off)*	***Now only appears if Enable SMB Always is OFF***

Enable SMB After Carbs	*toggle*	*(On/Off)*	Enable SMB After Carbs	*toggle*	*(On/Off)*	***Now only appears if Enable SMB Always is OFF***
Enable SMB With High Glucose	*toggle*	*(On/Off)*	**Enable SMB With High BG**	*toggle*	*(On/Off)*	**Name** ***Now only appears if Enable SMB Always is OFF***
High Glucose Target	**picker**	*decimal (110 mg/dL / 6.1 mmol/L)*	**... When Blood Glucose is Over (mg/dl)**	**typed value**	*decimal (110)*	**Name** **Entry Type** **Format** ***Now only appears when Enable SMB With High Glucose is ON, mmol/L entry***
Allow SMB with High Temptarget	*toggle*	*(On/Off)*	Allow SMB With High Temptarget	*toggle*	*(On/Off)*	--
Enable UAM	*toggle*	*(On/Off)*	Enable UAM	*toggle*	*(On/Off)*	--
Max SMB Basal Minutes	***picker***	*decimal (30 min)*	Max SMB Basal Minutes	***typed value***	*decimal (30)*	***Entry Type***
Max UAM Basal Minutes	***picker***	*decimal (30 min)*	**Max UAM SMB Basal Minutes**	***typed value***	*decimal (30)*	**Name,** ***Entry Type***
Max Allowed Glucose Rise for SMB	***picker***	***percentage (20%)***	**Max Delta-BG Threshold SMB**	***typed value***	***decimal (0.2)***	**Name,** ***Entry Type, Format***

Target Behavior

Trio 0.5+ Name	***Entry Type***	***Format (example)***	**Trio 0.2 Name**	***Entry Type***	***Format (example)***	**Change**
High Temp Target Raises Sensitivity	*toggle*	*(On/Off)*	**High Temptarget Raises Sensitivity/Exercise Mode**	*toggle*	*(On/Off)*	**Name**

Low Temp Target Lowers Sensitivity	*toggle*	*(On/Off)*	**Low Temptarget Lowers Sensitivity**	*toggle*	*(On/Off)*	**Name**
Sensitivity Raises Target	*toggle*	*(On/Off)*	Sensitivity Raises Target	*toggle*	*(On/Off)*	--
Resistance Lowers Target	*toggle*	*(On/Off)*	Resistance Lowers Target	*toggle*	*(On/Off)*	--
Half Basal Exercise Target	***picker***	*decimal (160 mg/dL / 8.9 mmol/L)*	Half Basal Exercise Target	***typed value***	*decimal (160)*	***Entry Type, mmol/L entry***

Additionals

Warning

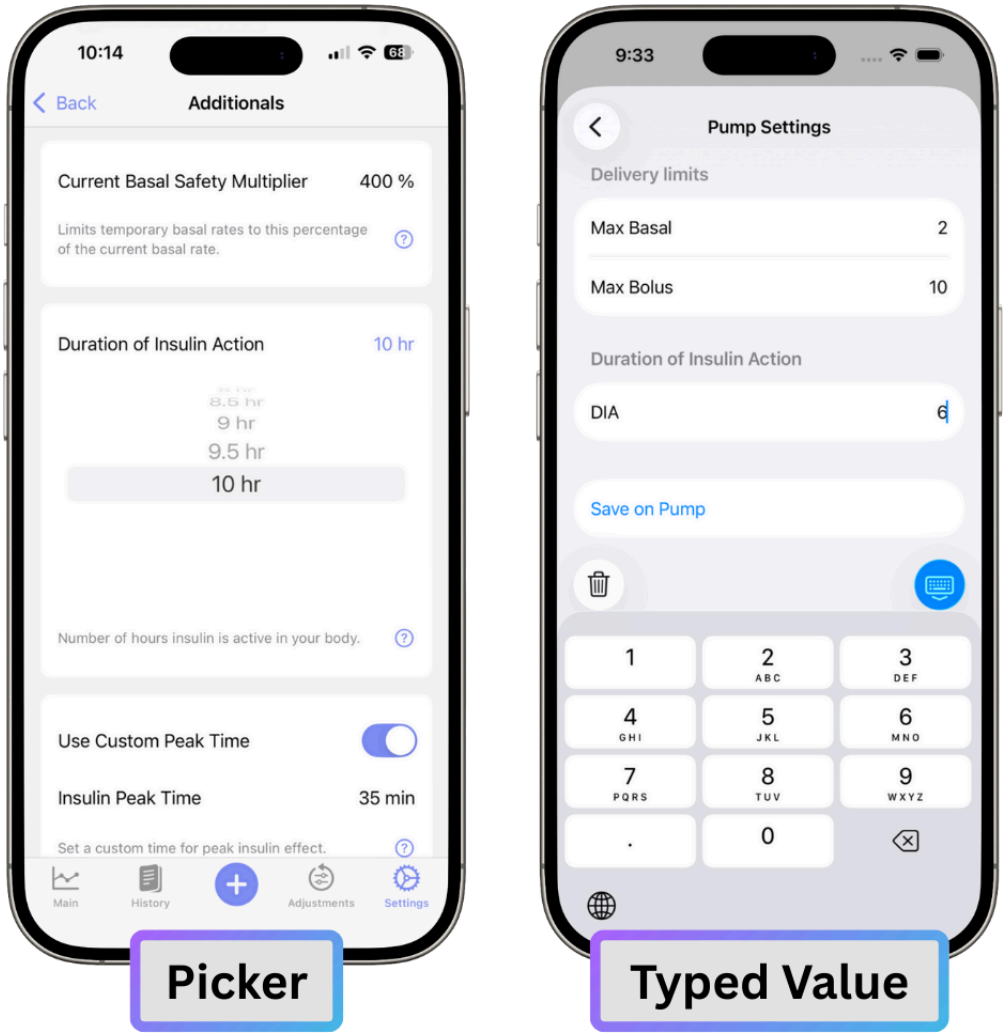
The settings in this section typically do not require any modifications.
Do not alter them without a solid understanding of what you are changing and the full impact it will have on the algorithm.

Trio 0.5+ Name	***Entry Type***	***Format (example)***	**Trio 0.2 Name**	***Entry Type***	***Format (example)***	**Change**
Max Daily Safety Multiplier	***picker***	***percentage (300%)***	Max Daily Safety Multiplier	***typed value***	***decimal (3)***	***Entry Type, Format***
Current Basal Safety Multiplier	***picker***	***percentage (400%)***	Current Basal Safety Multiplier	***typed value***	***decimal (4)***	***Entry Type, Format***
Duration of Insulin Action	***picker***	*decimal (10 hr)*	Duration of Insulin Action	***typed value***	*decimal (10)*	***Entry Type***
Use Custom Peak Time	*toggle*	*(On/Off)*	Use Custom Peak Time	*toggle*	*(On/Off)*	--
Insulin Peak Time	***picker***	*decimal (65 min)*	Insulin Peak Time	***typed value***	*decimal (65)*	***Entry Type***

Skip Neutral Temps	*toggle*	*(On/Off)*	Skip Neutral Temps	*toggle*	*(On/Off)*	--
Unsuspend If No Temp	*toggle*	*(On/Off)*	Unsuspend If No Temp	*toggle*	*(On/Off)*	--
SMB Delivery Ratio	***picker***	***percentage (50%)***	**SMB DeliveryRatio**	***typed value***	***decimal (0.5)***	**Name, *Entry Type, Format***
SMB Interval	***picker***	*decimal (3 min)*	SMB Interval	***typed value***	*decimal (3)*	***Entry Type***
Min 5m Carb Impact	***picker***	*decimal (8 mg/dL)*	**Min 5m Carbimpact**	***typed value***	*decimal (8)*	**Name, *Entry Type***
Remaining Carbs Percentage	***picker***	***percentage (100%)***	**Remaining Carbs Fraction**	***typed value***	***decimal (1)***	**Name, *Entry Type, Format***
Remaining Carbs Cap	***picker***	*decimal (90 g)*	Remaining Carbs Cap	***typed value***	*decimal (90)*	***Entry Type***
Noisy CGM Target Increase	***picker***	***percentage (130%)***	**Noisy CGM Target Multiplier**	***typed value***	***decimal (1.3)***	**Name, *Entry Type, Format***

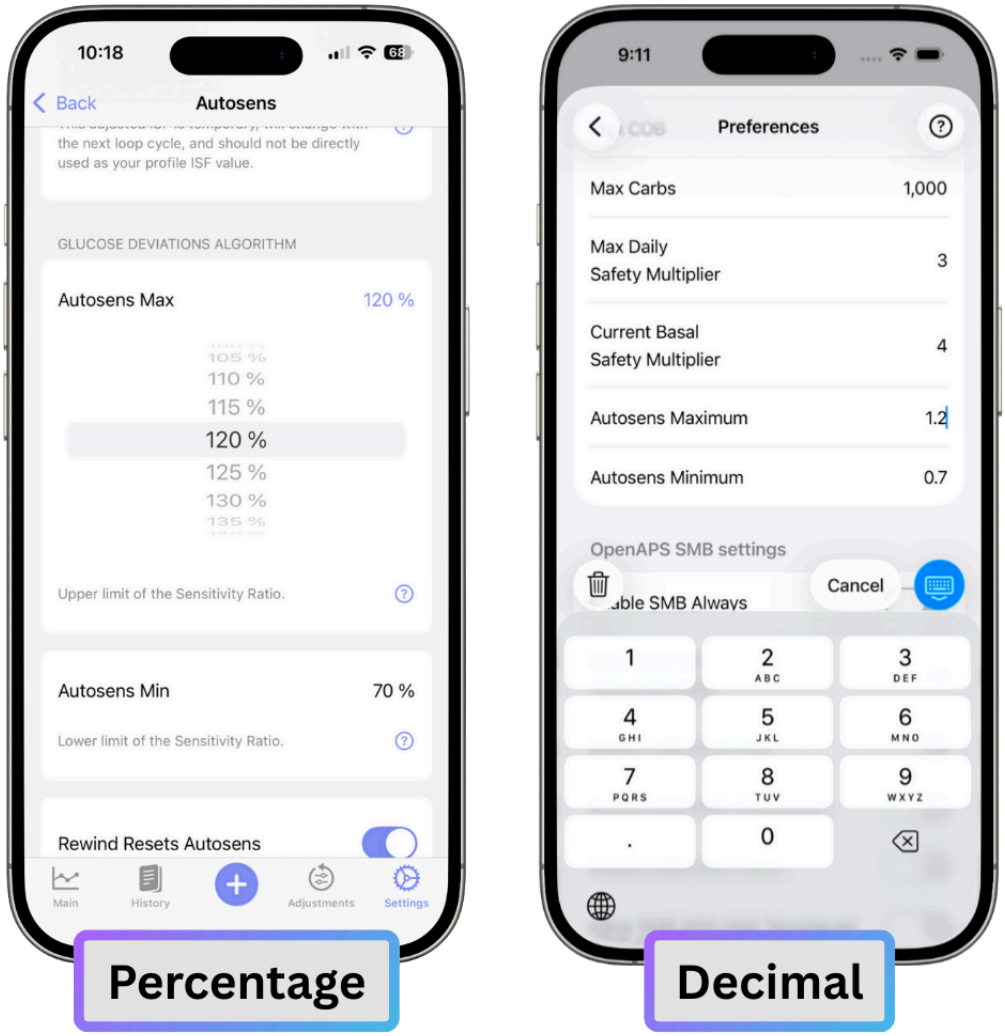
What are the reasons for these changes?

Entry Type: Picker vs Typed Entry



- A typed entry can result in mistyping a value and could result in unintended consequences within the algorithm.
- A picker ensures the setting is within the guardrails of the algorithm. Both Trio 0.2 and 0.5 (or higher) have guardrails, however when typed in 0.2, those values were ignored and the closest valid number was used. !!! example If you set your DIA to 2, Trio 0.2 would use a DIA of 5 without notifying you.

Format: Percentage vs Decimal



- Percentages are easier to comprehend than decimals when trying to make decisions on your settings adjustments.

Name Changes

- Some settings names in Trio 0.2 were the labels used in the Trio code and were really only clear to developers. This often times made those settings difficult to understand.
- Certain settings names in Trio 0.2 were unclear. We changed those names to make them easier to understand.
- In addition to updating names to be easier to understand, we also added clearer settings explanations in the app for every setting.

Coming from iAPS

What to Expect After Install

- **Onboarding Wizard**
  Trio will walk you through the initial setup of the app. Learn more in the New User Setup Guide.
- **User Interface**
  You can find more about the Trio User Interface in this walk-through of the app.
- **Remote Commands & iOS Shortcuts**
  Trigger actions like carb entry, bolus, overrides, or temporary targets remotely or through shortcut automation.
  - **Use Loop Follow v4.0 or Later for remote commands**
    More info at LoopFollowDocs
- **Bolus Calculator**
  The Bolus Calculator in Trio offers simple dosing with clear breakdowns and safety logic.
- **In-App Statistics**
  There are multiple statistics and graphs available in the Trio app. Click the link to learn more about the Statistics available in the app.
- **Important Re: Autotune**
  Autotune has not performed well with the addition of Dynamic ISF and many users using multiple ISF and CR in Therapy settings. For this reason, we have removed it until an improved Autotune-like feature is built.
- **No Dynamic Carb Ratio**
  There is no evidence to support that your carb ratio varies as ISF does with increasing or decreasing glucose levels. For this reason, Dynamic CR has not been implemented in Trio 0.5+.

What to Expect in Your First 3 Hours

Here are some things to keep in mind after moving from iAPS to Trio v0.5+ in the first 3 hours:

- **Dynamic ISF**
  Dynamic ISF is disabled for the first **7 days**.

  > ⚠ **Dynamic ISF & Closed Loop**
  >
  > Trio must be in **Closed Loop** for 7 days in order to enable Dynamic ISF.

- **Autosens**
  Autosens will require **8–24 hours** of data before it starts making adjustments. *You may need to enter carbs and bolus during this time if you aren't already doing so.*

> **Initial Limitation with Autosens**
>
> There's a long standing issue with Autosens where it will likely be stuck at 100% for the first 24 hours when there are carb entries or SMBs.
>
> - When carb entries or SMBs **are not** present, Autosens will kick in after 8 hours.
> - When carb entries or SMBs **are** present, Autosens will kick in after 24 hours.
>
> Whether or not to enable SMBs or enter carbs is a personal choice based on what you prioritize in your first 24 hours.

- **Onboarding Wizard**
  Take this time to go through the Onboarding Wizard. Reference the New User Setup Guide or ask questions on Facebook or Discord if you have any questions.
- **Test Settings**
  This is a great time to test your settings so you can start fresh on solid footing.
- **New User Interface**
  Use this time to learn the New User Interface.

> **Do not use a pump or cgm simulator**
>
> - Doing so will result in false data being stored and you will have to delete the app with all the data and reinstall before you can use it on a live pump.
> - This will also reset your 7-day waiting period for Dynamic ISF.
> - More information on simulator use

What to Expect After 24 Hours

A full day on Trio! Congratulations! Here's what to look for:

- **Autosens**
  Autosens now has enough data to make adjustments.
- **Review & Test Settings**
  This is a good time to review your last 24 hours of results to see if your core settings performed as expected. If you had unexpected challenges with your settings, look at how to test them over the next 7 days:
  - Basal Testing
  - ISF Testing
  - Carb Ratio Testing
- **Review Dynamic ISF Documentation**
  Review the documentation on Using Dynamic ISF in preparation for Day 7.

What to Expect After 7 Days

You've completed a week on Trio! Here's what you should see:

- **Dynamic ISF**
  Dynamic ISF now has enough data. You now have the option to enable it.

  This is a great time to refresh Using Dynamic ISFand check your settings for Logarithmic Dynamic ISF or Sigmoid Dynamic ISF.

  > **Can't Enable Dynamic ISF?**
  >
  > If Dynamic ISF does not give you the option to enable, you may have experienced one or more of the following:
  >
  > - You were not in closed loop for the full 7 days.
  >   - *Trio needs 7 days of closed loop to safely enable Dynamic ISF*
  > - You had significant loss of connection with your CGM and/or pump.
  >   - *Check your last week of Looping Statistics in the Stats tab in the Trio app. You need both **7 days** and an **85% success rate** to enable Dynamic ISF.*
  > - You enabled a significant number of manual Temp Basals with long run times.
  >   - *When a manual Temp Basal is set, Trio is unable to complete a loop cycle for the duration of the temp basal. This will cause a reduction in your success rate. If you do not have **85% success rate** for **7 days**, you cannot enable Dynamic ISF.*

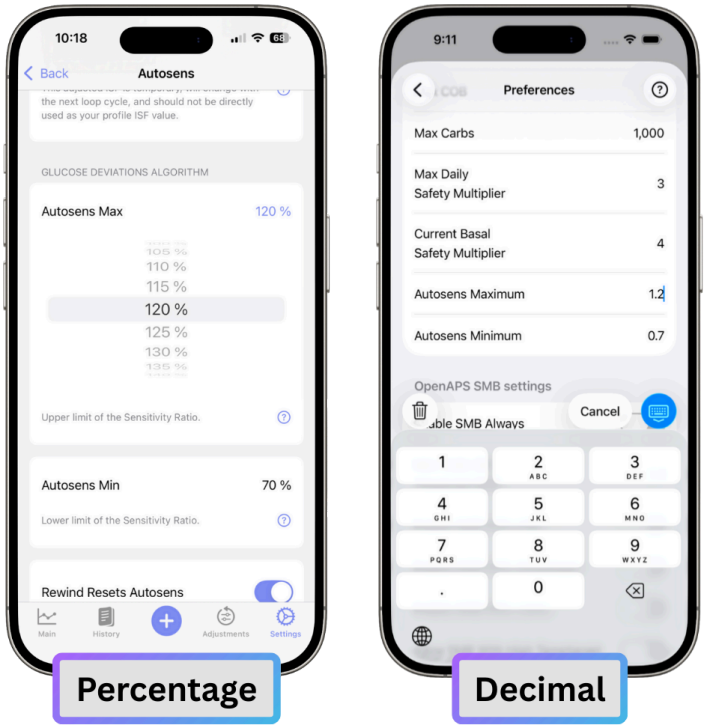
- **Dynamic CR**
  Dynamic CR has been removed due to no known scientific validity to CR changing with an increase or decrease in glucose.

Convert iAPS Settings to Trio Settings

When you move your settings from iAPS to Trio, some of them might look different even if they have the same or similar names. That’s because we some of the numbers went from decimals to percentages to make them easier to understand.

Example

In iAPS you might set "Autosens Maximum" to 1.2. In Trio, we show "Autosens Max" as 120%, which means the same thing, but is easier to understand.



Tip

- To convert a value from a decimal to a percentage, multiply it by 100.
- To convert from a percentage to a decimal, divide by 100.

Some setting names have also changed. We did this to make things less confusing. To help you out, we added definitions right in the app. Just tap the question mark icon next to any setting to see what it means. Healthcare Professionals and users can also find those settings and explanations in this section of the docs.

In iAPS, some settings had limits (called guardrails) that weren't always visible, and others had no limits at all. In Trio, we've made those guardrails easier to see and added a few more where needed, to help prevent settings that could lead to unsafe or unexpected outcomes.

The charts below will help you see which setting names or formats have changed, and where to find them in both iAPS and Trio.

Therapy Settings

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS

Glucose Targets	decimal (100 mg/dL | 5.5 mmol/L)	Settings → Therapy → Glucose Targets	**Target Glucose**	decimal (100 mg/dL | 5.5 mmol/L)	Settings → Target Glucose
Basal Rates	decimal (1.0 U/hr)	Settings → Therapy → Basal Rates	**Basal Profile**	decimal (1.0 U/hr)	Settings → Basal Profile
Carb Ratios	decimal (10 g/U)	Settings → Therapy → Carb Ratios	**Carb Ratios**	decimal (10 g/U)	Settings → Carb Ratios
Insulin Sensitivities	decimal (54 mg/dL/U | 3.0 mmol/L/U)	Settings → Therapy → Insulin Sensitivities	**Insulin Sensitivities**	decimal (54 mg/dL/U | 3.0 mmol/L/U)	Settings → Insulin Sensitivities

Units and Limits

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Glucose Units	selection (mg/dL or mmol/L)	Settings → Therapy → Units and Limits	**Glucose Units**	selection (mg/dL or mmol/L)	Settings → OpenAPS → Glucose units
Maximum Insulin On Board (IOB)	decimal (2 U)	Settings → Therapy → Units and Limits → Maximum Insulin On Board (IOB)	**Max IOB**	decimal (2)	Settings → OpenAPS → Max IOB
Maximum Bolus	decimal (10 U)	Settings → Therapy → Units and Limits → Maximum Bolus	**Max Bolus**	decimal (10)	Settings → Pump Settings → Max Bolus
Maximum Basal Rate	decimal (2 U/hr)	Settings → Therapy → Units and Limits → Maximum Basal Rate	**Max Basal**	decimal (2)	Settings → Pump Settings → Max Basal
Maximum Carbs on Board (COB)	decimal (120 g)	Settings → Therapy → Units and Limits → Maximum Carbs on Board (COB)	**Max COB**	decimal (120)	Settings → OpenAPS → Max COB

Minimum Safety Threshold	decimal (60 mg/dL)	Settings → Therapy → Units and Limits → Minimum Safety Threshold	**Threshold Setting**	decimal (65 mg/dL)	Settings → Dynamic ISF → Threshold Setting

Algorithm Settings

Autosens

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Autosens Min	percentage (70%)	Settings → Algorithm → Autosens → Autosens Min	**Autosens Minimum**	decimal (0.7)	Settings → OpenAPS → Autosens Minimum
Autosens Max	percentage (120%)	Settings → Algorithm → Autosens → Autosens Max	**Autosens Maximum**	decimal (1.2)	Settings → OpenAPS → Autosens Maximum
Rewind Resets Autosens	toggle (On/Off)	Settings → Algorithm → Autosens → Autosens Max	**Rewind Resets Autosens**	toggle (On/Off)	Settings → OpenAPS → Rewind Resets Autosens

SMB (Super Micro Bolus)

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Enable SMB Always	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always	**Enable SMB Always**	toggle (On/Off)	Settings → OpenAPS → Enable SMB Always
Enable SMB With COB	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) →			

		Enable SMB With COB	**Enable SMB With COB**	toggle (On/Off)	Settings → OpenAPS → Enable SMB With COB
Enable SMB with Temptarget	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB With TempTarget	**Enable SMB With COB**	toggle (On/Off)	Settings → OpenAPS → Enable SMB with Temptarget
Enable SMB After Carbs	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB After Carbs	**Enable SMB After Carbs**	toggle (On/Off)	Settings → OpenAPS → Enable SMB After Carbs
Enable SMB With High Glucose	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB With High Glucose	**Enable SMB With High BG**	toggle (On/Off)	Settings → OpenAPS → Enable SMB With High BG
High Glucose Target	decimal (110 mg/dL / 6.1 mmol/L)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB With High Glucose → High Glucose Target	**...When Blood Glucose is Above:**	decimal (110 mg/dL / 6.1 mmol/L)	Settings → OpenAPS → ...When Blood Glucose is Above:
Allow SMB with High Temp Target	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Allow SMB With High Temptarget	**Allow SMB With High Temptarget**	toggle (On/Off)	Settings → OpenAPS → Allow SMB With High Temptarget
Enable UAM	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable UAM	**Enable UAM**	toggle (On/Off)	Settings → OpenAPS → Enable UAM
Max SMB Basal Minutes	decimal (30 min)	Settings → Algorithm → Super Micro Bolus (SMB) → Max SMB Basal Minutes	**Max SMB Basal Minutes**	decimal (30)	Settings → OpenAPS → Max SMB Basal Minutes

Max UAM Basal Minutes	decimal (30 min)	Settings → Algorithm → Super Micro Bolus (SMB) → Max UAM Basal Minutes	**Max UAM SMB Basal Minutes**	decimal (30)	Settings → OpenAPS → Max UAM SMB Basal Minutes
Max Allowed Glucose Rise for SMB	percentage (20%)	Settings → Algorithm → Super Micro Bolus (SMB) → Max Allowed Glucose Rise for SMB	**Max Delta-BG Threshold SMB**	decimal (0.2)	Settings → OpenAPS → Max Delta-BG Threshold SMB

Dynamic Settings

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Dynamic ISF	selection (Disabled / Logarithmic / Sigmoid)	Settings → Algorithm → Dynamic Settings → Dynamic ISF	**Activate Dynamic Sensitivity (ISF)**	toggle (On/Off)	Settings → Dynamic ISF → Activate Dynamic Sensitivity (ISF)
Adjustment Factor	percentage (80%)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Logarithmic) → Adjustment Factor	**Adjustment Factor**	decimal (0.8)	Settings → Dynamic ISF → Activate Dynamic Sensitivity (ISF) (ON) → Use Sigmoid Function (OFF) → Adjustment Factor
Sigmoid Adjustment Factor	percentage (50%)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Sigmoid) → Sigmoid Adjustment Factor	**Adjustment Factor**	decimal (0.5)	Settings → Dynamic ISF → Activate Dynamic Sensitivity (ISF) (ON) → Use Sigmoid Function (ON) → Adjustment Factor

Weighted Average of TDD	percentage (35%)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Logarithmic/Sigmoid) → Weighted Average of TDD	**Weighted Average of TDD. Weight of past 24 hours:**	decimal (0.35)	Settings → Dynamic ISF → Activate Dynamic Sensitivity (ISF) (ON) → Weighted Average of TDD. Weight of past 24 hours:
Adjust Basal	toggle (On/Off)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Logarithmic/Sigmoid) → Adjust Basal	***No Equivalent Setting***	N/A	N/A

Target Behavior

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
High Temp Target Raises Sensitivity	toggle (On/Off)	Settings → Algorithm → Target Behavior → High Temp Target Raises Sensitivity	**High Temptarget Raises Sensitivity**	toggle (On/Off)	Settings → OpenAPS → High Temptarget Raises Sensitivity
Low Temp Target Lowers Sensitivity	toggle (On/Off)	Settings → Algorithm → Target Behavior → Low Temp Target Lowers Sensitivity	**Low Temptarget Lowers Sensitivity**	toggle (On/Off)	Settings → OpenAPS → Low Temptarget Lowers Sensitivity
Sensitivity Raises Target	toggle (On/Off)	Settings → Algorithm → Target Behavior → Sensitivity Raises Target	**Sensitivity Raises Target**	toggle (On/Off)	Settings → OpenAPS → Sensitivity Raises Target
Resistance Lowers Target	toggle (On/Off)	Settings → Algorithm → Target Behavior → Resistance Lowers Target	**Resistance Lowers Target**	toggle (On/Off)	Settings → OpenAPS → Resistance Lowers Target

Half Basal Exercise Target	decimal (160 mg/dL | 8.9 mmol/L)	Settings → Algorithm → Target Behavior → Half Basal Exercise Target	**Half Basal Exercise Target**	decimal (160 mg/dL | 8.9 mmol/L)	Settings → OpenAPS → Half Basal Exercise Target

Additionals

> **Warning**
>
> The settings in this section typically do not require any modifications.
> Do not alter them without a solid understanding of what you are changing and the full impact it will have on the algorithm.

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Max Daily Safety Multiplier	percentage (300%)	Settings → Algorithm → Additionals → Max Daily Safety Multiplier	**Max Daily Safety Multiplier**	decimal (3)	Settings → OpenAPS → Max Daily Safety Multiplier
Current Basal Safety Multiplier	percentage (400%)	Settings → Algorithm → Additionals → Current Basal Safety Multiplier	**Current Basal Safety Multiplier**	decimal (4)	Settings → OpenAPS → Current Basal Safety Multiplier
Duration of Insulin Action	decimal (10 hr)	Settings → Algorithm → Additionals → Duration of Insulin Action	**Duration of Insulin Action**	decimal (10)	Settings → Pump Settings → Duration of Insulin Action
Use Custom Peak Time	toggle (On/Off)	Settings → Algorithm → Additionals → Use Custom Peak Time	**Use Custom Peak Time**	toggle (On/Off)	Settings → OpenAPS → Use Custom Peak Time

Insulin Peak Time	decimal (65 min)	Settings → Algorithm → Additionals → Use Custom Peak Time (ON) → Insulin Peak Time	**Insulin Peak Time**	decimal (65)	Settings → OpenAPS → Insulin Peak Time
Skip Neutral Temps	toggle (On/Off)	Settings → Algorithm → Additionals → Skip Neutral Temps	**Skip Neutral Temps**	toggle (On/Off)	Settings → OpenAPS → Skip Neutral Temps
Unsuspend If No Temp	toggle (On/Off)	Settings → Algorithm → Additionals → Unsuspend If No Temp	**Unsuspend If No Temp**	toggle (On/Off)	Settings → OpenAPS → Unsuspend If No Temp
SMB Delivery Ratio	percentage (50%)	Settings → Algorithm → Additionals → SMB Delivery Ratio	**SMB DeliveryRatio**	decimal (0.5)	Settings → OpenAPS → SMB DeliveryRatio
SMB Interval	decimal (3 min)	Settings → Algorithm → Additionals → SMB Interval	**SMB Interval**	decimal (3)	Settings → OpenAPS → SMB Interval
Min 5m Carb Impact	decimal (8 mg/dL)	Settings → Algorithm → Additionals → Min 5m Carb Impact	**Min 5m Carbimpact**	decimal (8)	Settings → OpenAPS → Min 5m Carbimpact
Remaining Carbs Percentage	percentage (100%)	Settings → Algorithm → Additionals → Remaining Carbs Percentage	**Remaining Carbs Fraction**	decimal (1)	Settings → OpenAPS → Remaining Carbs Fraction
Remaining Carbs Cap	decimal (90g)	Settings → Algorithm → Additionals → Remaining Carbs Cap	**Remaining Carbs Cap**	decimal (90)	Settings → OpenAPS → Remaining Carbs Cap
Noisy CGM Target Increase	percentage (130%)	Settings → Algorithm → Additionals → Noisy CGM Target Increase	**Noisy CGM Target Multiplier**	decimal (1.3)	Settings → OpenAPS → Noisy CGM Target Multiplier

Features

Bolus Calculator

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Display Meal Presets	toggle (On/Off)	Settings → Features → Bolus Calculator → Display Meal Presets	***No Equivalent Setting***	N/A	N/A
Recommended Bolus Percentage	percentage (80%)	Settings → Features → Bolus Calculator → Recommended Bolus Percentage	**Override With a Factor Of**	decimal (0.8)	Settings → Bolus Calculator → Override With A Factor Of
Enable Reduced Bolus	toggle (On/Off)	Settings → Features → Bolus Calculator → Enable Reduced Bolus	**Apply factor for fatty meals**	toggle (On/Off)	Settings → Bolus Calculator → Apply factor for fatty meals
Reduced Bolus Percentage	percentage (30%)	Settings → Features → Bolus Calculator → Enable Reduced Bolus (ON) → Reduced Bolus Percentage	**Override With a Factor Of**	decimal (0.7)	Settings → Bolus Calculator → Apply factor for fatty meals (ON) → Override With A Factor Of
Enable Super Bolus	toggle (On/Off)	Settings → Features → Bolus Calculator → Enable Super Bolus	***No Equivalent Setting***	N/A	N/A
Super Bolus Percentage	percentage (100%)	Settings → Features → Bolus Calculator → Enable Super Bolus (ON) → Super Bolus Percentage	***No Equivalent Setting***	N/A	N/A

Very Low Glucose Warning	toggle (On/Off)	Settings → Features → Bolus Calculator → Very Low Glucose Warning	***No Equivalent Setting***	N/A	N/A

Meal Settings

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Max Carbs	decimal (250 g)	Settings → Meal Settings → Max Carbs	**Max Carbs**	decimal (1,000)	Settings → OpenAPS → Max Carbs
Max Protein	decimal (250 g)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Max Protein	***No Equivalent Setting***	N/A	N/A
Max Fat	decimal (250 g)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Max Fat	***No Equivalent Setting***	N/A	N/A
Max Meal Absorption Time	decimal (6 hr)	Settings → Meal Settings → Max Meal Absorption Time	***No Equivalent Setting***	N/A	N/A
Enable Fat and Protein Entries	toggle (On/Off)	Settings → Meal Settings → Enable Fat and Protein Entries	**Display and allow Fat and Protein entries**	toggle (On/Off)	Settings → UI/UX → Display and allow Fat and Protein entries
Fat and Protein Delay	decimal (60 min)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Fat and Protein Delay	**Delay in Minutes**	decimal (60)	Settings → Fat and Protein Conversion → Delay In Minutes
Maximum Duration	decimal (8 hr)	Settings → Meal Settings → Enable Fat and Protein			

		Entries (ON) → Maximum Duration	**Maximum Duration In Hours**	decimal (8)	Settings → Fat and Protein Conversion → Maximum Duration in Hours
Spread Interval	decimal (30 min)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Spread Interval	**Interval In Minutes**	decimal (30)	Settings → Fat and Protein Conversion → Interval In Minutes
Fat and Protein Percentage	percentage (50%)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Fat and Protein Percentage	**Override With A Factor Of**	decimal (0.5)	Settings → Fat and Protein Conversion → Override With A Factor Of

Other Notable Settings

Trio Name	Setting Format (example)	Location in Trio	iAPS Name	Setting Format (example)	Location in iAPS
Temp Target: Sensitivity Adjustment	selection/percentage (Standard/Custom) (100%)	Adjustments → Add Temp Target → Sensitivity Adjustment	**Temp Target: Experimental**	toggle/percentage (On/Off) (100%)	Temp Target → Experimental (ON)

Coming from Loop

What to Expect After Install

- **Onboarding Wizard**
  Trio will walk you through the initial setup of the app. Learn more in the New User Setup Guide.
- **User Interface**
  You can find more about the Trio User Interface in this walk-through of the app.
- **Remote Commands & iOS Shortcuts**
  Trigger actions like carb entry, bolus, overrides, or temporary targets remotely or through shortcut automation.
  - **Use Loop Follow v4.0 or Later for remote commands**
    Loop Follow is the companion app for Trio. More info at LoopFollowDocs.

> **Note**
>
> Loop Caregiver will not work with Trio

- **Bolus Calculator**
  The Bolus Calculator in Trio offers simple dosing with clear breakdowns and safety logic.
- **In-App Statistics**
  There are multiple statistics and graphs available in the Trio app. Click the link to learn more about the Statistics available in the app.

What to Expect in Your First 3 Hours

Here are some things to keep in mind after moving from Loop to Trio v0.5+ in the first 3 hours:

- **Onboarding Wizard**
  Take this time to go through the Onboarding Wizard. Reference the New User Setup Guide or ask questions on Facebook or Discord if you have any questions.
- **Test Settings**
  Your Loop settings may be a great starting point for Trio, but often times they need some adjustment. This is a great time to test your settings so you can start fresh on solid footing.
- **User Interface**
  Use this time to learn the User Interface.
- **Dynamic ISF**
  Dynamic ISF is disabled for the first **7 days**.

> **Dynamic ISF & Closed Loop**
>
> Trio must be in **Closed Loop** for 7 days in order to enable Dynamic ISF.

- **Autosens**
  Autosens will require **8–24 hours** of data before it starts making adjustments. *You may need to enter carbs and bolus during this time if you aren't already doing so.*

  > **Initial Limitation with Autosens**
  >
  > There's a long standing issue with Autosens where it will likely be stuck at 100% for the first 24 hours when there are carb entries or SMBs.
  >
  > - When carb entries or SMBs **are not** present, Autosens will kick in after 8 hours.
  > - When carb entries or SMBs **are** present, Autosens will kick in after 24 hours.
  >
  > Whether or not to enable SMBs or enter carbs is a personal choice based on what you prioritize in your first 24 hours.

- **Nightscout Changes**
  - You will need to update your Nightscout page to use Trio plugins rather than Loop plugins.
  - Replace `loop` with `openaps` in the variables `ENABLE`, `SHOW_PLUGINS`, and `SHOW_FORECAST`.
  - If you pay for Nightscout through a provider and do not know how to update your variables, reference your provider's documentation or reach out to them for assistance.

> **Do not use a pump or cgm simulator**
>
> - Doing so will result in false data being stored and you will have to delete the app with all the data and reinstall before you can use it on a live pump.
> - This will also reset your 7-day waiting period for Dynamic ISF.
> - More information on simulator use

What to Expect After 24 Hours

A full day on Trio! Congratulations! Here's what to look for:

- **Autosens**
  Autosens now has enough data to make adjustments.
- **Review & Test Settings**
  This is a good time to review your last 24 hours of results to see if your core settings performed as expected. If you had unexpected challenges with your settings, look at how to test them over the next 7 days:
  - Basal Testing
  - ISF Testing
  - Carb Ratio Testing
- **Review Dynamic ISF Documentation**
  Review the documentation on Using Dynamic ISF in preparation for Day 7.

- **Meal Time Struggles**
  Sometimes, users coming from Loop will notice stability during fasting periods, but meal times are more challenging than they were with Loop. This is due to a shift in algorithms. Loop is more carb-centric, causing users to place a heavy reliance upon their Carb Ratio over their ISF value. Trio is more reliant upon your ISF value and as such you may find you need to adjust your ISF and possibly your CR when you switch from Loop to Trio.

What to Expect After 7 Days

You've completed a week on Trio! Here's what you should see:

- **Dynamic ISF**
  Dynamic ISF now has enough data. You now have the option to enable it.

  This is a great time to refresh Using Dynamic ISFand check your settings for Logarithmic Dynamic ISF or Sigmoid Dynamic ISF.

  > **Can't Enable Dynamic ISF?**
  >
  > If Dynamic ISF does not give you the option to enable, you may have experienced one or more of the following:
  >
  > - You were not in closed loop for the full 7 days.
  >   - *Trio needs 7 days of closed loop to safely enable Dynamic ISF*
  > - You had significant loss of connection with your CGM and/or pump.
  >   - *Check your last week of Looping Statistics in the Stats tab in the Trio app. You need both **7 days** and an **85% success rate** to enable Dynamic ISF.*
  > - You enabled a significant number of manual Temp Basals with long run times.
  >   - *When a manual Temp Basal is set, Trio is unable to complete a loop cycle for the duration of the temp basal. This will cause a reduction in your success rate. If you do not have **85% success rate** for **7 days**, you cannot enable Dynamic ISF.*

Convert Loop Settings to Trio Settings

When you start on Trio, you will notice there are significantly more settings than in Loop. A few can be manually added. Below are the settings that are in both Loop and Trio as well as where to locate them in each app.

Therapy Settings

Trio Name	Setting Format (example)	Location in Trio	Loop Name	Setting Format (example)	Location in Loop

Glucose Targets	schedule (100 mg/dL — 12:00 AM)	Settings → Therapy → Glucose Targets	**Correction Range**	decimal range (100 – 115 mg/dL)	Settings → Therapy Settings → Correction Range
Basal Rates	schedule (1.0 U/hr — 12:00 AM)	Settings → Therapy → Basal Rates	**Basal Rates**	schedule (12:00 AM — 1.0 U/hr)	Settings → Therapy Settings → Basal Rates
Carb Ratios	schedule (15 g/U — 12:00 AM)	Settings → Therapy → Carb Ratios	**Carb Ratios**	schedule (12:00AM — 15 g/U)	Setting → Therapy Settings → Carb Ratios
Insulin Sensitivities	schedule (50 mg/dL/U — 12:00 AM)	Settings → Therapy → Insulin Sensitivities	**Insulin Sensitivities**	schedule (12:00 AM — 50 mg/dL/U)	Settings → Therapy Settings → Insulin Sensitivities

Units and Limits

Trio Name	Setting Format (example)	Location in Trio	Loop Name	Setting Format (example)	Location in Loop
Glucose Units	selection (mg/dL or mmol/L)	Settings → Therapy → Units and Limits	**Apple Health: Blood Glucose Unit**	selection (mg/dL or mmol/L)	Health App → Browse → Vitals → Blood Glucose → Unit
Maximum Bolus	decimal (10 U)	Settings → Therapy → Units and Limits → Maximum Bolus	**Maximum Bolus**	decimal (8 U)	Settings → Therapy Settings → Maximum Bolus
Maximum Basal Rate	decimal (2 U/hr)	Settings → Therapy → Units and Limits → Maximum Basal Rate	**Maximum Basal Rate**	decimal (2 U/hr)	Settings → Therapy Settings → Maximum Basal Rate
Minimum Safety Threshold	decimal (60 mg/dL)	Settings → Therapy → Units and Limits → Minimum Safety Threshold	**Glucose Safety Limit**	decimal (80 mg/dL)	Settings → Therapy Settings → Glucose Safety Limit

Algorithm Settings

SMB (Super Micro Bolus)

Trio Name	Setting Format (example)	Location in Trio	Loop Name	Setting Format (example)	Location in Loop
Enable SMB Always	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always	**Automatic Bolus**	selection (Temp Basal Only/Automatic Bolus)	Settings → Dosing Strategy → Automatic Bolus

Coming from AAPS

What to Expect After Install

- **Onboarding Wizard**
  Trio will walk you through the initial setup of the app. Learn more in the New User Setup Guide.
- **User Interface**
  You can find more about the Trio User Interface in this walk-through of the app.
- **Remote Commands & iOS Shortcuts**
  Trigger actions like carb entry, bolus, overrides, or temporary targets remotely or through shortcut automation.
  - **Use Loop Follow v4.0 or Later for remote commands**
    More info at LoopFollowDocs
- **Bolus Calculator**
  The Bolus Calculator in Trio offers simple dosing with clear breakdowns and safety logic.
- **In-App Statistics**
  There are multiple statistics and graphs available in the Trio app. Click the link to learn more about the Statistics available in the app.
- **Important Re: Autotune**
  Autotune has not performed as it was intended for years with the addition of Dynamic ISF and the change from a single ISF and CR in Therapy settings. For this reason, we have removed it until it can be rewritten to work with Trio or a new Autotune-like feature can be built.

What to Expect in Your First 3 Hours

Here are some things to keep in mind after moving from AndroidAPS to Trio 0.5+ in the first 3 hours:

- **Onboarding Wizard**
  Take this time to go through the Onboarding Wizard. Reference the New User Setup Guide or ask questions on Facebook or Discord if you have any questions.
- **Test Settings**
  This is a great time to test your settings so you can start fresh on solid footing.
- **User Interface**
  Use this time to learn the User Interface.
- **Dynamic ISF**
  - Dynamic ISF is disabled for the first **7 days**.
  - If you were using Dynamic ISF in AAPS, you will have to wait a week before enabling dynamic ISF in Trio.
  - Data import from AAPS/Nightscout is not available.
- **Settings** When migrating from AAPS to Trio, many users will achieve similar results by using similar settings. The core algorithm is the same between AndroidAPS and Trio.

> **Dynamic ISF & Closed Loop**
>
> Trio must be in **Closed Loop** for 7 days in order to enable Dynamic ISF.

- **Autosens**
  Autosens will require **8–24 hours** of data before it starts making adjustments. *You may need to enter carbs and bolus during this time if you aren't already doing so.*

> **Initial Limitation with Autosens**
>
> There's a long standing issue with Autosens where it will likely be stuck at 1 for the first 24 hours when there are carb entries or SMBs.
>
> - When carb entries or SMBs **are not** present, Autosens will kick in after 8 hours.
> - When carb entries or SMBs **are** present, Autosens will kick in after 24 hours.
>
> Whether or not to enable SMBs or enter carbs is a personal choice based on what you prioritize in your first 24 hours.

> **Do not use a pump or cgm simulator**
>
> - Doing so will result in false data being stored and you will have to delete the app with all the data and reinstall before you can use it on a live pump.
> - This will also reset your 7-day waiting period for Dynamic ISF.
> - More information on simulator use

What to Expect After 24 Hours

A full day on Trio! Congratulations! Here's what to look for:

- **Autosens**
  Autosens now has enough data to make adjustments.
- **Review & Test Settings**
  This is a good time to review your last 24 hours of results to see if your core settings performed as expected. If you had unexpected challenges with your settings, look at how to test them over the next 7 days:
  - Basal Testing
  - ISF Testing
  - Carb Ratio Testing
- **Review Dynamic ISF Documentation**
  Review the documentation on Using Dynamic ISF in preparation for Day 7.

What to Expect After 7 Days

You've completed a week on Trio! Here's what you should see:

- **Dynamic ISF**
  - Dynamic ISF now has enough data. You now have the option to enable it.
  - The Logarithmic formula in Trio is the most similar to AAPS dynamic ISF. The Sigmoid formula behaves very differently (informative links below). Which one you use is up to you.
  - This is a great time to refresh Using Dynamic ISF and check your settings for Logarithmic Dynamic ISF or Sigmoid Dynamic ISF.

Can't Enable Dynamic ISF?

If Dynamic ISF does not give you the option to enable, you may have experienced one or more of the following:

- You were not in closed loop for the full 7 days.
  - *Trio needs 7 days of closed loop to safely enable Dynamic ISF*
- You had significant loss of connection with your CGM and/or pump.
  - *Check your last week of Looping Statistics in the Stats tab in the Trio app. You need both **7 days** and an **85% success rate** to enable Dynamic ISF.*
- You enabled a significant number of manual Temp Basals with long run times.
  - *When a manual Temp Basal is set, Trio is unable to complete a loop cycle for the duration of the temp basal. This will cause a reduction in your success rate. If you do not have **85% success rate** for **7 days**, you cannot enable Dynamic ISF.*

Convert AAPS Settings to Trio Settings

When you move your settings from AAPS to Trio, there are multiple settings that have similar names. We've created this chart to help you identify which settings are equivalent between the two apps and where to find them.

Some Settings May Need Conversion

- To convert a value from a decimal to a percentage, multiply it by 100.
- To convert from a percentage to a decimal, divide by 100.

Therapy Settings

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Glucose Targets	decimal (100 mg/dL | 5.5 mmol/L)	Settings → Therapy → Glucose Targets	**TARG**	decimal range (100-120 mg/dL)	LP → TARG
Basal Rates	decimal (1.0 U/hr)	Settings → Therapy → Basal Rates	**BAS**	decimal (1.0 U/hr)	LP → BAS
Carb Ratios	decimal (10 g/U)	Settings → Therapy → Carb Ratios	**IC**	decimal (10 g/U)	LP → IC
Insulin Sensitivities	decimal (54 mg/dL/U | 3.0 mmol/L/U)	Settings → Therapy → Insulin Sensitivities	**ISF**	decimal (54)	LP → ISF

Units and Limits

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Glucose Units	selection (mg/dL or mmol/L)	Settings → Therapy → Units and Limits	**Units**	selection (mg/dL or mmol/L)	Setup Wizard → Units
Maximum Insulin On Board (IOB)	decimal (2 U)	Settings → Therapy → Units and Limits → Maximum Insulin On Board (IOB)	**Maximum total IOB OpenAPS can't go over**	decimal (2)	Preferences → OpenAPS SMB → Maximum total IOB OpenAPS can't go over
Maximum Bolus	decimal (10 U)	Settings → Therapy → Units and Limits → Maximum Bolus	**Max allowed bolus**	decimal (10.0)	Preferences → Treatments safety → Max allowed bolus

Maximum Basal Rate	decimal (2 U/hr)	Settings → Therapy → Units and Limits → Maximum Basal Rate	**Max U/h a Temp Basal can be set to**	decimal (4.0)	Preferences → OpenAPS SMB
Maximum Carbs on Board (COB)	decimal (120 g)	Settings → Therapy → Units and Limits → Maximum Carbs on Board (COB)	**Max allowed carbs**	decimal (120)	Preferences → Treatments safety → Max allowed carbs
Minimum Safety Threshold	decimal (60 mg/dL)	Settings → Therapy → Units and Limits → Minimum Safety Threshold	**BG level below which low glucose suspend occurs**	decimal (65)	Preferences → OpenAPS SMB → Enable dynamic sensitivity (ON) → BG level below which low glucose suspend occurs

Algorithm Settings

Autosens

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Autosens Min	percentage (70%)	Settings → Algorithm → Autosens → Autosens Min	**Min autosens ratio**	decimal (0.7)	Preferences → Absorption Settings → Advanced Settings Preferences → Min autosens ratio
Autosens Max	percentage (120%)	Settings → Algorithm → Autosens → Autosens Max	**Max autosens ratio**	decimal (1.2)	Preferences → Absorption Settings → Advanced Settings → Max autosens ratio
Rewind Resets Autosens	toggle (On/Off)	Settings → Algorithm → Autosens → Autosens Max	**Log a Pump Site Change**	*Always On*	*Always On*

SMB (Super Micro Bolus)

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Enable SMB Always	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always	**Enable SMB always**	toggle (On/Off)	Preferences → Enable SMB → Enable SMB always
Enable SMB With COB	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB With COB	**Enable SMB with COB**	toggle (On/Off)	Preferences → Enable SMB → Enable SMB with COB
Enable SMB with Temptarget	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB With TempTarget	**Enable SMB with temp targets**	toggle (On/Off)	Preferences → Enable SMB → Enable SMB with temp targets
Enable SMB After Carbs	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB After Carbs	**Enable SMB after carbs**	toggle (On/Off)	Preferences → Enable SMB → Enable SMB after carbs
Enable SMB With High Glucose	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB Always(OFF) → Enable SMB With High Glucose	***No Equivalent Setting***	N/A	N/A
High Glucose Target	decimal (110 mg/dL / 6.1 mmol/L)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable SMB With High Glucose			

		→ High Glucose Target	***No Equivalent Setting***	N/A	N/A
Allow SMB with High Temp Target	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Allow SMB With High Temptarget	**Enable SMB with high temp targets**	toggle (On/Off)	Preferences → OpenAPS SMB → Enable SMB → Enable SMB with high temp targets
Enable UAM	toggle (On/Off)	Settings → Algorithm → Super Micro Bolus (SMB) → Enable UAM	**Enable UAM**	toggle (On/Off)	Preferences → OpenAPS SMB → Enable UAM
Max SMB Basal Minutes	decimal (30 min)	Settings → Algorithm → Super Micro Bolus (SMB) → Max SMB Basal Minutes	**Max minutes of basal to limit SMB to**	decimal (30)	Preferences → OpenAPS SMB → Enable SMB → Max minutes of basal to limit SMB to
Max UAM Basal Minutes	decimal (30 min)	Settings → Algorithm → Super Micro Bolus (SMB) → Max UAM Basal Minutes	**Max minutes of basal to limit SMB to for UAM**	decimal (30)	Preferences → OpenAPS SMB → Enable UAM → Max minutes of basal to limit SMB to for UAM
Max Allowed Glucose Rise for SMB	percentage (20%)	Settings → Algorithm → Super Micro Bolus (SMB) → Max Allowed Glucose Rise for SMB	***No Equivalent Setting***	N/A	N/A

Dynamic Settings

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Dynamic ISF	selection (Disabled / Logarithmic / Sigmoid)	Settings → Algorithm → Dynamic Settings → Dynamic ISF	**Enable dynamic sensitivity**	toggle (On/Off)	Preferences → OpenAPS SMB → Enable dynamic sensitivity

Adjustment Factor	percentage (80%)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Logarithmic) → Adjustment Factor	**DynamicISF Adjustment Factor %**	percentage (100)	Preferences → OpenAPS SMB → Enable dynamic sensitivity (ON) → DynamicISF Adjustment Factor %
Sigmoid Adjustment Factor	percentage (50%)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Sigmoid) → Sigmoid Adjustment Factor	***No Equivalent Setting***	N/A	N/A
Weighted Average of TDD	percentage (35%)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Logarithmic/Sigmoid) → Weighted Average of TDD	***No Equivalent Setting***	N/A	N/A
Adjust Basal	toggle (On/Off)	Settings → Algorithm → Dynamic Settings → Dynamic ISF (Logarithmic/Sigmoid) → Adjust Basal	**Enable TDD based sensitivity ratio for basal and glucose target modification**	toggle (On/Off)	Preferences → OpenAPS SMB → Enable dynamic sensitivity (ON) → Enable TDD based sensitivity ratio for basal and glucose target modification

Target Behavior

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
High Temp Target Raises Sensitivity	toggle (On/Off)	Settings → Algorithm → Target Behavior → High Temp Target Raises Sensitivity	**High temp-target raises sensitivity**	toggle (On/Off)	Preferences → OpenAPS SMB → High temp-target raises sensitivity

Low Temp Target Lowers Sensitivity	toggle (On/Off)	Settings → Algorithm → Target Behavior → Low Temp Target Lowers Sensitivity	**Low temp-target lowers sensitivity**	toggle (On/Off)	Preferences → OpenAPS SMB → Low temp-target lowers sensitivity
Sensitivity Raises Target	toggle (On/Off)	Settings → Algorithm → Target Behavior → Sensitivity Raises Target	**Sensitivity raises target**	toggle (On/Off)	Preferences → OpenAPS SMB → Sensitivity raises target
Resistance Lowers Target	toggle (On/Off)	Settings → Algorithm → Target Behavior → Resistance Lowers Target	**Resistance lowers target**	toggle (On/Off)	Preferences → OpenAPS SMB → Resistance lowers target
Half Basal Exercise Target	decimal (160 mg/dL | 8.9 mmol/L)	Settings → Algorithm → Target Behavior → Half Basal Exercise Target	**Activity Target Value**	decimal (160 mg/dL | 8.9 mmol/L)	Preferences → Overview Default → Temp-targets → Activity target value

Additionals

Warning

The settings in this section typically do not require any modifications.
Do not alter them without a solid understanding of what you are changing and the full impact it will have on the algorithm.

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Max Daily Safety Multiplier	percentage (300%)	Settings → Algorithm → Additionals → Max Daily Safety Multiplier	**Max daily safety multiplier**	decimal (3.0)	Preferences → OpenAPS SMB → Advanced Settings → Max daily safety mulitiplier

Current Basal Safety Multiplier	percentage (400%)	Settings → Algorithm → Additionals → Current Basal Safety Multiplier	**Current basal safety multiplier**	decimal (4.0)	Preferences → OpenAPS SMB → Advanced Settings → Current basal safety multiplier
Duration of Insulin Action	decimal (10 hr)	Settings → Algorithm → Additionals → Duration of Insulin Action	**DIA**	decimal (10)	LP → DIA
Use Custom Peak Time	toggle (On/Off)	Settings → Algorithm → Additionals → Use Custom Peak Time	**Free-Peak Oref**	selection (Free-Peak Oref)	Config Builder → Profile → Insulin → Free-Peak Oref
Insulin Peak Time	decimal (65 min)	Settings → Algorithm → Additionals → Use Custom Peak Time (ON) → Insulin Peak Time	**Free-Peak Oref**	decimal (45)	Config Builder → Profile → Insulin → Free-Peak Oref → Peak Time
Skip Neutral Temps	toggle (On/Off)	Settings → Algorithm → Additionals → Skip Neutral Temps	***No Equivalent Setting***	N/A	N/A
Unsuspend If No Temp	toggle (On/Off)	Settings → Algorithm → Additionals → Unsuspend If No Temp	***No Equivalent Setting***	N/A	N/A
SMB Delivery Ratio	percentage (50%)	Settings → Algorithm → Additionals → SMB			

		Delivery Ratio	***No Equivalent Setting***	Always 50%	Always 50%
SMB Interval	decimal (3 min)	Settings → Algorithm → Additionals → SMB Interval	**How frequently SMBs will be given in min**	decimal (3)	Preferences → Enable SMB → How frequently SMBs will be given in min
Min 5m Carb Impact	decimal (8 mg/dL)	Settings → Algorithm → Additionals → Min 5m Carb Impact	**min_5m_carbimpact**	decimal (8)	Preferences → Absorption settings → Advanced → min_5m_carbimpact
Remaining Carbs Percentage	percentage (100%)	Settings → Algorithm → Additionals → Remaining Carbs Percentage	***No Equivalent Setting***	N/A	N/A
Remaining Carbs Cap	decimal (90g)	Settings → Algorithm → Additionals → Remaining Carbs Cap	***No Equivalent Setting***	N/A	N/A
Noisy CGM Target Increase	percentage (130%)	Settings → Algorithm → Additionals → Noisy CGM Target Increase	***No Equivalent Setting***	N/A	N/A

Features

Bolus Calculator

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Display Meal Presets	toggle (On/Off)	Settings → Features → Bolus Calculator →			

		Display Meal Presets	**Quick Wizard**	N/A	Preferences → Overview → QuickWizard Settings → Add / edit Quick Wizard
Recommended Bolus Percentage	percentage (80%)	Settings → Features → Bolus Calculator → Recommended Bolus Percentage	**Deliver this part of bolus wizard result**	percentage (100)	Preferences → Overview → Deliver this part of bolus wizard result
Enable Reduced Bolus	toggle (On/Off)	Settings → Features → Bolus Calculator → Enable Reduced Bolus	***No Equivalent Setting***	N/A	N/A
Reduced Bolus Percentage	percentage (30%)	Settings → Features → Bolus Calculator → Enable Reduced Bolus (ON) → Reduced Bolus Percentage	***No Equivalent Setting***	N/A	N/A
Enable Super Bolus	toggle (On/Off)	Settings → Features → Bolus Calculator → Enable Super Bolus	**Enable super bolus functionality in wizard**	toggle (On/Off)	Preferences → Overview → Advanced → Enable super bolus functionality in wizard
Super Bolus Percentage	percentage (100%)	Settings → Features → Bolus Calculator → Enable Super Bolus (ON) → Super Bolus Percentage	***No Equivalent Setting***	N/A	N/A
Very Low Glucose Warning	toggle (On/Off)	Settings → Features → Bolus Calculator → Very Low Glucose Warning	***No Equivalent Setting***	N/A	N/A

Meal Settings

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Max Carbs	decimal (250 g)	Settings → Meal Settings → Max Carbs	**Max allowed carbs**	decimal (100)	Preferences → Treatments safety → Max allowed carbs
Max Protein	decimal (250 g)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Max Protein	***No Equivalent Setting***	N/A	N/A
Max Fat	decimal (250 g)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Max Fat	***No Equivalent Setting***	N/A	N/A
Max Meal Absorption Time	decimal (6 hr)	Settings → Meal Settings → Max Meal Absorption Time	**Meal max absorption time**	decimal (6)	Preferences → Absorption Settings → Advanced → Meal max absorption time
Enable Fat and Protein Entries	toggle (On/Off)	Settings → Meal Settings → Enable Fat and Protein Entries	***No Equivalent Setting***	N/A	N/A
Fat and Protein Delay	decimal (60 min)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Fat and Protein Delay	***No Equivalent Setting***	N/A	N/A
Maximum Duration	decimal (8 hr)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Maximum Duration	***No Equivalent Setting***	N/A	N/A

Spread Interval	decimal (30 min)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Spread Interval	***No Equivalent Setting***	N/A	N/A
Fat and Protein Percentage	percentage (50%)	Settings → Meal Settings → Enable Fat and Protein Entries (ON) → Fat and Protein Percentage	***No Equivalent Setting***	N/A	N/A

Other Notable Settings

Trio Name	Setting Format (example)	Location in Trio	AAPS Name	Setting Format (example)	Location in AAPS
Smooth Glucose Value	toggle (On/Off)	Settings → Devices → CGM → Smooth Glucose Value	**Smoothing**	selection (No smoothing / Exponential smoothing / Average smoothing)	Config Builder → Smoothing

Custom Automations

Trio does not offer certain custom automations that are available in AAPS, such as automating a hypo temp target when glucose drops below a certain value. This can, however, be achieved one of two ways:

- Enable "Sensitivity Raises Target" which will automatically increase your target glucose when sensitivity ratio is below 100%.
- Enable "High Temp Target Raises Sensitivity" and save a custom temp target of <100 mg/dL (5.5 mmol/L) under adjustments that you can manually enable when recovering from a low glucose.

Trio Settings

Welcome to the Trio Settings homepage! Here you'll find explanations of all the settings in Trio. They are organized by their appearance in the Settings Menu in the Trio app.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Devices

Set up your pump, cgm, and smart watch

Therapy

Set your "Trio" of core settings (Basal, ISF, and CR), Glucose Target, and Units & Limits

Algorithm

Configure Trio's algorithm, SMBs, and dynamic options

Features

Set up bolusing options, shortcuts, remote control, and personalize the app

Notifications

Customize app notifications, including live activity and calendar events

Services

Connect Nightscout, Tidepool, and Apple Health

Close The Loop

Devices

Welcome to the Devices homepage! Here you'll find information on which devices are compatible with Trio and how to add your compatible devices to your Trio app.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Verify Device Compatibility

Phones

List of compatible phones

Watches

List of compatible watches

Pumps

List of compatible insulin pumps

CGMs

List of compatible continuous glucose monitors (CGMs)

Add Devices

Insulin Pump

Step by step guide to add your insulin pump

Continuous Glucose Monitor

Step by step guide to add your CGM

Smart Watch

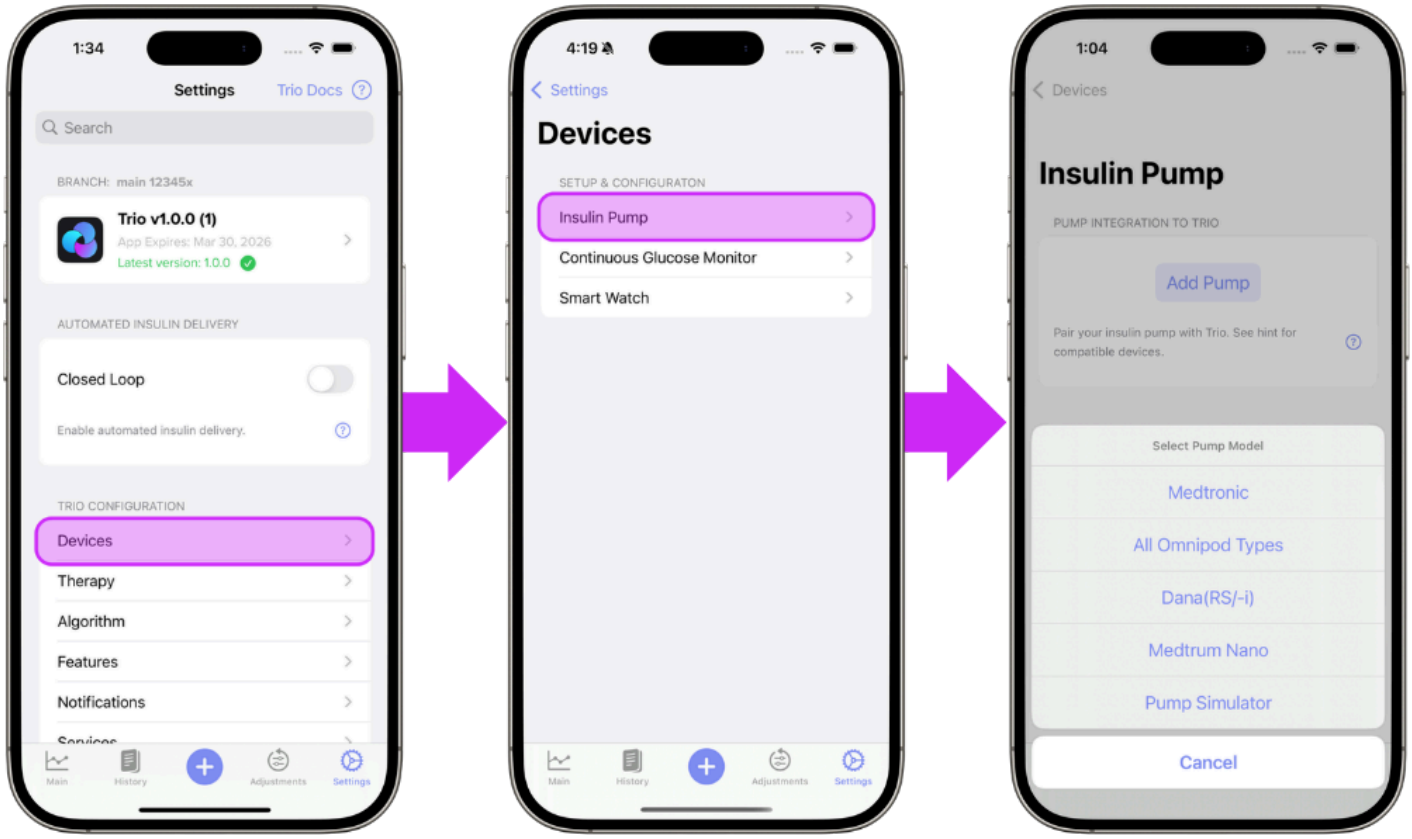
Information on adding an Apple watch or Garmin watch

Insulin Pump

This section allows you to choose your insulin pump. Most options are self-explanatory. For more information on compatible insulin pumps, please see the following link

Step 1: Add Pump

The first step in setting up your insulin pump on Trio is to tap the "Add Pump" button in the Devices Menu



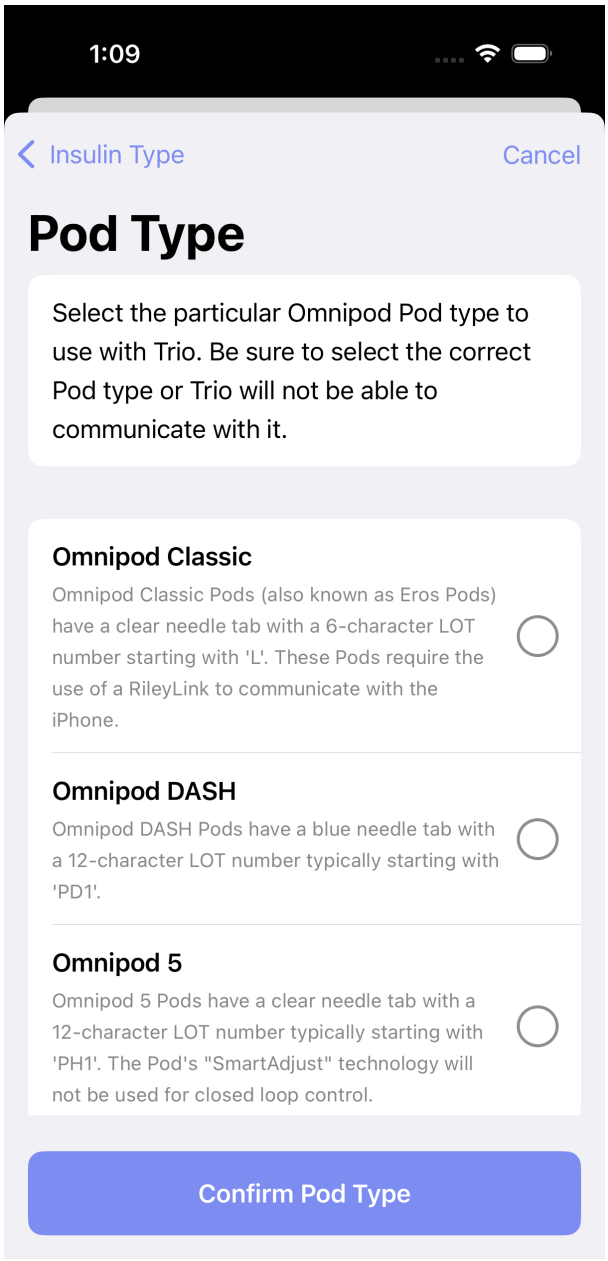
Step 2: Select Your Pump

Select your pump from the in-app menu and from the options below for step-by-step instructions. The links below will guide you through the connection instructions for your specific pump.

- All Omnipod Types
  - Omnipod Eros
  - Omnipod Dash
  - Omnipod 5
- Dana(RS/-i)
- Medtrum Nano
- Medtronic
- Pump Simulator

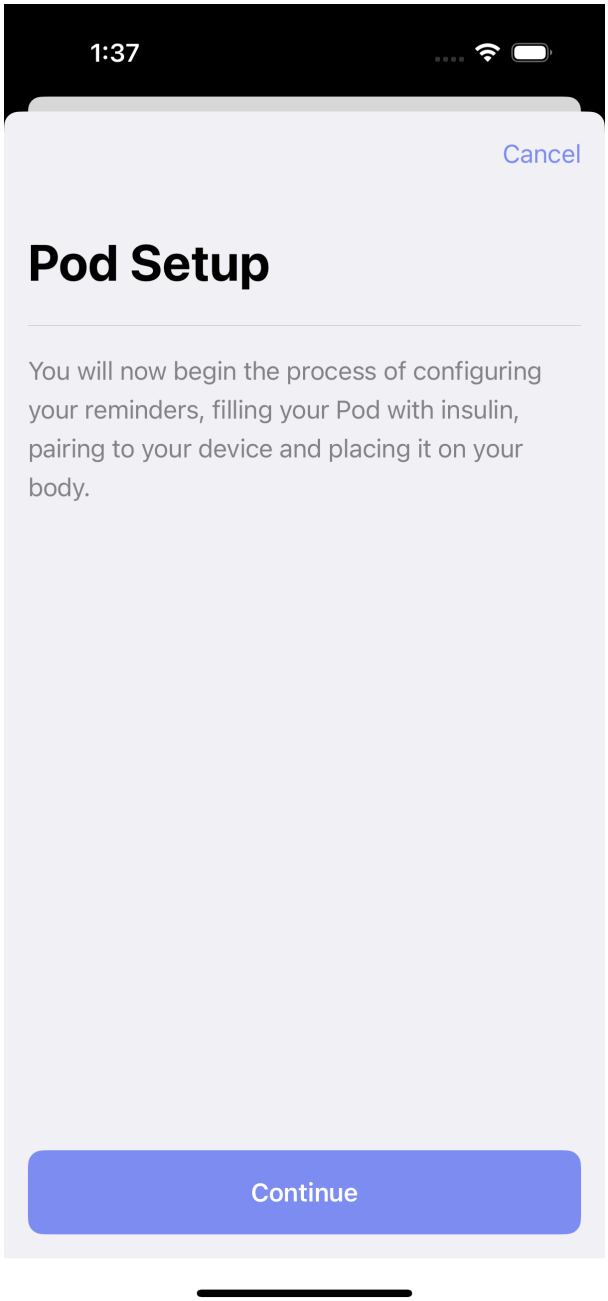
All Omnipod Types

Step 3 Select which Omnipod type you will use with Trio

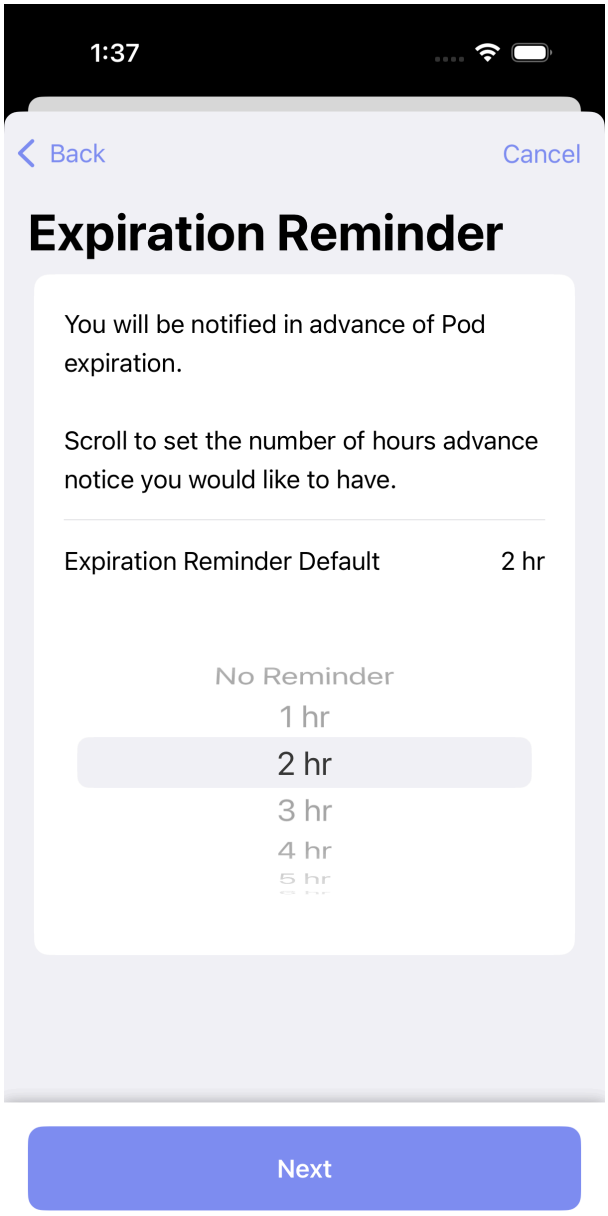


Omnipod Eros

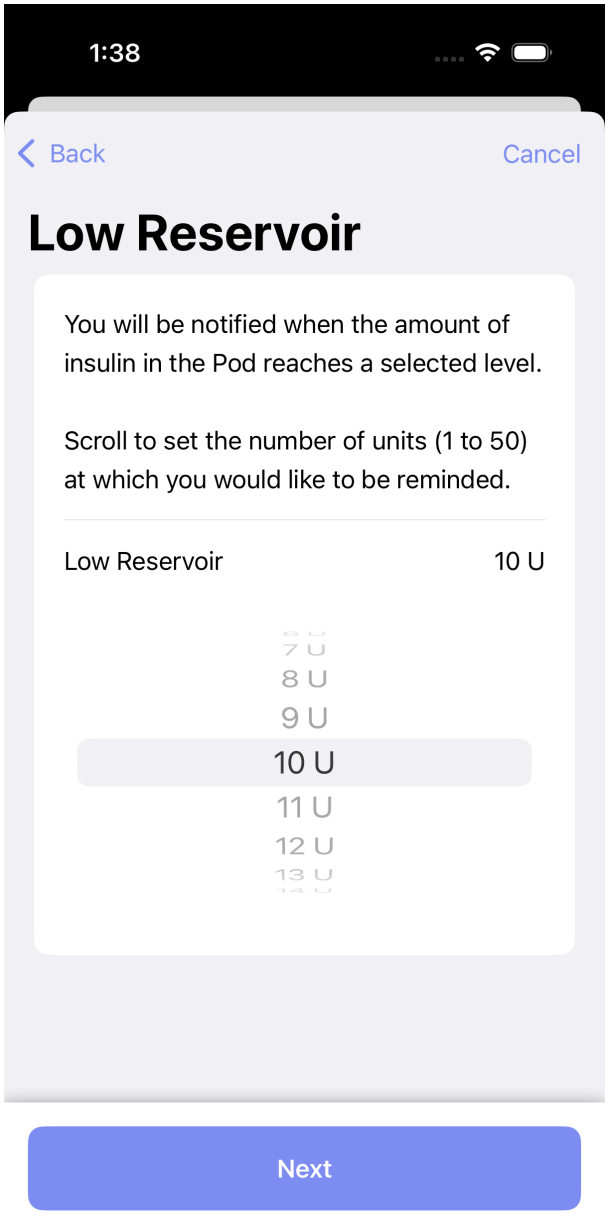
Step 4 Begin the process of configuring your Eros pod by tapping the blue "Continue" button at the bottom



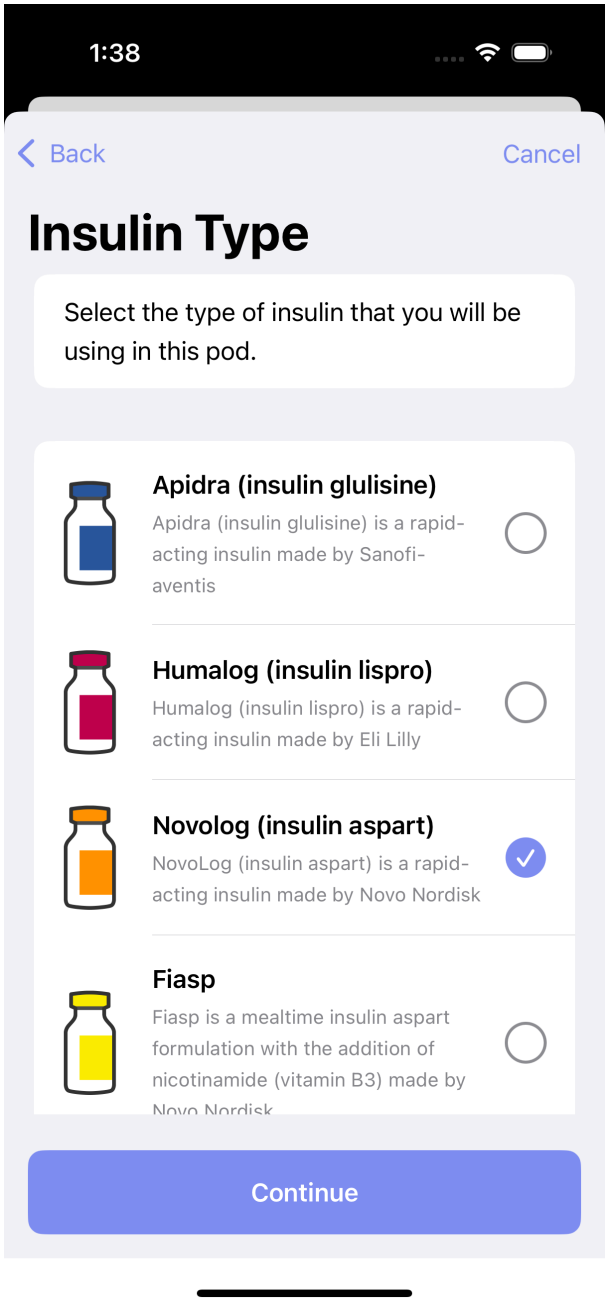
Step 5 Choose the expiration reminder that you prefer and tap "Next"



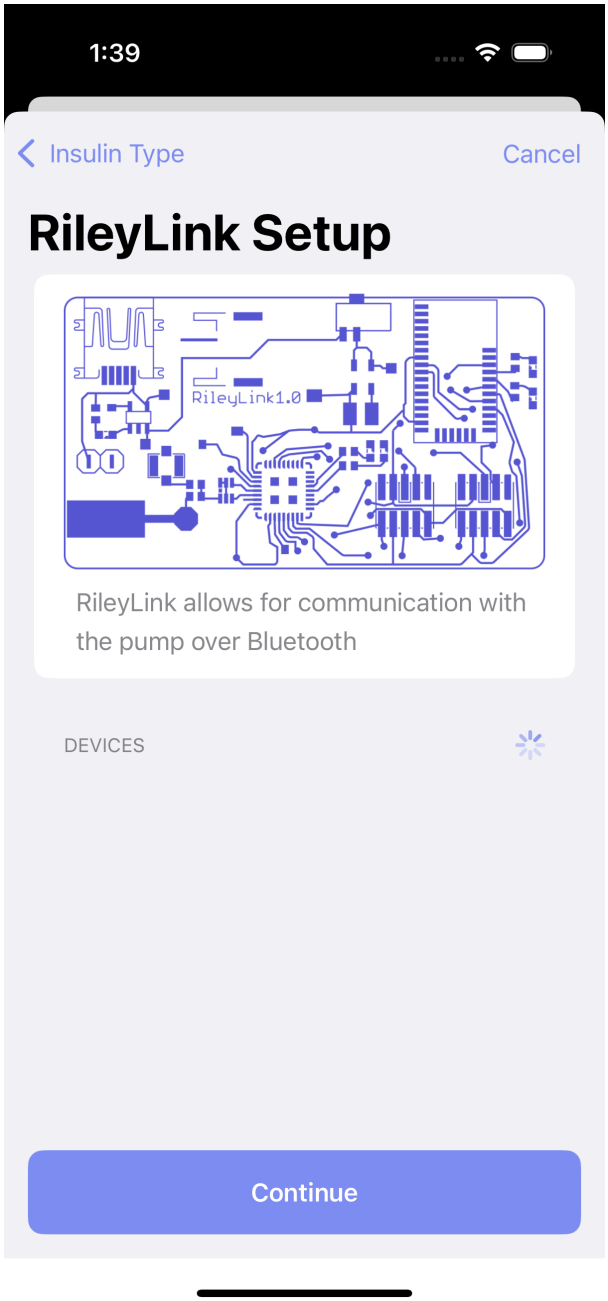
Step 6 Choose the low reservoir reminder that you prefer and tap "Next"



Step 7 Choose your insulin type and tap "Continue"



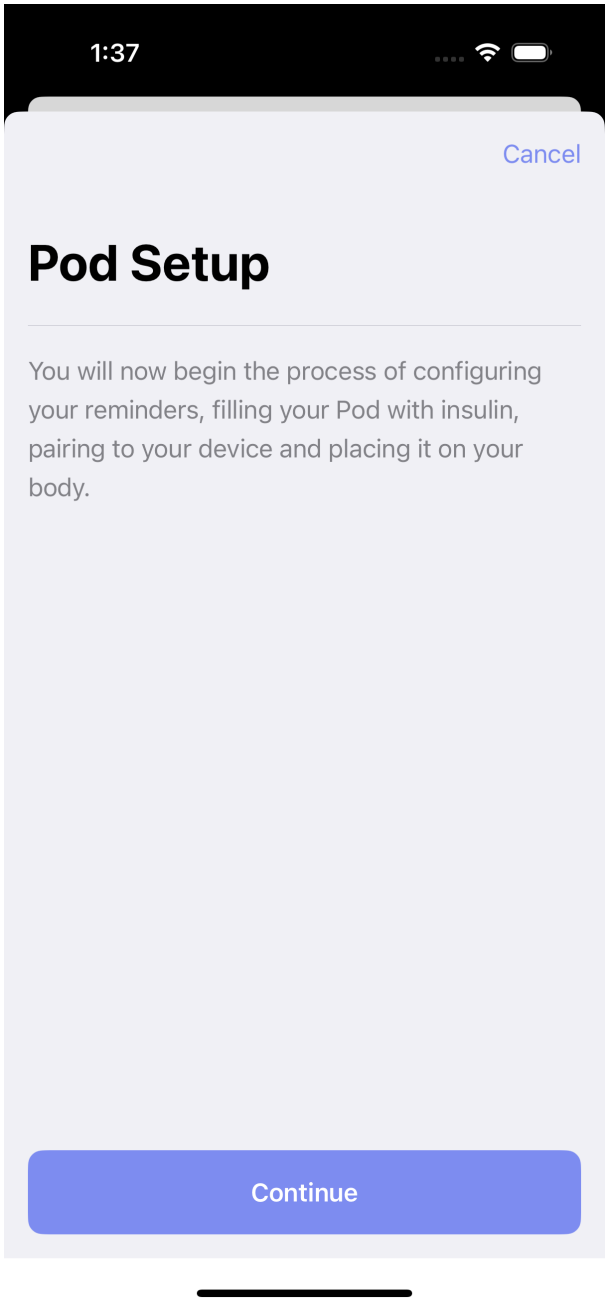
Step 8 Connect your RileyLink, OrangeLink, or EmaLink and tap "Continue"



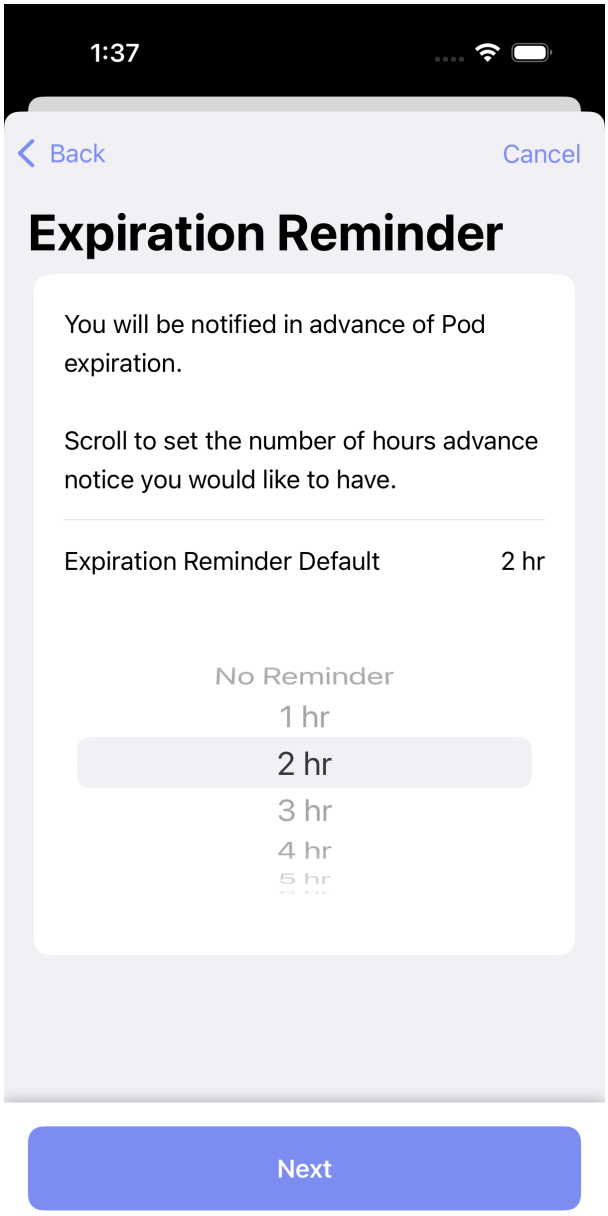
Step 8 Continue to Connect CGM *OR* return to New User Setup

Omnipod Dash

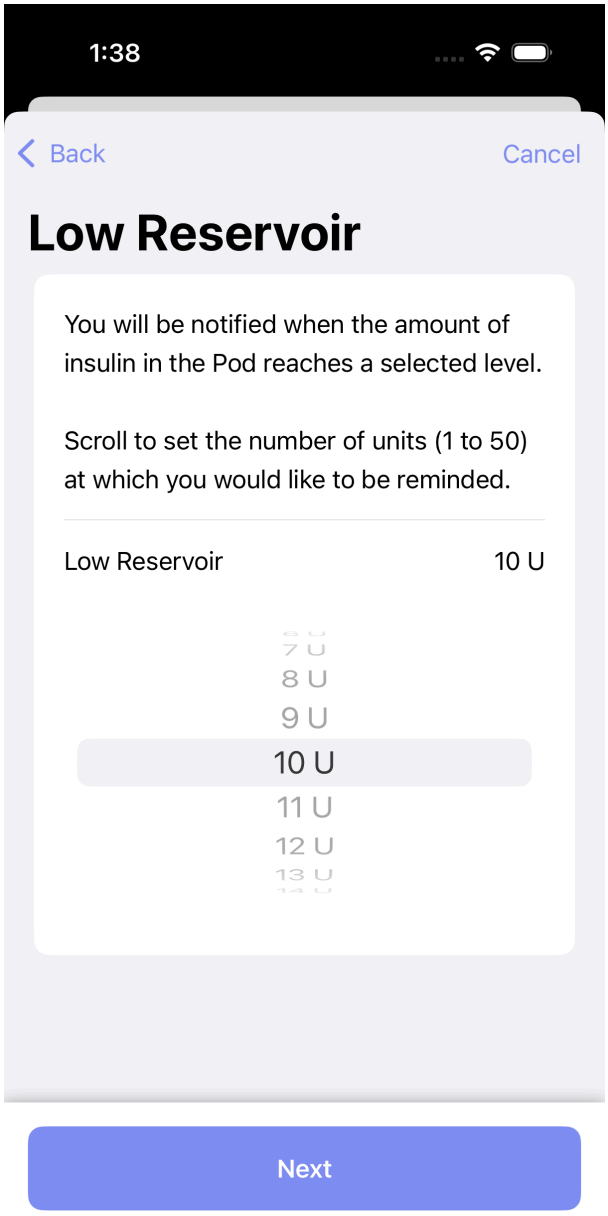
Step 4 Begin the process of configuring your Dash pod by tapping the blue "Continue" button at the bottom



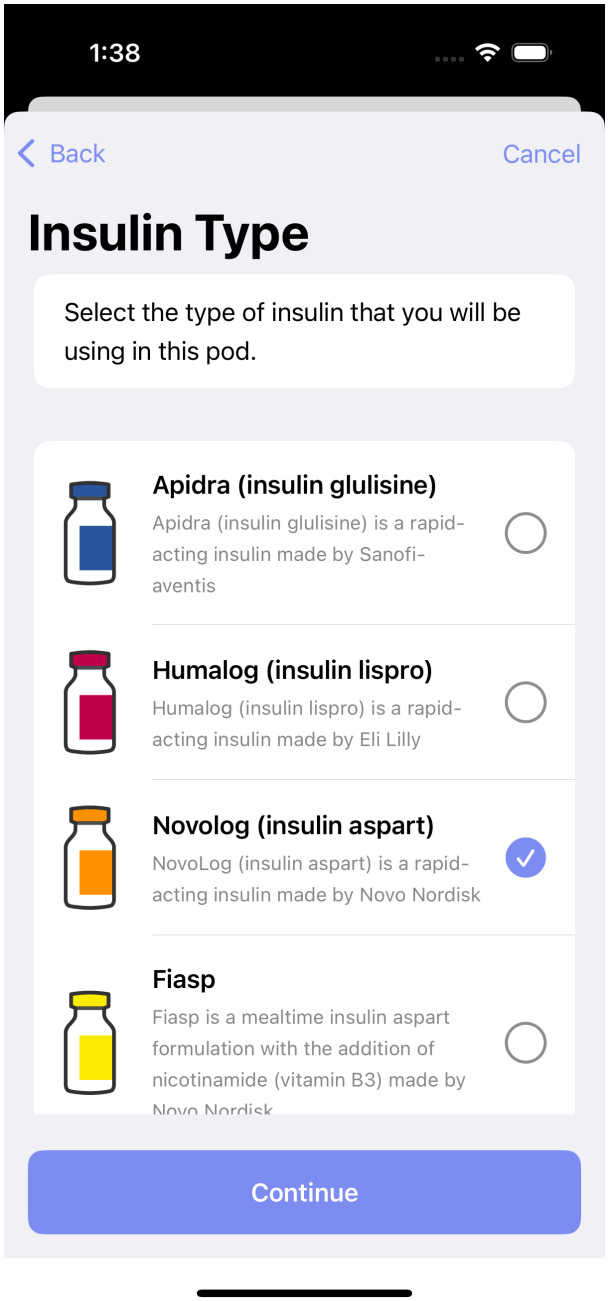
Step 5 Choose the expiration reminder that you prefer and tap "Next"



Step 6 Choose the low reservoir reminder that you prefer and tap "Next"



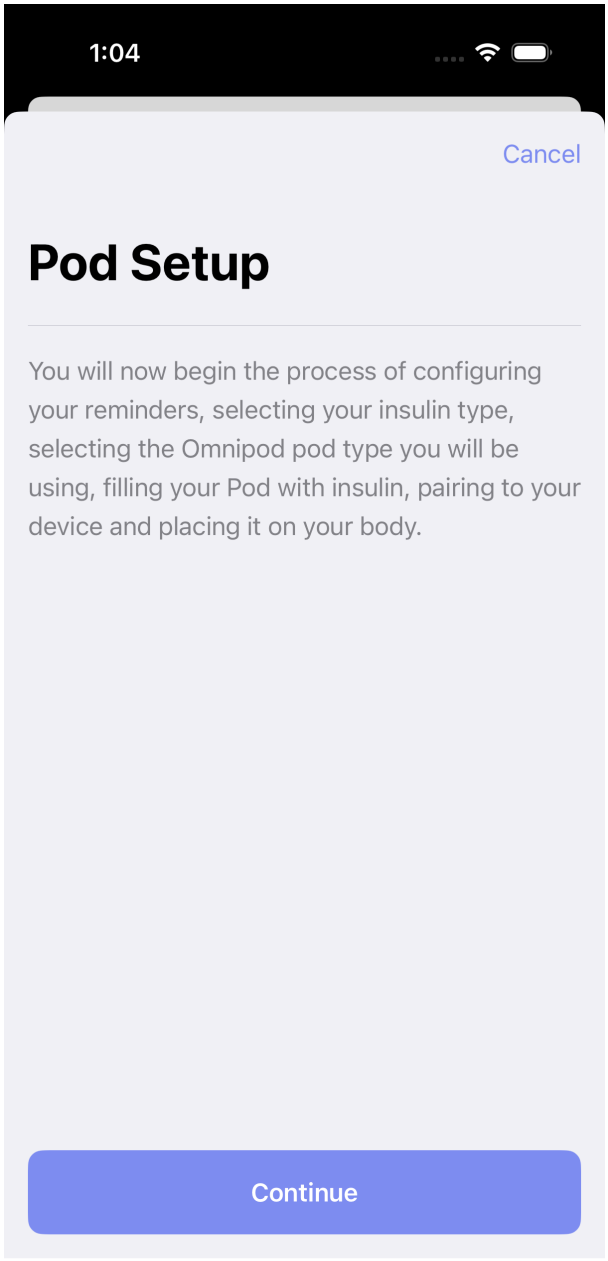
Step 7 Choose your insulin type and tap "Continue"



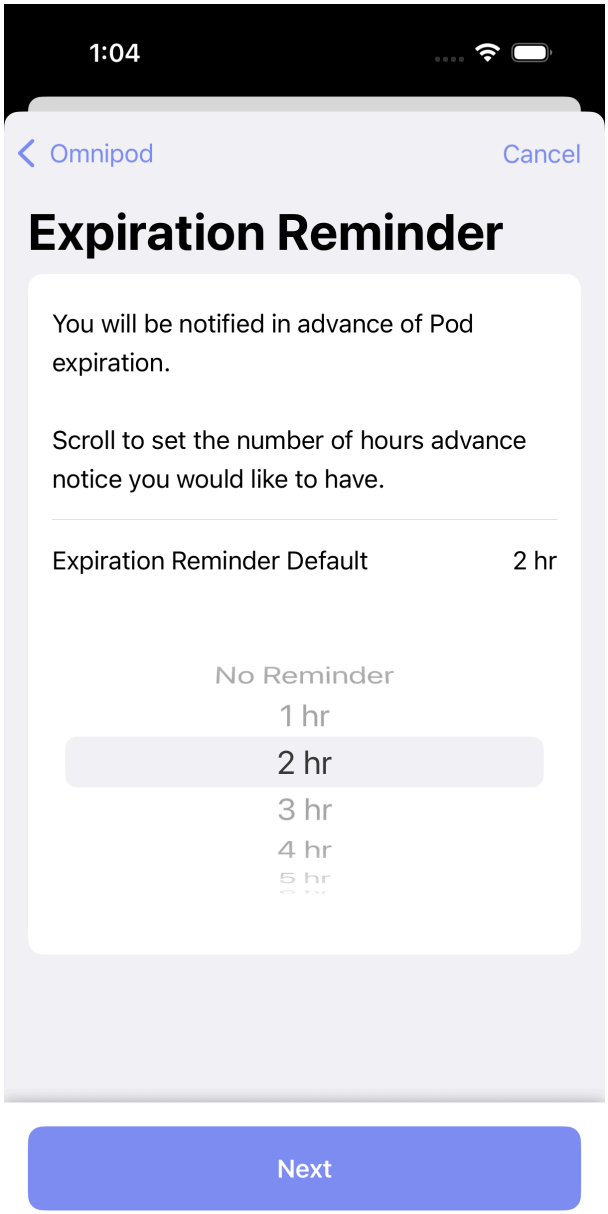
Step 8 Continue to Connect CGM *OR* return to New User Setup

Omnipod 5

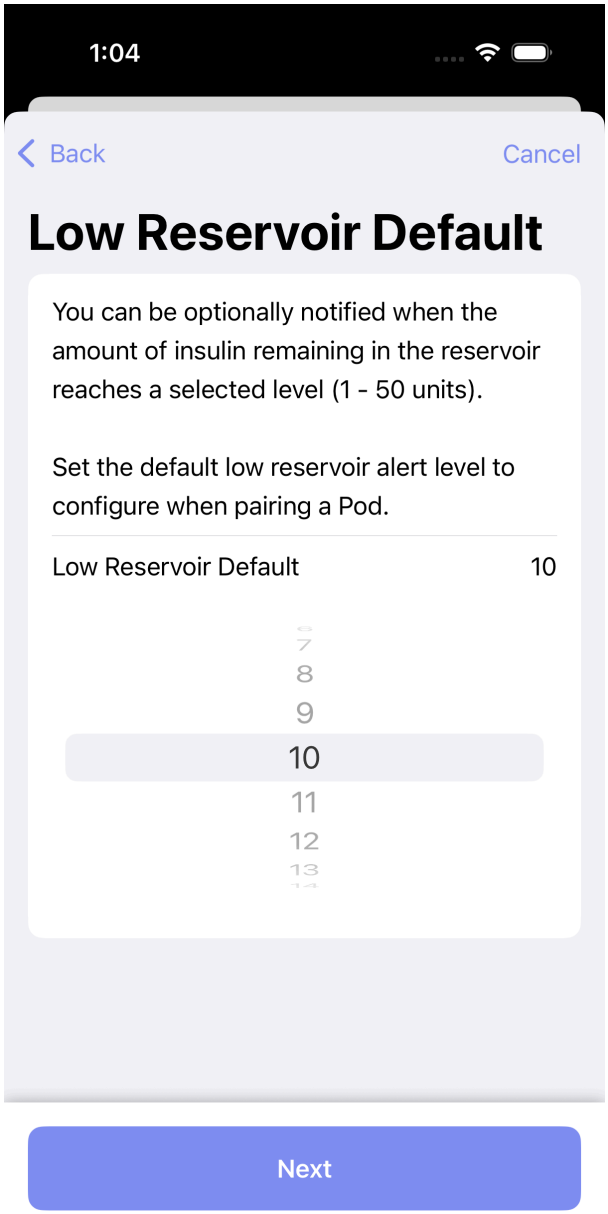
Step 4 Begin the process of configuring your Omnipod 5 pod by selecting Omnipod 5 and tap the purple "Continue" button



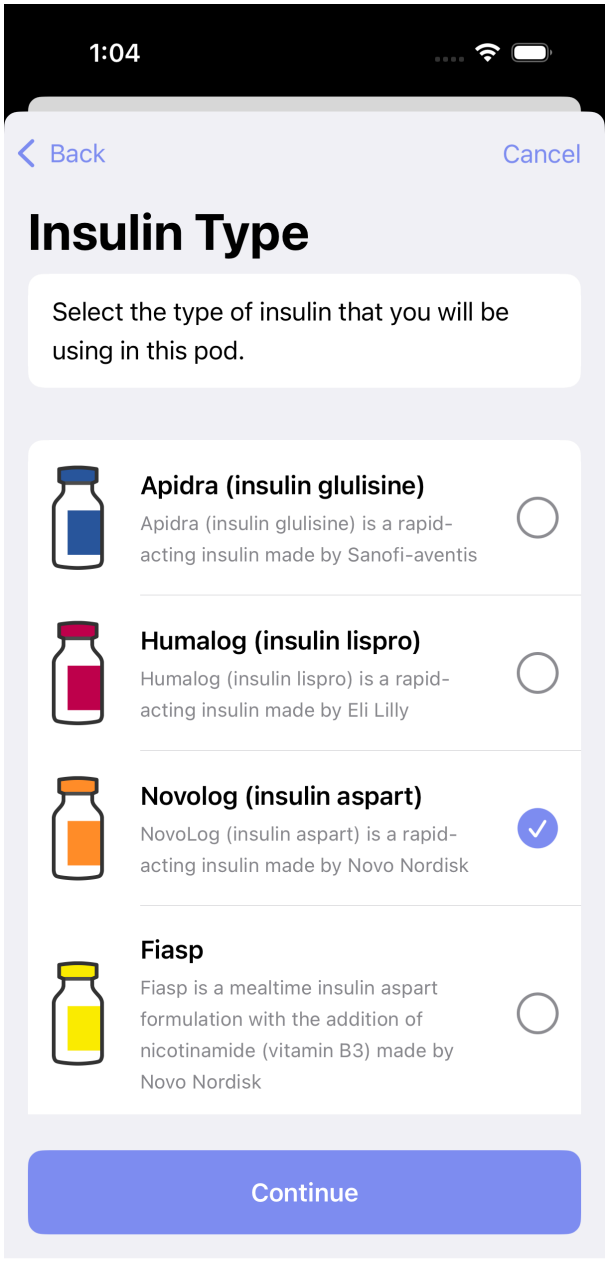
Step 5 Choose the expiration reminder that you prefer and tap "Next"



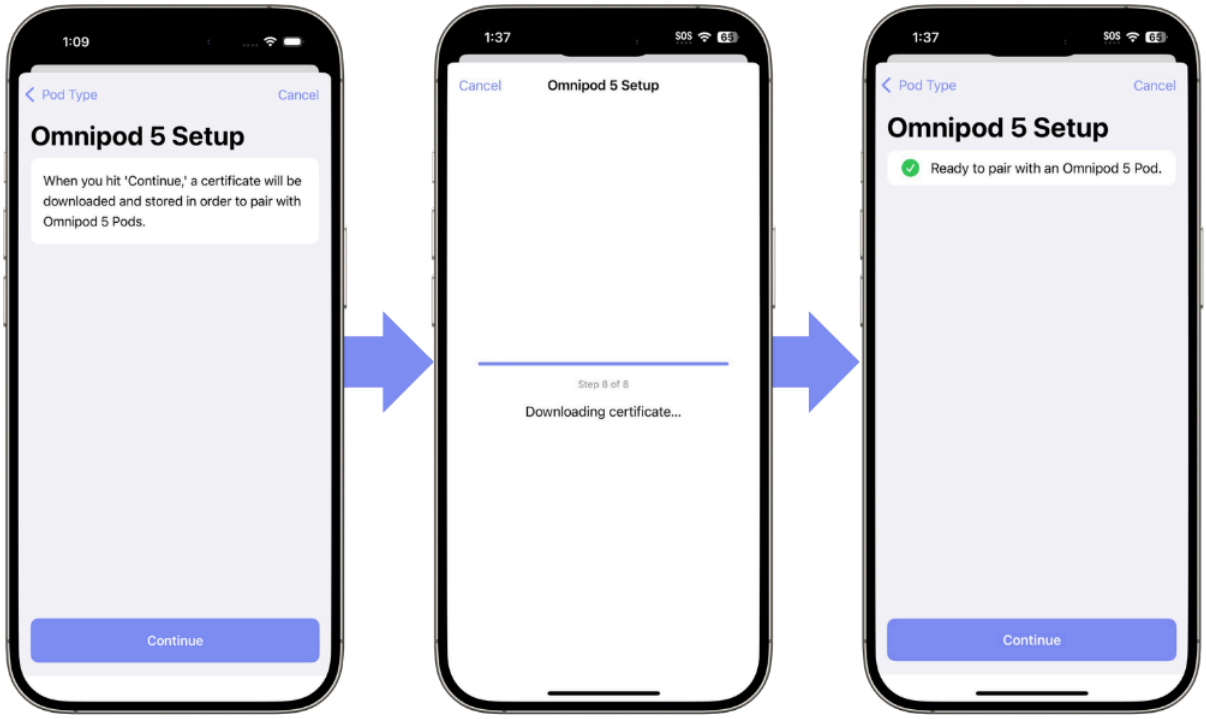
Step 6 Choose the low reservoir reminder that you prefer and tap "Next"



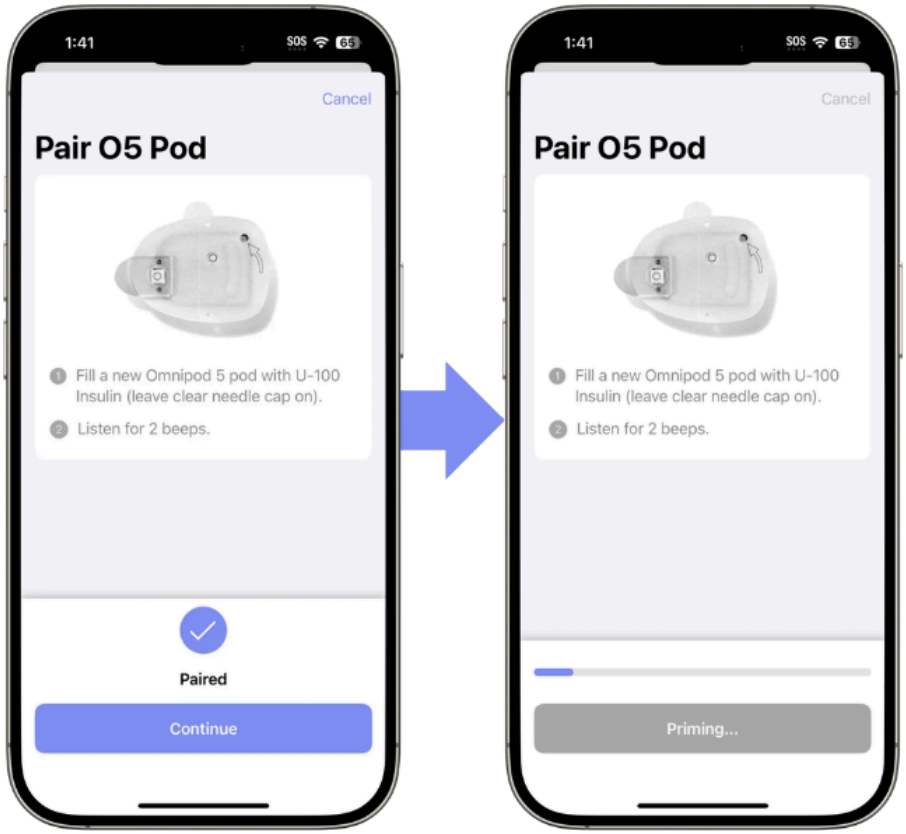
Step 7 Choose your insulin type and tap "Continue"



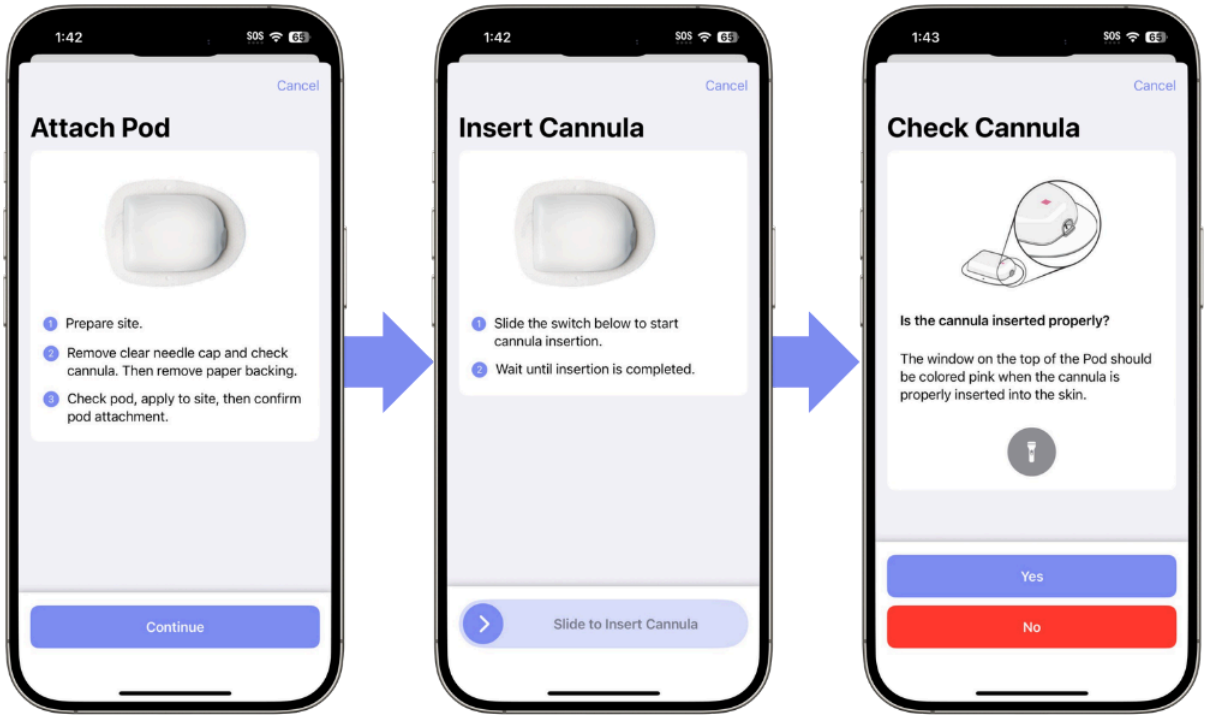
Step 8 Omnipod 5 requires a certificate to be pulled from a database online. This step requires internet access, but only for the very first Omnipod 5 pod. All future pods will not need this step or internet connection.



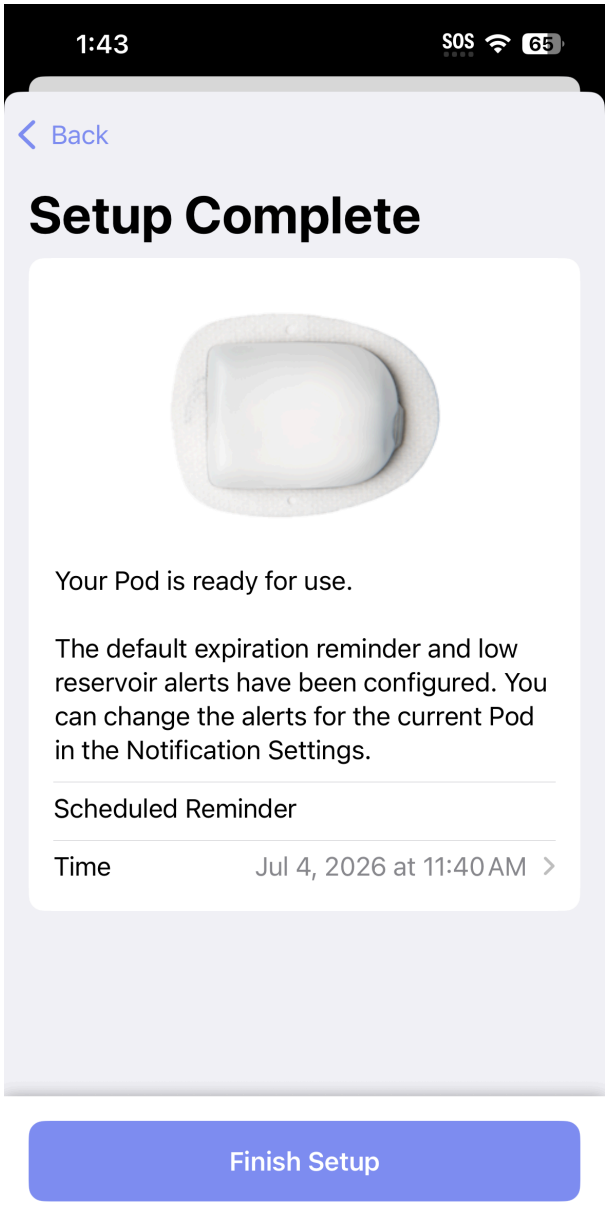
Step 9 Follow the on-screen instructions to pair your Omnipod 5 pod



Step 10 Proceed through the steps to insert the cannula



Step 11 Setup complete!



Step 12 Continue to Connect CGM *OR* return to New User Setup

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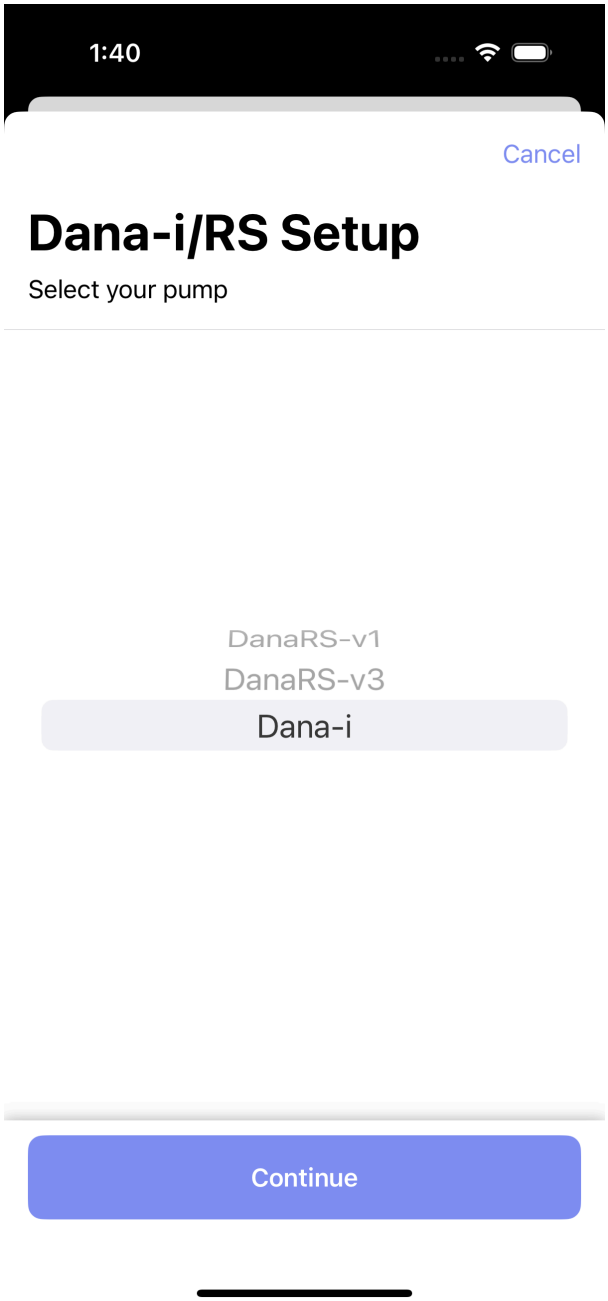
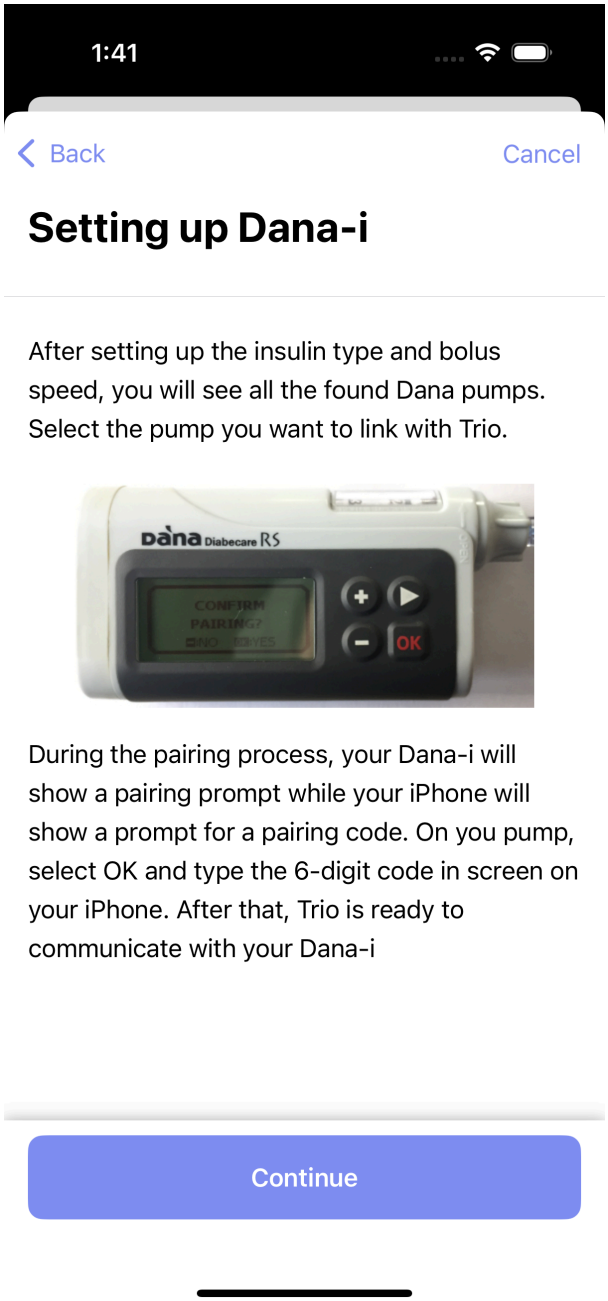
Dana(RS/-i)

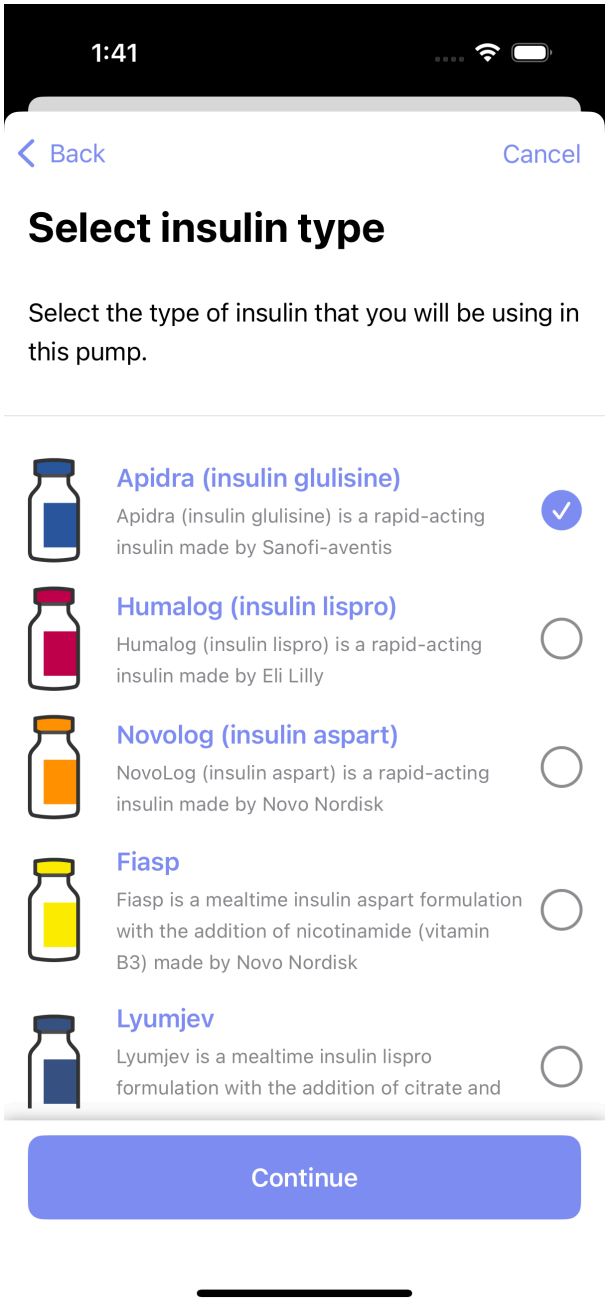
Step 3 Select your Dana pump model from the list and tap "Continue"



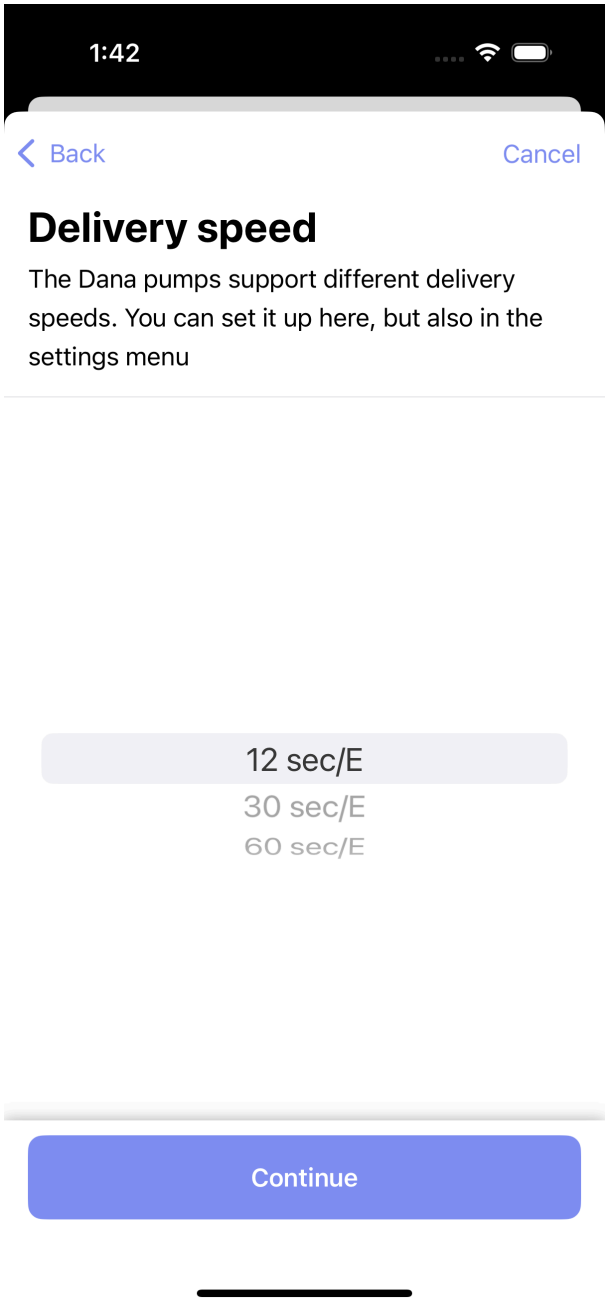
Step 4 Read the in-app information and tap "Continue" when ready to proceed



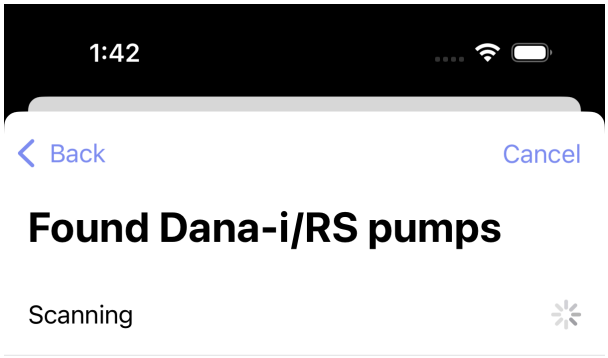
Step 5 Choose your insulin type and tap "Continue"



Step 6 Choose your insulin delivery speed and tap "Continue"



Step 7 Select your pump from the list of found pumps



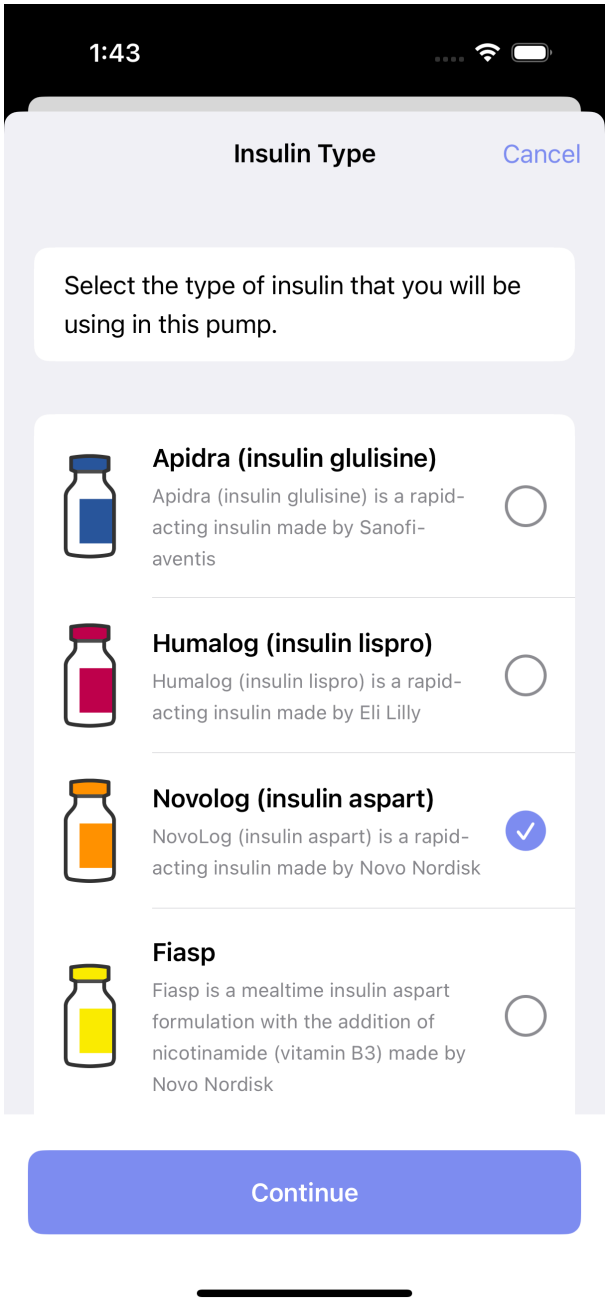
Step 8 Continue to Connect CGM *OR* return to New User Setup

Medtrum Nano

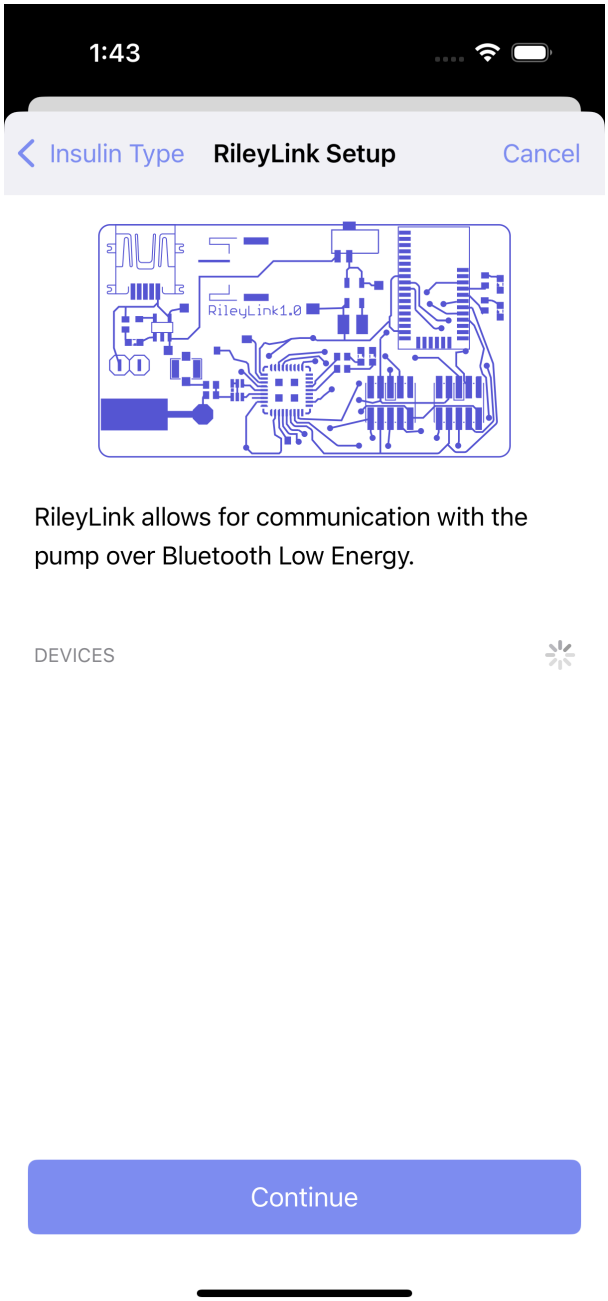
Setup Instructions Coming Soon

Medtronic

Step 3 Choose your insulin type and tap "Continue"



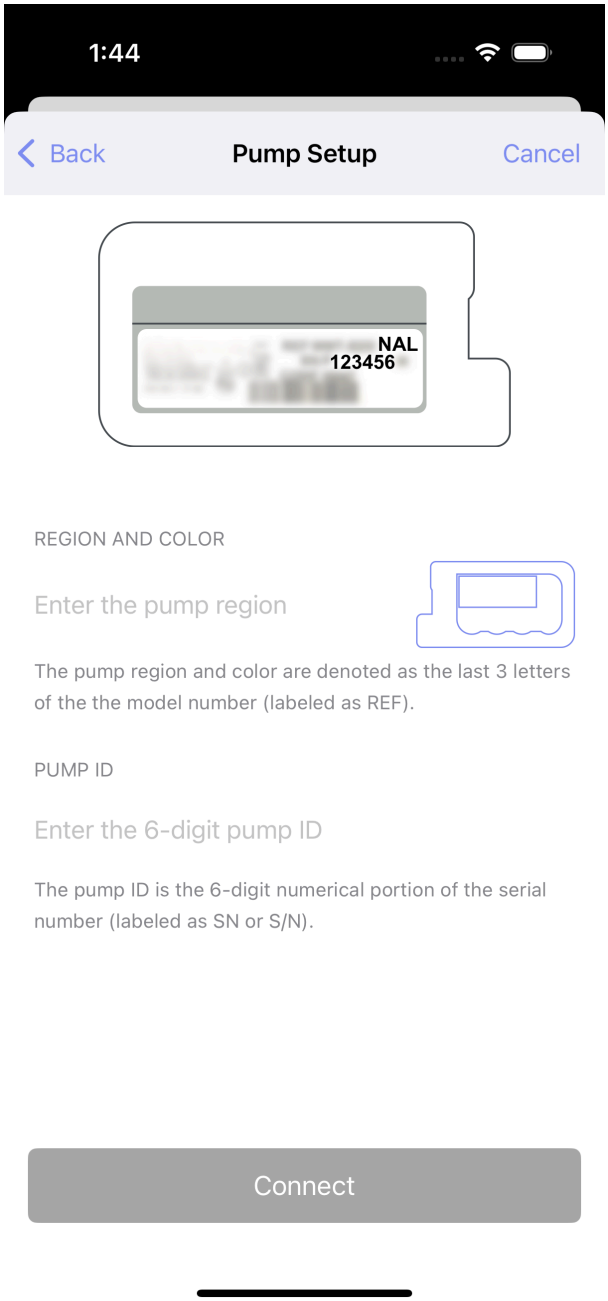
Step 4 Connect your RileyLink, OrangeLink, or EmaLink and tap "Continue"



Step 5a Enter your pump region and color

Step 5b Enter your 6-digit pump ID

Then, tap the "Connect" button when it turns blue at the bottom.



Step 6 Continue to Connect CGM *OR* return to New User Setup

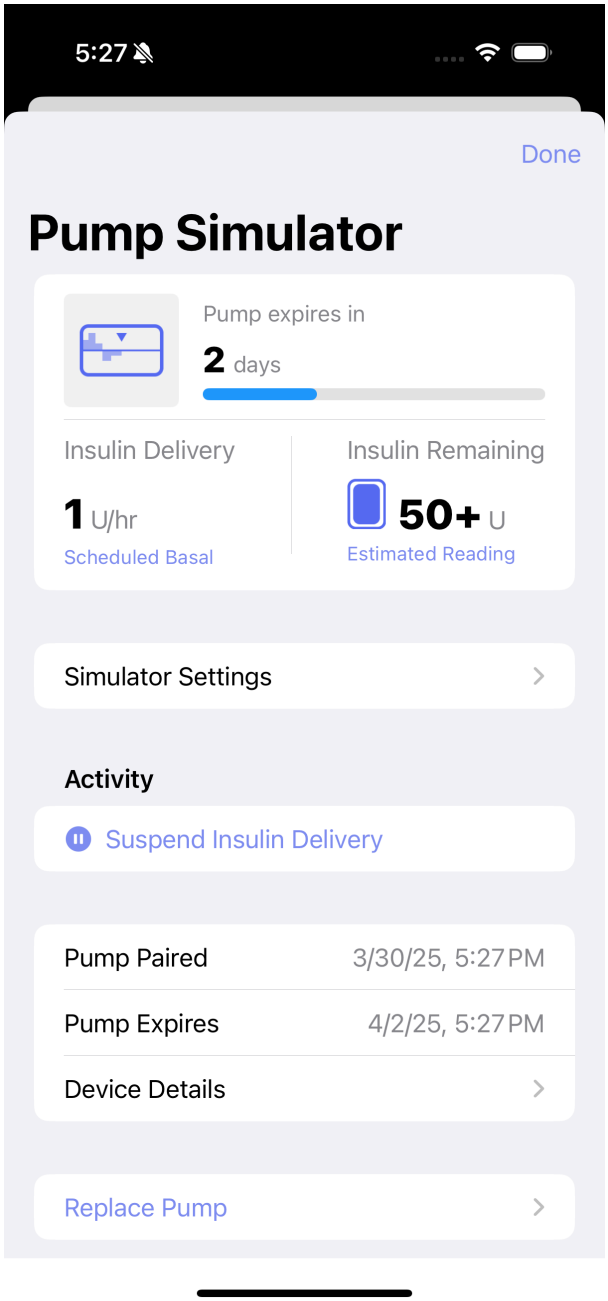
Pump Simulator

Warning

If using a pump simulator, it is important to understand:

- You will only experience the user interface of Trio.
- Using a pump simulator does not indicate how the app will perform, nor will it give accurate guidance or suggestions for insulin dosing.
- Trio may not run consistently in the background when using a pump simulator.
- If you use a pump simulator and switch to a live pump, you must delete all insulin dosing history by uninstalling the Trio app and reinstalling it.
- **Only use a pump simulator if you understand the conditions above.**

Step 3 Choose from the variety of simulator pump options and tap "Done" in the upper right corner to activate the pump simulator



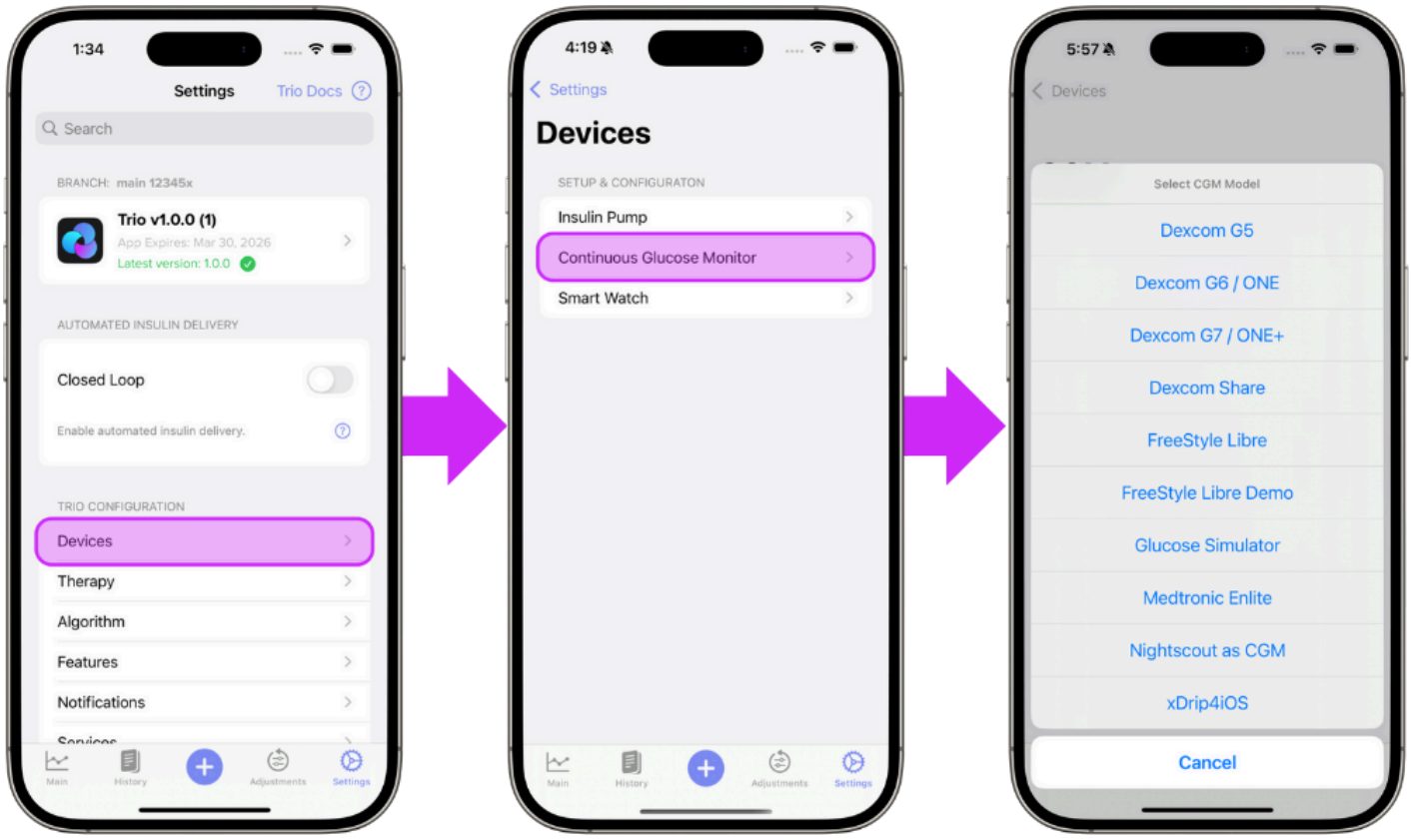
Step 4 Continue to Connect CGM *OR* return to New User Setup

CGM

This section allows you to choose your glucose data source. Most options are self-explanatory. For more information on compatible CGMs, please see the following link.

Step 1: Add CGM

The first step in setting up your continuous glucose monitor (CGM) on Trio is to tap the "Add CGM" button in the Devices Menu



Step 2: Select Your CGM

Select your CGM from the in-app menu and from the options below for step by step instructions. The links below will guide you through the connection instructions for your specific CGM:

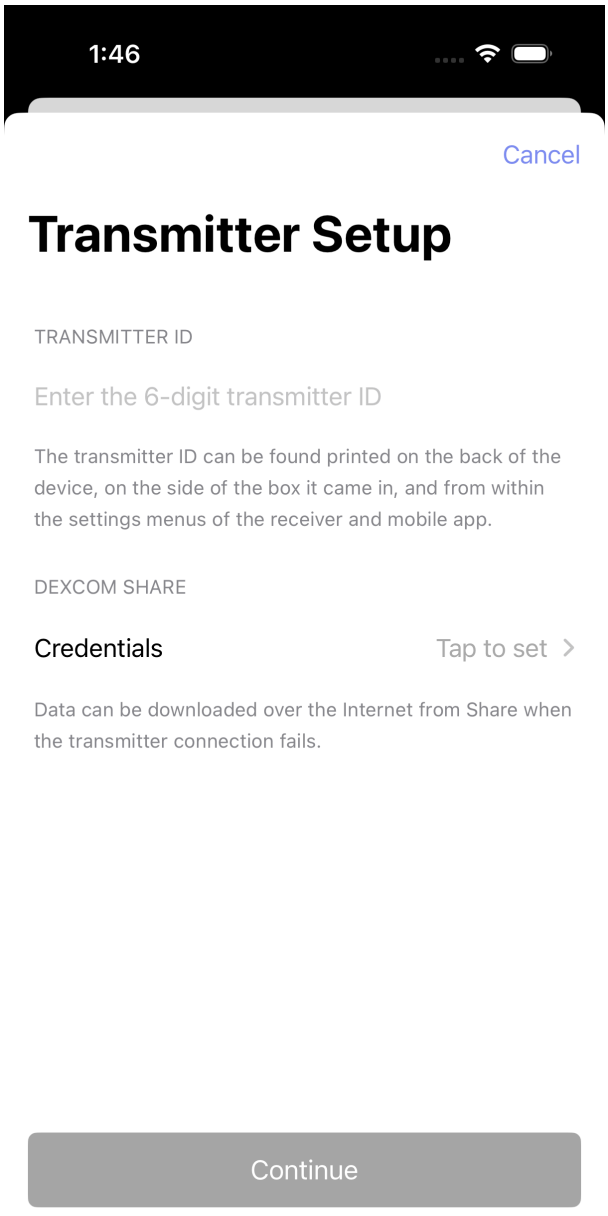
- Dexcom G5 / G6 / ONE
- Dexcom G7 / ONE+
- Dexcom Share
- FreeStyle Libre 1 / 2 / 2 Plus
- FreeStyle Libre 3 / 3 Plus
- Eversense E3 / 365
- Glucose Simulator
- Nightscout as CGM

- xDrip4iOS

Dexcom G5 / G6 / ONE

Trio will intercept glucose readings between the transmitter and the Dexcom app. If you are using a Dexcom G5, G6, or ONE sensor, tap Configuration CGM to enter your transmitter's 6-digit ID. *Dexcom Share Credentials are not necessary*. When you switch transmitters, you must delete your current transmitter from Trio by tapping Configuration CGM, scrolling down, and tapping Delete CGM. Once you do this, you can add the new transmitter with its Transmitter ID.

Step 3 Enter your 6-digit Dexcom transmitter ID.



Step 4 Continue to Connect Watch *OR* return to New User Setup

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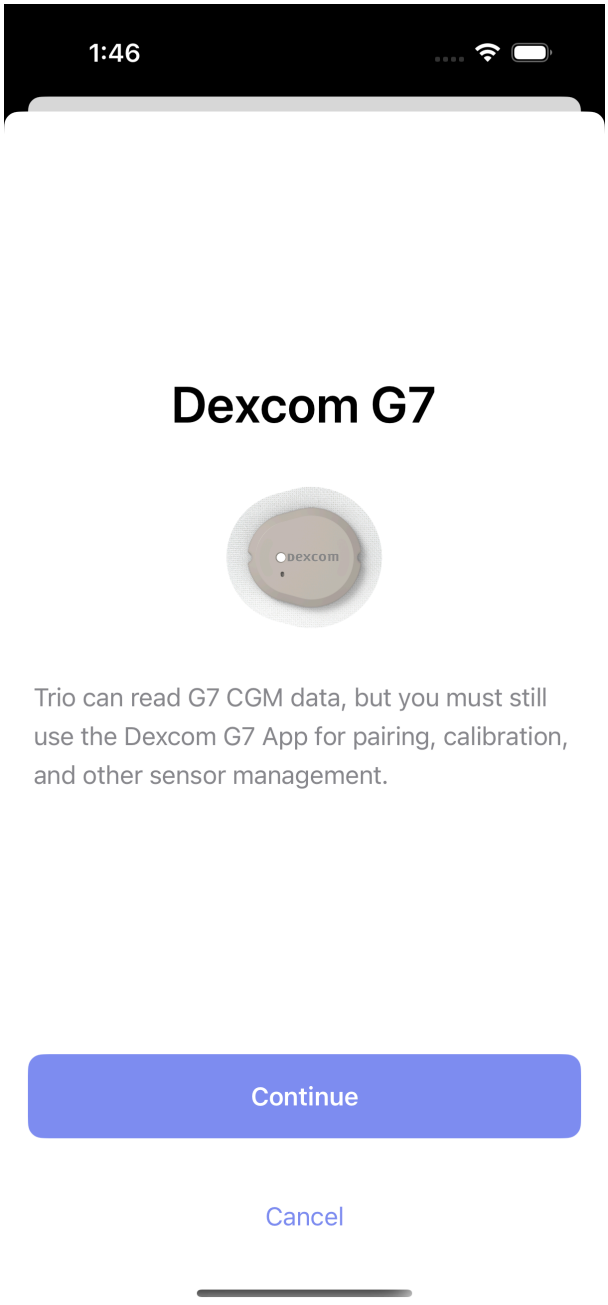
Dexcom G7 / ONE+

As long as the Dexcom G7 or ONE+ app is installed on the same phone, Trio can intercept its glucose readings. When a new G7 or ONE+ sensor is paired to the Dexcom app, Trio will automatically start reading it.

Step 3 Tap "Continue" to use the G7/ONE+ as your CGM source



Step 4 Continue to Connect Watch *OR* return to New User Setup

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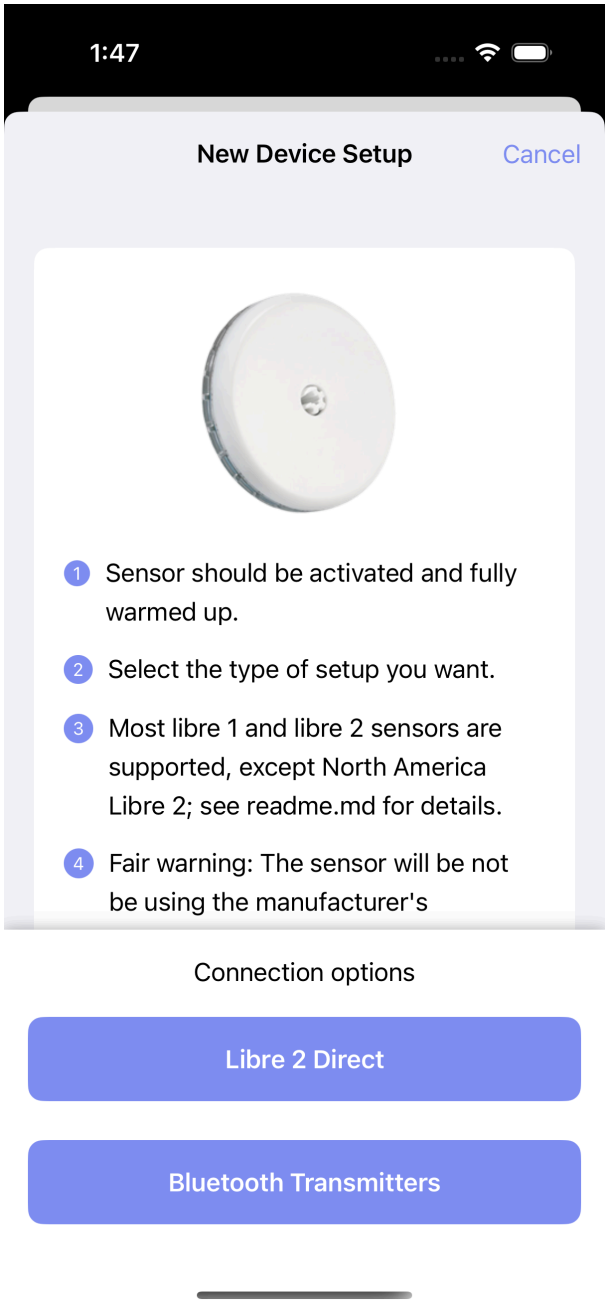
Dexcom Share

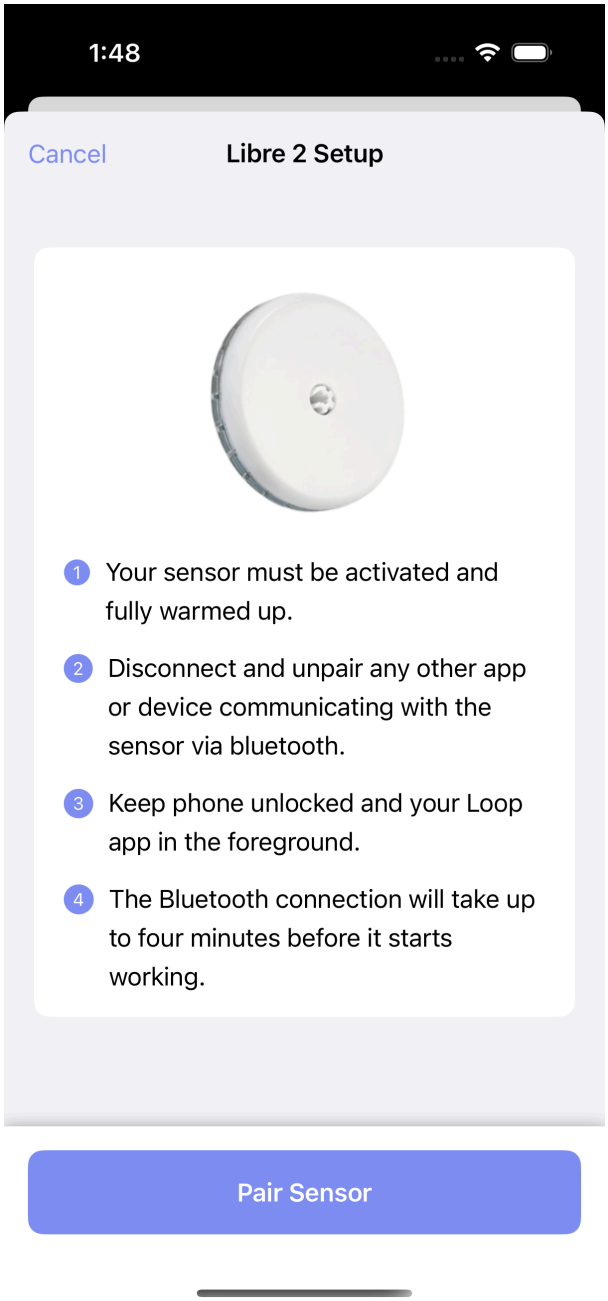
FreeStyle Libre 1 / 2 / 2 Plus

This option can be used to pair a compatible Libre 1, Libre 2, or Libre 2 Plus sensor directly to Trio without going through a separate app like xDrip4iOS. In the in-app menu, it is listed as "FreeStyle Libre 1 / 2 / 2+". For Libre 3 and Libre 3 Plus sensors, see FreeStyle Libre 3 / 3 Plus instead.

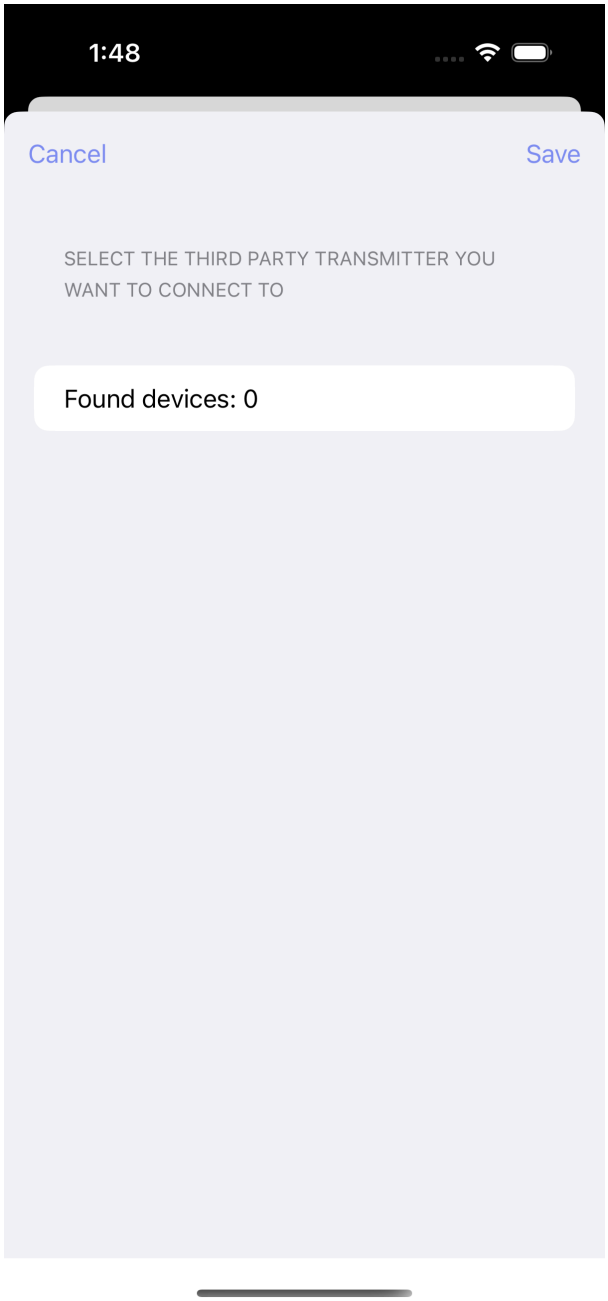
Step 3 Tap "Libre 2 Direct" to use Libre 2 sensors or "Bluetooth Transmitters" to use Libre 1 sensors with a Miao Miao or other 3rd party transmitter.



Step 4:Libre 2 Direct Tap "Pair Sensor" to connect your Libre 2/2+ sensor to Trio



Step 4:Bluetooth Transmitters (Libre 1) Select your 3rd party transmitter from the list of found devices and tap "Save"



Step 5 Continue to Connect Watch *OR* return to New User Setup

---

FreeStyle Libre 3 / 3 Plus

> ⚠ **You must build the `dev` branch to use the FreeStyle Libre 3 / 3 Plus CGM at this time**
>
> The FreeStyle Libre 3 / 3 Plus CGM is in `dev` and is in open-beta for now. In the in-app menu, it is listed as "FreeStyle Libre 3 / 3+ (Beta)".

Trio starts the sensor, and only one app can read it

Trio starts (activates) the sensor itself while pairing, so pair a new sensor that you have not started in another app. A Libre 3 sensor answers only to the app that started it. A sensor you start in Trio can no longer be read by the FreeStyle Libre 3 app, and a sensor you already started in the FreeStyle Libre 3 app cannot be paired to Trio.

The FreeStyle Libre 3 app's alarms will not reach you

Because Trio reads the sensor directly, the FreeStyle Libre 3 app's low and high glucose alarms are not active while the sensor is paired to Trio. Set up Trio's own Glucose Alarm Limits before you rely on this CGM, and consider a remote monitoring option such as Nightscout so someone else can see a low or high that you miss.

Wait until warm-up is completed

A newly started sensor warms up for about 60 minutes and reports no glucose during that time. Trio shows the remaining warm-up time on the home screen. Before using automated insulin delivery, be sure the warm-up has completed and the readings look reasonable to you.

A paid Apple Developer account is required

Pairing uses Near Field Communication (NFC), the short-range technology that lets you hold your phone against the sensor to talk to it. NFC is not available when Trio is built with a free Apple Developer account. See Apple Developer Account.

Step 3 Apply a new sensor to the back of your upper arm. Tap "HOW TO APPLY A SENSOR" if you would like Trio to walk you through it, then tap "Next".

Step 4 Tap "Start pairing", then hold the back of your phone against the sensor and keep it still. Your phone vibrates once the sensor is paired. If the phone moves away too early, the scan fails and you can try again.

Step 5 Wait for Trio to finish pairing. Trio starts the sensor, connects to it over Bluetooth, and then receives a glucose reading every 5 minutes once warm-up is finished.

Step 6 Write down your Receiver ID, which you will find in the Debug Info section of the Libre 3 CGM settings page. The Receiver ID is the identifier the sensor remembers Trio by, and it is the only way to re-pair a sensor that is already running, for example after you delete the CGM from Trio or reinstall the app. To use it, tap "Recover existing sensor" on the pairing screen and enter the ID. When you re-pair a sensor that is already running, Trio shows "Stabilizing" for a while and does not display glucose readings the sensor itself flags as unreliable.

Step 7 Continue to Connect Watch *OR* return to New User Setup

---

Eversense E3 / 365

You must build the `dev` branch to use the Eversense CGM at this time

The Eversense CGM is in `dev` and is in open-beta for now

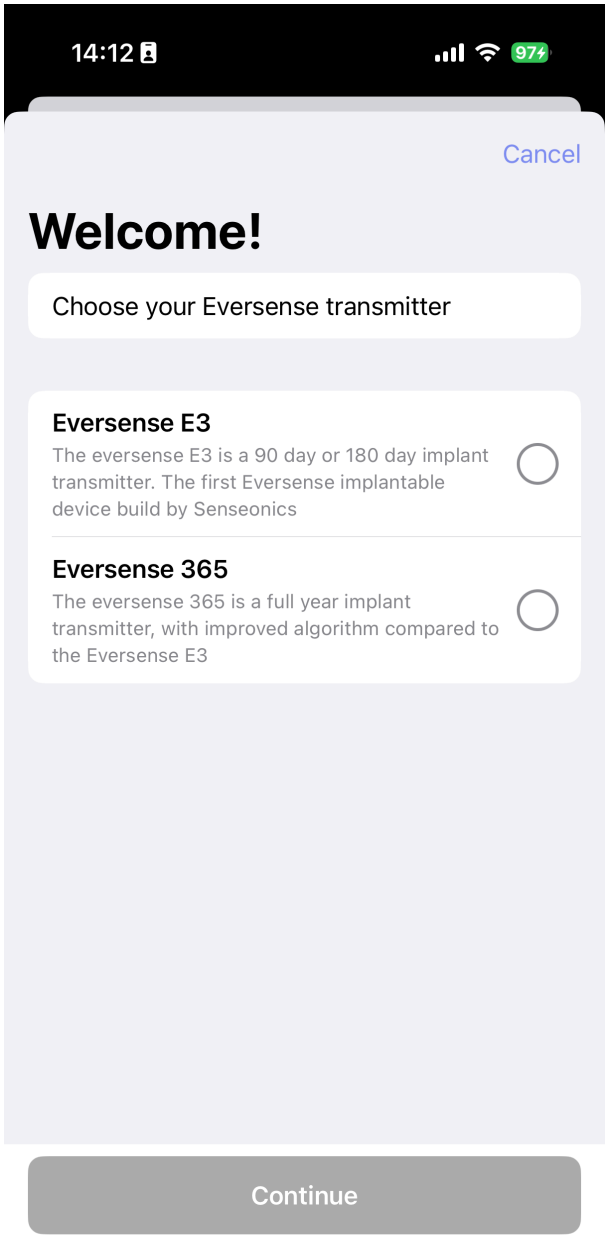
Wait until the initialization phase is completed

During the initialization phase of this system, the glucose reading might be incorrect. Before using automated insulin delivery, be sure to complete the initialization phase using the original Eversense app.

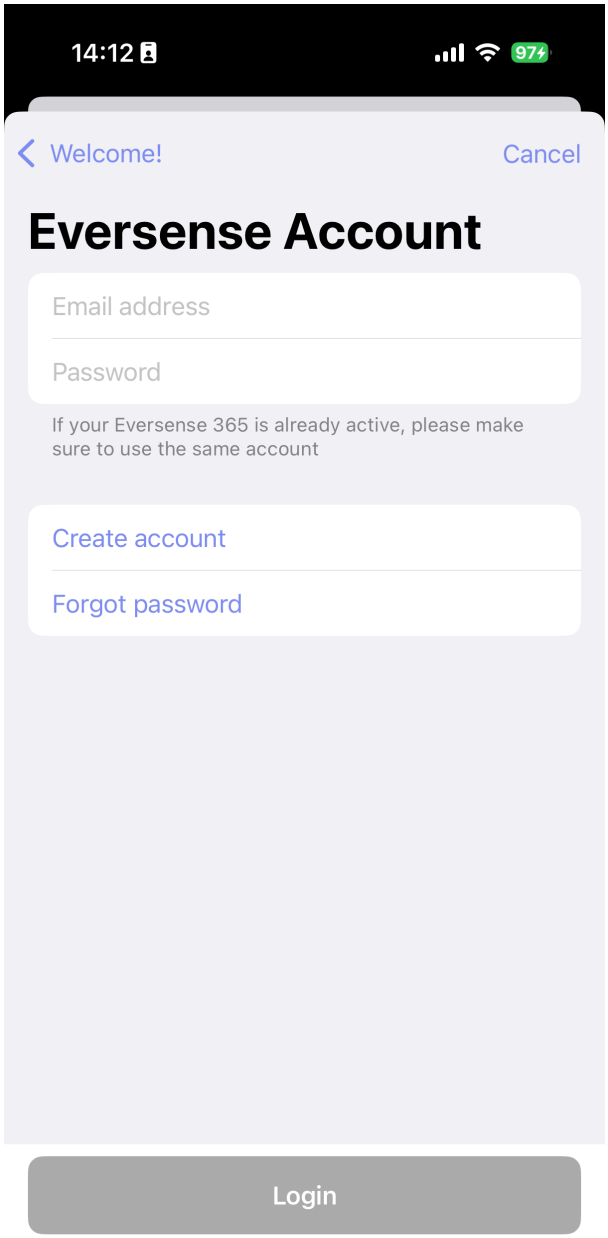
Do not use the Eversense 365 app and Trio concurrently

The 365 transmitter is not able to connect to multiple apps or devices at the same time, due to the security protocol. Make sure you either use the Eversense app or Trio, but not both at the same time.

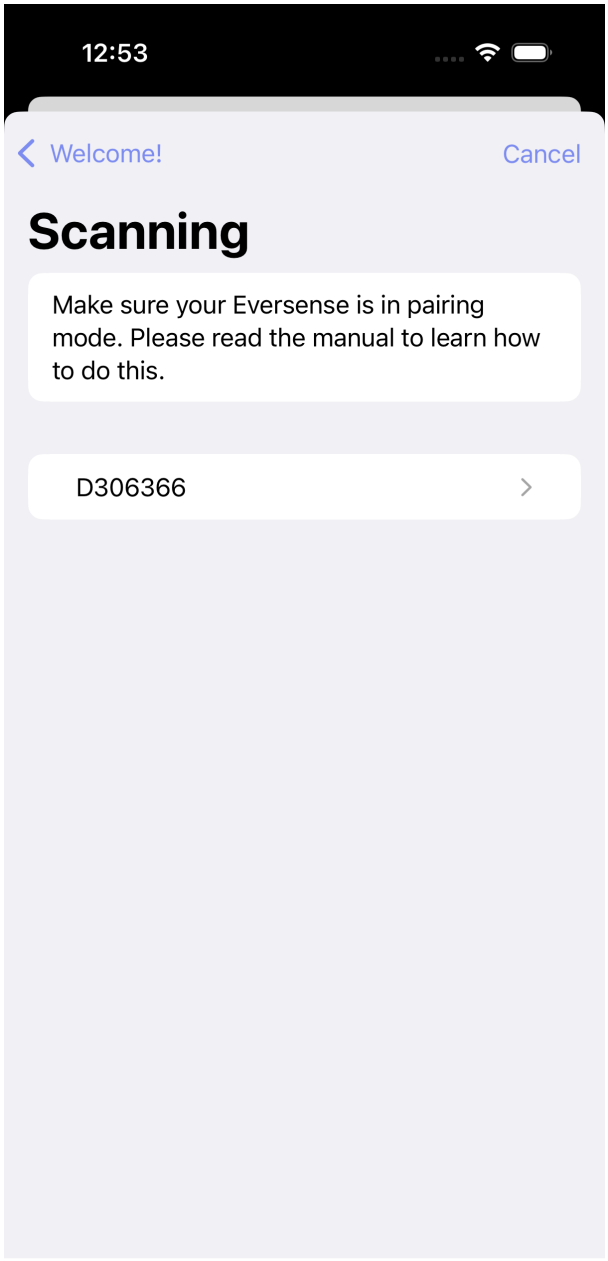
Step 3 Select whether you want to pair the Eversense E3 or the Eversense 365.



Step 4 (only for Eversense 365) Log in using your Eversense account. If you do not have an accunt or you forgot your password, the appropriate links are included in this step



Step 5 Now put your Eversense transmitter in pairing mode (done by tapping the button 3 times), and wait till your Serial Number shows up on screen. Click on the found transmitter, accept the pairing prompt (if show), and wait untill the pairing process is completed!



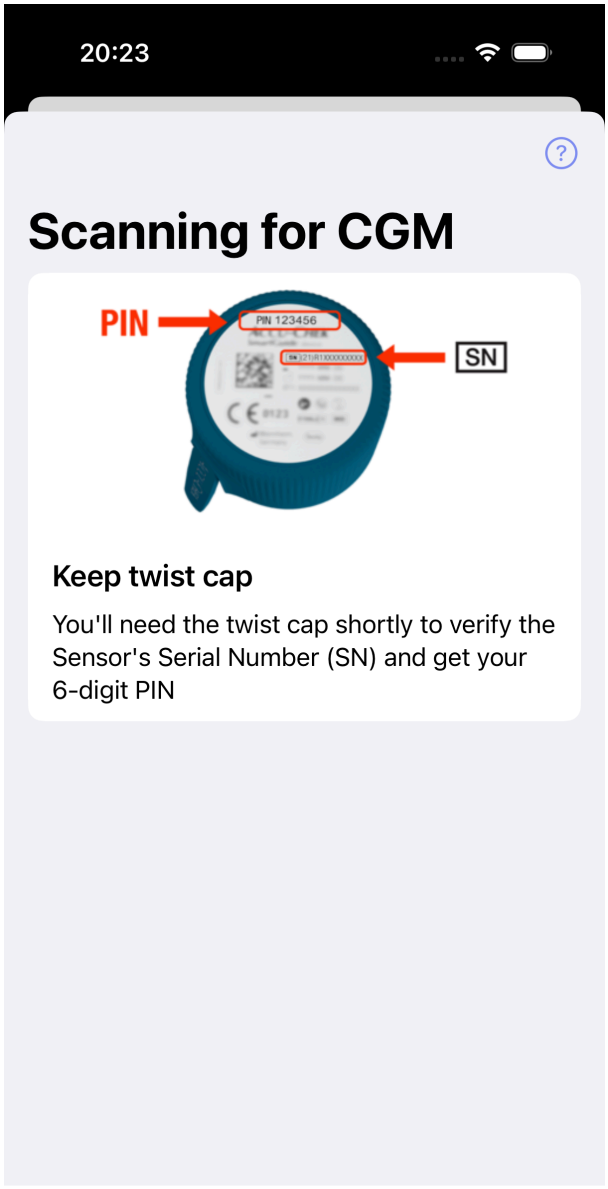
Step 6 Continue to Connect Watch *OR* return to New User Setup

---

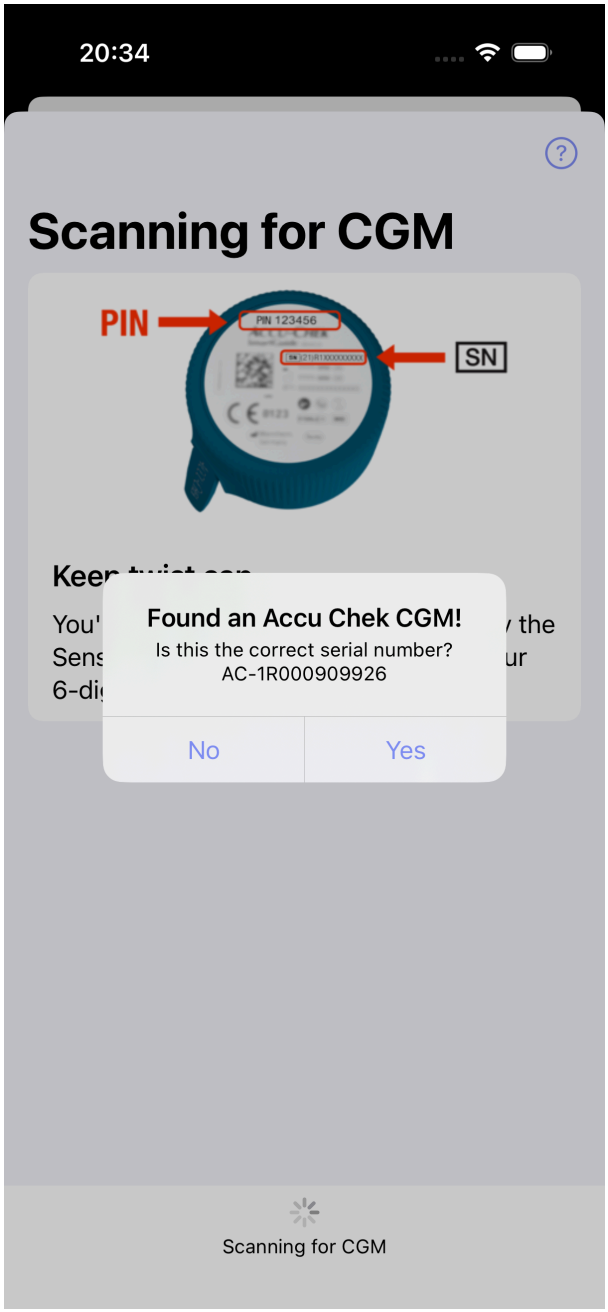
Accu-Chek SmartGuide

> **You must build the `dev` branch to use the Accu-Chek SmartGuide CGM at this time**
>
> The Accu-Chek SmartGuide CGM is in `dev` and is experimental for now

Step 3 Wait a minute while Trio is scanning for your Accu-Chek SmartGuide

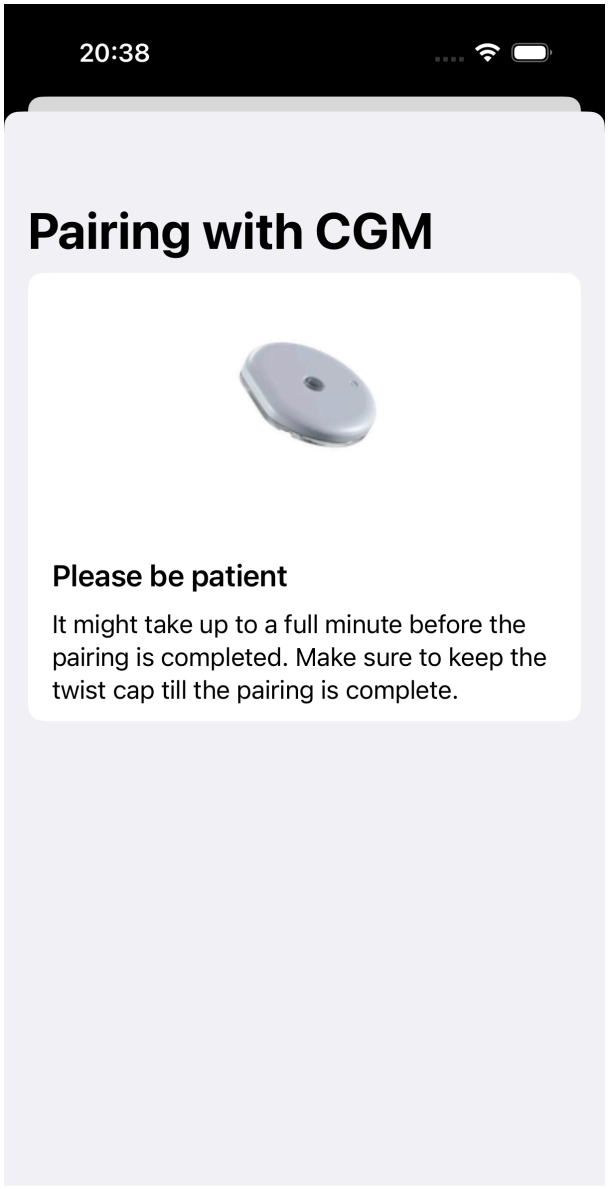


Step 4 Once Trio has found your Accu-Chek SmartGuide, it prompts you to confirm the Serial Number. This Serial Number can be found on the twist cap of your applicator.



Step 5 After you've confirmed the Serial Number, Trio will pair and configure your CGM. It might be the case that iOS asks for a Bluetooth pairing code, this code can be found at the top of the twist cap. After pairing, be sure to either never remove your CGM from the Bluetooth list or keep the twist cap/pincode somewhere safe.

Note you will need this pincode if you want to migrate to a new phone



Step 6 Continue to Connect Watch *OR* return to New User Setup

---

Glucose Simulator

> **Warning**
>
> If using a Glucose Simulator, it is important to understand:
>
> - You will only experience the user interface of Trio.
> - Using a Glucose Simulator does not indicate how the app will perform, nor will it give accurate guidance or suggestions for insulin dosing.
> - A Glucose Simulator should **NEVER** be used with a live pump connected to a living person or pet.
> - **Only use a Glucose Simulator if you understand the conditions above.**

Step 2 Continue to Connect Watch *OR* return to New User Setup

---

Nightscout as CGM

While using Nightscout as a CGM is an option, it should be avoided if possible because it does not keep Trio running in the background like other CGM options.

Step 2 Continue to Connect Watch *OR* return to New User Setup

---

xDrip4iOS

To use xDrip4iOS as a CGM source, you must build it yourself with the same Apple Developer account you used to build your Trio app. You cannot use Shuggah or a version distributed by someone else's TestFlight. Please see the following for instructions on how to build xDrip4iOS yourself: link

However, if you are using Dexcom G6 or ONE with xDrip4iOS, you can choose the Dexcom G6 option in Trio rather than xDrip4iOS, and Trio will intercept the glucose readings even if you're using Shuggah or someone else's TestFlight of xDrip4iOS.

Step 2 Continue to Connect Watch *OR* return to New User Setup

---

Smooth Glucose Value

Default: *OFF*

Smooths CGM readings using Savitzky-Golay Filtering

Smart Watch

Apple Watch

On an Apple Watch, you can add carbs and boluses, set a temporary target, and display BG, IOB, COB, etc.

Trio has a companion app for the Apple Watch.
The companion app shows the loop status and current insulin and carbs onboard. It also lets you add carbs, do a manual bolus, and set a temporary target. However, the companion app only serves as a remote control for the iPhone app; connection to the CGM and pump is still done through your iPhone, and you will have to bring it with you.

Apple Watch Series 3 and newer are supported.
Newer versions of the Apple Watch have better battery life and support an always-on display.

Confirm Bolus Faster

Contacts Configuration

Garmin

Garmin watches only display data and do not allow interaction for adding carbs or insulin. Two components can be installed on Garmin watches: a watch face that continuously displays information, and a data field that shows glucose information during activities. Both components are independent of each other and need to be sideloaded onto the watch as described below. No need for Garmin Connect in the background — the watch just needs to be connected to the phone that runs Trio. The watch face updates with Trio loop cycles.

Releases

As long as nothing is published in Garmin Connect IQ, every Garmin watch needs to be individually compiled. You can do this yourself using VS Code with the monkeyC plugin and a Garmin SDK. If not, see the link below for the released watches.

How to do it

1. Choose a folder on your Mac and get the latest release from this GitHub repo.
2. You need `*.prg` files for your watch model from the release and OpenMTP from OpenMTP to transfer them to the watch.
3. Install and start OpenMTP.
4. Connect your Garmin watch via a data USB cable to the MacBook and allow the USB connection. The Garmin watch USB mode should be set to MTP.
5. Transfer `*.prg` files for your watch model to the `/GARMIN/APPS` folder.
6. Wait for the transfer to finish and disconnect the watch.

7. Add the Garmin watch in Trio settings: `Settings -> Watch -> Garmin Watch -> Add Devices`.
8. Choose the Trio watch face on the watch and enjoy.

Hints

The watch face will not display information after another has been used. In that case, you have to uninstall the Trio watch face and reinstall it. No similar side effects occur with the data field.

If your watch is not listed in the Release section, please ask about it in the Garmin channel on Discord.

Therapy

Welcome to the Therapy Settings homepage! Here you will find the "Trio" of core settings, your basal rates, ISF, and CR. You'll also find the units and targets essential for a functioning Trio closed loop.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Units and Limits

Set your preferred glucose units and main dosing limitations

Glucose Targets

Step by step guide on setting your glucose target(s)

Basal Rates

Step by step guide on setting your basal rates

Carb Ratios

Step by step guide on setting your carb ratio(s)

Insulin Sensitivities

Step by step guide on setting your insulin sensitivity factor(s)

Units and Limits

Highlights

- Max IOB = Increase above 0 *before closing the loop*
- Max Bolus = Max bolus you give for a meal
- Max Basal = 4x highest hourly basal
- Max COB = Maximum carbs active at any given time

Glucose Units

Select either mg/dL or mmol/L.
All settings descriptions and screen information will be adjusted to match your selection here.

Max IOB

Default: *0 units*

Important Information

This must be set to a value greater than 0 for any insulin to be administered above your current set basal rate.
If this remains at 0, Trio can suspend your insulin to prevent lows, but can only administer enough insulin to return you to your current basal rate amount.

This is the maximum amount of Insulin on Board (IOB) above your profile basal rates from all sources - positive temporary basal rates, manual or meal boluses, and SMBs - that Trio is allowed to accumulate to address an above target glucose.

If a calculated amount exceeds this limit, the suggested and/or delivered amount will be reduced so that active Insulin on Board (IOB) will not exceed this safety limit.

Tip

You can still manually bolus above this limit, but the suggested bolus amount will never exceed this in the bolus calculator

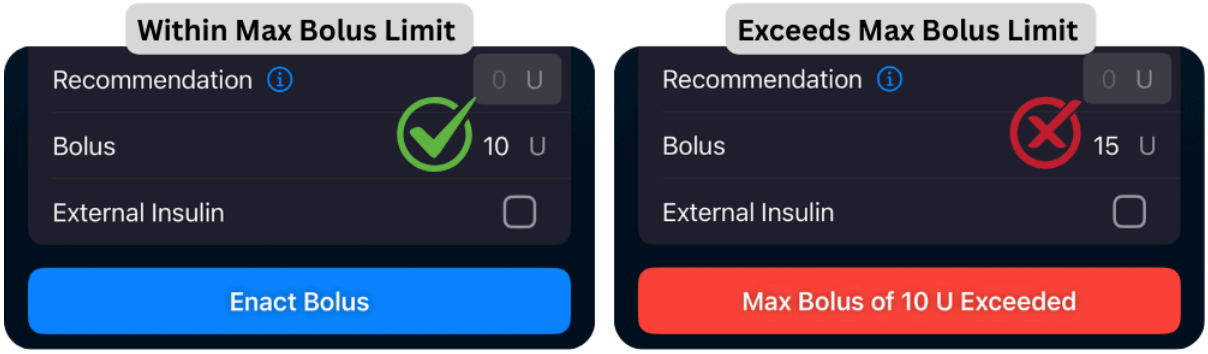
---

Max Bolus

Default: *10 units*
Setting Limits: *1-30 units*

This is the maximum bolus allowed to be delivered at one time. This only limits manual boluses given on the Treatments screen.

If you attempt to request a bolus larger than this, the bolus will not be accepted or proceed. So if your max bolus is set to 5U but you enter a 6U bolus, the "Enact Bolus" button will turn red and display a warning message. You will be unable to proceed until you lower your bolus to within this limit.



> **Tip**
>
> Most set this to their largest meal bolus, then adjust if needed.

Max Basal

Default: *2 units*
Setting Limits: *0.5-30 units*

This is the maximum basal rate allowed to be set or scheduled. This applies to bothe automatic temp basal rates, profile basal rates, and manual temp basal rates.

> **Tip**
>
> Generally, users enter a value that is 4 times their highest hourly basal rate

> **For Medtronic Users:**
>
> You must also manually set the max basal rate on the pump to match this value

---

Max COB

Default: *120 grams of carb*
Setting Limits: *0-300 g*

This setting defines the maximum amount of Carbs On Board (COB) allowed at any given time for Trio to use in dosing calculations. If more carbs are entered than allowed by this limit, Trio will cap the current COB in calculations to this Max COB setting and remain at this max until all remaining carbs have shown to be absorbed or 6 hours has passed, whichever comes first.

For example, if Max COB is 120g and you enter a meal containing 150g of carb, your COB will remain at 120g until the remaining 30g have replaced absorbed carbs.

Minimum Safety Threshold

Default: *Set By Algorithm*
Setting Limits: *60 - 120 mg/dL*

Trio uses a Safety Threshold to prevent insulin dosing when your current glucose reading is too low. This threshold is always active, but you can increase this threshold from the system-determined value by entering a value into this setting in Trio.

The system-determined value is based on this calculation:

$$Target\ Glucose - \frac{Target\ Glucose - 40}{2}$$

The threshold is a safety limiter function. If blood sugar at any point is predicted to go below this value, Trio will suspend insulin delivery (SMBs are halted and Temp Basal of 0 U/hr set) and wait until forcasting says otherwise. Increasing this setting can be useful if you are experiencing a high number of hypoglycemia events. Please review the OpenAPS documents if you want a better understanding of how it is used.

This setting allows you to choose a higher threshold setting than the default. Note that you cannot choose something lower than the default setting for a certain blood glucose target.

Bill has set a BG target of 110 mg/dl. In his Trio Dynamic Settings, he has set his threshold to 65 mg/dl. Will Trio use the default threshold or the minimum safety threshold he set?

Here are the formulas you'll need:

Default Safety Threshold:

$$Target\ Glucose - \frac{Target\ Glucose - 40}{2}$$

Then, compare that value to the Minimum Safety Threshold

Default Safety Threshold $>$ or $=$ or $<$ Minimum Safety Threshold

Now, enter in Bill's values

$$110 - \frac{110 - 40}{2} =$$

$$110 - \frac{70}{2} =$$

$$110 - 35 =$$

$$65\ mg/dL$$

$$75\ mg/dL > 65\ mg/dL$$

Answer

Because Trio's default threshold setting is 75 mg/dL for a 110 mg/dL blood glucose target, and that is greater than Bill's Minimum Safety Threshold, Trio will use the higher target of ***75 mg/dL*** and ignore this setting.

Bonus Question: Assuming Bill's target stays at 110 mg/dL, what would Bill have to set his Minimum Safety Threshold to for it to be used by Trio?

$$\geq 75\ mg/dL$$

Tip

Basal may be resumed if there is negative IOB and glucose is rising faster than the forecast

Return to New User Setup

Glucose Targets

Trio will target this value when calculating insulin needs. It should be set to the blood glucose you would like to reach when corrections are made. A recommended value is between 90-110mg/dL or 5-6mmol/L

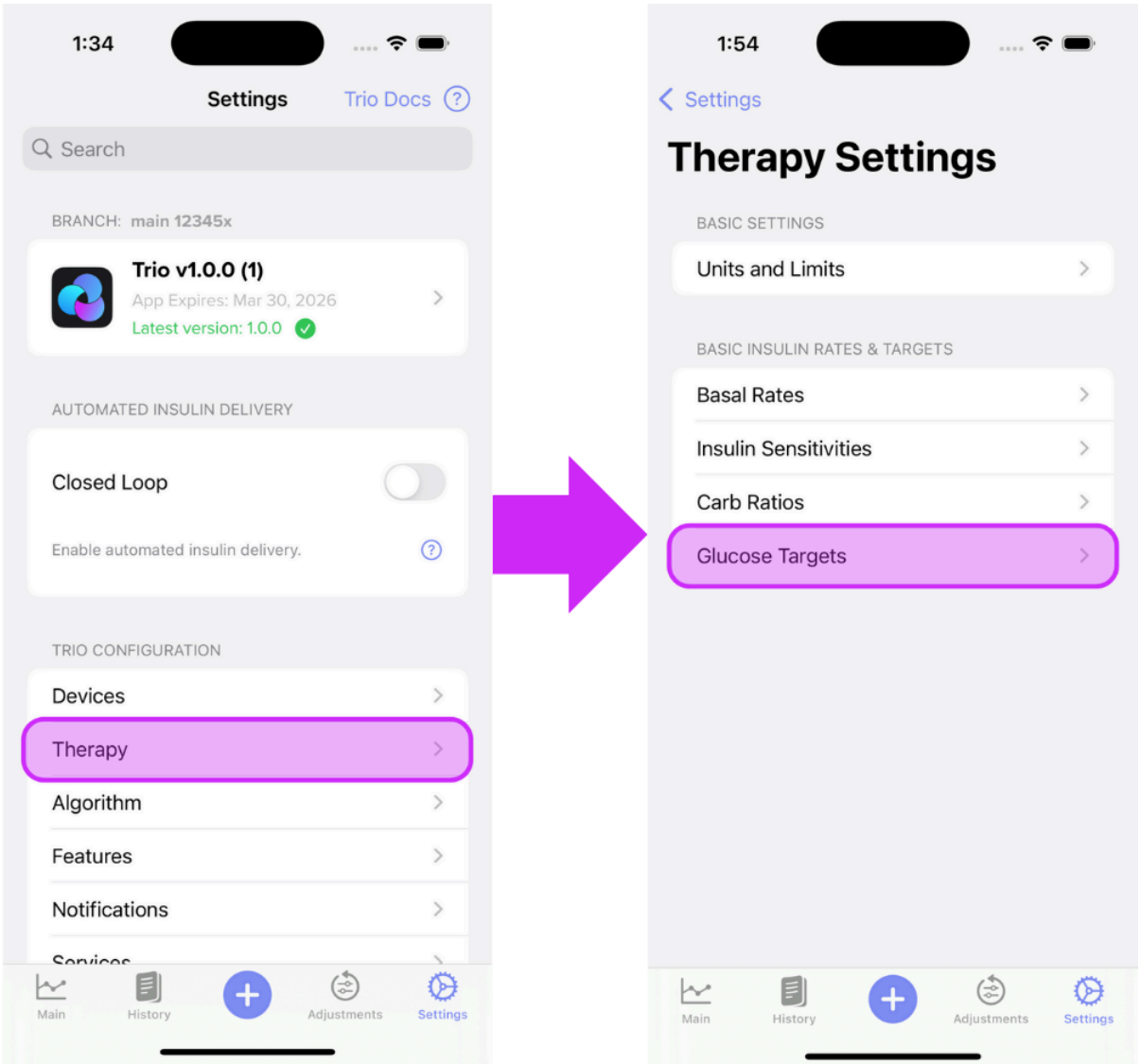
Tip

You can set different targets at different times.
Some users only have one target.
Others have a lower target during the day and a higher target during the night to avoid lows.

How to Enter Your Glucose Targets Into Trio

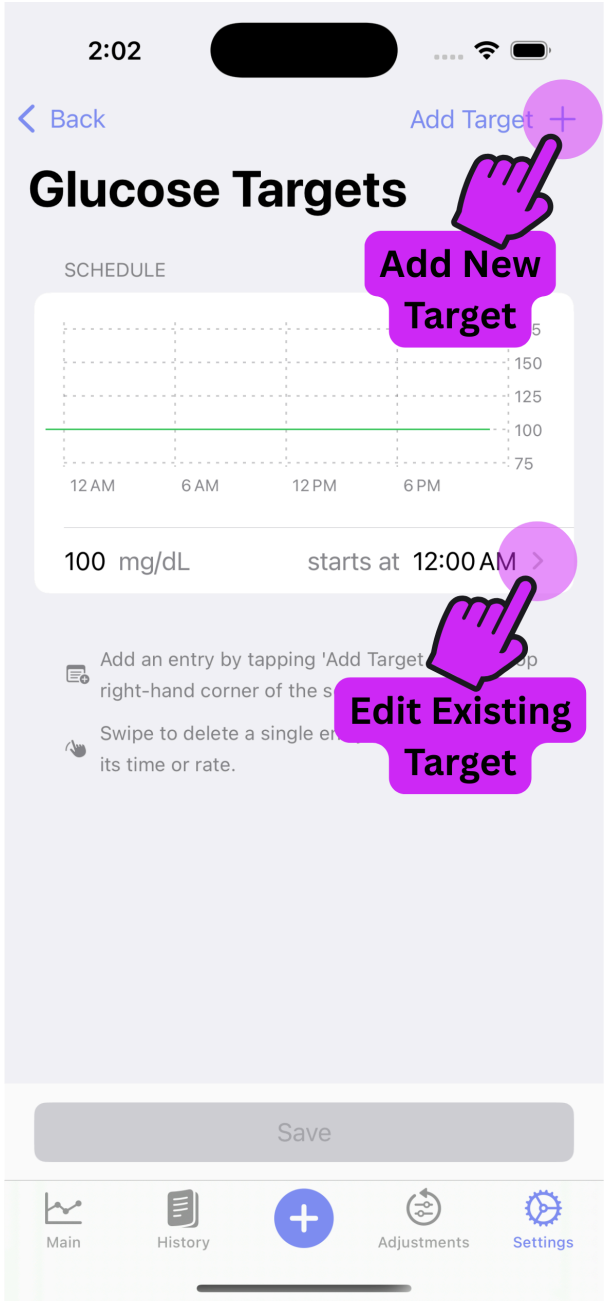
Step 1

Enter the Glucose Targets screen



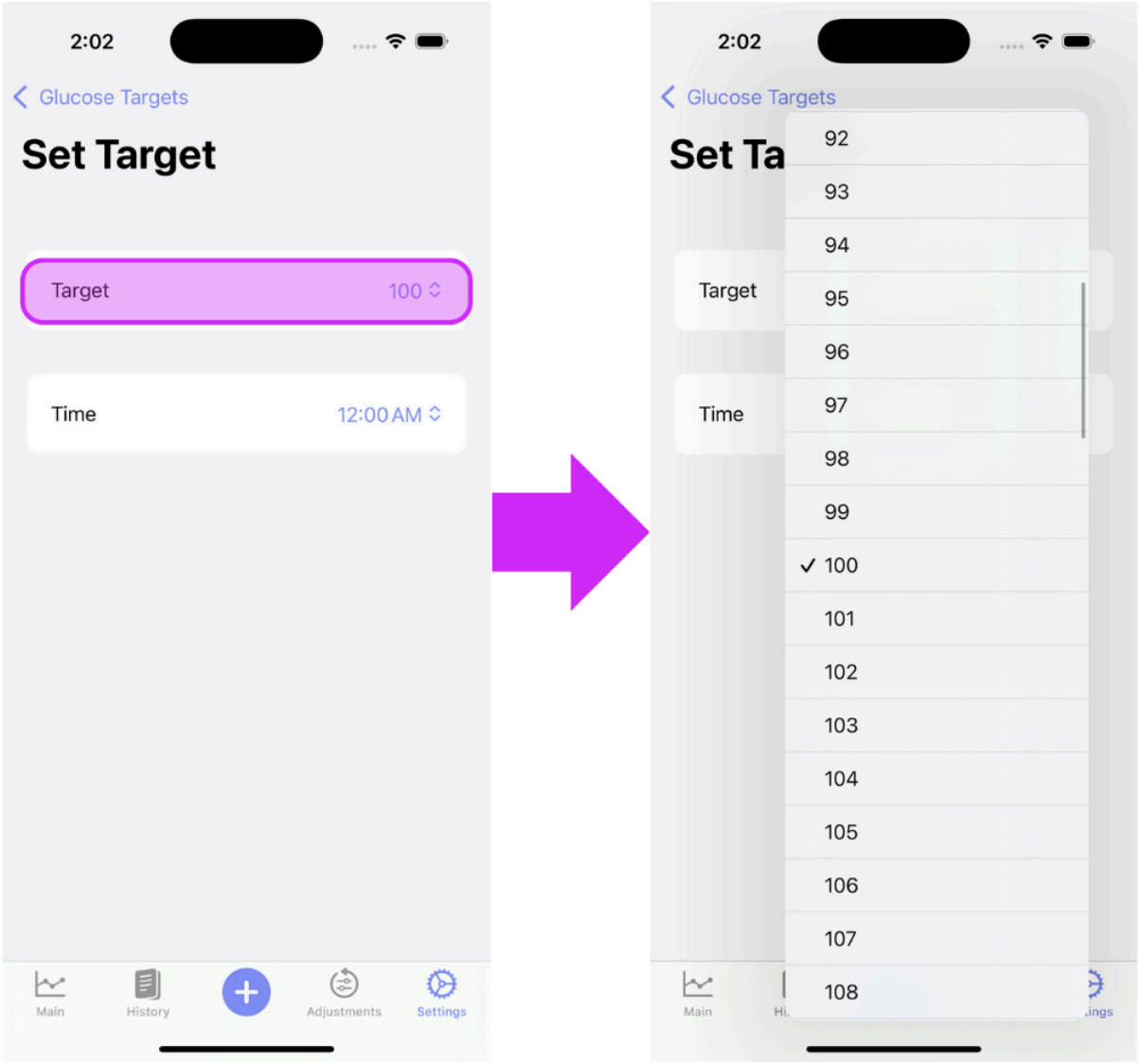
Step 2

Tap the "Add Rate +" on the top right of the screen until you have the number of targets you require. Then, edit each target by tapping the arrow to the right of the Glucose Target.



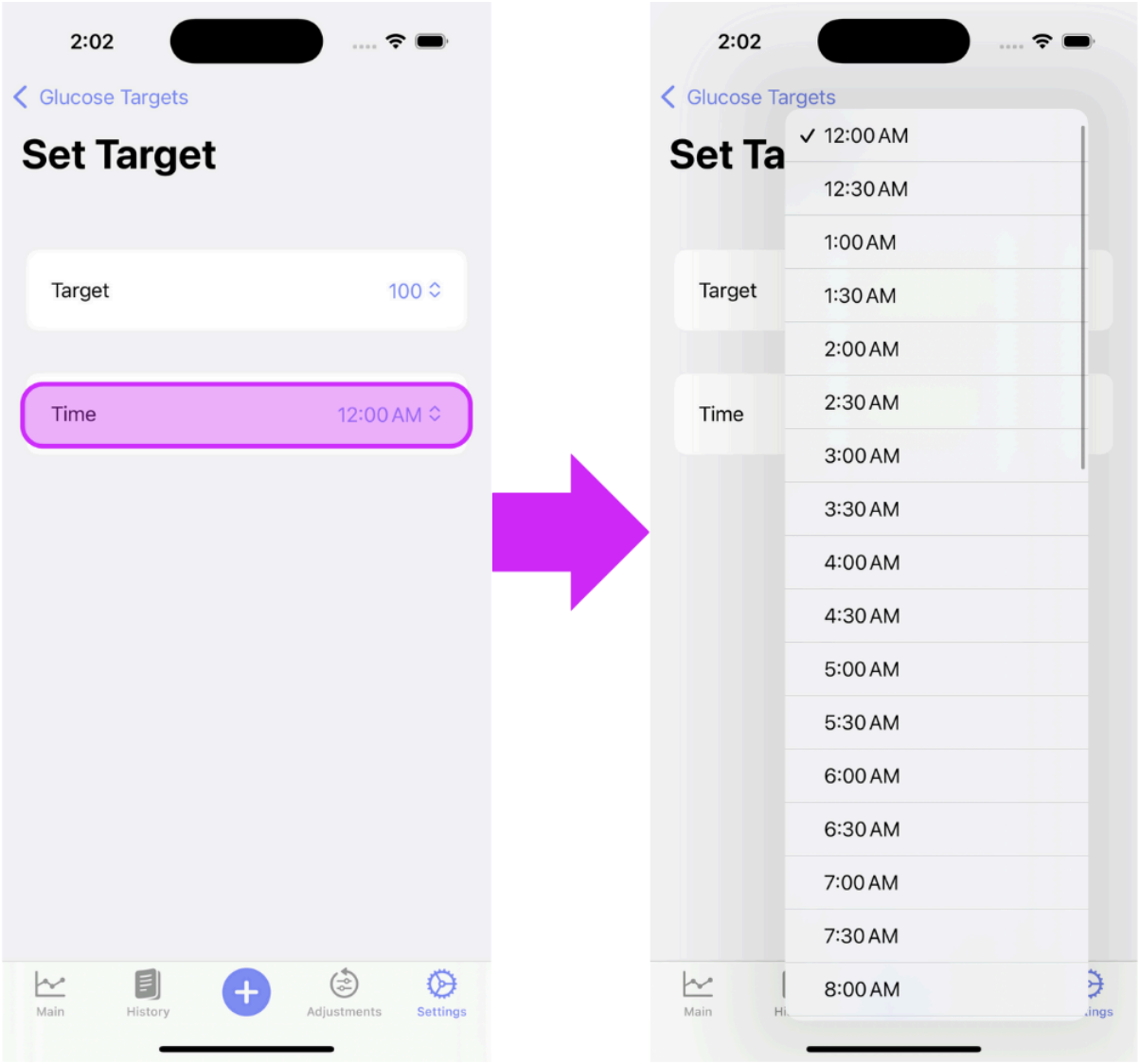
Step 3

Adjust the target glucose value



Step 4

Adjust the time

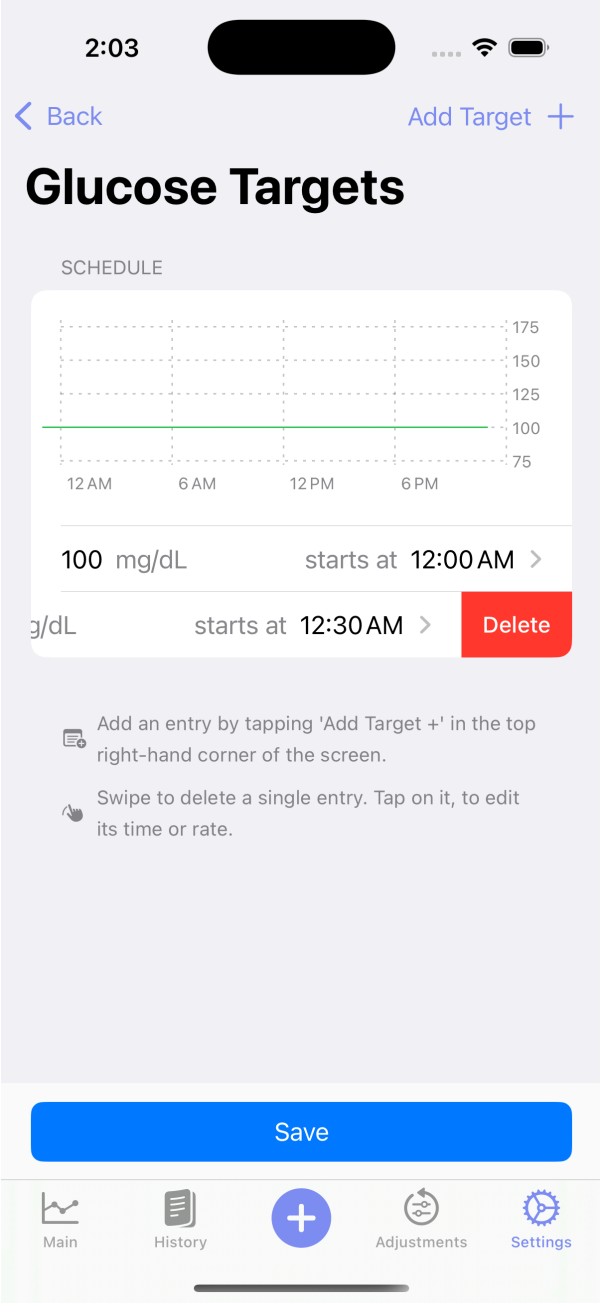


Step 5

Repeat Steps 2, 3, and 4 until all CRs are set

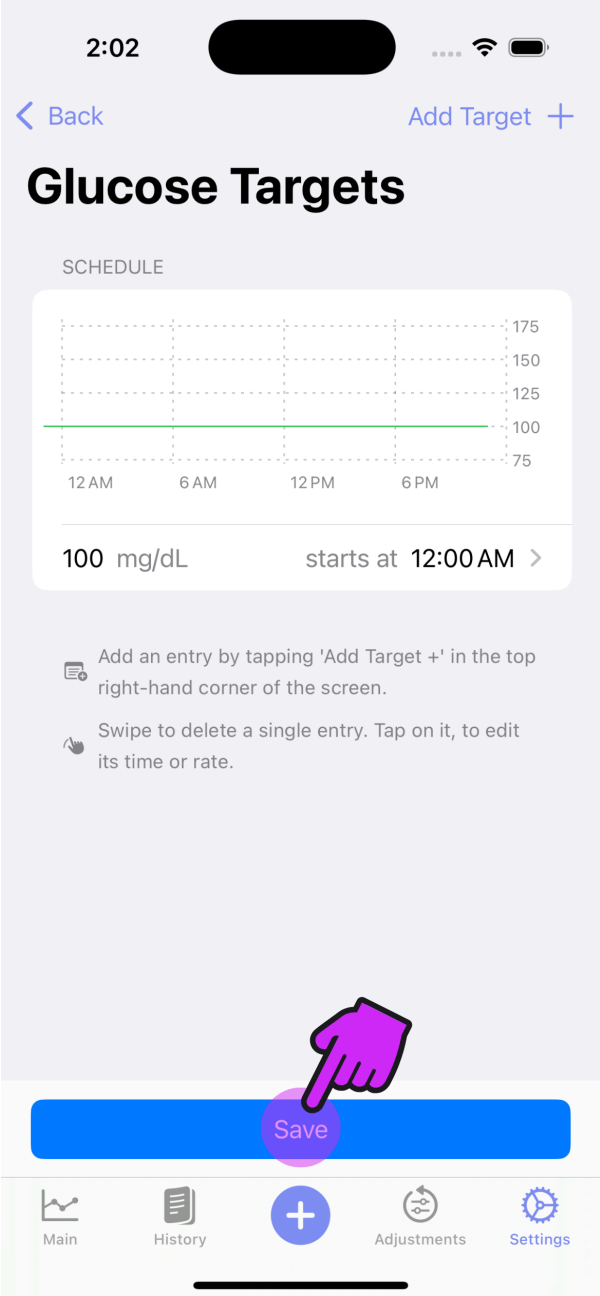
Delete a Glucose Target Entry

Should you need to delete an CR entry, just swipe left on the rate you need to remove.



Step 6 **IMPORTANT**

Save your changes!



Step 7

Proceed to Basal Rates or return to New User Setup

Basal Profile

Highlights

- Basal profile can be imported from a Medtronic pump or Nightscout account
- Walsh calculation may help you with a starting basal rate to adjust from
- Adjust your basal profile by doing fasting experiments and reviewing your IOB at times of fasting

Basal profile consists of your scheduled basal rates, which determine how much insulin is being delivered at each hour of the day. It is important to understand that these settings are not taken verbatim when looping with Trio. Based on your current blood sugar reading, they are adjusted every loop cycle and replaced with temporary basal rates. Your set values are altered by autosens or dynamic ISF based on your historical data.

Your basal profile values should be near your true value. Basal profiles are also important for insulin on board (IOB) calculations. Trio treats your scheduled basal profile as the zero point. The calculated IOB increases if you receive additional insulin on top of your basal rates, either as boluses or high temporary basal rates. Likewise, if you receive low temporary basal rates for a set period, your IOB decreases and can even become negative.

If you are coming from a pump, transferring basal profiles from your pump settings should be done with consideration and caution. They might not be entirely accurate for Trio. If you are going low or high while fasting, consider adjusting according to the instructions below

---

Testing/Adjusting Your Basal Rate

Baseline Calculation

If your current basal rates are close, but need some testing and adjustment, skip to the next section.

If your current basal rates are inaccurate or you are unsure where to even start, the formulas developed by Walsh, et.al. may help you find a starting point to then test or adjust your basal rates. Autosens and dynamicISF use these calculations as their foundation for making adjustments, so it stands to reason that a similar balancing of these settings would assist Trio in a more optimal performance.

Warning

This calculation is to be used as a starting point for testing and is not considered definitive or exact.

Basal Testing

The standard method is to test your basal by having a relaxing 4-6 hours without eating at least two hours before you begin the test. Does your blood sugar stay steady? Or do you climb and need a correction? Or do you go low and need to eat? Setting accurate basal rates is crucial for Trio success. They determine how much of the insulin delivered (from basal and bolus) is counted as insulin on board (IOB).

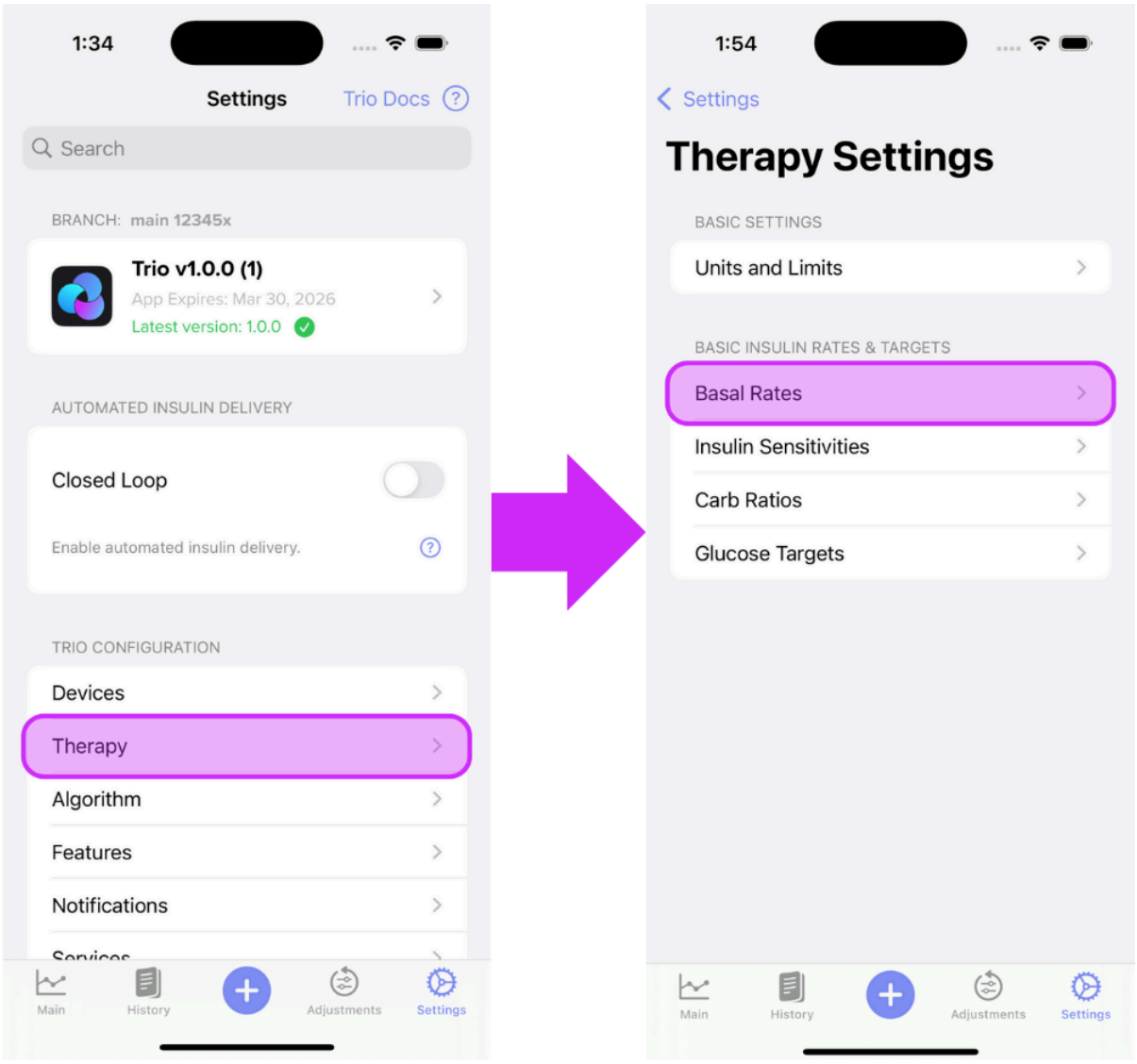
Basal Adjustment

You can also monitor your IOB to determine if your basal profile is accurate. Consistently negative or positive IOB during times of fasting *may* suggest that these hours need to decrease or increase their basal rate, respectively.

How To Enter Your Basal Profile(s) Into Trio

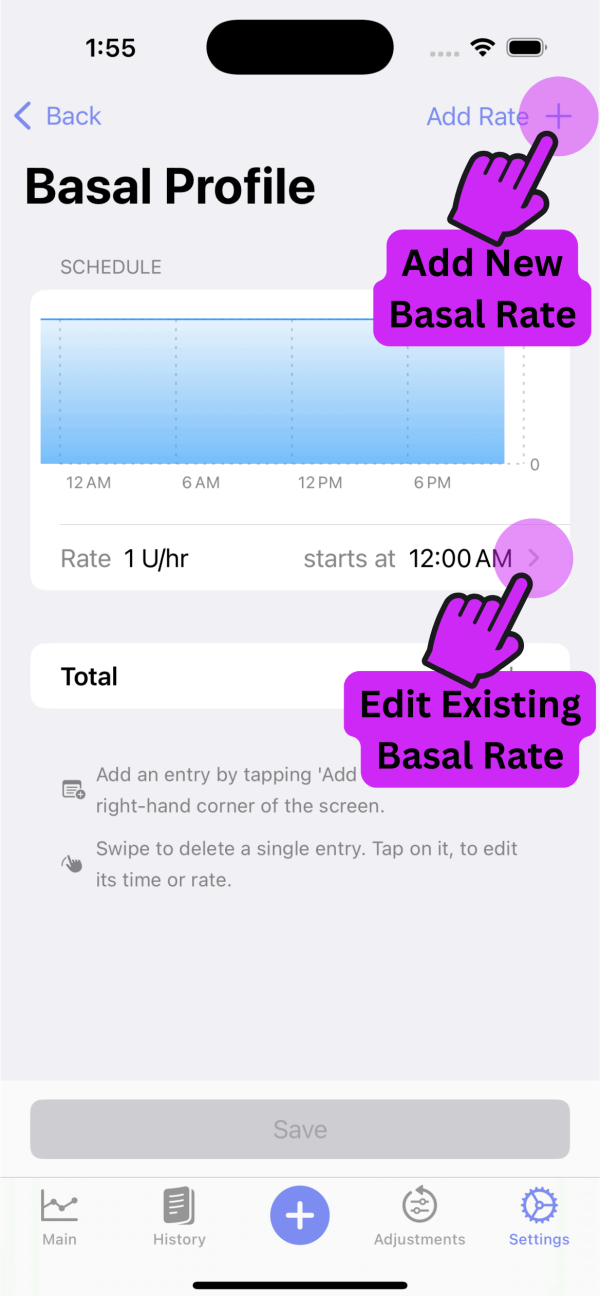
Step 1

Enter the Basal Profile screen



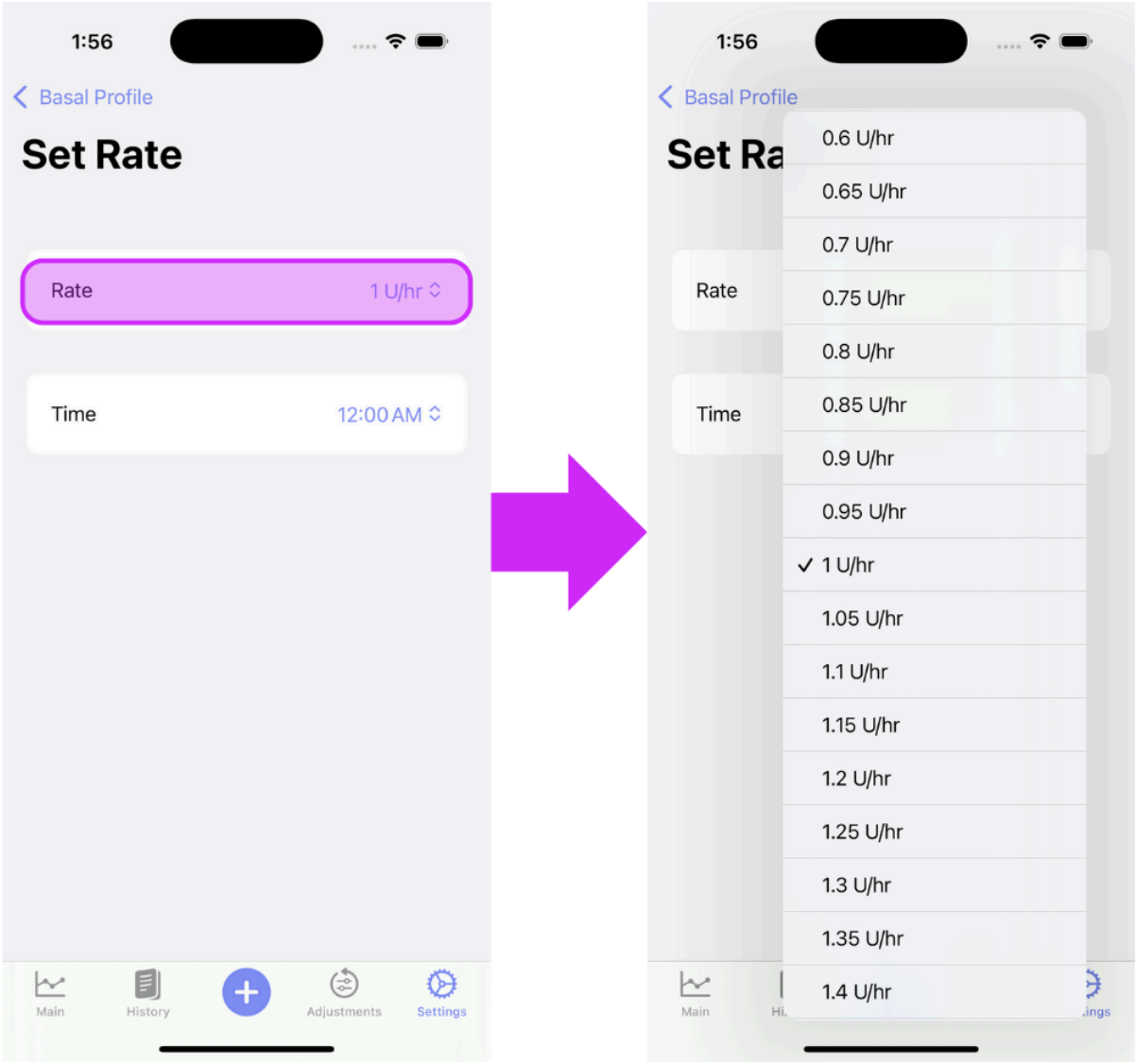
Step 2

Tap the "Add Rate +" on the top right of the screen until you have the number of basal rates you require. Then, edit each rate by tapping the arrow to the right of the basal rate.



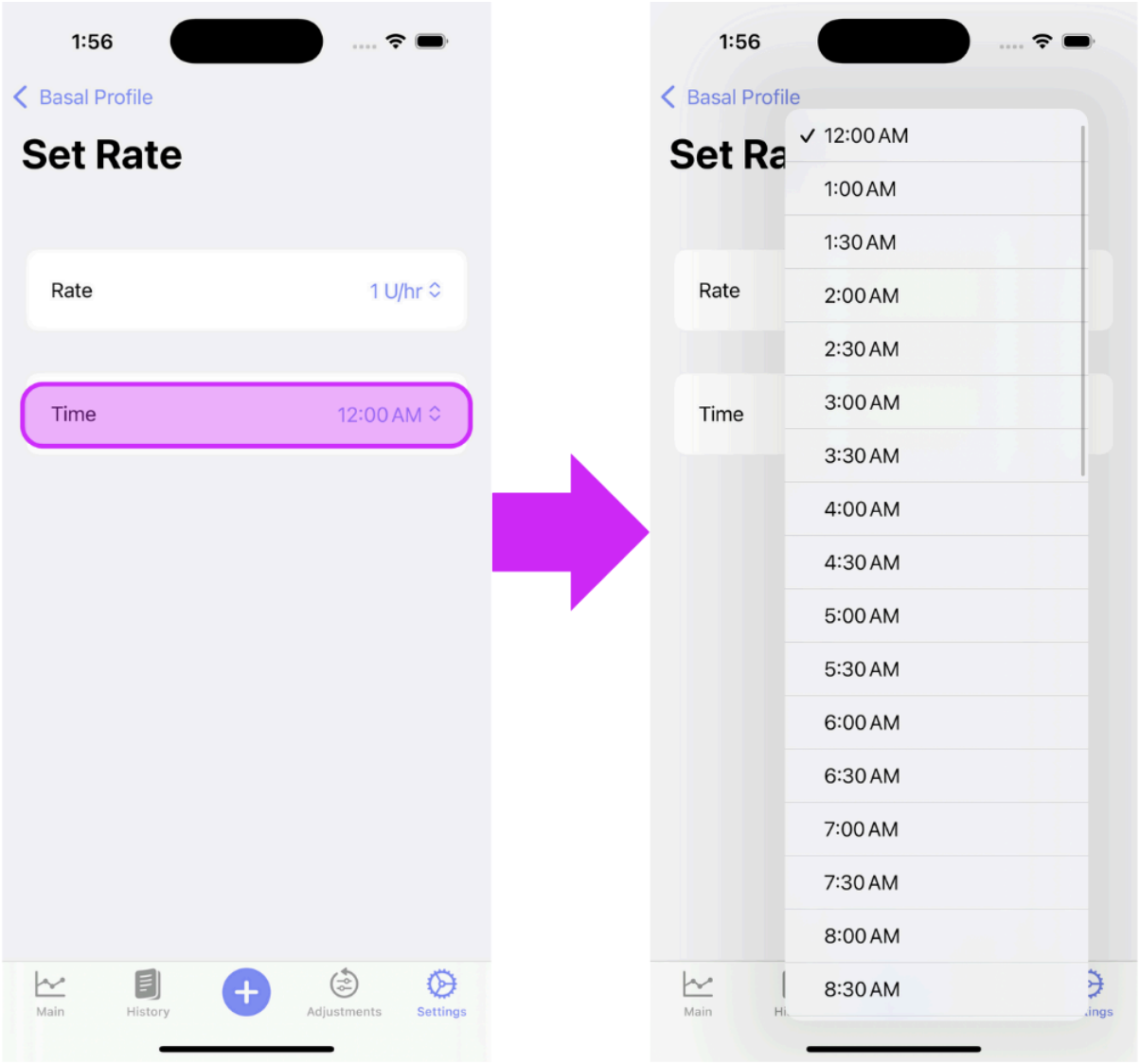
Step 3

Adjust the rate



Step 4

Adjust the time

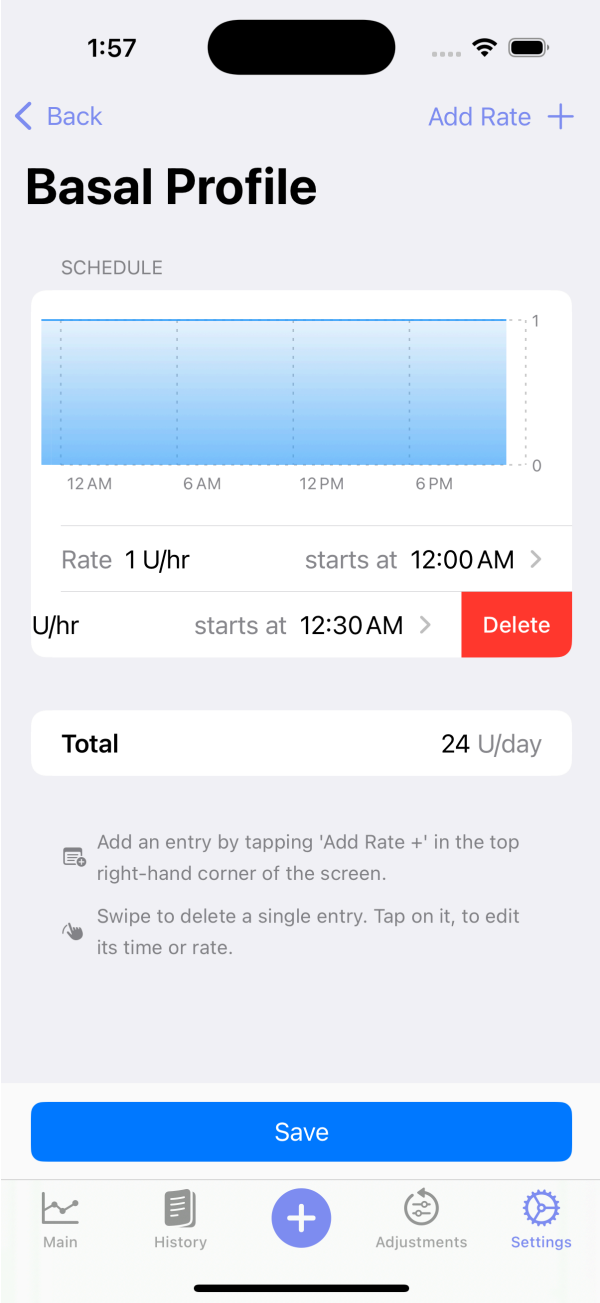


Step 5

Repeat Steps 2, 3, and 4 until all basal rates are set

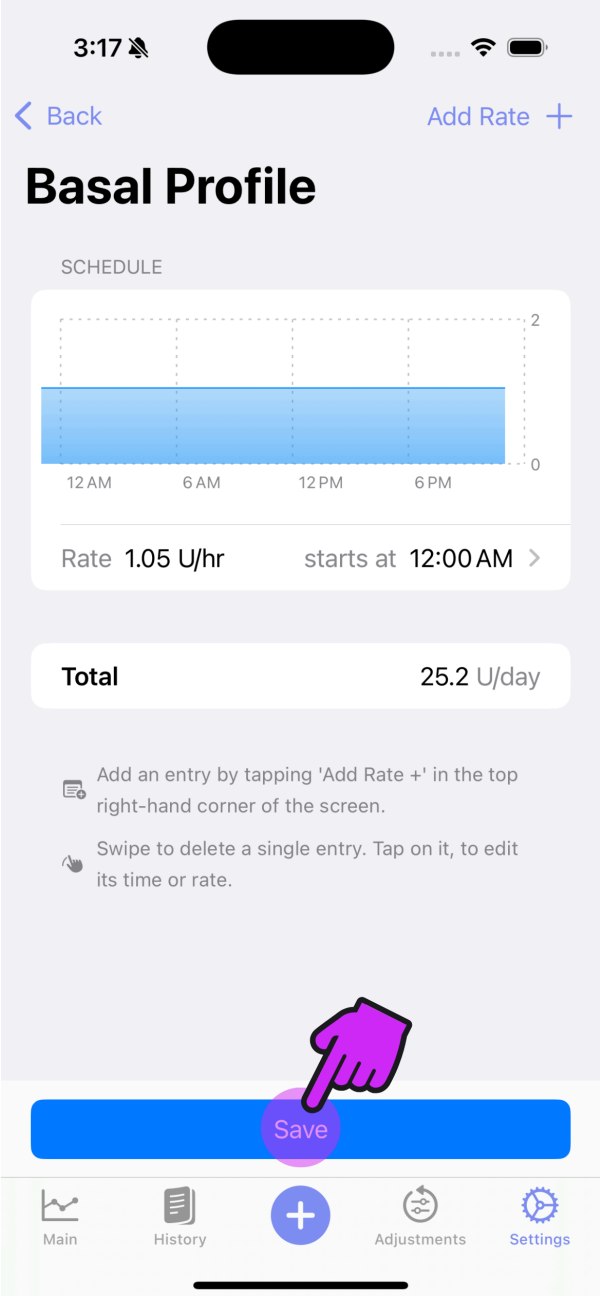
Delete a Basal Rate

Should you need to delete a basal rate, just swipe left on the rate you need to remove.



Step 6 **IMPORTANT**

Save your changes!



Step 7

Proceed to Carb Ratios or return to New User Setup

Carb Ratio (CR)

Highlights

- CR can be transferred from your pump
- Adjust your CR by performing a test meal experiment

CR refers to the amount of carbohydrates one unit of insulin is able to neutralize.

Bill has a CR of 10g/U. If Bill eats a meal containing 50g of carbs, how much insulin does Bill need for the entire meal?

Here is the formula you will need:

$$\frac{grams\ of\ carb\ eaten}{CR}$$

Calculate Bill's Insulin Dose:

$$\frac{50}{10} =$$

$$5\ units$$

Answer

Bill needs 5 units of insulin for his meal.

CR is not changed like basal rates or ISF. Your CR must be as accurate as possible for proper Trio function.

It is safe to transfer your CR from your pump settings. However, your settings may not be accurate if you are experiencing high peaks with meals or lows three hours afterward. If you have SMB/UAM on and are experiencing sharp drops, you may also need to optimize your ISF.

Testing/Adjusting Your Carb Ratio (CR)

Baseline Calculation

If your current carb ratio is close, but needs some testing and adjustment, skip to the next section.

If your current carb ratio is inaccurate or you are unsure where to even start, the adjustments in Trio are based on formulas developed by Walsh, et.al. and may help you find a starting point to then test or adjust your carb ratio.

> **Warning**
>
> This calculation is to be used as a starting point for testing and is not considered definitive or exact.

CR Testing

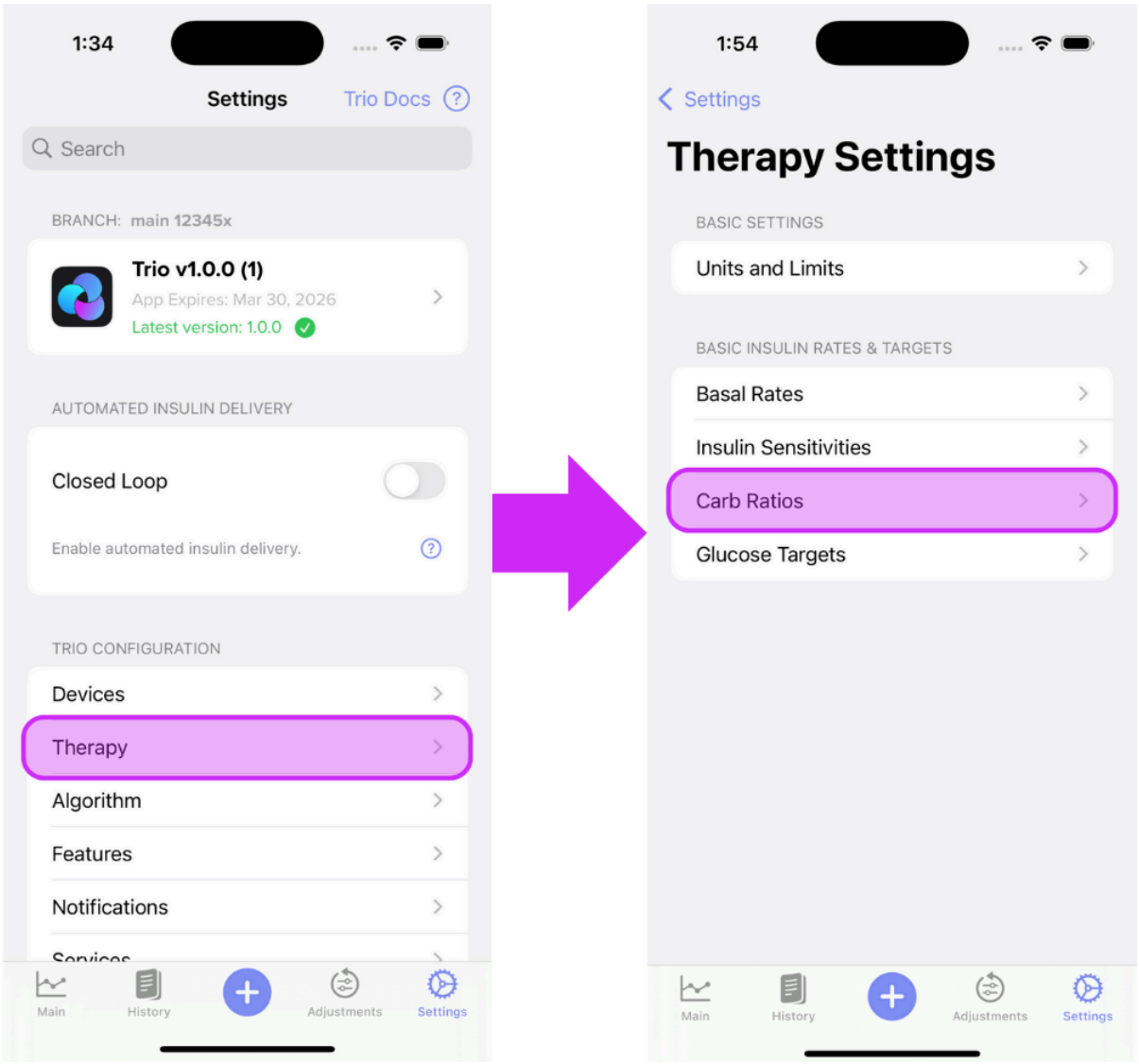
CR Adjustment

The standard way to adjust your CR is a test meal experiment, which can be done while not looping. Have a meal with a known amount of carbohydrates and bolus according to your current CR. Monitor your blood sugar at the three-hour mark; did you go high, low, or end up where you started prior to the meal? If you end up high, you can make your CR more aggressive by *DECREASING* the value. If you were low, make your CR less aggressive by *INCREASING* the value. You may also look to increase or decrease your adjustment factor if you have dynamic CR on.

How To Enter Your Carb Ratios (CR) Into Trio

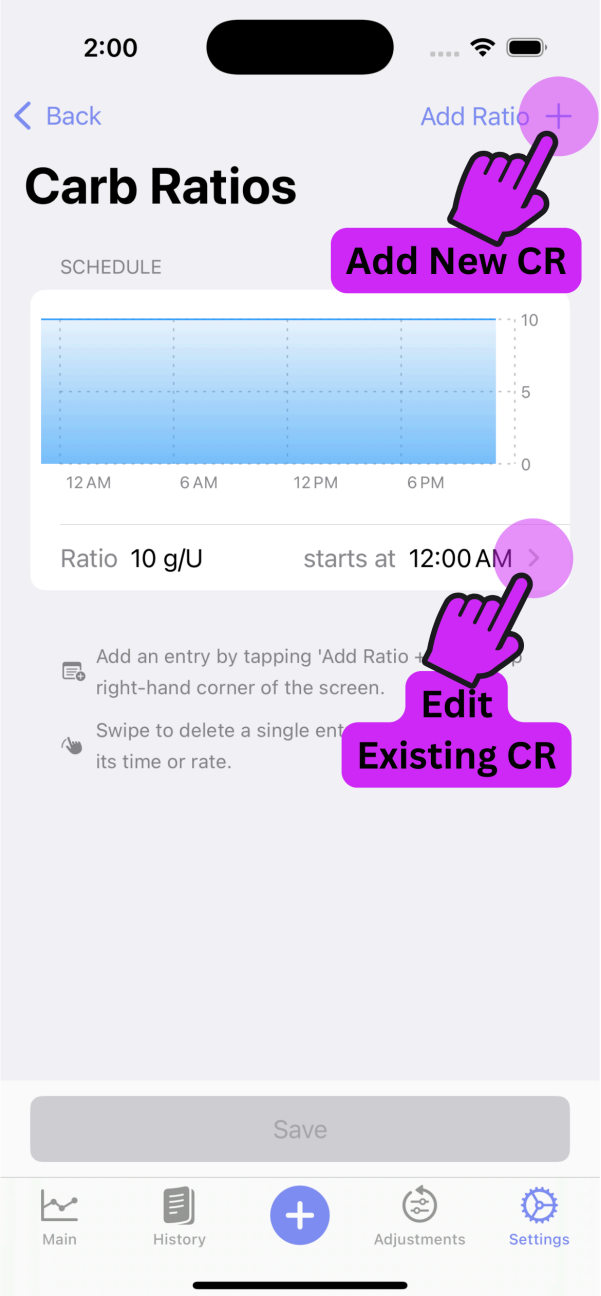
Step 1

Enter the CR Profile screen



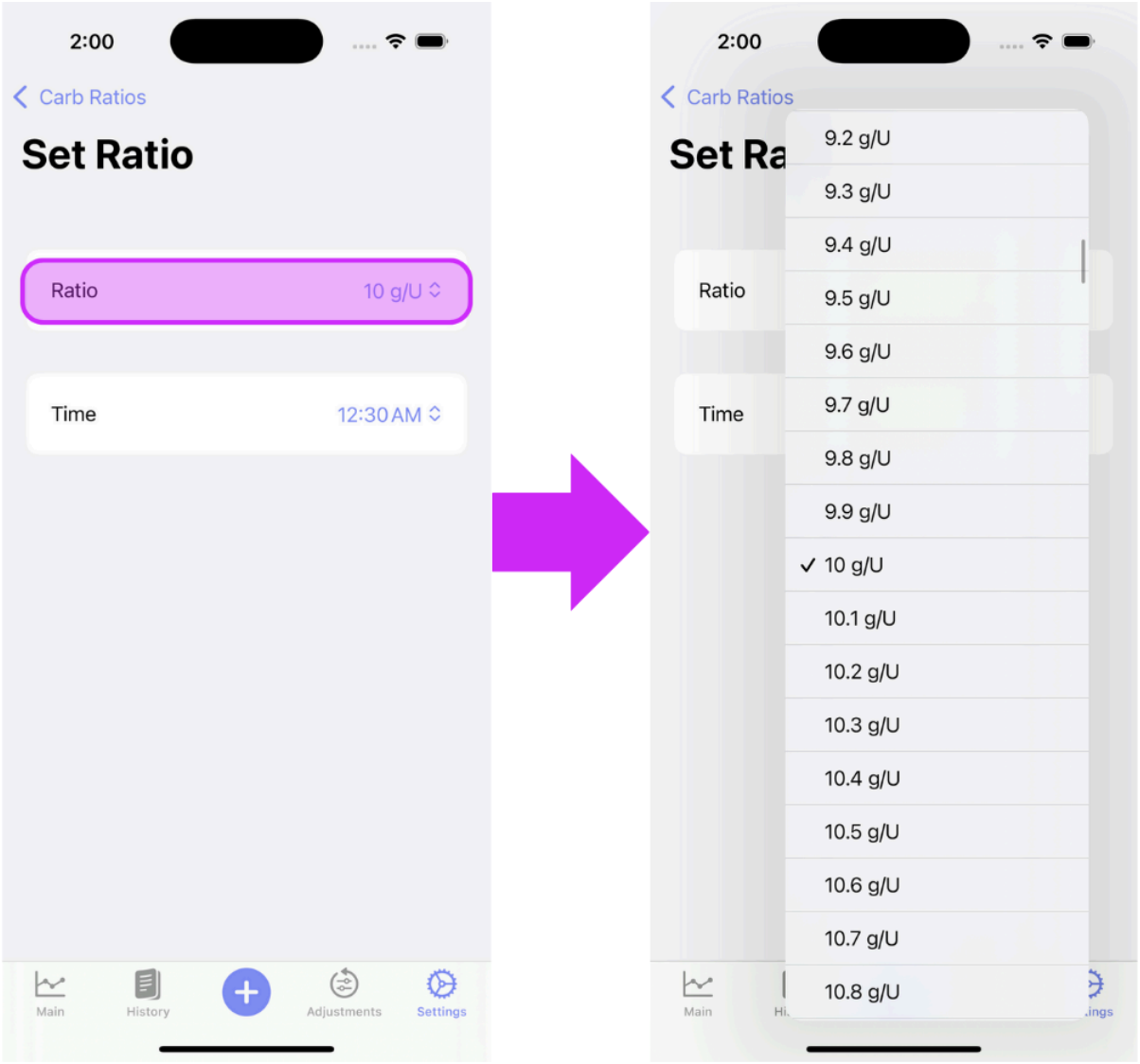
Step 2

Tap the "Add Rate +" on the top right of the screen until you have the number of CRs you require. Then, edit each rate by tapping the arrow to the right of the CR.



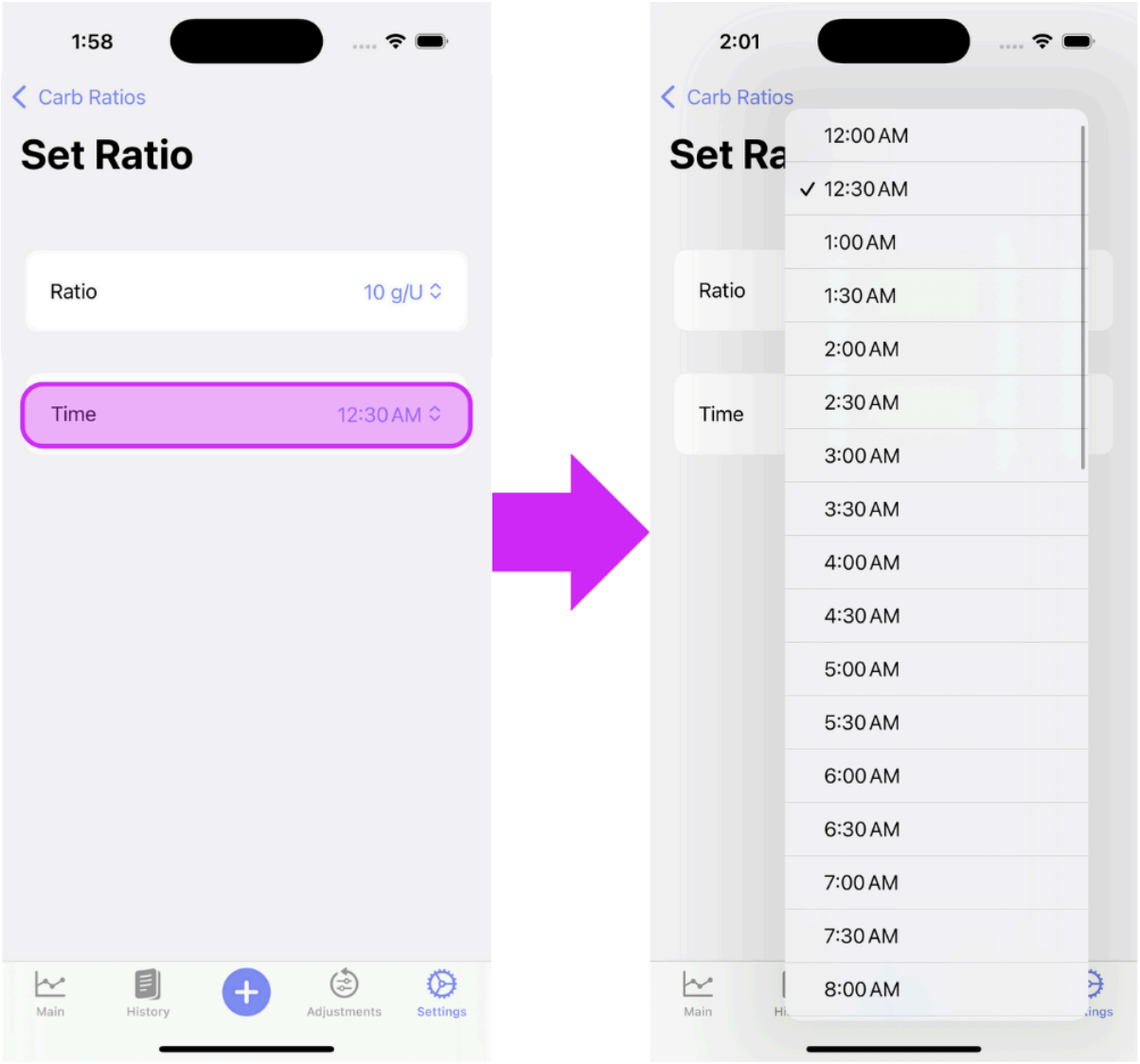
Step 3

Adjust the rate



Step 4

Adjust the time

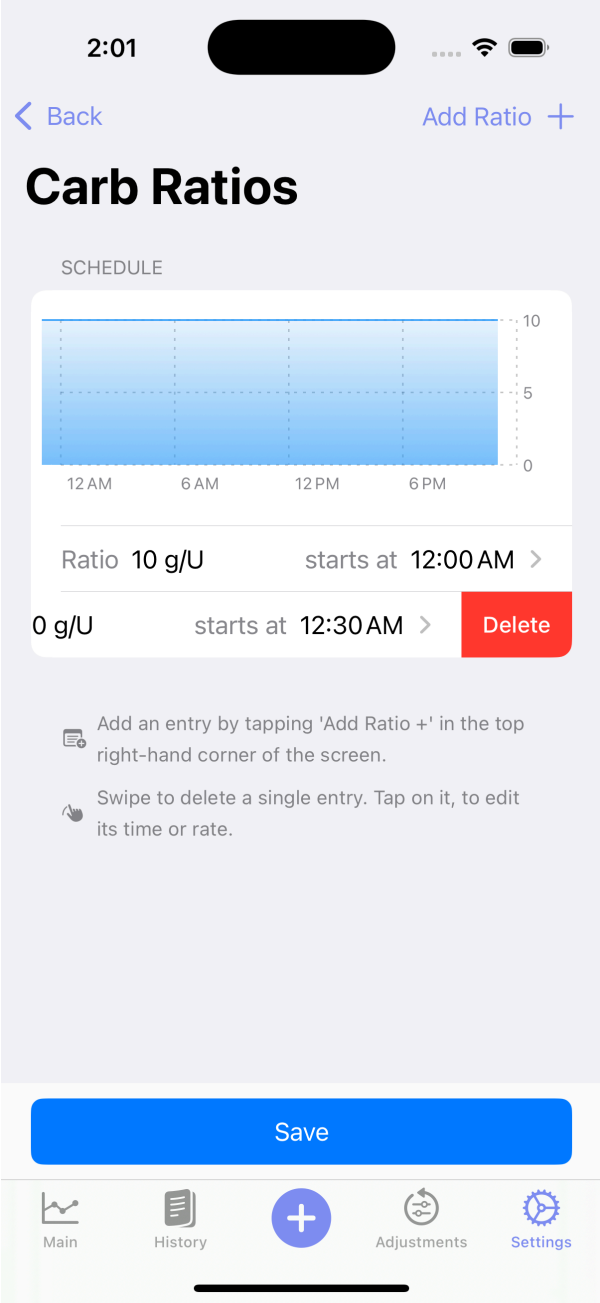


Step 5

Repeat Steps 2, 3, and 4 until all CRs are set

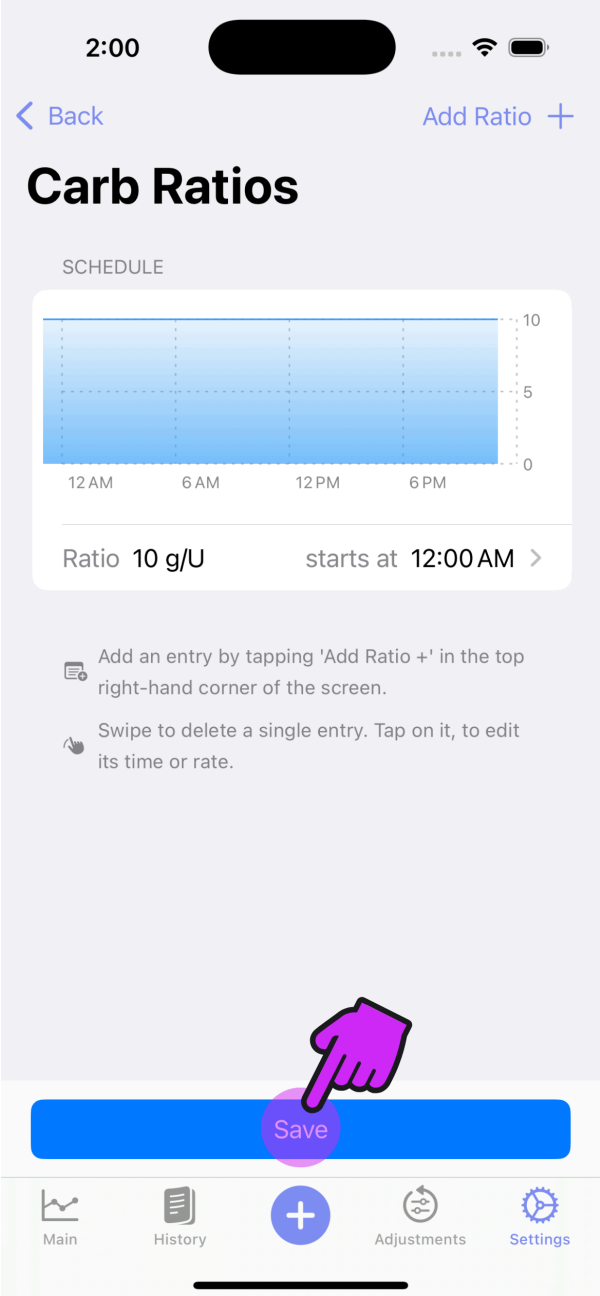
Delete an ISF Entry

Should you need to delete an CR entry, just swipe left on the rate you need to remove.



Step 6 **IMPORTANT**

Save your changes!



Step 7

Proceed to Insulin Sensitivities or return to New User Setup

Insulin Sensitivity Factor (ISF)

Highlights

- ISF is the most problematic setting for new loopers.
- ISF can be transferred from your pump to start.
- Adjust ISF by performing a correction and seeing how far your glucose drops in 4 hours.

ISF, also called insulin correction factor (ICF), refers to the amount of blood glucose in mmol/L (or mg/dL, depending on your settings) that one unit of insulin can neutralize.

Bill has an ISF of 50 (this is also written in shorthand as an ISF of 50 mg/dL/U (2.8 mmol/L/U)). This means 1 U of rapid insulin will bring Bill's sugar down by 50 mg/dL (2.8 mmol/L). How many units of insulin would Bill need to reach his target glucose of 100 if his current glucose reading is 250 mg/dL (13.9 mmol/L)?

Here is the formula you will need:

$$\frac{Current\ Glucose - Glucose\ Target}{ISF}$$

Calculate Bill's Insulin Dose:

$$\frac{250 - 100}{50} =$$

$$\frac{150}{50} =$$

$$3\ units$$

Answer

Bill needs 3 units of insulin to reach his target glucose of 100 mg/dL.

Like basal rates, ISF is not used verbatim by Trio but is modified over time as data on the patient is collected. Still, setting ISF as close to accurate as possible is important for Trio to function well.

It is safe to transfer your ISF from your pump settings. Note that almost all issues when starting with Trio are a result of an improperly set ISF. If you find you have lows with corrections, or you have SMB/UAM on and the application provides too much insulin at any time, resulting in a rollercoaster pattern, your ISF is likely to blame.

Testing/Adjusting Your ISF

Baseline Calculation

If your current ISF is close, but needs some testing and adjustment, skip to the next section.

If your current ISF is inaccurate or you are unsure where to even start, the adjustments in Trio are based on formulas developed by Walsh, et.al. and may help you find a starting point to then test or adjust your ISF.

> **Warning**
>
> This calculation is to be used as a starting point for testing and is not considered definitive or exact.

ISF Testing

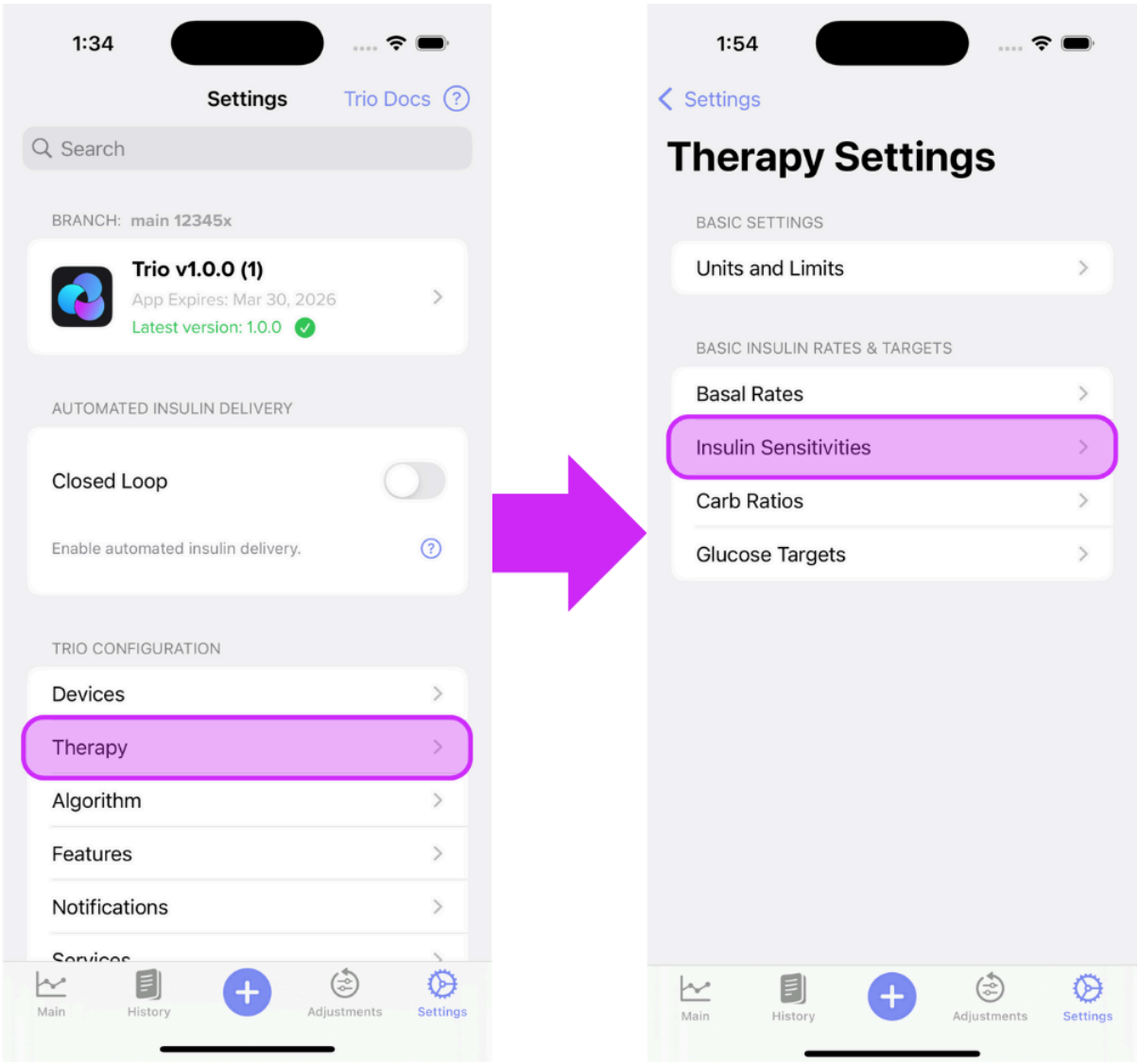
ISF Adjustment

There are a few ways you can work to adjust your ISF. The easiest method is simply bringing yourself to a higher glucose with a glucose tab or choosing a time when you are "stuck" higher than your target, then correcting based on your ISF. If you are higher than your target after 4 hours, make your ISF more aggressive by *DECREASING* the value. If you are lower than your target after 4 hours, make your ISF less aggressive by *INCREASING* the value.

How To Enter Your ISF Into Trio

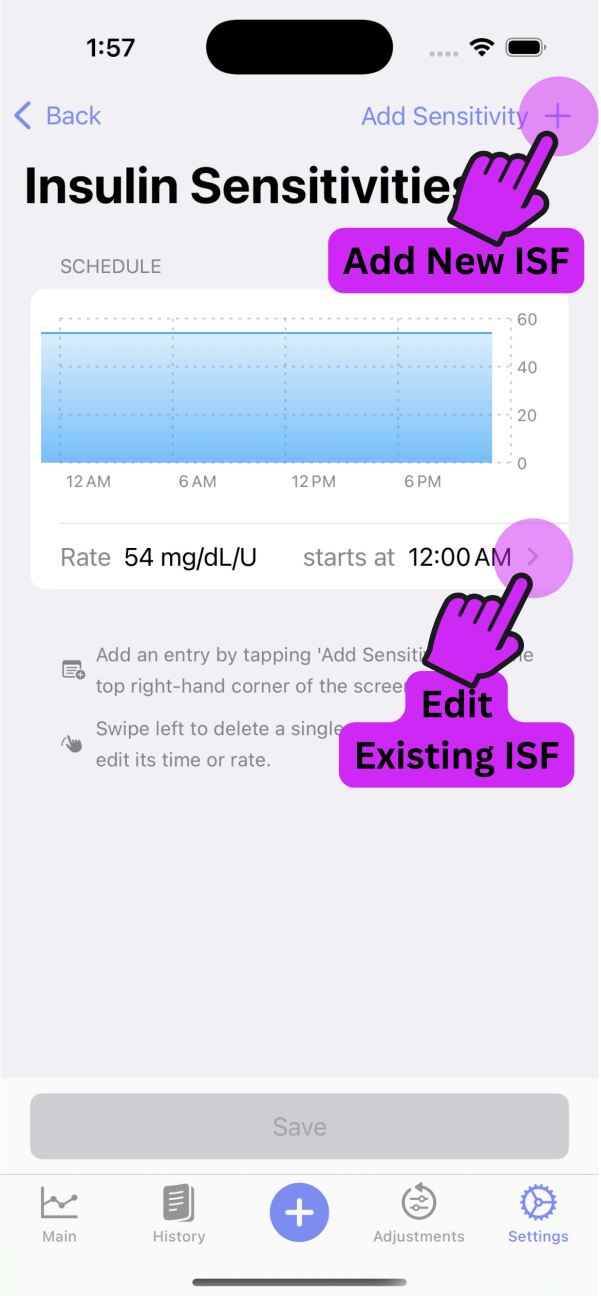
Step 1

Enter the ISF Profile screen



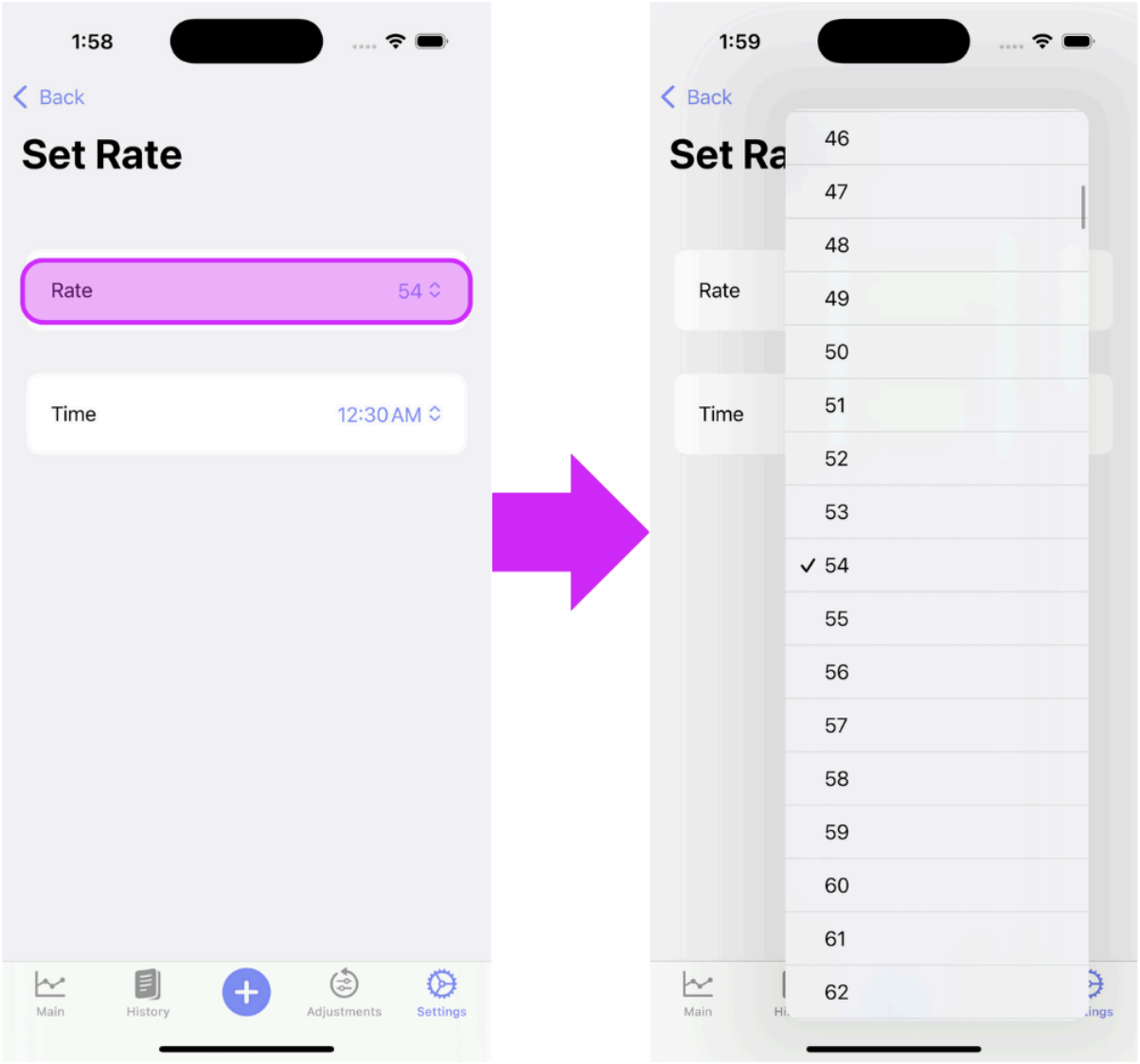
Step 2

Tap the "Add Rate +" on the top right of the screen until you have the number of ISFs you require. Then, edit each rate by tapping the arrow to the right of the ISF.



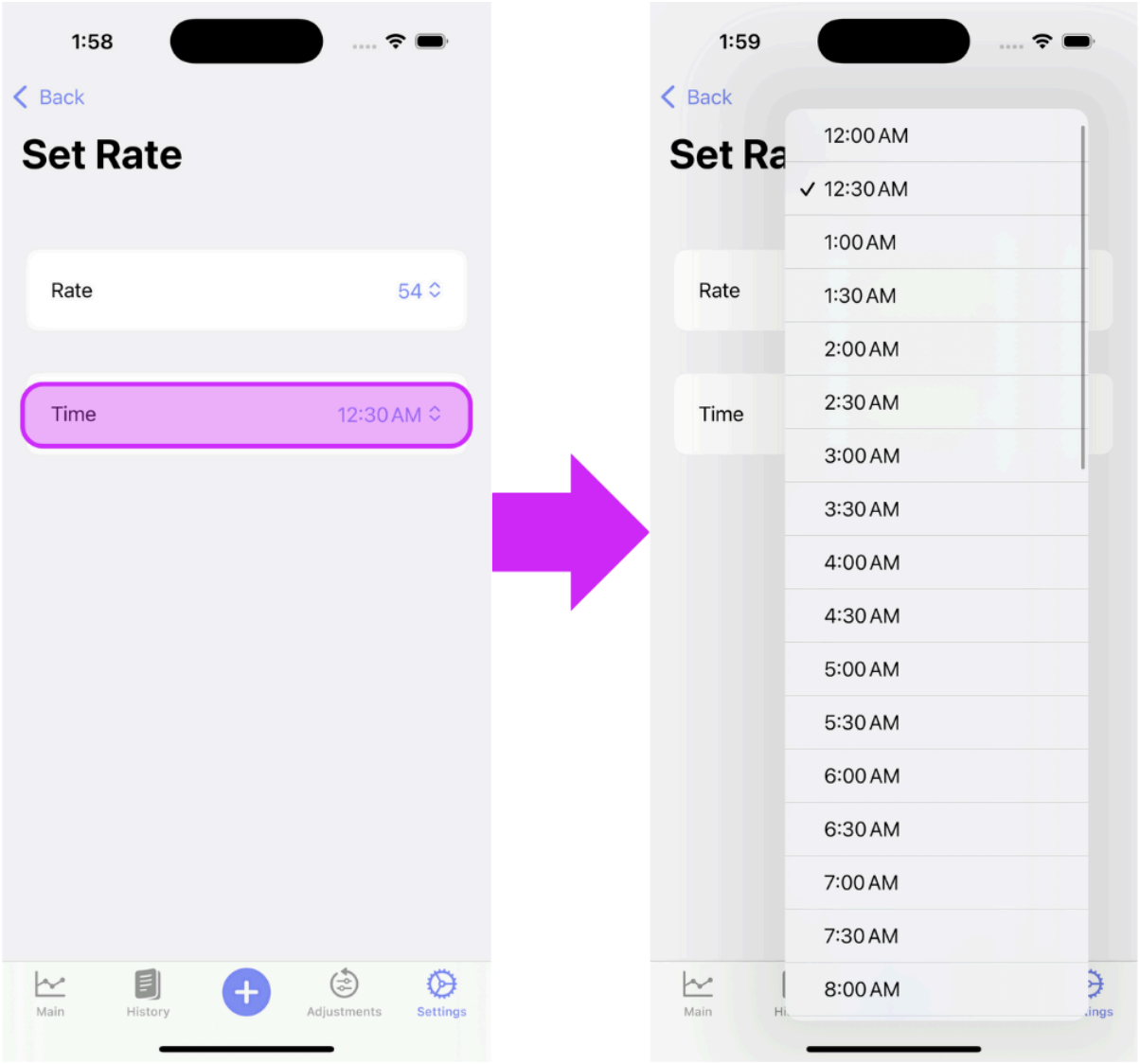
Step 3

Adjust the rate



Step 4

Adjust the time

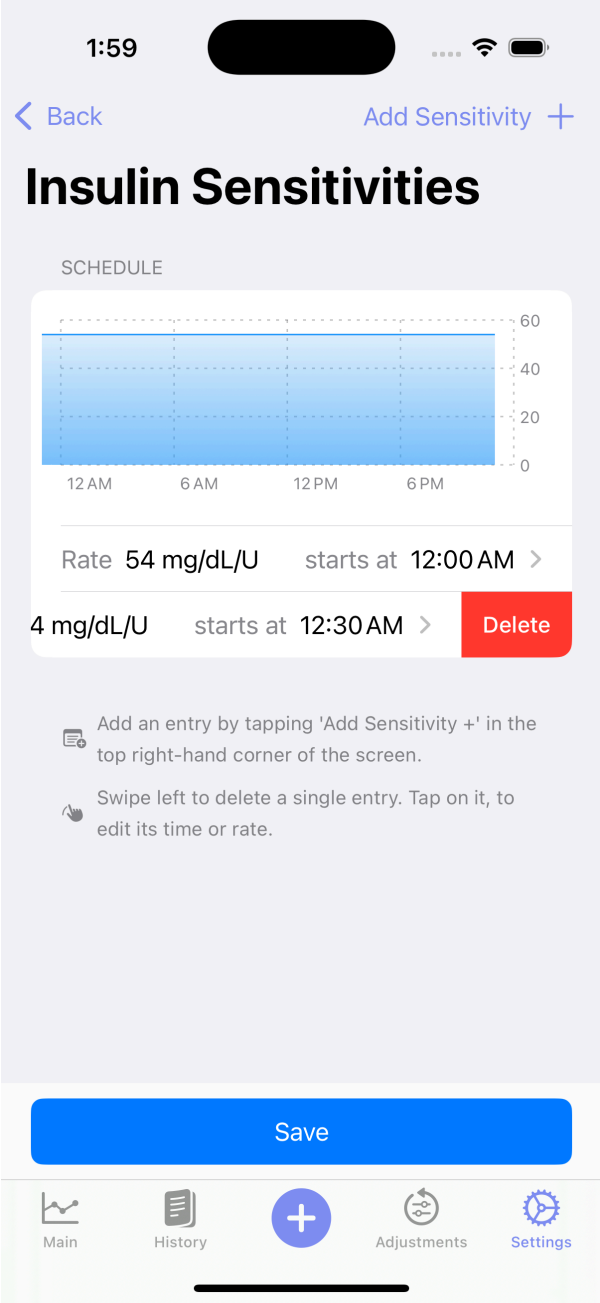


Step 5

Repeat Steps 2, 3, and 4 until all ISFs are set

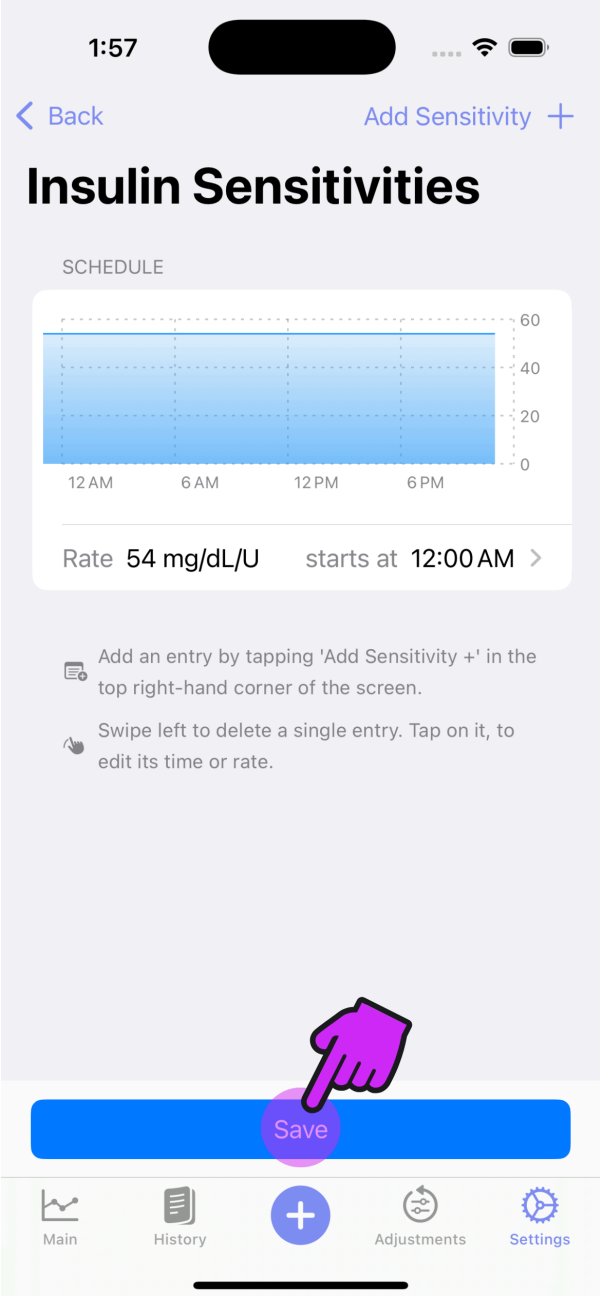
Delete an ISF Entry

Should you need to delete an ISF entry, just swipe left on the rate you need to remove.



Step 6 **IMPORTANT**

Save your changes!



Step 7

Return to New User Setup

Algorithm

Welcome to the Algorithm Settings homepage! Here you'll find information on the settings that influence the algorithm that runs Trio.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Autosens

Explanations of the Autosens settings

Super Micro Bolus (SMB)

Details on Super Micro Bolus settings

Dynamic Settings

Information on Dynamic Settings, including the mathematical equations behind them

Target Behavior

Explanations of each setting found in Target Behavior

*Additionals

More information on settings found in the Additionals section.
***Most users do not need to adjust these settings.**

Autosens Settings

Highlights

- Autosens Ratio has been renamed to Sensitivity Ratio
- Bias the calculated ISF/CR/Basal rates towards more or less aggressive values by increasing or decreasing the adjustment factor accordingly
- Autosens Max/Min limits sensitivity adjustments

Sensitivity Ratio

The Sensitivity Ratio (previously `Autosens Ratio`) is used to calculate the amount of adjustment needed to your basal rates and ISF.

Tip

This previously was shown as a decimal value as `Autosens Ratio` and is now shown as a percentage as `Sensitivity Ratio`.

If your Sensitivity Ratio is showing 120%, this means Trio has calculated that your insulin needs are 120% of your standard needs.

If your Senstivitity Ratio is 100%, Trio is using your entered Therapy Settings without any adjustments.

There are 3 options for you to select from for how your Sensitivity Ratio is calculated:

- Autosens: Adjusts your Sensitivity Ratio based on a series of comparisons of the last 8 hours versus the last 24 hours of insulin usage. **This is the default calculation.**
- Logarithmic Dynamic ISF: Adjusts your Sensitivity Ratio based on your Total Daily Dose (TDD) from the last 24 hours vs the last 10 days to adjust your Sensitivity Ratio.
- Sigmoid Dynamic ISF: Adjusts your Sensitivity Ratio based on how far above or below target your current glucose reading is.

Note

All 3 options are limited by your Autosens Max and Autosens Min settings. These settings prevent over-adjusting.

Autosens

Auto-sensitivity (Autosens) adjusts insulin delivery based on how sensitive or resistant you are to insulin at the time of the current loop cycle by analyzing past data to keep glucose levels stable.

Tip

This is the default calculation used in Trio. If you want to disable Autosens and use no other dynamic adjustments, **set both your Autosens Max and Autosens Min to 1**

How it Works

It looks at the last 8-24 hours of data, excluding meal-related changes, and adjusts insulin settings like basal rates and targets when needed to match your sensitivity or resistance to insulin.

What it Adjusts

Autosens modifies your current Insulin Sensitivity Factor (ISF), current basal rate, and current target glucose. Autosens does not examine meals or adjust your CR. It only assesses your sensitivity to insulin and adjusts ISF/basal rates/glucose targets accordingly.

Autosens finds Bill has been running more sensitive to insulin lately. In the last 24 hours, he has been 2X more sensitive to insulin, whereas, in the last 8 hours, he has been 3X more sensitive to insulin. Which adjustment will Autosens use?

Autosens takes the more conservative calculated sensitivity. In this example, the more conservative value is obtained from the 8-hour window because by assuming Bill is 3X more as opposed to 2X more sensitive to insulin, the system will give less insulin.

Calculated Sensitivity

Important Information

This adjusted ISF is temporary, will change with the next loop cycle, and should not be directly used as our profile ISF value.

Enabling and/or disabling Logarithmic Dynamic ISF and Sigmoid Dynamic ISF determines how your Sensitivity Ratio is calculated. Once calculated, Trio uses the following formula to determine your temporary, adjusted ISF:

$$\frac{Profile ISF}{Sensitivity\ Ratio} = Calculated\ Sensitivity$$

Bill has a Profile ISF of 54 mg/dL/U (3 mmol/L/U). Trio detects Bill has been more resistant to insulin lately and needs to increase his insulin. It calculates Bill has a Sensitivity Ratio of 110% using Autosens. What Calculated Sensitivity, aka temporary ISF, will Trio use for this loop cycle?

Here is the formula used for Calculated Sensitivity:

$$\frac{Profile ISF}{Sensitivity\ Ratio}$$

Calculate the temporary ISF, Calculated Sensitivity, used in this loop cycle:

$$\frac{54\ mg/dL/U}{110\%} =$$

$$49\ mg/dL/U$$

Answer

Bill now temporarily has an ISF of ***49 mg/dL/U (2.73 mmol/L/U)***. (Note that a larger Sensitivity Ratio results in a lower, more aggressive ISF.)

Autosens Max

Default: *120%*
Setting Limits: *50%-200%*

Autosens Max sets the maximum Sensitivity Ratio used by Autosens, Logarithmic Dynamic ISF, and Sigmoid Dynamic ISF.

Tip

Increasing this value allows automatic adjustments of basal rates to be higher, ISF to be lower, and CR (if enabled) to be lower.

Autosens Min

Default: *70%*
Setting Limits: *50%-100%*

Autosens Min sets the minimum Sensitivity Ratio used by Autosens, Logarithmic Dynamic ISF, and Sigmoid Dynamic ISF.

> **Tip**
>
> Decreasing this value allows automatic adjustments of basal rates to be lower, ISF to be higher, and CR (if enabled) to be higher.

Rewind Resets Autosens

Default: *ON*
This setting applies to Medtronic Users only

This feature resets the Sensitivity Ratio to neutral when you rewind your pump on the assumption that this corresponds to a site change.

Autosens will begin learning sensitivity anew from the time of the rewind, which may take up to 6 hours.

> **Tip**
>
> If you usually rewind your pump independently of site changes, you may want to keep this setting **OFF**.

SMB Settings

Highlights

- UAMs should be enabled when SMBs are enabled.
- If entering carbs and/or bolusing: Max UAM Basal Minutes should be set **lower** than Max SMB Basal Minutes.
- If **NOT** entering carbs and/or bolusing (FCL): Max UAM Basal Minutes should be set **higher** than Max SMB Basal Minutes.
- Super micro boluses (SMB) often include your basal insulin. Your basal rate will be temporarily reduced after an SMB is delivered to prevent overdosing insulin.
- SMBs reduce blood sugar more quickly than temporary basal rates.
- If you want to configure SMBs only to run in certain conditions, do not turn on 'Enable SMB Always'.
- For a detailed look at when SMBs are used, see the chart in Are SMBs Allowed? section.
- For help with setup, see the New User Guide.

Warning

All of the SMB options listed below are ***inclusive***, not exclusive. This means that if more than one option is enabled, only *one* of those options needs to apply in order for SMBs to be utilized.

Enable SMB Always

Default: *OFF*
When this setting is enabled, Super Micro Boluses (SMBs) will always be allowed if dosing calculations determine insulin is needed via the SMB delivery method. The exception is when a high temp target is set. Enabling SMB Always will remove the other redundant "Enable SMBs if..." options.

SMBs will remain on if you have a low temporary target set but will be fully disabled if a high temporary target is set (unless "Allow SMB With High Temptarget" is enabled).

The size of SMBs is limited by Max SMB Basal Minutes.

Enable SMB with COB

Default: *OFF*
When the carbs on board (COB) forecast line is active, enabling this feature allows Trio to use SMBs to deliver the insulin required.

> **Tip**
>
> Allowing SMBs when you have carbs on board can help with mealtime spikes in glucose.

Enable SMB with TempTarget

Default: *OFF*
Enabling this feature allows Trio to deliver insulin required using SMBs at times in which a manual temporary target (temp target) under **100 mg/dL (5.5 mmol/L)** is set.

> **Tip**
>
> Low temp targets are often set when struggling with high glucose readings. Allowing SMBs at this time can help counter elevated glucose levels faster.

Enable SMB After Carbs

Default: *OFF*
Enabling this feature allows Trio to deliver insulin required using SMBs for **6 hours** after any carb entry, regardless of whether there are active carbs on board (COB).

> **Tip**
>
> Carbs do not digest at the same rate. Allowing SMBs for the 6 hours after entering a meal will help Trio counter those carbs faster.

Enable SMB with High Glucose

Default: *OFF*
Enabling this feature allows Trio to deliver insulin required using SMBs when your glucose reading is above the value set as your 'High Glucose Target'. This additional setting will appear when you enable this feature.

High Glucose Target

Default: *110 mg/dL | 6.1 mmol/L*
If 'Enable SMB with High BG' is enabled, SMBs will be allowed if your glucose is above this value.

> **Important**
>
> - This setting was often misunderstood. It was never a restriction on SMBs. To help clarify it's purpose for users, it has moved under the only setting it applies to, `Enable SMB with High BG`.
> - Its only function is to allow SMBs if glucose is above this number and `Enable SMB with High BG` is turned ON.
> - **It will not *prevent* SMBs or UAMs below this number if other SMB settings are turned ON.**

Allow SMB with High TempTarget

Default: *OFF*
Enabling this feature allows Trio to deliver insulin required using SMBs when a manual Temp Target above **100 mg/dL (5.5 mmol/L)** is set.

> **Important**
>
> High Temp Targets are often used to recover from lows. If you use a high temp target for that purpose, this feature should remain disabled.

Enable UAM

Default: *OFF*

> **Important**
>
> This setting should be *ON* if other SMBs are enabled.

Enabling Unannounced Meal SMBs (UAMs) allows Trio to detect and respond to unexpected rises in glucose readings caused by unannounced or miscalculated meals, meals high in fat or protein, or other factors like adrenaline or cortisol.

It uses SMBs to deliver insulin in small amounts to correct glucose spikes. UAM also works in reverse, reducing or stopping insulin if glucose levels drop unexpectedly.

The size of UAMs is limited by Max UAM Basal Minutes

Tip

Enabling UAM will give minor adjustments in your insulin dosing to account for the difference in expected glucose and actual glucose readings.

- If you are entering carbs and/or bolusing for meals, `Max UAM Basal Minutes` should be set lower than `Max SMB Basal Minutes`.
- If you are **not** entering carbs or bolusing for meals, `Max UAM Basal Minutes` should be set higher than `Max SMB Basal Minutes`.

Max SMB Basal Minutes

Default: *30 minutes*
Setting Limits: *30-180 minutes*

If any of the SMB options are enabled, this limit will apply to all SMBs except UAM SMBs. This is a limit on the size of a single SMB. One SMB can only be as large as this many minutes of your current profile basal rate.

To calculate the maximum SMB allowed based on this setting, use the following formula:

$$\frac{Max\ SMB\ Basal\ Minutes}{60} \times Current\ Basal\ Rate$$

Bill's current basal rate is 2.0 units/hr. His `Max SMB Basal Minutes` is set to 30 minutes. What is the largest SMB he can receive?

Here is the formula you will need:

$$\frac{Max\ SMB\ Basal\ Minutes}{60} \times Current\ Basal\ Rate$$

Calculate Bill's largest SMB:

$$\frac{30}{60} \times 2.0 =$$

$$\frac{1}{2} \times 2.0 =$$

$$1.0\ unit$$

Answer

The largest SMB Bill can receive is **1.0 unit** every 5 minutes.

Warning

Increasing this value above 90 minutes may impact Trio's ability to effectively zero temp and prevent lows.

Tip

If you are entering carbs or bolusing for meals, your **Max SMB Basal Minutes should be set *higher*** than your Max UAM Basal Minutes

---

Max UAM Basal Minutes

Default: *30 minutes*
Setting Limits: *30-180 minutes*

If UAM is enabled, this setting limits the size of each UAM SMB. One UAM SMB can only be as large as this many minutes of your current profile basal rate.

To calculate the maximum UAM allowed based on this setting, use the following formula:

$$\frac{Max\ UAM\ Basal\ Minutes}{60} \times Current\ Basal\ Rate$$

> **Warning**
>
> Increasing this value above 90 minutes may impact Trio's ability to effectively zero temp and prevent lows.

> **Tip**
>
> If you are **NOT** entering carbs or bolusing for meals and using Trio as a Fully Closed Loop (FCL), your **Max UAM Basal Minutes should be set *higher*** than your Max SMB Basal Minutes

---

Max Allowed Glucose Rise for SMB

Default: *20%*
Setting Limits: *10%-40%*

This safety limiter looks at the difference between your last two blood glucose readings. If the difference is above this percent, Trio suspects them to be incorrect and will suspend all SMB delivery accordingly (including UAM). You can adjust the amount of change that should be allowed before SMBs are delivered.

Bill's last CGM reading was 90 mg/dL. The very next reading is 115 mg/dL. Will Bill receive the insulin needed as an SMB?

Here are the formulas you'll need:

Rise Percentage:

$$\frac{Current\ Glucose - Previous\ Glucose}{Previous\ Glucose}$$

Compare to Max Allowed Glucose Rise for SMB

$$Rise\ Percentage > \ or \ = \ or \ < Max\ Allowed\ Glucose\ Rise\ for\ SMB\ Setting$$

No SMB: $>$
Yes SMB: $<$ or $=$

Calculate if an SMB will be used:

$$\frac{115 - 90}{90} =$$

$$\frac{25}{90} =$$

$$+0.28\ or\ +28\%$$

His rise, or increase in glucose, is an increase of ***28%***

$$28\% > 20\% = No\ SMBs$$

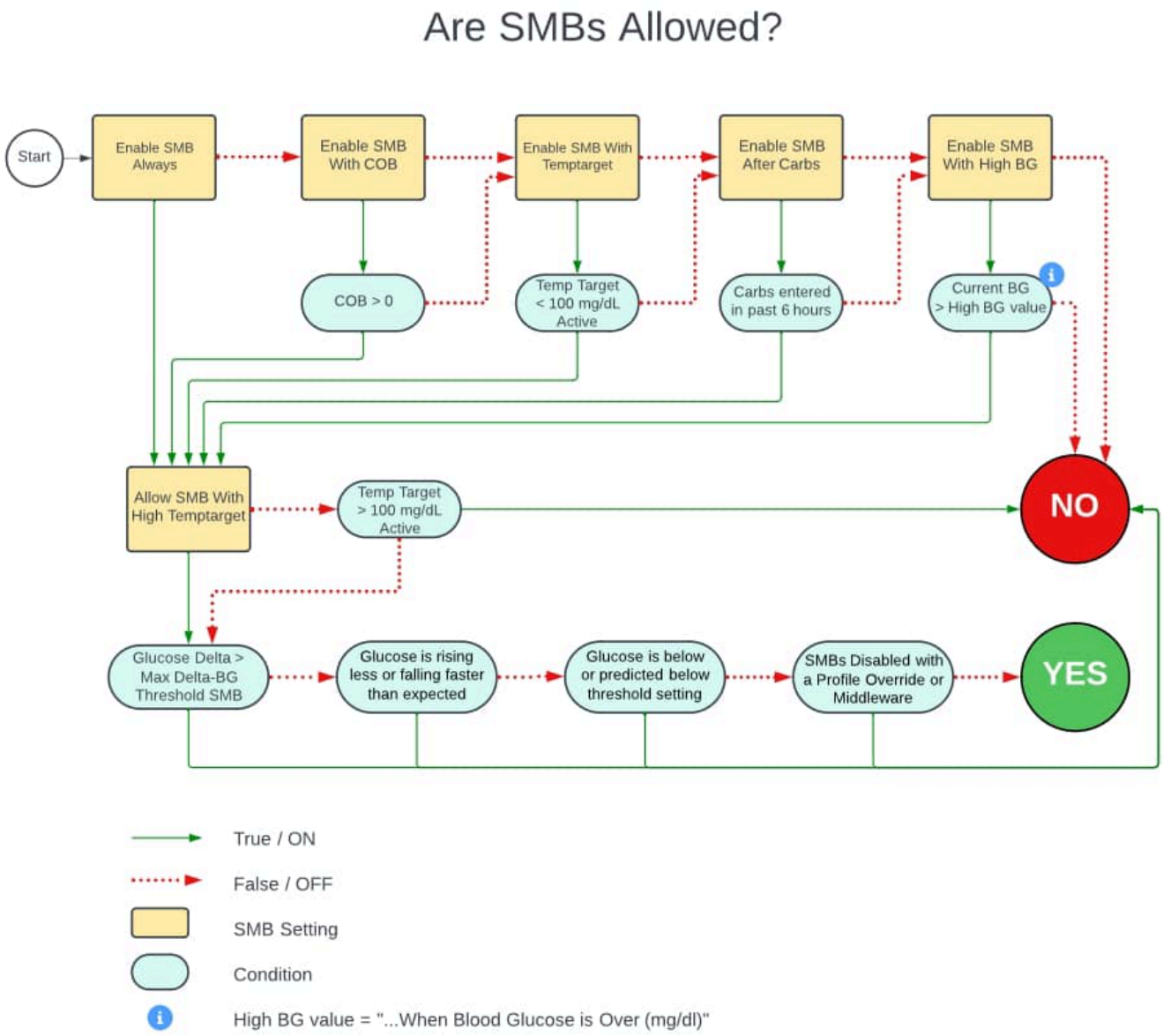
Answer

This increase is larger than the threshold, so **no SMBs will be given**. Trio will administer needed insulin via temp basal adjustment.

Tip

For Fully Closed Loop (FCL), 30% is advised.

Are SMBs Allowed?



By following the flow chart above, you can see which combination of settings will allow SMBs.

- If a setting in the top row is toggled off, look at the next box to the right. If no box in the top row is toggled on, then SMBs will not be allowed.
- If any of the settings in the top row are toggled on and their condition is true, follow the green line down to the "Allow SMB with High Temptarget" box.
- If "Allow SMB with High Temptarget" is toggled on (NOT the default), then continue to follow the green line to the bottom conditions.
- If "Allow SMB with High Temptarget" is toggled off (which IS the default), it will then check if you've set a Temp Target (not a custom profile) above 100 mg/dL (5.5 mmol/L). If you have a Temp Target set above 100 mg/dL, then SMBs are DISABLED and not allowed.

If you've made it to the bottom row, it checks all those conditions, and if none of them are true, then SMBs are allowed.

Here is the order of settings Trio uses when deciding whether to enable or disable SMBs:

- Disable when a High Temptarget is set (unless "Allow SMB with High Temp Target" is enabled)
- Enable if "Enable SMB Always" is set (unless disabled for "High Temp Target")
- Enable while there are COB
- Enable for a full 6 hours after any carb entry
- Enable if a Low Temp Target is set

Dynamic Settings

Dynamic ISF: Logarithmic

Default: *Disabled*

> **Important**
>
> It is important to enter your information into the Desmos graphs found here **before** turning on Logarithmic Dynamic ISF. The default settings work for a majority of users, but not all.
>
> Use the sliders in Desmos to determine what your Adjustment Factor (AF) should be so that your ProfileISF is used when your glucose is at ***150 mg/dL (8.3 mmol/L)***.

> **Tip**
>
> You must first choose `Logarithmic` or `Sigmoid` before any other dynamic features will appear

Activating this feature allows Trio to calculate your Sensitivity Ratio using the logarithmic dynamic formula, rather than the Autosens formula with each loop cycle by considering factors such as: your current glucose, the weighted total daily dose of insulin (TDD), your adjustment factor setting (AF), and a few other data points. Using Logarithmic Dynamic ISF allows you to customize your Sensitivity Ratio calculation beyond what is allowed with Autosens.

Below is the formula used for calculating the Sensitivity Ratio using Logarithmic Dynamic ISF for mg/dL:

$$Sensitivity\ Ratio = ProfileISF \times AF \times TDD \times \ln\left(\frac{\left(\frac{Glucose}{Peak}\right)+1}{1800}\right)$$

Here is that same formula adjusted for mmol/L:

$$Sensitivity\ Ratio = ProfileISF \times AF \times TDD \times \ln\left(\frac{\left(\frac{Glucose}{0.0555 \times Peak}\right)+1}{100}\right)$$

> **Info**
>
> This formula considers your profile ISF, current glucose, total daily dose (TDD over the last 24 hours), insulin peak effect (Peak), and Adjustment Factor (AF) that allows for user tuning of Dynamic ISF.

After the Sensitivity Ratio is calculated, your Calculated Sensitivity is then determined by using the same formula as Autosens:

$$Calculated\ Sensitivity = \frac{Profile ISF}{Sensitivity\ Ratio}$$

Dynamic ISF: Sigmoid

Default: *Disabled*

Turning on the Sigmoid Formula setting replaces the default logarithmic formula used to determine your Sensitivity Ratio. Your Calculated Sensitivity is calculated using a sigmoid curve rather than the logarithmic function.

The curve's steepness, reflecting how big adjustments are from one reading to another, is influenced by the Adjustment Factor, while Autosens Max and Min settings determine the limits of the ratio adjustment. Autosens Max and Min can also influence the curve's steepness with the Sigmoid Formula.

When using the Sigmoid Formula, Total Daily Dose (TDD) has much less of an impact on adjustments to sensitivity. Sigmoid is more reliant upon how far from the target your glucose readings are.

> **Warning**
>
> It is not recommended to set Autosens Max higher than 150% when using Sigmoid

> **Info**
>
> As of the publication of this documentation, there has been no empirical data analysis to support the use of Sigmoid for dynamic sensitivity determination.

Adjustment Factor (Logarithmic)

Default: *80%*
Setting Limits: *30%-150%*

Adjustment Factor (AF) allows you to control how quickly and effectively Dynamic ISF responds to changes in glucose levels.

Adjusting this value shifts and steepens the curve of logarithmic Dynamic ISF. Increasing this setting will cause Trio to respond faster to changes in sensitivity, but can also shift the response to new values.

> **Warning**
>
> Please enter your adjustments in the Desmos graphs before you change them in the app to verify your adjustments are changing as intended.

> **Caution**
>
> ***Adjustment Factor (AF) is not a safety limiter***
>
> - Increasing AF means you are telling the system that ALL dynamically calculated ISF values have not been aggressive enough, and you want the system to make them more aggressive.
> - Decreasing AF means you are telling the system that ALL dynamically calculated values are too aggressive, and to make them less so.

Sigmoid Adjustment Factor

Default: *50%*
Setting Limits: *10%-200%*

The Sigmoid Adjustment Factor (AF) allows you to control the rate in which Trio responds to changes in insulin sensitivity and adjusts at which glucose value you will reach your Autosens Max and Min limits.

Sigmoid Adjustment Factor influences both how much your ISF values change between 2 glucose readings and how quickly you will reach the limits you've set. Increasing this setting increases the rate of change and reduces the range of glucose values between your Autosens Max and Autosens Min limits.

Due to how the curve is calculated, increasing this setting has a different impact than it's Logarithmic counterpart. Please use caution when adjusting this setting.

> **Warning**
>
> Please enter your adjustments in the Desmos graphs before you change them in the app to verify your adjustments are changing as intended.

Weighted Average of TDD

Default: *35%*
Setting Limits: *5%-100%*

This setting adjusts how much weight is given to your recent daily total insulin dose (TDD) when calculating your Sensitivity Ratio using either the Logarithmic or Sigmoid Formulas.

At the default setting, Trio weights your TDD used with 35% of your last 24 hours and 65% of the last 10 days of data.

- Set at **100%** = TDD is composed of 100% the last 24 hours of TDD data + 0% the last 10 days of TDD data
- Set at **35%** (default) = TDD is composed of 35% the last 24 hours of TDD data + 65% the last 10 days of TDD data
- Set at **0%** = TDD is composed of 0% the last 24 hours of TDD data + 100% the last 10 days of TDD data

Bill has a TDD of 55 U over the last 24 hours. He has had a TDD of 48 U over the last 10 days. His `Weighted Average of TDD` is set at 35%. What TDD is used to calculate his Sensitivity Ratio?

Here's the formula to calculate Weighted Average of TDD:

$$(24\ hours\ TDD \times Weighted\ Avg\ of\ TDD\ \%) + (10\ days\ TDD \times (100 - Weighted\ Avg\ of\ TDD$$

Calculate Bill's Weighted Average of TDD:

$$(55 \times 35\%) + (48 \times (100 - 35)\%) =$$

$$(55 \times 35\%) + (48 \times 65\%) =$$

$$19.25 + 31.2 =$$

$$50.45\ U$$

Answer

Trio will use a TDD of ***50.45 U*** in Bill's Sensitivity Ratio calculation.

As you increase the default to a higher number, the adjusted basal rates will be more influenced by your last 24-hour insulin usage, resulting in more variable changes.

Adjust Basal

Default: *OFF*

Adjust Basal replaces the sensitivity-based formula normally used by Trio for adjusting your basal rates with a new formula based on your total daily insulin use (TDD). Use this if the current Trio adjustments of basal rates are not adequate.

Turn this setting on to give basal adjustments more agility. Keep this setting off if your basal needs are not highly variable.

Calculation used if this setting is OFF:

$$New\ Basal\ Rate = Current\ Basal\ Rate \times Sensitivity\ Ratio$$

Calculation used if this setting is ON:

$$Adjust\ Basal\ Ratio = \frac{Weighted\ Average\ of\ TDD}{10\ day\ Average\ of\ TDD}$$

$$New\ Basal\ Rate = Current\ Basal\ Rate \times Adjust\ Basal\ Ratio$$

See Weighted Average of TDD setting to understand how this variable is calculated.

Bill's TDD has been 55 U over the last 24 hours, and his 10-day average is 48 U. He has set his `Weighted average of TDD` in preferences to 35%. His current profile basal rate is 1.0 U/h. What will his new basal rate be with `Adjust Basal` turned ON?

Here are the formulas you'll need:

Weighted Average of TDD:

$$(24\ hours\ TDD \times Weighted\ Avg\ of\ TDD\ \%) + (10\ days\ TDD \times (100 - Weighted\ Avg\ of\ TDD$$

Adjust Basal Ratio:

$$\frac{Weighted\ Average\ of\ TDD}{10\ day\ Average\ of\ TDD}$$

New Basal Rate:

$$Current\ Basal\ Rate \times Adjust\ Basal\ Ratio$$

First, calculate the Weighted Average of TDD:

$$(55 \times 35\%) + (48 \times (100 - 35)\%) =$$

$$(55 \times 35\%) + (48 \times 65\%) =$$

$$19.25 + 31.2 =$$

$$50.45\ U$$

Next, calculate the Adjust Basal Ratio:

$$\frac{50.45}{48} =$$

$$1.05$$

Finally, calculate the New Basal Rate

$$1.0 \times 1.05 =$$

$$1.05\ U/hr$$

▼ Details

Answer

Adjust Basal will replace Bill's profile basal rate with a new basal rate of ***1.05 U/hr*** for this loop cycle.

Logarithmic Desmos Graphs

Click here to view a graph depicting the logarithmic formula in mg/dL

Click here to view a graph depicting the logarithmic formula in mmol/L

Sigmoid Desmos Graphs

Click here to view a graph depicting the sigmoid formula in mg/dL

Click here to view a graph depicting the sigmoid formula in mmol/L

Target Behavior

High Temp Target Raises Sensitivity

Default: *OFF*

When this feature is enabled, manually setting a temporary target above 100 mg/dL (5.5 mmol/L) will decrease the Sensitivity Ratio used for ISF and basal adjustments, resulting in less insulin delivered overall. This scales with the temp target set; the higher the temp target, the lower the Sensitivity Ratio used.

If Half Basal Exercise Target is set to 160 mg/dL (default), a temp target of 120 mg/dL uses a Sensitivity Ratio of 0.75. A temp target of 140 mg/dL uses a Sensitivity Ratio of 0.6.

> **Note**
>
> The effect of this can be adjusted using the Half Basal Exercise Target setting

---

Low Temp Target Lowers Sensitivity

Default: *OFF*

When this feature is enabled, setting a temporary target below 100 mg/dL (5.5 mmol/L) will increase the Sensitivity Ratio used for ISF and basal adjustments, resulting in more insulin delivered overall. This scales with the temp target set; the lower the temp target, the higher the Sensitivity Ratio used.

If Half Basal Exercise Target is set to 160 mg/dL (default), a temp target of 95 mg/dL uses a Sensitivity Ratio of 1.09. A temp target of 85 mg/dL uses a Sensitivity Ratio of 1.33.

> **Note**
>
> The effect of this can be adjusted using the Half Basal Exercise Target setting

---

Sensitivity Raises Target

Default: *OFF*

Enabling this feature causes Trio to automatically raise the targeted glucose if it detects an increase in insulin sensitivity. If Sensitivity Ratio is > 1, target glucose will be increased.

Resistance Lowers Target

Default: *OFF*

Enabling this feature causes Trio to automatically lower the targeted glucose if it detects resistance to insulin. If Sensitivity Ratio is < 1, target glucose will be decreased.

Half Basal Exercise Target

Default: *160 mg/dL*
Setting Limits: *100-300 mg/dL*

The Half Basal Exercise Target allows ou to scale down your basal insulin during exercise or scale up your basal insulin when eating soon by setting a manual temp target. To scale down for exercise, set a temp target above 100 mg/dL. To scale up for eating soon, set a temp target below 100 mg/dL.

For example, at a temp target of 160 mg/dL, your basal is reduced by 50%, but this is scaled depending on how far away from 100 your temp target is set.

- Temp Target $= 160$ mg/dL: 50% reduction in profile basal
- Temp Target $= 140$ mg/dL: 60% reduction in profile basal
- Temp Target $= 120$ mg/dL: 75% reduction in profile basal

You can see how adjusting this setting will influence your profile basal when you set a temp target in Trio. On the screen, you will be able to set a custom Half Basal Exercise Target for the duration of the temp target you set.

Additionals

Read This Before Proceeding

This settings in this section typically do not require ***any*** modifications. Do not alter them without a solid understanding of what you are changing and the full impact it will have on the algorithm.

Max Daily Safety Multiplier

Default: *300%*
Setting Limits: *100%-500%*

This setting restricts the maximum automated temporary basal rate Trio can set. At the default of 300%, it caps temporary basal rates at 3 times your highest programmed basal rate of ***all your basal rates***.

This limit works in conjunction with Current Basal Safety Multiplier. Trio will use the smaller of the two limits. It serves as a safety limit, ensuring no temporary basal rates exceed safe levels.

Warning

Increasing this setting is not advised

Current Basal Safety Multiplier

Default: *400%*
Setting Limits: *100%-500%*

This setting also restricts the maximum automated temporary basal rate Trio can set. At the default of 400%, it caps temporary basal rates to 4 times your ***current*** programmed basal rate.

This limit works in conjunction with Max Daily Safety Multiplier. Trio will use the smaller of the two limits. It serves as a safety limit, ensuring no temporary basal rates exceed safe levels.

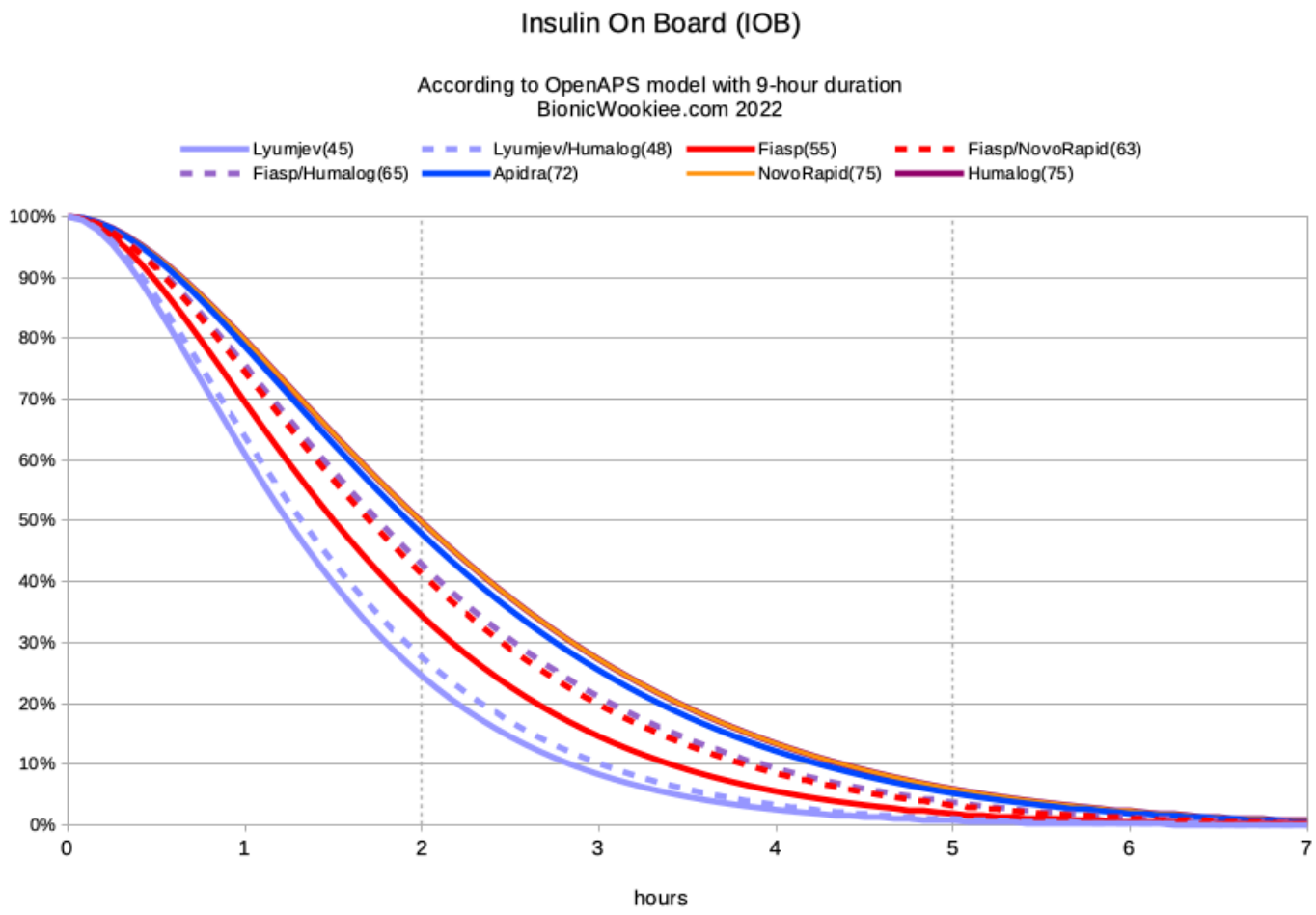
Warning

Increasing this setting is not advised

Duration of Insulin Action

Default: *10 hours*
Setting Limits: *5-10 hours*

The Duration of Insulin Action (DIA) defines how long your insulin continues to lower glucose readings after it is given. This helps the system accurately track Insulin on Board (IOB), avoid over- or under- corrections by considering the tail end of insulin's effect.



Trio, as with other Oref-based systems like AndroidAPS, uses an exponential decay model for DIA. Working in collaboration with David Burren (Bionic Wookie), we found the DIA that most accurately reflects actual insulin activity in the body, is a 10 hour DIA. You can read more about insulin decay models used in Oref-based systems at bionicwookie.com

> **Warning**
>
> Trio will not allow you to set a DIA below 5 hours. However, 5 hours is an extremely low DIA and it is not advised.

Use Custom Peak Time

Default: *OFF*

Insulin Peak Time reflects when insulin is most effective in lowering glucose, aka the peak of insulin activity. Toggling this setting on exposes the Custom Peak Time setting and allows you to customize that peak.

Typically, this setting does not need to be adjusted. The one common case for adjusting insulin peak time is for Lyumjev users, where the programmed ultra-rapid insulin peak matches Fiasp's peak time rather than Lyumjev's.

Insulin Peak Time

Default: *Determined By Insulin Selection*
Setting Limits: *35-120 min*

The default settings are determined by your insulin selection:
- Rapid-Acting (Humalog, Novolog/Novorapid, Apidra) - Default: 75 minutes - Setting Limits: 50-120 minutes - Ultra-Rapid (Fiasp, Lyumjev) - Default: 55 minutes - Setting Limits: 35-100 minutes

> **Lyumjev Users**
>
> Lyumjev has a peak time of 45 minutes

Skip Neutral Temps

Default: *OFF*

When Skip Neutral Temps is enabled, Trio will not set neutral basal rates shortly before the hour, minimizing hourly pump alerts on Medtronic pumps. This can help light sleepers avoid alerts, but will delay basal adjustments. This will also only come into effect if SMBs are *disabled*.

For most users, leaving this *OFF* is recommended to ensure consistent basal delivery and loop calculation. If this option is enabled, loops will be skipped during the last 5 minutes of the hour.

Unsuspend If No Temp

Default: *OFF*

Enabling this setting allows Trio to resume your pump after a *manual* zero temp basal was set. This feature ensures insulin delivery restarts if you forget to manually unsuspend, adding a safeguard for pump reconnections.

Suspend Zeros IOB

Default: *ON*

This is defaulted to on because it resets any active temporary basal rate to a 0 U/hr temp basal rate when you suspend your pump, either on the pump with Medtronic or on the pump screen with Omnipod. This setting ensures no undelivered temp basal insulin is recorded while insulin delivery is suspended on the pump.

Info

This setting will be removed from Trio in the near future

SMB Delivery Ratio

Default: *50%*
Setting Limits: *30%-70%*

This is a safety limiter. Trio determines how much insulin is required to get you back to your target glucose. If SMB is enabled, Trio then delivers an SMB that defaults to half the required insulin.

Because SMBs can occur every 5 minutes with each loop cycle, it is important to set this value to a conservative level that will allow Trio to safely decrease insulin should needs suddenly change.

Trio determines Bill needs 2.4 units to return him to his target glucose. His `SMB Delivery Ratio` is set to 50%. What amount of insulin will Trio deliver?

$$2.4 \times 50\% = 1.2\ units$$

***Bonus Question*: Based on Bill's `Max SMB Basal Minutes` setting above, will he get an SMB of 1.2 units?**

$$1.0 < 1.2$$

No, Bill will only get **1.0 unit** because after Trio calculated his needed insulin, it reduced it by his `SMB Delivery Ratio`, and *then* Trio limited the amount to his `Max SMB Basal Minutes` because it was higher than this setting.

Tip

Allowed range for this setting is 30% - 70%

SMB Interval

Default *3 min*
Setting Limits: *1-10 min*

The minimum minutes since the last SMB or manual bolus that an automated SMB will be permitted.

> **Tip**
>
> Keep this setting at the default of 3 minutes to help Trio enact needed SMBs without interruption

---

Min 5m Carb Impact

Default: *5 mg/dL*
Setting Limits: *1-20 mg/dL*

This setting is used only when carb absorption isn't reflected in glucose data.

The default is an expected 8 mg/dL/5min. This affects how fast COB decays when there is little to no variation in glucose readings. The default setting correlates to a carb absorption rate of 24g/hr at a Carb Sensitivity Factor (CSF) of 4 mg/dL/g.

> **Info**
>
> This setting is ***only*** used when glucose is not changing after a meal entry. When glucose *is* changing, your CSF is used to determine carb decay and this setting is ignored.

Remaining Carbs Percentage

Default: *100%*
Setting Limits: *50%-100%*

Trio will reserve unabsorbed carb entries for 4 hours. This setting determines what percentage of the entered carbs will be in reserve for future glucose fluctuations.

At the default of 100%, all unabsorbed carbs will remain in reserve until glucose reflects carb absorption or 4 hours has elapsed.

> **Tip**
>
> Trio will not dose for these carbs until glucose reflects their absorption

---

Remaining Carbs Cap

Default: *90g*
Setting Limits: *0-200g*

In combination with the previous setting, Trio will limit the number of carbs that can remain in reserve for future carb absorption. This setting is the maximum grams of carbs that can be held over the 4 hours following a meal entry.

Noisy CGM Target Increase

Default: *130%*
Setting Limits: *100%-200%*

This setting raises your glucose target by this percent of your current target glucose when CGM readings are fluctuating widely.

This helps reduce the risk of incorrect insulin dosing based on inaccurate sensor data, ensuring safer insulin adjustments during periods of poor CGM accuracy.

Bill has a Dexcom G7 and his CGM readings are very jumpy for the first 24 hours. His current glucose target is 110 mg/dL (6.1 mmol/L). What will Trio adjust his target glucose to in order to prevent extra, unnecessary insulin?

Here is the formula you will need:

$$Target\ Glucose \times NoisyCGM\ Target\ Increase$$

Calculate Bill's new, temporary target:

$$110 \times 130\%$$

$$143\ mg/dL\ (7.9 mmol/L)$$

Answer

Trio will increase his target glucose to ***143 mg/dL*** while a noisy CGM is indicated.

Features

Welcome to the features homepage! Here you'll find information on a variety of customization features available to you in the Trio app.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Bolus Calculator

Personalize the bolus calculator using these settings

Meal Settings

Adjust maximum meal absorption time and Fat/Protein settings

Shortcuts

Enable and set up shortcuts

Remote Control

Allow remote control and find the shared secret used in Loop Follow

User Interface

Personalize the User Interface with a variety of options and preferences

App Icons

Change the Trio app icon

Bolus Calculator

Display Meal Presets

Default: *ON*

Enabling this feature allows you to create and save preset meals.

Recommended Bolus Percentage

Default: *80%*
Setting Limits: *5%-150%*

Recommended Bolus Percentage is a safety feature built into Trio. Trio first calculates the insulin required for your meal. That dosage is then multiplied by your Recommended Bolus Percentage. The adjusted dosage is shown in the bolus calculator as your `Suggested Insulin Dose`.

$$Bolus\ Required \times \frac{Recommended\ Bolus\ Percentage}{100}$$

Because Trio utilizes SMBs and UAMs to help you reach your target glucose and other AID systems do not bolus for COB the same way, this is initially set below the full calculated amount at 80%.

> **Tip**
>
> - When SMBs and UAMs are enabled, you may find your current CR results in lows and needs to be increased before you increase this setting
> - ***New Trio Users*:** It is not advised to set this to 100% until you have verified that your core settings (basal rates, ISF, and CR) do not need adjusting

---

Reduced Bolus and Super Bolus Options

> **Warning**
>
> Do not enable these features until you have verified and optimized your carb ratio (CR) setting

Enable Reduced Bolus Option

Default: *OFF*

Enabling this setting adds a "Reduced Bolus" option to the bolus calculator. Once this feature is enabled, a Reduced Bolus Percentage setting will appear for you to select. The default for this setting is 70% of the full calculated bolus.

Reduced Bolus Percentage

Default: *70%*
Setting Limits: *5%-100%*

When entering a meal into the bolus calculator, select the Reduced Bolus option to utilize this lower percentage rather than your Recommended Bolus Percentage set in the setting above. This calculation is used in conjunction with your Recommended Bolus Percentage using the formula below:

$$Bolus\ Required \times \frac{Recommended\ Bolus\ Percentage}{100} \times \frac{Reduced\ Bolus\ Percentage}{100}$$

This setting is useful for meals that require less insulin up front and more later, like pizza.

Bill has a Recommended Bolus Percentage of 80% and sets his Reduced Bolus Percentage to 50%. Trio has calculated that his insulin required for this meal is 1.5 U. What will his insulin dose be when he uses the Reduced Bolus Option in the bolus calculator?

Here is the formula:

$$Bolus\ Required \times \frac{Recommended\ Bolus\ Percentage}{100} \times \frac{Reduced\ Bolus\ Percentage}{100}$$

Enter Bill's numbers to calculate the actual percentage used:

$$1.5 \times \frac{80}{100} \times \frac{50}{100} =$$

$$1.5 \times 0.8 \times 0.5 =$$

$$1.5 \times 0.4 =$$

$$0.6\ units$$

Answer

Bill will receive a reduced bolus recommendation of 0.6 units, or **40%** of the full bolus calculation.

Enable Super Bolus Option

Default: *OFF*

Enabling this setting adds a "Super Bolus" Option to the bolus calculator. Once this feature is enabled, a Super Bolus Percentage setting will appear for you to set.

This option adds this set percentage of your current basal rate to your suggested bolus amount in the bolus calculator. If you want to receive all of your current basal rate in addition to your suggested meal bolus, use 100% for your Super Bolus Percentage.

Super Bolus Percentage

Default: *100%*
Setting Limits: *5%-200%*

When entering a meal into the bolus calculator, select the Super Bolus option to utilize this higher bolus amount rather than your Recommended Bolus Percentage set in the setting above. This calculation is used in conjunction with your Recommended Bolus Percentage and current basal rate using the formula below:

$$(Bolus\ Required \times \frac{Recommended\ Bolus\ Percentage}{100}) + (Current\ Basal\ Rate \times \frac{Super\ Bolus\ P}{100}$$

This setting is useful for meals that require more insulin up front, for example, Cinnamon Toast Crunch or candy.

Bill has a Recommended Bolus Percentage of 80% and sets his Super Bolus Percentage to 200%. His current basal rate is 0.8 U/hr. Trio has calculated that his insulin required for this meal is 1.5 U. What will his insulin dose be when he uses the Super Bolus Option in the bolus calculator?

Here is the formula:

$$(Bolus\ Required \times \frac{Recommended\ Bolus\ Percentage}{100}) + (Current\ Basal\ Rate \times \frac{Super\ Bolus}{10($$

Enter Bill's numbers to calculate the insulin dose:

$$(1.5 \times \frac{80}{100}) + (0.8 \times \frac{200}{100}) =$$

$$(1.5 \times 0.8) + (0.8 \times 2) =$$

$$1.2 + 1.6 =$$

$$2.8\ units$$

Answer

Bill will receive a super bolus recommendation of 2.8 units.

Very Low Glucose Warning

Default: *Off*

This setting triggers a confirmation dialog if you attempt to bolus when glucose is <54 mg/dL (3 mmol/L). It is also triggered when the lowest forecasted glucose (minPredBG) is <54 mg/dL (3 mmol/L).

Note

The forecast used for this warning does not include carbs or insulin that are entered for the current bolus calculation.

Meal Settings

Limits

Max Carbs

Default: *250 g*
Setting Limits: *0-300 g*

The maximum value of carbs in grams allowed per meal entry in the bolus calculator.

Max Fat

Default: *250 g*
Setting Limits: *0-300 g*

The maximum value of fat in grams allowed per meal entry in the bolus calculator.

> **Tip**
>
> This will only be visible if Enable Fat and Protein Entries is turned on.

Max Protein

Default: *250 g*
Setting Limits: *0-300 g*

The maximum value of protein in grams allowed per meal entry in the bolus calculator.

> **Tip**
>
> This will only be visible if Enable Fat and Protein Entries is turned on.

Max Meal Absorption Time

Default: *6 hours*
Setting Limits: *4-10 hours*

Carb entries will be fully decayed by the number of hours specified as Max Meal Absorption Time. Meals that are high in fat and/or protein can have long lasting effects on BG levels. To allow such late meal effects to be considered by the carb decay model, a longer Max Meal Absorption Time than the default 6 hours can be set.

If carb entries decay too slowly, it is possible to set a lower than default setting. But this should typically be adressed by tuning ISF and CR settings instead, which in combination determines the rate of carb decay.

Gastroparesis

If you have gastroparesis, increasing this setting may help keep carbs around for delayed digestion. Please also adjust Recommended Bolus Percentage to a lower percent to avoid lows due to delayed digestion.

Enable Fat and Protein Entries

Default: *OFF*

Enabling this setting allows you to log fat and protein, which are then converted into future carb equivalents (FPUs) using the Warsaw Method.

Warsaw Method

The Warsaw Method helps account for the delayed glucose spikes caused by fat and protein in meals. It uses Fat-Protein Units (FPU) to calculate the carb effect from fat and protein. The system spreads insulin delivery over several hours to mimic natural insulin release, helping to manage post-meal glucose spikes.

For more information, click to see the Omnicalculator.

Fat Conversion

$$F = fat(g) \times 90\%$$

Protein conversion

$$P = protein(g) \times 40\%$$

FPU Conversion

$$F + P = FPU(g)$$

or

$$(fat(g) \times 90\%) + (protein(g) \times 40\%) = FPU(g)$$

You can personalize the conversion calculation by adjusting the following settings that will appear when this option is enabled:

- Fat and Protein Delay
- Maximum duration
- Spread Interval
- Fat and protein percentage

Fat and Protein Delay

Default: *60 min*
Setting Limits: *15-120 min*

This setting defines the time between when you log fat & protein and when the system start delivering insulin for the Fat-Protein Unit Carb Equivalents (FPUs).

This delay accounts for the slower absorption of fat and protein, as calculated by the Warsaw Method, ensuring insulin delivery is properly times to manage glucose spikes caused by high-fat, high-protein meals.

Increasing this setting results in FPU entries that begin later.

Decreasing this setting results in FPU entries that begin earlier.

Maximum Duration

Default: *8 hours*
Setting Limits: *5-12 hours*

This sets the maximum length of time that Fat and Protein Carb Equivalents (FPUs) will be extended over from a single Fat and/or Protein bolus calculator entry.

It is one factor used in combination with the Fat and Protein Delay, Spread Interval, and Fat and Protein Factor to create the FPU entries.

Increasing this setting may result in ***more*** FPU entries with ***smaller*** carb values.

Decreasing this setting may result in ***fewer*** FPU entries with ***larger*** carb values.

Spread Interval

Default: *30 min*
Setting Limits: *10-60 min*

This determines how many minutes will be between individual Fat-Protein Unit Carb Equivalent (FPU) entries from a single Fat and/or Protein bolus calculator entry.

The shorter the interval, the smoother the correlating dosing result.

Increasing this setting may result in ***fewer*** FPU entries with ***larger*** carb values.

Decreasing this setting may result in ***more*** FPU entries with ***smaller*** carb values.

Fat and Protein Percentage

Default: *50%*
Setting Limits: *10-120%*

This setting changes how much effect the fat and protein entry has on the FPU calculations.

At the default setting of 50%, this reduces the converted amount by half.

Use the grid below to see how adjusting this percentage changes the fat and protein conversion rates:

10%

Minimum Adjustment

This is the smallest allowed FPU conversion in Trio

$$(F \times 10\%) + (P \times 10\%) =$$

$$(fat(g) \times 90\% \times 10\%) + (protein(g) \times 40\% \times 10\%) =$$

10% Adjusted FPU Calculation:

$$(fat(g) \times 9\%) + (protein(g) \times 4\%) = FPU(g)$$

30%

Reduced Adjustment

$$(F \times 30\%) + (P \times 30\%) =$$

$$(fat(g) \times 90\% \times 30\%) + (protein(g) \times 40\% \times 30\%) =$$

30% Adjusted FPU Calculation:

$$(fat(g) \times 27\%) + (protein(g) \times 12\%) = FPU(g)$$

50%

Default

$$(F \times 50\%) + (P \times 50\%) =$$

$$(fat(g) \times 90\% \times 50\%) + (protein(g) \times 40\% \times 50\%) =$$

50% Adjusted FPU Calculation:

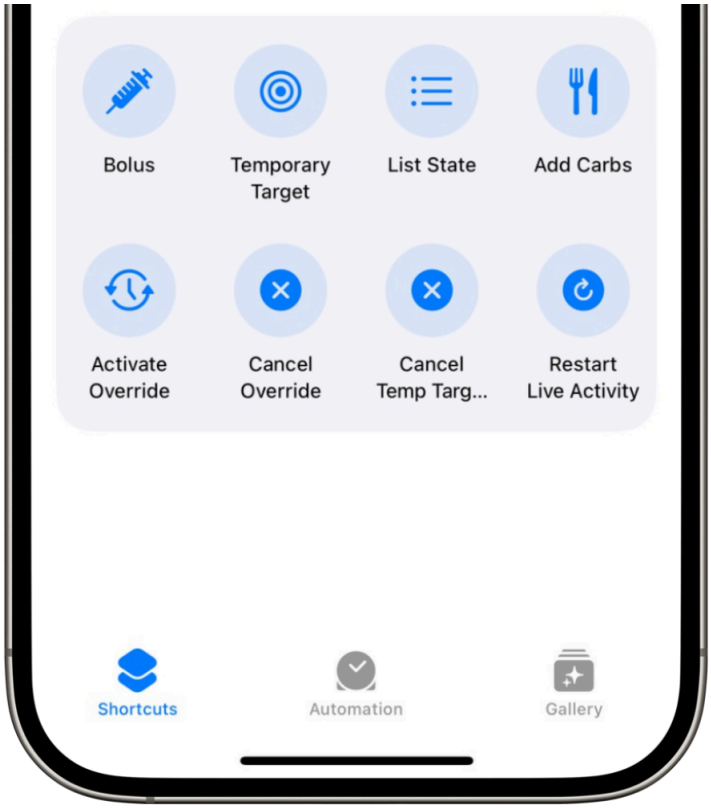
Tip

You may find your normal carb ratio needs to increase to a larger number (weaker) when you begin adding fat and protein entries. For this reason, it is best to start with a factor of 50% and adjust as you find the right balance between carb ratio and FPU conversion.

How Fat and Protein Entry Settings Work Together

Coming Soon!

Shortcuts



Allow Bolusing with Shortcuts

Default: *OFF*

Enabling this setting allows the iOS Shortcuts App to send bolus commands to Trio.

> **Tip**
>
> Disabling this setting will still allow other Shortcut commands, like Temp Targets, Add Carbs, and Start/End Overrides.

Available Shortcut actions:

- Bolus (if enabled)
- List State
- Add Carbs
- Activate Override
- Cancel Override
- Temporary Target
- Cancel Temporary Target

- Restart Live Activity

Bolus

If Allow Bolusing with Shortcuts is enabled, you will be able to use shortcuts to deliver a bolus.

List State

Shows the last glucose reading, glucose trend arrow, time since last reading, and glucose delta. It also shows current IOB and COB.

Add Carbs

Adds a carb entry into Trio.

> **Tip**
>
> You can also add fat or protein as part of the same entry.

Activate Override

Activates a chosen preset override.

> **Tip**
>
> You must first save the Preset Override on the Adjustments screen before you can activate an override using Shortcuts.

Cancel Override

Cancels an already set override.

Temporary Target

Activates a chosen preset temp target.

> **Tip**
>
> You must first save the Preset Temp Target on the Adjustments screen before you can activate a temp target using Shortcuts.

Cancel Temporary Target

Cancels and already active temp target.

Restart Live Activity

Restarts Trio's live activity.

Remote Control

Remote Control Overview

Trio can accept remote commands from *Nightscout* or from *LoopFollow*. There are a variety of options, but the final control of whether remote commands will be enacted rests with the Trio user. They can enable or disable remote control.

> ⚠ ***Nightscout* version must be 15.0.2 or newer**
>
> To properly display the OpenAPS pill with Trio 0.5.x (or newer), your *Nightscout* version must be 15.0.2 (or newer). If you do not see the expected treatments or pills in the *Nightscout* dashboard, follow the steps to Configure for OpenAPS.

The most powerful arrangement, for Trio 0.5.x (or newer), is to configure the *LoopFollow* app to use the Trio Remote Control (TRC) setting.

The limited use of remote control with *Nightscout*, i.e., entry of Carb Correction and Temporary Targets when Careportal is authenticated, continues to be supported with Trio.

***Nightscout*URL or App**	**Options**
Careportal	Carb Correction Temporary Target Temporary Target Cancel

Additional remote capabilities are offered for Trio using the *LoopFollow* app with these versions:

- Trio 0.5.x (or newer)
- *LoopFollow* version 2.4.0 (or newer)

***LoopFollow* Remote Type**	**Options**
Nightscout	Set and Cancel Temp Target
Trio Remote Control	Meal (Carbs or Carbs & Bolus) Bolus Temp Target Overrides

How does this differ from Trio 0.2.x?

Trio can use *Nightscout* Careportal to enter `Carb Correction`, and start and cancel `Temporary Target`.

- This was available in Trio 0.2.x and continues to be available in Trio 0.5.x (or newer).

Trio 0.2.x supported other remote options (using announcements via Careportal).

- Those options were replaced by the more secure Trio Remote Control for Trio 0.5.x (or newer)
- **Using announcements to provide remote control of the Trio phone is no longer supported**

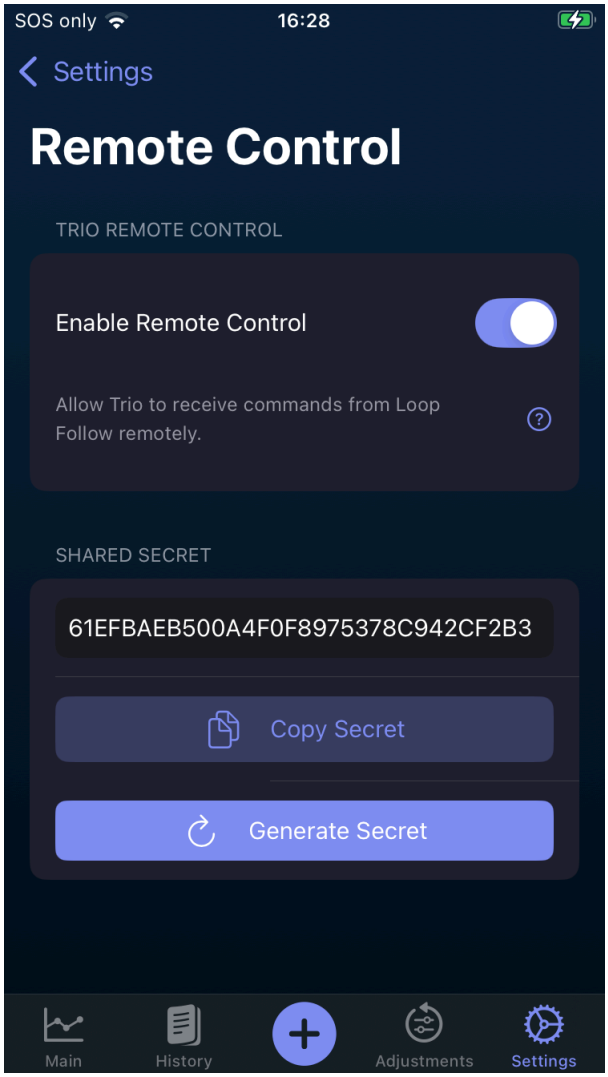
Trio Remote Control

Default: *OFF*

Remote control must be enabled on the Trio phone or no remote information is accepted by the Trio phone.

> You can search for this screen in Trio settings or go through the sequence: Trio, Settings, Features, Remote Control.

Once Remote Control is enabled, a Shared Secret is available. This is only used if you want to Configure *LoopFollow* Trio Remote Control.



When Remote Control is enabled on the Trio app and the *LoopFollow* phone is properly configured, you can add carbs, send boluses, set or cancel overrides or temporary targets from the *LoopFollow* phone to the Trio phone via *Apple* push notifications.

The `SHARED SECRET` should be copied from the Trio phone and added to the `Shared Secret` row of the *LoopFollow* Remote Settings screen as part of the configuration for using *LoopFollow*.

> **Important**
>
> The ability for the Trio app to be remotely controlled will be **disabled** when `Enable Remote Control` is turned OFF, even if you have *LoopFollow* configured with the correct shared secret or your *Nightscout* URL has Careportal access. This is for the protection of the Trio user, so that they **always** are the primary controller of their insulin dosing app.

LoopFollow Overview

The most flexible remote control for a caregiver to help their loved one is to use the *LoopFollow* app on the caregiver's phone. It is available only for iOS devices: iPhone, iPad or Mac.

The remote commands are sent from the *LoopFollow* phone to the *Trio* phone using Apple Push Notifications. A real-time response is returned to the *LoopFollow* phone for most cases with either success or failure, with the reason.

The documentation for *LoopFollow* can be reached by typing `loopfollowdocs.org` in a browser.

To configure *LoopFollow* for *Trio Remote Control*, read these two pages in detail:

- LoopFollowDocs: Remote Control Overview
- LoopFollowDocs: Trio Remote Control

Use *LoopFollow* Trio Remote Control

Once the *LoopFollow* phone is configured, and while the Trio phone is handy, test sending Remote Commands. It is good to also have a browser open with the *Nightscout* site displayed.

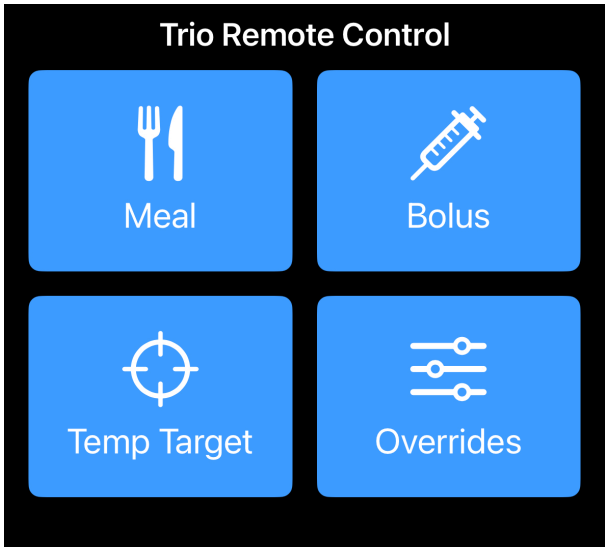
Remember to give the system time to update.

The sequence for a remote command is

- *LoopFollow* sends the command via *Apple Push Notifications* (APNS)
- *Apple Push Notifications* passes the command to *Trio*
    - *Trio* performs additional safety checks and either accepts or rejects the command
- *Trio* sends a real-time response via *Apple Push Notifications*
- *Apple Push Notifications* sends the response to the *LoopFollow* phone

Trio also uploads the treatments to *Nightscout* and *LoopFollow* presents those treatments on the *LoopFollow* mainb screen.

LoopFollow also provides logs and reports for these treatements.



Remote Meal

The Remote Meal command allows you to log carbohydrates (and optionally fat and protein) to Trio from LoopFollow, with or without an accompanying bolus.

How to Send a Remote Meal

1. Open LoopFollow and tap the Remote Control button
2. Select "Meal" from the command options
3. Enter the meal details:
    - **Carbs** (required): Grams of carbohydrates
    - **Fat** (optional): Grams of fat
    - **Protein** (optional): Grams of protein
    - **Bolus** (optional): Units of insulin to deliver with the meal
    - **Schedule** (optional): Time to log the carbs (for pre-bolusing)
4. Tap "Send" to transmit the command

What Happens When Trio Receives a Meal Command

1. **Validation**: Trio checks that:
    - At least one macronutrient (carbs, fat, or protein) is provided
    - Carbs don't exceed your Max Carbs setting
    - Fat doesn't exceed your Max Fat setting
    - Protein doesn't exceed your Max Protein setting
    - No newer carb entries exist (prevents accidental duplicates)
2. **Carb Entry Creation**: Trio creates a carb entry with:
    - The macronutrients you specified
    - A note: "Remote meal command"
    - FPU (Fat Protein Units) enabled if fat OR protein is present
    - Scheduled time (if provided)
3. **Bolus Delivery** (if bolus amount was included):
    - The bolus is delivered immediately (even if the meal is scheduled)
    - Bolus validation checks are performed (see Remote Bolus section)
4. **Response Notification**: *Trio* sends a success or failure notification to *LoopFollow* via APNS
5. **Upload to Nightscout**: The meal entry is logged to Nightscout for your records

> ⚠ **Scheduled Meals with Bolus**
>
> When entering meals and choosing to schedule the meal, any bolus included in the meal is enacted immediately. Only the carb entry is entered according to the schedule.

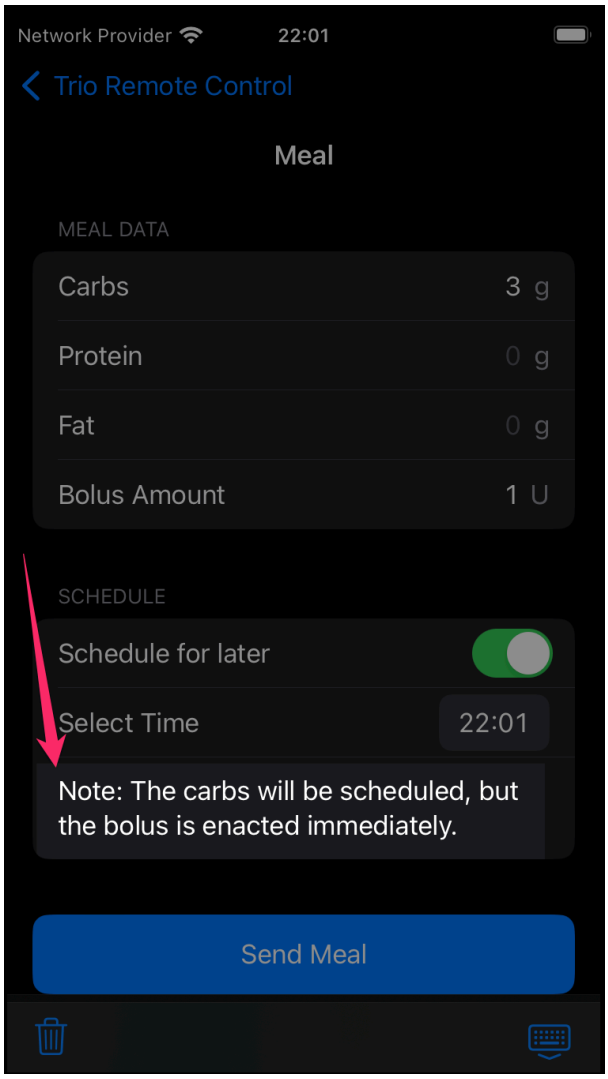
Safety Features

- **Duplicate Prevention**: Won't log carbs if a newer entry already exists
- **Limit Enforcement**: Respects your Max Carbs, Max Fat, and Max Protein settings
- **FPU Calculation**: Automatically enables Fat Protein Units for low-carb, high-fat/protein meals
- **Nightscout Logging**: All remote meals are logged for audit trail

Use Cases

- **Pre-bolusing**: Schedule a meal for 15-30 minutes in the future while delivering insulin now
- **Macronutrient Tracking**: Log fat and protein for better extended bolus calculations
- **Meal + Bolus**: Deliver a complete meal bolus remotely in one command
- **Caregiver Support**: Parents can log meals for children at school

When entering meals and choosing to schedule the meal, any bolus included in the meal is enacted immediately. Only the carb entry is entered according to the schedule.



Remote Bolus

The Remote Bolus command allows you to deliver insulin remotely via LoopFollow. This is one of the most powerful remote features and includes multiple safety checks.

How to Send a Remote Bolus

1. Open LoopFollow and tap the Remote Control button
2. Select "Bolus" from the command options
3. Enter the bolus amount in units (U)
4. Tap "Send" to transmit the command

What Happens When Trio Receives a Bolus Command

1. **Validation**: Trio performs comprehensive safety checks:
    - Bolus amount is provided and greater than 0
    - Bolus amount doesn't exceed your Max Bolus setting
    - Current IOB (Insulin on Board) is calculable
    - Current IOB + bolus amount doesn't exceed Max IOB
    - No recent bolus >20% of the requested amount in the last 10 minutes
    - APSManager is available and functioning
2. **Insulin Delivery**: If all checks pass:
    - Trio sends the bolus command to your insulin pump
    - The pump delivers the insulin
    - Trio monitors the delivery for completion
3. **Response Notification**: *Trio* sends a success or failure notification to *LoopFollow* via APNS
4. **Upload to Nightscout**: The bolus is logged to Nightscout with:
    - Amount delivered
    - Timestamp
    - Note indicating it was a remote command

Safety Features

Max Bolus Protection Your Max Bolus setting (configured in Trio settings) prevents delivery of dangerously large boluses. A remote bolus request exceeding this limit will be rejected.

Example: If Max Bolus = 10 U, a remote request for 12 U will fail with error message.

Max IOB Protection Trio calculates your current Insulin on Board and ensures the new bolus won't exceed your Max IOB safety limit.

Formula: `Current IOB + Requested Bolus ≤ Max IOB`

Example: - Current IOB: 8 U - Max IOB: 12 U - Remote bolus request: 5 U - Result: REJECTED (8 + 5 = 13 U, which exceeds 12 U limit)

Duplicate Bolus Prevention Trio checks for recent boluses in the last 10 minutes. If a bolus greater than 20% of the requested amount was recently delivered, the remote command is rejected.

Example: - Remote request: 5 U - Recent bolus (8 minutes ago): 4.5 U - 20% of 5 U = 1 U - Since 4.5 U > 1 U, the request is REJECTED as a likely duplicate

Purpose: Prevents accidental double-dosing if LoopFollow doesn't immediately reflect a bolus you just delivered manually.

Time Window Validation All remote commands include a timestamp. Trio rejects commands that are: - More than 10 minutes old (prevents replay attacks) - From the future (prevents clock manipulation)

Error Messages

If a remote bolus fails, you'll receive a notification explaining why:

Error	Meaning	Action
"Bolus amount exceeds max bolus"	Requested bolus > Max Bolus setting	Reduce bolus amount or increase Max Bolus in settings
"IOB would exceed max IOB"	Current IOB + bolus > Max IOB	Wait for IOB to decrease, or increase Max IOB
"Recent bolus detected"	Similar bolus in last 10 minutes	Wait 10 minutes or verify this isn't a duplicate
"APSManager unavailable"	Trio can't access pump	Check Trio app is running and pump is connected
"Command too old"	Timestamp > 10 minutes ago	Check phone clocks are synchronized; resend command

Use Cases

- **Meal Boluses**: Deliver insulin for meals when away from phone
- **Correction Boluses**: Correct high blood sugar remotely
- **Caregiver Support**: Parents can dose insulin for children
- **Emergencies**: Deliver insulin if user can't access their phone

Important Safety Note

Remote bolus is a powerful feature that delivers real insulin. Always:

- Verify the bolus amount before sending
- Check current blood glucose and IOB before sending
- Ensure the user is aware a bolus is being sent
- Have emergency glucagon available
- Never use remote bolus as a prank or without authorization

Temp Target

The Temp Target command allows you to set or cancel temporary glucose targets remotely via LoopFollow.

How to Set a Remote Temp Target

1. Open LoopFollow and tap the Remote Control button
2. Select "Temp Target" from the command options
3. Enter the temp target details:
    - **Target**: Desired glucose target in mg/dL (or mmol/L)
    - **Duration**: How long the target should be active (in minutes)
4. Tap "Send" to transmit the command

What Happens When Trio Receives a Temp Target Command

1. **Validation**: Trio checks that:
    - Target value is provided and valid
    - Duration is provided and greater than 0
2. **Temp Target Creation**: Trio creates a temporary target with:
    - `targetTop` and `targetBottom` both set to the specified target
    - Duration in minutes
    - Custom name indicating it's a remote target
    - Marked as a "local" entry
3. **Storage**: The temp target is saved to Trio's temp target storage
4. **Sync to Nightscout**: The temp target is uploaded to Nightscout
5. **UI Update**: Trio posts notifications to update the UI immediately
6. **Response Notification**: LoopFollow receives confirmation

How to Cancel a Remote Temp Target

1. Open LoopFollow and tap the Remote Control button

2. Select "Cancel Temp Target" from the options
3. Tap "Send" to transmit the command

What Happens When Canceling

1. **Fetch Active Targets**: Trio retrieves all currently active temp targets
2. **Create End Records**: For each active temp target, Trio creates a `TempTargetRunStored` record marking it as ended
3. **Disable All**: All active temp targets are marked as disabled
4. **Sync to Nightscout**: Changes are uploaded
5. **UI Update**: Trio interface updates to show no active temp targets

Common Temp Target Values

Target	Use Case
80-100 mg/dL	Before meals (tighter control)
110-120 mg/dL	Sleeping/overnight (prevent lows)
140-160 mg/dL	Exercise (prevent lows)
120-130 mg/dL	After meals (less aggressive corrections)

Temp Targets and Algorithm Behavior

Temporary targets affect how Trio delivers insulin:

- **Higher targets** → Less aggressive insulin delivery, fewer/smaller SMBs
- **Lower targets** → More aggressive insulin delivery, more/larger SMBs
- Temp targets override your normal target glucose setting
- Trio respects your safety limits (Max IOB, Max SMB) regardless of temp target

Use Cases

- **Exercise**: Set higher target before/during exercise to prevent lows
- **Sleep**: Set higher target at bedtime for peace of mind
- **Pre-meal**: Set lower target before eating to achieve tighter control
- **Illness**: Adjust targets when sick
- **Cancel**: Return to normal targets anytime

Overrides

The Override command allows you to activate or cancel override presets remotely via LoopFollow. Overrides are powerful tools that simultaneously adjust multiple settings (insulin sensitivity, basal rates, carb ratios, and targets) based on predefined presets.

Prerequisites

Before using remote overrides, you must:

1. **Create Override Presets** in Trio:
    - Go to Trio Settings → Overrides
    - Create and name your override presets (e.g., "Exercise", "Sick Day")
    - Configure the percentage adjustments for each preset
2. **Configure LoopFollow** with the exact preset names

How to Start a Remote Override

1. Open LoopFollow and tap the Remote Control button
2. Select "Start Override" from the command options
3. Select the override preset from the dropdown (must match a preset name in Trio)
4. Tap "Send" to transmit the command

What Happens When Trio Receives a Start Override Command

1. **Validation**: Trio checks that:
    - Override name is provided and not empty
    - Override name matches an existing preset (exact match, case-sensitive)
2. **Fetch Presets**: Trio retrieves all override presets from storage
3. **Find Matching Preset**: Searches for a preset with the exact name specified
4. **Disable Other Overrides**: All currently active overrides are disabled first
5. **Enable Requested Override**: The matching override preset is:
    - Enabled
    - Timestamp set to current time
    - Marked as not yet uploaded to Nightscout
6. **Save to Core Data**: Changes are persisted
7. **UI Update**: Trio posts notifications to refresh the UI
8. **Upload to Nightscout**: The active override is logged
9. **Response Notification**: LoopFollow receives confirmation

How to Cancel a Remote Override

1. Open LoopFollow and tap the Remote Control button
2. Select "Cancel Override" from the options

3. Tap "Send" to transmit the command

What Happens When Canceling

1. **Fetch Active Overrides**: Trio retrieves all currently active overrides
2. **Create End Records**: For each active override, Trio creates an `OverrideRunStored` record marking it as ended
3. **Disable All**: All active overrides are marked as disabled
4. **Sync to Nightscout**: Changes are uploaded
5. **UI Update**: Trio returns to normal settings

How Overrides Work

Overrides modify your therapy settings by percentage:

Setting	Override Adjustment	Effect
Insulin Sensitivity (ISF)	Percentage multiplier	50% = half the insulin, 200% = double the insulin
Basal Rates	Percentage multiplier	150% = 1.5x normal basal
Carb Ratios	Percentage multiplier	75% = less insulin per carb
Target Glucose	Override target value	Completely replaces normal target

Example "Exercise" Override: - ISF: 150% (more sensitive, less insulin) - Basal: 75% (reduced basal delivery) - CR: 120% (less insulin for carbs) - Target: 140 mg/dL (higher target to prevent lows)

Example "Sick Day" Override: - ISF: 50% (more resistant, more insulin) - Basal: 150% (increased basal delivery) - CR: 80% (more insulin for carbs) - Target: 110 mg/dL (tighter control)

Override Preset Name Matching

> ⚠ **Exact Match Required**
>
> The override name sent from LoopFollow must **exactly match** a preset name in Trio:
>
> - Case-sensitive: "Exercise" ≠ "exercise"
> - Spacing matters: "Sick Day" ≠ "SickDay"
> - Spelling must be exact
>
> If no match is found, the command will fail with an error message.

Use Cases

- **Exercise**: Activate exercise mode remotely before/during physical activity
- **Illness**: Switch to sick day settings when user is unwell

- **Travel**: Adjust settings for time zone changes or schedule disruptions
- **Stress**: Modify insulin delivery during stressful periods
- **Menstrual Cycle**: Adjust for hormonal changes
- **Cancel**: Return to normal settings anytime

Safety Considerations

- **Overrides are powerful**: They affect multiple settings simultaneously
- **Test first**: Test override presets locally before using remotely
- **Monitor closely**: Watch glucose trends closely when an override is active
- **Duration**: Consider setting automatic duration limits on override presets
- **Communication**: Ensure the user knows when an override is activated remotely

Troubleshooting

Issue	Cause	Solution
"Override not found"	Name doesn't match any preset	Check spelling and capitalization in both Trio and LoopFollow
"Failed to enable override"	Trio can't access storage	Restart Trio app and try again
Override not taking effect	Override successfully activated but settings unchanged	Verify the override preset has non-zero percentage adjustments configured

---

Troubleshooting

This section covers known troubleshooting issues:

- *Nightscout* not displaying Trio data: Configure for OpenAPS
- Was able to select Trio Remote Control in *LoopFollow* but it is no longer working: Stop *Nightscout* access from the *Loop* app
- Cannot select Trio Remote Control in *LoopFollow*: Update Profile

Configure for OpenAPS

The *Nightscout* version must be 15.0.2 (or newer) to properly display the OpenAPS pill with Trio 0.5.x (or newer). Check your revision: *Nightscout* URL, Menu, scroll to bottom and examine the About section.

If you transitioned from the *Loop* app, you must make some modifications to *Nightscout* before you will be successful viewing your Trio data in your *Nightscout* site.

In *Nightscout*, you need to modify these config vars:

Config Var	`Loop`	`Trio`
`ENABLE`	`loop`	`openaps`
`SHOW_PLUGINS`	`loop`	`openaps`
`SHOW_FORECAST`	`loop`	`openaps`

Remember to restart the *Nightscout* server (restart dynos) after updating these variables.

Stop *Nightscout* access from the *Loop* app

If you were previously running the *Loop* app:

- Remove *Nightscout* from *Loop* Services
- Add *Nightscout* credentials to Trio
    - You need the URL and the API_SECRET.

In addition to this step, you may need to force the profile (from Trio) to upload to *Nightscout* and overwrite the one stored as the default profile in *Nightscout*.

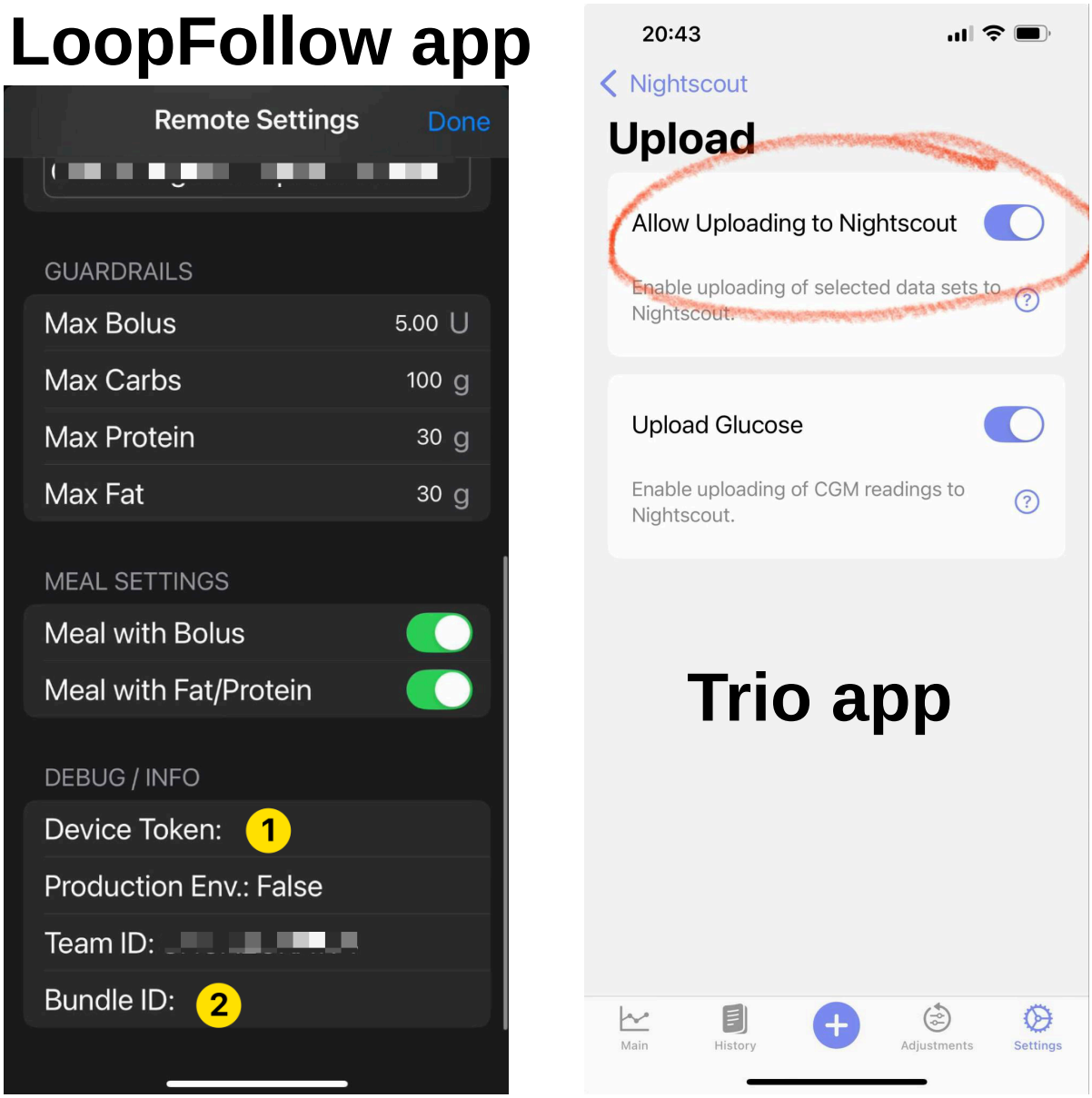
Update Profile

If you were previously running the *Loop* app, take the actions in the previous section and then force the profile to update.

To force a profile to update to *Nightscout*, go to the Trio app and toggle Allow Uploading to Nightscout off (disable) and then enable it again.

Once the user has toggled "Allow Uploading to Nightscout", *LoopFollow* needs to be refreshed (pull down glucose value to refresh) or re-started in order to fetch the correct information. *LoopFollow* will refresh eventually, but most users are impatient.

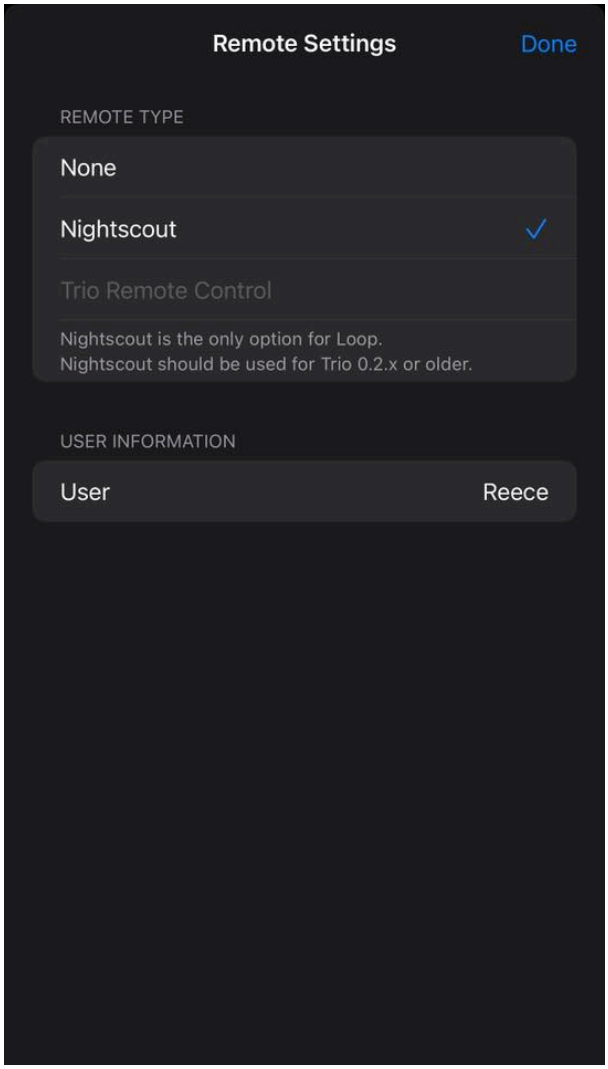
If the Debug Info in *LoopFollow* is missing a Device Token or a Bundle ID, as shown on the left side of the graphic, you need to make sure the *Loop* app is no longer uploading to *Nightscout* and force the profile to update.

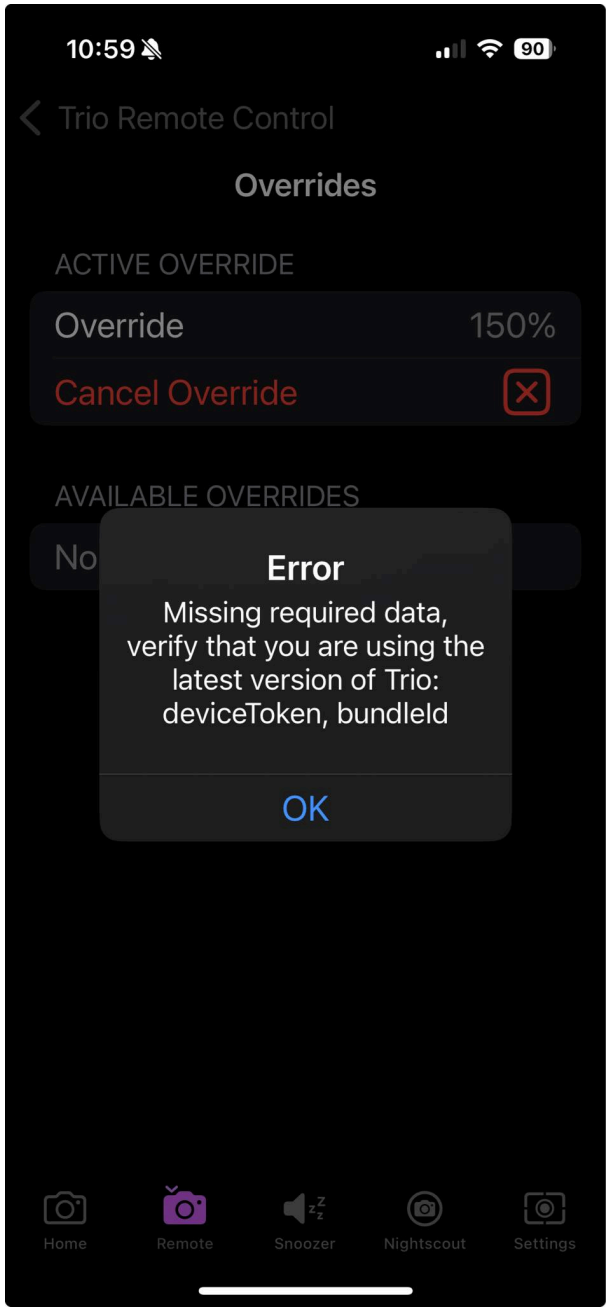


Trio Remote Control Stops Working

Other signatures that you need to force the update are shown in the graphics below - for both these instances, Trio Remote Control (TRC) was working with *LoopFollow* and then stopped working:

TRC Option Not Allowed	TRC Error





Build *LoopFollow*

The *LoopFollowDocs* have complete build instructions:

- LoopFollowDocs: Build Options

User Interface Settings Menu

Light Mode

Dark Mode

Appearance

Default: *System Default*

Options

System Default:
This option follows the phone's current color scheme.

Light:
Always in Light Mode

Dark:
Always in Dark Mode

Glucose Color Scheme

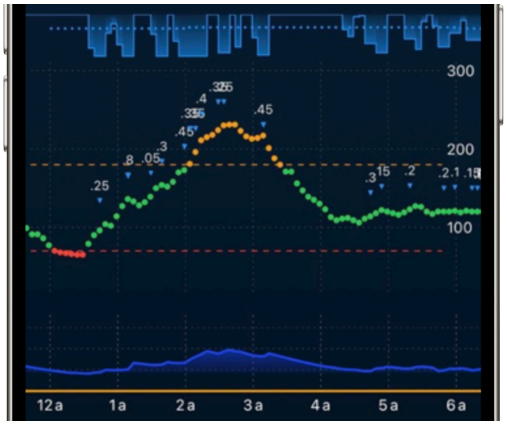
Default: *Static*

Options

Static:
This option uses orange, green, and red to indicate where the glucose reading falls on the range set.

Threshold Graph Legend

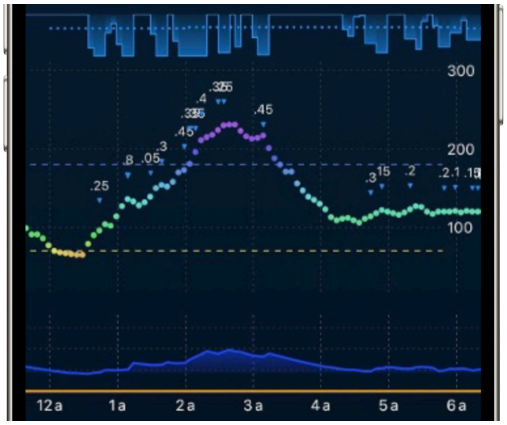
- High Threshold
- Low Threshold
- Above Range Glucose
- In Range Glucose
- Below Range Glucose



Dynamic:
This option uses the color wheel to indicate where the glucose reading falls in relation to your target glucose.
**The colors of the threshold lines are also dynamic, so the exact coloring will depend on your settings*

Threshold Graph Legend

- High Threshold (=180 mg/dL)
- Low Threshold (=70 mg/dL)
- Above Target
- At or Near Target
- Below Target



Axis Grid Lines

Show X-Axis Grid Lines

Default: *ON*
Choose whether or not to display the X-Axis (horizontal) grid lines on the glucose graph.

Show Y-Axis Grid Lines

Default: *ON*
Choose whether or not to display the Y-Axis (vertical) grid lines on the glucose graph.

Show Low and High Thresholds

Default: *ON*
This setting displays the upper and lower values for your glucose target range.

Low Threshold

Default: *70 mg/dL (3.9 mmol/L)*
The lower value of your displayed glucose target range.

High Threshold

Default: *180 mg/dL (10mmol/L)*
The upper value of your displayed glucose target range.

> **Tip**
>
> This range is for display and statistical purposes only and does not influence insulin dosing.

Forecast Display Type

Default: *Cone*
This setting allows you to choose between the Cone of Uncertainty (Cone) and OpenAPS Forecast Lines (Lines) for the glucose forecast display. Descriptions for each option found below.

Options

Cone (of Uncertainty):

Combines all possible forecasts from the OpenAPS lines into a single range. This option has been shown to reduce confusion and stress around algorithm forecasts by providing a clearer, less overwhelming visual representation.

Lines:

Uses the IOB, COB, UAM, and ZT forecast lines from the OpenAPS (Oref) Algorithm used in Trio. This option provides a more detailed view of the algorithm's forecast and may be more or less helpful depending on the user's preference.

---

eA1c Display Unit

Default: *Percent*
Choose which format you'd prefer the eA1c (estimated A1c) value to be displayed as. Choose between percentage (Ex: 6.5%) or mmol/mol (Ex: 48 mmol/mol).

Options

Percent or **mmol/mol**

Show Carbs Required Badge

Default: *OFF*
Turning this on will show the grams of carbs needed to prevent a low as a notification badge on the Trio home screen, located above the main icon.

Carbs Required Threshold

If `Show Carbs Required Badge` is enabled, this setting will appear. Set this to the lowest number of carbs you'd like to be recommended. A recommendation will not be given if the carbs required is below this number.

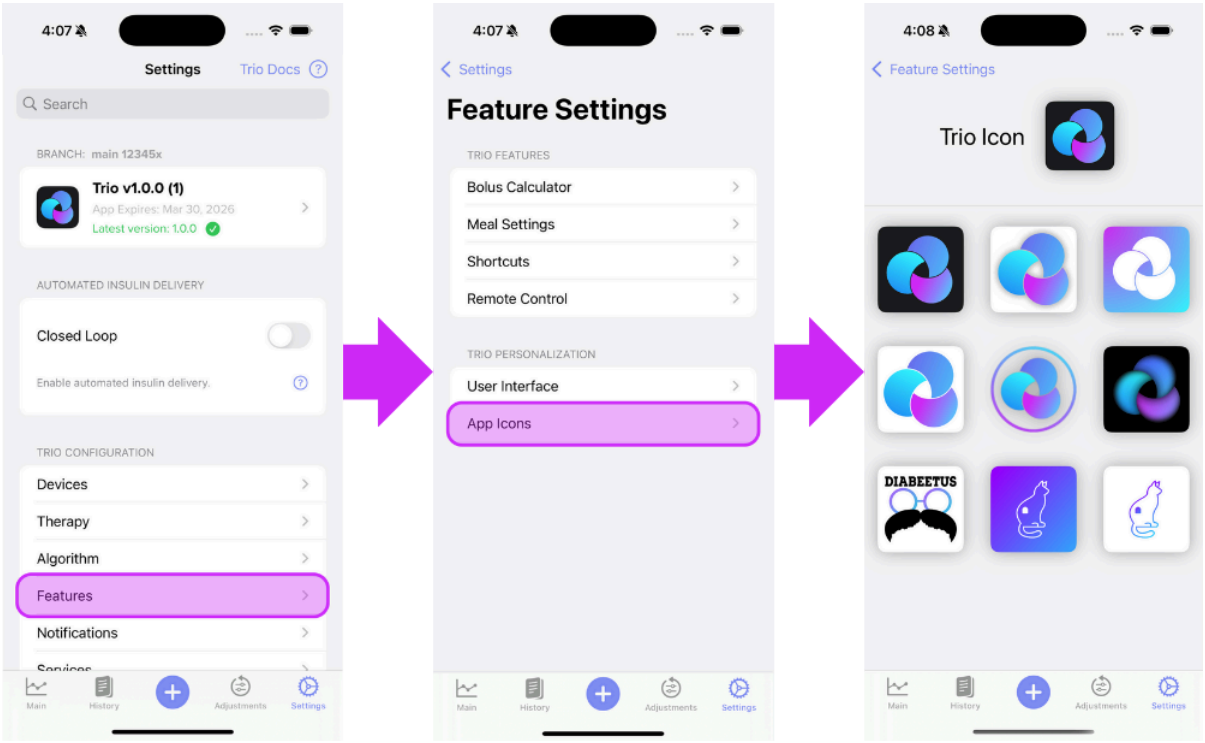
Tip

The carbs suggested with this feature are to be used as a recommendation, not as a requirement. Depending on the current accuracy of your sensor and the accuracy of your settings, the suggested carbs can vary widely from what is actually required. Ultimately, use your best judgment before ingesting the suggested quantity of carbs.

App Icons

Trio has multiple different icons to choose from.

To change the Trio icon to suit your homescreen you can go to Settings -> Features -> App Icons and select your prefered Trio icon.



App Diagnostics ✏️

Crash Reporting

Default: *ON*

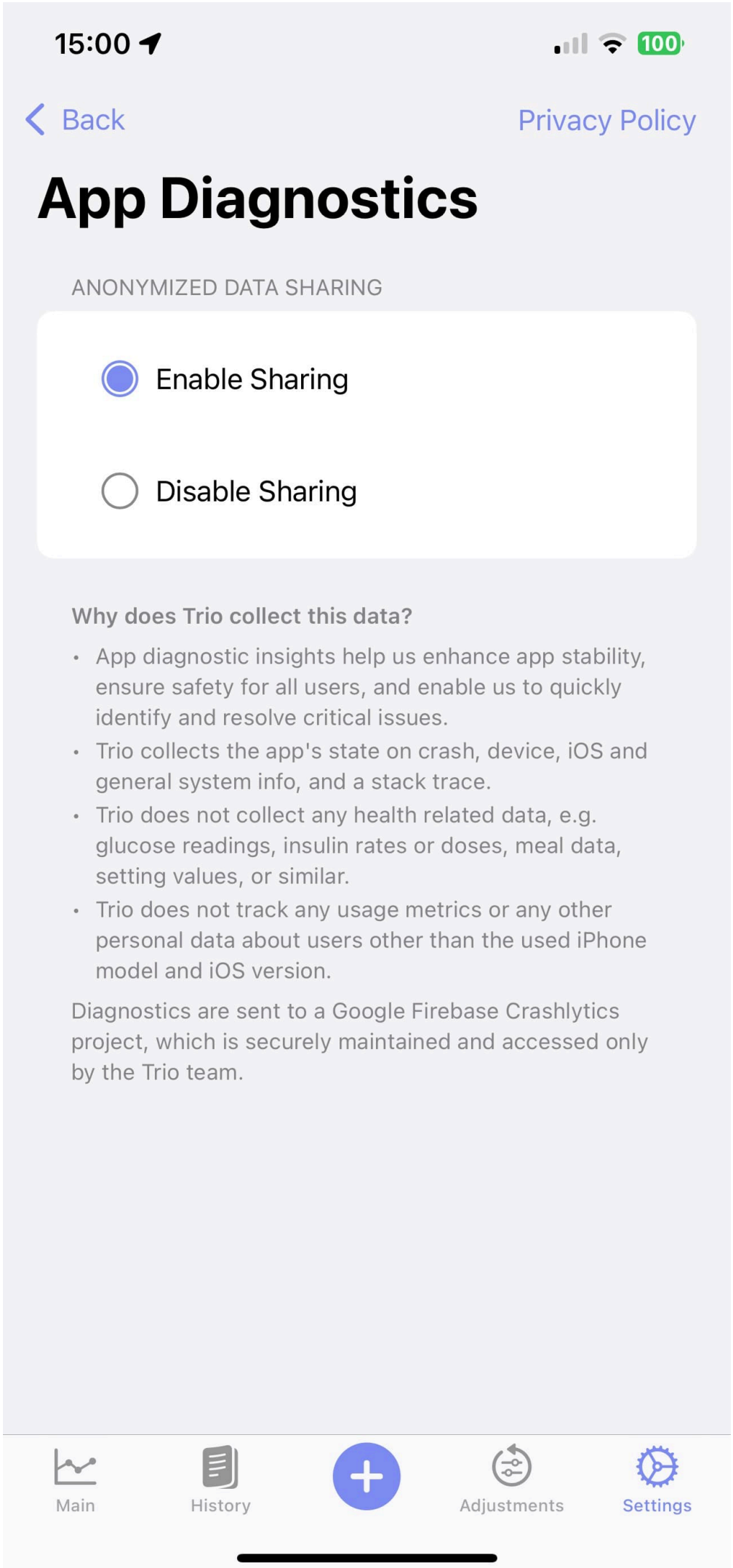
Crash Reporting (Opt-In by default, with ability to Opt-Out).

Read the Trio Privacy Policy.

When activated, Trio's App Diagnostics feature automatically captures anonymized crash reports so the developers can keep the app stable and safe.

The captured data **never** includes personal or health information.
No glucose values, insulin data, meals, settings, or usage analytics are collected.

Crash reports are sent securely to the team's private **Google Firebase Crashlytics** project, where **only the Trio developers can view them and use them to identify and fix critical issues quickly**.


15:00
100
Back
Privacy Policy
App Diagnostics
ANONYMIZED DATA SHARING
Enable Sharing
Disable Sharing
Why does Trio collect this data?
• App diagnostic insights help us enhance app stability, ensure safety for all users, and enable us to quickly identify and resolve critical issues.
• Trio collects the app's state on crash, device, iOS and general system info, and a stack trace.
• Trio does not collect any health related data, e.g. glucose readings, insulin rates or doses, meal data, setting values, or similar.
• Trio does not track any usage metrics or any other personal data about users other than the used iPhone model and iOS version.
Diagnostics are sent to a Google Firebase Crashlytics project, which is securely maintained and accessed only by the Trio team.
Main
History
Adjustments
Settings

Notifications

Welcome to the Notifications homepage! Here you'll find information on the various methods of keeping you updated on Trio's status and important information while you are not actively using the app.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Trio Notifications

Set up and customize Trio's alerts and notifications

Live Activity

Configure live activity widgets

Calendar Events

Use calendar events to display data

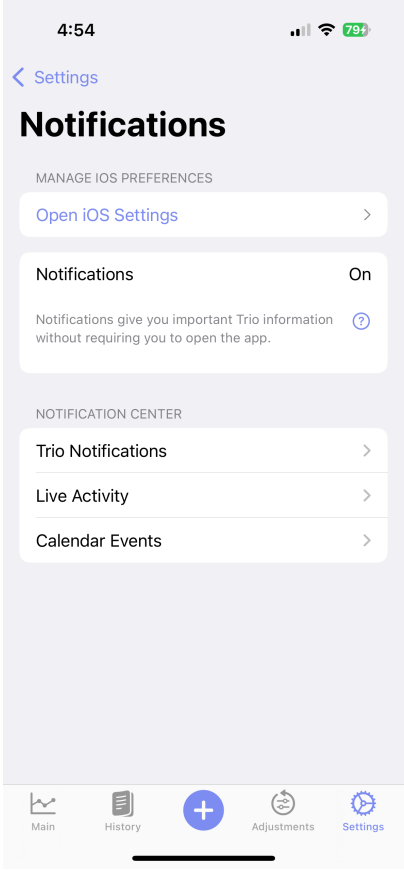
Notifications

Notifications give you important Trio information without requiring you to open the app. It is essential that these are allowed in your iPhone system settings. You can tailor them to your needs in this section.

You may notice a few different alerts and warnings in regards to notifications. Click on the tabs below to see what they are and how to remedy them.

Notifications ON

You are good to go! Your notifications are turned on. Proceed to the 3 notification options on the app screen to customize your app notifications.



Notifications OFF

Your notifications are off. Open iOS Settings and enable them

Play Alarm Sound

Default: *OFF*

This will cause a sound to be triggered by Trio notifications for Carbs Required and Glucose Low/High Alarms.

Always Notify Pump

Default: *ON*

With iOS Trio Notifications enabled, you can let Trio display most Pump Notifications in iOS Notifications Center as a Banner, List, and/or on the Lock Screen. It allows you to refer to Trio Information at a glance and troubleshoot any informational issue. Setting iOS Notifications Banner Style to *Persistent* displays alert banners in the app until you dismiss them.

> **Note**
>
> If iOS Trio Notifications are disabled, Trio will display these messages in-app as a banner only

---

Always Notify CGM

Default: *ON*

With iOS Trio Notifications enabled, you can let Trio display most CGM Notifications in iOS Notifications Center as a Banner, List, and/or on the Lock Screen. It allows you to refer to Trio Information at a glance and troubleshoot any informational issue. Setting iOS Notifications Banner Style to *Persistent* displays alert banners in the app until you dismiss them.

> **Note**
>
> If iOS Trio Notifications are disabled, Trio will display these messages in-app as a banner only

---

Always Notify Carb

Default: *ON*

With iOS Trio Notifications enabled, you can let Trio display most Carb Notifications in iOS Notifications Center as a Banner, List, and/or on the Lock Screen. It allows you to refer to Trio Information at a glance and troubleshoot any informational issue. Setting iOS Notifications Banner Style to *Persistent* displays alert banners in the app until you dismiss them.

> **Note**
>
> If iOS Trio Notifications are disabled, Trio will display these messages in-app as a banner only

---

Always Notify Algorithm

Default: *ON*

With iOS Trio Notifications enabled, you can let Trio display most Algorithm Notifications in iOS Notifications Center as a Banner, List, and/or on the Lock Screen. It allows you to refer to Trio Information at a glance and troubleshoot any informational issue. Setting iOS Notifications Banner Style to *Persistent* displays alert banners in the app until you dismiss them.

> **Note**
>
> If iOS Trio Notifications are disabled, Trio will display these messages in-app as a banner only

---

Show Glucose App Badge

Default: *OFF*

This will add your current glucose readings on the top right of your Trio app icon as a red notification badge.

> **Note**
>
> Changing this setting takes effect on the *next* glucose reading

---

Glucose Notifications

Default: *DISABLED* Set the Glucose Notifications Option. Descriptions for each option found below:

Disabled

No glucose notifications will be triggered

Always

A notification will be triggered every time your glucose is updated in Trio

Glucose Alarm Limits

Low Glucose Alarm Limit

Default: *70 mg/dL (3.9 mmol/L)*
Range: *40 - 100 mg/dL (2.2 - 5.5 mmol/L)*

This setting determines the low value for your alert range. Set this to the low value in which you'd like to receive alerts.

High Glucose Alarm Limit

Default: *180 mg/dL*
Range: *100 - 400 mg/dL (5.5 - 22.2 mmol/L)*

This setting determines the high value for your alert range. Set this to the high value in which you'd like to receive alerts.

Add Glucose Source to Alarm

Default: *OFF*

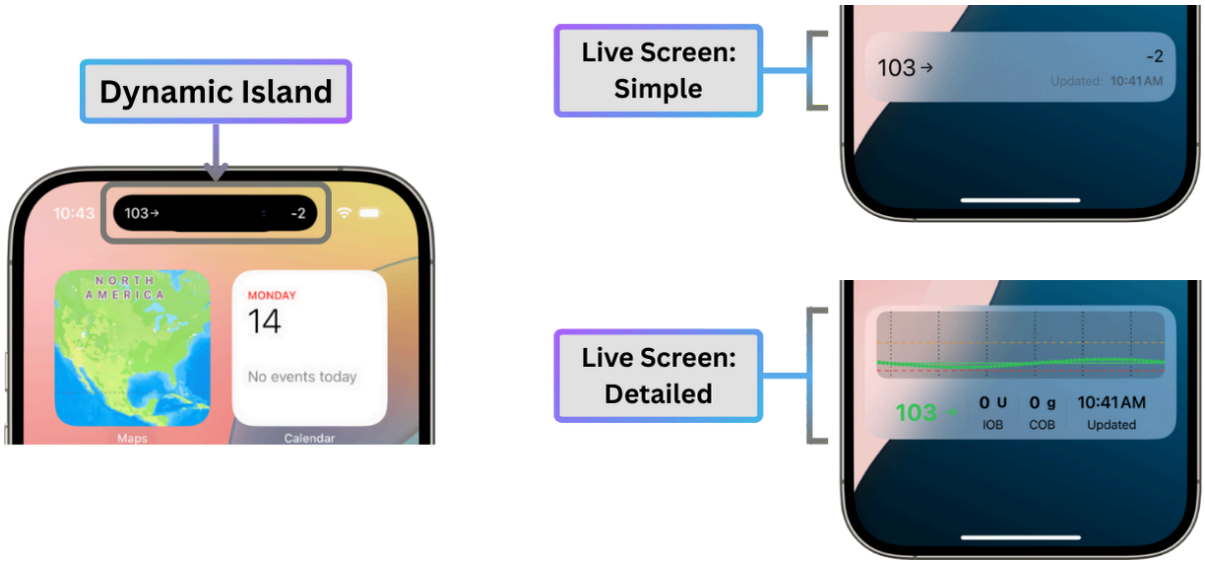
The source of your glucose reading will be added to the notification.

Live Activity

Enable Live Activity

Default: *OFF*

With Live Activities enabled, Trio displays your current data on your phone's Lock Screen and the Dynamic Island. Once this is enabled, you will have the option to choose either a "Simple" Display or "Detailed" Display with the setting Live Screen Widget Style.

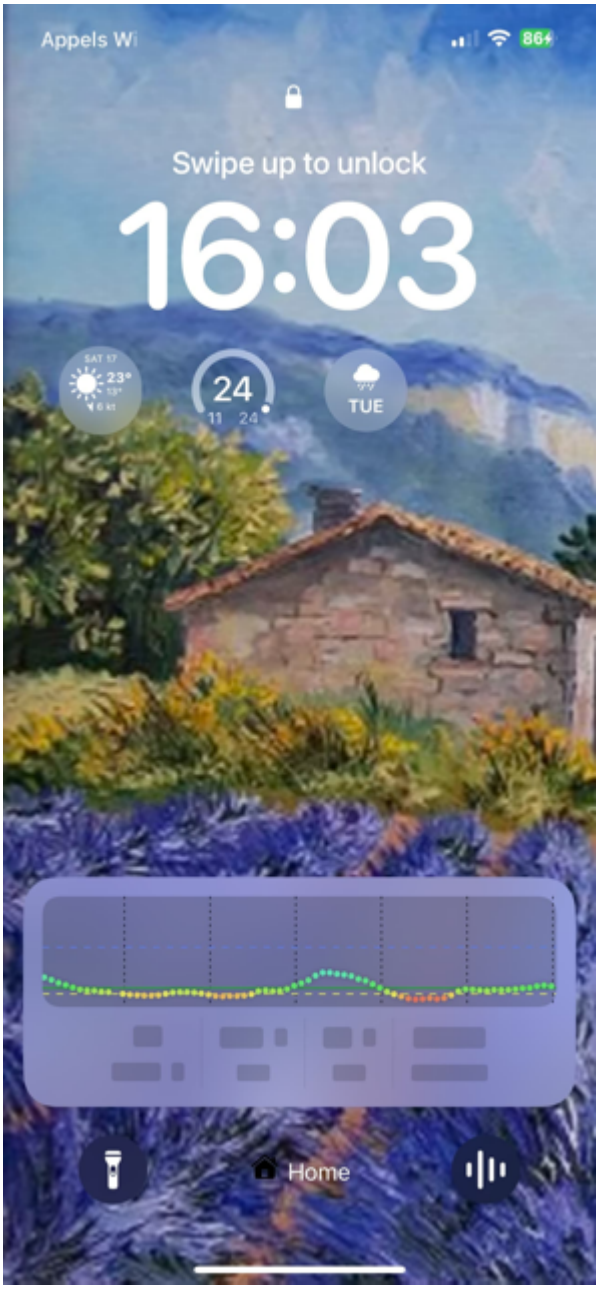


Fix hidden Live Activity values on lock screen

If Trio Live Activities appear greyed out when your iPhone is locked and you want them to remain visible at all times, you need to adjust this setting.

Show Live Activity Values on Locked iPhone

If **Trio Live Activities appear greyed out (blurred/hidden) when your iPhone is locked**, this is caused by a privacy setting in iOS.



To show all Live Activities even when the phone is locked:

- Open the Apple **Settings** app.
- Tap `Face ID & Passcode`.
- Enter your passcode if prompted.
- Scroll down to the `Allow Access When Locked` section.
- Enable the toggle for **Live Activities**.

> **This setting affects all apps**
>
> Once enabled, Live Activities from Trio—and other apps—will be shown on the Lock Screen.

Live Screen Widget Style

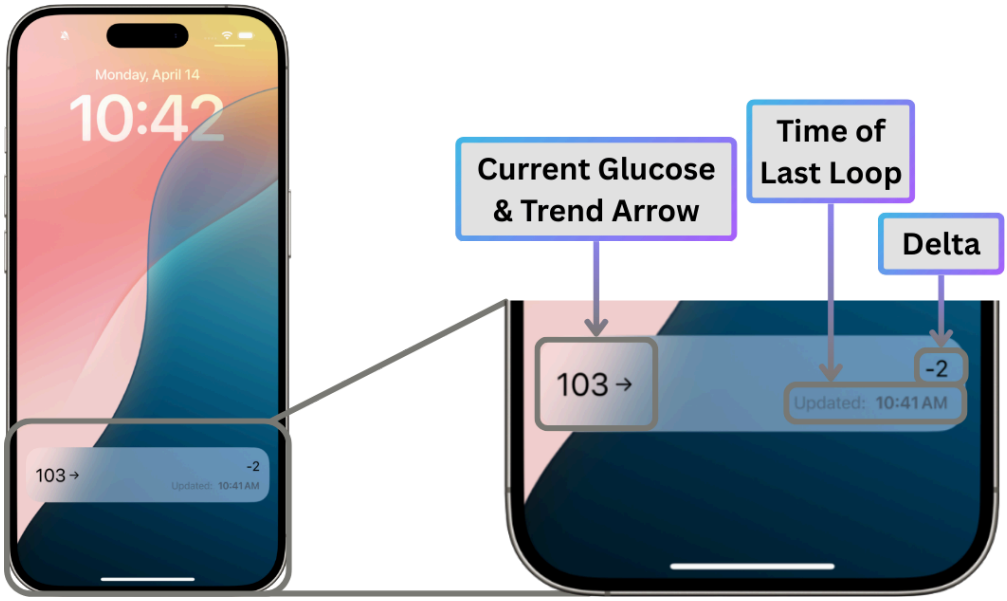
Default: *Simple*

Customize your Live Activity Widget by choosing between "Simple" or "Detailed"

You'll find more details on both "Simple" and "Detailed" styles below:

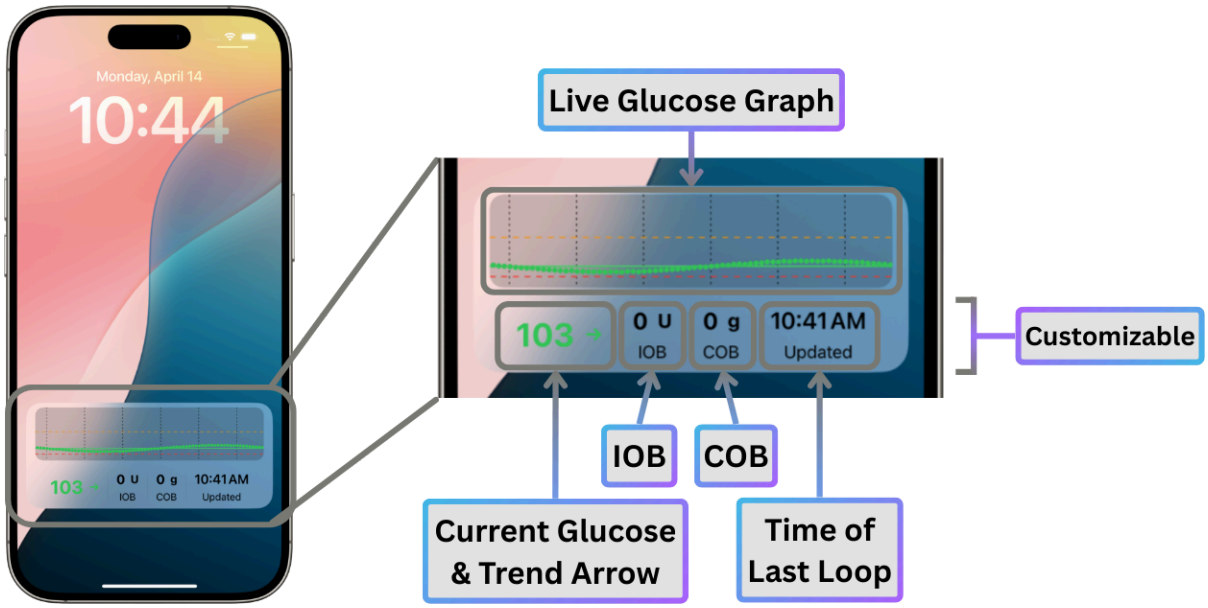
Simple

Trio's simple widget displays your current glucose reading, trend arrow, delta, and the timestamp of the current reading.



Detailed

The detailed widget displays your live glucose graph.



Calendar Events

Create Events in Calendar

Default: *OFF*

When enabled, Trio will create a customizable calendar event to keep you notified of your current glucose reading with every successful loop cycle.

After this setting is turned on, the available customizations will appear. You can customize with the calendar of your choosing, use of emoji labels, and the inclusion of IOB and COB data.

> **Info**
>
> Once a new calendar event is created, the previous event will be automatically deleted

> **Note**
>
> This is useful if you use CarPlay or a variety of other external services that limit the view of most apps, but allow the calendar app.

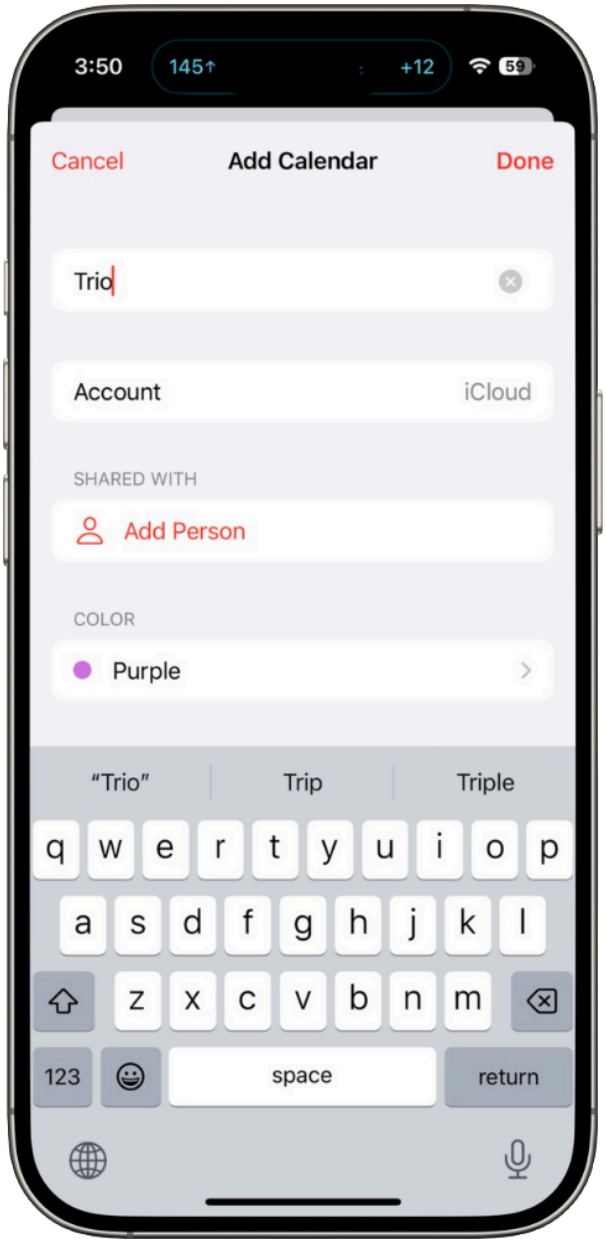
Choose Calendar

Default: *OFF*

Here you see all the available calendars as a drop down menu that you can use for Calendar Events.

It is recommended that you start a specific calendar to be used only for Trio Calendar Events.

To create a Trio calendar:
Step 1: In your phone's settings, open the calendar app. At the bottom, tap "Calendars" and then "Add Calendar" also at the bottom. Follow the prompts to create a calendar labeled "Trio". And tap "Done" to create a Trio calendar.



Step 2: Once you've created a Trio calendar, you can return to this setting in the Trio app and select the Trio calendar from the drop down menu.

Important

Selecting a calendar that you use for other events can result in invitations being auto-declined. It is best to have a separate calendar for only Trio Calendar Events to prevent any unintended conflicts.

Display Emojis as Labels

Default: *OFF*

When enabled, the calendar event created will indicate where the current glucose reading is in relation to your statistical target range.

🟠 = Above Range Glucose
🟢 = In Range Glucose
🔴 = Below Range Glucose

If Display IOB and COB is also enabled, the words "IOB" and "COB" will be replaced with emojis.

💉 = IOB
🥨 = COB

Display IOB and COB

Default: *OFF*

When enabled, Trio will include the current IOB and COB values below the current glucose reading with each calendar event created.

Services

Welcome to the Services homepage! This section provides you with additional information on the services built into the Trio app. There are also Companion Apps that will work alongside Trio.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

Nightscout ✏️

🚧Documentation Under Construction🚧 Connect your nightscout, import settings, backfill glucose, and more

Tidepool ✏️

🚧Documentation Under Construction🚧 Connect Tidepool

Apple Health ✏️

🚧Documentation Under Construction🚧 Connect Apple Health

Visualization and Monitoring ✏️

Highlights

- *Nightscout* is optional but encouraged
- *Nightscout* is a useful tool for data analysis and followers
- Create a *Nightscout* server and input your URL and `API_SECRET` in *Trio*.
  Test the connection and enable `Allow Uploading to Nightscout`.

Nightscout Introduction

Nightscout (NS) is an open-source project that allows real-time access to CGM data via a personal website, smartwatch viewers, or apps and widgets available for smartphones. Setting up a *Nightscout* web app is recommended to visualize your *Trio* closed loop.

Nightscout allows a user to upload CGM data from various sources to an online database and cloud computing service. The information is then processed and displayed visually as a graph. Plugins also allow more information to be shown about *Trio*.

Even if you don't choose to share your *Nightscout* site with another person, it will be helpful for you to visualize what the loop is doing and what it's been doing, plus generate helpful reports for understanding your data.

For help making a Nightscout account, please see the OpenAPS documentation.

Trio Setup

To enable your *Nightscout* connection:

- input your ***Nightscout* URL**, including the `https://` and your `API_SECRET`.
- Toggle on `Allow Uploading to Nightscout` so *Trio* can share its predictions and settings with *Nightscout*.

A Nightscout URL starts with httpS://

Your *Nightscout* URL must start with `https://`.
To set this up correctly, do not forget the letter `s` between `http` and `://`.

Tidepool ✏️

Apple Health ✏️

> **Info**
>
> Apple Health is optional

Enabling Apple Health allows Trio to write information on your blood sugar readings, insulin, and carbohydrates to Apple Health and read blood sugar readings in Apple Health. You will also need to give permission in iPhone Settings, Health, Data Access & Devices to write data to Apple Health. This is useful for those who like to analyze their data with TidePool, Cockpit, or Glooko.

Closing The Loop

Closed loop is turned off by default. This means Trio cannot make adjustments autonomously. Until you enable closed loop, the system relies on your approval of any and all actions it takes.

What is Closed Loop?

"Closed Loop" means Trio will use the settings and parameters you have inputted into the system to make automated insulin dosing decisions for you every 5 minutes.

> **Tip**
>
> Trio works best as a closed loop system. If you need to test your settings, it's best to close the loop and follow the configuration here.

What is Open Loop?

"Open Loop" means Trio cannot make any adjustments automatically. In open loop, the system relies solely on you to make any adjustments.

You can control your pump and bolus with the Trio app, but nothing can be done without your approval. This is often referred to as running in open loop. You will get no Trio-initiated protection from lows or highs while in open loop. Trio essentially becomes a fancy non-AID PDM with a bolus calculator.

Want to stay in Open Loop?

Many users feel the need to stay in open loop initially for a variety of reasons, such as testing settings or weariness about a new algorithm. It is not advised to keep the loop open with Trio due to the significant loss in functionality. Rather, configure the following settings as such:

- Set Max IOB to **0**
- Set Autosens Max to **100%**
- Set Autosens Min to **100%**
- All other Algorithm Settings should remain at their defaults if you are a new user

> **Existing Users**
>
> If you are an existing user making these adjustments for testing your settings, Algorithm Settings can remain as you had them previously configured. Keep in mind: you may need to adjust them after your settings test is completed if there are significant changes made to your Core Trio Settings (basal, ISF, CR).

This will allow Trio to continue communications with your pump and CGM uninterrupted. Setting Max IOB this way will allow Trio to prevent lows by reducing basal, but will not allow Trio to treat highs automatically. Setting Autosens Max and Min this way will stop all algorithm sensitivity adjustments, and Trio will only utilize your profile settings as

you entered them. This will significantly hinder the algorithm, turning it into a non-dynamic system with only low-prevention. This is much safer than attempting to run in open loop.

Former Loop Users

Loop and Trio function differently in open loop due to the differences in the underlying algorithms. It is rarely advised to stay in open loop with Trio, as most of the functionality of the algorithm depends on a closed loop system.

Tip

Remember that everyone's diabetes is different, and there is no one-size-fits-all, so take the time to optimize your settings for the Oref algorithm.

Help

Welcome to the help homepage! Here you will find a variety of resources that will help in your use of Trio.

Please use the navigation menu ☰, search bar 🔍, or the menu below to find the section you are looking for.

FAQs

Frequently Asked Questions

Safety ✏️

🚧 Documentation Under Construction 🚧

Glossary

A compilation of Trio terms and definitions

Troubleshooting ✏️

🚧 Documentation Under Construction 🚧

Share Logs ✏️

Export Trio's application logs through the iOS share sheet for quick support.

- Submit an Issue

Translate ✏️

🚧 Documentation Under Construction 🚧
Documentation in other languages

Support Communities

Discord
Facebook

Frequently Asked Questions

Various resources, including Facebook, and Discord, are available to assist you with setting up, maintaining, and succeeding in your Trio journey. Below is a series of questions we see asked daily. We hope it helps.

Trio FAQs

What is Trio?

Trio is an open-source artificial pancreas system based on the OpenAPS (Oref0) algorithm. It uses your entered settings, carbohydrates, and historical data to automate insulin delivery and reduce the time spent managing your diabetes.

!!! important "Important" *Trio is a DIY, open-source community project. As such, the user must build and maintain their app with the support of the Trio community. Trio has not undergone evaluation by any regulatory agency and is for use at your own risk.*

Trio History

The project started with Ivan Valkou's FreeAPS X implementation of the OpenAPS algorithm for iPhone and was later forked and rebranded as iAPS. Since then, many developers have contributed substantial work, leading to new features and enhancements.

There were differing views on development, open source, and peer review that led to the separation from the iAPS repository and the birth of Trio as a distinct entity. This transition marks the project's new phase, symbolizing its evolution and the collaborative development's dynamic nature.

Trio continues to leverage a variety of frameworks from the DIY looping community and remains at the forefront of DIY diabetes management solutions, constantly evolving with valuable contributions from its community.

Can I download Trio from the App Store?

Trio is not available to download from the App Store. Trio is an open-source, do-it-yourself automated insulin delivery application. Community members who wish to utilize Trio must build it themselves.

While building your application may initially seem daunting, information and support are abundant and available to help you progress down this pathway. Each step required to build Trio can be found within this documentation. Additional support can be found on the Trio Facebook group and Discord group.

Do I have to be "tech-smart" to build Trio?

No. You do not need any experience in code or computers to build Trio. If you own a computer or tablet and an iPhone, you already have the required experience. Beyond that, read the directions slowly and diligently. All the information you will need is in these documents.

Often, the non-tech people do better than the tech people in building Trio. Why? Because the non-tech people take the time to read slowly and look at the screenshots in the directions. The tech people often skim and miss steps,

which then leads to build errors that have to be retraced and fixed.

How long does it take to build Trio?

The answer to this question is very user-dependent. If you are new to building and need to immerse yourself in the documentation, then a few hours. However, some users like to break this down into smaller, more manageable chunks and complete the task over 24 hours. If you are a repeat builder familiar with the documentation, build, and update process, then 10 - 15 minutes. There are two options for building at present:

- The Browser Build method using GitHub.
- The Mac Build method using Xcode.

Do I need to own an Apple computer?

No, you do not need to own an Apple computer. The "Browser Build" method can be completed on any computer with access to the Internet and GitHub. However, if you chose the "Mac Build" method, while you do not necessarily need to own an Apple computer, you must at least borrow one.

How often will I need to access the computer to maintain Trio?

When you use the "Browser Build" method, you must access a browser at least once every 90 days to update with the Browser. This is simple enough to do the steps on your phone in just a few minutes.

When you use Build with Mac: 1. When you first build. 2. Once per year, at the minimum, after that. However, if you decide to use a free Apple Developer account, you must rebuild every 7 days.

Important

Periodically, developers will release updates to Trio. These updates may include new features or important safety updates. It is recommended that as updates are released, users update their Trio app.

Do I have to rebuild Trio if I change insulin pumps or CGMs?

No. Trio lets you move between different pumps and CGM configurations within the app.

I am moving from another closed-loop system. Will my settings transfer over?

If you've been using other closed-loop solutions, it's essential to understand the differences between the one you're used to and Trio. Before making the transition, we recommend that you take the time to read through the documentation to build a basic understanding of Trio. Detailed information on transitioning from other closed-loop systems can be found here

Depending on which system you are transitioning from, whether DIY or commercial, some settings may directly transfer across, and others may not. Before commencing on Trio, it is crucial that users test and refine their basals, carb ratios, and insulin sensitivity factors... Thats right... Take a minute and read that again... Before commencing on Trio, it is crucial that users test and refine their basals, carb ratios, and insulin sensitivity factors.

Why? Trio is only as good as the information you give it. Basal, carb ratios, and insulin sensitivity factors form the basis for all computing by Trio's algorithms. If you have given Trio poor-quality information, you will see poor-quality output. This will likely present as unexpected behavior to the everyday user and result in a poor user experience.

> **Important**
>
> *Before commencing on Trio, users must test and refine their basals, carb ratios, and insulin sensitivity factors.*

What about other pumps? When can I use them with Trio?

Cracking of pumps for use with the Trio app is a significant undertaking. At the same time, developers are constantly monitoring pumps that could be integrated into Trio. Ultimately, cracking and creating drivers for these pumps requires multiple developers to devote significant amounts of their time. Realistically, it can take upwards of 12 months for new pump integration. Information on supported pumps can be found under Compatible Pumps.

I regularly go places without phone service or an internet connection. Will Trio work without these?

Yes. Trio does not require internet or cell coverage to work. So long as the Trio user has Bluetooth enabled on the iPhone, the CGM and insulin pump will continue working in conjunction with Trio.

> **Warning**
>
> *If you have chosen a CGM source that requires the internet (i.e., Nightscout or Dexcom Share), users must maintain an internet connection. If you enter an area without internet, you will lose your CGM readings and Trio will stop looping.*

Can I have someone follow my Trio app?

There are multiple options available for someone to follow your loop data.

Nightscout
Nightscout is an open-source cloud application used by people with diabetes and parents of kids with diabetes to visualize, store, and share the data from their Continuous Glucose Monitors in real-time. Once set up, Nightscout acts as a central repository of blood glucose and insulin dosing/treatment data for a single person, allowing you to view the CGM graph and treatment data anywhere using just a Web browser connected to the internet.

Details of Trio's **integration** with Nightscout can be found here.
Information on **creating** your Nightscout **site** can be found here.

LoopFollow
LoopFollow is an open-source, do-it-yourself iPhone app that combines your T1D's display information and alerts into a single, easy-to-use interface. LoopFollow pulls data from your Nightscout or Dexcom Share server. You can set up more than one app on a phone if you follow more than one person with diabetes. Information on creating and using your LoopFollow app can be found LoopFollowDocs.

> **Warning**
>
> *All options for following your Trio app rely on an active internet connection. If you or the person you are following loses connection to the internet, your LoopFollow app will stop receiving data.*

Algorithm FAQs

What algorithm does Trio use?

Trio is an open-source artificial pancreas system based on the OpenAPS (Oref0) algorithm.

Where can I learn more about the Oref algorithm?

The first place to start if you want to learn more about the underlying algorithm is the extensive OpenAPS documentation. OpenAPS is the underlying algorithm Trio implements, with minor modifications mostly pertaining to the Dynamic settings.

Why does the `Recommended Bolus Percentage` default to 80%?

`Recommended Bolus Percentage` is a safety feature built into Trio. By default, Trio first calculates an "insulin recommended" value when bolusing for carbs, which is the total dosage. That dosage is multiplied by your `Recommended Bolus Percentage` to display your suggested insulin dose. Trio then delivers the remaining insulin via Temp Basals and/or SMBs as the blood sugar rises.

`Recommended Bolus Percentage` is a setting that allows the user to alter the amount initially delivered. By default 80% of the required meal bolus is delivered before the meal. You can increase or decrease this to alter the insulin delivered before the meal.

How does the `Max IOB` limit impact Trio's insulin dosing?

The maximum amount of insulin on board (i.e., in the body). Including insulin from all sources (basal and bolus) automatically delivered by Trio.

The default is zero, meaning Trio can only set temporary basal rates lower than your profile basal rate. It cannot set temporary basal rates that exceed your profile basal rate in cases of high blood sugar, and it cannot use super micro boluses to control blood sugar.

You will need to increase this value to allow Trio to start automating your insulin delivery. If you are nervous, you can start by setting this value to your average meal bolus. Then, over time, you can begin to increase the value.

> **Tip**
>
> *The default recommendation is "average meal bolus + 3 x max hourly basal" when using super micro boluses.*

How does the `Max COB` limit impact Trio's algorithm?

The maximum amount of carbs that Trio is allowed to bolus or set high-temp targets for. This safety feature protects against erroneous carbohydrate entries that could lead to hypoglycemia episodes.

What is `Autosens`? How do `Autosens Max` and `Autosens Min` impact Trio's decision-making?

Autosens allows the system to analyze historical data on the go and make adjustments if it recognizes that you are reacting more sensitively or resistant to insulin than usual. Autosens will then make temporary adjustments to the basal, ISF, and targets used for calculating temp basals in order to keep BG closer to your configured target.

`Autosens Max` and `Autosens Min` are the safety limiters the algorithm places on Autosens. `Autosens Max` determines the maximum ratio autosens can use for its adjustments. Increasing this value allows autosens to increase insulin through your basal profile, ISF, and target blood glucose. Conversely, `Autosens Min` determines the minimum ratio autosens can use for adjustments. Decreasing this value allows autosens to reduce insulin given through your basal profile, ISF, and target blood glucose.

Various blog posts have been written over the years regarding Autosens, and links to some can be found below:

- Dana Lewis, Autosensitivity.
- Tim Street, Understanding Autosens.

What are SMBs?

Super micro bolus, or SMB, is designed to give you reasonable amounts of bolus insulin upfront and use reduced temporary basal rates to balance out the peak insulin timing safely. Why? The reason for using SMB over a higher temporary basal rate is that it allows insulin to be delivered sooner. Thus, avoiding glucose spikes by front-loading insulin delivery.

Various blog posts have been written over the years regarding SMBs, and links to some can be found below:

- Dana Lewis, Introducing oref1 and super-microboluses
- Tim Street, Understanding SMB and oref1.

How does `Max SMB Basal Minutes` limit SMBs?

`Max SMB Basal Minutes` is one of the major limits on how much insulin a single SMB can deliver. The amount of insulin an SMB can deliver is related to the amount of scheduled basal insulin in your "Basal Profile" settings.

Example: Bill is using Trio and has SMBs turned on in the settings. Bill's current basal rate is 1 unit per hour. Bill has `Max SMB Basal minutes` set to the default of 30. Therefore, a single SMB can deliver no more insulin than would be delivered by your basal in 30 minutes, i.e., 0.5 units.

What is UAM?

UAM is a way for the algorithm to recognize that either there is carb absorption or an unaccounted-for increase in glucose levels as a result of more carbs being onboard than entered or as a result of hormonal variation and then

dose with it effectively.

To understand the logic behind UAM, have a read of Time Streets blog post Understanding SMB and oref1.

How does `Max UAM SMB Basal Minutes` limit SMBs?

`Max UAM SMB Basal Minutes` limits the size of SMBs that Trio can deliver when it detects an unannounced meal (UAM). The amount of insulin a UAM SMB can deliver is related to the amount of scheduled basal insulin in your "Basal Profile" settings. You can configure this setting to make UAM more or less aggressive in correcting meal spikes.

Example: Bill has SMBs and UAM turned on in the settings. The algorithm has detected unexpected carb absorption and would like to deliver an SMB in response. Bill's current basal rate is 1 unit per hour. Bill has `Max UAM SMB Basal minutes` set to the default of 30. Therefore, a single SMB can deliver no more insulin than would be delivered by your basal in 30 minutes, i.e., 0.5 units.

I've turned on SMBs, but every time I receive an SMB, Trio cuts back or zero-temps my basal. Is that normal?

Yes. SMBs effectively borrow insulin from the future to front load insulin delivery to assist in avoiding glucose spikes. Long, low, or zero temporary basal rates are expected to be observed with SMBs activated. This allows the algorithm to balance out the peak insulin timing and avoid hypoglycemia secondary to the increased insulin on board.

How do I make SMBs more responsive?

SMBs are limited by your Max `Max SMB Basal Minutes` and `Max UAM SMB Basal Minutes`. The default for both of these settings is 30 minutes. This default is conservative for most users, and it is suggested that this value be increased while monitoring the effect gradually. Raising this value will allow for bigger SMBs, thus making the dosing stronger. You need to decide how much you want to limit (or not limit) your SMBs.

Why isn't Trio looping when CGM glucose is 40 mg/dL (2.2 mmol/L) or lower?

Trio can’t determine the glucose value when under 40mg/dL (2.2 mmol/L). This is why the CGM apps from Dexcom and Libre are displaying “Low.” A “Low” reading is also sometimes due to a sensor error or a faulty sensor needing replacement. Trio can't safely make a glucose prediction without knowing the glucose value or if it’s rising or falling.

Why isn't Trio looping or delivering insulin when CGM glucose is 400 mg/dL (22.2 mmol/L) or higher?

Trio can’t determine the glucose value when over 400mg/dL (22.2 mmol/L). This is why the CGM apps from Dexcom and Libre display “High.” A “High” reading is also sometimes due to a sensor error or a faulty sensor needing replacement. Trio can't safely make a glucose prediction without knowing the glucose value or if it’s rising or falling.

Omnipod FAQs

What pods work with Trio?

All 3 pod types are compatible with Trio. You can use Eros, DASH, or Omnipod 5 pods. Note that Omnipod 5 requires building a feature branch at this time.

Do I still need a PDM/Controller with Trio?

You do not need the Omnipod Personal Diabetes Manager (PDM) or Omnipod 5 Controller; you need the pod supplies. They will pair with only one device at a time. So, a pod is either paired with a PDM/Controller or your Trio app on your iPhone.

In other words, your PDM/Controller can stay in the diabetes closet while you are Looping. You cannot use the PDM/Controller for a pod activated with the Trio app. That doesn't mean you should get rid of your PDM/Controller if you have one. Instead, keep it for backup situations if you lose your phone.

Does Trio cause an increase in pod failures?

There is more communication between your phone and the pod than the PDM/Controller, which increases the load on the pod's battery. Most people have no increase in pod failures. However, it does occasionally happen.

CGM FAQs

What CGMs are supported by Trio?

The following CGM can be used with Trio:

- Dexcom G5
- Dexcom G6 / ONE
- Dexcom G7 / ONE+
- FreeStyle Libre 1 / 2 / 2 Plus (limited Models)
    - US Libre 1 10-day sensors via Transmitters
    - US Libre 1 14-day sensors via Transmitters
    - International Libre 1 sensors via Transmitters
    - European Libre 2 / 2 Plus sensors via Transmitters
    - European Libre 2 / 2 Plus sensors directly via Bluetooth
- Freestyle Libre 3 / 3 Plus, all models, directly via Bluetooth (open beta, `dev` branch only)
- Eversense E3 / 365 (open beta, `dev` branch only)
- Accu-Chek SmartGuide (open beta, `dev` branch only)

What CGM apps can be used with Trio?

xDrip4iOS
To use xDrip4iOS as a CGM source, you must build it yourself with the same Apple Developer account you used to build your Trio app. Please see the following for instructions on how to build xDrip4iOS yourself here.

> **Tip**
>
> *If you use xDrip4iOS with the Dexcom G6/ONE, in Trio select* `Dexcom G6` *as the CGM source. Trio will intercept glucose readings directly from the transmitter.*

Nightscout

Yes. Nightscout can be used as a remote CGM source in Trio. However, this option should be avoided because it does not keep Trio running in the background like other CGM options. You will have to open Trio manually to make it run loop cycles.

> **Credit**
>
> Much of the information in this FAQ has been adapted from Trio Docs, Loop Docs, and OpenAPS Docs, and I would like to acknowledge their contribution to this page's success.

Safety ✏️

Glossary ✏️

Each item in the glossary is also a Tooltip. The word or phrase is repeated in parentheses to assist those using Google Translate.

When Google Translate is selected:

- Tooltips, indicated by dashed underline, continue to be shown in English
- On the Glossary page
  - The first instance (in bold) remains in English
  - The second instance (in parentheses) is translated to your selected language

API (API): Application Programming Interface

APN (APN): Apple Push Notification service

App Group (App Group): a unique identifier that Apple users for a given app, yours has your TEAMID embedded in it

BLE (BLE): Bluetooth low energy, used for communication by phones, CGM and some pumps

branch (branch): version of code within a single repository or workspace repository

CAGE (CAGE): cannula (or pump site) age on Nightscout site

Certificate (Certificate): Apple certificate is used to sign your iOS or Mac apps - tied to but different from your permanent Developer ID

CGM (CGM): continuous glucose monitor, wearable medical device that measures and reports glucose in interstitial fluid

clone (clone): create a copy of a repository on your computer including revision history and ability to update using git commands

COB (COB): Carbs on Board, affects automated insulin delivery

commit (commit): a formal change to files in a repository; each commit has an alphanumeric identifier (SHA-1)

Correction Factor (Correction Factor): how many points your blood sugar will drop for each unit of insulin

CR (CR): Carb Ratio; how many grams of carbs are covered by one unit of rapid-acting insulin

DIA (DIA): Duration of Insulin Action, the full time insulin is active including a long, low-level tail

DIY (DIY): Do it yourself, a common acronym for the open-source software community (and the maker community)

fork (fork): a copy of code in a github repository other than the original

GIF (GIF): Graphics Interchange Format (GIF) can be used for small animations and low-resolution video clips

git (git): a tool for version control

GitHub (GitHub): an online service for storing repositories, accessible from a browser

github.com (github.com): an online service for storing repositories, accessible from a browser

Glucose Chart (Glucose Chart): Display of measured and predicted glucose values

GMT (GMT): Greenwich Mean Time is mean (average) solar time at 0 degrees longitude, see UTC

Guardrails (Guardrails): limits in the code for user selected settings, recommended and absolute limits are provided

Hamburger Menu (Hamburger Menu): three parallel lines that, when tapped, open a new menu

Identifiers (Identifiers): names of modules found on your Apple Developer Identifiers page that are required for GitHub build method

IOB (IOB): Insulin on Board, affects automated insulin delivery: the current active insulin (above or below the basal rate)

iOS (iOS): operating system used by Apple Mobile devices (iPhone, iPod, iPad)

ISF (ISF): Insulin Sensitivity Factor; how many points your blood sugar will drop for each unit of insulin; sometimes called Correction Factor

Issue (Issue): On github - a formal method to report a problem, either code behavior or documentation

JSON (JSON): JavaScript Object Notation; a standard data interchange format that is text-based and human readable

macOS (macOS): operating system for Mac computer

Loop (Loop): With a capital L, Loop is one of several do-it-yourself artifical pancreas systems

Match-Secrets (Match-Secrets): a private repository you must create in your github account, stores keys required to build with github Build Actions

MTB (MTB): Manual Temporary Basal: user initiated temporary basal, Omnipod Common feature

MDT (MDT): common abbreviation for Medtronic pumps

modal (modal): message or alert appearing in front of app that must be acknowledged to return to app

NFC (NFC): Near-Field Communication is used for scanning devices such as Libre sensors

Nightscout (Nightscout): a personal website used to view your glucose and diabetes management data.

Omnipod (Omnipod): Insulet tubeless insulin pump

PR (PR): Pull Request - a formal method to request changes to a repository

Provisioning Profile (Provisioning Profile): associates your app with your Developer ID and limits app lifetime to 1 year (paid) or 1 week (free)

Pull Request (Pull Request): formal method to request changes to a repository

QR (QR): a machine-readable code consisting of an array of black and white squares

repository (repository): contains project files and each file's revision history

TEAMID (TEAMID): One of 6 Secrets: Apple Developer account member number

FASTLANE_ISSUER_ID (FASTLANE_ISSUER_ID): One of 6 Secrets: the issuer ID is associated with your Apple Developer ID and never changes

FASTLANE_KEY_ID (FASTLANE_KEY_ID): One of 6 Secrets: Key ID provided when you create an API key in App Store Connect; it is associated with the FASTLANE_KEY

FASTLANE_KEY (FASTLANE_KEY): One of 6 Secrets: Really long key (several lines); it and FASTLANE_KEY_ID are generated together

GH_PAT (GH_PAT): One of 6 Secrets: Generated with your GitHub account; set it to never expire

MATCH_PASSWORD (MATCH_PASSWORD): One of 6 Secrets: Git repo encryption passphrase, used to encrypt the Match-Secrets repository when it is created. You must save and use the same MATCH_PASSWORD for every repository if you build multiple apps.

SHA-1 (SHA-1): Secure Hash Algorithm 1; used to generate an alphanumeric code for commits in git (github)

Terminal (Terminal): interface for entering commands to the computer

TestFlight (TestFlight): a method to distribute apps without direct connection

Tooltip (Tooltip): brief definitions provided for important terms and abbreviations on the website

URL (URL): website address (Uniform Resource Locator)

watchOS (watchOS): Apple watch operating system; must be compatible with phone iOS

workflow (workflow): a set of instructions to GitHub to perform an action; the instruction files are found in the .github/workflows folder of the repository

Xcode (Xcode): program used to build an app

Troubleshooting Common Issues ✏️

Adapted from looptips

Site Change and Site Failures

Tips

- Pump site changes can temporarily impact control.
- Pump site failures can negatively impact control for hours.
  Turn off closed loop when failure is discovered, and only turn it back on when a new pump is installed.

Site Change

Pump site changes can negatively impact your immediate glycemic control. Different delivery sites' ability to absorb insulin varies, impacting your perceived insulin sensitivity. Trio should be able to accommodate this change by resetting its Autosens measurements (assuming "Rewind Resets Autosens" is on) and adapting as needed.

Site Failure

Pump failures can result from kinks or improper insertion of the cannula. As a result of a pump failure, Trio may assume that more insulin is on board the patient than what was delivered, leading to sustained highs even after the pump is replaced.

When you find out your pump has failed, immediately open loop to prevent Trio from trying to deliver more insulin. Upon your next pump installation, you can then close loop and be patient. Trio will wait for the calculated insulin on board to drop before delivering insulin.

Low Treatment

Tips

- Enter carbs for treatment into Trio, but do not bolus for it
- Consider reducing your profile ISF or Adjustment Factor if Dynamic ISF is enabled

Low blood sugar events can happen from time to time, but you will typically not require as many carbs to treat your lows as opposed to MDI or using a pump. When you treat a low, you should enter the carbs consumed into the Trio system so its autotune functionality can make better choices on your ICR, ISF, and basal rates.

If you experience a roller coaster pattern of highs and lows after treatment, your insulin sensitivity factor (ISF) is most likely to blame. Consider making your insulin sensitivity less aggressive by 10% or reducing your adjustment factor by 0.1 increments.

Rebound Lows

Tips

- Treating lows with fast carbs can give a fast BG rise
- Under certain circumstances, Trio might react to the fast BG rise by giving insulin, leading to another low

To avoid a second low, you can instruct Trio not to give any SMBs for a while after a low, giving your BG time to stabilize. Set up a Temporary Target preset that is a bit higher than your usual target (e.g., 7,5 mmol/L | 136 mg/dl). Ensure that Allow SMB With High Temptarget is disabled. Make the temporary target last for a while (e.g., 30-45 min).

Warning

Setting a temporary target is not the same as disconnecting the pump. Trio will still be able to give you insulin during the temporary target, but only by adjusting basal rates, resulting in a less aggressive treatment.

Highs after meals

Tip

It is *normal* for BG to rise after a meal. This happens to everyone, with or without diabetes. The key is to avoid very high BG levels and bring them back into range as soon as possible.

Fast Rise

Scenario: *Your BG rises fast when eating and ends up too high.*

If your BG rises fast and ends up too high, it is usually because the carbs were absorbed before the insulin peaked. If we rule out site issues, illness, etc, the logical approach is to ensure you get enough insulin ahead of the carbs. You can achieve this by increasing your pre-bolus timing for the meal. This gives the insulin time to absorb before the carbs kick in. You can find more information on the mealtime strategy page.

Persistent High After Meals

Scenario: *Your BG gets high after a meal and stays high for a long time. Trio gives you some SMBs, but not enough to bring you back down.*

Trio calculates the amount of insulin needed to bring blood glucose back into range. Additional insulin is needed when blood glucose remains high after a meal, but Trio needs the correct settings to make those adjustments.

- The first adjustment should be to reduce your CR. Reducing your carb ratio will result in **more** insulin for the carbs entered.

Fast Rise, Then Low

Scenario: *Your BG rises and ends up too high after a meal, then falls and ends up too low.*

In this scenario, you are not getting enough insulin upfront to deal with the carb absorption, but the total amount of insulin you're getting is too much - causing a low. If this is a repeating scenario, you should look into **all of** the following:

1. If all the delivered insulin is from one bolus, you should consider adjusting your CR setting so that the bolus calculator gives you less insulin. You could also consider pre-bolusing, giving the insulin some time to absorb before you start eating.
2. If the delivered insulin is part bolus and part SMBs, you should consider adjusting your ISF setting so that the SMBs give you less insulin. It would help if you also considered pre-bolusing, giving the insulin more time to absorb before you start eating.
3. In both (1) and (2), if you use the dynamic ISF feature, you should consider lowering the Adjustment Factor and adjusting the Autosens max/min settings.

Please read through the chapters on Autosens and Dynamic settings.

Sharing Trio Logs

The "*Share Logs*" feature lets you export the app's logs (from today and yesterday) and opens the iOS share sheet so you can transmit them to support channels, developers, or storage services for troubleshooting and record-keeping.

> **Check existing solutions**
>
> When you run into an issue and want to share your logs, please take a moment to check if it's already been solved by browsing through Facebook or Discord first.

Follow the steps below to save and share the logs.

Export Logs

- From within the **Trio** app,
  tap `⚙️ Settings › Support & Communities › Share Logs`
- Use the standard **iOS share sheet** to send these files via Mail, cloud-storage apps, or any other destination available in the share sheet.

Once you've exported the logs, you can share them where needed.

Share Logs

Logs are most commonly shared in the following cases:

- **Community troubleshooting**:
  Post logs to the Trio `#bug-reports` channel on **Discord** or relevant **Facebook** groups when volunteers ask for them to help diagnose an issue.

  > **Overcoming file size limits: Compress Logs**
  >
  > If Discord or Facebook complains about the file size when you upload the logs, **compress them** into an archive before sending.

- **Developer bug reports**: When requested, attach logs to a **GitHub issue** so that developers can use them to reproduce, investigate, and resolve the problem.

Compressing Logs

Discord and *Facebook* will **reject** your **file upload** if it is **larger than 25 MB**.
In this case, you first need to compress the log files into an archive before sending.

The following sections describe how to do that on Windows and macOS.

Compress Logs on Windows

To compress the log files on *Windows*, you can use either a graphical application or the terminal.

Compress Logs on Windows with a Graphical Application

1. Launch the Windows **File Explorer**
2. **Select Log Files**
   - Navigate to the folder containing the log files.
   - Hold `^ Ctrl` and **click** `log.txt`.
   - Hold `^ Ctrl` and **click** `log_prev.txt`.
3. **Compression**
   - Right-click one selected file
   - Select `Send to` in the contextual menu.
     - Select `Compressed (zipped) folder`.
4. **Name Archive**
   - Type `triologs.zip` and press `Enter ⏎`.

Compress Logs on Windows in the Terminal

If you are not comfortable using the **command line in a terminal** on *Windows*, consider using a graphical application instead; otherwise, keep reading.

Choose your favorite Windows Shell below **PowerShell** or "**Command Prompt (CMD)**", then follow the instructions.

PowerShell

1. **Open PowerShell**:
   - Right-click the `Start menu`
   - Select `Windows PowerShell`.
2. **Navigate to Log Folder**:

   ```
   cd $HOME\Downloads
   ```

   Replace `$HOME/Downloads` with the folder where you downloaded the log files.
3. **Run Compression Command**:

   ```
   Compress-Archive -Path log.txt, log_prev.txt -DestinationPath triologs.zip
   ```

   where:
   - `-Path` is followed by the files to compress.
   - `-DestinationPath` is followed by the name of the ZIP archive file.

Compress Logs on macOS

To compress the log files on a *Mac*, you can use either a graphical application or the terminal.

Compress Logs on macOS with a Graphical Application

1. Launch the `Finder` application
2. **Select Log Files**:
   - Hold `⌘ Cmd` and **click** `log.txt`
   - Hold `⌘ Cmd` and **click** `log_prev.txt`.
3. **Compression** :
   - Right-click the blue highlighted lines.
   - Select `Compress` in the contextual menu.
4. **Name Archive**:
   - An `Archive.zip` file appears.
   - Rename it to `triologs.zip`.

Compress Logs on macOS in the Terminal

If you are not comfortable using the **command line in a terminal** on your Mac, consider using a graphical application instead; otherwise, keep reading:

1. **Launch Terminal**:
   - Press `⌘ Cmd` + `Space`
   - Type `Terminal`
   - Press `Enter ⏎`
2. **Navigate to Downloads Folder**:

   ```
   cd ~/Downloads
   ```

   Replace `~/Downloads` with the folder where you downloaded the log files.
3. **Create Archive**: To create the archive, you can use `zip` or `tar`.

   ```
   zip triologs.zip log.txt log_prev.txt
   ```

   If this method does not work because `zip` is not installed, you can install it like so:

   ```
   xcode-select --install
   ```

   You can also use the *tar gzip* **alternative** method to compress the log files:

   ```
   tar -czf triologs.tar.gz log.txt log_prev.txt
   ```

Translation ✏️

Language List